



QUALIFICATION PROGRAMME

Associate **Module 7**

Financial Management

FOURTH EDITION



Hong Kong Institute of
Certified Public Accountants
香港會計師公會



Hong Kong's
CPA Qualification
香港會計師專業資格



MODULE 7
**FINANCIAL
MANAGEMENT**

Qualification Programme

Fourth Edition



Hong Kong Institute of
Certified Public Accountants
香港會計師公會

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DIRECTOR'S MESSAGE

Congratulations on choosing the Qualification Programme ('QP') of the Hong Kong Institute of Certified Public Accountants ('HKICPA') as your pathway to becoming a CPA! You have joined thousands of others on this exciting and important journey to develop the knowledge, skills and perspectives you need to succeed in your career and becoming a valued member of the Institute.

The world is evolving rapidly, so too is the business environment. The Accounting profession faces a number of challenges and trends including technological enhancement, regulatory development, changing societal expectations and more.

Professional accountants are no longer left only to deal with numbers, but also to analyse and advise. We are also expected to be highly strategic, collaborative, and building trust by demonstrating relevance and value to many aspects of society.

The QP of the HKICPA aims at qualifying accountants with the agility needed to embrace the changing environment. You will grow and discover a plethora of relevant competencies through QP by completing training programmes, passing professional examinations and acquiring practical experience under an authorised employer or supervisor. In the longer term, we hope that you will succeed not only in accountancy but also in enhancing your employability and portability so that you will be able to help business and society move forward.

We are delighted to partner with you on your development journey.

The QP consists of three levels of designations:

- The Associate Level aims to build a solid foundation of technical accounting knowledge.
- The Professional Level aims to deepen your technical capabilities and develop core enabling competences in the workplace.
- The Capstone integrates your knowledge, skills and experiences to resolve business problems and emerge as a top tier accounting professional.

We have designed this Learning Pack to provide you with the valuable resources for your development on attaining your CPA designation under the QP. I trust you will be successful and enjoy your QP journey!

Should you require any assistance at any time, please feel free to contact us on (852) 2287 7228.

Kit Wong
Director of Education and Training
Hong Kong Institute of Certified Public Accountants



INTRODUCTION

Successfully preparing for a career in accounting is a significant undertaking. To better prepare you for this challenging profession, the Hong Kong Institute of Certified Public Accountants (HKICPA) provides a qualification programme (QP) comprising three progressive levels: A 10-module Associate Level, a four-module Professional Level including workshops and a Capstone that includes three-day workshops and a final examination.

The Associate Level of the QP comprises 10 modules. Each of these modules involves approximately 80 hours of self-study and a closed-book module examination (except for Module 10: Business and Company Law in which the examination is open-book).

- **Module 1:** Accounting
- **Module 2:** Management Accounting
- **Module 3:** Business Economics
- **Module 4:** Business Management
- **Module 5:** Information Management
- **Module 6:** Financial Accounting
- **Module 7:** Financial Management
- **Module 8:** Principles of Auditing
- **Module 9:** Principles of Taxation
- **Module 10:** Business and Company Law

While each of the Associate Level and Professional Level modules stands on its own, the modules are also arranged in a series of ‘verticals’ that map to the CPA competence blueprint. These verticals are designed to develop an area of knowledge, through two or three modules, from basic understanding to professional excellence.

The Financial Accounting and Reporting vertical runs from Module 1, through Module 6 to Professional Level Module 11: Financial Reporting. A second vertical, Management Accounting, runs from Module 2, via Module 7 to Professional Level Module 12: Business Finance. The third vertical, Audit and Assurance, develops from Module 8 to Professional Level Module 13: Business Assurance. The fourth vertical, Taxation, takes students from Module 9 to Professional Level Module 14: Taxation.

Each Associate Level module of the new QP requires students to sit a three-hour examination. Two exam sittings are held each year in June session and December session. Please refer to the Student Handbook for the examination structure and the cut-off rule on the examinable content.



HKICPA PROFICIENCY LEVELS AND TAXONOMY

The proficiency level indicated in the table below reflects the level at which the topics covered in particular learning outcome is tested. There are three levels of proficiency:

- Level 1 is the foundational level, covering the skills of knowledge and comprehension.
- Level 2 is the intermediate level, covering the skills of application and analysis.
- Level 3 is the advanced level, covering the skills of integration and evaluation.

You are expected to understand which skill is exercised based on the taxonomy verbs with which it is associated.

Please note that the list of taxonomy verbs below is for reference only and does not represent an exhaustive list.

LEVEL 1: FOUNDATION		
Skill	Verb	Definition
Knowledge The remembering of previously learned material (recall of facts)	<i>Define</i>	Give the accepted meaning of
	<i>Identify</i>	List or ascertain possibilities before analysis; Point to the essential part or parts
	<i>List</i>	Provide a concise summary of the relevant points, often in bullet point format
	<i>Outline</i>	Give the main facts about something
	<i>State</i>	Accurately articulate established principles, concepts, terms etc.
Comprehension Demonstrative understanding of facts and ideas by organising, comparing, translating, interpreting, giving descriptions and stating main ideas	<i>Describe</i>	Communicate the key features of something, present a detailed account of something focusing on depth of knowledge
	<i>Explain</i>	Make clear the details of something; or show the reason for, or underlying cause of, or the means by which something occurs
	<i>Illustrate</i>	Offer examples to show how something happens, that something happens, or make concrete a concept by giving examples
	<i>Interpret</i>	Make clear the meaning of something and its implications
	<i>Summarise</i>	Describe something concisely; bring together the main facts

LEVEL 2: INTERMEDIATE

Skill	Verb	Definition
Application Using new knowledge. Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way	<i>Account for / Demonstrate</i>	Give details of accounting entries to be made for in the context of financial reporting or justify (if used in a more general context); Demonstrate the accounting treatment by using a set of accounts
	<i>Apply</i>	Demonstrate knowledge, concepts or techniques; Use established methods / tools / procedures to resolve relatively straightforward scenarios or problems
	<i>Calculate / Compute</i>	Determine by computation or arrive at by mathematical means or processes
	<i>Prepare</i>	Follow established procedures / methods to create a report of financial information or commentary (e.g. using a proforma spreadsheet)
	<i>Solve</i>	To work out to a result or conclusion
	<i>Use</i>	Apply in a practical way
Analysis Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalisation	<i>Analyse</i>	To examine methodically by separating into parts and studying the interrelationships in order to discover essential features
	<i>Compare</i>	Critically consider two or more things, emphasising their similarities
	<i>Contrast</i>	Critically consider two or more things, emphasising their differences
	<i>Classify / Categorise</i>	Apply concepts to categorise information or groups into categories
	<i>Justify</i>	Explain the reason for recommendation made, or underlying cause of, based on an analysis of a range of available options
	<i>Prioritise / Determine</i>	Determine the order for dealing with a series of items or tasks according to their relative importance e.g. Determine the priorities / determine the level of importance

LEVEL 3: ADVANCED

Skill	Verb	Definition
Integration Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions	<i>Construct</i>	To form an idea, a process or procedure by bringing together various theoretical and conceptual elements
	<i>Design</i>	Develop a procedure/process or course of action based on selection of the optimum combination from a range of available options
	<i>Develop</i>	To bring something into existence that has not previously existed, or to reshape something from its initial position into something more refined; Use judgement to bring to a more advanced or effective state or to create a plan
	<i>Formulate</i>	Devise and put a plan into words
	<i>Integrate</i>	Combine one aspect of learning with another to form a holistic understanding of a process, procedure or course of action
	<i>Plan / Propose</i>	Formulate a detailed proposal for doing or achieving something
	<i>Produce</i>	Draw together similar or disparate items to form a report containing financial and/or non-financial information
Evaluation The ability to judge the value of material for a given purpose	<i>Advise</i>	Communicate appropriately the recommended course of action based on an analysis of specific circumstances in a manner suited to the recipient
	<i>Appraise</i>	Assess the value or quality of something; or to assess its performance
	<i>Consider</i>	Think carefully about something before making a decision; to look closely or attentively at something through a process involving critical thinking
	<i>Evaluate</i>	Assess and determine the value, importance or qualities of something, normally with reference to specific criteria and draw conclusions
	<i>Recommend</i>	Select the best course of action or choice; Advocate a particular outcome or course of action based on an analysis of a range of available options

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LEARNING OUTCOMES

Each module includes Principal Learning Outcomes and Supporting Learning Outcomes arranged along a series of proficiency levels.

Module 7

Syllabus area	Weighting (%)
Analyse the financial environment in which businesses operate	5–10
Apply financial risk framework and different methods of managing financial risks	5–10
Apply pricing strategies and decisions	5–15
Analyse the strategic management accounting framework	5–10
Produce financial analysis	15–30
Consider and apply information for effective performance measurement	5–15
Explain and analyse different sources of finance and capital structure	10–20
Apply appraisal techniques for capital investment projects	10–15
Produce financial forecasts and evaluate business plans	5–10

The syllabus weighting table indicates the relative weightings of the syllabus areas encompassed in this module. It serves as a guide to the percentage of study time spent on each syllabus area. In the long run, the marks allocation in the module examinations would conform to the weightings as shown above. The exact range of marks allocation in each module examination may deviate from the weightings for suitably robust questions to be set.

Learning Outcomes	Proficiency Level	Chapter where covered
PRINCIPAL LO1: ANALYSE THE FINANCIAL ENVIRONMENT IN WHICH BUSINESSES OPERATE		
LO1.01: Explain and analyse the financial environment in which a business entity operates	2	
1.01.01 Analyse the concept of market efficiency and the implications of the Efficient Market Hypothesis for issuers and investors	2	1
1.01.02 Describe the financial environment in which an entity operates	1	1
1.01.03 Describe the structure of Hong Kong equity and debt markets and its financial and banking system	1	1
1.01.04 Explain the function of financial markets and the role they play in the value-creation process	1	1
1.01.05 Explain the role of treasury management and describe its responsibilities and activities	1	1

Learning Outcomes	Proficiency Level	Chapter where covered
PRINCIPAL LO2: APPLY FINANCIAL RISK FRAMEWORK AND DIFFERENT METHODS OF MANAGING FINANCIAL RISKS		
LO2.01: Account for the nature and types of risk	2	
2.01.01 Illustrate financial risk with their respective measurements	1	16
2.01.02 Explain and compare different methods of managing key financial risks	2	16
PRINCIPAL LO3: APPLY PRICING STRATEGIES AND DECISIONS		
LO3.01: Calculate prices for products and services	2	
3.01.01 Explain the factors affecting pricing policies	1	12
3.01.02 Describe different pricing strategies: • All forms of cost-plus • Skimming • Penetration • Complementary product • Product-line • Volume discounting • Discrimination • Relevant cost	1	12
3.01.03 Explain the concept of cost of quality and environmental costs	1	12
3.01.04 Apply different approaches of pricing products and services considering external market factors and internal cost structures	2	12
3.01.05 Describe and explain possible pricing strategies for non-profit making entities	1	12
PRINCIPAL LO4: ANALYSE THE STRATEGIC MANAGEMENT ACCOUNTING FRAMEWORK		
LO4.01: Describe the key elements of a strategic management accounting framework	1	
4.01.01 Explain current issues in strategic management accounting	1	11
4.01.02 Explain the linkages between management accounting and the strategic management of an organisation	1	11
4.01.03 Describe the business planning processes including the development of corporate strategic plans	1	11
4.01.04 Describe strategic cost management techniques, including just-in-time (JIT) and total quality management (TQM) concepts, product life cycle costing, target costing, quality costing, theory of constraints and throughput analysis	1	11
LO4.02: Analyse proposed investment projects	2	
4.02.01 Calculate relevant cash flows and discount rates for investment appraisal purposes, taking into account inflation and tax	2	13
4.02.02 Apply various models to establish the value of a business	2	13

Learning Outcomes	Proficiency Level	Chapter where covered
4.02.03 Analyse a proposed investment project by applying appropriate investment appraisal techniques given the objectives and circumstances of an organisation	2	13
4.02.04 Analyse an appropriate investment strategy when capital is rationed	2	13
4.02.05 Explain other non-financial considerations for investment appraisal	1	13
4.02.06 Identify social responsibility, sustainability and environmental consequences of investment decisions and explain the impact on investment appraisal	1	13
LO4.03: Explain the results of a post-appraisal audit of projects	1	
4.03.01 Explain the role of a post-appraisal (or post-completion) audit in assessing the success of a project	1	14
4.03.02 Explain the steps in carrying out a post-appraisal audit of projects	1	14
4.03.03 Explain management behaviour in a post-appraisal audit of projects	1	14
4.03.04 Explain an entity's system for the post-appraisal audit of projects	1	14
PRINCIPAL LO5: PRODUCE FINANCIAL ANALYSIS		
LO5.01: Apply techniques of financial analysis	2	
5.01.01 Identify and calculate main stock exchange indicators	2	9
5.01.02 Analyse financial performance using accounting and finance ratios, and compare the result with other market players/competitors	2	9
LO5.02: Evaluate a business entity's financial strategy	3	
5.02.01 Evaluate the financial strategy of an entity and recommend financial actions that will add value to the entity	3	9
5.02.02 Evaluate the credit policy of company's suppliers and credit terms given to the company's customers to identify any substantial gap in between	3	9
LO5.03: Produce a forecast cash flow statement for a business entity	3	
5.03.01 Analyse the role of cash flow analysis in the evaluation of an entity's strategic and operational plans	2	9
5.03.02 Produce cash flow statements to analyse the financial strategy of an entity and its short- and medium-term financial requirements	3	9
5.03.03 Explain how an entity can use cash flow reporting systems to monitor and manage its financial strategy	1	9
LO5.04: Prepare profitability projections	2	
5.04.01 Explain how profitability projections can be used to examine the impact of an entity's financial strategy	1	9
5.04.02 Prepare projections of future profitability to evaluate an entity's financial strategy	2	9

Learning Outcomes	Proficiency Level	Chapter where covered
LO5.05: Analyse the liquidity and solvency positions for a business entity	2	
5.05.01 Analyse the impact of an entity's financial strategy on its liquidity and solvency through the use of ratios and other techniques	2	9
5.05.02 Describe the symptoms of and remedies for overtrading and excessive leverage	1	9
PRINCIPAL LO6: CONSIDER AND APPLY INFORMATION FOR EFFECTIVE PERFORMANCE MEASUREMENT		
LO6.01: Apply information for effective performance management and consider the relationship between non-financial measures and financial measures	3	
6.01.01 Illustrate the organisational structure affecting performance measurement e.g. cascade down KPIs from the top	1	15
6.01.02 Identify different types of responsibility centres, including cost centres, revenue centres, profit centres and investment centres	1	15
6.01.03 Explain how to evaluate the performance of responsibility centres	1	15
6.01.04 Prepare segment reports showing performance or profitability of different segments	2	15
6.01.05 Analyse how to apply economic value added (EVA) as a performance measure, its linkage to weighted average cost of capital (WACC) and its roles in capital management	2	15
6.01.06 Analyse the limitations of financial measures such as return on investments (ROI), residual income (RI) and EVA	2	15
6.01.07 Consider the relationship between non-financial measures and financial measures	3	15
6.01.08 State the advantages and disadvantages of decentralisation in a sizable organisation	1	15
PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE		
LO7.01: Explain and analyse the theories of major capital market models	2	
7.01.01 Explain the concept of time value of money involving simple interest and compound interest	1	3
7.01.02 Explain the different sources of finance (short-term and long-term) and methods of raising capital	1	2, 4, 5
7.01.03 Describe different types of bonds and their features	1	4
7.01.04 Calculate the value of a bond	2	4
7.01.05 Describe different types of equities with their features and different forms of dividends	1	5
7.01.06 Calculate the share prices based on different dividend growth patterns	2	5
7.01.07 Analyse the risk and return relationship for individual securities and a portfolio of securities, and apply the capital asset pricing model (CAPM) to the pricing of securities	2	6

Learning Outcomes	Proficiency Level	Chapter where covered
7.01.08 Explain and calculate the cost of equity, cost of debt and the weighted average cost of capital (WACC)	2	7
7.01.09 Explain the advantages and risks associated with external and internal financing	1	2
7.01.10 Explain and illustrate through calculations the theory of capital markets including: portfolio theory; capital asset pricing model; cost of capital; Modigliani Miller theories; and efficient market hypothesis	2	8
7.01.11 Explain, other than Modigliani Miller theories, various types of capital structure theory, including agency theory, trade-off theory, signaling theory and pecking order theory	1	8
7.01.12 Explain the key considerations for establishing a corporate capital structure	1	8
7.01.13 Describe the use of financial products, including derivatives, in relation to corporate financing and risk management	1	8
7.01.14 Explain the use of external experts for advisory on transaction services for fund raising	1	8
PRINCIPAL LO8: APPLY APPRAISAL TECHNIQUES FOR CAPITAL INVESTMENT PROJECTS		
LO8.01: Apply appropriate appraisal techniques to capital investment projects	2	
8.01.01 Apply different appraisal techniques to evaluate capital investment projects including payback period, discounted payback, accounting rate of return, net present value and internal rate of return	2	13
8.01.02 Describe the impact of taxation, inflation, risk and uncertainty in capital investment appraisal	1	13
8.01.03 Explain and compare the strengths and weaknesses of different project appraisal techniques	2	13
8.01.04 Apply the concept of incremental cash flows in capital investment appraisal	2	13
8.01.05 Calculate relevant discount rates for each capital investment projects, taking into account inflation and tax	2	13
PRINCIPAL LO9: PRODUCE FINANCIAL FORECASTS AND EVALUATE BUSINESS PLANS		
LO9.01: Formulate plans and forecasts for a business entity	3	
9.01.01 Formulate and evaluate cash flow and profit projections for a business	3	10
9.01.02 Formulate and evaluate business plans for an entity	3	10
9.01.03 Explain the role and composition of business plans in helping an entity achieve its objectives	1	10



STUDY TEXT KEY FEATURES

Each of the Associate Level and Professional Module texts include a series of pedagogical features designed to help QP candidates better absorb the material, reach the required proficiency levels and meet the outlined Learning Outcomes (LOs).

The aim of these features is to help students understand the content while regularly reinforcing concepts and building the skills necessary to successfully complete each of the modules and progress through the Associate Level, the Professional Level and the Capstone.

Each chapter includes these features:

- **Chapter topic list:** A succinct list of the specific topics covered in the chapter.
- **Learning outcomes:** Outlines the specific knowledge points covered in the chapter and the specific skills related to each learning outcome (LO) discussed in the chapter.
- **Opening case:** A case study that aims to relate the material covered in the chapter to a real-life situation. At times, this opening case may be linked to opening cases in other chapters.
- **Overview:** Provides a more detailed preview of the material covered in the chapter.
- **Exhibits and charts:** Through illustrations and examples, exhibits and charts aim to convey information in graphic fashion or actual examples of accounting, reporting or calculations that are likely to be used in actual practice.
- **Illustrative examples:** Case studies that explore specific issues related to the chapter topics and further understanding of the LOs.
- **Apply and analyse:** Exam questions with analysis provided to show how to approach answering the question and apply what was learned from the concepts presented in the chapter.
- **Ethics in practice:** Ethical discussions on issues that may arise during professional practice.
- **Key learning point:** A concise summary of a salient point that is key to achieving chapter LOs.
- **Knowledge check questions:** A set of questions geared at furthering students understanding of specific topics, work through problems and meet the chapter LOs.
- **Summary:** A list of the concepts and topics covered in the chapter in an easy-to-review format.
- **Mind map:** A graphic depiction of the knowledge conveyed in the chapter to facilitate understanding of the LOs.
- **List of formulas:** A compilation of the equations introduced in the chapter.
- **Exam practice questions:** Questions similar to those likely to be featured in the examination paper required for each QP module.



FORMULA TABLES

Chapter 3

Simple interest

$$\text{Interest} = p \times i \times n$$

where

- p = principal,
- i = interest rate, and
- n = time.

Compound interest

$$\text{Interest} = p(1 + i/n)^{nt}$$

where

- p = principal,
- i = interest rate,
- n = number of times interest is compounded, and
- t = number of years that apply.

Future value

$$\text{Future Value (FV)} = p \times (1 + i)^n$$

where

- FV = future value of a single amount,
- p = principal,
- i = interest rate for a period, and
- n = number of periods.

Present value

$$\text{Present Value (PV)} = \text{Future Value (FV)} / (1 + i)^n$$

Present value of ordinary annuity

$$PVA_n = \frac{CF}{i} \times \left[1 - \frac{1}{(1+i)^n} \right]$$

where

- PVA_n = PV of an n period annuity,
- CF = level and equally spaced cash flows,
- i = interest rate or discount rate, and
- n = number of periods.

Future value of ordinary annuity

$$FVA_n = CF \times \left[\frac{(1+i)^n - 1}{i} \right]$$

where

- FVA_n = FVA at the end of n periods,
- CF = cash flows that are constant and evenly spaced,
- i = interest rate, and
- n = time periods.

Present value of growing annuity

$$PV_0 = \frac{PMT_0(1+g)}{(1+k)^1} + \frac{PMT_0(1+g)^2}{(1+k)^2} + \frac{PMT_0(1+g)^3}{(1+k)^3} + \dots + \frac{PMT_0(1+g)^n}{(1+k)^n}$$

This equation can be reduced to the following expression:

$$PV_0 = \frac{PMT_1}{k-g} \times \left[1 - \left(\frac{1+g}{1+k} \right)^n \right]$$

where

- PV = present value,
- PMT = payment,
- k = effective annual rate,
- g = payout, and
- n = time period.

Present value of growing perpetuity

$$PV_0 = \frac{PMT_0(1+g)}{(1+k)^1} + \frac{PMT_0(1+g)^2}{(1+k)^2} + \frac{PMT_0(1+g)^3}{(1+k)^3} + \dots + \frac{PMT_0(1+g)^\infty}{(1+k)^\infty}$$

This equation can be reduced to the following expression:

$$PV_0 = \frac{PMT_0(1+g)}{k-g} = \frac{PMT_1}{k-g}$$

Effective annual rate (EAR)

$$EAR = (1 + i / m)^m - 1$$

where

- i = interest rate,
- m = number of compounding periods.

Chapter 4

Current yield

$$CY = \frac{\text{Annual interest}}{B}$$

where

- CY = Current yield,
- B = Market price.

Yield to maturity (YTM)

$$B = i \times \left[\frac{1 - \frac{1}{(1+k_b)^n}}{k_b} \right] + F \times \frac{1}{(1+k_b)^n}$$

where

- B = Market price,
- k_b = Discount rate or YTM,
- n = Number of periods to maturity,
- F = Amount to be paid at maturity, and
- i = Interest.

Bond valuation formula

$$P_B = \frac{C_1}{1+i} + \frac{C_2}{(1+i)^2} + \dots + \frac{C_n + F_n}{(1+i)^n}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i)^n} \right] + \frac{F_n}{(1+i)^n}$$

where

- P_B = price of the bond,
- C_t = coupon payment in period t ,
- F_n = par value at maturity,
- i = interest rate, and
- n = periods to maturity.

Semi-annual compounding formula

$$P_B = \frac{C/m}{1+i/m} + \frac{C/m}{(1+i/m)^2} + \dots + \frac{C/m + F_{mn}}{(1+i/m)^{mn}}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i/m)^{mn}} \right] + \frac{F_n}{(1+i/m)^{mn}}$$

where

- C = the annual coupon payment,
- F_{mn} = the amount of the cash payment at maturity (face value),
- m = number of coupon payments per year,
- n = years to maturity, and
- i = market interest rate.

Pricing of zero coupon bonds

$$P_B = \frac{F_{mn}}{\left(1 + \frac{i}{m} \right)^{mn}}$$

where

- P_B = price of a bond,
- F_{mn} = amount of payment at maturity,
- i = annual market interest rate,
- n = years until payment is due, and
- m = number of times interest is compounded per year.

Effective annual yield (EAY)

$$EAY = \left(1 + \frac{\text{Quoted interest rate}}{m} \right)^m - 1$$

where

- Quoted interest rate = simple annual yield (semi-annual yield × 2),
- m = the number of compounding periods per year.

Chapter 5

Present value of a stock

$$PV(\text{stock}) = PV(\text{dividend}) + PV(\text{sale price})$$

Single-period valuation

$$P_0 = \frac{D_1 + P_1}{1 + R}$$

where

- P_0 = current price of a stock,
- D_1 = dividend paid at the end of the period,
- P_1 = stock price at the end of the period,
- R = required return on common stock in a particular risk class.

Zero-growth valuation

$$P_0 = D / R$$

Where:

- P_0 = Current price of the share,
- D = Constant cash dividend in each period,
- R = Required return.

Constant-growth valuation

$$P_0 = \frac{D_1}{R - g}$$

Where:

- P_0 = Current price of share,
- D_1 = Dividend in the next period ($t = 1$),
- g_0 = Constant dividend growth rate,
- R = Required return.

Mixed-growth valuation

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \cdots + \frac{D_t}{(1+R)^t} + \frac{P_t}{(1+R)^t}$$

where:

- P_0 = Current price of share,
- D_t = Dividend in period t ,
- R = Required return.

Preference share with fixed maturity

$$P_0 = \frac{D/m}{1+i/m} + \frac{D/m}{(1+i/m)^2} + \frac{D/m}{(1+i/m)^3} + \cdots + \frac{D/m + P_{mn}}{(1+i/m)^{mn}}$$

where:

- D = Annual preference share dividend payment,
- P = Stated value of preference share,
- i = Yield to maturity,
- m = Number of times dividend payments are made per year,
- n = Number of years to maturity.

Preference share with no maturity

$$P_0 = \frac{D}{R}$$

where:

- P_0 = Current price of share,
- D = constant cash dividend, and
- R = interest rate or required rate of return.

Chapter 6

Capital appreciation component of a return

$$R_{CA} = \frac{\text{Capital Appreciation}}{\text{Initial Price}} = \frac{P_1 - P_0}{P_0} = \frac{\Delta P}{P_0}$$

Income component of a return

$$R_I = \frac{\text{Cash Flow}}{\text{Initial Price}} = \frac{CF_1}{P_0}$$

Total holding period return

$$R_T = R_{CA} + R_I = \frac{\Delta P}{P_0} + \frac{CF_1}{P_0} = \frac{\Delta P + CF_1}{P_0}$$

where:

- P_1 = Price at period 1
- P_0 = Price at period 0
- ΔP = Change in price
- CF_1 = Cash flow at period 1

Arithmetic mean (AM)

$$AM = \frac{\sum_{i=1}^n R_i}{n}$$

Geometric mean (GM)

$$GM = [(1+R_1)(1+R_2)(1+R_3)\dots(1+R_n)]^{\frac{1}{n}} - 1$$

where:

- R_i = Interest rate of i th period of time
- n = Number of periods

Total return

Total return = expected return + systematic portion + unsystematic portion

$$R = E(R) + m + \varepsilon$$

Expected return with different probability

$$E(R_{Asset}) = \sum_{i=1}^n (p_i \times R_i) = (p_1 \times R_1) + (p_2 \times R_2) + \dots + (p_n \times R_n)$$

Expected return with same probability

$$E(R_{Asset}) = \frac{\sum_{i=1}^n (R_i)}{n} = \frac{R_1 + R_2 + \dots + R_n}{n}$$

where:

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R_{Asset})$ = Expected return of an asset.

Variance with different probability

$$\text{Var}(R) = \sigma_R^2 = \sum_{i=1}^n \left\{ p_i \times [R_i - E(R)]^2 \right\}$$

where:

$\text{Var}(R)$ = variance of return of an asset,

σ_R^2 = variance of return of an asset,

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R)$ = Expected return of an asset.

Variance with same probability

$$\sigma_R^2 = \frac{\sum_{i=1}^n [R_i - E(R)]^2}{n}$$

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

Coefficient of variation

$$CV_{Asset} = \frac{\sigma_{R_{Asset}}}{E(R_{Asset})}$$

Modified coefficient of variation

$$CV_{Asset}^* = \frac{\sigma_{R_{Asset}}}{E(R_{Asset}) - R_{rf}}$$

Portfolio's expected return

$$E(R_p) = \sum_{i=1}^m w_i E(R_i)$$

where:

- $E(R_p)$ = Expected return on the portfolio,
- m = Number of assets in the portfolio,
- w_i = Proportion of the portfolio invested in asset i ,
- $E(R_i)$ = Expected return on asset i .

Variance of a two-asset portfolio

$$\sigma_{R_{2 Asset portfolio}}^2 = w_A^2 \sigma_{R_A}^2 + w_B^2 \sigma_{R_B}^2 + 2w_A w_B COV_{R_{A,B}}$$

where:

- $\sigma_{portfolio}$ = Variance of return of the portfolio,
- w_{R_A} = Weight of return of asset A in the portfolio,
- w_{R_B} = Weight of return of asset B in the portfolio.

Covariance of two securities

$$COV_{R_{AB}} = \sum_{i=1}^n (p_i \times [(R_{Ai} - E(R_A))] \times [R_{Bi} - E(R_B)])$$

where:

- p_i = Probability of event i ,
- R_{Ai} = Return of asset A under event i ,

- R_{Bi} = Return of asset B under event i ,
- $E(R_A)$ = Expected return of asset A, and
- $E(R_B)$ = Expected return of asset B.

Standard deviation of a two-asset portfolio

$$\sigma_{\text{portfolio}} = \sqrt{w_{R_A}^2 \sigma_{R_A}^2 + w_{R_B}^2 \sigma_{R_B}^2 + 2w_{R_A} w_{R_B} \text{COV}_{R_{AB}}}$$

where:

$\sigma_{\text{portfolio}}$ = Standard deviation of return of the portfolio

w_{R_A} = Weight of return of asset A in the portfolio

w_{R_B} = Weight of return of asset B in the portfolio

Correlation coefficient of two securities

$$\rho_{R_{AB}} = \frac{\text{COV}_{R_{AB}}}{\sigma_{R_A} \sigma_{R_B}}$$

Beta of asset

$$\beta_i = \frac{\rho_{R_{Mi}} \times \sigma_i}{\sigma_M} = \frac{\text{COV}_{R_{Mi}}}{\sigma_M^2}$$

Capital asset pricing model

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

where:

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

Portfolio's expected return under CAPM

$$E(R_{n \text{ Asset Portfolio}}) = R_{rf} + \beta_{n \text{ Asset Portfolio}} [E(R_m) - R_{rf}]$$

while

$$\begin{aligned} E(R_{n \text{ Asset Portfolio}}) &= \sum_{i=1}^n (x_i \times E(R_i)) \\ &= (x_1 \times E(R_1)) + (x_2 \times E(R_2)) + \dots + (x_n \times E(R_n)) \end{aligned}$$

$$\beta_{n \text{ Asset Portfolio}} = \sum_{i=1}^n x_i \beta_i = x_1 \beta_1 + x_2 \beta_2 + x_3 \beta_3 + \dots + x_n \beta_n$$

where:

- x_i = Weighting of asset i ,
- $E(R_i)$ = Expected return of asset i ,
- β_i = Beta of asset i .

Chapter 7

Dividend growth model (DGM)

$$K_{cs} = \frac{D_1}{P_0} + g$$

where:

- k_{cs} = Required returns on common stock,
- D_1 = Dividend paid in the next period ($t = 1$),
- P_0 = current price per share,
- g = Expected growth rate of dividends.

Capital asset pricing model (CAPM)

$$E(R_i) = R_{rf} + \beta_i (R_m - R_{rf})$$

where:

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

Pricing of preference shares

$$k_{ps} = \frac{D_{ps}}{P_{ps}}$$

Where

- D is dividend,
- P is current price,
- k is cost, or discount rate.

Weighted average cost of capital (WACC)

$$k_{\text{Firm}} = \sum_{i=1}^n x_i k_i = x_1 k_1 + x_2 k_2 + x_3 k_3 + \dots + x_n k_n$$

$$k_{\text{Firm}} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{Equity}} k_{\text{Equity}}$$

Where

- k = The cost of funds;
- x = The weight of the amount within the total amount.

Chapter 8

Firm's enterprise value

$$V_{\text{Firm}} = V_{\text{Assets}} = V_{\text{Debt}} + V_{\text{Equity}}$$

Weighted average cost of capital (WACC)

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt pretax}} (1 - t) + x_{\text{ps}} k_{\text{ps}} + x_{\text{cs}} k_{\text{cs}}$$

Weighted average cost of capital (WACC) – simplified (assuming no taxes and preference shares)

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{cs}} k_{\text{cs}}$$

Proportion of debt and equity in a firm's capital structure

$$x_{\text{Debt}} = \frac{V_{\text{Debt}}}{V_{\text{Debt}} + V_{\text{cs}}}$$

$$X_{cs} = \frac{V_{cs}}{V_{Debt} + V_{cs}}$$

Required return on common stock

$$k_{cs} = k_{Assets} + \left(\frac{V_{Debt}}{V_{cs}} \right) (k_{Assets} - k_{Debt})$$

Gains and losses from naked short position

$$\text{Profit (loss) from short position} = (F - S_T) \times n$$

Where

- F = Forward price
- S_T = Spot price
- n = number of contracts

Arbitrage pricing theory

$$ER_i = a_0 + b_{i1}F_1 + b_{i2}F_2 + \dots + b_{in}F_n$$

Where

- ER_i = expected return on security i ,
- a_0 = expected return on a security with zero systematic risk,
- b_i = sensitivity of security i to risk factor 1 to n ,
- F = risk premium for risk factor 1 to n .

Chapter 9

Price-to-book ratio

$$\text{Price-to-book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

Earnings per share ratio

$$\text{Earnings per share} = \frac{\text{Net income}}{\text{Shares outstanding}}$$

Price-earnings ratio

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

Current ratio

$$\text{Current ratio} = \frac{\text{Current assets (CA)}}{\text{Current liabilities (CL)}}$$

Quick ratio

$$\text{Quick ratio} = \frac{\text{Cash (C)} + \text{Marketable securities (MS)} + \text{Accounts receivable (AR)}}{\text{Current liabilities (CL)}}$$

Total debt ratio

$$\text{Total debt ratio} = \frac{\text{Total liabilities (TL)}}{\text{Total assets (TA)}}$$

Debt-to-equity ratio

$$\text{Debt-to-equity ratio} = \frac{\text{Total debt}}{\text{Total equity}}$$

Equity multiplier

$$\text{Equity multiplier} = \frac{\text{Total assets}}{\text{Total equity}}$$

Equity multiplier related to debt-to-equity ratio

$$\text{Equity multiplier} = 1 + \text{Debt-to-equity ratio}$$

Financial position identity

$$\text{Total assets} = \text{Total liabilities (debt)} + \text{Total shareholders equity}$$

Times-interest-earned ratio

$$\text{Times-interest-earned} = \text{Earnings before interest and tax (EBIT)} / \text{Interest}$$

Cash coverage ratio

$$\text{Cash coverage ratio} = \frac{\text{Earnings before interest and tax (EBIT) + Depreciation and/or Amortisation}}{\text{Interest}}$$

Receivables turnover

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable average / ending balance}}$$

Days' receivables

$$\text{Days' receivables} = \frac{365}{\text{Receivables turnover}}$$

Payables turnover ratio

$$\text{Payables turnover} = \frac{\text{Net credit purchases}}{\text{Account payable (average/ending balance)}}$$

Alternative formula of Payables turnover ratio

$$\text{Payables turnover} = \frac{\text{Cost of goods sold}}{\text{Account payable (average/ending balance)}}$$

Days' payables

$$\text{Days' payables} = \frac{365}{\text{Payables turnover}}$$

Inventory turnover ratio

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory average / ending balance}}$$

Days' inventory ratio

$$\text{Days' inventory ratio} = \frac{365}{\text{Inventory turnover}}$$

Fixed asset turnover ratio

$$\text{Fixed asset turnover} = \frac{\text{Sales}}{\text{Net fixed assets}}$$

Total asset turnover ratio

$$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$$

Gross profit margin

$$\text{Gross profit margin} = \frac{\text{Net sales} - \text{Cost of goods sold}}{\text{Net sales}}$$

Operating profit margin

$$\text{Operating profit margin} = \frac{\text{Earnings before interest and taxes (EBIT)}}{\text{Net sales}}$$

Net profit margin

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Net sales}}$$

EBIT return on assets

$$\text{EROA} = \frac{\text{EBIT}}{\text{Total assets}}$$

Return on assets

$$\text{ROA} = \frac{\text{Net income}}{\text{Total assets}}$$

Return on equity

$$\text{ROE} = \frac{\text{Net income}}{\text{Total equity}}$$

Average value

$$\text{Average asset or equity value} = \frac{\text{Beginning value} + \text{Ending value}}{2}$$

Book value per share

$$\text{Book value per share} = \frac{\text{Shareholders' equity (SE)}}{\text{Number of shares}}$$

Dividend yield

$$\text{Dividend yield} = \frac{\text{Dividend per share (DPS)}}{\text{Current share price (P)}}$$

Dividend payout ratio

$$\text{Dividend payout} = \frac{\text{Dividend per share (DPS)}}{\text{Earnings per share (EPS)}}$$

Price-earnings (P/E) ratio

$$\text{P/E ratio} = \frac{\text{Price per share (P)}}{\text{Earnings per share (EPS)}}$$

Forward P/E ratio

$$\text{Forward P/E ratio} = \frac{\text{Price per share (P)}}{\text{Estimated earnings per share (EEPS)}}$$

Price-to-book ratio

$$\text{Price-to-book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

EBITDA multiple

$$\text{EBITDA multiple} = \frac{\text{Total enterprise value (TEV)}}{\text{Earnings before interest, taxes, depreciation and amortisation (EBITDA)}}$$

Derived from the ROA equation

$$\begin{aligned} \text{ROA} &= \text{Net income/Total assets} \\ &= (\text{Net income/Net sales}) \times (\text{Net sales/Total assets}) \\ &= \text{Net profit margin} \times \text{Total asset turnover} \end{aligned}$$

Net profit margin ratio can be written as:

$$\text{Net profit margin} = \text{Net income/Net sales} = (\text{EBIT/Net sales}) \times (\text{EBT/EBIT}) \times (\text{Net income/EBT})$$

ROE equation, we start with the formula discussed earlier:

$$\begin{aligned}\text{ROE} &= \text{Net income/Total equity} \\ &= (\text{Net income/Total assets}) \times (\text{Total assets/Total equity}) \\ &= \text{ROA} \times \text{Equity multiplier}\end{aligned}$$

DuPont equation

$$\text{ROE} = \text{Net profit margin} \times \text{Total asset turnover} \times \text{Equity multiplier}$$

DuPont equation ratio

$$\text{ROE} = (\text{Net income/Net sales}) \times (\text{Net sales/Total assets}) \times (\text{Total assets/Total equity})$$

Operating cycle

$$\text{Operating cycle} = \text{Days' inventory} + \text{Days' receivables}$$

Cash conversion cycle (CCC)

$$\text{CCC} = \text{Days' inventory} + \text{Days' receivables} - \text{Days' payables}$$

Internal growth rate

With opening asset balance

$$\text{Internal growth rate} = \text{Return on asset (ROA)} \times \text{Retention rate } (b)$$

With closing asset balance

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - (\text{ROA} \times b)}$$

Sustainable growth rate

With opening equity balance

$$\text{Sustainable growth rate} = \text{Return on equity (ROE)} \times \text{Retention rate (} b \text{)}$$

With closing equity balance

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - (\text{ROE} \times b)}$$

Profitability index

$$\text{Profitability index} = \frac{\text{Present value of future cash flows}}{\text{Initial cost}}$$

Degree of financial leverage (DFL)

$$\text{DFL} = \frac{\text{Percentage change in earnings per share (EPS)}}{\text{Percentage change in EBIT}}$$

Degree of combined leverage (DCL)

$$\text{DCL} = \frac{\text{Percentage change in EPS}}{\text{Percentage change in sales}}$$

Degree of operating leverage (DOL)

$$\text{DOL} = \frac{\text{Percentage change in EBIT}}{\text{Percentage change in sales}}$$

Chapter 10

Free cash flows (FCF)

$$\text{FCF} = [(\text{Revenue} - \text{Op Ex} - \text{D \& A}) \times (1 - t)] + \text{D \& A} - \text{Cap Exp} - \text{Add WC}$$

Retention ratio

$$\text{Retention ratio} = \text{Addition to retained earnings} / \text{Net income}$$

Dividend payout ratio

$$\text{Dividend payout ratio} = \text{Cash dividends} / \text{Net income}$$

Retention ratio

$$\text{Retention ratio} = \text{Addition to retained earnings} / \text{Net income}$$

Capital intensity ratio

$$\text{Capital intensity ratio} = \text{Total assets} / \text{Net sales}$$

Pre-tax operating cash flow

$$\text{EBITDA Break-even} = \frac{\text{FC}}{\text{Price} - \text{Unit VC}}$$

Chapter 12

Cost-plus formula

$$\text{Cost} + \text{Mark-up percentage} \times \text{Cost} = \text{Target selling price}$$

Chapter 13

Valuation under dividend growth model

$$P_0 = \frac{D_1}{R - g}$$

Free cash flow

$$\text{FCF} = [(\text{Revenue} - \text{Op Ex} - \text{D \& A}) \times (1 - t)] + \text{D \& A} - \text{Cap Exp} - \text{Add WC}$$

Valuation under price-to-earnings ratio

$$P_0 = \text{Estimated EPS}_1 \times \text{Justified P/E}$$

Accounting rate of return (ARR)

$$\text{ARR} = \text{Average net income} / \text{Average book value}$$

Internal rate of return

$$\begin{aligned} \text{NPV} &= \text{NCF}_0 + \frac{\text{NCF}_1}{1+\text{IRR}} + \frac{\text{NCF}_2}{(1+\text{IRR})^2} + \dots + \frac{\text{NCF}_n}{(1+\text{IRR})^n} \\ &= \sum_{t=0}^n \frac{\text{NCF}_t}{(1+\text{IRR})^t} = 0 \end{aligned}$$

Modified internal rate of return

$$\text{MIRR} = \sqrt[n]{\frac{\text{FV}(\text{Positive cash flows} \times \text{Cost of capital})}{\text{PV}(\text{Initial outlays} \times \text{Financing cost})}} - 1$$

Profitability index

$$\text{PI} = \text{PV}(\text{cash inflows}) / \text{PV}(\text{cash outflows})$$

Chapter 15

Return on investment (ROI)

$$\text{ROI} = \text{Net profit} / \text{Average operating assets}$$

Residual income

$$\text{Residual income} = \text{Operating income} - (\text{Required rate of return} \times \text{Average operating assets})$$

Economic value added (EVA)

$$\begin{aligned} \text{EVA} &= \text{Adjusted after-tax operating income} - [\text{Weighted average cost of capital (WACC)} \\ &\quad \times (\text{Adjusted total assets} - \text{Current liabilities})] \end{aligned}$$

Weighted average cost of capital (WACC)

$$\text{WACC} = E/V \times R_e + D/V \times R_d \times (1 - T_c)$$

Part A

Financial Environment



Chapter 1 Financial Environment

Chapter 2 Sources of Finance

1

Financial Environment

CHAPTER TOPIC LIST

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LEARNING OUTCOMES

PRINCIPAL LO1: ANALYSE THE FINANCIAL ENVIRONMENT IN WHICH BUSINESSES OPERATE

LO1.01: Explain and analyse the financial environment in which a business entity operates

1.01.01 Analyse the concept of market efficiency and the implications of the Efficient Market Hypothesis for issuers and investors

1.01.02 Describe the financial environment in which an entity operates

1.01.03 Describe the structure of Hong Kong equity and debt markets and its financial and banking system

1.01.04 Explain the function of financial markets and the role they play in the value-creation process

1.01.05 Explain the role of treasury management and describe its responsibilities and activities



OPENING CASE

ABC ENTERTAINMENT LTD

The senior executives of ABC Entertainment Ltd, who intend to take the company public in a few years, have reviewed the company's statement of financial position as at 31 March 2018 and the cash flow forecast for 2019 as part of their financial and treasury management. ABC Entertainment's primary business activities are in hospitality and entertainment in Hong Kong. Sometimes ABC Entertainment also invests in marketable securities and subsidiary companies outside the core business.

ABC Entertainment Ltd's management are facing multiple financial challenges, including how to maximise shareholders' wealth, how to improve liquidity in both the short- and long-term, and how to best use the company's excess cash.

They would like to make sure the company has sufficient short-term liquidity not only for operations, but also for debt service, so they analyse the company's working capital. Management also pays close attention to the company's cash flow forecast for 2019.

Based on this, they assess expected cash flows and how they would impact the company's abilities to honour its liabilities on the statement of financial position as they become due.

The illustrative example in this chapter would outline in greater detail how the review is done by the executives. 



OVERVIEW

Financial management focuses on how to best utilise the assets that a company has to meet the objectives of management. This can be a complex undertaking and, as this chapter explores, one that requires thorough understanding of financial markets.

Finance is the study of how and under what terms money is allocated between lenders and borrowers. Funds typically flow from buyers to issuers through different types of financial contracts. Companies are often involved in both sides of this relationship with several goals, including to maximise both profits and the value of a company or its shares.

Financial accounting is an important practice within companies. Its focus is typically external and generates information for review by various **stakeholders**, including investors, financial institutions and the board of directors, as opposed to **management accounting** that aims to generate information for use by internal management. Top C-suite executives, such as chief executive officer (CEO), chief financial officer (CFO) and chief risk officer (CRO), hold overarching responsibility for financial accounting in firms.

As this chapter explores, understanding financial accounting requires thorough understanding of the functions and operations of the various financial markets, including primary and secondary markets as well as the various markets for foreign exchange, securities, derivatives and other financial instruments. For practical purposes, Hong Kong Exchanges and Clearing Ltd (HKEx) operates primary and secondary markets for various securities and instruments with the most famous security – stock.

Another important consideration for financial managers is treasury or cash management which is important to understand the cash needs of a company. These needs are, in broad strokes, the underlying motivation for financial management.

Finally, this chapter concludes with an explanation and discussion of the Efficient Market Hypothesis and its role in driving the performance of markets.



1.1 ROLES AND FUNCTIONS OF FINANCE

Finance refers to the study of how money flows between lenders and borrowers. The terms 'lenders' and 'borrowers' are, in their broadest sense, used to refer to people or companies or other institutions that have money to invest or need to borrow.

Finance examines how resources are allocated and also examines under what terms and through what channels the allocations are made.

Money is simply a means of exchanging value between parties. The efficient, timely and reliable transfer of money between parties underpins economic activity. In order for the exchange of money to occur, markets are required to facilitate the interaction between buyers and sellers, set terms of a transaction and execute that transaction.

The financial system covers quite a number of markets, like markets for cash, shares, bonds or foreign exchange. Each of these markets have their own purpose, rules and trading mechanisms.

The term, finance, refers to both the study of how money is managed and the process of acquiring money. Financial management refers to the efficient and effective use of company resources to accomplish its objectives.

Finance is key to the success of any business. Managers should thoroughly understand how finance works. This knowledge can help businesses:

- *Start up*: Including paying for expenses, such as rent, or purchasing equipment to produce products and services.
- *Operate*: Ensuring businesses have sufficient cash on hand to operate.
- *Expand*: Such as purchasing equipment to boost production capacity, research and development of new products and services, do marketing to reach new customer base.

A major concern for all businesses is the way they are financed. It is important for managers to select appropriate sources of funding, as all entities need funding, no matter how small or large their turnover or asset base.

Smaller businesses tend to look to the financial institutions, in the first instance, as suppliers of intermediated finance, in other words they look to banks for loans. Larger entities with standing in the community are able to access the financial markets and financial institutions for funds.

1.1.1 Debt Financing

Large entities wanting to raise debt finance can issue corporate bonds, notes and debentures as methods of financing. The differences between the various methods of financing are related to their historical roles.

Borrowers issue bonds, notes or debentures as proof that debts exist. When securities are traded, the security itself – the piece of paper or electronic record – is basically proof of ownership.

Lenders or owners of bonds at the time of maturity are the entities that receive payment from the issuers of bonds.

1.1.2 Equity Financing

Owners of companies may want to expand or sell the stake of the company. To this end, ownership stakes could be sold to investors, which is the basis of raising equity financing. Ownership rights are transferred by buying and selling shares, either ordinary or preference. Ordinary shares are the most common.

The finance options available to a company depend on its ownership structure. Sole proprietorships or partnerships of two to twenty people may have to take a different approach to finance than listed companies that can tap into financing from thousands of individual shareholders or even investment institutions.

Every business has reasons for being. Because of their different sizes and ownership structures, it is to be expected that there are a range of goals amongst businesses. For example, a family partnership might want to earn enough to live comfortably, put away some funds to educate the children or not work on Saturdays or Sundays. In contrast, the ownership of a large corporation is much more removed from the operations of the company, they are the shareholders. Because the owners are not closely connected with the everyday operations of the business, it is likely that their goals are simplified and focused largely on financial metrics, such as profit maximisation and shareholder returns.

► Knowledge Check Questions

Question 1

Identify which of the following is true about finance.

- A** Finance refers to the study of how money moves between lenders and borrowers.
- B** Finance is different from economics. Economics does not study how resources are allocated.
- C** The different parts of finance are integrated amongst themselves.
- D** The only important part of finance is business finance.

Question 2

State which of the following is the most accurate description of finance.

- A** The study of how entities acquire, spend and manage financial assets.
- B** The study of how entities manage financial assets.
- C** The study of how government and businesses acquire, spend and manage financial assets.
- D** The study of how money is used to purchase goods and services.

1.2 SCOPE AND OBJECTIVES OF FINANCIAL MANAGEMENT

The financial system offers a range of ways to finance activities. The job of the finance manager is to work out the best way to structure finances and make effective financial decisions.

Whenever funds are transferred, a financial contract comes into existence. These contracts are called financial securities. Operating in the financial space requires a basic grasp of securities, corporate law, financial institutions and how markets operate.

One goal for financial decision-making is profit maximisation, while another goal is to maximise cash flows.

Although profit maximisation may seem a logical goal for a business, it has some serious drawbacks.

It is not hard to pin down what is meant by 'profit'. To a business person, profits are just revenues minus expenses. To an accountant, accounting rules make the definition more elusive. Much relies on how the rules are applied and interpreted. In the extreme case where rules are distorted to exaggerate results favourably, profits may be misrepresented. This is the origin of the term **creative accounting**. In addition, costs can be minimised today by not investing new equipment or delaying maintenance, but this may not be in the best interest of the company or its owners.

A second problem is that accounting profits are not necessarily the same as cash flows. For example, many companies recognise revenues at the time a sale is made, typically done before receiving the cash payment for the sale. Ultimately, the owners of a business are more accepting of cash for payment because making investments or buying goods and services require cash.

A third problem with profit maximisation as a goal is that it does not distinguish between getting a dollar today and getting a dollar sometime in the future. In finance, the timing of cash flows is extremely important. For example, the longer a person goes without paying a credit card balance, the higher the interest costs. The interest accrues because of the **time value of money**; the longer an entity or person has access to money, the higher the cost. The time value of money is one of the most important concepts in finance.

Profit maximisation ignores the risks associated with cash flows. There is a trade-off between risk and expected return. When a choice has to be made between two investments with similar returns but different levels of risk, the obvious choice is to go with the less risky option. This is an intuitive choice as few people like to take risks, meaning they are risk averse, and must be compensated for taking it. The profit maximisation goal ignores differences in value caused by differences in risk.

Finance managers aim to maximise the share price of their companies and, in turn, maximising the wealth of shareholders. They do this by making investment and financing decisions that ensure the total present value of cash inflows exceeds the total value of cash outflows. The focus is to maximise cash flows, not profits. A better strategy can help make companies nimbler and in a better position to execute plans that can, in turn, lead to higher share prices than other companies that fail to get the balance right.

1.2.1 Financial Managers – Chief Executive Officer, Chief Financial Officer, Corporate Treasurer and Chief Risk Officer

Financial managers constitute a firm's central nervous system. At the top of the decision-making chain of command, they use financial information to determine the overall direction their firm will take.

The **chief executive officer (CEO)** is the top manager in a company and has the final decision-making authority. The most important responsibility of the CEO is to set the strategic direction of a company and guide the management team. The CEO reports to the board of directors and the board oversees top management and is accountable to shareholders.

Senior financial managers at large companies may hold the rank of deputy chief executive or a senior executive and can hold the title of **chief financial officer (CFO)**. In smaller companies, the job is typically more focused on accounting, with the top financial officer carrying a title of financial controller or chief accountant.

The CFO reports to the CEO and focuses on managing the financial operations while working on strategy with the CEO. A number of positions report to the CFO. The CFO is not to be confused with the corporate treasurer, who looks after collecting and disbursing cash, manage investments to earn return and raise new capital. The corporate treasurer is also in charge of managing excess cash and superannuation arrangements as well as financial relationships.

A final key position is the **chief risk officer (CRO)** whose job is to identify, analyse and manage risks, generated internally and externally, that could threaten the operations of a company. Amongst other duties, the CRO is also responsible to ensure compliance with the myriad regulations that companies have to follow as well as reviewing both the investments that companies make and the operations of various business units.

Key Learning Point

Finance refers to both the study of how money is managed and the process of acquiring money. *Financial management* refers to the efficient and effective use of company resources to accomplish its objectives.

► Knowledge Check Questions

Question 3

Identify the primary goal of the financial manager of a profit-seeking organisation.

- A Maximise market share
- B Maximise the owners' wealth
- C Increase sales and profit
- D Have healthy cash flow

Question 4

Identify which of the following refers to the study of how money is managed.

- A Marketing
- B Finance
- C Taxation
- D Accounting

1.3 FINANCIAL ENVIRONMENT

A complex economy such as Hong Kong's includes many different types of financial markets and institutions involved in direct and indirect financing. Each of these markets plays a different role in the economy.



1.3.1 Functions and Roles of Financial Markets in the Value-creation Process

Financial markets are characterised by direct and indirect flows of funds. But the term ‘financial market’ is very broad and covers all markets that deal with short- and long-term funds. The **money market** deals with short-term funds while the **capital market** deals with long-term funds. Companies may be involved in both markets.

There are also **derivative securities markets** to deal in financial contracts or instruments valued on the basis of underlying debt or equity securities. One commonly used derivative security is the option to buy or sell shares for a specific price and within a certain amount of time. Derivatives may be used to speculate on future price directions of underlying financial assets. Organised exchanges handle standardised derivative security contracts, while electronic markets handle negotiated contracts that may involve commercial banks or other institutions.

Foreign exchange markets (also called FOREX markets) are electronic markets in which banks and institutional traders buy and sell various currencies on behalf of businesses and other clients. Businesses that sell their products in foreign countries usually receive payment in foreign currencies. However, because the relative values of currencies may change, it is normal for firms to use the foreign exchange markets to reduce the risk of holding too much of particular currencies.

A **primary market** is any market where companies initially sell new security issues (debt or equity). For example, suppose ABC Entertainment Ltd wants to raise HK\$500 million for a business expansion and decides to raise the money through the issue of ordinary shares. The company will issue new equity (ordinary shares) in the primary market for corporate shares, probably with the help of an underwriter like a large bank or investment bank.

Initial public offerings (IPOs) is the process of first-time issuance of shares to the public.

A **secondary market**, however, is the market for the buying and selling of securities amongst investors. When securities are bought and sold in the secondary market, the original issuers (i.e. the companies that issued these securities in the primary market) do not receive any money.

When the public trades stocks on the Hong Kong Stock Exchange, it is engaging in trades in the secondary market. Secondary markets are like the markets for used cars. Car manufacturers do not receive any of the money when a person buys or sells a car.

Secondary markets enable investors to buy and sell securities. Investors are typically willing to pay more for securities for which there is an active secondary market. In turn, companies whose securities have active secondary markets typically have lower funding costs than similar companies whose securities do not have active secondary markets.

Marketability, which is the ease with which a security can be sold and converted into cash, is an important characteristic of a security for investors. The marketability of a security depends on availability of investors, on transaction costs, the costs of trading and searching for information. Securities with lower transaction costs have greater marketability because they are easier to trade in secondary markets.

A concept closely related to marketability is **liquidity**. Liquidity is the ability to convert an asset into cash quickly without loss of value. The terms marketability and liquidity are often

used interchangeably in normal use, but there are differences. Liquidity implies that when the security is sold, its value will be preserved but this implication does not apply to marketability.

There are two main types of specialists that facilitate transactions in secondary markets. **Brokers** bring buyers and sellers together, execute transactions for clients and charge fees. They do not bear any risk of ownership. **Dealers** make markets for securities and have to bear risk. They buy and sell securities that they own and profit in the same way that retailers benefit from selling inventory.

1.3.2 Hong Kong Equity and Debt Markets

Hong Kong is one of Asia's – and the world's – largest, most active and most liquid securities markets. It is a popular destination for Mainland Chinese companies to make their IPOs.

Hong Kong Exchanges and Clearing Limited (HKEx) operates the Hong Kong Exchange via its wholly owned subsidiary, the Stock Exchange of Hong Kong Limited (SEHK). Mainland IPOs aside, the SEHK has become increasingly connected to mainland Chinese markets and investors with the launch of the Shanghai-Hong Kong Stock Connect in 2014 and the Shenzhen-Hong Kong Stock Connect in 2016. In 2015, a scheme enabling the mutual recognition of Hong Kong and Chinese exchange-traded funds (ETFs) went into effect.

More than 2,000 companies are listed on the SEHK, which does not feature control over capital movement, nor do capital gains or dividend income taxes apply.

Bond Connect, a mutual bond access programme between Hong Kong and mainland China, was launched in July 2017. It was another significant breakthrough in the opening of the mainland China bond market and is also an important step in closer interaction and cooperation between the Mainland and Hong Kong. Bond Connect deepens mutual access between the Mainland and Hong Kong's capital markets. This marks the first time that eligible foreign investors have been allowed to invest in mainland China's interbank bond market through a link connecting the domestic and overseas bond markets.

Hong Kong has one of the world's freest debt markets, with international investors allowed to invest in Hong Kong-issued debt instruments without limit. More than one-third of debt issued in Hong Kong is private, while the local government is an active issuer of bonds under its Government Bond (GB) Programme that targets the non-banking investor base.

1.3.3 Major Indexes

There are thousands of securities listed on the major stock markets of the world, of which the Hong Kong stock market is one. It would be impractical to track the performance of every single one of these securities to arrive at a picture of market performance. Rather, various stock market operators and other information providers have developed stock markets indexes.

The indexes outlined below represent some of the major indexes used to track the performance of stock markets in Hong Kong and the United States. While these indexes are important for every investor to follow, they are only a few examples in a universe of indexes used to track all manner of financial products. There are indexes to track the performance of ETFs and bonds, to name but two examples. Some of these indexes are produced by the major indexes and others by independent index providers. For instance, the Hang Seng Markit iBoxx Offshore RMB Bond Index family tracks different types of bonds. There are ETFs that are

basically unit trusts or mutual funds with shares that are traded on stock exchanges. There are also index providers that track all manner of products and combination of financial products, such as industries or sectors. S&P Dow Jones Indices, FTSE Russell and MSCI all provide different types of indexes.

Below is an exploration of some of the larger indexes associated with stock markets in Hong Kong and the United States, including the Hang Seng Index, the S&P 500, the NASDAQ Composite and the Dow Jones Industrial Average.

1.3.3.1 Hang Seng Index

Published by a subsidiary of Hang Seng Bank, the Hang Seng Index (HSI) is an index of the largest and most liquid companies that are traded on the Main Board of the Hong Kong Stock Exchange. It is the most widely quoted gauge of the Hong Kong stock market.

Stocks included in the index are free-float adjusted for investable representation. A 10% capping is applied to avoid single stock domination. The HSI also features four sub-indexes that organise members according to primary business areas: commerce and industry, finance, properties and utilities.

The HSI was first launched in 1969 and now includes 50 stocks.

1.3.3.2 Standard & Poor 500

The Standard & Poor's 500 Index (S&P 500) is a market-cap-weighted index that follows 500 large-capitalisation US companies and captures around 80% of available US market capitalisation.

More than US\$7.8 trillion is benchmarked to the index, with approximately US\$2.2 trillion of that total represented by index assets. It serves as a benchmark for the US stock market as well as a barometer for the world's largest publicly traded companies. All constituents must be US companies.

At least 50% of shares outstanding must be available for trading. S&P 500 members are determined by Standard & Poor's analysts and economists according to market size, liquidity and other criteria.

1.3.3.3 NASDAQ Composite

NASDAQ refers to both the global electronic market for trading securities and the benchmark index for US technology stocks launched in New York City in 1971 by the National Association of Securities Dealers (NASD).

The index, the NASDAQ Composite, is a grouping of over 3,000 NASDAQ-listed stocks with a strong technology focus. Now separated from NASD, NASDAQ combined with European exchange group OMX in 2007 to form NASDAQ OMX Group, the world's largest exchange company.

1.3.3.4 Dow Jones Industrial Average

One of the oldest indexes in the world, the Dow Jones Industrial Average (DJIA) is an average of 30 blue-chip companies listed in the United States. The index is weighted by price and includes all industries except transportation and utilities.

Created by Charles Dow in 1896, it is often simply referred to as 'the Dow' and is intended to serve as an indicator of the health of the US economy at large. The DJIA now represents about a quarter of the value of the US stock market.

1.3.4 Financial and Banking Systems

While the financial and banking systems often work together, they are two different systems that serve different purposes.

The financial system is the system that allows lenders and borrowers to exchange funds, in essence connecting the two, often through intermediaries such as banks or financial companies. Financial systems operate at national, global and firm-specific levels.

At the heart of the financial system is the financial market in which new security issues are sold for the first time (i.e. primary market), while a financial market in which the owners of outstanding securities can resell them to other investors is a secondary market.

The banking system, in contrast, is the network of institutions that provide financial services. These institutions facilitate payments, make loans, take deposits and help with investments, amongst other services for retail and corporate clients.

Brokers are market specialists that bring buyers and sellers together in secondary markets. Dealers 'make markets' for securities and bear risk. They make a market for a security by buying and selling from an inventory of securities that they own, bearing the risk that they may not be able to sell a security for more than they paid for it.

Organised exchanges consist of physical place and facilities for participants to carry out their business under a set of rules and regulations. Only members can use the exchange and each exchange has a limited number of seats.

Over-the-counter (OTC) markets have no central trading location. Instead, investors can execute OTC transactions by visiting or calling an OTC dealer, or by linking to the dealer via computer.

Short-term debt instruments – debt instruments with maturities of less than a year – are traded in money markets. Capital markets are where equity and debt instruments with maturities exceeding one year are traded.

Public markets are organised financial markets where the general public buys and sells securities through their stockbroker. Private markets involve direct transactions between two parties, with those transactions called private placements.

Futures and options markets are also called derivatives markets because the securities sold derive their value from some underlying asset. Futures contracts are made for future delivery of securities, foreign currencies, interest rates, or commodities. Options contracts requires one party – the option writer – to perform a specific act if requested by the option buyer or owner.

All of these, brokers, dealers, organised exchanges, OTC markets, money and capital markets, public and private markets, futures markets and options markets make up the financial system.

The banking system is different from the financial system in that its focus is to facilitate indirect market transactions. Banks often act as financial intermediaries between borrowers and lenders.



Commercial banks are the most prominent and largest financial intermediaries in the economy and offer the widest range of financial services to businesses. The major services that commercial banks offer customers are checking accounts and a variety of credit and loan arrangements. They also do a significant amount of equipment lease financing.

Life insurance companies obtain funds by selling life insurance policies that protect individuals against the loss of income from a family member's premature death, while **casualty insurance companies** sell protection against loss of property from fire, theft, accidents, negligence and other causes.

Pension funds provide retirement programmes for businesses as part of their employee benefits programmes. They obtain money from employee and employer contributions during the employee's working years, and they provide monthly payments upon retirement.

Investment funds sell shares to investors and use the funds to purchase securities.

Business finance companies sell short-term debt – known as commercial paper – to investors in the direct credit markets, where these funds are used to make a variety of short- and intermediate-term loans and leases to small and large businesses.

► Knowledge Check Questions

Question 5

Identify which of the following are intermediaries, such as banks, insurance companies and investment companies, that engage in financial activities to aid the flow of funds from savers to borrowers or investors.

- A Financial institutions
- B Financial market organisations
- C Federal agencies
- D International financial organisations

Question 6

Identify where the issuing of new securities, mortgages and other claims to wealth takes place.

- A Secondary market
- B Money market
- C Primary market
- D Securities market

Question 7

Identify the term that describes the ease with which a security can be sold and converted into cash.

- A Book value
- B Marketability
- C Convertibility
- D Market value

1.4 TREASURY MANAGEMENT

Treasury management is also known as cash management. Cash is an umbrella term referring to all currency, coins, cheques, money orders and other instruments that are either held on-hand or demand deposit accounts in financial institutions such as banks.

Given that cash provides negligible or no return and is subject to losing value as a result of inflation, an enterprise will seek to hold a minimum cash balance sufficient to address its debt and other obligations as they become due. Holding excessive cash is not ideal, as it will lead to a loss of potential returns.

Cash budgets are used to determine an enterprise's cash needs. A firm's cash budget consists of expected cash receipts and disbursements for the period of time specified under the budget. When a firm's projected cash balance exceeds what is necessary, company management can elect to invest the cash or pay down debt that has already been incurred.

Likewise, if cash on hand is insufficient to cover debt and requirements, management can choose to decrease the firm's cash requirements, prepare to borrow or make other contingencies to deal with the shortfall.

Amongst larger firms, daily cash reports are a preferred method for monitoring cash balances in order to identify and deal with cash surpluses and shortages. Given that expected cash inflows and outflows and their timing may not bear out as originally expected, firms maintain a store of cash that serves as a hedge for such uncertainty.

Viewed through the lens of their target cash balance, firms will speed up cash inflows or defer outflows when possible so as to have cash on hand longer for investment in projects with higher returns.

1.4.1 Roles of Treasury Management

The treasurer, the person responsible for treasury management, has a wide range of roles. These include:

- Maintaining relations with banks.
- Ensuring sufficient cash liquidity.
- Investing surplus funds.
- Understanding the firm's funding requirements.
- Understanding the firm's available sources of finance.
- Keeping the firm abreast of potential currency and interest rate risks.
- Monitoring taxes and policies relevant to the firm.

Guiding these and other duties assigned to the treasurer's job is fiduciary responsibility, which entails the obligation to always act in the firm's best interest.

1.4.2 Responsibilities of Treasury Management

A firm's treasury department has a number of vital responsibilities to the firm, including:

- *Cash forecasting:* Aggregating information from different departments throughout the firm in order to maintain a continuously updated cash forecast. Sources of such information can include the firm's budget, accounting records, board minutes regarding dividend payments and elsewhere.
- *Monitoring working capital:* Reviewing company policies that touch upon working capital, modelling the impact of these policies on the firm's cash flows.
- *Cash concentration:* Creating a system for channelling cash into a centralised investment account from which it can be funnelled into investments. This may be done via cash sweeps or notional pooling.
- *Investments:* Using the firm's investment policy to assign surplus cash to a range of different investments, keeping in mind their rates of return and how quickly they can be liquidated.
- *Risk management:* Using various strategies to hedge and reduce risks that the firm may be exposed to due to fluctuations in asset values, interest rates and currency values.

In addition to prudent investment of excess cash there are other aspects of cash management. These include managing the cash conversion cycle by speeding up cash collections to receive cash as quickly as possible from customers, reducing the accounts receivable period and keeping cash as long as possible, or lengthening the accounts payable period. Ethically, there are concerns with lengthening the payables period by paying bills late – it is also bad business.

That said, firms use several tools to speed up collections and slow down disbursements.

Most of them use the concept of **float**, or the delay that occurs when funds are sent amongst participants in a transaction. The collection float is the time in between the moment when a payer sends payment and the funds are actually credited in the bank account of the payee. A disbursement float is the time period between the moment when a payer sends payment and the funds are deducted from its bank account.

No matter dealing with collections or disbursements, float involves three components.

The first component is delivery or **transmission float** – the delay in transferring the means of payment from the payer or customer to the payee or provider of goods or services.

The second component is **processing float** – delays in processing incoming payments from customers by the receiving firm. Once a payment reaches its destination, 'in-house' delays can arise.

The third component is **clearing float** – the delay in transferring funds between payer and payee that arises from the banking system's clearing processes.

A firm's treasurer should seek ways to reduce the components of collection float. Pre-authorised cheques are one means of reducing delivery and processing delays. An example of a pre-authorised cheque is when you have monthly expenses automatically deducted from your checking account. The cheques allow firms to regularly deduct payments from customers' bank accounts and are often used to pay utility and insurance bills.

Slowing disbursements is not ideal for a firm's reputation. When credit gets tight, slow-paying customers are often the first to be terminated. Most disbursement systems involve selected banks where funds to be disbursed are deposited to maximise float. Many firms use a zero-balance account in which just enough funds are transferred into an account to cover that day's cheques presented for payment.

The growth of communications systems has resulted in firms using electronic data interchange as a means to send invoices and electronic payment for goods and services. Internet trading exchanges and web-based funds transfer (i.e. by smartphone) are the next steps in eliminating float.

1.4.3 Activities of Treasury Management

Treasury departments are responsible for managing the liquidity of a company. Liquidity refers to current and projected cash inflows and outflows. The goal of treasury management is to ensure that there is sufficient cash at hand to fund the operations of a company while also ensuring that any excess cash is invested properly.

Treasury management involves a number of varied activities that include:

- Cash forecasting that includes compiling information from accounting records, budgets and estimates.
- Working capital monitoring through which the treasury reviews corporate policies associated to working capital and estimate cash flows.
- Cash concentration to ensure effective investment of extra cash.
- Manage a company's investment of excess cash.
- Give credit to customers and manage the terms under which credit is given.
- Raise funds when additional cash is needed.
- Manage risk through various strategies to deal better with fluctuations in values, interest rates and foreign currency.
- Manage credit rating agency relations and manage relationships with banks.
- Report to senior management on issues such as market conditions, funding, returns on investment, risks related to cash and other topics.
- Advise on mergers and acquisition.

Illustrative Example 1

While the management of companies like ABC Entertainment Ltd. is always concerned with long term health of business and finance, it also has to be alert to potential short-term risks in finance. In addition, by analysing the cash revenue and cash costs expected in 2019, additional cash resources may be expected to be available to the company in 2019.

As shown in the following extract of its statement of financial position, ABC Entertainment has sufficient short-term liquidity, whereas total current assets are higher than total current liabilities.

Illustrative Example 1 (continued)

ABC Entertainment Ltd. At 31 December 2018		At 31 December 2018	
	HK\$ million		HK\$ million
CURRENT ASSETS		CURRENT LIABILITIES	
Inventories	169	Trade payable	7,426
Trade Receivable	9,719	Short Term Bank Loans	3,000
Prepayments Deposits	1,166	Accruals and Other Payables	1,033
Cash and Cash Equivalent	6,170	Total Current Liabilities	11,459
Total Current Assets	17,224		

Trade receivables of HK\$9,719 million are greater than trade payables, accruals and other payables of HK\$8,459 million (i.e. HK\$7,426 million + HK\$1,033 million). Assuming they all have similar payment terms, and assuming receivables have no default issues, receivable collection should cover payable settlements.

Cash balance of HK\$6,170 million also provides additional cushion to short term working capital and reduces risks. The extract above also shows that ABC Entertainment's short-term debts are sufficiently covered by cash.

The following is the extract of the statement of financial position and cash flow forecast:

ABC Entertainment Ltd. At 31 December 2018		2019	
	HK\$ million		HK\$ million
NONCURRENT LIABILITIES		CASH FLOW FORECAST	
Other Payables	1,094	Revenue	41,875
Long Term Bank Loans	8,265	Cost of Sales	(16,671)
	9,359	Gross Profit	25,204
CAPITAL AND RESERVES		Marketing Expenses	(13,500)
Share Capital	11,241	Administrative Expenses	(8,731)
Reserves	14,228	Finance Costs	(282)
Total Equity	25,469	Profit before Tax	2,691
		Income Tax	(444)
		Profit	2,247

Short-term bank loans of HK\$3,000 million are not only covered by cash of HK\$6,170 million. Expected profit in 2019 cash flow forecast of HK\$2,247 million could also be used for short-term loan repayments, as can be seen below.

The company also has sufficient financial liquidity beyond one year. In 2019, expected profit (HK\$2,247 million) is substantially greater than expected finance costs (HK\$282 million). The company could service bank interests with ease.

Equity raising would appear unnecessary at this point as expected profit and existing cash balance would fully cover short-term bank loans, and also cover a portion of long-term bank loans. Debt maturities of long-term bank loans of HK\$8,265 million would dictate when the company should consider refinancing of its long-term bank loans.

The company's strong profitability, in addition to its substantial cash balance, should make loan refinancing relatively easy.

The company should even consider whether it actually has too much cash given its strong profitability. Excess cash not needed for operation, investments or debt repayments could be paid back to shareholders as dividends.

► Knowledge Check Questions

Question 8

Identify which of the following describes the time between when a payer sends payment and the funds are credited to the payee's bank account.

- A Collection float
- B Disbursement float
- C Total float
- D Transmittal float

Question 9

Identify which of the following is not a basic strategy that should be used by a business firm in managing cash.

- A Paying accounts payable as late as possible without damaging the firm's credit rating.
- B Turning over inventory as slowly as possible while avoiding stock outs.
- C Collecting accounts receivable as quickly as possible without damaging customer relationships.
- D Tightening credit policies for customers.

➤ 1.5 EFFICIENT MARKET HYPOTHESIS

The **Efficient Market Hypothesis (EMH)** is a theory in finance which states that all known information is taken into stock prices immediately upon being known to the public. As a result, the discounting of price due to information is efficient and, in the strictest sense, this implies that prices accurately reflect or 'price in' all available information at any given time. However, 'efficiency' in this respect includes all information, whether accurate or not, and makes no distinction between valid and factual information versus rumour or gossip.

This implies that it is impossible to beat the market because prices are efficient. Therefore, price is expected to react only to new information. In the belief that stocks always trade at fair market value, an investor who subscribes to EMH will assume that there is no point in trying to time transactions as a consequence of this efficiency. Also, under this set of beliefs is the assumption that the market can be beaten only by purchasing higher-risk investments. This also increases the chance of market risk and loss.

1.5.1 Weak Form

The weak form EMH states that security prices are reflections of all past and publicly known market data, namely all past price and volume trading information.

There are some implications to this definition. To be efficient, weak form efficient markets should have all historical trading data reflected or discounted in current prices but these have no use in predicting future price changes.

Under this form of EMH, the theory cautions investors that future prices cannot be forecasted by studying past price behaviour. The theory states that there are no patterns in stock prices, meaning that technical analysis is not reliable as a timing system for entering and exiting positions. In other words, price follows a random series of movements.

1.5.2 Semi-strong Form

The semi-strong EMH states that security prices reflect all publicly known and available information. It also assumes that the weak form set of information is incorporated as well as all public information relevant to the security in question, including earnings, dividends, corporate investments and management changes.

Semi-strong form efficient markets will have all publicly available information reflected or discounted in current prices. As a result, it is considered pointless to analyse publicly available information in order to identify securities that are mispriced.

The semi-strong variation suggests that new information is incorporated into the price very quickly (efficiently) and without bias; thus, no returns beyond market averages are possible through timing or analysis.

1.5.3 Strong Form

Strong form EMH states that all public and private information is priced into security prices. This includes weak form and semi-strong form. All shares prices are efficient so that there are no situations in which traders can beat the market. On average, even managers and investors who beat the market simply reflect statistical ranges of outcomes, but exceptional performance does not contradict strong form EMH.

As a result, neither publicly available information nor insider information have value in identifying mispriced securities.

1.5.4 Implications of Efficient Market Hypothesis

The EMH has significant implication for both investors and issuers of securities. While the considerations are similar, both are coming at them from opposite points of view.

1.5.4.1 For Investors

The first implication of the EMH for investors is that technical analysis of the performance of securities is not likely to be rewarded as it is the market that ultimately makes decisions and these decisions are not necessarily based on past performance. Also, fundamental analysis of companies is also unlikely to be successful at generating abnormal profits after transaction costs and taxes for similar reasons.

Another implication for investors is that active trading strategies are unlikely to outperform passive or buy-and-hold strategies. However, many of the behaviours of stock prices (especially individual stocks versus overall markets) may be attributed to overconfidence and overreactions resulting from information bias. Some companies are in favour, others out of favour; this may cause movement that appears inefficient, but subject to retracement and self-correction.

Strong form EMH has been contradicted in studies. For example, one study published in 1995 claimed that low price-to-earnings (P/E) stocks have greater returns to investors than high P/E stocks. This demonstrates that EMH is not absolute or certain (Dreman and Berry, 1995).

Instead of accepting the inevitable belief that markets cannot be beaten, investors may observe the theory behind EMH, but focus on the basics of good investing by defining investment goals in terms of expected returns and acceptable risk levels.

1.5.4.2 For Issuers

For issuers, the implications of EMH could guide some decision-making.

One implication is that the timing of security issues or share repurchases is unimportant because prices are correct over a period of time and determined by the market.

However, management should monitor the price of a firm's securities to see whether price changes reflect new information or short-run momentum or any overreaction by the market, which sets the prices of securities.

Ethics in Practice 1

The EMH has three forms: weak form, semi-strong form and strong form. Legislation, law enforcement and the court adhere to the semi-strong form, which holds that publicly available information is already reflected in the security prices, but private information is not.

Those who make trades based upon private information, therefore, gain unfair advantages over other market participants. Thus, insider trading is banned and penalised.

At a recent court ruling in Hong Kong, it was made clear that acting on insider information is illegal regardless of whether the securities are publicly traded inside or outside Hong Kong.

The Court of First Instance found on 15 January 2016 that solicitors Eric Lee and Betty Young, along with Eric Lee's sister, Patsy Lee, contravened the Securities and Futures Ordinance (SFO) by engaging in fraud or deception in transactions involving securities of Hsinchu International Bank Company Ltd (Hsinchu Bank), a bank listed in Taiwan. The court did not find enough evidence against Eric Lee's other sister, Ms. Stella Lee. However, her illicit profit would still be seized.

The court's landmark ruling provided an interpretation of Section 300 of the SFO leading to a prohibition on the use of fraudulent or deceptive schemes in securities transactions.

The Securities and Futures Commission (SFC) initiated civil proceedings in the court against the four defendants in December 2010. The SFC alleged that Betty Young obtained information about a tender offer for Hsinchu Bank shares while working as a lawyer seconded by her employer Slaughter & May to the client Standard Chartered Bank (SCB), the acquirer in Hong Kong. The information about the offer was non-public as well as confidential and price-sensitive.

Afterwards, Betty Young bought Hsinchu Bank shares and told Eric Lee and his sisters to also buy shares before the tender offer was announced. This sharing of information added up to a fraud or deception under Section 300 of the SFO. The rationale was that Betty Young had responsibilities to her employer, the law firm, and the firm's client SCB. Such responsibility was to refrain from using information for personal gain.

Between 21 and 29 September 2006, the four defendants made a total of six purchases of Hsinchu Bank shares at an average price of NT\$16.99 per share (HK\$4.07 at the time).

Ethics in Practice 1 (*continued*)

On the afternoon of 29 September 2006, SCB announced the tender offer priced at NT\$24.50 (HK\$5.88), resulting in the defendants making a combined profit of HK\$2.9 million on these transactions.

The court ordered the seizure of illicit profit from all the defendants. The court also ordered that a copy of the judgement be sent to the Law Society of Hong Kong, because both Eric Lee and Betty Young were members.

Eric Lee and his two sisters appealed the court decision.

On 9 November 2017, the Court of Appeal upheld the decision of the lower court in favour of the SFC. The Court of Appeal held that Hong Kong courts could hear this case because many of the activities that contravened the rules happened in Hong Kong.

While the fraud was not aimed at counterparties on the Taiwanese Stock Exchange, the activities involved in the case were related to elements of Section 300 of the SFO to a significant degree. The court therefore took the view that any fraudulent or deceptive conduct employed in making an offer to buy securities in Hong Kong would fall within the scope of Section 300, regardless of whether the securities were traded overseas.

As the market becomes ever more globalised, cross-border trading and takeovers will only increase. Enforcement agencies around the world have also put more emphasis on joint enforcement actions across jurisdictions. The court's interpretation of Section 300 of the SFO is a pointed warning to market participants that the SFC may investigate and prosecute insider dealing, even with respect to shares listed overseas. The court's message is that it will not allow anyone to generate trading profit in public securities by using private information, a reflection of its embrace of the semi-strong form of the EMH.

1.5.5 Criticisms of Efficient Market Hypothesis

One of the assumptions for EMH is that all available information is perceived in exactly the same manner by all investors. The various approaches for stock analysis and valuations constitute doubts for the validity of the EMH. Different investors would have different evaluation perspectives when searching for investment opportunities; for example, one investor may like to invest in undervalued stocks in the market, as opposed to another who might like high potential growth stocks that might result in a different evaluation assessment of the fair market value of the stocks. Thus, one criticism against EMH is that a stock's value cannot be determined, even under an efficient market, as different investors value stocks differently.

Another assumption under EMH is that no one can beat the market and achieve a higher profitability than another investor, given the same investment amount. Given the identical information, all market participants can only arrive at the same investment returns. The simplest method to reach the market return is to invest in an index fund. The index fund would rise or fall due to the overall level of corporate profitability. However, in the real world, investors attain wide ranges of investment returns. There are many investment experts who

are able to beat the market consistently. An example is Warren Buffett, who has managed to outperform the market averages year after year. According to the EMH, if one investor earns excess returns, this implies that every investor can earn the excess return, but this is far from the truth.

In addition, observations under behavioural finance studies also contradict the assumptions of EMH about the investor rationality. The EMH presumes that all market participants make rational decisions after taking into consideration all of the objective factors that may affect the stock price, which is information generated from technical and fundamental analyses. Behavioural finance points out that the market may not be efficient as not all investors make rational decisions and psychological factors, such as human emotion and bias, also influences the investment decisions.

► Knowledge Check Questions

Question 10

Identify which of the following infers that information is quickly embedded in prices making it difficult for investors to 'beat the market'.

- A Stock investing
- B Efficient markets
- C Portfolio management
- D Asset allocation

Question 11

Identify which of the following happens when, in a business transaction, during which one party has information that is not available to the other parties.

- A Financial advantage
- B Inside information
- C Information efficiency
- D Profits

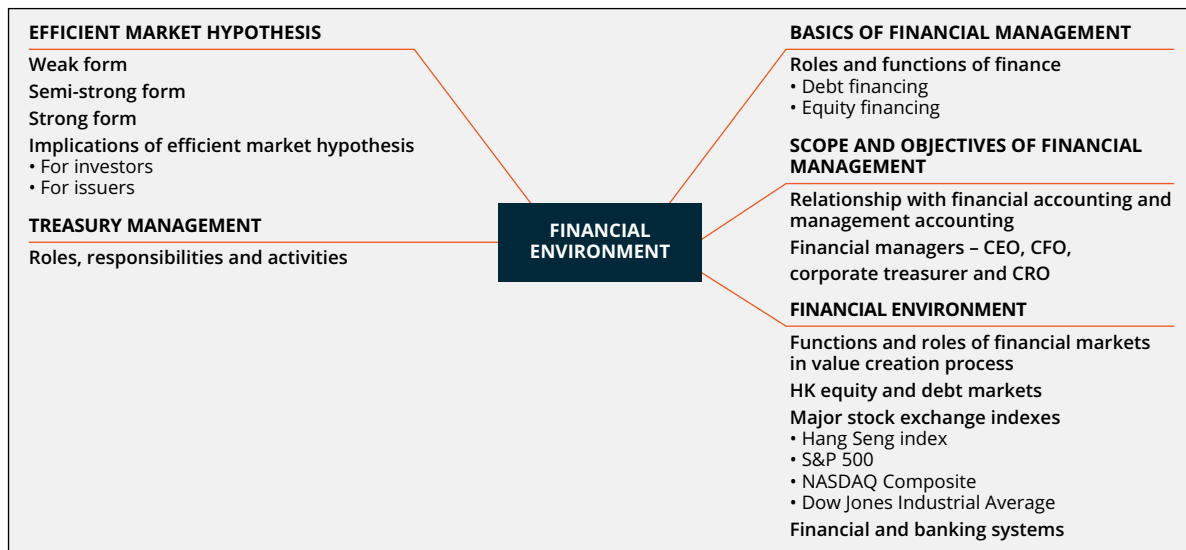


SUMMARY

- In its broadest terms, finance is the study of how and under what terms money is allocated between lenders and borrowers.
- In order for the exchange of money to occur, markets are required to facilitate the interaction between buyers and sellers, set terms of a transaction and execute that transaction.
- Finance is a key factor in the success or otherwise of any business. Businesses need finance to start up, operate and expand.
- The size of a business and the nature of its ownership often determine the finance options available to it.
- The financial system offers a range of ways to finance activities. The job of the finance manager is to work out the best way to structure finances and make effective financial decisions.
- One goal for financial decision-making that seems reasonable is profit maximisation. However, although profit maximisation may seem a logical goal for a business, it has some serious drawbacks including difficulties defining profit, the difference between profits and cash flows, costs created by the time value of money and cash flow risks.
- The goal for financial managers is to make decisions that maximise the company's share price. By maximising share price, management will help to maximise shareholders' wealth.
- The differences between financial and management accounting are on areas of aggregation, efficiency, information, reporting focus, standards, systems, time periods, timing and valuation.
- Financial managers constitute a firm's central nervous system. The CEO – the chief executive officer – has the final decision-making authority amongst all the company's executives. The CFO reports directly to the CEO and focuses on managing all aspects of the company's financial side, as well as working closely with the CEO on strategic issues. A final key position is the CRO or chief risk officer whose job is to identify, analyse and mitigate risks, internally and externally, that could threaten the operations of a company.
- A complex economy such as Hong Kong's includes many different types of financial markets and institutions involved in direct and indirect financing.
- Direct and indirect flows of funds occur in financial markets. When the focus is on short-term funds it is called a money market. In contrast, when the focus is on the long-term funds it is called a capital market. There are also derivative securities markets, which are markets for financial contracts or instruments that derive their values from underlying debt and equity securities. Foreign exchange markets (also called FOREX markets) are markets for various currencies.
- A primary market is a market where companies initially sell new security issues. Issues that are open to the public are called initial public offerings (IPOs). A secondary market is a market where securities are bought and sold between owners and investors.

- Marketability is the ease with which a security can be sold and converted into cash, an important characteristic of a security to investors.
- A concept closely related to marketability is liquidity, the ability to convert an asset into cash quickly without loss of value.
- Hong Kong Exchanges and Clearing Limited (HKEx) operates the Hong Kong Exchange via its wholly owned subsidiary, the Stock Exchange of Hong Kong Limited (SEHK).
- The largest and most commonly tracked stock indexes in Hong Kong are the Hang Seng Index of Hong Kong stocks and the US-focused S&P 500, NASDAQ Composite and Dow Jones Industrial Average.
- While Composite the financial and banking systems often work together, they are two different systems that serve different purposes. At the heart of the financial system is the financial market in which new security issues are sold for the first time (i.e. primary market). The banking system is different from the financial system in that its focus is to facilitate indirect market transactions. Banks often act as financial intermediaries between borrowers and lenders.
- Treasury management is also known as cash management. Cash is an umbrella term referring to all currency, coins, cheques, money orders and other instruments that are either held on-hand or demand deposit accounts in financial institutions such as banks.
- The treasurer is the person responsible for treasury management.
- In addition to prudent investment of excess cash, management is also focused on the conversion cycle by speeding up cash collections to receive cash as quickly as possible from customers, or reducing the accounts payable period, and keeping cash as long as possible, or lengthening the accounts payable period.
- The EMH states that markets are efficient and, in the strictest sense, this implies that prices accurately reflect or 'price in' all available information at any given time. The EMH has three forms: weak form EMH, semi-strong form EMH and strong form EMH.
- The EMH has different implications for investors and issuers.

MIND MAP



► Answers to Knowledge Check Questions

Question 1

Answer A is correct. By definition, finance is the study of how money moves between lenders and borrowers.

Answer B is incorrect. Economics also studies resource allocation.

Answer C is incorrect. All parts of finance must be integrated.

Answer D is incorrect. Finance has many important parts.

Question 2

Answer A is correct. By definition, finance is the study of how individuals, institutions, governments and businesses acquire, spend and manage money and other financial assets.

Answer B is incorrect. Finance also studies how individuals, institutions and governments acquire, spend and manage money and other financial assets.

Answer C is incorrect. Finance also studies how individuals and institutions acquire, spend and manage money and other financial assets.

Answer D is incorrect. Finance studies how money and other financial assets are acquired, spent and managed.

Question 3

Answer A is incorrect. There is usually trade-off between profit and market share, the business will not be sustainable with high market share but no profits, or even making a loss.

Answer B is correct. The goal for financial managers is to make decisions that maximise the company's share price. By maximising share price, management will help to maximise shareholders' wealth.

Answer C is incorrect. Profit maximisation ignores the uncertainty (or risk) associated with cash flows.

Answer D is incorrect. Instead of a healthy cash flow, the value cash flow is more important.

Question 4

Answer A is incorrect. Marketing is the study of sales and promotion.

Answer B is correct. By definition, managing money is the major study of finance.

Answer C is incorrect. Taxation is the study of laws regarding paying taxes to government.

Answer D is incorrect. Accounting is the study of recording transactions.

Question 5

Answer A is correct. By definition, financial institutions are intermediaries, such as banks, insurance companies, and investment companies that engage in financial activities to aid the flow of funds from savers to borrowers or investors.

Answer B is incorrect. Financial market organisations only provide facilities that enable intermediaries to function.

Answer C is incorrect. Federal agencies are regulators, not intermediaries.

Answer D is incorrect. They are trade groups that promote the interests of market participants.

Question 6

Answer A is incorrect. The secondary market only trades in the securities that already sold in the primary market where new securities are not available.

Answer B is incorrect. The money market trades in high liquidity assets, where mortgages are not available.

Answer C is correct. The issuing of new securities, mortgages, and other claims to wealth takes place in the primary market.

Answer D is incorrect. The securities market trades in securities but not the other assets.

Question 7

Answer A is incorrect. It describes accounting value of the security.

Answer B is correct. By definition, the securities with high marketability will have relative active trade in the market and can be sold for a price at which similar items are selling.

Answer C is incorrect. It describes legal ability of the security to be changed into another instrument of value.

Answer D is incorrect. It describes the value that the security should be able to fetch in ideal conditions.

Question 8

Answer A is correct. By definition, the collection float is the time between when a payer sends payment and the funds are credited to the payee's bank account.

Answer B is incorrect. Disbursement float is the time between when a payer sends payment and when the funds are deducted from the payer's bank account.

Answer C is incorrect. Total float is the combination of all three components of float.

Answer D is incorrect. Transmittal float is the delay in transferring the means of payment from the payer or customer to the payee or provider of goods or services.



Question 9

Answer A is incorrect. Paying the accounts payable as late as possible can increase the cash in hand and the liquidity.

Answer B is correct. The business firms that keep too many inventories will lower the firm's liquidity and lengthen the time of cash collection.

Answer C is incorrect. Collecting the receivable as quickly as possible can increase the cash in hand liquidity.

Answer D is incorrect. Tightening credit policies for customers can receive the payment earlier, shortening the time of cash collection.

Question 10

Answer A is incorrect. Stock investing describes the process of picking stocks.

Answer B is correct. When the market is efficient, all news is immediately reflected by the stock price; technical and fundamental analysis cannot be used to beat the market.

Answer C is incorrect. It describes the process of managing all the securities in a portfolio.

Answer D is incorrect. It describes the process of distributing funds between different categories of assets.

Question 11

Answer A is incorrect. It has nothing to do with information advantage.

Answer B is correct. Inside information means that all the parties in the transactions do not share the same information.

Answer C is incorrect. It deals with how quickly information is reflected in security price.

Answer D is incorrect. It deals with generating economic return.

EXAM PRACTICE

QUESTION 1

Identify on which of the following the size of the cash buffer depends.

- A. The ability to easily acquire more inventory.
- B. The predictability of cash inflows.
- C. Total fixed asset investment.
- D. The ability to accelerate asset depreciation.

QUESTION 2

Identify what the goal of a company's cash management policy is.

- A. To increase the cash conversion cycle.
- B. To increase the payment period.
- C. To optimise the cash requirement.
- D. To maximise cash outflows.

QUESTION 3

State why profit maximisation is not the best goal for a company and what an appropriate goal is.

QUESTION 4

Describe the most important governing body within a business organisation and what responsibilities it has.

QUESTION 5

State what the appropriate goal of financial managers is. Explain whether managers' decisions can affect this goal in any way, and how.

QUESTION 6

Explain the economic role of brokers and dealers and how each of them makes a profit.

QUESTION 7

Identify the main difference between money markets and capital markets.

QUESTION 8

Explain what marketability of a security means and how it is determined.

QUESTION 9

Explain what over-the-counter markets (OTC) are and how they differ from organised exchanges.

QUESTION 10

State what a float is and explain why it is important to cash management.

QUESTION 11

List the three components of float and identify which are under the control of the firm seeking to reduce collection float.

QUESTION 12

Identify the ethical conflict that is present in insider trading.

QUESTION 13

You believe that you can make abnormally profitable trades by observing that the CFO of a company always wears a green suit on days that the company is about to release positive information about itself. State which form of market efficiency is consistent with your belief.

QUESTION 14

Describe the informational differences that separate the three forms of market efficiency.

QUESTION 15

Explain the roles of lenders and borrowers in the world of finance, and how they interact with one another.

QUESTION 16

List the primary reasons that a company will borrow money and describe how borrowed funds are used.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. It depends on the abilities of suppliers to provide.

Answer B is correct. By definition, the size of a cash buffer is large when the company expects a slow cash inflow and vice versa.

Answer C is incorrect. Total fixed asset investment has nothing to do with cash cushioning.

Answer D is incorrect. Acceleration of asset depreciation depends on tax regulation.

QUESTION 2

Answer A is incorrect. The increase of a cash conversion cycle actually increases the use of cash.

Answer B is incorrect. It is only one of the methods used in cash management.

Answer C is correct. By definition, optimising the cash requirement is the major goal of cash management in order to reduce outstanding debt.

Answer D is incorrect. It actually increases use of cash.

QUESTION 3

Profit maximisation may seem to be the logical goal for any company but this has many drawbacks. First, profit can be defined in a number of different ways, and variations in profit for similar companies can vary widely. Second, accounting profits and cash flows are not exactly equal. Third, profit maximisation does not consider timing and ignores risk associated with cash flows. An appropriate goal for financial managers who do not have these concerns is to maximise the value of the company's current share price. To achieve this goal, management must make financial decisions so that the total value of cash inflows exceeds the total value of cash outflows.

QUESTION 4

The board of directors is the most important governing body within an organisation. Its main role is to represent the shareholders. The board also hires (and occasionally fires) the CEO and advises on major decisions.

QUESTION 5

The goal for financial managers is to maximise the value of a company's shares. Managers make decisions that affect share prices, which are determined by the future cash flows that a company can generate. Managers can affect cash flows in different ways. They can, for example, choose products or services to produce, assets to purchase and advertising campaigns to undertake.

QUESTION 6

Both brokers and dealers play a similar economic role by bringing together buyers and sellers of a commodity in a market. While brokers only facilitate a transaction by helping the two parties make a transaction and are therefore only compensated for taking on that role, dealers, on the other hand, take on risk by purchasing (or selling) a commodity from a seller (buyer) without another buyer (seller) necessarily being available, and are compensated for taking the risk.

QUESTION 7

Money markets are where short-term debt instruments with maturities of less than one year are traded while capital markets are where equity securities and debt instruments with maturities of more than one year are traded.

QUESTION 8

The marketability of a security refers to the ease or difficulty with which it can be sold in the market. There are many factors that affect the level of marketability of a security, such as the security's coupon and maturity, the total transaction cost (that includes the cost of trading, the physical transfer cost and the search and information cost). As the costs become lower, the security's marketability becomes greater.

QUESTION 9

OTC is a market for securities that are not listed on one of the organised exchanges. Unlike organised exchanges, there is no central trading location. The securities transactions are made via phone or computer instead of the floor of an exchange. Small companies usually trade their shares OTC because they are not able to qualify for one of the organised exchanges.

QUESTION 10

The float is the delay between when funds are sent by a payer and when they are actually received. The collection float is the time between when a payer sends payment and the funds are credited to the payee's bank account. The disbursement float, instead, refers to the lag between the moment when a payer sends payment to when the funds are actually deducted from its bank account. Float periods are important considerations for firms because they impact cash balances and the amounts that firms can invest, while also reducing financing needs.

QUESTION 11

The three components of float are transmission float, processing float and clearing float. Firms can most directly impact the transmission float and processing float. The banking system controls the clearing float through its check-clearing process. Firms can try to reduce float periods through, for example, electronic payment forms or pre-payments.

QUESTION 12

Insider trading is an example of information asymmetry. The main concept is that investment decisions should be made on an even playing field. Insider trading is morally wrong and is illegal.

QUESTION 13

You believe that the share price of this company is influenced by the CFO's wardrobe choice and that this private information will grant you abnormally high returns. For that reason, your belief is consistent with the semi-strong form of market efficiency, according to which it is likely to earn you abnormally high returns by trading on private information.

QUESTION 14

The strong form of market efficiency suggests that the price of a security reflects all information and that there is no private or inside information that could change prices. The semi-strong form states that all public information made available to investors is reflected in the security's price but insiders with access to private information could profit



from trading on knowledge before it is public. The weak form of market efficiency holds that there is both public and private information that may not be reflected in the price of a security and having access to this information can lead to profits that are abnormal.

QUESTION 15

Lenders play a critical role in finance, and represent individuals or organisations with funds available to invest. Rather than taking an equity stake in a company, lenders provide capital in the form of debt and earn profits from interest that borrowers pay. Lenders of bonds have an advantage over equity stakeholders; they have priority of repayment in the event of liquidation, making debt safer than equity (when issued as a bond contract).

Borrowers are those needing capital and who borrow money from lenders. Borrowers must decide on a balance between equity and debt capitalisation for their business. Equity capitalisation is found in the form of shares of stock (in a corporation) or partnerships; equity investors share in the profits of the business and are involved to some degree in decisions made by management about how the organisation will grow. Debt capitalisation is found in the form of short-term and long-term notes, contractual obligations defining the amount borrowed, and interest rate and compounding method. It also describes a repayment schedule and periodic payment amounts. Debt also is found in the form of bonds, longer-term obligations entered between borrower and lender and is more formal than a note. Bonds are also subject to valuation at a premium or discount, depending on current market rates versus the fixed contractual rate of the bond.

QUESTION 16

A company will borrow money for three reasons. First, borrowed money is used as start-up capital. This is the most difficult borrowing to obtain, because many new businesses fail and a lender will view start-up lending as a high risk. This will be reflected in the interest rate a lender expects to earn from start-up lending.

The second reason for borrowing is to continue operations once the organisational phase has been completed. The company needs money for marketing, advertising and payment of overheads; to build inventory levels; and to finance sales made on credit. When a company cannot issue stock or take on new investors without losing too large a share of controlling interest, borrowing money is an alternative that the business is likely to pursue. However, in operating the company, cash flows become a major concern. Borrowed money must be repaid. The initial borrowed funds ease cash flows in the moment; but this must be turned around quickly to generate adequate cash to continue operations, which includes repayment of borrowed funds plus interest.

The third reason to borrow is to expand operations. This may be represented by larger markets, greater range of products or services or acquisitions of competing companies. All of these initiatives demand financing, and borrowing money is the most obvious way to accomplish this. However, the business must ensure that the expansion plan will generate cash flows to repay borrowed funds, and in a timely enough manner to make sense out of acquiring more debt.

2

Sources of Finance

CHAPTER TOPIC LIST

2.1 Short-Term Financing

- 2.1.1 Commercial Paper
- 2.1.2 Bankers' Acceptance
- 2.1.3 Line of Credit
- 2.1.4 Bank Overdrafts
- 2.1.5 Term Loans
- 2.1.6 Family/Friends Loans
- 2.1.7 Leasing Financing

2.2 Long-Term Financing

- 2.2.1 Common Stock
- 2.2.2 Preference Stock
- 2.2.3 Long-Term Loans
- 2.2.4 Bonds
- 2.2.5 Hybrid Securities

2.3 Raising Capital

- 2.3.1 Venture Capitalists
- 2.3.2 Initial Public Offering
- 2.3.3 General Cash Offer by a Public Company
- 2.3.4 Private Market and Bank Loans

2.4 External Financing versus Internal Financing

- 2.4.1 Internal Financing
- 2.4.2 Advantages and Disadvantages of External Financing
- 2.4.3 Advantages and Disadvantages of Internal Financing



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE
--

LO7.01: Explain and analyse the theories of major capital market models
--

7.01.02 Explain the different sources of finance (short-term and long-term) and methods of raising capital

7.01.09 Explain the advantages and risks associated with external and internal financing



OPENING CASE

ABC INVESTMENT LTD

ABC Investment Ltd is a fully owned subsidiary of ABC Entertainment Ltd. ABC Investment Ltd has wide ranging mandates to invest in various instruments and sectors. It regularly invests in a number of listed companies. Financial accounting professionals manage the company's portfolio of stocks and work with the company's many subsidiaries to help them raise funds.

The work involves considering and analysing a wide range of equity and debt instruments to determine the most suitable instruments for investments and to find options for the lowest funding costs in a number of fundraising situations.

Financial accountants work with subsidiaries on debt funding costs after careful analyses of market information. There are many short-term debt instruments to choose from, each with their own market niche.

It is also necessary to evaluate long-term debt instruments to fund long-term investments of subsidiaries.

Furthermore, financial accounting professionals evaluate equity and hybrid securities as potential investment opportunities. These hybrid securities look like equities, but may also carry debt characteristics. The different features require careful calculation.

Ultimately, financial accountants provide the company and its subsidiaries with advice based on the results of analyses and calculations. 

OVERVIEW

Access to enough cash to operate and cover any liabilities is a key requirement for any business. Without enough cash, businesses wither and die.

Companies can raise cash through **debt** – essentially borrowing – or by issuing equity. Both approaches have their advantages and disadvantages. This chapter focuses on debt financing and the different factors that business should take into consideration.

Financing can be short-term options – such as commercial papers, bankers' acceptance, lines of credit, term loans, loans from family or friends, bank overdrafts and lease financing – or long-term options – including both preference share and common stock, long-term loans, hybrid securities, bonds and government loans.

This chapter deals with the various options available to companies to raise funds as well as the multiple vehicles to channel funds such as venture capitalists, initial public offerings (IPOs), general cash offers, private markets and bank loans.

For companies, the effective use of loans and various debt financing tools is part of the efficient management of working capital. Working capital is focused on determining the right mix of short- and long-term assets that a firm should hold and how to finance these assets. A key consideration of this practice is cash management, which this chapter discusses at length. Just as managing cash is important, so is managing credit and inventory, both of which impact the cash flow of a company.

Companies sometimes rely on long-term funding to raise cash. Long-term loans are often tied to some form of collateral and carry risks of their own, such as the possibility of firms acquiring too much of it.

Companies also have to make choices, depending on their needs, between internal financing and external financing. There are different types of both as well as advantages and disadvantages. Internal financing is cheaper but may come in smaller amounts while external financing can be more expensive and more complicated but generate large amounts of financing.



2.1 SHORT-TERM FINANCING

Companies may want to borrow or lend for the short term or the long term. If they borrow or lend in the short-term, meaning less than one year, they are using the money market, which is the market for short-term financial obligations.

For investing purposes, a company may choose to invest in short-term instruments that are issued by the government, or it may choose to invest in instruments issued by financial or non-financial companies. For borrowing purposes, a company may borrow from a financial institution, such as a commercial bank, or issue short-term securities that can be bought by institutional investors, other companies, or individuals.

2.1.1 Commercial Paper

Commercial paper (CP) is a name for short-term promissory notes issued by companies, normally with maturities of 1 to 270 days. Typical commercial paper issue is for 30 days and the amounts are generally very large. Commercial paper is a promissory note and has credit risk because the issuer may fail to pay investors back, even though they are typically secured by lines of credit.

Most often, CP is issued at a discount from its face value but issuers may instead issue it as an interest-bearing obligation. When CP is issued at a discount, the investor buys the paper at less than its face value and receives the face value at maturity. The difference represents the interest that the investor receives.

Interest on CP is calculated using the bond equivalent yield and a 360-day year. For example, if a company issues CP with a face value of HK\$100 million for HK\$99.55 million, and the maturity is 30 days, the bond equivalent yield on this paper is:

$$\begin{aligned} i_{bcy} &= \frac{\text{HK\$100 million} - \text{HK\$99.55 million}}{\text{HK\$99.55 million}} \times \frac{360}{30} \\ &= 0.004520 \times 12 \\ &= 0.05424 \text{ or } 5.424\% \end{aligned}$$

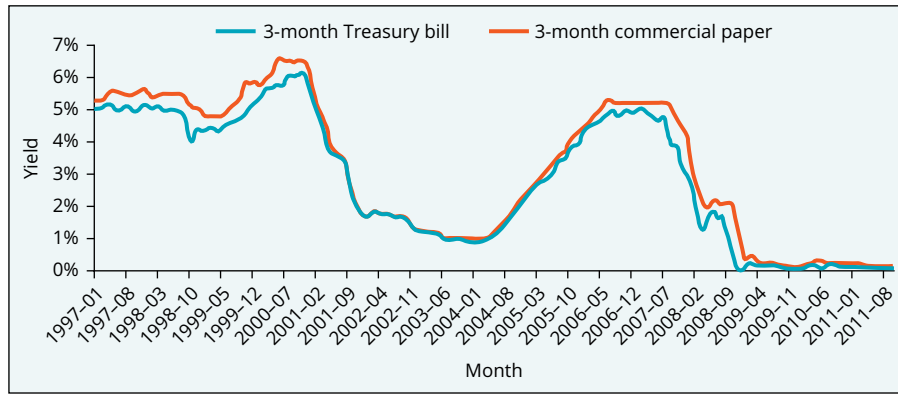
Money market instruments are quoted based on an approximate interest rate. A 5.424% annual rate is equivalent to 0.4520% per month. So, under normal circumstance, the investor would receive HK\$100 million after one month but if the issuer defaults, the investor gets nothing. The interest rate quoted represents a promised yield or the rate investors receive if they are paid as promised.

For issues where there is virtually no default or credit risk, the promised yields are the expected rates of return but when there is default risk, the expected rate of return is lower than the promised yield.

The promised yield is the maximum rate that the investor can hope to earn by holding the security to maturity.

Because commercial paper has default risk, the yields on commercial paper are higher than what they would be on US Treasury bills (T-bills). The difference between the yield on CP and on T-bills is the yield spread (Exhibit 2.1).

The yield spread represents compensation for the possibility of default on CP. It is important to note that default risk affects promised yields, the higher the default risk is, the higher the required promised yield.



► **EXHIBIT 2.1 Yield spreads** (Source: data from Federal Reserve Bank of St. Louis, Series TB3MS and CPN3M.)

The actual risk of investing in CP is very low because only companies that are creditworthy raise funds this way and CP is typically secured by lines of credit. As a result, the yield spread tends to be small, as Exhibit 2.1 shows.

At times of economic distress, yield spreads tend to be larger but are still typically small. The largest yield spread in Exhibit 2.1 happened in May 2000 when it rose to 75 basis points. The financial crisis of 2007–2008 had an impact on yields and yield spreads. Spreads widened early on in the crisis in 2008 and narrowed again through 2010.

Despite the low risk, assessing default risk requires both time and analytic skills, a process that may be worthwhile when investing for a long period but one that few investors consider worthwhile on 30-day investments. Still, default risk affects yields and credit rating agencies have developed a basic measure to provide investors with information that may do away with the need for individual analysis.

The major credit rating firms are Standard & Poor's (S&P), Moody's Investors Service and Fitch Ratings, and their classifications for CP are outlined in Exhibit 2.2.

Description	Standard & Poor's	Moody's Investors Service	Fitch Ratings
Highest short-term credit quality	A-1	Prime-1 or P-1	F1
Good short-term credit quality	A-2	Prime-2 or P-2	F2
Fair short-term credit quality	A-3	Prime-3 or P-3	F3
Speculative short-term credit quality	B	NP	B
High short-term default risk	C		C
Default	D	Default	D

► **EXHIBIT 2.2 Rating classifications** (Source: data from Standard & Poor's, Moody's Investor Service, Fitch Ratings.)

Investors are often wary of investing in CP of companies without enough liquidity to ensure money will be available to pay off CP if the companies cannot roll it over by selling to new investors. Liquidity support is usually a dedicated backup line of credit from a bank, which means companies can use the credit line to pay off CP if the need arises. The commitment fee to establish a line of credit for liquidity support is about 0.125% per year, which adds to the costs of issuing CP.

Another form of CP is asset-backed CP. Asset-backed commercial paper (ABCP) is secured with specific assets like receivables and is generally issued by special purpose vehicles (SPV). Companies create SPVs to buy assets and issue CP and dedicate the cash flows of these assets to repay the CP. But, because SPVs are separate entities, the creditworthiness of the ABCP depends on the credit quality of the assets that back the ABCP. Credit enhancements can affect risk as well as both the rating and yield of the ABCP.

There are a number of possible credit enhancements such as over collateralisation of the paper (value of secured assets being greater than the face value of CP) or external enhancements like insurance. The maturities of ABCP are not longer than 270 days and typically between 90 and 140 days.

The issuer of commercial paper can either sell the paper directly to the investors or use a dealer. Issuers that sell directly to investors are usually financial institutions or captive finance companies. Commercial paper sold through a dealer is first sold to the dealer, and then the dealer uses its resources to sell the commercial paper. Usually, the investment bank charges five basis points (that is, 0.05%) as a selling commission, but this is often bid down, depending on market conditions.

2.1.2 Bankers' Acceptance

Bankers' acceptance (BA) is used in cross-border trading. Exporters receive a 'bill of exchange' in return for the export of goods. This bill of exchange is, basically, a promissory note issued by the foreign importer and guaranteed by a bank. The exporter's bank then accepts the note, which can be sold for cash. BAs are also issued at a discount from their face value with maturities of between 1 and 180 days and typically 90 days.

The cost of issuing BAs is the yield plus the bank's stamping or acceptance fee, usually 0.50–0.75%, depending on credit risk.

Commercial banks are critical links in short-term financing. For instance, when issuing CP, companies are typically required to have a backup line of credit to ensure there is always money available to pay off CP. Because credit lines may not be available if the creditworthiness of a company deteriorates, banks typically review the credit conditions of their clients.

Banks are just as important for the BA market because the bank's credit, but not the company's, supports BAs. However, the CP and BA markets are only useful for companies that need large and continuous financing because both are sold in large amounts of at least US\$1 million. For smaller, irregular-sized financing requirements, companies tend to rely on traditional bank financing such as loans and lines of credit.

2.1.3 Line of Credit

Banks provide **lines of credit** and borrowers pay interest on the amount borrowed. In addition to interest costs, banks typically charge a fee of around 0.50% for setting up the line of credit and this fee is also charged on the unused balance.

The standard line of credit is an operating or demand line of credit that is made available by a bank for the company's operating purposes and is not meant to be used to back up CP. Operating lines of credit are, technically, demand loans that can be cancelled at any time but banks rarely do anything that could trigger financial problems for companies, which might leave them legally liable for damages suffered by the companies. A line of credit represents

a maximum amount that companies can draw on and transfer into a checking or current account. The standard cost of the operating line is the prime lending rate, which is a floating interest rate and changes regularly.

Banks make lines of credit available to its most creditworthy customers at rates below prime, particularly if they are large and can access the money market. Still, most companies are 'prime plus' borrowers, with rates quoted as prime plus a particular percentage.

A second type of line of credit, the term line of credit or revolving line of credit, gives companies additional stability. The amounts owed on the line of credit 'revolve' depending on the changes in the cash position of the company as it receives payment from customers or makes payments to suppliers.

The term is usually at least 364 days and generally as long as 5 years, renewable every 6 months. Generally, a five-year 'revolver' is renewed at the end of each year, making it an 'evergreen' five-year line of credit. A revolving line of credit is a credit commitment, so the bank has to provide capital against these commitments to ensure the liquidity of the bank. The 364-day line of credit happens because more capital is required against a 1-year or 365-day line.

Revolving lines of credit are flexible financing tools for both the bank and the company. They can be used for straight borrowings, to back up CP or other commitments like forward foreign currency contracts. Banks have tightened up on the potential uses for lines of credit; focusing on their uses related to liquidity and not credit enhancement. This means banks may withdraw or adjust lines of credit if there is a material adverse change to the credit quality of a company. The line of credit is outstanding to provide a company with liquidity that it can draw down only if its credit quality is constant.

To retain access to funds in a line of credit, companies have to meet a series of restrictions. The maximum value of the line of credit is determined using a standard formula. For example, a line may be set up so that it cannot exceed 75% of a company's receivables plus 50% of its inventory. Companies have to provide periodic abbreviated financial statements, sometimes as often as monthly. The company may also have to abide by financial restrictions such as maintaining a minimum current ratio, a net shareholders' **equity** or minimum interest coverage ratio or asset coverage ratio.

Banks can pull revolving lines of credit or prevent companies from borrowing beyond a certain limit based on the restrictions, but the restrictions can have only so much effectiveness because companies may have more than one bank account and may be aware before the bank knows if they are going to break the conditions. So, they may draw down their lines of credit before telling the bank of any violations and transfer the funds to other bank accounts. However, this could hurt a company's ability to borrow in the future or lead to higher fees.

2.1.4 Bank Overdrafts

In short-term financing, the chartered banks are important. Most firms rely on backup line of credit or **bank overdrafts** to ensure that funds are always available to pay off the liabilities. However, the line of credit is not available if the credit condition of the company (or security to which it is tied) deteriorates, and the credit condition is reviewed continually by the bank.

A bank overdraft is an extension of credit by a bank that kicks in when the balance of an account reaches zero. A bank overdraft makes it possible for the company (or individual) that holds the account to continue withdrawing money up to a certain limit. Often, the interest in an overdraft is higher than the interest in a bank loan.

2.1.5 Term Loans

Term loans differ from lines of credit because they have a fixed maturity, require repayment to be made on a fixed schedule and are made to finance longer-term requirements such as equipment purchases.

These loans are structured in a variety of ways and can qualify as both short-term and long-term financing. Some have bullet or balloon payments with interest payments only paid until maturity when the entire principal is due. Such loans generally provide permanent financing for company operations. By comparison, amortising loans are similar to conventional loans with monthly payment of both interest and principal, both valued as annuities.

Term loans may be based on the prime lending rate which is a floating rate, but they tend to be offered with a fixed interest rate. Insurance companies and business finance companies sometimes also offer term loans, and usually for longer periods of time, than banks.

Term loans have some advantages. One is that they are easy to arrange, particularly given that companies have relationships with banks, which means banks also have knowledge of the activities of the company. Also, a term loan can be structured to fit with operating and revolving lines of credit.

2.1.6 Family/Friends Loans

Loans from family or friends are amongst the most common sources of financing for start-ups.

Loans from family or friends often come with benefits that other types of financing, such as bank loans, do not provide. For starters, the lender may be more flexible about payment arrangements. Family and friends may also lend money at low or no interest at all.

Another benefit, one that applies to most loans, is that a loan only has to be paid back whereas investors have a say in the operations of the business.

The downsides of loans from family or friends for businesses may be personal disputes in the event of non-repayment.

In all cases, however, loans and repayments for loans to family and friends are a source of finance that is commonly used and has to be accounted for and reported. Basic terms for loan agreements with family or friends should include:

- the principal (amount borrowed);
- interest, if any; and
- repayment terms.

2.1.7 Leasing Financing

Firms do not purchase all assets they use. Many assets that businesses use are leased (rented). A **lease** is an agreement between a lessor, who is the legal owner of an asset, and a lessee, who obtains the rights to use an asset through a lease agreement. It is generally considered as long-term financing if it goes for more than a year.

Also, since business lease agreements generally require no deposit, they essentially represent 100% debt financing. For these reasons, the decision to purchase or lease an asset is directly related to the capital structure choices that managers make.

Leasing provides an alternative to financing an asset. A lease agreement is a financial arrangement in which the lessee pays the owner of an asset, the lessor, to use the asset for a period of time. A lease divides the right to use an asset into the right to use it during the term of the lease and the right to use it after the lease ends (known as salvage right).

For example, an oil company might lease a drilling rig for a few months or a transport company might lease trucks. Once the lease period is over, control goes back to the owner. Companies may lease all types of assets, including office space, furniture, computers, copy machines, cars, trucks, rail cars, airplanes, ships or drilling rigs, to name a few examples. And the length of the leases can vary widely, from a few minutes for a supercomputer to years for office space.

To calculate the value of an asset, lessors add the present value of the lease payments that will be received and the salvage value of the asset at the end of the lease. By way of example, consider a company that rents cars for one day and up to several years. The value of a one-day rental fee will be small compared to the total value of the car. If the daily rental fee for a HK\$200,000 car is HK\$500, this fee represents only 0.25% of the total value.

Alternatively, the lease payments on a seven-year lease might have a present value that equals the entire HK\$200,000 value of the car. In such a case, the lessee is effectively paying the same as it would pay to buy the car because the salvage value is likely to be small.

Illustrative Example 1

ABC Investment Ltd has to arrange for a line of credit. The following is taken from the company's most recent statement of financial position.

	HK\$
Cash	1,271,987
Accounts receivable	18,536,000
Inventory	74,196,000
Total current assets	94,003,987
Property and equipment (net)	126,323,000
Goodwill	46,888,000
Total assets	267,214,987

The bank provides credit of up to 75% of the value of receivables and 50% of the value of inventory. So, the maximum credit limit for ABC Investment is:

	HK\$
75% of receivables	13,902,000
50% of inventory	37,098,000
Credit limit	51,000,000

Illustrative Example 2

Great Stationary Ltd has a revolving line of credit of HK\$50 million. One restriction imposed with this financing arrangement is that Great Stationary has to maintain a minimum interest coverage ratio of 1.75. If this is the only borrowing, and the annual interest rate is 9.75%, how much does the company have to earn to abide by this restriction?

Amount borrowed	HK\$50,000,000
Interest rate	9.75%
Interest	HK\$4,875,000
Coverage ratio	1.75 times
Required earnings before interest	HK\$8,531,250

Key Learning Point

Short-term financing options for companies include commercial paper, bankers' acceptance, lines of credit, term loans, loans from family or friends, bank overdrafts and leasing.

► Knowledge Check Questions

Question 1

State the best way to describe short-term debt instruments, usually unsecured, issued by companies.

- A T-bills
- B Corporate bonds
- C Commercial paper
- D Asset-backed commercial paper

Question 2

Identify the best way to describe the quoted interest rate, which is the rate received if the issuer does not default and the investor is paid off on time.

- A Yield-to-call
- B Discount yield
- C Promised yield
- D Bond equivalent yield

► Knowledge Check Questions (*continued*)

Question 3

Identify which of the following statements is incorrect.

- A** Default risk affects promised yields on commercial paper.
- B** Investors can earn more than the promised rate if they hold the security to maturity.
- C** The yield spread is compensation for the risk that the commercial paper issuer defaults.
- D** The yield spread is the difference between the yield on one debt instrument and the yield on another.

2.2 LONG-TERM FINANCING

Long-term financing is debt issued for more than one year. It is sometimes called funded debt. Short-term debt is not permanent capital. Companies can use long-term debt to retire an accumulation of short-term debt.

Term loans are considered private financing and may be offered by banks, insurance companies or financial companies. Companies may rely on existing banking relationships for bank loans or rely on other information for a variety of other type of loans.

For public offerings, the Hong Kong Securities and Futures Commission (SFC) requires a prospectus. Companies issue an offering memorandum, a less detailed document, used for private placements of securities.

For public financing, it is important to realise that securities laws and regulations, and the required filing of a prospectus, apply to debt as well as equity offerings.

2.2.1 Common Stock

Common equity or **common stock** (also called ordinary stock) represents the residual ownership of the company. Common shareholders are the last in line if a company is liquidated, after creditors and preference shareholders.

Common shareholders may receive dividends only after preference shareholders receive theirs and after any interest and principal due on debt is paid. However, if a company generates income or value beyond what it needs to pay creditors and preference shareholders, common shareholders receive this income or value.

A company's statement of financial position shows common equity in the accounts of:

- Common stock, which is the par value of any issued shares although 'par value' is not used in Hong Kong.
- Additional paid-in capital, which is the difference between what investors paid for the shares at issuance and their value.

- Retained earnings.
- Accumulated comprehensive income or loss, which is any gain or loss that bypassed the statement of profit or loss and went directly to equity in the statement of financial position.

If the company has bought its own shares back from investors, the value of these shares is represented in treasury stock, which is a contra-equity account (that is, it is a subtraction from equity). The share bought back may be cancelled in accordance with the requirement of local regulations.

2.2.1.1 Rights

Generally, the owners' right to common equity interests are:

- vote at any shareholders' meeting of the company;
- receive any dividend pay-out declared by the company; and
- receive the remaining property of the company on dissolution.

These three rights then define the basic rights of common shareholders.

Common shareholders have the right to vote on important decisions at the annual shareholders' meeting. Shareholders can participate in decisions such as electing members of the board of directors, appointing auditors, making changes to articles of incorporation and bylaws and voting on major changes to company operations. It is hard to say what a major change is but it normally could include major acquisitions of other companies or the disposition of major assets.

Even these rights can be severely circumscribed. In the past, equity holders had a pre-emptive right, to maintain proportional ownership when new shares were issued. However, many companies changed their corporate structure to remove the pre-emptive right, allowing them to sell shares to new investors. Most companies have also increased the number of authorised shares and no longer need shareholders' approval to issue more shares. Companies could issue new shares quickly to major institutional investors, and thus dilute the ownership share of existing investors.

Two remaining rights that shareholders still have are to receive dividends and any remaining property when a company is dissolved. Equity holders do not have a right to a dividend until it is declared by the board of directors.

There are restrictions on when companies can declare a dividend, including the consideration that the company has to have the resources to pay the dividend without impairing its ability to meet contractual commitments. The payment of dividends reflects the fact that equity holders are last in line after all other claims on earnings streams.

Shareholders have the right to limited liability, meaning they cannot be held responsible for obligations beyond a specified amount. Equity holders are legally protected from being forced to put more money into a company. As a result, if a company cannot make interest payments, equity holders cannot be forced to inject more money into a company so that it can meet its commitments. If creditors put a company into bankruptcy and there is not enough money to pay off all claims, equity holders cannot be forced to put up more money. Limited liability ensures shareholders can only lose their initial investment, so their downside risk is limited but they have unlimited upside potential.

2.2.1.2 Multiple Class of Shares

While most corporations have one class of common stock, some have two. One class is publicly traded and another may be owned by a small group of investors. For example, the automotive company Ford has 71 million class B shares that trade on the New York Stock Exchange while the Ford family owns 5% of the company's stock but controls 40% through the founding family class of stock. Another example is Berkshire Hathaway, which has classes A and B shares with different ownership, voting rights and convertibility.

The articles of incorporation determine the rights of owners of a corporation. These are the rules of the governance of the corporation.

There is no particular meaning to the designation of the class of stock as 'A', 'B' or otherwise. This is arbitrary and at the discretion of the issuing company. In some cases, a class of shares may actually not have any voting rights and they are known as non-voting shares. Voting rights translate into control and the ability to impact decisions, such as electing the board of directors. Control is important. A company can only be taken over by buying the voting shares, which are more valuable than non-voting shares in this case. Many family-run businesses try to maintain control by putting in place two different kinds of shares for their companies, which allow for access to capital without giving up much control. With this in mind, limited or non-voting shares usually have a slight premium in terms of the right to a dividend to offset the lost value of control.

2.2.2 Preference Stock

Preference equity or **preference stock** is an equity interest in a company that is senior to common equity.

On a company's statement of financial position, preference equity includes preference stock, which is the par value of issued preference shares, and additional paid-in capital from preference shares, which is the difference between what investors paid for shares when they were issued and the par value of the shares. The most basic type of preference share is the traditional straight preference share, which is without a maturity date and pays a fixed dividend at regular intervals. These shares are typically valued as perpetuities.

2.2.2.1 Maturity

Straight preference stock is a perpetual security, which means it never matures. Some variations in preference shares have a limited life.

For instance, retractable preference stock – or term preference shares or redeemable preference shares – gives the investor the right to sell the stock back to the issuer, which creates an early maturity date.

Retractable preference shares allow for early retirement. The typical retraction date is five years. So, while preference shares may last longer, they are typically valued as if they mature at the end of that period.

One variation on retractable preference shares are soft-retractable preference shares, which give issuers the right to pay cash or in shares of common stock at redemption.

Mandatory redeemable preference stock is another type of preference shares with a maturity. These shares mature at a specified date or at a specified event, such as a merger.

Financial institutions issue these shares because they are considered equity capital for regulatory purposes.

Yet another type of preference stock is a trust preference security, through which a company creates a trust, issues debt to the trust, and then the trust issues preference stock.

2.2.2.2 Dividend

The basic type of preference stock has a fixed dividend stated as a fixed dollar amount or as a percentage of the par value of the preference stock for every period.

Not all preference stock has a fixed dividend. Some has a floating rate and this type of stock generally has a long maturity date and the dividend is reset every three or six months through an auction mechanism so that the dividend yield is in line with current market interest rates.

Alternatively, many floating rate preference shares issued by major banks have their dividend rate float with some base rate, such as the Hong Kong Interbank Offer Rate (HIBOR), usually to some maximum rate, but specify a floor or minimum dividend rate. This form of floating rate stock is generally known as adjustable rate preference stock (ARPS). ARPS have market values very close to their par value because their dividend rate is always very close to the current market rate.

Participating preference stock is a preference share that pays a fixed dividend and investors also receive an additional dividend if the dividend on common stock exceeds a specified amount. Preference stock is frequently used in venture capital financing of start-ups and may have a 'cap' or limit to the participation.

Preference shares typically have a cumulative provision, which means dividends on common shares are not paid until after they are paid in full on preference shares, both current and arrears. This ensures that common shareholders do not use their voting rights to suspend payment on preference share dividends. At times, some preference shares come with a provision that gives them some voting rights as arrears accumulate.

Companies in financial difficulties often suspend dividend payments. When they recover, however, they may have to make significant cash payments for the arrears of the preference dividends. In practice, what often happens is the arrears are paid through the issue of common shares or a combination of cash and shares.

A summary of the different types of dividends can be seen in Exhibit 2.3.

Dividend type	Description
Fixed	Fixed for the entire life of the preferred stock
Floating	Based on a specified benchmark
Adjustable rate	Based on periodic auctions
Participating	A fixed rate plus a share of income
Cumulative	Any missed dividends must be paid before current dividends to preferred or common shareholders

► **EXHIBIT 2.3** Types of dividends

2.2.2.3 Other Features

Preference stock may have other features.

One such feature is embedded options that allow investors to convert the preference shares for a predetermined number of common shares or another security.

Most convertible preference shares are callable, allowing the issuer to force conversion when the call price is unattractive relative to what the stock value investors would get at the time of conversion.

For instance, if a share of convertible preference stock is trading for HK\$150 and can be converted into common stock at one preference share for five common shares, then the value of the convertible preference stock likely differs from straight preference stock because investors may receive both dividends and, possibly, profits from the conversion. This leads to a conversion premium being built into the market value of the preference stock that moves in tandem with the value of the common stock.

Preference stock may be mandatory convertible preference stock, which requires conversion at or before a specified date.

Most preference stock is non-voting, so the investors do not have representation in the issuer's board of directors. Yet, preference shares may have a contingent voting right, which is a voting right that kicks in when the issuer does not pay the specified dividend.

2.2.3 Long-Term Loans

Long-term financing generally refers to any debt issued with a term longer than one year. When companies accumulate 'too much' short-term debt, they may fund this debt through long-term debt.

Banks and other financial institutions provide some long-term financing through term loans. Other forms of financing may involve public financing.

It is also common practice to use notes payable in long-term debt financing. Long-term notes payable are similar to short-term interest-bearing notes payable but for periods longer than a year.

A long-term note may be secured by a mortgage linked to a particular asset. Mortgages are commonly used to buy homes and companies often use them to acquire plant assets. The mortgage may have a fixed or adjustable rate of interest and the terms typically call for instalments to be paid throughout the term of the mortgage. Each payment includes both interest and principal on the loan. Payments do not change but the amount of interest paid in each instalment decreases and the amount of principal payment increases.

2.2.4 Bonds

A key source of long-term financing for large companies are **bond** issues. Bond issues can be very different from one to another and these differences are often based on a number of factors, which affect the nature of all bond issues.

2.2.4.1 Maturity

Debts mature when borrowers repay the amount borrowed and the length to maturity is specified at the beginning of the debt issuance.

One source of confusion is often terminology related to government debt. In general, a debt obligation that matures in less than 10 years is a 'note' or 'medium-term note' (MTN). An obligation with maturity between 4 and 10 years is an intermediate-term bond. Debt obligations with terms longer than 10 years are generally referred to as a bond or a long-term bond.

A note or bond generally requires a fixed payment in the future, which may be called 'face value', 'maturity value' and 'par value'. Some bonds include interest payments, with a fixed coupon rate specified as a percentage of the par value of the note or bond, or a floating rate specified relative to a reference rate such as the HIBOR. The fixed amount at maturity and the interest may be in the same currency as the issuer's home country or may be issued in another country's currency.

2.2.4.2 Security

Sometimes, issuers of debt provide collateral to back up their debt in case of a default. This type of debt is known as secured debt. A term debenture, for instance, is a debt obligation secured by the general credit of the issuer.

Bonds backed by specific collateral include mortgage bonds backed by real estate or asset-backed notes or bonds backed by assets, groups of assets or cash flows. Investors often expect greater returns on unsecured debt.

2.2.4.3 Seniority

Claims on debt security have an order of seniority that is made clear by how the debt is identified as senior or junior, unsubordinated or subordinated. Senior debt has preference over other obligations in terms of issuer's capitalisation, while junior debt ranks below other securities. Subordinated debt is unsecured debt that is ranked below other debts of the issuer and unsubordinated debt has no unsecured debt that ranks higher than it.

Seniority is a key consideration in determining the risk of a debt and its value. The more junior or subordinated a security, the greater the compensation that investors demand for risk.

2.2.4.4 Indenture Agreements

Debt may have an indenture agreement, part of a legal agreement between an issuer and an investor. The indenture agreement describes the interest, maturity, collateral and other features of the debt like callability and convertibility.

There is a great deal of flexibility in how the covenants are structured in an indenture agreement. A covenant is a provision that issuers must satisfy. Bond indentures have covenants. An affirmative covenant, refers to the promises an issuer makes, such as timely payments to investors. A negative covenant is a prohibition or restriction, such as prohibiting issuers from mortgaging or pledging assets to secure other debts. Restriction on paying dividends is another type of negative covenant.

2.2.4.5 Embedded Options

In addition to provisions with respect to maturity, seniority and collateral, notes and bonds may also have features that give the issuer flexibility in managing their financing costs and give investors flexibility to convert their investments.

These features are options embedded in the debt contract.

An example of an embedded option is a callable option on a bond. A bond may be callable according to a schedule of call prices, whereby the call price depends on the specific date the security is called from the investor.

Issuers gain flexibility in financing when a debt obligation is callable. A callable bond is a bond in which the issuer has the option to buy the bond back from the investor at a specified price, the call price.

The terms of a callable bond may specify call protection, a period of time in which the bond may not be called back. The call features add risk to investors because as interest rates fall, the probability that the security may be called away increases. As a result, investors may expect greater return on a callable security.

Investors may find greater flexibility investing in convertible debt, which can be changed into another security. The conversion value represents the value if it is converted immediately into common stock or other securities. This value establishes the lowest price for the value of the bond, which is the minimum of the conversion value or the value of the bond without the conversion.

2.2.4.6 Bond Rating

Ratings are a key component of the commercial paper market since this type of paper has a short maturity that makes it expensive to undertake credit analysis. On the other hand, institutions often do their own credit analysis for longer-term debt.

Bond ratings are important. Bond ratings from the major rating agencies are summarised in Exhibit 2.4.

Description	Standard & Poor's	Moody's Investors Service	Fitch Ratings	DBRS
Prime, maximum safety	AAA	Aaa	AAA	AAA
High grade, high quality	AA	Aa	AA	AA
Upper medium grade	A	A	A	A
Lower medium grade	BBB	Baa	BBB	BBB
Speculative	BB	Ba	BB	BB
Highly speculative	B	B	B	B
Substantial risk; poor standing	CCC	Caa	CCC	CCC
Extremely speculative		Ca		CC
May be in default		C		C
Default	D		DDD	D
			DD	
			D	

► EXHIBIT 2.4 Bond ratings

The rating of AAA/Aaa is the highest rating possible and each rating may be modified with a high or low rating to communicate further its creditworthiness within that class. The lowest

investment-grade bond rating is BBB/Baa, and below this, a bond is commonly referred to as a junk bond, although it is more politely referred to as a high-yield bond.

Bond ratings are the result of extensive consultation and the examination of at least five years of financial statements. On top of this research, rating firms also generally have prior knowledge of the company from their detailed industry study. Two basic principles used to determine ratings: the stable rating philosophy and the hierarchy principle.

The stable rating philosophy is based on the idea that ratings are based on structural factors, not cyclical ones. The fortunes of most companies will fluctuate with the business cycle: When the economy is hot, most companies will make money, and conversely, when it is in recession, most companies will struggle.

Holders of 20- or 30-year debt can expect the company to operate through many ups and downs of the business cycle. The rating firm aims to see through these predictable effects and change ratings only when a clear structural change occurs in the company's credit.

The hierarchy principle relies on the reality that companies do not issue ratings but rating agencies do. Ratings apply to specific debt, not to companies.

Rating firms consider six basic factors to determine ratings:

1. Core profitability: Based on standard profit measures.
2. Asset quality: Rating agencies look at the importance of intangibles like goodwill, the market value of assets, how it uses derivatives and risk management.
3. Strategy and management strength: Companies have assets and management, so it is important to consider the quality of senior management, particularly important for companies involved in mergers and acquisitions (M&A) activities.
4. Statement of financial position strength: To understand the financial position of a company, rating agencies may look at debt ratios, coverage tests, and the amount of financial flexibility available to a company. Rating agencies assess the company's reliance on short-term debt, its commitment to capital expenditure and support from other parties. Size is important because larger companies are less risky and may have more market power.
5. Business strength: Including market share, growth prospects, diversified operations, management information systems, key intangibles and industry issues.
6. Miscellaneous issues: A broad category such as quality of accounting statements, the structure of the bond indenture, and the importance of the company and industry to the economy.

Overall, rating debt instruments combines standard financial analysis based on the ratios with a broader company and industry analysis. The result is a mixture of quantitative and qualitative factors.

2.2.5 Hybrid Securities

A last major class of securities issued by companies, which includes warrants and convertible securities, are known as hybrid securities. Hybrid securities include both debt and equity components and may also include preference shares.

2.2.5.1 Warrants and Rights Issues

Rights issues give the 'right' or opportunity to buy or sell a particular security once a set of conditions are met. There are two main types of rights issues:

- Direct rights offerings that are sold directly to investors without any backstops, which means that no large institution investor is willing to purchase any unexercised offerings.
- Insured or standby rights offerings are more expensive but allow for institutions such as investment banks to purchase any unexercised rights.

A **warrant** is a security that provides the investor with the option to convert or exchange the warrant into another security, such as a common stock. A call option is the right to buy a specific asset (the underlying) at a specified price (the exercise price) within a specified period of time (the time to expiration). Warrants are the corporate finance equivalent of call options.

A warrant is a type of rights offering. Rights offerings give existing shareholders the right, but not necessarily the obligation, to buy shares in a company. Companies use rights issues to raise capital at lower prices. Investors often tap into them because they give them the option to buy shares in a company at discounted prices and then sell them in the market.

There are two major differences between warrants and call options. Call options are transactions between two external investors, so that what one gains the other loses without any impact on the company, and are usually issued with very short maturities, unlike warrants that are longer term.

As long-term options, warrants are extremely valuable and are frequently used as sweeteners to facilitate access to financing. For example, some early state mining companies use warrants to attract venture capital and avoid problems making interest payments. Because warrants have low interest rates, they make it easy for companies to manage payments over time.

Even though warrants are often granted along with another security, issuers can issue 'naked' warrants, which are not linked to another security. Warrants issued along with another security may also be detachable; which means investors may sell them on. Most detachable warrants are traded over the counter but they can also be listed on a stock exchange. Warrants linked to bonds are similar to convertible bonds.

A few considerations are important regarding call options:

- The lower the exercise price, the more valuable the option because the cost of buying the underlying asset is less, which makes it more likely that the value of the asset will be greater than the exercise price.
- Call options with longer expiration times are more valuable, because there is more time for the value of the underlying asset to exceed the exercise price.
- Call options linked to assets with more volatile values are more valuable because it is more likely that the price of the underlying asset will top the exercise price.
- Interest rates also have an impact. The higher the rate of interest, the greater the value of the option to put off acquiring or investing in the underlying stock.

2.2.5.2 Convertible Bonds and Contingent Convertible Bonds (Coco Bonds)

Corporate **convertible bonds** can be converted into shares of common stock at a predetermined ratio, based on the bondholder's discretion.

For example, a HK\$1,000 face-value bond may be convertible into 100 ordinary shares. Conversion features are valuable because investors benefit if stock prices rise.

In most cases, conversion ratios are set so that stock prices must rise at least 15–20% before it makes sense to convert bonds to stock. Conversion feature is valuable to bondholders, so that convertible bonds are issued at lower rates of interest, which also means companies can pay less interest.

Contingent convertible bonds, also known as **coco bonds**, resemble convertible bonds, which also have a strike price, or the price of the stock when the bond is converted into stock. There is an additional bar to cross, which is the conversion being triggered once specific capital conditions are reached. Coco bonds tend to benefit firms more than investors.

Advantages and Disadvantages of Convertible Bond and Coco Bond

Convertible bonds and coco bonds are hybrids, with features of both bonds and equity.

Some of the advantages of convertible bonds and coco bonds are:

- Lower cost.
- Lower interest rate than standard bonds because of their potential convertibility.
- The lower interest rate means companies have to use less cash.
- Typical conversion ratios require stock prices to rise at least 15–20% for it to make sense to convert bonds to stock.
- The costs and interest may be tax deductible, which offers some savings and lowers costs.
- When bonds are converted into equity, the company has no further obligations.

There are disadvantages to these types of bonds, however:

- These types of bonds may be converted into shares, which may lead to a dilution of the company's share base.
- A dilution of the share base could lower share prices.
- This dilution may also impact any future fundraising through seasoned share sales.

Illustrative Example 3

The hot Internet stock that ABC Investment Ltd bought last year does not look so hot anymore. In fact, the Internet company has just declared bankruptcy. At the time, ABC Investment bought shares equivalent to a 5% stake in the Internet company for HK\$5 million.

Creditors, such as equipment suppliers and employees, are looking to recover some of their investment. They are owed HK\$15 million.

Illustrative Example 3 (continued)

The question for ABC Investment Ltd is how much it will receive, as an equity holder, if the Internet company's assets are, when liquidated, worth HK\$17 million.

The calculation is relatively simple.

The amount that will be distributed amongst all shareholders will be, roughly, the HK\$17 million in assets minus the HK\$15 million owed to creditors. That leaves HK\$2 million to be distributed amongst all shareholders.

The amount that ABC Investment Ltd will receive depends on its shareholding, which is 5%. So, in broad strokes, ABC Investment Ltd is entitled to just HK\$100,000, a big loss for the company.

Key Learning Point

Long-term financing refers to debt with a maturity of more than one year. This type of debt is sometimes called funded debt. Short-term debt is not considered permanent capital, which means that when a company has too much short-term debt, it funds this debt using long-term debt.

► Knowledge Check Questions

Question 4

Identify which of the following statements about debt is correct.

- A** Interest payments are not tax deductible.
- B** Interest payments and principal payments must be the same amounts in all periods.
- C** Bond indenture is a legal document, specifying identity of bondholder, and so on.
- D** Bond buyers are paid a series of fixed periodic amounts before the maturity date.

Question 5

Identify which of the following is a certificate of ownership representing a proportionate share of the equity interest in a corporation.

- A** Preference share
- B** Warrant
- C** Hybrid security
- D** Common share



2.3 RAISING CAPITAL

Companies need to raise capital to fund their operations, launch new products or expand. While this capital may come from the initial investment of founders, from short-term or long-term borrowing, it may also come from other external sources that may inject funds into the company.

There are differences between capital that is raised from financing. Sources of capital fundraising include venture capital, tapping public funds through initial public offerings (IPO), cash offers, the private market and loans. Each of these has different characteristics and processes.

2.3.1 Venture Capitalists

Venture capital firms started raising capital through limited partnerships or funds in the 1960s and in doing so revolutionised the industry. Venture capital gives entrepreneurs, with limited access to traditional sources of funding, access to capital.

Traditional sources of funding are often not accessible for new businesses because:

1. there is too much risk;
2. they have few productive assets such as machinery, equipment, and physical inventory; and
3. they have issues with information asymmetry, because entrepreneurs know more about their companies than lenders.

Consequently, only a handful of new companies survive. To reduce their risk, venture capitalists use a number of tactics, such as providing funding in stages, requiring entrepreneurs' personal investments, creating syndicates and knowing about the industry they work in.

Staged funding allows venture capitalists to reassess the performance of the companies that they invest in at every stage and make changes to the investment or the company as needed. Companies usually go through three to seven funding stages.

Venture capitalists typically get an equity interest in the company for their investment, often in the form of preference stock convertible into common stock.

Syndication is common in seed- and early-stage venture capital investments, which happens when a venture capitalist sells part of a deal to other investors. Syndication can lower risk through diversification of the originating venture capitalist's investment portfolio.

Knowledge of an industry and technology also helps venture capitalists lower their risk. Specialisation can give venture investors an advantage over generalist lenders or investors.

However, venture capitalists are not long-term investors. They generally stay with a firm until it is successful, usually between three and seven years, and then exit. Most venture capital agreements include provisions on the exit process, such as the timing, method and price. In

general, venture capitalists exit firms by selling to a strategic buyer, selling to a financial buyer or selling through an initial public offering (IPO).

2.3.2 Initial Public Offering

Successful businesses eventually outgrow the ability of private sources of equity to fund their growth, so they may turn to the public through an **initial public offering (IPO)** of common stock. An IPO is the first sale of common stock by a company in the public market.

An IPO is the first sale of common stock by a company in the public market. The marketing and pricing of IPOs is somewhat different from seasoned public offering. A **seasoned public offering** is a sale of securities that has existing publicly traded securities. A public offering refers to securities registered with a regulator, as opposed to those sold in the private market.

The decision for an IPO depends on a balance of the advantages and disadvantages.

Potential advantages on going public:

- It makes it possible to raise larger amounts of capital.
- It allows firms to raise more capital through follow-on seasoned offerings.
- It allows entrepreneurs to retain control.
- There is a secondary market for stockholders to buy and sell shares.
- An IPO may make it easier to attract top management talent.

Potential disadvantages on going public:

- IPOs can be expensive because the stock is not seasoned and has no established record. It also includes multiple costs such as legal fees, accounting expenses, printing, travel expenses, filing fees, consultant fees and taxes.
- The costs of complying with ongoing disclosure can be a significant disadvantage. In Hong Kong, the compliance costs may be lower for companies listed on the GEM, compared to the Main Board.
- Disclosures can create transparency issues such as more disclosures than other private firms which could lead to a competitive disadvantage.
- Regulatory requirements for quarterly filings may lead to a focus on short-term profits.

Firms work with investment bankers to complete an IPO. Investment bankers provide three basic services:

1. Origination, including giving advice on IPO process.
2. Underwriting, i.e. bearing risk.
3. Distribution, i.e. reselling securities to the public.

2.3.3 General Cash Offer by a Public Company

Companies continue to need funding after going public and they want to find the cheapest possible funding. A general cash offer may be the cheapest source of funding for financially sound firms.

The procedures in a general cash offer are similar to those involved in an IPO and are summarised here:

1. Type of security and amount to be raised.
2. Approvals from the board of directors.
3. Registration statement with the regulator.
4. Offer price.
5. Closing of the offering.

The issuer has flexibility in how securities are sold and how they are registered, two factors that affect cost.

In a general cash offer, management may sell securities on a competitive or a negotiated basis.

In a competitive sale, the issuer hires an investment bank to do the origination for selling a specific type and number of securities. The issuer then invites underwriters to bid on the issue and the winning underwriter pays for the securities and makes them available to investors at offer price.

In a negotiated sale, the issuer works with a specific underwriter to start the origination process. The origination and underwriter's fees are negotiated, and the issuer and underwriter work closely on the issue, including assessment of market demand and determining the offer timing and offer price.

It is not entirely clear which method leads to lower funding costs.

The argument for competitive sale is that more competition should lead to lower costs. The argument for negotiated sale is that the investment banker working closely with the issuer has an intimate knowledge of the company which reduces uncertainty and results in ultimately lower costs. Also, a negotiated sale may also create competition from underwriters who were not chosen but try to underwrite the potential next issue, and may also provide the same benefits as a competitive sale.

Ultimately, the best method depends on different factors such as the type of securities, the complexity of the issue and market condition.

Most experts say a competitive sale is cheaper to sell vanilla bonds – bonds with standardised and simple terms which are well-known to investors – during stable market conditions. However, when the market is unstable or there are complex circumstances to consider, negotiated sales might provide a less costly method of sale. For equity securities, negotiated sales tend to have lower costs because of the complexity of equity issue which can be better handled under negotiated sales. Virtually all equity issues involve negotiated sales.

Preparing an SFC registration statement is also an expensive process.

Issuers have to consider costs like the underwriting spread and out-of-pocket expenses as well.

Issuing common stock is the costliest option and issuing non-convertible corporate bonds is the cheapest. For instance, for a security issue of US\$500 million or more, the cost of issuing common stock is about 3.63% of the amount raised compared to just 0.65% for corporate bonds. The difference is due to underwriting risk, sales commissions and higher administrative expenses.

There are also significant economies of scale in both underwriting spreads and out-of-pocket expenses.

2.3.4 Private Market and Bank Loans

Firms may need – or prefer – to tap private markets through private placements of equity or debt. Smaller firms or firms with lower credit ratings may not have access to public markets and have to rely on private markets for the lowest-cost funding.

Another factor that may affect whether firms can sell securities to the public is market conditions. During times of unstable conditions, firms may find it difficult to sell securities to the public because investors want to avoid risk during these times, a phenomenon called ‘flight to quality’ on Wall Street.

2.3.4.1 Private Placements

Large private companies, often owned by entrepreneurs or families, sometimes choose to sell securities in private markets, even though they can access public markets. Firms may avoid public markets to avoid the regulatory costs and transparency requirements or believe their business structures are too complex and difficult to explain to the public at large.

A private placement, in the private market, happens when firms sell unregistered securities to investors. Sales of both debt or equity can be placed privately. Roughly half of all corporate debt is sold in private markets.

Banks often help firms with private placements by finding potential buyers, arranging deals and doing origination work and helping negotiate terms and price, but not underwriting issues. Compared to public offerings, private placements may have lower cost of funds and private lenders may be more willing to negotiate changes to a bond contract, if needed.

Additionally, firms in financial distress may be able to solve problems without going to bankruptcy court. Firms can tap into funds faster with more flexible structures, but private placements may also come with restrictions on the resale of the securities.

Private placement securities have limited marketability. Because the issues are not marketable, so investors often require a higher yield than comparable public offerings.

2.3.4.2 Private Equity

Private equity (PE) companies bring together funds from a variety of sources, including financial and insurance firms, pension funds, individuals, families, corporations, foundations and endowments. Private equity firms are more inclined to invest in more mature companies. Private equity managers usually look to increase the value of the companies they purchase, and then sell them for a profit. PE companies tend to hold investments for three to five years.

PE firms may also provide funds for established businesses with operations that they may be interested in selling because they no longer fit their strategies or for which they are offered a good price. PE firms typically set up funds organised as limited partnerships or limited liability companies that include both general partners who manage investments and limited partners who invest money.

Limited partners share in the income, capital gains and tax benefits from PE funds, while general partners earn income, capital gains and tax benefits. General partners collect management fees and a percentage of income and capital gains in return for their management of PE funds. PE funds have historically focused on investing in small and medium-size firms with stable cash flows and with potential to scale up, but an increase in the amount of capital available to invest has led PE firms to do large deals.

A PE firm may borrow three or four times as much money as it invests. By using debt, PE firms free up their own cash to make other investments. The acquisition of a company using a large amount of debt, or leverage, is called a leveraged buyout.

Private equity firms rely on a number of techniques to improve the performance of the firms in which they invest. One is to put in place the best possible management teams. A second is to closely monitor performance and advise management. A third is to facilitate mergers and acquisitions to improve the competitive positions of a company.

In certain instances, there might be conflicts of interest arising from the separation of owners and management but these problems are typically smaller in companies owned by PE firms than in public firms.

In public firms, managers make most day-to-day decisions as agents of stockholders, but managers may pursue their own self-interest, a misalignment that may result in agency costs. Still, PE funds may have lower agency costs than the publicly held firms because general partners in PE funds are owners and directly benefit from creating value.

A final consideration is that PE companies face a much smaller regulatory burden and fewer reporting requirements compared to public companies.

Private investments in public equity, or PIPE, are transactions in which a public company sells unregistered stock to an investor, such as a hedge fund, or to other institutional investors. Since the 1990s, PIPE transactions have become more common. In a PIPE transaction, investors buy securities from a public company in a private placement, almost always at a discount that makes up for limits on the marketability of securities, because the securities cannot be resold in public markets.

As part of the PIPE contract, the company often agrees to register the restricted securities, usually within 90 days of the PIPE closing. Once the securities are registered, they can be resold in the secondary markets. If the issuer cannot register the security, it do be required to pay damages of as much as 1.5% per month as compensation.

Nevertheless, a PIPE transaction gives issuers faster access to capital at lower costs than public offerings.

2.3.4.3 Bank Loans

Another important source of funds for businesses are simple bank loans. Most small and medium-size companies regularly borrow from commercial banks.

The most common type of loan is a prime-rate loan, with an interest rate based on the prime rate, which is used to finance working capital. Banks often require a zero-loan balance for a short time each year to make sure this type of loan is not used as long-term funding.

Banks may provide a number of services with their loans, including financial advices, which may add to the cost of a loan.

Term loans are typically business loans with maturities greater than one year. Term loans are the most common form of long-term financing provided by commercial banks. There is significant variation in how loans are structured, with maturities ranging between 1 and 15 years, although typically in the 1-to-5-year range.

Bank term loans may be secured or unsecured with the funds used for a variety of purposes.

► Knowledge Check Questions

Question 6

Identify which of the following makes venture capital funding unique.

- A The degree of risk
- B The reality that primary assets in most new firms are intangibles
- C Information asymmetry
- D All of the above

Question 7

Identify which of the following is true for investment banks.

- A They do audits
- B They cannot issue IPOs
- C They resell securities
- D They do not participate in secondary markets



2.4 EXTERNAL FINANCING versus INTERNAL FINANCING

Funding (money) is required to undertake virtually any business activity, so finance is the lifeblood of any company. It is what keeps it going. Depending from where the finance or funding comes, it can be classified as internal financing or external financing.

Internal financing covers finance that comes from within the business, such as retained earnings or working capital that is effectively managed. External financing comes from external sources, such as financial institutions, investors or public fundraising, as discussed in the above sections.

Both have their own advantages and disadvantages as well as requirements.

2.4.1 Internal Financing

Internal sources of finance are generated within the business, typically from existing assets or activities. Activities can include the sale of stock or fixed assets, debt collection, working capital or, frequently, retained earnings.

2.4.1.1 Retained Earnings

Perhaps the most common source of internal financing is retained earnings. Retained earnings is net income or profit that a company retains in the business.

While the balance in retained earnings noted in the financial statements is part of the shareholders' claim on the total assets of the company, these retained earnings are not a claim on any specific asset and business can use them to fund further growth or operations rather than pay them out to shareholders as dividends. Companies generally disclose retained earnings restrictions in the notes to the financial statements.

2.4.1.2 Effective Working Capital Management

Another important source of internal financing is working capital. And to be effective as a source of finance, working capital has to be managed efficiently.

Most working capital management activities are focused on the following working capital accounts:

1. Cash (including marketable securities)
2. Receivables, prepaid assets
3. Inventory
4. Payables

A decision to increase working capital should lead to increased sales, better relationships with vendors and suppliers, and fewer work stoppages. However, there is a cost to extra working capital and there is no easy way to determine the optimal level for any one firm. Management makes a strategic choice.

One important concept in working capital management is the **cash conversion cycle**, or the amount of time from when a company pays for raw materials until it receives cash from sales. The length of the cash conversion cycle is directly connected to how much capital a company needs.

There are a series of steps to follow from when a company pays for raw materials to when it receives payment for finished goods:

1. The company purchases raw materials and incurs conversion costs to convert raw materials into finished goods.
2. Finished goods are held in finished goods inventory.
3. Finished goods are sold, often on credit.
4. Customers repay the credit and the company receives the cash.

There are several goals for managers to manage the cash conversion cycle, including:

- Delay paying accounts payable without penalties as long as possible.
- Maintain the minimum required level of raw material inventories to support production without any production delays.

- Use as few inputs as possible while maintaining quality.
- Maintain an optimal level of finished goods inventory to minimise investment in the inventory but avoid losing sales.
- Offer attractive credit to boost sales, but minimise financing costs and risk of non-payment.
- Collect receivables as quickly as possible.

The aim for managers is to maximise value by managing the length of the cash conversion cycle. To do this, managers have to consider two key questions. First, what is the appropriate amount and mix of short-term assets for the firm? Second, how should the firm finance these short-term assets?

Firms should hold some short-term assets to operate smoothly. Without enough cash on hand, companies may not be able to pay their bills. Without a raw materials inventory, production might stop. Without a finished goods inventory, sales might be lost.

A number of factors are key considerations in any discussion of working capital management:

1. **Current assets:** Cash and other assets that are to be converted into cash in a year or less, such as cash, marketable securities, accounts receivable, inventory and prepaid expenses. These assets are usually listed on the statement of financial position in order of liquidity.
2. **Current liabilities:** Also called short-term liabilities, these include obligations to be repaid in a year or less. Some may bear interest, like short-term notes and long-term debt maturing during the current period while accounts payable, accrued expenses, accrued taxes and wages may not carry interest.
3. **Working capital:** Or gross working capital, includes funds in cash and marketable security accounts, accounts receivable, inventory and other current assets. Working capital is defined in terms of current assets, which means the two terms are the same.
4. **Net working capital (NWC)** refers to the difference between current assets and current liabilities. NWC measures liquidity. It is the amount of working capital a company has left after paying off its short-term liabilities. The larger the NWC, the more liquidity a firm has. Most firms have a positive NWC.
5. Working capital management focuses on managing current assets and their financing. Financial managers determine the optimum balance for each current asset account and determine the mix of short-term debt, long-term debt and equity.
6. Working capital efficiency refers to how efficiently working capital is used and is measured by the cash conversion cycle. Firms with shorter cash conversion cycles are using their capital more efficiently.
7. **Liquidity** refers to how easily a company can convert assets into cash without losses.

Cash Management

The timing and size of cash inflows and outflows do not always match and this leads to a requirement to manage working capital accounts and make sure the company can operate with the smallest investments of working capital. This management is based on analysis of cost-benefit trade-offs.

Cash, which produces no interest, is important for three reasons.

The first is to make transactions possible. Cash is required to make payments to suppliers, staff and the government and comes in from customers. Companies often hold on to cash because the inflows and outflows of cash do not always match. Firms that run out of cash have to sell investments or borrow.

The second reason is to make sure the firm can overcome unexpected crises or benefit from unexpected opportunities.

The third reason is that banks generally require companies to hold minimum cash balances for loans and other services, which are known as compensating balances.

Managers tend to focus on transaction and precautionary motives when making decisions on how much cash to keep on hand. This amount should also satisfy any compensating balance requirements that the company has. The compensating balance is the basic level of cash a firm needs to hold.

Another factor to consider when deciding how much cash to keep on hand is how a company collects payments. The collection time, or float, refers to the time between the moment a company makes a payment and when cash becomes available to the company.

Collection time can be broken down into three parts. First is delivery or mailing time. Second is processing delay. Third, a delay between the time of deposit and the time cash is available for withdrawal.

Different types of payments have different cash collection cycles. Cash payments at the point of sale have zero cash collection time. Cheques or credit cards at the point of sale have no mailing time but could be associated with some processing and availability delays.

Any moves to reduce the total collection time also reduces the total cash requirements of a firm. The ability of a firm to implement policies that lower collection times may depend on the industry that they operate in and customer expectations.

Credit Management

Working capital must be funded somehow. Options include short-term debt, long-term debt, equity or a mixture. Financial managers use various strategies to manage credit.

A key consideration is that some working capital needs are short term while others are long term, or even permanent. The amount of working capital also fluctuates over time. For instance, finished goods inventories might increase before the holiday season; working capital will remain high after finished goods inventories are sold and converted into accounts receivable, and will then decline after receivables are collected.

The minimum level of working capital needed even for the slowest part of the year, is permanent working capital which is the level of working capital that the firm will always need for operations.

As businesses expand, they would require more working capital and more long-term productive assets. There are three basic strategies to finance working capital and fixed assets:

1. A **maturity matching strategy**: In this strategy, all seasonal working capital needs are funded with short-term borrowing. Short-term borrowing fluctuates with the level of

sales and working capital. Long-term financing is used to fund permanent working capital and fixed assets. This is one of the most basic techniques financial managers use to reduce risk when financing assets.

2. A **long-term funding strategy**: This strategy uses long-term debt and long-term equity to finance fixed assets, permanent working capital and seasonal working capital.
3. A **short-term funding strategy**: In this strategy, short-term debt is used to fund all seasonal working capital and part of the permanent working capital and fixed assets. This strategy can lower the costs of funding for a firm but may also require a company to periodically refinance long-term assets.

Every strategy has benefits and costs, and financial managers tend to use variations of any of the strategies to meet their objectives.

Many financial managers try to match the maturities of assets and liabilities, using short-term financing for short-term assets and long-term financing for long-term assets. At the same time, however, firms also look for the lowest financing costs. Sometimes using short-term financing may lead to lower rates but leave the firm exposed to any large increases in interest rates. This also means that managers prefer to use long-term assets with long-term debt to lock in the cost of capital and eliminate risk. Long-term debt is also frequently used to fund permanent working capital, but some managers prefer to use a mix of short-term and long-term debt.

Inventory Management

Another consideration for managers is to determine how much inventory to carry. There are two sides to inventory management. Management typically wants to have variety and quantity of inventory to give customers choice but this may lead to high carrying costs. However, low inventory levels may lead to lost sales (see Module 2, Chapter 3). The inventory turnover ratio and days in inventory are common measures used to determine inventory levels.

Inventory turnover measures how many times inventory is sold during an accounting period. It is a measure of the liquidity of the inventory. Inventory turnover is calculated by dividing the cost of goods sold by the average amount of inventory. This can be done using beginning and ending inventory balances.

Days in inventory is another measure related to inventory turnover but measures the average number of days that inventory is held. It is calculated by dividing 365 by the inventory turnover and gives managers an estimate of how long it takes a company to sell its inventory.

Companies that are able to keep inventory low and turnover high while meeting customer needs are most likely to succeed.

Manufacturing companies generally carry three types of inventory:

1. Raw materials
2. Work in process
3. Finished goods

Another important consideration, one linked to how much inventory to hold, is how much to order. A larger inventory gives companies more time before they have to order again, which can translate into fewer orders through the year.

The **economic order quantity** (EOQ) model is a tool that helps managers to determine the appropriate quantity for a particular type of inventory to order. This model mathematically determines order quantities to minimise costs associated with ordering and holding inventory. The model accounts for inventory reorder and carrying costs. Determining the optimal amount of inventory requires finding the trade-off between the two costs.

Larger orders increase average inventory size and the carrying cost of inventory but lower reorder cost. Therefore, the optimal order size strikes a balance between the two costs (see Module 2 Chapter 3).

Many companies now use just-in-time inventory management techniques, first developed by Japanese companies like the Toyota Motor Company. The system can significantly reduce costs and is based on daily (or even hourly) needs of companies, with suppliers delivering raw materials just as they are needed, which basically eliminates raw material inventory costs and the risks associated with obsolescence or theft. The system does increase risks associated with suppliers, which could cause a shutdown if they fail to deliver in time.

2.4.2 Advantages and Disadvantages of External Financing

External financing comes from outside the company. This includes sources of funds such as financial institutions (banks), commercial paper, sales of stock, trade credit and others.

Unlike internal financing, external financing may be expensive and may require some collateral. However, the amounts raised that may be generated from external financing – debt financing, equity financing or a combination – tend to be much larger than those generated through internal financing.

Companies may choose external financing through debt financing or equity financing. The former is simply borrowing while the latter involves raising money in exchange for a partial ownership of the company. Both approaches have their advantages and disadvantages.

For companies, debt financing has a number of benefits compared to equity financing including:

- The lender has no claim on the equity of the business. A loan (debt) does not dilute ownership of a company.
- Companies repay the principal plus interest but the lender has no further claim on any future profits of a business.
- In general (except with variable rate loans) payment of principal and interest are known and predictable.
- Interest is often tax deductible, which lowers the cost of the loan.
- Debt financing is easier than equity financing.
- Companies do not have to provide regular reports to lenders as they do with investors.

However, debt financing has a number of disadvantages:

- Debt must be repaid, equity financing does not.
- Because interest costs are fixed, the break-even point becomes higher and high interest costs could become a significant burden.
- Cash flow is required to pay both the principal and interest.

- Some loans restrict the activities of a company or the financing options that management can pursue.
- The higher the amount of debt, the riskier the company may be considered.
- Sometimes, companies have to put up assets as collateral.

Equity financing also has its benefits, compared to debt financing. These benefits include:

- Equity financing does not have to be repaid, so it does not add to the costs of the business.
- For companies with poor or no credit history, equity financing may facilitate access to funding.
- Equity financing may also provide access to other benefits, such as the expertise and networks of investors.

However, equity financing has a number of disadvantages:

- Equity financing is tied to an ownership stake in the company, which means future profits may be shared through dividend payment.
- Investors hold an ownership stake in the company, which dilutes ownership stakes and can lead to a loss of control.

2.4.3 Advantages and Disadvantages of Internal Financing

A key benefit of internal financing is that it costs less than external financing and does not require any collateral. A key downside is that the amounts raised tend to be lower than external financing and often limited to what the business can generate on its own.

The main sources of internal financing are retained earnings and working capital.

Companies may choose to use the cash resulting from the excess of revenues over expenses to purchase buildings, equipment, and other assets.

The balance in retained earnings is generally available for dividend declarations or further investment in the business.

A part of the retained earnings may not be available for dividends as a result of retained earnings restrictions. Restrictions are some of the key disadvantages of using internal financing and result from any of the following:

- Legal. Many governments require companies to restrict retained earnings for the cost of treasury shares purchased, which keeps the legal capital of the corporation intact until the treasury shares are sold.
- Contractual. Long-term debt contracts may include clauses to restrict earnings, limiting the use of corporate assets for dividends and increasing the likelihood that the corporation will meet its obligations.
- Voluntary restrictions. The board of directors may create retained earnings restrictions. By reducing the portion of retained earnings available for dividends, the company can make more capital available for other purposes.

Using internal sources of financing also has a number of advantages, including:

- Lower costs of funds. Without the use of intermediaries, companies may be liable for lower or no interest payments.
- Fewer restrictions on the use of funds. Banks or investors may put conditions or restrictions on how companies use the capital that they provide, regardless of whether it is a loan or an investment.
- No need for collateral. When using internal funds, companies do not have to guarantee those funds with assets, a common requirement of banks or other investors.
- Ownership is not diluted. The use of internal funds means companies do not have to issue additional shares or give up any ownership stake in exchange for funds. In turn, this means that ownership is not diluted.

In the end, firms make decisions on whether to use internal or external financing based on their current circumstances, the cost of funds, the levels of working capital and the future plans for the company.

► Knowledge Check Questions

Question 8

Identify which of the following statements is true.

- A** The cash conversion cycle begins when a firm purchases raw materials to be used in the production of goods.
- B** The cash conversion cycle begins when a firm uses cash to purchase raw materials and ends when it collects payments on sales.
- C** The cash conversion cycle is the longer the better.
- D** The cash conversion cycle does not end with the finished goods being sold to customers and the cash collected on sales; but the time taken by a firm to pay for its purchases.

Question 9

Identify which of the following statements is true of economic order quantity (EOQ).

- A** The EOQ mathematically determines the minimum total inventory cost.
- B** The EOQ takes into account reorder costs and inventory carrying costs.
- C** The optimal order size is determined by the EOQ model.
- D** All of the above.



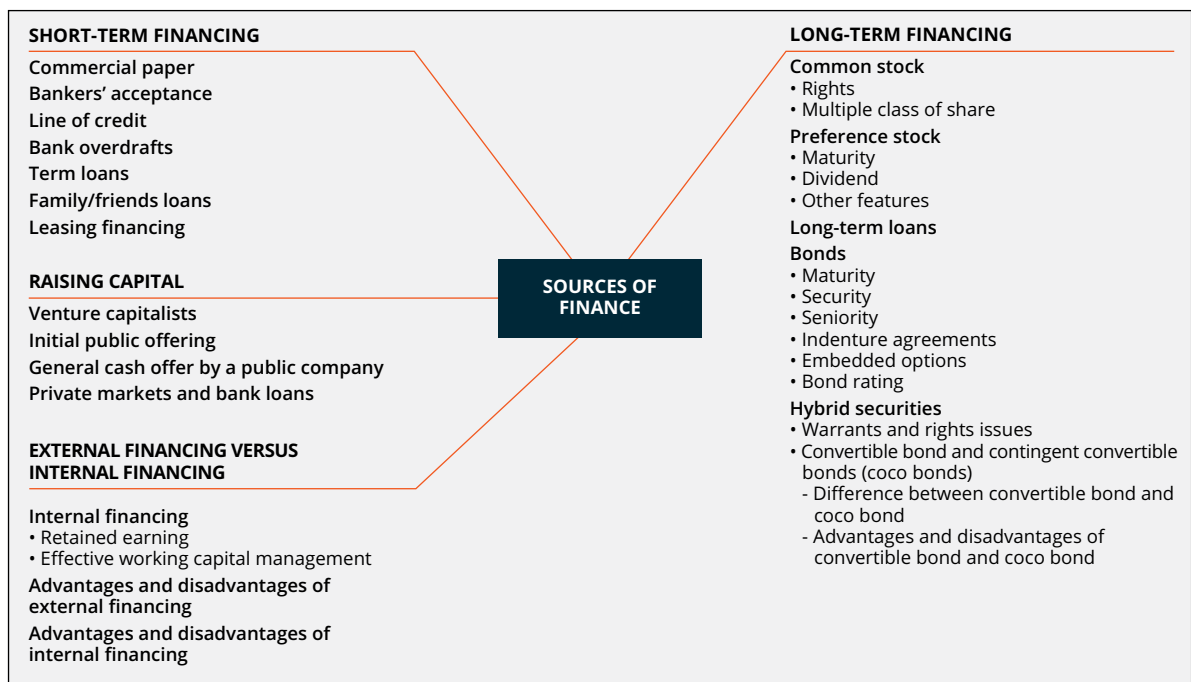
SUMMARY

- For investing purposes, a company may choose to invest in short-term instruments that are issued by the government, or it may choose to invest in instruments issued by financial or non-financial companies.
- Companies can issue commercial paper (CP), which is a name for short-term promissory notes. Commercial paper is normally issued with maturities from 1 to 270 days, but the typical commercial paper issue is for a maturity of 30 days.
- Commercial paper is most often issued at a discount from its face value, though issuers may choose to issue the paper as an interest-bearing obligation.
- The quoted interest rate is a promised yield. The difference between the promised yield on commercial paper and the yield on the equivalent-maturity T-bill is called the yield spread.
- The actual risk of investing in commercial paper is extremely low.
- Another form of commercial paper is asset-backed commercial paper. Asset-backed commercial paper (ABCP) is commercial paper secured with specific assets, such as receivables, and is typically issued by special purpose vehicles (SPV).
- A bankers' acceptance (BA) is created as a result of international trade. Typically, an exporter will receive a 'bill of exchange' in return for the export of its goods. Bankers' acceptances are usually issued at a discount from their face value, and the maturity may range from 1 to 180 days, though typically the maturity is 90 days.
- The line of credit is a basic financing tool provided by banks. In general, the borrower pays interest on the amount borrowed under the line of credit.
- Term loans differ from lines of credit because they have a fixed maturity, require repayment to be made on a fixed schedule, and are made to finance longer-term requirements such as equipment purchases.
- Loans from family or friends may be more flexible about payment arrangements. Family and friends may also lend money at low or no interest at all.
- In short-term financing, the chartered banks are important. Most firms rely on backup lines of credit or bank overdrafts to ensure that funds are always available to pay off the liabilities.
- A lease is an agreement between a lessor, who is the legal owner of an asset, and a lessee, who obtains the rights to use an asset through a lease agreement.
- Long-term financing generally refers to any debt issued with a term longer than one year and is often called funded debt.
- Common equity or common stock is the residual ownership of the company. As such, common shareholders are the last in line if the company is liquidated, waiting in line behind creditors and preference shareholders.
- Generally, owners of common equity interests have the right to vote at any shareholders' meeting of the company, receive any dividend payout declared by the company and receive the remaining property of the company on dissolution.

- Most corporations have a single class of common stock. However, some corporations will have two classes, with one class publicly traded and another class owned by a small group of investors, such as the founding family or founding investors.
- Preference equity or preference stock is an equity interest in a company that is senior to common equity.
- Straight preference stock is a perpetual security; it never matures. However, some variations in preference shares provide a finite life to the stock.
- The basic type of preference stock has a dividend that is fixed in amount, which is stated as a fixed dollar amount per period or as a percentage of the par value of the preference stock.
- Preference stock may also have other features, such as an embedded option. Convertible preference stock has an embedded option that permits the investor to convert the preference stock for a specified number of common shares
- A warrant is a security that provides the investor with the option to convert or exchange the warrant into another security, such as a common stock.
- Similar to a warrant, a call option is the right to buy a specified asset (the underlying) at a specified price (the exercise price) within a specified period of time (the time to expiration).
- Corporate convertible bonds can be converted into common shares at a predetermined ratio set by the bondholder.
- Contingent convertible bonds, also known as coco bonds, resemble convertible bonds, which also have a strike price, or the price of the stock when the bond is converted into stock
- The maturity of a debt obligation is the point in time at which the borrower repays the amount borrowed. This maturity is generally established when the debt obligation is issued.
- In general use, a debt obligation that matures in less than 10 years is often referred to as a note or a medium-term note (MTN). An obligation with a maturity between 4 and 10 years is often referred to as an intermediate-term bond. Debt obligations with terms longer than 10 years are generally referred to as a bond or a long-term bond.
- We refer to a debt obligation that is backed by collateral as secured debt, where this collateral may be a specific asset, a group of assets, or a specific cash flow.
- Senior debt has preference over other debt in terms of the issuer's capitalisation. A junior debt ranks behind other securities. A subordinated debt is unsecured and ranks behind other debts of the issuer. Unsubordinated debt has no other unsecured debt that ranks higher than it.
- Debt obligations may have an indenture agreement, which is part of the legal agreement between the issuer and the investor.
- Ratings are essential in the commercial paper market because investments have such a short maturity. The rating of AAA/Aaa is the highest rating possible and the lowest investment-grade bond rating is BBB/Baa, and below this, a bond is commonly referred to as a junk bond.
- Companies sometimes receive funding from governments in the form of loans and assistance.
- Venture capital allows entrepreneurs to access funding when traditional sources are inaccessible due to the high degree of risk, lack of assets and information asymmetry.
- Companies can raise larger sums of cash or facilitate the exit of venture capital through an initial public offering (IPO).

- The advantages of going public include larger amounts of capital, the possibility of raising follow-on capital, retaining control and attracting top management talent. However, the costs can be high.
- A general cash offer is a sale of debt or equity open to all investors. It is made by public companies that have sold stock in the past
- Private equity firms establish private equity funds to make investments.
- Small- and medium-size companies can have difficulties raising money in the public markets. It may be more efficient or cheaper to sell stock privately.
- The most common type of business loan is a prime-rate loan. Term loans are typically business loans with maturities greater than one year.
- Depending on where the finance comes from, it can be classified as internal financing or external financing.
- The most common source of internal financing is retained earnings. Retained earnings is net income that a company retains in the business. Another important source of internal financing is working capital.
- Cash is an important element of working capital. Cash (and many demand deposits) do not generate interest but firms still hold positive cash balances for various reasons.
- External financing comes from outside the company. This includes sources of funds such as financial institutions (banks), commercial paper, sales of stock, trade credit and others.

MIND MAP



► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. T-bills are the short-term debts issued by US government.

Answer B is incorrect. Corporate bonds are long-term debts.

Answer C is correct. Commercial paper (CP) are short-term debt instruments, usually unsecured, issued by companies.

Answer D is incorrect. Asset-backed commercial paper is secured by assets.

Question 2

Answer A is incorrect. It is the yield to the call date.

Answer B is incorrect. Discount yield is used to calculate the yield on municipal notes, commercial paper and treasury bills sold at a discount.

Answer C is correct. The promised yield is the quoted interest rate received if the issuer does not default and the investor is paid off on time, as promised.

Answer D is incorrect. It is the effective rate of commercial paper.

Question 3

Answer A is incorrect. A higher default risk will increase the promise and vice versa.

Answer B is correct. The promised yield is the maximum an investor can earn by holding the security to maturity.

Answer C is incorrect. The yield spread or credit spread is the difference between the quoted rates of return on two different investments, usually of different credit qualities but similar maturities. It is often an indication of the risk premium for one investment product over another.

Answer D is incorrect. The yield spread or credit spread is the difference between the quoted rates of return on two different investments, usually of different credit qualities but similar maturities. It is often an indication of the risk premium for one investment product over another.

Question 4

Answer A is incorrect. Interest payment is tax deductible.

Answer B is incorrect. Interest payment depends on companies' policies.

Answer C is incorrect. The indenture agreement describes a debt obligation's interest, maturity, any collateral, and other features, such as callability and convertibility.

Answer D is correct. Bond issuers are the parties that have to pay bond buyers a series of fixed payments before the maturities.

Question 5

Answer A is incorrect. A preference share has no claim on a firm's residual value.

Answer B is incorrect. A warrant has no claim on ownership until it is converted into a common share.

Answer C is incorrect. Hybrid security has no claim on ownership until it is converted into a common share.

Answer D is correct. By definition, common share is a certificate of ownership in a corporation.

Question 6

Answer A is incorrect. It is only one of the characteristics of venture capital funding.

Answer B is incorrect. It is only one of the characteristics of venture capital funding.

Answer B is incorrect. It is only one of the characteristics of venture capital funding.

Answer D is correct. Venture capital funding carries a high risk. Most new firms' assets are primarily intangible. Information asymmetry is a serious issue in venture capital funding.

Question 7

Answer A is incorrect. Usually accounting firms provide auditing service instead of investment banks.

Answer B is incorrect. Investment banks can be listed, e.g. JP Morgan is listed in NYSE.

Answer C is correct. One of the job duties of investment banks is to bring new securities to market by helping private firms to go public.

Answer D is incorrect. Investment banks are involved in the secondary market without restrictions.

Question 8

Answer A is incorrect. Cash conversion is concerned with actual receiving and paying cash, rather than purchases or sales on credit.

Answer B is correct. By definition, the cash conversion cycle is the length of time from the point at which a company actually pays for raw materials until the point at which it receives cash from the sale of finished goods made from those materials.

Answer C is incorrect. A shorter cash conversion cycle means a higher liquidity of the firm.

Answer D is incorrect. The cash conversion cycle ends with the finished goods sold to customers and the cash collected on the sales.

Question 9

Answer A is incorrect. The other two statements are also true.

Answer B is incorrect. The other two statements are also true.

Answer C is incorrect. The other two statements are also true.

Answer D is correct. The formula of EOQ is made up by the recorder costs, sales per period and the carrying costs and to find the optimal level of inventory that has to be ordered which is also the minimum total inventory cost.

EXAM PRACTICE

QUESTION 1

Identify the best way to describe equity which is senior with respect to claims on income and assets.

- A. Debt
- B. Preference share
- C. Hybrid security
- D. Common share

QUESTION 2

Identify which of the following is not part of shareholders' equity.

- A. Treasury stock
- B. Accumulated earnings less dividends
- C. Accumulated comprehensive income or loss
- D. Capital that owners paid in to buy the initial shares of stock

QUESTION 3

Identify which of the following is an advantage to a firm that goes public.

- A. Low cost of going public
- B. It is easier for the firm to attract top talent
- C. It gives entrepreneurs more control over the firm
- D. It is cheaper than issuing debt

QUESTION 4

State the three basic rights of common shareholders.

QUESTION 5

Explain the differences between a call option and a warrant.

QUESTION 6

Explain why a company would consider issuing a long-term debt obligation if it can borrow funds by issuing commercial paper.

QUESTION 7

Once its bonds are issued, a company should not be concerned about its debt rating. Analyse this statement and explain whether it is true or not.

QUESTION 8

A majority of firms choose a firm-commitment underwriting arrangement rather than a best-effort arrangement for their IPO. Justify why this is the case.

QUESTION 9

Describe the four primary types of short-term financing in the money market (commercial paper, banker's acceptance, line of credit and term loans).

QUESTION 10

Identify the three primary classifications of common equity found on a corporate statement of financial position.

QUESTION 11

Explain why a company might reveal more than one class of common stock.

 ANSWERS TO EXAM PRACTICE**QUESTION 1**

Answer A is incorrect. Debt has no claim on income and assets.

Answer B is correct. By definition, preference share is a certificate of ownership of a fixed claim on the equity of a corporation. Preference equity is senior to common equity with respect to claims on income and assets.

Answer C is incorrect. Hybrid security has no claim on income and assets until it is converted into equity.

Answer D is incorrect. Common share's claim on profit and assets comes after preference share.

QUESTION 2

Answer A is correct. The shares bought back by the company are treasury shares and are a contra account to equity; the amount paid for these shares is subtracted from equity.

Answer B is incorrect. Accumulated earnings less dividends is part of equity.

Answer C is incorrect. Accumulated comprehensive income or loss is part of equity.

Answer D is incorrect. They will transfer to shareholders' equity after their accounts were closed.

QUESTION 3

Answer A is incorrect. The cost of going public is high, e.g. fee to underwriters and other professionals.

Answer B is correct. Going public reduces entrepreneurs' control over the firm, but going public makes it easier to attract top management talent and to better motivate current managers.

Answer C is incorrect. The ownership will be distributed to shareholders once it goes public.

Answer D is incorrect. IPOs have high costs.

QUESTION 4

The right to vote.

The right to receive any declared dividends.

The right to the residual property in the event of the dissolution of the corporation.

QUESTION 5

A call option is an option held by the issuer of a security that permits the issuer to buy back the security at a specified price within a specified period of time. A warrant is an option held by a bondholder (or bought separately, if separable) that permits the bondholder to exercise the warrant to buy a specified number of common shares at a specified price.

QUESTION 6

If the company has an ongoing need for the funds, issuing a long-term debt may be preferred if there is risk that the company may not be able to roll over its commercial paper (e.g. a credit crunch). In addition, there may be instances in which the yield curve is abnormal, such that short-term rates are higher than long-term rates.

QUESTION 7

Even though a company has already issued its bonds, it should be concerned about its debt rating, because this will determine its cost of capital on future money needs whether these are from bank loans or additional bond offerings. A lowered credit rating can even affect the performance of a company's stock. The debt rating also impacts the company's ability to do things beyond just borrow, such as recruit and compensate employees, negotiate with suppliers, and position itself against the competition.

QUESTION 8

With a firm-commitment arrangement, the issuing firm is guaranteed to receive a specific amount of money from proceeds of the IPO. The underwriter bears the price risk of the issue. Price risk is the risk that the resale price to the market might be lower than that of the agreed price to the issuing firm.

QUESTION 9

In the money market, commercial paper refers to one form of borrowing. It is a short-term promissory note and may be used to bridge periods between transaction and payment; or to provide financing for activities not requiring permanent lending.

The bankers' acceptance is a document offering a promise to pay. It serves as a bill of exchange and is often used for financing of exported goods.

A line of credit in its most common form is a specific maximum level of funds that a borrower can access as needed. Interest is charged only on outstanding balances taken out, rather than on a larger fixed sum of loan. The repayment schedule is also flexible, so that a borrower can move in and out of varying levels of debt and keep interest expense to a minimum.

A term loan has a fixed maturity and repayment schedule. It is more formal than a line of credit because the entire amount loaned is given to the borrower at the point of origination. Periodic payments (usually monthly) include a calculated interest as well as principal, and the entire loan must be retired by the final due date.

QUESTION 10

Common equity is part of the 'net worth' section of the statement of financial position, which is the net difference between assets and liabilities. The common portion of net worth has three major parts. Common stock is the par value of shares issued and outstanding. This may be reduced if the company purchases some of its own shares and retires them as treasury stock, which reduces the balance of common stock.

A second category is paid-in capital, which is new capital provided in the form of equity beyond the initial par value of common stock.

The third category is retained earnings, the accumulated value of net profit or loss reported each year. As the books are closed at the end of the fiscal year, the previous retained earnings balance is increased by net profits or decreased by net losses.

QUESTION 11

Multiple classes of common stock are possible, although in most cases a single class is shown on the corporation's books. This is the par value of common shares issued and outstanding.

A second class of common shares may be held by a small group of investors, often the founders of the company.

A third form of common stock may specify distinctions from other classes by voting rights or levels of declared dividends.

Part B

Investing Fundamentals



Chapter 3 Time Value of Money and Investing Basics

Chapter 4 Introduction to Bond Investments

Chapter 5 Equities Valuation

3

Time Value of Money and Investing Basics

CHAPTER TOPIC LIST

3.1 Interest

- 3.1.1** Simple Interest
- 3.1.2** Compound Interest

3.2 Time Value of Money

- 3.2.1** Future Value and Compounding
- 3.2.2** Present Value and Discounting

3.3 Discounted Cash Flows and Valuation

- 3.3.1** Multiple Cash Flows
- 3.3.2** Level Cash Flows
- 3.3.3** Constant Growth Cash Flows

3.4 Annualised Interest Rates

- 3.4.1** Annual Percentage Rate
- 3.4.2** Effective Annual Rate



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE

LO7.01: Explain and analyse the theories of major capital market models

7.01.01 Explain the concept of time value of money involving simple interest and compound interest



OPENING CASE

ABC INVESTMENT LTD

ABC Investment Ltd, an investment subsidiary of ABC Entertainment Ltd, is presented with a series of opportunities in bonds and annuities. They have different interest rates that are compounded in different ways. At first glance, these instruments may look deceptively similar, but they vary hugely in values based on how investment returns are computed or discounted.

Compound interest and simple interest may look similar in calculation, but their results are a world apart. ABC Investment Ltd's management takes into consideration various opportunities, such as the time value of money, and an essential concept for valuing stocks and bonds: 'One dollar today is always worth more than one dollar tomorrow'.

ABC Investment Ltd's Chief Investment Officer (CIO) reviews how the investment opportunities are packaged and presented using varying interest computing methods and compound periods.

Various sellers naturally try to induce the CIO into buying an investment that does not provide the best possible return. This makes it imperative for the CIO to have clear information about the present and future values that can be derived from each of the opportunities.

The nuances of interest computing methods and compound periods can be difficult to grasp. The difference between annual percentage rate (APR) and effective annual rate (EAR) has been routinely used by credit card companies to wring maximum income from credit card holders.

Understanding these concepts is key for the CIO's advisers to present accurately and make it possible for the CIO to make the best and most profitable decision possible. 

OVERVIEW

Interest income represents an important source of return on investment. The two types of interest are simple interest and compound interest. Simple interest is consistent throughout the period of investment, as it is based solely on the initial investment, or principal. Compound interest, which is more commonly used, increases over time, as it is based on both the principal and any previously accrued interest.

Understanding interest and its relationship with money is key to understanding another fundamental concept for financial managers: the time value of money. Just like the saying 'a bird in the hand is worth two in the bush', HK\$100 held today is worth more than HK\$100 that will arrive tomorrow.

This chapter explores other concepts associated with this axiom. Another important one is future value, a measure of how much an investment or investments will be worth after earning interest for one or more time periods, that is, after the interest has compounded. Present value, in contrast, is a measure of how much a future investment or investments are worth now and is calculated by discounting.

Calculating the present value of a series of equal amounts to be received (or paid) at equal intervals over some future period of time is a common necessity. An annuity is a series of equal dollar amounts. An ordinary annuity – as compared to an annuity due – means that the series of equal amounts is received (or paid) at the end of each period. In an annuity due, the series of equal amounts is received (or paid) at the beginning of each period.

Interest rates can be annualised. The annual percentage rate (APR) is the annualised effective interest rate without compounding on loans that are for a fraction of a year. The effective annual rate (EAR), also known as the annual percentage yield, is the annual percentage rate with compounding on loans that are for a fraction of a year.

3.1 INTEREST

Interest refers to the charge assessed for borrowing money, or what investors may expect to receive from investing (or lending) money. It is a fundamental component of investing and financing. Understanding the complexities of how interest is calculated and how it plays into value creation is key to the work of a financial accountant.

In broad strokes, there are two types of interest – although each is affected by other variables. The first is simple interest and the second is compound interest.



In essence, interest is a payment for the use of money belonging to another party. The difference between an amount borrowed, called **principal**, and the total that is repaid is the interest. How much interest is paid or collected is typically set as a rate covering a period of time and stated as an annual rate.

There are three elements involved in the amount of interest in a transaction:

1. *Principal (P)*
2. *Interest Rate (i)*
3. *Time (n)*

3.1.1 Simple Interest

Simple interest is interest paid or received on only the initial investment (the principal).

Although, in practice, simple interest is used for a limited number of applications, it is worth introducing it first to contrast it with compound interest, which is used extensively.

The equation for calculating simple interest can be used to find the value of an investment accruing simple interest at any time. That equation is:

$$\text{Interest} = P \times i \times n$$

where 'P' is the amount of the principal, 'i' is the interest rate and 'n' is the amount of time in question.

3.1.2 Compound Interest

Compound interest is interest earned on the principal and on future interest payments and can lead to rapid growth of an investment. The growth is related to both the returns earned and the length of the period for which the money is borrowed. Compound interest is the return on the principal for two or more time periods.

Compounding computes interest on the principal and on any interest earned as long as the interest is left on deposit.

The formula to compute compound interest is:

$$\text{Interest} = P \left(1 + \frac{i}{n} \right)^{nt}$$

where:

- P = principal,
- i = interest rate
- n = number of times interest is compounded, and
- t = number of years that apply.

To explain the difference between simple and compound interest, assume that ABC Entertainment Ltd deposits HK\$1,000,000 at a simple interest of 2% per year. Using simple interest, that principal will earn ABC Entertainment Ltd HK\$20,000 per year for as long as it is deposited. After three years, ABC Entertainment will have HK\$1,060,000.

At the same time, ABC Entertainment Ltd deposits another HK\$1,000,000 also at 2% per year with another institution, but this time earning compound interest of 2% per year, compounded annually. After the same three years, ABC Entertainment will have HKD\$1,000,000 $\times (1 + 2\%)^3 = \text{HK\$1,061,208}$.

Compound interest uses the accumulation of both principal and interest at the end of each year to calculate interest for the next year. This helps explain why a compound interest account is larger and the difference grows larger over time.

Compounding assumes that unpaid interest earned converts to form part of the principal. The accumulated balance at the end of a period converts to form the new principal on which interest is earned during the next period. Most business situations use compound interest. Simple interest is commonly applicable only to short-term situations of one year or less.

Illustrative Example 1

Two bonds of HK\$100 million each are presented to ABC Investment Ltd to consider as possible investments.

One pays 10% simple interest, the other 10% compound interest. The maturity for both is 20 years.

The question is which one will generate higher returns? The chart below shows the interest rates and ending balances of both simple and compound rate bonds.

Value of HK\$100 million invested at 10% simple and compound interest rate						
Year	Simple interest			Compound interest		
	Beginning balance	Interest	Ending balance	Beginning balance	Interest	Ending balance
1	100	10	110	100.0	10.0	110.0
2	110	10	120	110.0	11.0	121.0
3	120	10	130	121.0	12.1	133.1
10	190	10	200	235.8	24.0	259.8
20	290	10	300	611.6	61.0	672.6

As the chart shows, the investment value grows much faster with the bond that offers compound interest at the same 10% rate. The longer the investment horizon, the greater the differential in the return on investment.

By the end of year 10, the compounded return is already 30% greater than the simple return (HK\$259.8 million vs HK\$200 million). By the end of 20 years, the difference is more than double (HK\$672.6 million vs HK\$300 million).

The bond that pays 10% compound interest produces the superior investment return.

Illustrative Example 2

The chief financial officer (CFO) of a company has undertaken to update the forecasts for the investment portfolio. Under the current estimate, the company will earn 4% on its portfolio over the coming year.

However, this forecast is expressed as simple interest.

What would be the return on investment based on an interest rate of 4% compounded quarterly?

Answer

The year's starting balance is HK\$1,215,800.

Simple return was calculated as:

$$\text{HK\$1,215,800} \times 4\% = \text{HK\$48,632}$$

Converting this total to compound a return, the 4% per year represents 1% per quarter. The recalculated return is:

$$\text{First quarter: } 1,215,800 \times 1\% = 12,158, \text{ total HK\$1,227,958}$$

$$\text{Second quarter: } 1,227,958 \times 1\% = 12,280, \text{ total HK\$1,240,238}$$

$$\text{Third quarter: } 1,240,238 \times 1\% = 12,402, \text{ total, HK\$1,252,640}$$

$$\text{Fourth quarter: } 1,252,640 \times 1\% = 12,526, \text{ total HK\$1,265,166}$$

$$\text{Total compound return: } 1,265,166 - 1,215,800 = \text{HK\$49,366}$$

Key Learning Point

Compound interest can result in dramatic growth in the value of an investment over time.

► Knowledge Check Questions

Question 1

Mr. Wong invested HK\$12,000 in a security that pays 2% annual simple interest. Calculate how much interest Mr. Wong will earn in the third year.

- A HK\$160
- B HK\$240
- C HK\$720
- D HK\$12,735

► Knowledge Check Questions (*continued*)

Question 2

Mr. Tang makes two investments: GHI and RST. Both pay interest at the end of each year and neither interest rate changes over time. The interest earned each year is:

Year	GHI	RST
1	50	50.00
2	50	52.50
3	50	55.13
4	50	57.88

State which of the following is most correct.

- A Both GHI and RST earn simple interest.
- B Only GHI earns simple interest while RST earns compound interest.
- C GHI earns compound interest but RST earns simple interest.
- D Both GHI and RST earn compound interest.

Question 3

State what the interest earned on both the original investment and the accumulated interest over time is called.

- A Growth rate
- B Compound interest
- C Simple interest
- D Cost of capital

3.2 TIME VALUE OF MONEY

Virtually every area of financial management requires understanding and use of the **time value of money**. Although money is a unit of measure used in financial management, not all money has the same value. This is in large part due to the time value associated with money.

Money currently held is worth more than money that will be received in the future. In order to analyse investment proposals and other opportunities, money that relates to different points or periods in time must be adjusted to comparable values. This is done by converting all dollars (or whatever currency is being used) into their current or present value.

3.2.1 Future Value and Compounding

The value of an investment after earning interest for one or more time periods is called future value (FV) and this process of turning an initial investment into FV is called **compounding**. The concepts of future value and compounding are illustrated with an example.

The **future value** is the value at a future date of a given amount invested, assuming compound interest. The formula to calculate future value is:

$$\text{Future Value (FV)} = P \times (1 + i)^n$$

where:

- FV = future value of an amount,
- P = principal,
- i = interest rate for one period, and
- n = number of periods.

Suppose HK\$100 is deposited in a bank savings account that pays interest at 10% a year. How much money will be in the account in three year? The calculation may look something like this:

$$\begin{aligned} \text{Future value at the end of year 3} &= \text{Principal} + \text{Interest earned} \\ &= 100 \times (1 + 0.10)^3 \\ &= 100 \times (1.331) \\ &= \text{HK\$133.10} \end{aligned}$$

Another way to compute the future value of a single amount includes a future value table (Exhibit 3.1). Exhibit 3.1 shows the compounding periods (n), periodic interest rates, and the future value factors (which are the five-digit decimal numbers in each column). To use the table, multiply the principal by the future value factor for the number of periods and the interest rate. For instance, the future value factor for two periods at 9% is 1.18810. If one multiplies this by HK\$1,000,000 equals HK\$1,188,100, which is the accumulated balance at the end of year 2 of an HK\$1,000,000 deposit at 9% interest.

(n) Periods	4%	5%	6%	7%	8%	9%	10%	11%	12%	15%
0	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
1	1.04000	1.05000	1.06000	1.07000	1.08000	1.09000	1.10000	1.11000	1.12000	1.15000
2	1.08160	1.10250	1.12360	1.14490	1.16640	1.18810	1.21000	1.23210	1.25440	1.32250
3	1.12486	1.15763	1.19102	1.22504	1.25971	1.29503	1.33100	1.36763	1.40493	1.52088
4	1.16986	1.21551	1.26248	1.31080	1.36049	1.41158	1.46410	1.51807	1.57352	1.74901
5	1.21665	1.27628	1.33823	1.40255	1.46933	1.53862	1.61051	1.68506	1.76234	2.01136
6	1.26532	1.34010	1.41852	1.50073	1.58687	1.67710	1.77156	1.87041	1.97382	2.31306
7	1.31593	1.40710	1.50363	1.60578	1.71382	1.82804	1.94872	2.07616	2.21068	2.66002
8	1.36857	1.47746	1.59385	1.71819	1.85093	1.99256	2.14359	2.30454	2.47596	3.05902
9	1.42331	1.55133	1.68948	1.83846	1.99900	2.17189	2.35795	2.55803	2.77308	3.51788
10	1.48024	1.62889	1.79085	1.96715	2.15892	2.36736	2.59374	2.83942	3.10585	4.04556
11	1.53945	1.71034	1.89830	2.10485	2.33164	2.58043	2.85312	3.15176	3.47855	4.65239
12	1.60103	1.79586	2.01220	2.25219	2.51817	2.81267	3.13843	3.49845	3.89598	5.35025

► **EXHIBIT 3.1** Future value table

(n) Periods	4%	5%	6%	7%	8%	9%	10%	11%	12%	15%
13	1.66507	1.88565	2.13293	2.40985	2.71962	3.06581	3.45227	3.88328	4.36349	6.15279
14	1.73168	1.97993	2.26090	2.57853	2.93719	3.34173	3.79750	4.31044	4.88711	7.07571
15	1.80094	2.07893	2.39656	2.75903	3.17217	3.64248	4.17725	4.78459	5.47357	8.13706
16	1.87298	2.18287	2.54035	2.95216	3.42594	3.97031	4.59497	5.31089	6.13039	9.35762
17	1.94790	2.29202	2.69277	3.15882	3.70002	4.32763	5.05447	5.89509	6.86604	10.76126
18	2.02582	2.40662	2.85434	3.37993	3.99602	4.71712	5.55992	6.54355	7.68997	12.37545
19	2.10685	2.52695	3.02560	3.61653	4.31570	5.14166	6.11591	7.26334	8.61276	14.23177
20	2.19112	2.65330	3.20714	3.86968	4.66096	5.60441	6.72750	8.06231	9.64629	16.36654

► **EXHIBIT 3.1 (Continued)**

3.2.2 Present Value and Discounting

The calculation of the present value of a single future amount is frequently used in assessing investment opportunities for which a portion of the cash flows relate to a single amount to be received – or paid – in the future.

In order to determine the present value of an amount of money to be received in the future, a discount – or interest – rate must be used. That rate measures the time value attached to money by the entity.

The **present value** (PV) is the value now of an amount that will be paid or received at a future time with compound interest. PV like FV is based on the following three variables:

1. the amount to be received in the future;
2. the number of periods until the amount is received; and
3. the interest or discount rate.

Determining PV is often called **discounting** the future amount.

These calculations of PV can be used to measure a variety of items such as the market price of a bond, notes payable and lease liabilities. Capital budgeting and other investments are also evaluated using PV. PV techniques are also used in rate of return and internal rate of return calculations.

The formula for present value is:

$$\text{Present Value (PV)} = \text{Future Value (FV)} / (1 + i)^n$$

Illustrative Example 3

A financial professional is asked to evaluate two bonds. One will pay HK\$100 million in five years. The other will pay HK\$96 million in four years. The asking price for both bonds is the same at HK\$64 million.

If the funding cost is 10%, which bond should the financial professional recommend for purchase?

Illustrative Example 3 (continued)

The first step is to consider the present value of both bonds, as seen in the present value table.

Present value of bonds based on 10% of discount rate						
Year	5-year maturity			4-year maturity		
	Pays HK\$100 million at maturity			Pays HK\$96 million at maturity		
	Ending balance	Discount factor	Beginning balance	Ending balance	Discount factor	Beginning balance
5	100	1.1	91			
4	91	1.1	83	96	1.1	87
3	83	1.1	75	87	1.1	79
2	75	1.1	68	79	1.1	72
1	68	1.1	62 Present value	72	1.1	66 Present value
Bond price			64			

By discounting the bond value at 10% per annum (discount factor of 1.1), the 5-year bond has a present value of HK\$62 million and the 4-year bond has a present value of HK\$66 million.

The present value is how much each bond is worth to your company based on its funding cost.

One should never pay more than what a financial product is worth today. The bond price is HK\$64 million for each.

Given the above, the clear recommendation should be to purchase the 4-year bond because the present value (HK\$66 million) is higher than the price that has to be paid today (HK\$64 million).

Key Learning Point

Money held today is worth more than the same amount held in the future.

► Knowledge Check Questions**Question 4**

If Mr. Tam is indifferent between receiving HK\$1,000 today and HK\$1,100 in one year, calculate the percentage of his opportunity cost.

- A 1.10%
- B 10%
- C 110%
- D 100%

► Knowledge Check Questions (*continued*)

Question 5

Lottery A pays HK\$1,000 today and Lottery B pays HK\$1,750 at the end of five years from now. If the discount rate is 5%, determine which lottery is the best choice.

- A** Lottery A, because it is available to me now.
- B** Lottery A, because its future value is HK\$1,276.
- C** Lottery B, because its present value is HK\$1,371 which is more than that of Lottery A.
- D** Lottery B, because it pays HK\$1,750 which is more than HK\$1,000 from Lottery A.

Question 6

An investor puts HK\$2,000 at 5% compounded annually. Compute the value of the investment in five years.

- A** HK\$500.00
- B** HK\$552.56
- C** HK\$2,500.00
- D** HK\$2,552.56

3.3 DISCOUNTED CASH FLOWS AND VALUATION

Calculating the value of different cash flows from investments made at different times is a key skill for financial managers.

The idea of the time value of money suggests that money today is worth more than money that will be received in the future. This concept flows into other important concepts of simple interest, compound interest and future value calculations. All of them flow into present value and discounted cash flow analysis.

All these concepts may be applied to multiple cash flows. Most business decisions include cash flows over time.

For example, if ABC Entertainment Ltd wants to consider adding a new product, it has to analyse the expected cash flows over time. At first, there will be a large outlay of cash to build production lines but the project should produce inflows for years into the future. These cash flows happen over time, so this analysis should take the time value of money into account and discount cash flows using the PV formula.

3.3.1 Multiple Cash Flows

Suppose ABC Entertainment Ltd is considering investing more than a single lump sum of money – as most companies do.

To make the numbers easier to work, the company puts HK\$1,000 in a bank savings account today and another HK\$1,000 in a year. How much money will the company have after two years if the bank pays 10% interest per year?

To solve this FV problem, it is possible to use the equation for future value:

$$\text{Future Value (FV)} = P \times (1 + i)^n$$

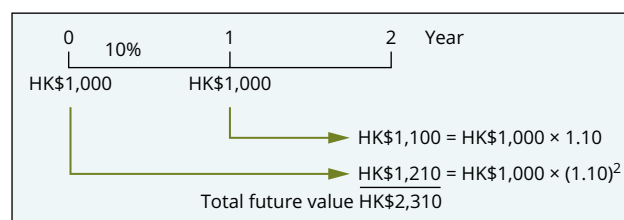
But first, it is possible to construct a timeline to clearly observe the size and the timing of cash flows. Exhibit 3.2 shows two cash flows. The first is invested for two years and the compounding is as follows:

$$\begin{aligned} FV_2 &= p \times (1 + i)^n \\ &= 1,000 \times (1 + 0.10)^2 \\ &= 1,000 \times 1.21 \\ &= \text{HK\$1,210} \end{aligned}$$

The second set of cash flows earn simple interest for one period and grow to:

$$\begin{aligned} FV_1 &= p \times (1 + i)^n \\ &= 1,000 \times (1 + 0.10) \\ &= 1,000 \times 1.10 \\ &= \text{HK\$1,100} \end{aligned}$$

Exhibit 3.2 shows a total amount of HK\$2,310 in the account after two years, which is made up of the sum of these two amounts HK\$1,100 + HK\$1,210.



► **EXHIBIT 3.2 Multiple cash flows** (Source: data from Kidwell 2018. Reproduced with permission of John Wiley & Sons.)

Using this approach, it is possible to expand the number of investments and cash flows and calculate the total return from multiple cash flows over time.

To summarise, solving FV problems with multiple cash flows is a process consisting of three steps.

1. Draw a time line of cash flows
2. Calculate the FV of each cash flow
3. Add up all FVs

It is necessary to calculate PV of a series of future cash flows, for example, to determine the price of a bond, make a decision to buy a new machine or determine how much a business is worth.

Solving PV linked to multiple cash flows is not unlike solving FV problems linked to multiple cash flows. A key consideration is determining the appropriate discount rate. When investing in capital projects, for example, it may be appropriate to use the project's cost of capital as the discount rate to evaluate a long-term investment project.

Many companies use the weighted average cost of capital (WACC) as the project's cost of capital. The WACC is the after-tax cost of a dollar of additional long-term financing to the company. Companies finance long-term investments by using long-term financing. This is appropriate if the project under consideration is an 'average' risk investment project for the company.

If a company is considering a project that is atypical, more or less risky than usual, it should reflect this in the discount rate chosen. For example, a company could be considering an investment that entails greater (or lower) risks than its traditional investments, say investing in a stock instead of a government bond. Under these circumstances, a higher discount rate should be used to account for the higher risk. If a company does not adjust the discount rate under these circumstances and merely uses a constant WACC to select investment projects, it will make inappropriate decisions.

3.3.2 Level Cash Flows

Contracts that require the payment of equal amounts of cash over several time periods are common in finance. Most term loans and insurance policies require a series of equal payments that are typically made monthly. Almost all consumer loans are paid in this fashion. An **annuity** is a financial contract based on regular cash flows over a limited number of periods. In a **perpetuity**, the payments continue forever.

Some cash flows, such as those from annuities and perpetuities, are more straightforward to calculate.

3.3.2.1 Annuities

Calculating the present value of a series of equal amounts to be received (or paid) at equal intervals over some future period of time is a common necessity.

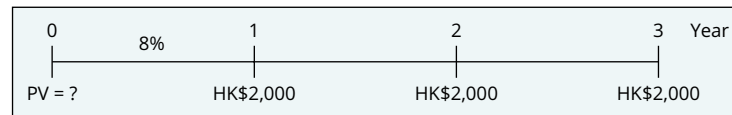
An annuity is a series of equal dollar amounts. An ordinary annuity means that the series of equal amounts is received (or paid) at the end of each period. In an annuity due, the series of equal amounts is received (or paid) at the beginning of each period. The present value of the series of amounts to be received (or paid) in the future will be less than the sum of the future nominal values of those amounts.

The calculation of the future value of an annuity due determines the value at some future date of a series of equal amounts to be paid (or received) at equal intervals over some future period of time with the amounts paid (or received) at the beginning of each period. The future value will be greater than the sum of the series of the amounts as a result of interest earned on each amount. Because the first amount is paid (or received) at the start of the

annuity, the amount of interest earned and the amount at the end of the annuity will be greater than the future value of an ordinary annuity.

Present Value of an Ordinary Annuity

It is often necessary to determine the PV of an annuity (PVA). For instance, if a financial contract pays HK\$2,000 at the end of each of three years and the discount rate is 8%, then the time line for this contract is:



To do this, it is necessary to calculate the PV of each cash flow (CF_n) and add them together.

This approach works if the number of cash flows is small. In situations involving annuities, however, the number of cash flows is large. Doing the calculations manually may be time consuming. For example, 25-year home loans are spread across 300 monthly payments. The problem can be simplified. All the cash flows (CF) in an annuity are the same, so the present value of an annuity (PVA_n) with n equal cash flows (CF) at interest rate i is the sum of the individual PV calculations:

$$PVA_n = \left[CF \times \frac{1}{1+i} \right] + \left[CF \times \frac{1}{(1+i)^2} \right] + \cdots + \left[CF \times \frac{1}{(1+i)^n} \right]$$

It is possible to simplify this equation to the following, which is a useful formula for the present value of an annuity (PVA):

$$\begin{aligned} PVA_n &= \frac{CF}{i} \times \left[1 - \frac{1}{(1+i)^n} \right] \\ &= CF \times \left[\frac{1 - 1/(1+i)^n}{i} \right] \end{aligned}$$

where:

- PVA_n = PV of an n period annuity,
- CF = level and equally spaced cash flows,
- i = interest rate or discount rate, and
- n = number of periods.

Let's apply the equation to the example involving a 3-year annuity with a HK\$2,000 annual cash flow at 8%. To solve for PVA_n we first plug our values into the equation and then solve for PVA.

$$\begin{aligned}
 PVA_n &= CF \times \left[\frac{1 - 1/(1+i)^n}{i} \right] \\
 &= 2,000 \times \left[\frac{1 - 1/1.08^3}{0.08} \right] \\
 &= 2,000 \times \left[\frac{1 - 1/1.2597}{0.08} \right] \\
 &= 2,000 \times \left[\frac{1 - 0.7938}{0.08} \right] \\
 &= 2,000 \times \left[\frac{0.2062}{0.08} \right] \\
 &= 2,000 \times 2.5771 \\
 &= \text{HK\$}5,154.19
 \end{aligned}$$

One consideration is that rounding could lead to some differences and, when dealing with large values, these differences may be large. A financial calculator should lead to the same answer as doing the calculations without rounding.

Future Value of an Ordinary Annuity

When working with annuities, the focus is calculating their PV and, at times, it is necessary to calculate the FV of an annuity (FVA). This is often the case when, for example, working with monthly savings plans or calculating terminal values for retirement plans with regular contributions.

For instance, if a person might save HK\$1,000 by the end of every year for four years to buy a racing bicycle. The bicycle costs HK\$4,500. The bank pays 8% interest per year. The question is, will that generate enough money to buy the bicycle after four years?

The first step in finding the answer is to set out a timeline of cashflows. It is possible to calculate the FV for each cash flow using the following equation:

$$FV_n = PV \times (1+i)^n$$

The next step is to add up all the cash flows.

The total FV of the four payments is HK\$4,506.11, just enough.

Most business projects are longer and involve more payments. A way to solve more complex problems is to start by calculating PVA using the equation for the PV of an ordinary annuity and then use the equation of FV to calculate the FV of the PVA. An easier way, generally used, is to condense this process into just one step using the FVA formula, which we get by substituting PVA for PV in the FVA formula:

$$\begin{aligned}
 FVA_n &= PVA_n \times (1+i)^n \\
 &= \frac{CF}{i} \times \left[1 - \frac{1}{(1+i)^n} \right] \times (1+i)^n
 \end{aligned}$$

$$= \frac{CF}{i} \times [(1+i)^n - 1]$$

$$FVA_n = CF \times \left[\frac{(1+i)^n - 1}{i} \right]$$

where:

- FVA_n = FVA at the end of n periods,
- PVA_n = PV of an n period annuity,
- CF = cash flows that are constant and evenly spaced,
- i = interest rate, and
- n = time periods.

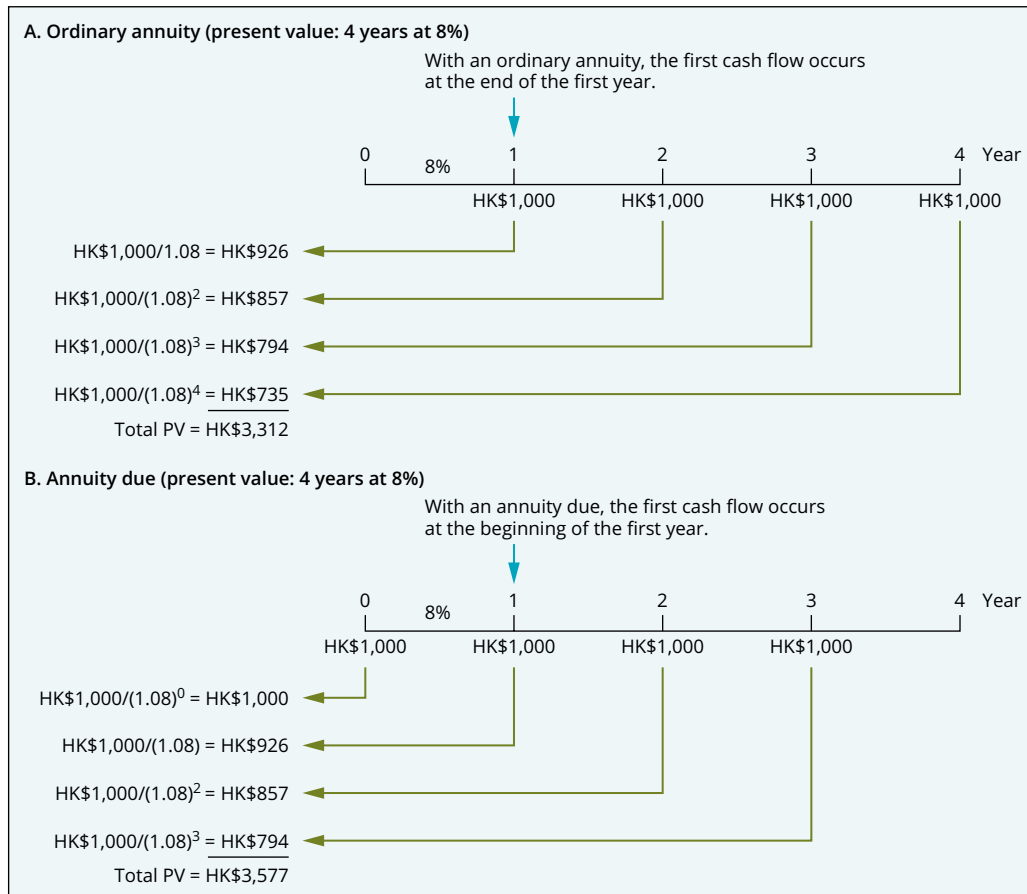
Using the above formula to calculate FVA for the bicycle problem is relatively straightforward and the calculation is similar to the one used for PVA problems. We plug our values into the equation:

$$\begin{aligned}
 FVA_n &= CF \times \left[\frac{(1+i)^n - 1}{i} \right] \\
 &= 1,000 \times \left[\frac{1.08^4 - 1}{0.08} \right] \\
 &= 1,000 \times \frac{0.36049}{0.08} \\
 &= 1,000 \times 4.5061 \\
 &= \text{HK\$}4,506.11
 \end{aligned}$$

Annuities Due

Another type of annuity often used in business is the annuity due. In this type of annuity, cash payments start right away at the beginning of the first period. The rental of an apartment is an example of an annuity due, with the first payment due immediately, the second payment due on the first day of the second month and so on.

The following figures compare the cash flows for an ordinary annuity and an annuity due. Both annuities are made up of four HK\$1,000 cash flows at 8% interest. The first set of numbers are for an ordinary annuity with cash flows at the end of the period. The second set refers to an annuity due with cash flows at the beginning of the period.



The numbers show that the PV of the annuity due is larger than the PV of an ordinary annuity because the cash flows of the annuity due are shifted forwards one year and discounted less.

3.3.2.2 Perpetuities

A perpetual annuity is called a perpetuity. The concept of a perpetuity – which has no end date – has somewhat limited application but it may be the most basic type of annuity.

A perpetuity is a series of regular cash flows that continues forever. Some cash flows have no determined limit. For example, a dividend continues to be paid forever unless a corporation changes its dividend policy.

Preference share issues are the most important perpetuities in modern securities markets. Issuers of preference shares promise to pay a fixed dividend forever or until a retirement date. Any unpaid dividends must be paid to preference shareholders before any dividends are paid to ordinary shareholders.

Because a company can have an indefinite life, cash flows might also have an indefinite life. Constant expected cash flows can be regarded as a perpetuity.

The formula to calculate the present value of a perpetuity is:

$$PVP = \frac{CF}{i}$$

where:

- PVP = present value of a perpetuity,
- CF = periodic cash flow, and
- i = interest rate.

The present value of a perpetuity is not very different from the present value of a very long annuity. The important point is that cash flows to be received far in the future may have a very small PV. For example, a specific type of bond known as a Consols, which are US and UK government debt issues of perpetual bonds. Another common example in the form of a perpetuity refers to the dividend of preference shares, yet despite its issue is not common in the Hong Kong Stock Exchange.

3.3.3 Constant Growth Cash Flows

Some cash flows are designed to grow over time. The cash flows discussed earlier are relatively constant and based on interest rates that do not change. What changes is the amount on which the interest is compounded.

Some cash flows, for example, from some manufacturing operations that expand capacity or from mining operations that generate less product over time, may grow or shrink over time. These changes are also important considerations for companies.

Growing annuities and perpetuities should be understood as part of a well-developed financial management strategy.

3.3.3.1 Growing Annuity

It is sometimes necessary to evaluate a stream of cash flows that will grow (or shrink) at a constant rate per period (g) over a given period of time, ending at some terminal point (n). Such a stream represents a growing annuity.

The present value of this stream of payments can be calculated as follows:

$$PV_0 = \frac{PMT_0(1+g)}{(1+k)^1} + \frac{PMT_0(1+g)^2}{(1+k)^2} + \frac{PMT_0(1+g)^3}{(1+k)^3} + \dots + \frac{PMT_0(1+g)^n}{(1+k)^n}$$

where:

- PV = present value,
- PMT = payment,
- k = effective annual rate,
- g = growth rate, and
- n = time period.

This equation multiplies the payment (PMT) by a growing factor for every period. This is a geometric series with a definite ending (as opposed to a perpetuity that continues forever, in theory).

With some work, this equation can be reduced to the following expression:

$$PV_0 = \frac{PMT_1}{k - g} \times \left[1 - \left(\frac{1+g}{1+k} \right)^n \right]$$

There are a few important considerations here.

The first is that this relationship holds only when the effective annual rate (k) is greater than the payout (g). Otherwise, the answer is negative, which is uninformative.

The second is that only future estimated cash flows and estimated growth in these cash flows are relevant.

The third is that the relationship holds only when growth in payments is expected to occur at the same rate for the duration of the annuity.

3.3.3.2 Growing Perpetuity

Sometimes we may want to evaluate a stream of cash flows that will grow (or shrink) at a constant rate per period (g) indefinitely. The present value of this stream of payments can be calculated as follows:

$$PV_0 = \frac{PMT_0(1+g)}{(1+k)^1} + \frac{PMT_0(1+g)^2}{(1+k)^2} + \frac{PMT_0(1+g)^3}{(1+k)^3} + \dots + \frac{PMT_0(1+g)^\infty}{(1+k)^\infty}$$

Notice that what we are doing here is multiplying the payment by a changing factor in every period. This is easily solved because it is a geometric series, so it can be reduced to the following expression:

$$PV_0 = \frac{PMT_0(1+g)}{k - g} = \frac{PMT_1}{k - g}$$

As with growing annuities, the relationship holds only when the effective annual rate (k) is greater than the payout (g), only future estimated cash flows and estimated growth in these cash flows are relevant and the relationship holds only when growth in payments is expected to occur indefinitely at the same rate.

Illustrative Example 4

ABC Investment Ltd is a generous corporate citizen. It has decided to promote the standard of accounting teaching in Hong Kong by endowing a professorship to a local university. One professor should receive HK\$1.6 million a year in salary.

If the university investment manager can earn 8% return annually in perpetuity, your company would have to donate HK\$20 million today to provide the professor holding the seat with HK\$1.6 million annual income for ever (HK\$1.6 million / 8%). The donation of HK\$20 million is the value of the annuity.

Illustrative Example 4 (continued)

After further consideration, your employer has come to realise that salary adjustment must be made each year to cover the cost of inflation. The annuity must be sufficient to give the professor 3% pay raise each year in perpetuity.

With the same assumption of 8% return, the required annuity has become HK\$32 million ($\text{HK\$1.6 million} / [8\% - 3\%]$). This generous annuity would ensure the professor would receive 3% pay raise to maintain the standard of living in perpetuity.

Key Learning Point

The concepts of present value and discounted cash flow analysis can be applied to multiple cash flows to help financial managers make more effective decisions when considering investments in annuities and perpetuities.

► Knowledge Check Questions**Question 7**

Sheung Wan Financial Services Company offers a perpetuity of HK\$5,000 per year with the first payment in one year. If the opportunity cost is 8% compounded annually, calculate the present value of the perpetuity today.

- A HK\$57,500
- B HK\$62,500
- C HK\$67,500
- D HK\$125,000

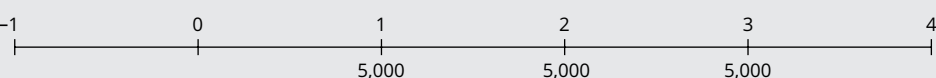
Question 8

Sheung Wan Financial Services Company offers a perpetuity of HK\$5,000 per year with the first payment immediately. If the opportunity cost is 8% compounded annually, calculate the present value of the perpetuity today.

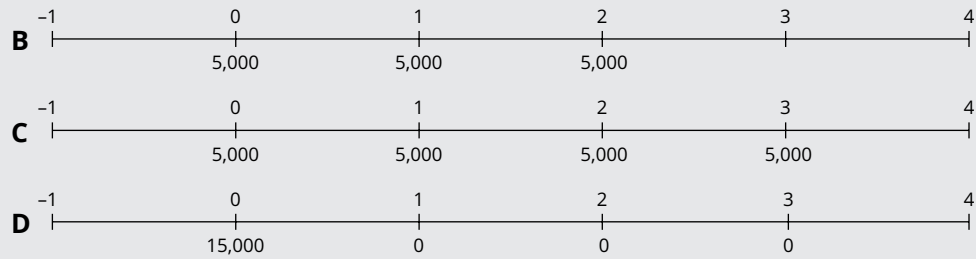
- A HK\$57,500
- B HK\$62,500
- C HK\$67,500
- D HK\$125,000

Question 9

Determine which of the following is the most appropriate timeline for the cash flows of a three-year annuity due with annual cash flows of HK\$5,000.

- A 

► Knowledge Check Questions (*continued*)



➤ 3.4 ANNUALISED INTEREST RATES

In most calculations discussed in this chapter, a single interest rate was supplied. However, when working with real market data, the situation is not so straightforward. Companies have to deal with interest rates that can be calculated in different ways.

There are two approaches most commonly used to calculate interest rates. The first is the annual percentage rate (APR), which is basically the rate without the effects of compounding. The second is the effective annual rate (EAR), which includes the compounding calculation.

3.4.1 Annual Percentage Rate

The **annual percentage rate (APR)**, sometimes called the nominal rate, is the annualised effective interest rate without compounding on loans that are for the fraction of a year. It is computed as the effective interest rate for the fraction of a year multiplied by the number of time fractions in a year, for example, 2 if semi-annual, 4 if quarterly, and so on.

A credit card, for example, may charge APR of 18% or 1.5% per month for 12 months.

This rate can be both simplistic and a little deceptive, as very few loans or investments are priced using simple interest.

3.4.2 Effective Annual Rate

The right way to annualise an interest rate is to calculate the effective interest rate per year. The **effective annual rate (EAR)** is the annual interest rate taking compounding into account.

The formula for EAR is:

$$\text{EAR} = (1 + i / m)^m - 1$$

where:

- i = interest rate,
- m = number of compounding periods.



The EAR, also known as the annual percentage yield, is the annual percentage rate with compounding on loans that are for a fraction of a year. The assumption is that interest is paid on interest that would accumulate for each period during the year.

In most cases, a simple annual interest rate, like the APR or nominal rate, is quoted and this means that the quoted rate has been annualised by multiplying the rate per period by the number of periods in a year.

The EAR conversion formula takes into account the number of compounding periods and adjusts the annualised rate for the time value of money. The EAR represents the true cost of borrowing and lending, which means that this is the rate that should be used in finance decisions.

Illustrative Example 5

ABC Investment Ltd has an investment in a credit card company. Your CIO has asked you for input on the structure of interest rate that should be imposed on credit card borrowers. You have been told that the annual percentage rate (APR) must be kept at no higher than 24% because of competition in the credit card business. What are the various ways to structure the interest rate to give your credit card company maximum interest income?

You know the power of compounding. You know the more frequent compounding occurs, the higher the effective annual interest rate that you can earn. Therefore, you have decided to tinker with the compounding frequency to determine the effects.

Annual percentage rate (APR) versus effective annual rate (EAR)

Interest compounding period	Number of compounding periods in a year	Annual percentage rate (APR)	Interest rate per compounding period	Effective annual rate (EAR)
Semi-monthly	24	24%	1%	26.97%
Monthly	12	24%	2%	26.82%
Bi-monthly	6	24%	4%	26.53%
Quarterly	4	24%	6%	26.25%
Semi-annually	2	24%	12%	25.44%
Annually	1	24%	24%	24.00%

Your tinkering clearly shows that with the same 24% APR, high frequency compounding generates more interest income for your credit card company while the consumers technically still pay the same 24% interest rate in terms of APR. You have also noticed the only time APR and EAR are the same is when interest rate is paid only once a year.

You will suggest the credit card company charge the customers based on semi-monthly compounding of interest rate to maximise interest income.

Illustrative Example 6

The nominal interest rate and annual percentage are different. What is the purpose of calculating compounded interest?

Analysis

Money gains value as compounding occurs. A compounded rate is higher on an annualised basis because more periods of interest are calculated. As a result of compounding, each period's interest increases due to interest on the previous period's interest.

Key Learning Point

There are two approaches most commonly used to calculate interest rates. The first is the annual percentage rate (APR), which is basically the rate without the effects of compounding. The second is the effective annual rate (EAR), which includes the compounding calculation.

► Knowledge Check Questions

Question 10

A credit card has a quoted rate of 18.5% compounded daily. Calculate the effective annual rate. (Assume 360 days a year.)

- A 66.60%
- B 20.32%
- C 51.39%
- D 18.50%

Question 11

Ms. Wang borrows HK\$10,000 and promises to repay a total of HK\$10,500 in one month. Calculate the effective annual interest rate she is paying.

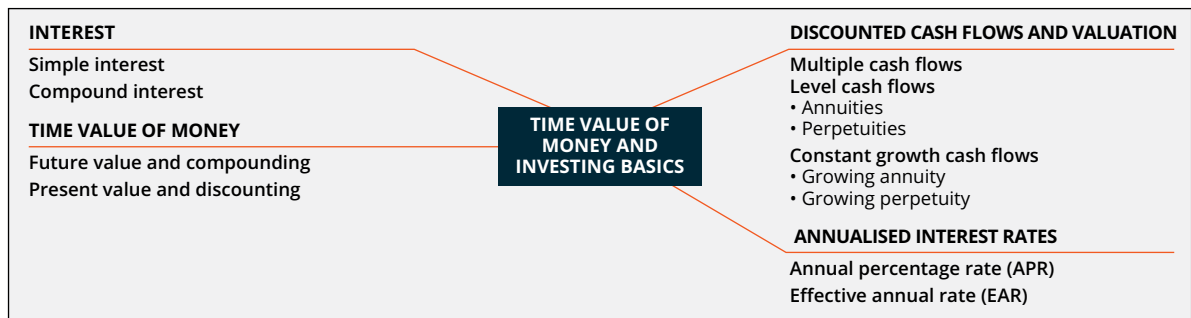
- A 5.00%
- B 60.00%
- C 79.59%
- D 179.59%



SUMMARY

- Interest refers to the charge assessed for borrowing money, or what investors may expect to receive from investing (or lending) money. It is a fundamental component of investing and finance.
- Simple interest is consistently applied to the original investment (the principal) throughout an investment period.
- Compound interest is based on both the principal and any previously accrued interest.
- Future value is a measure of how much an investment or investments will be worth after earning interest for one or more time periods, that is, after the interest has compounded.
- Present value (PV) is the value now of an amount that will be paid or received in the future and assuming compound interest. Both PV and FV are based on the future amount to be received, the amount of time until the amount is received, and the interest or discount rate.
- The concept of the time value of money – that money in hand today is worth more than money to be received in the future – flows into other important concepts of simple interest, compound interest and future value calculations. All of them flow into present value and discounted cash flow analysis. All these concepts have to be applied to multiple cash flows. Most business decisions include cash flows over time.
- In business situations, it is often necessary to calculate the present value of a series of future cash flows.
- A financial contract with cash flows that are spaced equally and level over a period of time is an annuity.
- An annuity is a series of equal dollar amounts. An ordinary annuity – as compared to an annuity due – means that the series of equal amounts is received (or paid) at the end of each period.
- A perpetual annuity is called a perpetuity. The concept of a perpetuity – which has no end date – has somewhat limited application but it may be the simplest type of annuity to understand.
- A stream of cash flows that will grow (or shrink) at a constant rate per period over a given period of time, ending at some terminal point represents a growing annuity. If there is no defined end point, it is called a growing perpetuity.
- The annual percentage rate (APR) is the annualised effective interest rate without compounding on loans that are for only the portion of a year.
- The effective annual percentage rate (EAR), also known as the annual percentage yield, is the annual percentage rate with compounding.

MIND MAP



LIST OF FORMULAS

Simple interest

$$\text{Interest} = p \times i \times n$$

where

- p = principal,
- i = interest rate, and
- n = time.

Compound interest

$$\text{Interest} = p(1 + i/n)^{nt}$$

where

- p = principal,
- i = interest rate,
- n = is the number of times per year that interest is compounded, and
- t = number of years.

Future value

$$\text{Future Value (FV)} = p \times (1 + i)^n$$

where

- FV = future value of a single amount,
- p = principal,
- i = interest rate for a period,
- n = number of periods.

Present value

$$\text{Present Value (PV)} = \text{Future Value (FV)} / (1 + i)^n$$

Present value of ordinary annuity

$$\text{PVA}_n = \frac{\text{CF}}{i} \times \left[1 - \frac{1}{(1 + i)^n} \right]$$

where

- PVA_n = PV of an n period annuity,
- CF = level and equally spaced cash flows,
- i = interest rate or discount rate, and
- n = number of periods.

Future value of ordinary annuity

$$\text{FVA}_n = \text{CF} \times \left[\frac{(1 + i)^n - 1}{i} \right]$$

where

- FVA_n = FVA at the end of n periods,
- CF = cash flows that are constant and evenly spaced,
- i = interest rate, and
- n = time periods.

Present value of growing annuity

$$\text{PV}_0 = \frac{\text{PMT}_0(1 + g)}{(1 + k)^1} + \frac{\text{PMT}_0(1 + g)^2}{(1 + k)^2} + \frac{\text{PMT}_0(1 + g)^3}{(1 + k)^3} + \dots + \frac{\text{PMT}_0(1 + g)^n}{(1 + k)^n}$$

This equation can be reduced to the following expression:

$$\text{PV}_0 = \frac{\text{PMT}_1}{k - g} \times \left[1 - \left(\frac{1 + g}{1 + k} \right)^n \right]$$

where

- PV = present value,
- PMT = payment,
- k = effective annual rate,
- g = payout, and
- n = time period.

Present value of growing perpetuity

$$PV_0 = \frac{PMT_0(1+g)}{(1+k)^1} + \frac{PMT_0(1+g)^2}{(1+k)^2} + \frac{PMT_0(1+g)^3}{(1+k)^3} + \dots + \frac{PMT_0(1+g)^\infty}{(1+k)^\infty}$$

This equation can be reduced to the following expression:

$$PV_0 = \frac{PMT_0(1+g)}{k-g} = \frac{PMT_1}{k-g}$$

Effective annual rate

$$EAR = (1 + i/m)^m - 1$$

where

- i = interest rate,
- m = number of compounding periods.

► Answers to Knowledge Check Questions

Question 1

Answer B is correct because $HK\$12,000 \times 0.02 = HK\240 .

Question 2

Answer A is incorrect. The two investments pay different interest since year 2.

Answer B is correct. GHI earns simple interest which is a fixed amount each period. RST earns compound interest which rises over time.

Answer C is incorrect. The interests paid by GHI remains unchanged for years, it earns simple interest, and the interest of RST is interest on interest, it earns compound interest.

Answer D is incorrect. The two investments pay different interest since year 2.

Question 3

Answer A is incorrect. Growth rate has nothing to do with interest rate.

Answer B is correct. By definition, compound interest calculates the return on that original investment and interest.

Answer C is incorrect. Simple interest rate is the interest earned on the original investment only and is a fixed amount each period.

Answer D is incorrect. Cost of capital is what the enterprise has to pay for the use of its capital.

Question 4

Answer B is correct. $HK\$1,100 \div HK\$1,000 - 1 = 0.1$

Question 5

Answer C is correct. Present value of lottery B = $HK\$1,750 \div 1.05^5 = HK\$1,371.17$, which is greater than the present value of lottery A.

Question 6

Answer D is correct. Future value of investment = $\text{HK\$}2,000 \times 1.05^5 = \text{HK\$}2,552.56$.

Question 7

Answer B is correct. $\text{HK\$}5,000 \div 0.08 = \text{HK\$}62,500$.

Question 8

Answer C is correct. $\text{HK\$}5,000 \div 0.08 + \text{HK\$}5,000 = \text{HK\$}67,500$.

Question 9

Answer A is incorrect. It is ordinary annuity.

Answer B is correct. The annuity due offers cash inflow at the start of each year, the investors will receive HK\$5,000 at the beginning of years 1, 2 and 3 which are the end of years 0, 1 and 2.

Answer C is incorrect. It is four-year annuity due.

Answer D is incorrect. It is one-year annuity due that pays HK\$15,000 in cash flow.

Question 10

Answer B is correct. $\text{EAR} = (1 + 0.185 / 365)^{365} - 1 = 20.32\%$.

Question 11

Answer C is correct. $1.05^{12} - 1 = 79.59\%$.

 EXAM PRACTICE**QUESTION 1**

An investor puts HK\$2,000 at 5% compounded annually. Determine how much interest is earned in the fifth year. (Round your answer to two decimals.)

- A. HK\$100.00
- B. HK\$121.55
- C. HK\$500.00
- D. HK\$552.56

QUESTION 2

Mr. Xiang invests HK\$25,000 every year for 20 years at an interest rate of 7%. Calculate the total value of the investment after 20 years. (Round your answer to two decimals.)

- A. HK\$1,096,629.42
- B. HK\$1,024,887.31
- C. HK\$283,389.88
- D. HK\$264,850.36

QUESTION 3

Tommy Lau has made an investment of HK\$25,000 that promises to pay him 8% simple interest per year for 10 years. Determine how much interest he will earn in the:

- (a) First year
- (b) Ninth year

QUESTION 4

Tommy Lau's summer job has given him HK\$1,200 more than he needs for his tuition this year. The local bank pays simple interest at a rate of 0.5% per month. Determine how much interest he will earn in one year.

QUESTION 5

History tells us that a group of Dutch colonists purchased the island of Manhattan from the Native American residents in the year 1626. Payment was made with various items valued at HK\$24. If the colonists had invested this money in Europe and earned average returns of 5% per year, calculate the value of the investment after 389 years (rounding to the dollar), using:

- (a) Simple interest
- (b) Compound interest

QUESTION 6

Tommy Lau was awarded a scholarship that will pay HK\$5,000 one year from now. However, he really needs the money today and has decided to take out a loan. If the interest rate is 6%, calculate how much he can borrow so that the scholarship will just pay off the loan.

QUESTION 7

Tommy Lau is buying a new car. His bank quotes a rate of 9.5% per year for a car loan. Calculate the effective annual rate if the compounding occurs:

- (a) Annually
- (b) Quarterly
- (c) Monthly

QUESTION 8

Ms. Lee, a retired librarian, would like to donate some money to her alma mater to endow a HK\$5,000 annual scholarship. The first scholarship will be awarded in five years. The university will manage the funds and expects to earn 3% per year. Compute how much Ms. Lee will have to donate so that the endowment fund never runs out.

QUESTION 9

Mrs. Lau has been offered two perpetuities: grow and shrink. Grow promises her HK\$100 in one year and an annual cash flow that will increase by 4% per year forever. Shrink, in contrast, promises her HK\$1,000 in one year but the annual cash flow will decline by 2% forever. If her opportunity cost is 5% per year and both annuities cost HK\$1,000, identify which annuity offers her the greater value.

QUESTION 10

Ms. Tam has to choose between two investments that have the same cost today. Both investments will ultimately pay HK\$1,300 but at different times, as shown in the table below. If Ms. Tam does not choose one of these investments, she could leave the funds in a bank account paying 5% per year. Determine which investment should Ms. Tam choose.

Year	Investment A	Investment B
1	HK\$0	HK\$200
2	HK\$500	HK\$400
3	HK\$800	HK\$700

QUESTION 11

Ms. Tam has now decided to use HK\$500,000 to put a down payment on an apartment, where she will live for next two years and then sell it. The bank has offered a mortgage rate of 5.1% compounded semi-annually for two years amortised over 25 years. The apartment costs HK\$2,800,000.

- If two friends rent rooms and pay Ms. Tam HK\$4,750 each at the end of each month, compute how much more money she will need to pay the mortgage every month.
- Ms. Tam wants to sell the apartment in two years for a price high enough to cover the remaining principal and recover her down payment. Calculate the minimum sale price.

QUESTION 12

Tommy Lau wants to buy another new car that will cost HK\$29,000.

- Calculate the monthly car payments for a loan amortised over 60 months at a nominal interest rate of 8.5% per year compounded monthly.
- Prepare an amortisation schedule for the car loan for the first ten months.
- Using the amortisation schedule, determine how much Tommy Lau owes on the loan after ten months on a five-year loan. Calculate the present value of this amount.

ANSWER TO EXAM PRACTICE**QUESTION 1**

Answer B is correct. $\text{HK\$}2,000 \times (1.05^5 - 1.05^4) = \text{HK\$}121.55$.

QUESTION 2

Answer A is correct. $n = 20$, $i/y = 7$, $\text{PMT} = \text{HK\$}25,000$, due, $\text{FV} = \text{HK\$}1,096,629.420$.

QUESTION 3

As this is simple interest, Tommy will earn the same amount of interest each year. The annual amount of interest is $8\% \times \text{initial investment} = .08 \times \text{HK\$}25,000 = \text{HK\$}2,000$.

- HK\$2,000
- HK\$2,000

QUESTION 4

Tommy Lau will be paid interest each month for 12 months, but without compounding. The total interest earned is $(n \times P \times k) = (12 \times \text{HK\$}1,200 \times 0.5\%) = \text{HK\$}72$.

QUESTION 5

(a) Value = $P + (n \times P \times k) = 24 + (389 \times 24 \times 5\%) = \text{HK\$}491$.

(b) $FV_{389\text{years}} = 24 \times (1 + 0.05)^{389} = \text{HK\$}4,196,126,573$.

QUESTION 6

Since the future value of the amount that is to be repaid is HK\$5,000, the amount that can be borrowed is the present value amount. This is calculated as follows:

$$\text{Present Value (PV)} = \text{Future Value (FV)} / (1 + i)^n$$

$$PV = 5,000 / (1 + 0.06)^1$$

$$= \text{HK\$}4,716.98$$

QUESTION 7

(a) For annual compounding, the effective annual rate will be the same as the quoted rate.

$$\text{To check this: } k = \left(1 + \frac{QR}{m}\right)^m - 1 = \left(1 + \frac{9.5\%}{1}\right)^1 - 1 = 9.5\%$$

(b) With quarterly compounding, set $m = 4$, $k = \left(1 + \frac{9.5\%}{4}\right)^4 - 1 = 9.84\%$

(c) With monthly compounding, set $m = 12$, $k = \left(1 + \frac{9.5\%}{12}\right)^{12} - 1 = 9.92\%$

QUESTION 8

The present value of the perpetual scholarship payment at the end of 4 years:

$$PV_4 = PMT \left[\frac{1}{k} \right] = 5,000 \times \left[\frac{1}{0.03} \right] = \text{HK\$}166,667$$

So present value today is $\text{HK\$}166,667 / (1.03)^4 = \text{HK\$}148,081$

Ms. Lee will need to endow HK\$148,081 today for the scholarship to start in 5 years.

QUESTION 9

Present value of grow: $PV_{GROW} = \frac{100}{0.05 - 0.04} = \text{HK\$}10,000$

Present value of shrink: $PV_{SHRINK} = \frac{1,000}{0.05 - (-0.02)} = \text{HK\$}14,285.71$

Grow exceeds the cost by HK\$9,000 while shrink exceeds the investment cost by HK\$14,285.71. Shrink is preferred, as it exceeds the investment cost by the most.

QUESTION 10

Find the present value of the money paid back to Ms. Tam by each investment, using the interest rate on the alternative (the bank account) as the discount rate.

For Investment A:

$$PV_0 = \frac{500}{(1+0.05)^2} + \frac{800}{(1+0.05)^3} = 453.51 + 691.07 = \text{HK\$1,144.58}$$

For Investment B:

$$PV_0 = \frac{200}{(1+0.05)^1} + \frac{400}{(1+0.05)^2} + \frac{700}{(1+0.05)^3} = 190.48 + 362.81 + 604.69 = \text{HK\$1,157.98}$$

Ms. Tam would prefer Investment B, because it has the higher present value.

QUESTION 11

- (a) The effective monthly interest rate is, $k_{\text{monthly}} = \left(1 + \frac{0.051}{2}\right)^{2/12} - 1 = 0.4206\%$

The amount of the mortgage loan will be $(\text{HK\$2,800,000} - \text{HK\$500,000}) = \text{HK\$2,300,000}$, and there will be $12 \times 25 = 300$ monthly payments, the value of which can be calculated with a financial calculator, $N = 300$, $PV = -2,300,000$, $i/y = 0.4206$, $FV = 0$, $CPT PMT = 13,500.89$. Ms. Tam's two friends will be paying $2 \times \text{HK\$4,750} = \text{HK\$9,500}$ in rent, so she will need an additional $\text{HK\$13,500.89} - \text{HK\$9,500} = \text{HK\$4,000.89}$ to cover the mortgage payment.

- (b) In two years, Ms. Tam will have made 24 payments, leaving 276 payments to go. The present value of these payments is the outstanding value of the mortgage loan. Use the calculator again: $N = 276$, $i/y = 0.4206$, $PMT = 13,500.89$, $FV = 0$, $CPT PV = 2,200,336.58$. To pay off the loan, and recoup her down payment, Ms. Tam would have to sell the house for at least $\text{HK\$2,200,336.58} + \text{HK\$500,000} = \text{HK\$2,700,336.58}$.

QUESTION 12

- (a) First, find the effective interest corresponding to the frequency of Tommy's car payments ($f = 12$); with monthly compounding, set $m = 12$.

$$k_{\text{monthly}} = \left(1 + \frac{QR}{m}\right)^{m/f} - 1 = \left(1 + \frac{8.5\%}{12}\right)^{12/12} - 1 = 0.70833\%. \text{ The 60 car payments form an}$$

'annuity' whose present value is the amount of the loan (the price of the car):

$$\text{HK\$29,000} = \text{PMT} \left[\frac{1 - \frac{1}{(1 + 0.0070833)^{60}}}{0.0070833} \right] \Rightarrow \text{PMT} = \text{HK\$594.98}$$

(b) Use the effective monthly interest rate from part A, $k = 0.70833\%$.

Period	(1) Principal outstanding	(2) Payment	(3) Interest = $k \times (1)$	(4) Principal repayment = (2) – (3)	Ending principal = (1) – (4)
1	29,000.00	594.98	205.42	389.56	28,610.44
2	28,610.44	594.98	202.66	392.32	28,218.12
3	28,218.12	594.98	199.88	395.10	27,823.01
4	27,823.01	594.98	197.08	397.90	27,425.11
5	27,425.11	594.98	194.26	400.72	27,024.40
6	27,024.40	594.98	191.42	403.56	26,620.84
7	26,620.84	594.98	188.56	406.42	26,214.42
8	26,214.42	594.98	185.69	409.29	25,805.13
9	25,805.13	594.98	182.79	412.19	25,392.94
10	25,392.94	594.98	179.87	415.11	24,977.82

(c) After ten months, the principal outstanding is HK\$24,977.82 (from the amortisation table). The present value of this amount is:

$$PV_0 = 24,977.82 \times \left(\frac{1}{(1 + 0.0070833)^{10}} \right) = \text{HK\$}23,275.58$$

4

Introduction to Bond Investments

CHAPTER TOPIC LIST

4.1 Types of Bonds

- 4.1.1 Types of Corporate Bonds
- 4.1.2 Types of Government Bonds

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- 4.2.1 Current Yield and Yield to Maturity
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4.3 Valuation of Bonds

- 4.3.1 Bond Valuation Formula
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4.4 Bond Yields

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- 4.5.1 Interest Rate
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4.6 Bond Agreement/Indenture and Key Features – Call Provision

4.7 Bond Rating

- 4.7.1 Issue Rating
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LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE

LO7.01: Explain and analyse the theories of major capital market models

7.01.02 Explain the different sources of finance (short-term and long-term) and methods of raising capital

7.01.03 Describe different types of bonds and their features

7.01.04 Calculate the value of a bond



OPENING CASE

ABC INVESTMENT LTD

Staff with a demonstrated knowledge of finance and investments are an important asset to any company, not least ABC Investment Ltd. Such expertise is integral in managing the company's sizeable bond portfolio, working with the chief investment officer (CIO).

ABC Investment Ltd owns a varied portfolio of bonds. Brokers contact the company on a daily basis to present more buying opportunities for the company to expand its investment portfolio. The challenge for investment managers is how to compare and contrast all the competing investing instruments and make recommendations that can be justified.

Doing this requires a methodical approach to assess investment opportunities. The savvy investment managers calculate different types of yields for potential bond investments, putting to work an understanding of discounted cash flow in order to compute bond prices.

This approach also requires handling issues arising from **bond theorems** and considering fluctuations in bond prices and how those prices relate to changing market discount rates. These considerations are key to selecting suitable bond investments in a changing interest rate environment. In so doing, investment managers develop an appreciation of the practical benefits of ratings of both issues and issuers.

The calculations may seem complex at times and the inputs can be multiple but familiarity with them is imperative to recommend or make the best decisions for ABC Investment Ltd's portfolio and the long-term success of the company. 



OVERVIEW

Bonds, whether issued by governments or the private sector, are an important element of the financial management for many companies. Bond markets represent a major source of financing for companies, financial institutions and governments.

This chapter explores the types, structure, pricing and risks associated with bonds and their roles in finance.

Companies and governments use bonds to raise funds by agreeing to pay bond buyers back over a certain period of time at a predetermined rate of interest. There are various structures for bond issues including fixed-income bonds, floating-rate bonds, zero coupon bonds and convertible bonds. Governments also issue bonds of different maturities, including bonds, notes and short-term bills – which are bonds with very short maturities.

This chapter also explores the key terminology associated with the bond market. To understand bonds, it is important to understand the terminology of the market.

With an understanding of the terminology and the types of bonds in the market, the chapter explores how different types of bonds are valued, including par value bonds, discount bonds and premium bonds as well as explaining the compounding formula for bonds and explaining zero coupon bonds.

Another important consideration is bond yields, and the different types of calculations that are required to determine them, including the yield to maturity, the effective annual yield and the realised yield.

Finally, this chapter explores issues associated with interest rate risks and the impacts on bond prices and their volatility, the requirements of bond agreements and the role of rating both issues and issuers.



4.1 TYPES OF BONDS

Bonds provide investors with predetermined future payments. As such, the analysis of bonds provides an important application of compounding and discounting.

Bond markets represent a major source of financing for companies, financial institutions and governments.

Bonds are generically considered to be long-term debt instruments. They hold the promise of fixed payments but the specific terms depend on various factors, including maturity. Short-term bonds, with a maturity of less than a year, are generally called bills or paper. Longer-term bonds with maturities of one to ten years are called notes. Those with maturities of longer than ten years are called bonds.

All of these are valued in similar fashion. In broad strokes, bonds may be divided into corporate bonds and government bonds.

The key feature of a **bond** is that the issuer agrees to pay the bondholder (investor) a regular series of cash payments and to repay the full principal amount by the maturity date.

These promises are stipulated in the bond contract and are a fixed contractual commitment.

There are multiple possible payment structures. Traditional 'coupon-paying' bonds result in equal payments at regular intervals, most often annual or semi-annual payments. The full principal is repaid at the date of maturity. The interest payment is called a '**coupon**'. The names come from the old practice of including actual coupons with the bonds that investors cut out from the bond certificate to claim payments. Bonds today are typically registered and paid directly into a bank account. The lump sum payment of principal at maturity is called a **bullet payment** or **balloon payment**.

Bonds are amongst the most basic financial instruments on the market. Still, it is important to understand the terminology associated with bonds to fully understand the value and potential profit associated with a particular issue.

The **face value**, or **par value**, for most bonds in the United States is US\$1,000, and it is the principal amount owed to the bondholder at maturity.

The **coupon payments** represent the interest payments bondholders receive. These payments are commonly made annually or semi-annually.

A bond's **coupon rate** is the annual coupon payment divided by the bond's face value. A bond that has a face value of US\$1,000 and pays US\$80 of coupon interest annually, has a coupon rate of 8%.

4.1.1 Types of Corporate Bonds

4.1.1.1 Fixed-income/Coupon Bonds

Bonds in which the interest paid to investors is fixed for the life of the contract are called **fixed-income bonds**.

The most common corporate bonds have fixed coupon payments and the entire original principal is paid at maturity when the bonds are retired.

The coupon payments – or interest payments – made at regular intervals are fixed for the life of the bond contract.

4.1.1.2 Floating-rate Bonds

A bond that pays a rate of interest that fluctuates over the life of the instrument is called a **floating-rate bond**.

Typically, the rate of interest paid is tied to a macroeconomic benchmark such as the prime rate, the US Treasury Bill rate, the London Interbank Offered Rate (LIBOR), the Euro Short Term rate (€STR) or a similar rate, plus a spread (which is a constant amount).

In Hong Kong, two HKD rates are currently available:

- Hong Kong Interbank Offer Rate (HIBOR) is the annualised rate charged for inter-bank lending on Hong Kong Dollar (HKD) denominated instruments for a specified period ranging from overnight to one year. It is calculated daily at 11:00 a.m. local time based on quotations from 20 banks designated by the Hong Kong Association of Banks.
- Hong Kong Dollar Overnight Index Average (HONIA) is the annualised rate charged and is the effective overnight (i.e., where repayment is made on the following Hong Kong business day) reference rate of HKD unsecured lending transactions in the Hong Kong interbank market, executed through the panel of contributing brokers. The rate reflects actual transactions and is calculated as a weighted average of deals executed. The rate reflects transactions undertaken between 8:00 a.m. and 4:00 p.m. Hong Kong time on the relevant business day and is reported at 5:00 p.m. each business day.

4.1.1.3 Zero Coupon Bond

A **zero coupon bond** is a debt security that does not pay interest during its life. There are no coupons to submit for interest payments.

Zero coupon bonds provide a profit to investors by selling at a deep discount (i.e. an amount less than the maturity value) and paying full face value when redeemed at maturity. Because they provide payment only at maturity, the market price tends to fluctuate more than that of coupon bonds.

4.1.1.4 Convertible Bonds

Convertible bonds provide the bond holder with the option of converting the bonds into a specified number of shares of equity (stock) of the issuing company. Investors in convertible bonds exercise an option to convert if the market price of the stock increases.

There are two features of convertible bonds that need consideration:

- Bonds issued with a convertible feature usually pay a lower rate of interest than comparable bonds without a convertible feature. This lower interest rate results from the value of the holder's option to convert the bonds into stock.
- Convertible bonds may also contain a call provision, which would permit the issuing company to call the bonds so as to limit the upside potential recognised by the bond holders.

4.1.2 Types of Government Bonds

Just as there are different types of corporate bonds that are differentiated by their various characteristics – fixed or floating rate, coupon or zero coupon, or convertibility – as well as the length of their maturity date, there are also different types of government bonds.

In general, they are categorised by their longevity. Bonds, notes and bills are the three main types of government bonds. They have similar characteristics but different maturities and coupon payments.

4.1.2.1 Treasury Bonds

A **Treasury bond** (or T-bond) is a debt security issued by a government, typically refers to issue by the US government. It typically has a maturity of more than 10 years and interest payments are made semi-annually. T-bonds are generally seen as risk free.

The maturity of Treasury bonds ranges from 10 to 30 years and they come in denominations of US\$1,000. Treasury bonds are typically sold at auctions.

There is also a secondary market for Treasury bonds, which makes this type of government bond very liquid but also means that the price of these bonds fluctuates over time. The price of Treasury bonds on the secondary market may drop when auction rates increase (meaning the rate that buyers are willing to accept when Treasury bonds are auctioned) because the value of the future cash flow of these bonds is discounted at a higher rate. However, prices increase when auction rates drop.

Hong Kong issues government bonds through a Government Bond Programme. Most Hong Kong government bonds are 10-year bonds and are also sold through tender in the primary market and in the secondary market.

4.1.2.2 Treasury Notes

A **Treasury note** is a debt instrument or security issued by a government – most famously the US federal government. Typical maturities range from two to ten years. Treasury notes or bonds are often used to raise funds to reconcile the imbalances between what the government earns in taxes and what it spends.

The risk of default on Treasury notes is very low and these securities are easily marketable in secondary securities markets. An investor may initially purchase a 20-year Treasury bond, for instance, and can easily sell it later in a secondary capital market.

4.1.2.3 Treasury Bills

Treasury bills (or T-bills) are basically short-term bonds. They are issued by governments, although the name typically refers to issues by the US government.

Treasury bills have short maturity dates of 4 weeks, 13 weeks, 26 weeks or 52 weeks and have no stated interest rate. Instead, the government sells T-bills at a discount to the face amount and buyers aim to profit from the difference between the selling prices and the redemption price. In other words, the interest earned is the difference between the buying price and the redemption price at maturity.

The discount amount varies from one Treasury bill to the next because these types of bonds are sold through a competitive bidding process.

Illustrative Example 1

The bond summary table shown in Exhibit 4.1 is typical type of table that a broker might send to an investor on a particular day. It lists some of the bonds available for sale on the day and demonstrates the variety of bonds available in the market.

Selected corporate bonds and US government bonds (as of 11 June 2018)				
Currency	Issuer	Coupon	Maturity	Price
USD	Edgesale Development	6.35%	11 January 2020	99.61
USD	Le Grandeur Property	6.75%	15 April 2021	106.80
USD	Magical Forest Holdings	7.25%	4 April 2021	102.49
USD	Marina Spot Group	6.25%	28 June 2021	96.07
USD	Mouse Fields Property Holdings	5.25%	23 February 2023	89.37
USD	USA	0.00%	23 May 2019	97.85
USD	USA	2.50%	31 May 2020	100.01
USD	USA	2.63%	15 May 2021	99.98
USD	USA	2.75%	31 May 2023	99.88
USD	USA	3.13%	15 May 2048	100.84

► **EXHIBIT 4.1** Bond summary table

The data presented in the table only shows the coupon, maturity and price. These are the most basic features used in bond valuation.

One of the bonds shown in the table has a zero coupon while others carry fixed coupon rates.

Illustrative Example 1 (continued)

Bonds issued by the US government are considered risk free and tend to offer lower coupon and higher prices than corporate bonds. Some prices are above par value (US\$100) while others trade below par, but they will all pay back par value at maturity.

These bond features impact the valuation of investment opportunities.

Key Learning Point

The key feature of a bond is that the issuer agrees to pay the bondholder (investor) a regular series of cash payments and to repay the full principal amount by the maturity date.

► Knowledge Check Questions**Question 1**

Identify which of the following statements is true of convertible bonds.

- A** The most significant disadvantage to a corporation of issuing convertible bonds is that they increase the cash that the firm must use to make interest payments.
- B** The typical conversion ratio is set so that the corporation's stock price must appreciate 5% or less before it is profitable for the holder to convert the bond to stock.
- C** Corporation that issue convertible bonds can raise funds at a lower interest rate.
- D** The typical issue of convertible bonds allows the holder of the bond to convert it to preferred stock.

Question 2

Identify which of the following is most accurate about zero coupon bonds.

- A** They sell at a premium when first issued.
- B** They are generally more expensive than similar coupon bonds.
- C** They can always be converted to common stock.
- D** They sell at a deep discount at first.

Question 3

Identify which of the following is NOT true with regard to zero coupon bonds.

- A** Zero coupon bonds have no coupon payments and include a single payment at maturity.
- B** Zero coupon bonds sell for less than other bonds with periodic payments.
- C** Zero coupon bonds pay no principal at maturity.
- D** All of the above.

4.2 YIELDS AND PRICE OF BONDS

Understanding the yields that bonds can generate and the price that a company should pay for a particular bond is a very important part of investing in bonds. It is important to understand the current yield and the yield to maturity of bonds as well as the clean price and dirty price of specific bonds.

4.2.1 Current Yield and Yield to Maturity

The ratio of the annual interest divided by the current market price of a bond is the **current yield (CY)**. This is not a true measure of the return bondholders can expect because it does not take into account the purchase price relative to future cash flows. Rather, it relies on next year's interest payment.

The current yield is sometimes called the cash yield or flat yield and can be calculated as:

$$CY = \frac{\text{Annual interest}}{B}$$

where:

- CY = current yield, and
- B = market price.

The discount rate used to evaluate bonds is referred to as the **yield to maturity (YTM)**.

It is the yield that an investor would realise if he or she bought the bond at the current price, held it to maturity, received all the promised payments on time, and reinvested the cash flows received at the yield to maturity.

With the price of a bond and information about interest and principal repayments (timing and amount), it is possible to use the following equation to estimate the YTM:

$$B = i \times \left[\frac{1 - \frac{1}{(1 + k_b)^n}}{k_b} \right] + F \times \frac{1}{(1 + k_b)^n}$$

where:

- B = market price,
- k_b = discount rate or YTM,
- n = number of periods to maturity,
- F = amount to be paid at maturity, and
- i = interest.

A discount rate, the YTM makes the present value of the coupon and principal payments equal to the price of the bond. The YTM is the annual yield that an investor earns if the bond is held to maturity and all payments associated with the bond are made.

As interest rates increase or decrease, a bond's YTM will change frequently. It is calculated based on the promise by the issuer to make the payments stipulated in the bond contract.

4.2.2 Dirty Price and Clean Price

A final piece of terminology that is useful to understand is the dirty price and clean price of a bond.

The **dirty price** refers to a quote for a bond that includes both the current value of all the future cash flows that a bond generates, including the interest on the next coupon payment. Most European markets quote bonds using the dirty price.

Because interest accrues 'in between' coupon payments, then interest increases as the payment date approaches. The further away the payment date is, the lower the interest that the coupon pays.

In the United States, however, it is more common to quote bonds using the clean price, which is the price of a bond immediately after a coupon payment.

The **clean price** does not include any accrued interest or interest that will be accrued on the next coupon payment date.

On any given day, brokers present a number of bond investment opportunities. Each of them is likely to have different coupons and maturities. All the bonds presented are in US dollars.

To analyse the yields on the bonds available for investment, divide annual coupon payments by current bond prices to calculate current yields. You may also compute yield to maturity by using the various formulas or a financial calculator.

To calculate the yield, the present value of the coupon and principal payments should be made equal to the price of the bond today. The use of a financial calculator facilitates the calculation greatly when the right inputs are entered.

One way to analyse the information is to produce a table that shows the bonds offered by the broker and their current yield and YTM.

Illustrative Example 2

Sky Mountain Inc bond and Strongest Enterprise bond are zero coupon bonds (Exhibit 4.2). In calculation, it is possible to assume that interest compounds semi-annually on similar coupon-paying bonds.

When comparing the current yield and yield to maturity, it becomes clear that they can vary widely.

Illustrative Example 2 (continued)

Selected bonds available today. Currency in US dollar										
Issuer	Years to maturity	Coupon	Coupon payment frequency	Price	Coupon frequency per year	No. of periods to maturity	Coupon payment per period	Principal repayment per bond	Current yield	Yield to maturity
Candy Floss Industries	3	7.25%	Annual	913.88	1	3	72.5	1,000	7.93%	10.75%
Hopefully Inc	10	8.00%	Semi-annual	1,100.00	2	20	40.0	1,000	7.27%	6.62%
Sky Mountain Inc	5	0.00%	Semi-annual	726.27	2	10	0.0	1,000	0.00%	6.50%
Sand Bench Corp	4	6.00%	Semi-annual	868.43	2	8	30.0	1,000	6.91%	10.08%
Strongest Enterprise	10	0.00%	Semi-annual	404.59	2	20	0.0	1,000	0.00%	9.25%

► EXHIBIT 4.2 Selected bonds available

The Strongest Enterprise bond has a current yield of zero because it is a zero coupon bond.

However, its YTM is still competitive at 9.25% when compared with what is available from other bonds.

The Sky Mountain Inc bond also offers zero current yield. Its YTM is also low at 6.5% when compared with other bonds.

A coupon, by itself, does not indicate how high or low YTM would be. Hopefully Inc for example, offers the highest coupon at 8%, but its YTM is rather low at 6.62%. Candy Floss offers a lower coupon at 7.25%, but it produces much higher YTM at 10.75%.

With this information, it is possible to update the CIO on the analysis of these bonds. If YTM is the only consideration, it would make sense to recommend Candy Floss and Sand Bench bonds for possible investments because they offer the highest YTM.

Key Learning Point

Yield to maturity (YTM) is the yield that an investor would realise if he or she bought the bond at the current price, held it to maturity, received all the promised payments on time, and reinvested the cash flows received at the yield to maturity.

► Knowledge Check Questions

Question 4

Identify which of the following statements is NOT true of bonds.

- A** It is necessary to calculate the present value of a bond's expected cash flows to calculate the price of a bond.
- B** The value of an asset is the future value of its cash flows.
- C** The required discount rate or rate of return of a bond is the market interest rate, sometimes called the bond's yield to maturity.
- D** Expected future cash flows are estimated using the payments issuers will make on bonds (the coupons) and the value to be received at maturity.

Question 5

State what the YTM of a bond does to the present value of the coupon and principal payments.

- A** Exceed the bond price
- B** Equal to zero
- C** The same as the price of the bond
- D** Lower than the price of the bond

➤ 4.3 VALUATION OF BONDS

With information about the par value, the term to maturity and the coupon rate, investors can make informed decisions about the amount and the timing of all the future promised payments on a bond.

The price of a bond is determined by discounting these future payments using an appropriate discount rate, often referred to as the 'market rate of interest'.

The discount rate depends on market conditions, most notably interest rates, as well as factors specific to the issue and issuer.

The price of a bond equals the present value of future payments on the bond. That is to say, the price is the same as the present value of the interest payments and the par value that will be repaid at maturity.

4.3.1 Bond Valuation Formula

The equation for the price of a bond can be shown as:

$$P_B = \frac{C_1}{1+i} + \frac{C_2}{(1+i)^2} + \dots + \frac{C_n + F_n}{(1+i)^n}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i)^n} \right] + \frac{F_n}{(1+i)^n}$$

where:

- P_B = price of the bond,
- C_t = coupon payment in period t ,
- F_n = par value at maturity,
- i = interest rate, and
- n = periods to maturity.

As a bond is a kind of annuity, we will receive a predetermined payment at the maturity date, the value of the bond will be the present value of the bond. The formula of the bond is the same with calculating the present value of an annuity (PVA), the cash flow will be substituted by the coupon payment, for the last payment, we will receive also the face value of the bond, so we have to add F_n .

4.3.1.1 Par Value Bonds

A property of the bond pricing equation is that each time the coupon rate equals the bond yield or the market rate of interest on similar bonds, the bond will sell at par value. These bonds are called **par value bonds**.

For example, consider a three-year bond with a face value of HK\$1,000 and an annual coupon rate of 5% during a time when the yield or market rate of interest on similar bonds is 5%. The price of this bond can be determined with the following:

$$P_B = \frac{50}{1.05} + \frac{50}{(1.05)^2} + \dots + \frac{1,050}{(1.05)^3} = \text{HK\$1,000}$$

As expected, the price of the bond equals its par value.

4.3.1.2 Discount Bonds

What happens to the price of the bond if the market rate of interest rises overnight to 8%? This is described in the following equation:

$$P_B = \frac{50}{1.08} + \frac{50}{(1.08)^2} + \dots + \frac{1,050}{(1.08)^3} = \text{HK\$922.69}$$

When i is equal to 8%, the price of the bond drops to HK\$922.69 and the bond sells at a below par value. These are called **discount bonds**. This happens because of the fixed nature of the coupon payments.

If the market rate of interest is 8% and the bond pays only 5%, it does not make sense for anyone to buy the bond at par value. This would be like choosing to get a lower return. It is not possible to change the coupon rate to 8% because it is fixed. This is why bonds are known as fixed-income securities.

The only way to increase the yield of the bond to 8% is to reduce its price to HK\$922.69. The bond's yield will be exactly 8% at this price, which is the current market rate for similar bonds.

Through the price reduction of HK\$77.31 (HK\$1,000 – HK\$922.69 = HK\$77.31), the new owner is given an additional 'interest' in the form of a capital gain.

4.3.1.3 Premium Bonds

What happens to the price of the bond if interest rates fall to 2% but the coupon rate remains at 5%? The price would go up to HK\$1,086.52.

At this higher price, the yield on the bond would be exactly 2%, or equal to the current market yield. The HK\$86.52 (HK\$1,086.52 – HK\$1,000 = HK\$86.52) premium adjusts the yield to 2%.

Bonds that sell at above par are **premium bonds**, and this happens when the coupon rate on a bond is higher than the market rate of interest.

Issues linked to bond pricing can be summarised as follows, where i is the market rate of interest:

1. $i >$ coupon rate: bond sells at a discount
2. $i <$ coupon rate: bond sells for a premium
3. $i =$ coupon rate: bond sells at par.

4.3.2 Semi-annual Compounding Formula

Bonds in Europe tend to pay coupons on an annual basis while in the United States most pay interest twice a year.

So, if a bond has a coupon rate of 8%, and the coupon rate paid semi-annually, each year the bondholder will receive two coupon payments of HK\$40 each, totalling HK\$80 (HK\$40 × 2 = HK\$80). The bond valuation formula can be modified as shown below to adjust for coupon payments made more than once a year:

$$P_B = \frac{C/m}{1+i/m} + \frac{C/m}{(1+i/m)^2} + \dots + \frac{C/m + F_{mn}}{(1+i/m)^{mn}}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i/m)^{mn}} \right] + \frac{F_{mn}}{(1+i/m)^{mn}}$$

where:

- C = the annual coupon payment,
- F_{mn} = the amount of the cash payment at maturity (face value),
- m = number of coupon payments per year,
- n = years to maturity, and
- i = market interest rate.

In bonds with semi-annual coupon payments, m equals 2. The calculation is the same regardless of whether bond prices are computed annually, semi-annually, quarterly or some

other period. It is only necessary to know that the yield, coupon payment and maturity are adjusted to be constant with the stated compounding period.

4.3.3 Zero Coupon Bonds

Zero coupon bonds do not make periodic payments but guarantee only a single payment at maturity. The yield of a zero coupon bond is a special case of a semi-annual compounding formula with all coupon payments equal to zero. So, the equation is as follows:

$$P_B = \frac{F_{mn}}{\left(1 + \frac{i}{m}\right)^{mn}}$$

where:

- P_B = price of a bond,
- F_{mn} = amount of payment at maturity,
- i = annual market rate of interest,
- n = years until payment is due, and
- m = number of times interest is compounded per year.

Key Learning Point

The price of a bond is determined by discounting those future payments using an appropriate discount rate, often referred to as the 'market rate of interest'.

► Knowledge Check Questions

Question 6

Identify what the bond price will be if its coupon rate is equal to the market rate of interest.

- A** For a price equal to its face value
- B** For more than its face value
- C** For less than its face value
- D** None of the above

Question 7

Determine when bonds sell at a discount.

- A** When the market rate of interest is less than the bond's coupon rate.
- B** When the market rate of interest is greater than the bond's coupon rate.
- C** When the market rate of interest is equal to the bond's coupon rate.
- D** None of the above.



► Knowledge Check Questions (*continued*)

Question 8

State when bonds sell at a premium.

- A When the market rate of interest is less than the bond's coupon rate.
- B When the market rate of interest is greater than the bond's coupon rate.
- C When the market rate of interest is equal to the bond's coupon rate.
- D None of the above.

Question 9

A company issues a 10-year bond with a coupon rate of 7%. The current market rate of interest is 9%. Assuming annual payments, calculate the present value of the bond. (Round your final answer only to the nearest dollar.)

- A HK\$872
- B HK\$1,066
- C HK\$990
- D HK\$945



4.4 BOND YIELDS

The return that an investor expects from a bond is known as the bond yield.

There are different types of bond yields such as **nominal yield**, which refers to interest paid divided by the face value of the bond; the **current yield**, which refers to the annual interest of a bond divided by its market price; and the **required yield**, which is the yield investors are looking for.

Determining the yield of a bond requires consideration of a few variables including the yield to maturity, the effective annual yield and the realised yield.

4.4.1 Yield to Maturity

The yield to maturity (YTM) of a bond makes the present value of the coupon and principal payments equal to the price of the bond. The YTM may be seen as the promised yield because it is what investors earn every year if they hold the bond to maturity and all the payments are made as promised.

The YTM of a bond changes as interest rates go up or down, but the calculation is based on the promise from issuers to make interest and principal payments as agreed through the bond contract.

4.4.2 Effective Annual Yield

So far, when pricing a bond with a semi-annual compounding period, the assumption was that the bond's annual yield is twice the semi-annual yield. Even though this convention is used on Wall Street and by other practitioners who deal in bonds, it does not consider compounding. As a result, the values do not represent the true cost of funds and using them can lead to the wrong decisions.

Annualised yields are computed by multiplying the yield per period by the number of compounding periods. This calculation is only appropriate for decision-making when comparing bonds with the same compounding frequencies. For instance, an investor needs to be cautious when comparing yields of European and US bonds because the interest on the former is compounded annually while the latter is compounded twice a year.

The accurate way to annualise an interest rate is to calculate the effective annual interest rate (EAR), sometimes called the **effective annual yield (EAY)**. The way to annualise the yield on a bond is as follows:

$$\text{EAY} = \left(1 + \frac{\text{Quoted interest rate}}{m} \right)^m - 1$$

where:

- Quoted interest rate = simple annual yield (semi-annual yield × 2),
- m = the number of compounding periods per year.

4.4.3 Realised Yield

The YTM points to the return on a bond if all payments are made and investors hold the bond to maturity. However, investors may sell their bonds before maturity. So, the **realised yield** is the return earned on a bond from cash flows that were actually received.

The realised yield represents the interest rate at which the present value of actual cash flows from the investment equal the price of the bond. The realised yield makes it possible for investors to understand what they actually earned on their investment.

Apply and Analyse 1

Apply

ABC Investment Ltd purchased a number of bonds for its fixed-income portfolio some time ago. The company's CIO is wondering whether this is the right time to liquidate some of them and take profit.

This requires analysing the bonds to calculate what yield would be realised if they are sold at today's market price.

To that end, the relevant information is compiled in the table in Exhibit 4.3 in order to compute realised yield. The mechanics of the calculation are identical to that of yield to maturity so it is possible to equate the present value of today's price and all coupon payments received to the price paid for the bond in the past.

Apply and Analyse 1 (continued)

Selected bonds in the portfolio. Currency in US dollar										
Issuer	Years held	Coupon	Coupon payment frequency	Par value	Price paid	Current price	Coupon frequency per year	No. of periods to maturity	Coupon payment per period	Realised yield
Cave Inc	3	12.000%	Annual	1,000.00	913.44	804.59	1	3	120.000	9.52%
Flightmaze Corp	4	5.500%	Annual	1,000.00	947.68	894.52	1	4	55.000	4.49%
Oats & Company	2	7.875%	Semi-annual	1,000.00	1,000.00	1,078.75	2	4	39.375	11.49%
Table Foods	5	8.400%	Semi-annual	1,000.00	936.05	1,048.77	2	10	42.000	10.85%
Flyhome Corp	4	7.375%	Semi-annual	1,000.00	974.33	1,023.56	2	8	36.875	8.65%

► EXHIBIT 4.3 Selected bonds in portfolio**Analysis**

The calculations show that the five bonds have produced a wide range of realised yields to date.

Cave and Flightmaze bonds trade at lower prices today than the purchase prices. This would tend to hold down realised yield. In the case of Flightmaze, you will notice that a low coupon of 5.5%, when combined with lower current price, has produced a particularly poor yield of 4.49%.

Oats and Table bonds have the highest realised yields to date. In both cases, the current price is higher than the purchase price. This capital gain adds to the yield. It also makes the realised yield higher than the coupon rate.

The information of realised yield alone may not be sufficient to make a decision on which bonds should be sold. Each bond issuer's credit quality should be reassessed, and potential future return should also be reassessed as of now, in order to make an informed decision.

On the basis of the calculation, Cave and Flightmaze bonds appear to be the best options to sell and they would more likely be sold and the sales proceeds would be deployed to other more profitable investments. The other three bonds are probably more likely to be kept at this point, without further information.

Key Learning Point

Determining the yield of a bond requires consideration of a few variables including the yield to maturity, the effective annual yield and the realised yield.

► Knowledge Check Questions

Question 10

Identify which one of the following is NOT true of a realised yield.

- A** The realised yield is the return that investors earn on a bond given cash flows that were actually received.
- B** The realised yield is equal to the yield to maturity, even when the bond is sold before maturity.
- C** It is the interest rate at which the present value of the actual cash flows that an investment generates are equal to the price of the bond at the time of its sale.
- D** All of the above.

Question 11

Ms. Lau wants to invest her bonus in a four-year bond. The interest that the bond pays is 11% and it is paid semi-annually. The bonds are selling at HK\$962.13 today. Calculate the yield of this bond if held to maturity. (Round to the nearest answer.)

- A** 11.5%
- B** 11.8%
- C** 12.5%
- D** 12.2%

Question 12

Five years ago, Ms. Lee bought a 10-year bond that pays 8% semi-annually for US\$981.10. Today, she sold it for US\$1,067.22. Calculate the realised yield. (Round to the nearest percentage.)

- A** 7%
- B** 8%
- C** 11%
- D** 10%

Question 13

An investor earns a semi-annual yield of 6.4% on a bond paying coupons twice a year. Calculate the effective annual yield. (Round to two decimal places.)

- A** 12.80%
- B** 6.40%
- C** 6.50%
- D** 13.21%



4.5 FACTORS AFFECTING BOND PRICE

Interest rate risk is unavoidable for anyone who owns bonds because interest rates are always changing in financial markets. Companies are particularly interested in the relationship among prices, maturity and coupon rates.

4.5.1 Interest Rate

Interest rate movements are negatively related to bond prices. As interest rates decline, the prices of bonds rise and vice versa. This negative relation occurs because the coupon rate on most bonds is fixed at the time the bonds are issued. The negative relation also affects all other financial claims that pay a fixed rate of interest to investors.

4.5.2 Maturity

Volatility refers to the range of change in the price of a bond. Volatility in bond prices is affected by its maturity and the coupon rate.

Bonds with longer maturity dates tend to be more volatile than bonds with shorter maturity dates.

The duration directly impacts the cash flow that a bond generates. Longer durations mean more cash flow but also more opportunities to re-evaluate the price of the bond.

4.5.3 Coupon Rate

In general, bonds with lower coupon rates tend to be more volatile than those with higher coupon rates.

As with the maturity, the coupon rates also affect the volatility in the price of a bond. If the coupon rate is lower, the price that investors are agreeable to pay for that bond in the secondary market is likely to fluctuate more depending on market conditions.

Conversely, bonds with higher coupon rates are more resilient to changes in market conditions as they promise higher cash flows throughout the duration of the bond.

Apply and Analyse 2

Apply

The CIO of ABC Investment Ltd is concerned about the direction that interest rates will move in. The global economy seems to be doing well, especially in the United States. A push by the US Federal Reserve to raise interest rates could impact the bonds in which the company invests.

With this in mind, the CIO asks finance staff to determine the impact of rising interest rates on a number of bonds that are under consideration for possible investment.

If the discount rates are 1% higher than current levels, how would the bond prices be different for those bonds that the CIO is considering buying? Making the decision will require calculating and classifying the potential impact from rising interest rates.

Apply and Analyse 2 (continued)

The information on the six bonds under consideration is on hand.

With this information, it is possible to analyse the impact of rising interest rates, assuming purchases are made today at this morning's market prices, and rising interest rates have also taken effect today.

All these considerations make it possible to calculate the results in Exhibit 4.4.

Selected bonds available today. Currency in US dollar									
Issuer	Years to maturity	Coupon	Coupon payment frequency	Discount rate	Principal repayment per bond	Coupon frequency per year	No. of periods to maturity	Coupon payment per period	Price this morning
Sustainable Corp	10	8.00%	Annual	6.000%	1,000	1	10	80.0	1,147.20
Lord Inc	3	6.10%	Semi-annual	5.800%	1,000	2	6	30.5	1,008.15
Sociability Group	15	8.50%	Annual	8.500%	1,000	1	15	85.0	1,000.00
Landmade Info System	10	8.00%	Semi-annual	8.125%	1,000	2	20	40.0	991.56
Landmade Info System	10	0.00%	Semi-annual	8.125%	1,000	2	20	0.0	450.94
Guardshield Company	8	6.50%	Semi-annual	8.000%	1,000	2	16	32.5	912.61

New bond prices with discount rate 1% higher. Currency in US dollar											
Issuer	Years to maturity	Coupon	Coupon payment frequency	Discount rate	Principal repayment per bond	Coupon frequency per year	No. of periods to maturity	Coupon payment per period	New price this afternoon	Change in bond price	Price change
Sustainable Corp	10	8.00%	Annual	7.000%	1,000	1	10	80.0	1,070.24	(76.96)	-6.7%
Lord Inc	3	6.10%	Semi-annual	6.800%	1,000	2	6	30.5	981.29	(26.86)	-2.7%
Sociability Group	15	8.50%	Annual	9.500%	1,000	1	15	85.0	921.72	(78.28)	-7.8%
Landmade Info System	10	8.00%	Semi-annual	9.125%	1,000	2	20	40.0	927.23	(64.33)	-6.5%
Landmade Info System	10	0.00%	Semi-annual	9.125%	1,000	2	20	0.0	409.71	(41.22)	-9.1%
Guardshield Company	8	6.50%	Semi-annual	9.000%	1,000	2	16	32.5	859.57	(53.04)	-5.8%

► EXHIBIT 4.4 Selected bonds available and new bond prices with 1% higher discount rate

Explain what would be an appropriate recommendation.

Analysis

Bond prices for all of the options have suffered losses without exception, as predicted by bond theorems which stipulate opposite relationship between bond price and interest rate movements. As discount rates go up, bond prices come down.

Apply and Analyse 2 (continued)

The Lord Inc bond, for example, suffered only 2.7% loss in price because it has the shortest maturity. The Guardshield Company bond suffered relatively modest losses of 5.8% because it has the next shortest maturity. Sustainable Corp, Sociability Group and Landmade Info have the worst losses because of their long maturities.

The comparison also confirms the bond theorems that long-term bonds have more price volatility than short-term bonds for any given changes in interest rates.

Finally, finance staff compare the changes in prices of the two Landmade Info bonds which have identical maturities and discount rates, but different coupon rates. Landmade Info zero coupon bond suffered bigger losses than Landmade Info 8% coupon bond. This also confirms the bond theorems that the prices of lower coupon bonds change more than the prices of higher coupon bonds for a given change in interest rates.

All this information can help the CIO make a better decision on the bond characteristics that are important when considering what bonds to buy in a rising interest rate environment.

Key Learning Point

Volatility refers to the range of change in the price of a bond. Factors affecting bond price include interest rate, maturity and coupon rate.

► Knowledge Check Questions

Question 14

Identify which one of the following is true.

- A** Long-term bonds are less price volatile than short-term bonds of similar risk.
- B** As interest rates decline, the prices of bonds rise and vice versa.
- C** Everything else being equal, short-term bonds have more risk than long-term bonds.
- D** All of the above.

Question 15

Identify which one of the following is NOT true.

- A** The risk that bond prices will change as interest rates change is interest rate risk.
- B** Bond prices and changes to interest rates are inversely related.
- C** When interest rates increase, bond prices increase.
- D** Long-term bonds have higher price volatility than short-term bonds of similar risk.



4.6 BOND AGREEMENT/INDENTURE AND KEY FEATURES – CALL PROVISION

Most bonds are sold to a variety of investors, and someone must make sure that the payments are made on time. This problem is solved by including all relevant details for a particular bond issue in a **bond indenture** that is held and administered by a trust company.

The bond indenture or **bond agreement** is a legal document that specifies the payment requirements, all other salient matters relating to the issue (such as any assets that might serve as security for the bond), protective provisions and other features.

The **trust company** ensures that the issuer follows the covenant provisions in the indenture. This makes the bond a legal contract between issuer and holder. The trust company ensures the terms of the contract are upheld.

Changes to the contract require approval from enough bondholders to represent between a half to two-thirds of the value of outstanding bonds.

Bond indentures include all the features attached to the cash payments.

The par value of most bonds in the United States is US\$1,000, although bond prices are typically quoted based on a par value of 100. In other words, if the price of a bond is quoted at US\$99.583, a US\$1,000 par value bond would be selling for US\$995.83. The term to maturity of the bond represents how much time is left until the maturity date.

The coupons or **regular interest payments** for bonds are determined by multiplying the coupon rate by the par value of the bond. For instance, a bond with a rate of 6% and a par value of US\$1,000 would pay coupons of US\$60 if they are paid annually, or US\$30 every six months if they are paid semi-annually.

There are also a number of provisions for security and protection.

Bonds are technically debt instruments secured by real assets. Sometimes, they are called mortgage bonds but not all of them are backed by real property.

Debentures, are generally unsecured or secured by a general floating charge over unencumbered assets.

Government bonds are debentures, for example, as they are not backed by a specific collateral but they are still generally referred to as bonds.

Collateral trust bonds are secured by other financial assets, like shares or treasury bills.

Equipment trust certificates are secured by equipment and the assets pledged as security are owned by investors through some kind of agreement until the loan has been retired.

Protective covenants restrict the actions of issuers. Negative covenants prohibit some actions, such as limiting dividend payment beyond a certain size or pledging assets to another lender. Positive covenants specify actions that companies have agreed to undertake, such as providing quarterly financial statements or maintain working capital at a certain level.

Some bonds have other features (or options) that may provide for some, or all, of the principal to be repaid before maturity.

Callable bonds give issuers the option to 'call' or repurchase outstanding bonds at predetermined prices and at specified times. A necessary condition for the call is that market interest rates decline.

A company may want to redeem its bonds and issue new ones to benefit from lower interest rates. This is not always enough of a reason to call bonds because of costs associated with a new issue, which may be greater than the savings from lower interest rates.

These types of bonds create higher risks to investors, so the call price generally exceeds the par value of bonds to provide an incentive for investors.

Alternatively, **retractable (or putable) bonds** allow bondholders to sell bonds back to issuers at predetermined prices earlier than the maturity date. This may retract or shorten the maturity of the bond. A necessary condition for bondholders to exercise their retractable option is that interest rates have gone up.

Extendible bonds allow bondholders to extend the maturity date of the bond.

Extendible and retractable both give bondholders flexibility to change the maturity of a bond.

Sinking fund provisions require issuers to set money aside every year so that there are enough funds to pay off bonds at maturity. There are two approaches to make these provisions. One is to repurchase a certain amount of debt every year. Another is for the firm to pay into a sinking fund to buy other bonds and make the capital available at maturity.

Issuers benefit from sinking fund provisions because they help them avoid having to pay the entire face value of an issue at maturity. However, if bonds are repurchased, sinking fund provisions are not always beneficial to investors because bonds may be called and repaid at random. In addition, bonds are normally repaid at par value and investors may suffer a loss if bonds are worth more. Finally, as the amount of an issue goes down, there may be a loss of liquidity because the bonds become less attractive. The operation of a sinking fund in which funds are used to buy government bonds may offset these disadvantages.

On the other hand, **purchase fund provisions** require the repurchase of an amount of debt only if it can be repurchased at or below a given price. These provisions tend to benefit the holders of debt because they provide liquidity and downward price support for the price of debt instruments.

Apply and Analyse 3

Apply

A broker is deciding whether to sell three callable bonds. To make a decision, the portfolio manager needs to compute realised returns by determining the investment outcomes when the bonds are called, or not called. Analyse the following three bonds:

- A company called Entertainment Inc issued eight-year bonds with a coupon rate of 6.375% and payments made semi-annually. The rate sought by the market is 7.65%. The bond is callable after five years at an 8.5% premium on the face value.
- Pen Corp has seven-year bonds that pay a coupon rate of 8.375% semi-annually with a current worth of US\$1,063.49. The bonds can be called in three years at a price of US\$1,075.

Apply and Analyse 3 (continued)

- Rain Corp has 18-year bonds that pay interest semi-annually and at a rate of 9.735%. The market yield to maturity of the bonds is 7.95%. The bonds can be called in five years at a premium of 13.5% over par value.

Analysis

After analysing the three bonds, the calculations lead to the results in Exhibit 4.5.

Calculation of realised yields of callable bonds										
Issuer	Years to maturity or call	Coupon	Coupon payment frequency	Discount rate	Maturity repayment or call price	Coupon frequency per year	No. of periods to maturity	Coupon payment per period	Bond price based on YTM	Realised yield
Entertainment Inc	8	6.375%	Semi-annual	7.650%	1,000	2	16	31.9	924.75	7.65%
Entertainment Inc if called	5	6.375%	Semi-annual		1,085	2	10	31.9	924.75	9.67%
Pen Corp	7	8.375%	Semi-annual		1,000	2	14	41.9	1,063.49	7.20%
Pen Corp if called	3	8.375%	Semi-annual		1,075	2	6	41.9	1,063.49	8.20%
Rain Corp	18	9.735%	Semi-annual	7.950%	1,000	2	36	48.7	1,169.34	7.95%
Rain Corp if called	5	9.735%	Semi-annual		1,135	2	10	48.7	1,169.34	7.83%

► EXHIBIT 4.5 Calculation of realised yields of callable bonds

The Entertainment Inc bond appears to have a discount rate which is the same as realised yield to maturity of 7.65%. Using a formula or financial calculator it is possible to compute the bond price of US\$924.75. This bond price, and the call price of US\$1,085, makes it possible to compute a realised yield of 9.67% if the bond is called in five years' time.

In the case of the Entertainment Inc bond, it might be a good option to call the bond in five years because it would give higher realised yield at 9.67% than if held to maturity at 7.65%.

For the Pen Corp bond, a bond price is in place. It is possible, then, to compute the yield to maturity of 7.2% based on the inputs. It is also possible to use the bond price and call price to compute realised yield to call of 8.2%. In the case of Pen Corp it may also be a good option to call the bond in three years to get the higher yield to call.

In the case of the Rain Corp bond, the discount rate of 7.95% is also the realised yield to maturity. It is possible to use other inputs to calculate the bond price of US\$1,169.34. This bond price and call price can be used to calculate realised yield to call of 7.83%. In the case of the Rain Corp bond, it makes sense for the bond to mature, and not be called, to generate the higher realised yield.

Key Learning Point

The bond indenture or bond agreement is a legal document that specifies the payment requirements, all other salient matters relating to the issue (such as any assets that might serve as security or collateral for the bond), any protective provisions and other additional features.

► Knowledge Check Questions

Question 16

Identify which of the following is NOT correct.

- A** Through callable bonds, issuer have the option to sell bonds at a predetermined price.
- B** Debentures are not secured bonds.
- C** Extendible bonds allow bondholders to extend maturity dates.
- D** Convertible bonds give bondholders an option to convert bonds into common shares at a predetermined conversion ratio.

➤ 4.7 BOND RATING

As most investors, particularly individuals and smaller businesses, do not have the knowledge to consider the probability of default of a specific bond issue or issuer, they rely on credit rating agencies to give them this information.

There are three well-known credit rating agencies: Moody's Investors Service, Standard & Poor's and Fitch. They rank bonds in order of the probability of default and publish the ratings.

4.7.1 Issue Rating

Most bond issuers purchase bond ratings from one or more agencies. For a one-time fee, the agency examines the credit quality of the firm (e.g. its ability to pay the promised coupon interest), the indenture provisions, covenants and the expected trends of firm and industry operations.

The agencies analyse and discuss with management to assign a bond rating (as shown in Exhibit 4.6) that indicates the likelihood of default (non-payment of coupon or par value, or violation of the bond indenture) on the bond issue.

On top of that, the rating agency also commits to frequently re-examine the issuer's risk.

For instance, should the financial position of the company change, the rating agency may put the issue on a credit watch list, either positive or negative. After that, the agency may downgrade, upgrade or reaffirm the original rating.

Moody's	Standard & Poor's	Fitch	
Aaa	AAA	AAA	Best quality, least credit risk
Aa1	AA+	AA+	High quality, slightly more risk than a top-rated bond
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper-medium grade, possible future credit quality difficulties
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Medium-quality bonds
Baa2	BBB	BBB	
Baa3	BBB-		
Ba1	BB+	BB+	Speculative issues, greater credit risk
Ba2	BB	BB	
Ba3	BB-	BB-	
Bl	B+	B+	Very speculative, likelihood of future default
B2	B	B	
B3	B-	B-	
Caa	CCC	CCC	Highly speculative, in default or high likelihood of going into default
Ca	CC	CC	
C	C	C	
	D	DDD	
		DD	
		D	

► **EXHIBIT 4.6 Bond ratings** (Source: data from Standard & Poor's, Moody's Investor Service, Fitch Ratings.)

Despite the cost and concerns associated with getting a lower-than-expected rating, having a rating makes issuer easier to sell bonds to the public. The rating signals that an independent agency has considered the qualities of the issuer and bond issue, and determined that the credit risk justifies the rating.

An unrated bond issue may not be well received by investors, who may wonder if the firm is trying to hide something and that the firm would have rated it if this was an attractive bond issue. At the same time, some investors such as pension funds and insurance companies may have restrictions against purchasing unrated debt.

A bond's security, or collateral provisions (discussed below), affect its credit rating. Bonds with junior or unsecured claims receive lower bond ratings, leading investors to demand higher yields to compensate for the higher risk. Thus, bond issues of a single firm can have diverse bond ratings if their security provisions are not the same.

Investment-grade bonds are those with ratings of Baa3, BBB-, or better. They are called 'investment grade' as, historically, investors (individuals and managed funds, such as bank trust department portfolios and pension funds) were allowed to invest in such bonds.



Bonds below an investment-grade rating were deemed too risky for such conservative portfolios and were not allowed to be held.

Times and regulations change, however, and bonds that are below investment grade, meaning they have ratings of Ba1, BB+, or lower, have gained a spot in many investment portfolios.

Known as **junk bonds** or (euphemistically) high-yield bonds, they have more risk but offer higher expected returns and have benefits in lowering unsystematic risk in diversified portfolios.

Formerly, issuing companies would seek the highest bond rating possible for new issues for two reasons. First, having high-rated debt added prestige to the company; an AAA-rated firm was viewed as being managed well, financially stable, and strong. In addition, such bonds gave the appearance of being a safe investment and, as they had an investment-grade rating, they would find demand amongst investors such as trust departments and pension funds.

The second main reason for seeking a high bond rating is that the higher rating saves the firm interest expense, as coupon rates on highly rated bonds are lower than those on lower rated bonds because of the risk-expected return trade-off.

Some firms prefer to issue debt rather than dividend-paying common stock as interest payments are tax deductible to the issuer. Others, as part of the firm's financial strategy, have issued bonds and used the funds to repurchase shares of common stock.

Investors have noted that having junk bonds in a portfolio offers potential return enhancements and may diversify risk. Over time, the stigma attached to junk bonds has diminished. Issuers and investors are more amenable to these securities.

For example, in 1980, less than a third of S&P-rated bonds were not investment quality and many of those were 'fallen angels'; that is, bonds that were initially issued with an investment-grade rating but then fell into the junk category because the issuing firm ran into financial difficulty.

By the late 1980s, more than half of rated debt was in the high-yield category, and by 2007 over 70% of S&P-rated firms had junk bonds outstanding.

The number of AAA-rated and AA-rated companies fell from 17% of issuers in 1980 to 2% in 2007; B-rated bonds, however, have grown from 7% of issues in 1980 to over 40% of issues in 2007.

Only two non-financial firms had a AAA rating in 2016: Johnson & Johnson and Microsoft.

ExxonMobil was a third firm with a AAA bond rating, but it was downgraded in early 2016 because of the implications of falling oil prices and the firm's commitment to large capital expenditures and dividends to shareholders.

4.7.2 Issuer Rating

The credit rating agencies also rate the issuers of bonds. The three most prominent ratings agencies rate the ability of issuers, whether companies or governments, to service their debts – meaning pay the coupons on all the bonds they issue as well as other obligations.

In addition to differences in terms to maturity, the yield on bonds will differ from the risk-free rate because of additional risks or features associated with these instruments. In other words, investors expect extra compensation for assuming additional risks, and therefore they require higher returns.

The most obvious difference in yield arises because of the different levels of **default risk** associated with the bond issuer. In other words, bondholders require higher yields to compensate them for the possibility that the borrower may default on the promised debt payments.

This factor is an obvious influence on yields. For example, three bond issues that have similar coupon rates and mature in about the same time may have different yields to reflect the different levels of default risk of their issuers.

Debt ratings are mainly assigned by the three debt-rating services, which perform detailed analyses of bond issuers to determine their ability to sustain their required payments of interest and principal.

The ratings may be modified by 'high' or 'low' to indicate the relative ranking within a category.

Illustrative Example 3

One of the practical uses of credit ratings is to make an estimate of what the market's required discount rate would be for bonds issued by a particular issuer of a particular rating band.

In bonds issued in US dollars, US Treasury bond yields are used as the risk-free rates. By knowing the issuer's bond rating or estimated bond rating (if not formally rated by credit rating agencies), and by using the current market risk premium for the relevant rating category, you could estimate the rough yields required by the market.

The table in Exhibit 4.7 shows the estimated required bond yields based on rating category and maturity as an illustration.

Estimated market yields by rating and maturity (12 June 2018)							
Rating	Risk premium ^a	Maturity	2 yr	3 yr	5 yr	7 yr	10 yr
		US Treasury yield	2.55%	2.67%	2.81%	2.92%	2.96%
AAA	0.62%		3.17%	3.29%	3.43%	3.54%	3.58%
AA	0.72%		3.27%	3.39%	3.53%	3.64%	3.68%
A	0.97%		3.52%	3.64%	3.78%	3.89%	3.93%
BBB	1.54%		4.09%	4.21%	4.35%	4.46%	4.50%
BB	2.35%		4.90%	5.02%	5.16%	5.27%	5.31%
B	3.61%		6.16%	6.28%	6.42%	6.53%	6.57%
CCC	6.78%		9.33%	9.45%	9.59%	9.70%	9.74%

^a ICE BofAML US corporate option-adjusted spread.

► **EXHIBIT 4.7** Estimated market yields by rating and maturity (Source: data from Intercontinental Exchange.)

Illustrative Example 3 (continued)

Actual yields required by bond investors are determined during pricing. There could also be significant variances in required discount rates even for issuers within the same rating category depending on company specific factors.

The rating agencies try to put companies with similar credit strengths within the same rating bands. Therefore, it would not be a surprise to see issuers of similar strengths facing similar discount rates in the market. As seen in the table, risk premium also goes up as rating deteriorates.

It is possible, then, to use credit ratings as a tool to estimate potential costs in raising funds through debt issues. It is also possible to estimate the impact on borrowing costs as the borrower's credit strengthens or deteriorates by sliding up or down the rating bands.

Key Learning Point

Investment-grade bonds are those with ratings of Baa3, BBB-, or better. Junk bonds or high-yield bonds, have more risk but offer higher expected returns.

► Knowledge Check Questions**Question 17**

Identify which of the following is incorrect.

- A** The liquidity preference theory says investors prefer short-term debt.
- B** Expectations theory suggests that a downward-sloping yield curve implies interest rates are expected to decline.
- C** The risk premium in the bond yield reflects the risk of default risk, liquidity risk and features specific to the issue.
- D** A debt rating of BB for S&P is better than AAA.

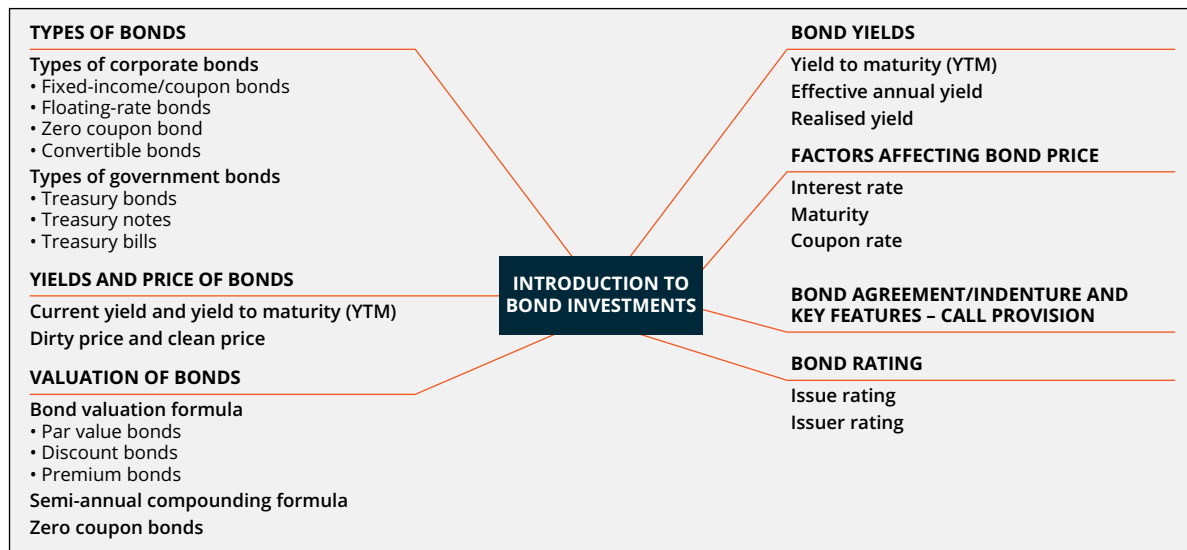


SUMMARY

- Bonds provide investors with predetermined future payments. Bond markets represent a major source of financing for companies, financial institutions and governments.
- Bonds are generically referred to as long-term debt instruments that promise fixed payments to their owners.
- The key feature of a bond is that the issuer agrees to pay the bondholder (investor) a regular series of cash payments and to repay the full principal amount by the maturity date.
- Bonds in which the interest paid to investors is fixed for the life of the contract are called fixed-income bonds.
- A bond that pays a rate of interest that fluctuates over the life of the instrument is called a floating-rate bond.
- A zero coupon bond is a debt security that does not pay interest during its life. There are no coupons to submit for interest payments.
- Convertible bonds provide the bond holder with the option of converting the bonds into a specified number of shares of equity (stock) of the issuing company.
- A Treasury bond (or T-bond) is a debt security issued by a government. It typically has a maturity of more than 10 years and interest payments are made semi-annually. A Treasury note has a typical maturity ranging from two years up to 30 years. Treasury bills have short maturity dates of four weeks, 13 weeks or 26 weeks and have no stated interest rate.
- The face value, or par value, for most bonds in the United States is US\$1,000. This is the principal owed to the bondholders at maturity.
- Coupon payments are the interest payments made to bondholders.
- The current yield (CY) is defined as the ratio of the annual coupon interest divided by the current market price.
- The dirty price refers to a quote for a bond that includes both the current value of all the future cash flows that a bond generates, including the interest on the next coupon payment. The clean price does not include any accrued interest or interest that will be accrued on the next coupon payment date.
- As a bond is a kind of annuity, we will receive a predetermined payment at the maturity date, the value of the bond will be the present value of the bond.
- Par value bonds are bonds in which the coupon rate is the same as the market rate of interest on similar bonds. These bonds sell at par value.
- In discount bonds the coupon rate is lower than the market rate of interest on similar bonds.
- Bonds that sell at prices above par are called premium bonds.

- The return that an investor expects from a bond is known as the bond yield.
- The yield to maturity or discount rate makes the present value of coupon and principal payments equal to the price of the bond.
- Annualised yields, when calculated by multiplying a yield per period by the number of compounding periods, are useful when comparing bonds with the same compounding frequencies.
- The realised yield is the actual return earned on a bond held over a period of an investment.
- Bond prices are negatively related to interest rate movements.
- Volatility refers to the range of change in the price of a bond. Volatility in bond prices is affected by its maturity and the coupon rate.
- Bonds with longer maturity dates tend to be more volatile than bonds with shorter maturity dates.
- In general, bonds with lower coupon rates tend to be more volatile than those with higher rates.
- The bond indenture or bond agreement is a legal document that specifies the payment requirements, all other salient matters relating to the issue (such as any assets that might serve as security or collateral for the bond), any protective provisions and other additional features.
- Most bond issuers purchase bond ratings from one or more agencies.
- Investment-grade bonds are those with ratings of Baa3, BBB-, or better. Junk bonds, or (euphemistically) high-yield bonds, have more risk but offer higher expected returns and have benefits in lowering unsystematic risk in diversified portfolios.

MIND MAP



LIST OF FORMULAS

Current yield

$$CY = \frac{\text{Annual interest}}{B}$$

where

- CY = current yield,
- B = market price.

Yield to maturity (YTM)

$$B = i \times \left[\frac{1 - \frac{1}{(1 - k_b)^n}}{k_b} \right] + F \times \frac{1}{(1 + k_b)^n}$$

where

- B = market price,
- k_b = discount rate,
- n = number of periods to maturity,
- F = amount to be paid at maturity, and
- i = interest.

Price of a bond

$$P_B = \frac{C_1}{1+i} + \frac{C_2}{(1+i)^2} + \dots + \frac{C_n + F_n}{(1+i)^n}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i)^n} \right] + \frac{F_n}{(1+i)^n}$$

where

- P_B = the price of the bond,
- C_t = coupon payment in period t , where $t = 1, 2, 3, \dots, n$,
- F_n = par value,
- i = market interest rate, and
- n = number of periods to maturity.

Semi-annual compounding formula

$$P_B = \frac{C/m}{1+i/m} + \frac{C/m}{(1+i/m)^2} + \dots + \frac{C/m + F_{mn}}{(1+i/m)^{mn}}$$

$$P_B = \frac{C}{i} \times \left[1 - \frac{1}{(1+i/m)^{mn}} \right] + \frac{F_n}{(1+i/m)^{mn}}$$

where

- C = annual coupon payment,
- F_{mn} = amount of the cash payment at maturity (face value),
- m = the number of coupon payments per year,
- n = years to maturity, and
- i = annual market interest rate.

Pricing of zero coupon bonds

$$P_B = \frac{F_{mn}}{\left(1 + \frac{i}{m} \right)^{mn}}$$

where

- P_B = price of a bond,
- F_{mn} = cash payment at maturity,
- i = annual market interest rate,
- n = years until payment is due, and
- m = number of times interest is compounded per year.

Effective annual yield

$$EAY = \left(1 + \frac{\text{Quoted interest rate}}{m} \right)^m - 1$$

where

- Quoted interest rate = simple annual yield (semi-annual yield × 2),
- m = the number of compounding periods per year.

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. There are no restrictions on the use of cash that is generated by issuing convertible bonds.

Answer B is incorrect. This is out of the scope.

Answer C is correct. Firm can convert the convertible bond into shares anytime, and share dividends are not fixed payment.

Answer D is incorrect. Convertible bonds are usually converted to common shares.

Question 2

Answer A is incorrect. They are typically sold at deep discount initially.

Answer B is incorrect. Zero coupon bonds are typically sold at a lower price than similar coupon bonds.

Answer C is incorrect. Only convertible bonds would convert to common shares.

Answer D is correct. As there is no interim cash flow, zero coupon bonds always sell at deep discount.

Question 3

Answer A is incorrect. Zero coupon pays full face value when redeemed at maturity.

Answer B is incorrect. The price of zero coupon bonds is fluctuated, it is not a must to sell for less than similar bonds that make periodic coupon payments.

Answer C is correct. Zero coupon bonds pay 'zero' coupon but the full value of bond at maturity.

Answer D is incorrect. Only C is the correct answer to this question.

Question 4

Answer A is incorrect. Bond price is always the present value of its expected cash flow.

Answer B is correct. By definition, the value of asset is calculated by the present value of all expected cash flow instead of the future value.

Answer C is incorrect. The required market interest rate is the yield to maturity.

Answer D is incorrect. Future cash flows are coupons and maturity value expected in the future.



Question 5

Answer A is incorrect. YTM must make the present value of the coupon and principal payments equal to the bond price.

Answer B is incorrect. YTM must make the present value of the coupon and principal payments equal to the bond price.

Answer C is correct. By definition, to calculate the bond value, we use the YTM to discount all the cash flow at present value.

Answer D is incorrect. YTM must make the present value of the coupon and principal payments equal to the bond price.

Question 6

Answer A is correct. By definition, when the interest rate is higher than the coupon rate, the bond sells for a discount. When the interest rate is lower than the coupon rate, the bond sells for a premium. When the interest rate is equal to the coupon rate, the bond sells at par value.

Answer B is incorrect. The bond should sell at face value.

Answer C is incorrect. The bond should sell at face value.

Answer D is incorrect. The bond should sell at face value.

Question 7

Answer A is incorrect. In this case the bond would sell at a premium.

Answer B is correct. When the coupon rate is lower than the market rate of interest, the bond will sell at a discount.

Answer C is incorrect. In this case the bond would sell at face value.

Answer D is incorrect. B is the correct answer.

Question 8

Answer A is correct. Whenever the coupon rate of a bond is higher than the market rate of interest, the bond sells at a premium.

Answer B is incorrect. In this case the bond would sell at a discount.

Answer C is incorrect. In this case the bond would sell at par value.

Answer D is incorrect. A is correct answer.

Question 9

Answer A is correct. Calculation:

Years to maturity = $n = 10$

Coupon rate = $C = 7\%$

Annual coupon = $\text{HK\$}1,000 \times 0.07 = \text{HK\$}70$

Current market rate = $i = 9\%$

$P = 871.65$

Question 10

Answer A is incorrect. This statement in this choice is true.

Answer B is correct. By definition, the yield to maturity is the return on a bond if the bond is held to maturity.

Answer C is incorrect. The statement in this choice is true.

Answer D is incorrect. Only B is the right answer to this question.

Question 11

Answer D is correct. Calculation:

Years to maturity = $n = 4$

Coupon rate = $C = 11\%$

Semi-annual coupon = $\text{HK\$}1,000 \times (0.11/2) = \text{HK\$}55$

Present value of bond = $P_B = \text{HK\$}962.13$

$C = 55$ ($110 / 2$) = PMT, $PV = 962.13$, $FV = 1,000$, $n = 4 \times 2 = 8$, *CPT i/y*, get $6.11 \times 2 = 12.2$ annual YTM.

Question 12

Answer D is correct. Calculation:

Purchase price of bond = $\text{HK\$}981.10$

Years investment held = $n = 5$

Coupon rate = $C = 8\%$

Payment frequency = $m = 2$

Semi-annual coupon = $\text{HK\$}1,000 \times (0.08/2) = \text{HK\$}40$

Realised yield = i

Selling price of bond = ' P_B ' = $\text{HK\$}1,067.22$

The realised rate of return is roughly 9.6% when rounded. The exact yield calculated with a financial calculator is 9.56%.

Question 13

Answer D is correct. Calculation:

Semi-annual yield = 6.4%

The effective annual yield can be computed as:

$$\text{EAY} = (1 + .128 / 2)^2 - 1 = 13.21\%$$

Question 14

Answer A is incorrect. Long-term bonds have longer horizon until maturity, the price volatility will be higher.

Answer B is correct. With the formula of bond price, holding other remains unchanged, interest rates decline, the prices of bonds rise and vice versa.

Answer C is incorrect. Long-term bonds involve more uncertainty and the risk will increase.

Answer D is incorrect. A and C are false.

Question 15

Answer A is incorrect. By definition, interest rate risk is the risk that bond prices will change as interest rates change.

Answer B is incorrect. Because of bond nature, interest rate changes and bond prices are inversely related.

Answer C is correct. When the interest increases, the bond price will decrease.

Answer D is incorrect. Long-term bonds involve more uncertainty and the risk will increase.

Question 16

Answer A is correct. Callable bonds give the issuer the option to 'call', or repurchase, outstanding bonds at predetermined call prices (generally at a premium over par) at specified times.

Answer B is incorrect. By definition, debentures are not secured by physical assets and collateral.

Answer C is incorrect. By definition, extendible bonds let bondholders to extend the maturity date.

Answer D is incorrect. By definition, convertible bonds give the bondholders an option to convert into common shares at a predetermined conversion ratio.

Question 17

Answer A is incorrect. Short-term debt has higher liquidity and less uncertainty.

Answer B is incorrect. By definition, a downward-sloping yield curve implies that interest rates are expected to decline in the future.

Answer C is incorrect. Risk premium is the reward for investors for bearing default risk, liquidity risk and issue-specific features.

Answer D is correct. The AAA grade in S&P is the best quality, least credit risk rating.

EXAM PRACTICE

QUESTION 1

Identify which of the following statements is true of convertible bonds.

- A.** The most significant disadvantage to a corporation of issuing convertible bonds is that they increase the cash that the firm must use to make interest payments.
- B.** The typical conversion ratio is set so that the firm's stock price must appreciate 5% or less before it is profitable for the holder to convert the bond to stock.
- C.** Firms that issue convertible bonds can do so at a lower interest rate.
- D.** The typical issue of convertible bonds allows the holder of the bond to convert it to preferred stock.

QUESTION 2

Identify which of the following is true.

- A.** The volatility of long-term bonds is lower than similar short-term bonds.
- B.** The prices of bonds rise as interest rates drop. As interest rates go up, bond prices drop.
- C.** Short-term bonds are riskier than long-term bonds, if everything else is the same.
- D.** Interest rate risk decreases as maturity increases.

QUESTION 3

Identify which of the following explains a bond's yield to maturity.

- A.** The discount rate, which ensures that the coupon and principal payments are higher than the price of the bond.
- B.** The discount rate, which makes the present value of the coupon and principal payments equal to zero.
- C.** The discount rate, which makes the price of the bond equal to the present value of the coupon and principal payments.
- D.** The discount rate, which makes the value of coupon and principal payments lower than the price of the bond.

QUESTION 4

Identify which of the following best describes marketability.

- A.** The ability of an investor to sell a security quickly, cheaply and at a fair market value.
- B.** The ability of an investor to sell at a profit under all circumstances.
- C.** The ability of an investor to sell the security above its par value.
- D.** None of the above.

QUESTION 5

You are offered a seven-year corporate bond for HK\$943.22. The bond has a coupon rate of 9% and pays a coupon semi-annually. Similar bonds yield 10%. Identify which of the following explains whether you should buy the bonds at the offered price. (Do not round intermediate computations. Round your final answer to the nearest dollar.)

- A.** Yes, the bond is worth more at HK\$1,015.
- B.** No, the bond is worth just HK\$921.
- C.** Yes, the bond is worth more than HK\$951.
- D.** No, the bond is worth HK\$912 only.

QUESTION 6

State the relationship between market rates and bond prices.

QUESTION 7

Calculate the yield of the following bond: FV = HK\$1,000; coupon rate = 6%, paid semi-annually; market rate = 5%; term to maturity = 10 years.

QUESTION 8

A bond is currently trading at HK\$841.70. It has 15 years to maturity. If you require a rate of return of 12%, calculate the bond's coupon rate if the bond pays semi-annual coupons.

QUESTION 9

Describe the effect on bond's price, if a bond-rating agency downgrades the rating of the bond.

QUESTION 10

Suppose that, several years ago, the Canadian government issued three very similar bonds; each has a HK\$1,000 face value and a 10% coupon rate and will mature in five years. The only difference between the bonds is the frequency of the coupon payments. The market yield is now 5.5%. Calculate the price of a bond paying coupons and describe the outcomes for payments.

- (a) Annually
- (b) Semi-annually
- (c) Monthly

QUESTION 11

You hold a convertible bond with a coupon rate of 8%. The market interest rate is 6%. The share price of the issuing company is going down, and you do not expect the company to recover quickly. Explain whether you should convert the bond into shares.

QUESTION 12

Identify price and yield to maturity of a bond paying an annual coupon of 7% and maturing in two years. The present value of a dollar to be received one year from today is 0.927644. The present value of a dollar to be received two years from today is 0.854172.

QUESTION 13

Calculate the current prices for zero coupon bonds with the following market yields which have a par value of HK\$1,000 and will mature in eight years.

- (a) 7.75%
- (b) 5.25%

In each case, calculate the percentage change in the price of the bond if the market yield rises by 1%.

QUESTION 14

You work for an asset management company. You are presented with a portfolio of bonds and loans that are available for you to invest. Brief terms of each instrument are shown as follows.

Three years ago, Company A issued 10-year bonds with a coupon of 6%. The current market rate is 8.5%. Coupon payments are annual.

Company B's 20-year zero coupon bonds compound interest semi-annually. The current market rate is 7.75%.

The five-year bonds of Company C are priced at HK\$1,065.23, with semi-annual coupon payments of 8.5%.

Company D is applying for a five-year term loan from your company. The borrower should pay a default risk premium of 1.75% over the treasury rate. The five-year treasury rate is 5.65%. The borrower must also pay a marketability risk premium of 0.80%.

Company E has issued seven-year bonds, they make semi-annual coupon payments of 7%. The opportunity cost is 8.25%.

Seven years ago, Company F issued 20-year bonds with a face value of HK\$1,000. The bonds paid interest annually with a coupon rate of 7%. The market rate of interest is 5.5% today.

Company G's 10-year zero coupon bond compounds interest semi-annually. The current market rate is 5.65%.

Company H has outstanding bonds that will mature in eight years. They pay interest semi-annually and have a coupon rate of 4.6%. The bonds are selling at HK\$888.92.

Given a change in market interest rates, you are considering between two bonds, a bond with 20 years until maturity and a bond with 5 years until maturity. Assume all the characteristics of these bonds are identical except the maturity dates.

Required:

- (a) Compute the current market value of one of these Company A bonds.
- (b) Compute the maximum price you should have paid for Company B bonds.
- (c) Compute the yield to maturity and effective annual yield that you would earn on Company C bonds.
- (d) Compute the total borrowing cost to Company D.
- (e) Compute the price that you should be willing to pay for Company E bonds.
- (f) Compute the current market price of a Company F bond. Are these bonds selling at a premium or discount?
- (g) Compute the maximum price you should have paid for Company G bonds?
- (h) Compute the yield to maturity on Company H bonds. Also compute the effective annual yield?
- (i) Explain which bonds will change more in price.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. Convertible bonds may decrease the cash that the firm must use by redeeming the bonds with shares.

Answer B is incorrect. The conversion ratio is set by negotiation between the issuer and the buyers and could be set at any percentage premium.

Answer C is correct. Firms that issue convertible bonds can generally do so at a lower interest rate.

Answer D is incorrect. The typical issue of convertible bonds allows the holder of the bond to convert it to common stock.

QUESTION 2

Answer A is incorrect. Long-term bonds have higher price volatility than short-term bonds of similar risk.

Answer B is correct. This is one of the bond theorems.

Answer C is incorrect. Short-term bonds are less risky than long-term bonds if everything else is the same.

Answer D is incorrect. As maturity increases, the interest rate risk rises.

QUESTION 3

Answer A is incorrect. It cannot exceed the price of the bond.

Answer B is incorrect. It cannot be equal to zero.

Answer C is correct. YTM equates the present value of the coupon and principal payments to the price of the bond.

Answer D is incorrect. It cannot be less than the price of the bond.

QUESTION 4

Answer A is correct. Marketability represents ability of an investor to sell a security quickly, at a low transaction cost and at a price close to its fair market value.

Answer B is incorrect. It does not guarantee the sale at a profit.

Answer C is incorrect. It does not guarantee the sale above par value.

Answer D is incorrect. Only A is correct.

QUESTION 5

Answer C is correct.

Calculation:

Years to maturity = $n = 7$

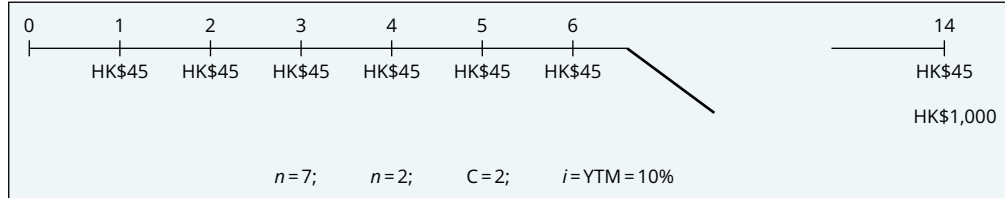
Coupon rate = $C = 9\%$

Frequency of payment = $m = 2$

Semi-annual coupon = $\text{HK\$}1,000 \times (0.09 / 2) = \text{HK\$}45.00$

Current market rate = $i = 10\%$

Present value of bond = PB



$$P_B = \frac{C}{2} \times \left[\frac{1 - \frac{1}{\left(1 + \frac{i}{2}\right)^{2n}}}{\frac{i}{2}} \right] + \frac{F}{\left(1 + \frac{i}{2}\right)^{2n}} = 45 \times \left[\frac{1 - \frac{1}{(1.05)^{14}}}{0.05} \right] + \frac{1000}{(1.05)^{14}}$$

$$= 445.44 + 505.07 = \text{HK\$}950.51$$

She should buy the bond at HK\$943.22 since it is worth HK\$950.51.

QUESTION 6

When market interest rates increase, prices of bonds decrease. When market interest rates decrease, prices of bonds increase.

QUESTION 7

Coupon = (HK\$1,000) (6%) / 2 = HK\$30, $n = 10 \times 2 = 20$, $k = 5\% / 2 = 2.5\%$

QUESTION 8

$n = 30$; $i/y = 6$; $PV = -841.7$; $FV = 1,000$; CPT PMT = 48.5

Semi-annual coupon = HK\$48.5

Annual = HK\$48.5 $\times 2$ = HK\$97

Coupon rate = $97 / 1,000 = 9.7\%$

QUESTION 9

Downgrading implies more risk, which in turn implies more return is required. Therefore the price of the bond is expected to fall.

QUESTION 10

(a) Using a financial calculator:

$n = 5$; $i/y = 5.5$; PMT = 100; $FV = 1,000$; CPT PV = -1,192.16

(b) Using a financial calculator:

$n = 10$; $i/y = 2.75$; PMT = 50; $FV = 1,000$; CPT PV = -1,194.40

(c) Using a financial calculator:

$n = 60$; $i/y = 0.458333$; PMT = 8.3333; $FV = 1,000$; CPT PV = -1,196.32

Notice that the value of the bond increases with more frequent coupon payments.

QUESTION 11

First, since the coupon is higher than the market rate of interest, it implies that the bond is trading at a premium. Then you also know that the company is not doing well so the price of the stock will most probably go further down. Therefore, there is no need to convert the bond into shares. So, you could keep the bond or sell it but not convert it into shares.

QUESTION 12

Bond price = HK\$70 $\times 0.927644$ + HK\$1,070 $\times 0.854172$ = HK\$978.90

$n = 2$; PMT = 70; $PV = -978.90$; $FV = 1,000$; CPT $i/y = 8.19$

Yield to maturity = 8.19%

QUESTION 13

Using a financial calculator:

(a) $n = 16$; $i/y = 3.875$; PMT = 0; $FV = 1,000$; CPT PV = -544.28

If the market yield increases to 8.75%, then $i/y = 4.375$ and we find PV = -504.03.

This is a price decrease of $(544.28 - 504.03) / 544.28 = 7.4\%$

(b) $n = 16$; $i/y = 2.625$; PMT = 0; $FV = 1,000$; CPT PV = -660.62

If the market yield increases to 6.25%, then $i/y = 3.125$ and we find PV = -611.19.

This is a price decrease of $(660.62 - 611.19) / 660.62 = 7.48\%$

The change in market yield was proportionately larger in part (b) than in (a) leading to a larger price impact.

QUESTION 14

- (a) Using a financial calculator:

Years to maturity = $n = 7$ Coupon rate = $C = 6\%$ Annual coupon = $\text{HK\$}1,000 \times 0.06 = \text{HK\$}60$ Current market rate = $i = 8.5\%$ Present value of bond = $\text{HK\$}872.04$

- (b) Years to maturity =
- $n = 20$

Coupon rate = $C = 0\%$ Market rate = $i = 7.75\%$ Payment frequency = $m = 2$

$$P_B = \frac{F_{mn}}{\left(1 + \frac{i}{m}\right)^{mn}} = \frac{1,000}{(1.03875)^{40}} = \text{HK\$}218.55$$

- (c) Years to maturity =
- $n = 5$

Coupon rate = $C = 8.5\%$ Market rate = i Semi-annual coupon payments = $\text{HK\$}1,000 \times (0.085 / 2) = \text{HK\$}42.50$ Present value = $P_B = \text{HK\$}1,065.23$ YTM = 6.934%

The effective annual yield can be calculated as:

$$\begin{aligned} \text{EAY} &= (1 + \text{Quoted rate} / m)^m - 1 \\ &= (1.03467)^2 - 1 \\ &= 0.07054 = 7.05\% \end{aligned}$$

- (d) Risk-free real rate of interest =
- 5.65%

Market risk premium = 0.80% Default risk premium = 1.75%

The company's borrowing cost is

$$5.65\% + 0.80\% + 1.75\% = 8.2\%.$$

- (e) Years to maturity =
- $n = 7$

Coupon rate = $C = 7\%$ Semi-annual coupon payments = $\text{HK\$}1,000 \times (0.07 / 2) = \text{HK\$}35$ Current market rate = $i = 8.25\%$ Price is $\text{HK\$}934.52$.

- (f) Years to maturity = $n = 13$

Coupon rate = $C = 7\%$

Annual coupon payment = $\text{HK\$}1,000 \times 0.07 = \text{HK\$}70$

Current market rate = $i = 5.5\%$

The bonds are selling at $\text{HK\$}1,136.75$

These bonds are selling at a premium.

- (g) Years to maturity = $n = 10$

Coupon rate = $C = 0\%$

Current market rate = $i = 5.65\%$

Frequency of payments = $m = 2$

$$P_B = \frac{F_{mn}}{\left(1 + \frac{i}{m}\right)^{mn}} = \frac{1,000}{\left(1 + 0.0565 / 2\right)^{2 \times 10}} = \text{HK\$}572.83$$

- (h) Number of years to maturity = $n = 8$

Coupon = $C = 4.6\%$

Market rate = i

Semi-annual coupon payment = $\text{HK\$}1,000 \times (0.046 / 2) = 23$

Present value = $P_B = \text{HK\$}888.92$

YTM = 6.395%

$$\begin{aligned} \text{EAY} &= (1 + \text{Quoted rate} / m)^m - 1 \\ &= (1 + 0.06395 / 2)^2 - 1 \\ &= (1.03198)^2 - 1 \\ &= 0.064972 = 6.497\% \end{aligned}$$

- (i) The price of the bond with 20 years to maturity will change more. The cash flows from the longer-term bond are paid further in the future. As a result, these cash flows are more heavily discounted and therefore more sensitive to interest rate changes.

5

Equities Valuation

CHAPTER TOPIC LIST

5.1 Types of Equities

- 5.1.1** Ordinary Shares
- 5.1.2** Preference Shares

5.2 Features of Dividends – Dividends and Taxes

5.3 Valuation of Equities

- 5.3.1** Common Stock Valuation
- 5.3.2** Constant Dividend: Preference Share Valuation

5.4 Other Forms of Dividends

- 5.4.1** Scrip/Stock Dividends

5.4.2 Stock Split

5.5 Types of Cash Dividends

- 5.5.1** Regular Cash Dividends
- 5.5.2** Extra Cash Dividends
- 5.5.3** Special Cash Dividends

5.6 Timeline for Dividends

- 5.6.1** Board Vote
- 5.6.2** Public Announcement
- 5.6.3** Ex-dividend Date
- 5.6.4** Record Date
- 5.6.5** Payable Date



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE
--

LO7.01: Explain and analyse the theories of major capital market models
--

7.01.02 Explain the different sources of finance (short-term and long-term) and methods of raising capital

7.01.05 Describe different types of equities with their features and different forms of dividends
--

7.01.06 Calculate the share prices based on different dividend growth patterns



OPENING CASE

ABC INVESTMENT LTD

ABC Investment Ltd has managed to grow successfully and the company's portfolio of investments has expanded. An important part of this portfolio is now equities.

The company's chief investment officer (CIO) has the final say on decisions involving the management of this equity portfolio. To make the best decisions possible, the CIO relies on the input and advice of finance professionals in his team.

Providing this advice requires careful analysis of stock valuations, work that is in large part based on a number of skills and approaches that are discussed in this chapter. To best advise the CIO, finance professionals build dividend models and apply discounted cash flow methodologies to compute the ideal current value of equity stock.

Determining the value of equities in a portfolio invariably brings finance professionals in contact with a variety of companies travelling along vastly different growth trajectories. Some are strong dividend payers upfront. Some require a large amount of patience before seeing the first dividend. Some have high growth rates, some do not. But they all need to be analysed and correct valuations must be placed on their equity stocks for the investments to be successful.

There are various considerations that go into this process, including the various forms of equities and dividends and how they are used in the financial world. Dividend payouts are more than a theoretical consideration, they are a key factor that investors take very seriously. In so doing, investors consider special dividends, stock splits and preference dividends and gain an appreciation of their real-life application. 

OVERVIEW

The use and valuation of equities may be one of the most common topics in financial management. It is the subject of this chapter.

There are different types of equities, which are typically characterised by their qualities. Common stock or ordinary shares may be the most frequently held by individual investors, but companies also issue preference shares. In all cases, investors also look to dividends to determine the return they may expect from a particular stock.

Companies take the decision of whether to pay dividends with care and follow a predetermined process to actually effect the payment.

For investors, the more significant consideration is how to value equities. This chapter outlines this process, which requires three steps: First, estimate expected future cash flows. Second, determine the required rate of return or discount rate. Third, compute the present value of a stock. The value of a stock (as with a bond) is the present value of the future cash flows that can be received from it.

But determining present value may require investors to take into consideration dividend payments, which are likely to fluctuate over time, as well as changes to the value of the stock. This chapter is focused on dividends.

There are several issues impacting dividends, including growth, frequency and type and these issues impact how investors calculate them in their projections. Because dividends can change from year to year, the calculations have to take these variations into account.

This chapter also discusses the timeline of dividend payouts and other types of dividends that companies sometimes pay.

5.1 TYPES OF EQUITIES

There are different types of equity securities but the most prevalent is common stock, which represents a basic ownership claim in a company. Another type is preference shares, which also represent an ownership interest in the corporation but get preferential treatment with preference shareholders taking precedence in dividends and in the event of a liquidation.

5.1.1 Ordinary Shares

Holders of **ordinary shares** or **common stock** have the right to vote on all important matters of the company, such as electing the board of directors or voting on mergers or acquisitions.

Owners of common stock are not entitled to any guaranteed dividend payments and have the lowest priority claim on the assets of the company.

The liability of common stockholders is limited to how much they originally paid for the shares, the original investment. The personal assets of common shareholders cannot be taken to satisfy any of the company's obligations.

Common stocks can also be considered perpetuities in that they do not mature. Ordinary shares are retired if the management of the company buys back the shares from investors or if the company is liquidated.

5.1.2 Preference Shares

Preference shares represent an ownership interest in a company and have preferential treatment compared to common stock. Preference shareholders take precedence over ordinary shareholders when it comes to dividends and in the distribution of corporate assets in the event of liquidation.

The board of directors declares the dividend that corporations pay to preference shareholders. If the dividend is not paid, the lack of payment is not legally seen as a default. Because preference shares are a form of equity, the dividends to this type of shares are paid with after-tax dollars.

Preference shares are an equity security but the owners do not have the privilege to vote unless the shares are convertible and converted into ordinary shares. Preference shares are also seen as perpetuities because they do not have a maturity. With that said, most preference shares are not true perpetuities because their share contracts often include call provisions or sinking fund provisions that call for management to retire a certain portion of preference shares every year.

It is still being debated in finance, whether preference shares are debt or equity.

There is an argument to be made that preference shares are a type of bond and there are several reasons for this argument:

- Regular (non-convertible) preference shares include no voting rights.
- Preference shareholders get a fixed dividend regardless of earnings. If the company is liquidated, preference shareholders receive a stated value and not a residual value.
- Preference shares have credit ratings similar to the ratings of those issued to bonds.
- Preference shares can sometimes be converted into common stock.
- Most of preference shares have a retirement date (or call and put date), which means they are not true perpetuities.

Preference shares are sometimes used in debt restructuring where the lenders receive preference shares in exchange for the loans they previously extended to the borrower. The borrower may have deteriorated into a potential default and preference share exchange is then used to replace unpaid loans to avoid triggering default.

Illustrative Example 1

Mitsubishi UFJ Financial Group (MUFG) and Mizuho Financial Group became preference shareholders in June 2015, when they supported Japanese electronics firm Sharp Corporation by swapping JPY200 billion of loans (JPY100 billion each) to an equivalent amount of preference share. As at the end of March 2015, the last full year before the debt-to-equity swap, Mizuho had JPY325 billion of loans outstanding to Sharp and MUFG had JPY318 billion.

The preference share dividend was set at 2.5% plus Japanese yen six months TIBOR (Tokyo Interbank Offered Rate).

The preference share has no voting right. It carries put options that require Sharp to redeem it with either cash or common stock (convertible into common stock). Sharp also has a call option to redeem it with cash.

The put option for which the preference share can be converted in common stock can be exercised only on or after 1 July 2019. The conversion into common stock would have created massive dilution of existing common shareholders, hence creating incentives for Sharp to redeem the preference share before July 2019.

The financial support from MUFG and Mizuho helped Sharp avoid a default and the company has been able to service its remaining debt, including loans from the two banks. Credit losses for the banks would have been substantial had Sharp failed because the average recovery rate of loans to defaulted large Japanese corporate borrowers is in the 10–19% range, much lower than the 40–60% global average recovery rate. The recovery rate of preference share if any, would have been close to zero.

In June 2018, Sharp Corporation announced its plan to redeem JPY200 billion of preference share held equally by MUFJ and Mizuho using the proceeds from a common equity offering. The preference share redemption is credit positive for both MUFG and Mizuho because the banks will recover substantially all of the loans that had been swapped to equity in 2015 to support then-distressed Sharp.

Under the announced terms of the preference share redemption, the two banks will receive a total of JPY185 billion, which is equivalent to the JPY200 billion face value of the preference share minus the dividends received to date. The redemption amount is lower than the face value but a much better outcome than if Sharp had defaulted.

Sharp's successful equity offering also suggests its improved business outlook, which reduces the credit risk on any remaining exposures to Sharp. Sharp reported a net income of JPY70 billion in the fiscal year that ended March 2018, after three consecutive years of losses totalling JPY503 billion.

Key Learning Point

Ordinary shares represent a basic ownership claim in a corporation. Preference shares also represent an ownership stake but receive preferential treatment.

 Knowledge Check Questions**Question 1**

Identify which of the following is a characteristic of common stock.

- A** Fixed dividends
- B** Fixed interest
- C** Limited liability
- D** Credit ratings

Question 2

Identify which of the following is not a characteristic of preference shares.

- A** Fixed dividend.
- B** With voting right.
- C** Equity of the company.
- D** Earlier claim of dividends than common shareholders.

 5.2 FEATURES OF DIVIDENDS – DIVIDENDS AND TAXES

A **dividend** is a pro rata distribution of cash or stock to the shareholders of a corporation. Pro rata means shareholders who, for instance, own 10% of the common shares of a corporation are entitled to receive 10% of the dividend being distributed.

All shareholders may receive dividends, regardless of whether they hold ordinary or preference shares. Companies are not obligated to pay dividends, which are not liabilities. Rather, the board of directors declares dividends for a period and the size of those dividends. The decision is based on the financial performance and plans of the company.

Dividends have to be paid from after-tax earnings and are not tax deductible, in contrast to interest payments on any debt, such as bonds, that are tax deductible.

Dividends can be cash, property, scrip or stock. The most common type are **cash dividends** but companies often also declare **stock dividends**.

Investors care deeply about the dividend practices of companies and dividends are also widely reported in the financial press. Some companies do not pay dividends, most famously Warren Buffett's Berkshire Hathaway.

A cash dividend is a pro rata distribution of cash to shareholders. Cash dividends are not paid on treasury shares (i.e. ordinary shares purchased back by the company). There are several conditions that companies have to meet to pay dividends:

1. **Retained earnings:** It is generally legal to pay dividends from retained earnings but cash dividend distributions from legal capital (using only the balance in common stock) are not. Dividends paid from paid-in capital are considered liquidating dividends.
2. **Adequate cash:** There is a difference between the legality of a dividend and a company's ability to pay for it. The board of directors of a company has to consider the current and future demands on the resources of a company before determining dividends. At times, current liabilities could make a cash dividend inappropriate while capital expenses could lead to smaller dividends as less cash is available.
3. **Declared dividends:** The board of directors makes the decision on whether to pay dividends. The board has full authority to 'declare' dividends. Dividends do not accrue like interest and are not a liability until they are declared. The amount and timing of a dividend are important issues for management to consider because dividend payments can impact the liquidity of the company and the company has to manage shareholders' expectation.

Any dividend payments reduce both current assets and current liabilities of a company but have no effect on the equity shareholders hold. However, the result of declaration and payment of cash dividend is decrease in both the equity and total assets of shareholders.

Preference shareholders have the right to receive dividends ahead of common shareholders. For instance, if the dividend rate on preference shares is HK\$5 per share, common shareholders do not and cannot receive dividends until after preference shareholders have been paid dividends. However, this first claim does not guarantee payment.

Most preference shares also have a preference on the assets of the company if the company fails, a feature that provides some security to this group of shareholders. The preference to assets may be for the par value of the shares or a specified liquidating value.

Preference shares often contain a **cumulative dividend** feature which stipulates that preference shareholders must be paid both current-year dividends and any unpaid prior-year dividends before common shareholders are paid. Preference share dividends not declared in a particular period are called dividends in arrears.

Companies cannot pay dividends to common shareholders while preference share dividends are in arrears, and companies should disclose in the notes to their financial statements the amount of dividends in arrears. This information allows investors to consider the impact of any commitment to pay dividends on the financial position of the company. Investors generally look unfavourably on companies that are unable to meet dividend obligations.

Key Learning Point

To pay out dividends, companies must have retained earnings, adequate cash and declared dividends.

 Knowledge Check Questions**Question 3**

Enrique holds 30% shares of Boo Ltd, the company declared HK\$60,000 cash dividend for all shareholders. Calculate how much dividend is payable to Enrique.

- A** HK\$18,000
- B** HK\$20,000
- C** HK\$60,000
- D** HK\$30,000

Question 4

Identify which of the following are forms of dividends:

- (i) Cash
- (ii) Scrip
- (iii) Property
- (iv) Stock

- A** (i) and (ii)
- B** (ii) and (iii)
- C** (i), (iii) and (iv)
- D** All of the above

 5.3 VALUATION OF EQUITIES

The value of any asset is the present value of its future cash flows. There are three steps needed to put a value on any asset:

1. Estimating expected future cash flows.
2. Determining the required rate of return, or discount rate. This rate reflects the riskiness of future cash flows.
3. Computing the present value of the future cash flows that helps determine the value of the asset.

Applying these steps to assets like bonds can be relatively easy because the cash flows are part of the bond contract and the required rate of return or discount rate is the yield to maturity on bonds with similar levels of risk (see Chapter 4). However, the valuation of equities is more difficult.

The valuation techniques described in this chapter assume that the entity is a going concern. 'Going concern' refers to the accounting assumption that the company will exist long enough to carry out its operations. In other words, the company is not on the verge of default or insolvency, which would require the use of different valuation techniques specifically designed for distressed companies.

5.3.1 Common Stock Valuation

Common stock valuation is difficult.

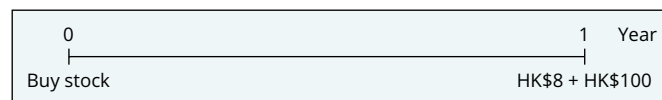
One complication is that, while the expected cash flows for bonds are clearly documented, common stock dividends are less certain. The board of directors declares dividends and is not necessarily obligated to pay dividends in cash, so the timing and size of future cash flows are not certain.

A second complications is that common stocks never expire, which means they are true perpetuities and, in theory, firms never have to redeem them.

A third complication is that the rate of return on common stock is not clearly visible, as opposed to the yield on bonds. This lack of clarity also makes it more difficult to group stocks into risk classes.

5.3.1.1 One-period Model

This model assumes that it is possible to tell the future with perfect certainty. And it considers the purchase of a share of stock to be sold after a year. If it were possible to know that in a year the price (P_1) that the stock can be sold for will be HK\$100 and that the stock will pay an HK\$8 dividend (D_1) at the end of the year, the timeline for the transaction is:



If the investment requires a 20% return, what is the price that an investor would be willing to pay for the stock today?

The value of stock is the present value of the future cash flows that it will generate, in this case:

- HK\$8 dividend,
- HK\$100 sale price.

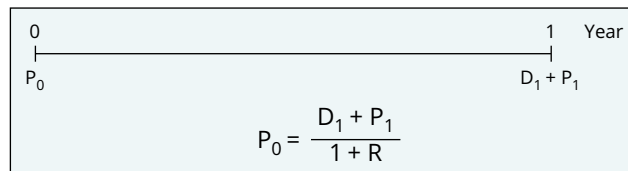
Using a 20% rate of return, the value of the stock equals the present value (PV) of the dividend plus the PV of the proceeds from the sale of the stock as per the following formula is:

$$PV(\text{stock}) = PV(\text{dividend}) + PV(\text{sale price})$$

$$\begin{aligned}
 &= \frac{8}{1+0.2} + \frac{100}{1+0.2} \\
 &= \frac{8+100}{1.2} = \frac{108}{1.2} \\
 &= \text{HK\$90}
 \end{aligned}$$

We can conclude that the value of the stock today is HK\$90.

If the investor pays HK\$90 for the stock, a one-year holding period return on the stock will be 20%. Another way to express this is to use a timeline to show the current value of the stock for a one-period model as follows:

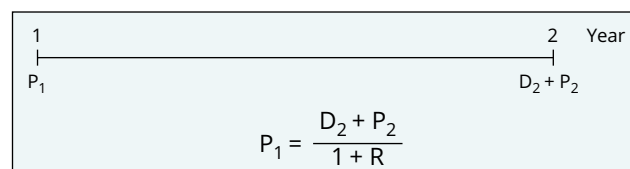


- P_0 = current price of the stock,
- D_1 = dividend paid at end of the period,
- P_1 = stock price at the end of the period,
- R = required return in a particular risk class.

P_0 shows time zero or the present day and P_1 is the price after one period. P_2 is the price after two periods, and so on. It is also worth noting that the price (P_t) in this context, means what the price should be as opposed to the actual market price.

Now, what if at the beginning of year 2 the investor wants to determine the price of a share of common stock that has a similar dividend pattern after one year?

The calculation is similar to the one above. The current price (P_1) is the PV of the dividend and the sale price of the stock. Both of these are received at the end of the year (P_2).



Repeating the process at the beginning of year 3 would lead to a similar result.

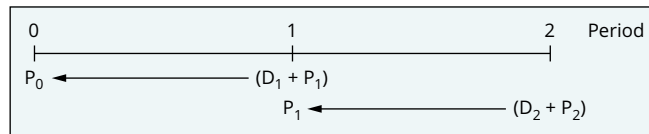
Every single-period model discounts the dividend and sale price at the end of the period by the required return.

5.3.1.2 Perpetuity Model

The one-period model is theoretically correct but does not seem very realistic.

Investors need a formula to value the stock that is useful for a **perpetuity** as opposed to one or two periods. Nevertheless, this can be worked around by stringing together a series of one-period stock pricing models to get a stock perpetuity model.

The first step in this process is to construct a two-period stock valuation model. The timeline for the two-period model is:



The process to construct a two-period model starts with the initial single-period valuation model:

$$P_0 = \frac{D_1 + P_1}{1 + R}$$

where:

- P_0 = Current price of a stock,
- D_1 = Dividend paid at the end of the period,
- P_1 = Stock price at the end of the period,
- R = Required return on common stock in a particular risk class.

The next step is to substitute into this equation the expression derived earlier for P_1 [$P_1 = (D_2 + P_2) / (1 + R)$] and obtain the following:

$$P_0 = \frac{D_1 + [(D_2 + P_2) / (1 + R)]}{1 + R}$$

By rearranging this equation, the result in a stock valuation model for two periods is:

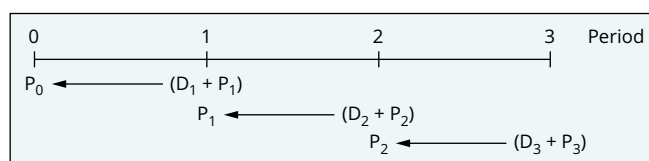
$$P_0 = \frac{D_1}{1 + R} + \frac{D_2}{(1 + R)^2} + \frac{P_2}{(1 + R)^2}$$

Finally, the second-period terms are combined to obtain:

$$P_0 = \frac{D_1}{1 + R} + \frac{D_2 + P_2}{(1 + R)^2}$$

The equation shows that the price of a share of common stock held for two periods equals the PV of the dividend in the first period (D_1) plus the PV of the dividend in the second period and the sale price (D_2 and P_2).

The next step is to construct a three-period model. The timeline for the three-period model is:



Substituting the equation for P_2 into the two-period valuation model, results in a three-period model, as shown below. Remember that $P_2 = (D_3 + P_3) / (1 + R)$.

$$\begin{aligned}
 P_0 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{P_2}{(1+R)^2} \\
 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{(D_3 + P_3) / (1+R)}{(1+R)^2} \\
 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{P_3}{(1+R)^3} \\
 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3 + P_3}{(1+R)^3}
 \end{aligned}$$

By now, it becomes clear that this process can continue for as many periods as is necessary in perpetuity by using the equation:

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \dots + \frac{D_t}{(1+R)^t} + \frac{P_t}{(1+R)^t}$$

In the above equation, t is the time period, which can be any number from one to infinity (∞). The formula uses t , instead of n , to denote the time period. This is because this variable is more commonly used in pricing equations while n denotes the number of periods.

In summary, the models show that the value of a share of common stock today (P_0) is the PV of future dividends and the future sale price of the stock. It is worth noting that this model is theoretically sound but not practical to apply as the number of dividends is infinite. Besides, it is not likely that anyone will be able to accurately forecast the dividend payments and sale price of the stock far into the future. Some realistic simplifying assumptions are required.

5.3.1.3 General Dividend Valuation Model

The last term in stock valuation models used earlier is the sale price of the stock in period t (P_t). In turn, t can be any number up to infinity.

One assumption of the model is that it is possible to forecast the future sale price of the stock. This is not practical.

As P_t moves towards infinity, its value moves towards zero because the discount factor approaches zero. As the model moves towards infinity, it is possible to ignore the $P_t / (1 + R)^t$ term. The final equation can be written as:

$$\begin{aligned}
 P_0 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{D_4}{(1+R)^4} + \frac{D_5}{(1+R)^5} + \dots + \frac{D_\infty}{(1+R)^\infty} \\
 &= \sum_{t=1}^{\infty} \frac{D_t}{(1+R)^t}
 \end{aligned}$$

This equation is an expression of the value of a share and states that the price of the share is the PV of all expected future dividends. The equation makes no assumption about future dividend patterns, such as a constant-growth rate. At the same time, it makes no assumption about when the stock will be sold. At the same time, the model says that to compute the current value of a stock it is necessary to forecast an infinite number of dividends.

What the equation does is to provide insight into what drives the constant changes to stock prices and drastic changes that these prices experience. The formula suggests that market expectations of future dividends determine the value of the stock, so stock prices change constantly as information becomes available.

5.3.1.4 Zero-growth Dividend Model

The simplest assumption is that the growth rate of dividends will be zero and that the dividend payment pattern will remain constant:

$$D_1 = D_2 = D_3 = \dots D_\infty$$

In this case the dividend discount model discussed above becomes:

$$P_0 = \frac{D}{1+R} + \frac{D}{(1+R)^2} + \frac{D}{(1+R)^3} + \frac{D}{(1+R)^4} + \frac{D}{(1+R)^5} + \dots + \frac{D}{(1+R)^\infty}$$

This cash flow pattern is a perpetuity with a constant cash flow.

An equation for such would show that the PV of a perpetuity with a constant cash flow is CF / i . In this case, CF is the constant cash flow and i is the rate of interest. This same relation can be shown in terms of the model for stock valuation as:

$$P_0 = D / R$$

where:

- P_0 = Current price of the stock,
- D = Constant cash dividend in each period,
- R = Required return.

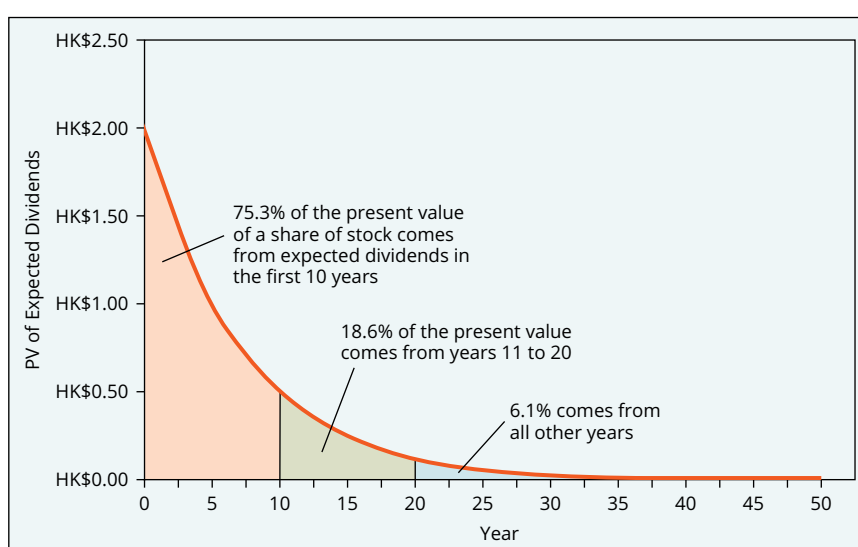
The model fits the pattern of dividends for a share of common stock, particularly for companies that are not growing or for preference shares.

5.3.1.5 Constant-growth Dividend Model

The dividend growth assumption suggests cash dividends are not constant but grow at a rate g . This rate of growth can be either positive or negative. A constant-growth rate provides a reasonable approximation of the pattern of dividends some companies pay out, particularly mature companies.

This assumption of an infinite time horizon creates some theoretical concerns but is not problematic in practice because most companies have a finite lifetime. Cash flows that are further into the future have a smaller PV. Dividends far into the future are very small and have little impact on the price of a stock.

For instance, Exhibit 5.1 shows constant dividends and a 15% discount rate. It clearly shows how the first 10 years of dividends make up 75% of the value of a share while dividends after year 20 contribute about 6% of the value.



► **EXHIBIT 5.1** The future value of a stock (Source: data from Parrino 2017.
Reproduced with permission of John Wiley & Sons.)

The constant-growth dividend can be easily identified and applied, which makes it a fairly straightforward model.

The first step is to determine a model to calculate the value of dividend payments in future periods. Assuming a constant rate of growth of g for dividends, it is possible to use the future value formula:

$$FV_n = PV \times (1 + i)^n$$

This formula can be used to calculate dividend payments by replacing FV_n with D_t , PV with D_0 , i with g , and n with t . The result is:

$$D_t = D_0 \times (1 + g)^t$$

where:

- D_t = Dividend payment in period t ,
- D_0 = Dividend paid in current period, $t = 0$,
- g = Constant-growth rate for dividends.

This equation makes it possible to calculate dividend payments for particular periods. The next dividend, for instance, paid at time $t = 1$, is D_1 . This is the current dividend (D_0) multiplied by the growth rate of $(1 + g)$. As a result, $D_1 = D_0 \times (1 + g)$.

It is worth noting that to compute dividend payments for future periods it is possible to multiply D_0 by the growth rate factor to a particular power. Still, the starting point is D_0 .

This makes it possible to develop a constant-growth dividend model in which the PV of a growing perpetuity is equal to the cash flow value from period 1 (CF_1) divided by the difference of the discount rate (i) and the rate of growth (g) of the cash flow:

$$PVP = \frac{CF_1}{i - g}$$

It is now possible to represent this same relation for stock valuation as follows:

$$P_0 = \frac{D_1}{R - g}$$

where:

- P_0 = Current price of stock,
- D_1 = Dividend in the next period ($t = 1$),
- g = Constant dividend growth rate,
- R = Required return.

The constant-growth dividend model states that the price of a share today equals the dividend in the next period divided by the difference of the discount rate and the dividend growth rate. It is important to remember that PVP represents the current value of the stock (P_0) and that this current value equals the PV of cash flows from dividends. It is worth noting that the growing perpetuity model is only valid if the growth rate is smaller than the required rate of return. The results are meaningless if R is equal to or less than g ($R \leq g$).

5.3.1.6 Computing Future Stock Prices

Another way to use the constant-growth dividend model is to modify it to determine the price of a share. In broad strokes, the price of a share of stock (P_t) can be expressed as the next

period dividend (D_{t+1}), g , and R , when the dividends from D_{t+1} forward are likely to grow at a constant rate. The modified equation then looks as follows:

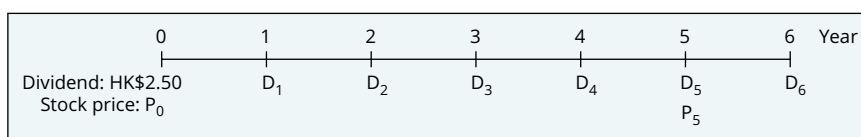
$$P_t = \frac{D_{t+1}}{R - g}$$

Notice that the equation to determine the value of a stock is just a special case of this equation in which $t = 0$.

One approach to understand better how this works is to set up the equation to calculate the share price at $t = 0$, which should create an equation that looks the same as the equation which determines the value of a stock.

For example, suppose a firm has a current dividend (D_0) of HK\$2.50, R is 15%, and g is 5%. What is the price of the stock today (P_0)? What will the price be five years from now (P_5)?

A timeline that identifies some of the crucial variables essential to solve the problem may be useful:



To find the current stock price, it is necessary to first calculate the dividend for the next period (D_1), which is at $t = 1$. The first step is to calculate the firm's dividend for next year:

$$D_1 = D_0 \times (1 + g) = 2.50 \times 1.05 = \text{HK\$}2.625$$

Now it is possible to use the equation to determine the value of a stock and find the price of the stock today:

$$P_0 = \frac{D_1}{R - g} = \frac{2.625}{0.15 - 0.05} = \frac{2.625}{0.10} = \text{HK\$}26.25$$

The next step is to find the value of the stock in five years. In this situation the equation is expressed as:

$$P_5 = \frac{D_6}{R - g}$$

Again, it is necessary to compute D_6 , and this is done with the equation discussed above:

$$D_6 = D_0 \times (1 + g)^6 = 2.50 \times (1.05)^6 = 2.50 \times 1.34 = \text{HK\$}3.35$$

The price of the stock in five years is therefore:

$$P_5 = \frac{3.35}{0.15 - 0.05} = \frac{3.35}{0.10} = \text{HK\$}33.50$$

Finally, note that $\text{HK\$}33.50 / (1.05)^5 = \text{HK\$}26.25$, which is the value today.

5.3.1.7 Mixed (Supernormal) Growth Dividend Model

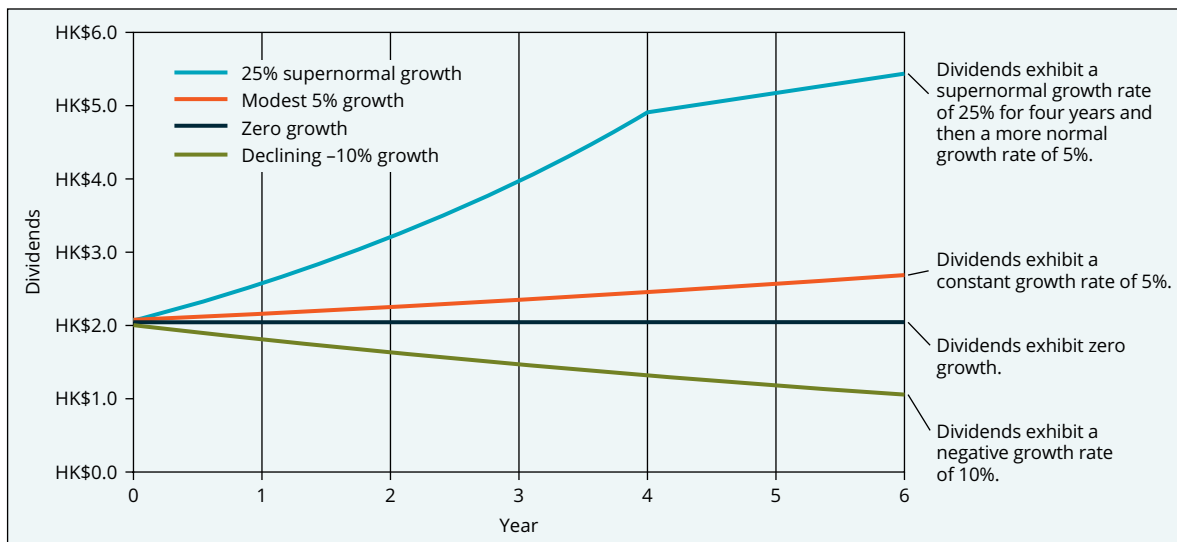
Rarely do dividends grow at a constant rate, rather, companies go through cycles and dividends vary over time.

For many firms, there are two stages of growth.

Successful companies tend to grow quickly in their early years, a time during which they may pay low dividends or even no dividends. Investors are more likely to receive returns from the capital appreciation of the stock, from a cash or stock payout in the case of an acquisition, or from a special cash dividend.

As companies mature, their growth may be seen to stabilise at a normal constant rate. This rate could be similar to the long-term inflation rate plus the long-term growth rate of the overall economy. During this stage, companies are more likely to pay more predictable dividends.

Exhibit 5.2 below shows various dividend growth patterns, a supernormal rate of 25% for four years at the top, a more sustainable growth rate of 5% in the middle, a no-growth rate and a negative growth rate. Companies that experience negative growth may also pay dividends.



► **EXHIBIT 5.2 Dividend growth patterns** (Source: data from Parrino 2017. Reproduced with permission of John Wiley & Sons.)

It is normal for successful companies to experience supernormal growth at the beginning of their life cycles. Periods of supernormal growth can also happen in older companies that reinvent themselves with new products or strategies.

It is not necessary to develop new equations to value a share in a company with a supernormal dividend growth pattern. Rather, it is possible to apply the general dividend model and the equation for the share price of stock with constant dividend growth at any point in time.

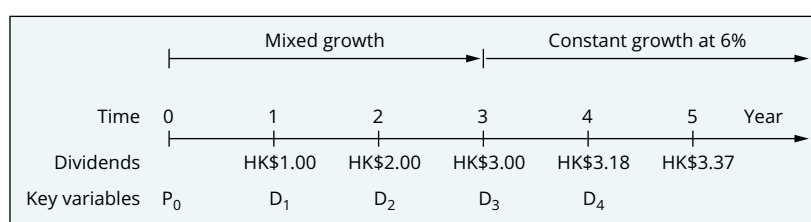
For instance, consider a company with an expected dividend pattern for three years of:

$$D_1 = \text{HK\$}1, D_2 = \text{HK\$}2, D_3 = \text{HK\$}3$$

The dividends are expected to grow at 6% a year constantly after three years.

What is the current price (P_0) of a share if the required rate of return is 15%?

The first step is to draw a timeline, as below:



It is a good idea to prepare a timeline to solve problems with complex dividend patterns and to place cash flows in the proper time periods. It is critical to find out when constant growth starts and to show the proper value on this growth.

There are, as the timeline shows, two different dividend patterns:

1. There is a mixed dividend growth pattern in D_1 through D_3 . This dividend can be valued using the general dividend valuation model.
2. Afterwards, dividends grow at 6% and a constant-growth dividend valuation model can be used to value them.

This means that the valuation model is:

$$P_0 = \text{PV (mixed dividend growth)} + \text{PV (constant dividend growth)}$$

The formula using these values is:

$$P_0 = \underbrace{\frac{D_1}{(1+R)} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3}}_{\text{PV of mixed-growth dividend payments}} + \underbrace{\frac{P_3}{(1+R)^3}}_{\text{PV of constant-growth dividend payments}}$$

The constant-growth dividend stream has a value of P_3 , which is the price at time $t = 3$. P_3 is the value of the future cash dividends discounted to this time period and with the required rate of return of 15%. This can be calculated as:

$$\begin{aligned} D_4 &= D_3 \times (1 + g) = 3.00 \times 1.06 = \text{HK\$}3.18 \\ P_3 &= \frac{D_4}{R - g} = \frac{3.18}{0.15 - 0.06} \\ &= \frac{3.18}{0.09} \\ &= \text{HK\$}35.33 \end{aligned}$$

The equation makes it possible to arrive at the value on year 3 of a constant-growth perpetuity that begins in year 4.

Because P_3 is at time period $t = 3$, there is a need to discount it back to the present ($t=0$), which would be done by dividing P_3 by $(1 + R)^3$.

Using these values, the equation looks like:

$$\begin{aligned} P_0 &= \frac{1.00}{1.15} + \frac{2.00}{(1.15)^2} + \frac{3.00}{(1.15)^3} + \frac{35.33}{(1.15)^3} \\ &= 0.87 + 1.51 + 1.97 + 23.23 \\ &= \text{HK\$}27.58 \end{aligned}$$

Thus, the value of the stock is HK\$27.58 per share.

A more general equation for a mixed or supernormal growth situation would look like:

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \dots + \frac{D_t}{(1+R)^t} + \frac{P_t}{(1+R)^t}$$

When the mixed- or high-growth period ends and dividends grow at a more constant rate of g , P_t can be calculated as follows:

$$P_t = \frac{D_{t+1}}{R - g}$$

5.3.2 Constant Dividend: Preference Share Valuation

Preference shares are hybrid securities, somewhere between bonds and common stock. A preference share has a higher priority claim on the assets of a company, but a lower claim than creditors should the company default. When calculating the value of preference shares, the key issue is whether they have a fixed maturity. If the contract for the preference shares requires the mandatory retirement of the stock at some point, analysts will likely treat it as a bond with a fixed maturity.

The most important difference between fixed maturity preference shares and bonds is the risk of default. Bond coupon payments are a legal obligation while preference share dividends are declared by the board and there is no default if they are not declared.

Still, a failure to pay a preference share dividend is a noteworthy breach that can point to serious financial troubles. As a result, managers work hard to pay preference share dividends.

5.3.2.1 Preference Share with a Fixed Maturity

Because preference share with a fixed maturity is considered similar to a bond, it is possible to use the bond valuation model to determine its price, or value.

Applying the model requires recognition of coupon payments (C) as being equivalent to dividend payments (D) and the frequency of preference share dividend payment. So, the equation can be restated as:

$$\begin{aligned} &\text{Preference stock price} = \text{PV (Dividend payments)} + \text{PV (Par value)} \\ P_0 &= \frac{D/m}{1+i/m} + \frac{D/m}{(1+i/m)^2} + \frac{D/m}{(1+i/m)^3} + \dots + \frac{D/m + P_{mn}}{(1+i/m)^{mn}} \end{aligned}$$

where:

- D = Annual preference share dividend payment,
- P = Stated value of preference share,
- i = Yield to maturity,
- m = Number of times dividend payments are made per year,
- n = Number of years to maturity.

By way of example, consider a utility company that has preference shares with an annual dividend payment of HK\$10 that is paid every quarter – four payments per year – as well as a stated (or par) value of HK\$100 along with 20-year maturity due to a sinking fund requirement. Similar issues have yields of 8%.

The first step to calculate the value of the preference share is to convert the data to quarterly compounding periods:

1. The market yield is 8% per year / 4 quarters = 2% per quarter.
2. The dividend payment is HK\$2.50 quarterly.
3. There are four dividend payments per year for 20 years or 80 payments altogether.

Working this into the equation makes it possible to find the value of the preference share:

$$\begin{aligned} P_0 &= \frac{2.50}{1.02} + \frac{2.50}{(1.02)^2} + \dots + \frac{102.50}{(1.02)^{80}} \\ &= \text{HK\$119.87} \end{aligned}$$

It is also possible to solve this problem with a financial calculator. The keystrokes are:

Enter	80	2	2.50	100
	N	i	PV	FV
Answer			-119.87	

5.3.2.2 Preference Share with No Maturity

Some preference share issues do not have a maturity, which means that they have dividends that are constant over time ($g = 0$) and go on into infinity. These preference shares can be valued as perpetuities. The equation is:

$$P_0 = \frac{D}{R}$$

where D is a constant cash dividend and R is the interest rate, or required rate of return.

For example, suppose an airline has perpetual preference shares that pay HK\$5 in dividend per year. Investors are looking for a rate of return of 8%. What is the value of the preference share? Using the equation, it is possible to determine the value:

$$P_0 = \frac{D}{R} = \frac{5.00}{0.08} = \text{HK\$}62.50$$

Apply and Analyse 1

Apply

Your CIO has expanded your work responsibilities to include analysis of the equity portfolio. Similar to bonds, brokers have no shortage of stock ideas to sell you every day. Brokers have sent you a portfolio of potential stocks all of which have 'buy' recommendations. Your CIO wants you to analyse their valuation based on expected dividends.

Brokers provide the following stock investment ideas:

- Nynet Inc paid a dividend of HK\$4.18 last year. Management is not expecting to increase dividends. The required rate of return is 18.5%. Nynet share price is HK\$23.
- Moriband Corp paid a dividend of HK\$2.15 yesterday and dividends are expected to grow at a 5% for the foreseeable future. The required rate of return is 15%. Moriband share price is HK\$24.
- Nyeil Inc is growing at 6.5% and the company's last dividend was HK\$3.36. The required rate of return is 18%. Nyeil's share price is HK\$30.
- Zhang Corp is growing quickly at 35% per year and this rate is expected to hold for seven years. The company plans to pay a first dividend in three years. This dividend will be HK\$5. After seven years, growth is expected to slow to 8.5%. The required rate of return is 14%. The company's share price is HK\$150.

Apply and Analyse 1 (continued)

Based on this information, identify which stocks are worthwhile investments and which ones the company should pass on.

Analysis

Regardless of the growth rate, each company's valuation is driven by the discounted cash flow of dividends. Even for companies that pay no dividends in the initial years such as Zhang Corp, the valuation can still be computed based on dividends much further into the future. As long as you have projected dividend payments and appropriate discount rates, you can value any stocks the brokers show you.

After valuation has been determined, it is compared with the current share price to determine if a stock is overvalued or undervalued.

Nynet Inc.

$$P_0 = \frac{D}{R} = \frac{4.18}{0.185} = \text{HK\$}22.59$$

Nynet valuation is lower than its share price of HK\$23. Nynet stock is overvalued.

Moriband Corp.

$$\begin{aligned} P_0 &= \frac{D_1}{R - g} = \frac{D_0(1 + g)}{R - g} \\ &= \frac{2.15(1.05)}{0.15 - 0.05} = \text{HK\$}22.58 \end{aligned}$$

Moriband valuation is lower than its share price of HK\$24. Moriband stock is overvalued.

Nyeil Inc.

$$\begin{aligned} P_0 &= \frac{D_1}{R - g} = \frac{3.36(1.065)}{0.18 - 0.065} \\ &= \text{HK\$}31.12 \end{aligned}$$

Nyeil valuation is higher than its share price of HK\$30. Nyeil stock is undervalued and should be purchased.

Zhang Corp.

$$g_{1-7} = 35\%; D_3 = 5.00; g = 8.5\%; R = 14\%$$

$$D_1 = D_2 = 0; D_3 = 5$$

$$D_4 = 5 \times 1.35 = 6.75$$

$$D_5 = 6.75 \times 1.35 = 9.11$$

$$D_6 = 9.11 \times 1.35 = 12.30$$

Apply and Analyse 1 (continued)

$$D_7 = 12.30 \times 1.35 = 16.61$$

$$D_8 = 16.61 \times 1.085 = 18.02$$

$$P_7 = \frac{D_8}{R - g} = \frac{18.02}{0.14 - 0.085} = 327.64$$

$$\begin{aligned} P_0 &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{D_4}{(1+R)^4} + \frac{D_5}{(1+R)^5} + \frac{D_6}{(1+R)^6} + \frac{D_7 + P_7}{(1+R)^7} \\ &= 0 + 0 + \frac{5}{(1.14)^3} + \frac{6.75}{(1.14)^4} + \frac{9.11}{(1.14)^5} + \frac{12.30}{(1.14)^6} + \frac{(16.61 + 327.64)}{(1.14)^7} \\ &= 3.38 + 4.00 + 4.73 + 5.60 + 137.58 \\ &= \text{HK\$}155.29 \end{aligned}$$

Diaz valuation is higher than its share price of HK\$150. Diaz stock is undervalued and should be purchased.

Key Learning Point

There are three steps to value an asset: (1) estimating future cash flows, (2) determining the required rate of return and (3) computing the present value of future cash flows.

► Knowledge Check Questions

Question 5

The One Corporation recently paid a dividend of HK\$3.00 per share. Management expects dividends to grow at a constant rate of 10% per year. Calculate the value of stock three years from today if the required rate of return from company stock is 14%. (Do not round intermediate calculations.)

- A HK\$82.50
- B HK\$109.81
- C HK\$21.42
- D HK\$99.82

Question 6

Identify which of the following is a simplifying assumption used to apply the dividend discount model on the valuation of a share of common stock.

- A Dividends vary each year.
- B Dividends are constant until the tenure of the firm.
- C Dividend growth rate is not equal to the required rate of return.
- D Dividends rate grows less than 0.

► Knowledge Check Questions (continued)**Question 7**

You want to estimate the price of a preference share that receives an annual dividend of HK\$6.25. The stock is likely to be called in eight years and investors will be paid HK\$110 per share. If the required rate of return is 7%, calculate the maximum price that investors are likely to pay for the stock.

- A** HK\$74.14
- B** HK\$89.29
- C** HK\$101.34
- D** HK\$116.25

Question 8

New Territories Bank has issued preference shares without a maturity date. The par value on the shares is HK\$100 and dividends of HK\$3 per share are paid quarterly. Calculate the current value of stock if the required rate of return is 10%.

- A** HK\$30
- B** HK\$120
- C** HK\$300
- D** HK\$100



5.4 OTHER FORMS OF DIVIDENDS

The types of dividends described in previous sections – regular cash dividends, extra cash dividends and special cash dividends – are the types of dividends most commonly used but they are not the only ones. There are other types of dividends that investors often consider.

5.4.1 Scrip/Stock Dividends

A stock dividend does not include the distribution of value, rather, a stock dividend involves the pro rata distribution of new shares to existing shareholders. For instance, a 10% stock dividend involves giving each shareholder new shares equal to 10% of the number of shares that the shareholder has – an investor with 100 shares would get 10 new ones. Stock dividends are less common than regular cash dividends but still happen occasionally. Scrip dividends are slightly different. It involves offering shareholders an option to receive dividends either as cash or additional stock.

Even though companies distribute shares they do not actually distribute value with a stock dividend. To explain, consider a company that pays a 10% stock dividend and has total assets worth HK\$1,100,000 with 10,000 shares outstanding and no debt. Because there is no debt, shareholders own all of the assets and a single share is worth HK\$110. The stock dividend results in an increase of 10% in the number of shares outstanding from 10,000 to 11,000.

The value of the company's assets does not change but the value of each share decreases from HK\$110 to HK\$100 when the additional shares are issued ($\text{HK\$1,100,000} / 11,000 \text{ shares} = \text{HK\$100 per share}$). In other words, the number of shares goes up but their value goes down, while shareholders are left with the same total value.

5.4.2 Stock Split

A **stock split** involves the distribution of a larger multiple of the outstanding shares. However, net value does not change. As the name suggests, a stock split leads to every share being divided into more than one. Stockholders often get one additional share for every share that they have, this type of split is known as a 2-for-1 stock split. For example, in a 2-for-1 split, the investor ends up with twice as many shares, but at half the previous value. So, an investor who owns 100 shares at HK\$50 per share (HK\$5,000) sees the stock split declared. After the split, the investor owns 200 shares at HK\$25 per share (still HK\$5,000).

There are other types of stock splits with different ratios, such as a 3-for-1 stock split.

An important difference between stock dividends and splits is that the former are generally scheduled events while the latter are infrequent. As is the case with a stock dividend, little of substance changes when a company undertakes a stock split. Stockholders might end up with more shares but with the same proportional ownership as before the split.

Illustrative Example 2

For highflying share prices, stock splits are sometimes used to make the share price more affordable to investors. One of the most successful stories is iPhone-maker Apple.

Apple (AAPL) has had four splits. The first was a 2-for-1 split on 16 June 1987. For every share of AAPL, shareholders ended up with two after the split.

The second split on 21 June 2000, was also a 2-for-1. Shareholders with two shares found themselves with four shares after the split.

AAPL's third split was on 28 February 2005 and it was another 2-for-1. So, the owner of four shares would finish with eight shares after the split.

A fourth split happened on 9 June 2014 and it was a 7-for-1 split. So, a shareholder with eight shares would have 56 shares after the split.

So, a shareholder that started with 1,000 shares of Apple stock before the first split would have 56,000 shares after the 2014 split; however, the proportional ownership in the company would not change.

Key Learning Point

Two other types of dividends are scrip and stock dividends and stock splits.

► Knowledge Check Questions

Question 9

Identify which of the following is true.

- A** Companies undertake stock splits to bring share prices down.
- B** Share prices go up after a stock split is announced.
- C** Share prices go down after a stock split is announced.
- D** Sometimes stock splits represent another approach to return value to shareholders.

Question 10

An investor owns 1,000 shares of Company E with a price per share of HK\$24. Company E announced a 3-for-1 stock split. Calculate how many shares the investor would have after the split and the total value of the holdings.

- A** 333 shares worth HK\$8,000
- B** 330 shares worth HK\$24,000
- C** 3,000 shares worth HK\$24,000
- D** 4,000 shares worth HK\$32,000

5.5 TYPES OF CASH DIVIDENDS

Cash dividends are cash distributions of accumulated earnings by a corporation to its shareholders. Cash dividends reduce the shareholder's equity in a company.

5.5.1 Regular Cash Dividends

The regular cash dividend is the most common form of dividend. This type of dividend is paid by cash on a regular basis.

Regular cash dividends are sometimes paid annually and sometimes paid quarterly and are most frequently used as a way for companies to distribute profits to shareholders.

5.5.2 Extra Cash Dividends

The management of a company always has the option of paying out extra cash dividends. This can happen when, for example, earnings are higher than expected.

Companies often pay extra cash dividends at the same time as regular dividends. In turn, some companies use these extra dividends to make sure shareholders receive a minimum portion of earnings each year.

For instance, management may want to distribute 40% of net income every year. If a company earns HK\$2 per share and the regular cash dividend is HK\$0.60 per share, management can pay an extra HK\$0.20 dividend at the end of the year to ensure that the company hits its 40% payout target $[(\text{HK\$}0.60 + \text{HK\$}0.20) / \text{HK\$}2.00 = 0.40, \text{ or } 40\%]$.

5.5.3 Special Cash Dividends

Special cash dividends and extra cash dividends are both one-time payments. Special cash dividends are generally much larger than extra dividends. These special dividends are used to distribute amounts of cash that are unusually large. A special dividend could also help a company distribute proceeds received after the sale of a major asset or to effect a change in the capital structure of a company. For example, in August 2017, China Mobile announced a special dividend of HK\$3.20 per share. This was the first ever special dividend from China Mobile since listing. Regular dividend for 2017 totalled HK\$3.205 per share. The special dividend amounted to RMB55,621 million (HK\$64,093 million or US\$8.2 billion).

Key Learning Point

There are three main types of cash dividends: (1) regular cash dividends, (2) extra cash dividends and (3) special cash dividends.

► Knowledge Check Questions

Question 11

Identify which of the following can define regular cash dividends.

- A** A common way for a company to return profits to shareholders.
- B** They are paid when company earnings are higher than anticipated.
- C** They are dividends that companies pay to shareholders after a liquidation.
- D** They are predetermined fixed payments to common shareholders.

Question 12

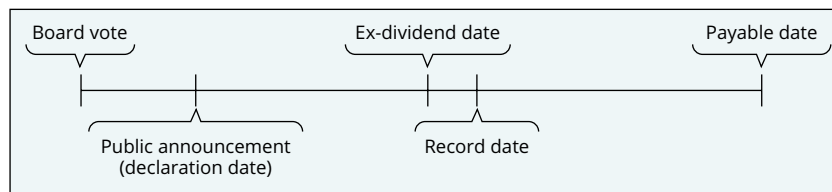
A company has a large amount of extra cash after a particular quarter after selling a significant asset. Identify the type of dividend the company is likely to use to distribute the proceeds to shareholders.

- A** Regular cash dividend
- B** Extra dividend
- C** Special dividend
- D** Liquidating dividend

5.6 TIMELINE FOR DIVIDENDS

The events leading up to the payment of a dividend is standard, more so for publicly listed companies than private companies.

The sequence of events at a publicly traded company for the payment of a dividend is:



5.6.1 Board Vote

The first step in the process is when the board of directors votes on whether to pay a dividend and how large the dividend should be. Directors are representatives of the shareholders and have to approve any distribution of value.

5.6.2 Public Announcement

As a second step, the company announces that it will pay a dividend. The actual date of the announcement is the **declaration date**, sometimes known as the announcement date. The announcement generally includes how much value shareholders will get and other dates related to the process of paying dividends.

Stock prices are affected by dividend announcements because such announcements inform the market about the performance of the company. If the signal that the dividend announcement sends is different than what investors expect they will react accordingly by buying or selling the stock.

For instance, a company announcement of a dividend that is larger than expected could suggest optimism about future profits and higher cash flows and lead to higher stock prices. An announcement that dividends will be cut or eliminated could have the opposite effect.

5.6.3 Ex-dividend Date

The **ex-dividend date** is another important date and is included in the public announcement. This is the first date on which the stock will be traded without rights to the dividend. Investors who buy shares before the ex-dividend date are entitled to the dividend but investors are not entitled to the dividend if they buy the stock after the ex-dividend date. Before the ex-dividend date, a stock is trading cum dividend, or with dividend, and it is trading ex dividend on or after the ex-dividend date.

The ex-dividend date can have a significant impact on a variety of investment-related factors, such as taxes and transaction costs, depending on the jurisdiction where the investment is made. In Hong Kong, individual investors are not liable for taxes on dividends, so the ex-dividend date is important because it may have an almost immediate impact on stock prices.

When they return value to shareholders, companies may end up reducing the investment of shareholders below what investors preferred, which may lead them to buy more shares, which may lead to incurring transaction costs.

The price of shares changes on the ex-dividend date even without any new information about the company. For instance, a company announces a dividend of HK\$1 per share

on a stock trading at HK\$10 per share and the ex-dividend date is tomorrow. The HK\$10 price includes the value of the dividend because investors have bought the shares with the expectation of receiving the dividend. Alternatively, an investor who buys the stock after the ex-dividend date knows that there is no dividend forthcoming and will likely be willing to pay less for the stock. It is worth keeping in mind that research has shown that stock prices drop on the ex-dividend date but by an amount smaller than the full dividend.

5.6.4 Record Date

The **record date** usually comes about two business days after the ex-dividend date. This is the date when an investor must be officially recorded as a shareholder to receive the dividend. The board of directors sets the record date and the company then informs the exchange, which sets the ex-dividend date. The ex-dividend date comes before the record date to account for the time it takes to update the shareholder list.

5.6.5 Payable Date

This is the final date process that leads to the payment of a dividend. The **payable date** is when investors receive dividends, generally a couple of weeks after the record date.

In short, there are three important dates in relation to dividends: (1) the declaration date, (2) the record date and (3) the payment date. Accounting entries are made on the declaration and payment dates.

Key Learning Point

The timeline for a dividend payout is standard. It starts with a board vote, followed by a public announcement, then an ex-dividend date, then a record date and finally a payable date.

► Knowledge Check Questions

Question 13

Identify on which date the stock price will adjust according to the dividend paid.

- A Declaration date
- B Ex-dividend date
- C Record date
- D Payable date

Question 14

Identify the last date before which an investor has to purchase the stock to receive a dividend.

- A Dividend declaration date
- B Ex-dividend date
- C Payable date
- D Record date

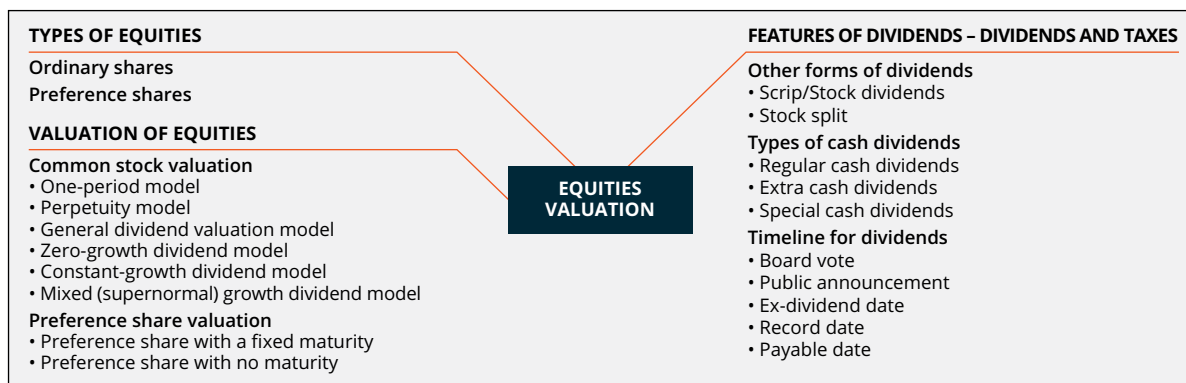


SUMMARY

- The most prevalent type of equity security is common stock. Common stock represents the basic ownership claim in a corporation. Preference share also represents an ownership interest in the corporation but receives preferential treatment.
- A dividend is a distribution of cash or stock to shareholders of a company proportional to ownership of shares. Preference share dividends – to holders of preference shares – must be paid before dividends are paid to ordinary shareholders.
- Cash dividends predominate in practice although companies also declare stock dividends with some frequency.
- For a corporation to pay cash dividends, it must have retained earnings, adequate cash and declared dividends.
- The value of any asset is the present value of its future cash flows. There are three steps that go into putting a value on any asset: (1) estimate expected future cash flows, (2) determine the required rate of return, and (3) compute the present value of the future cash flows to determine what the asset is worth.
- The value of the stock is the present value of the future cash flows that can be received from it. The formula to calculate present value is: $PV(\text{stock}) = PV(\text{dividend}) + PV(\text{sale price})$.
- Investors need a stock valuation formula for a perpetuity for more than one or two periods. This is possible by stringing together a series of one-period stock pricing models to get a stock perpetuity model.
- A zero-growth dividend model assumes dividends remain equal in perpetuity – they do not grow.
- A constant-growth dividend model is often an appropriate assumption for mature companies with a history of stable growth.
- The constant-growth dividend model can be changed to determine the price of a share at any point in time. The dividend growth model provides valid solutions only when g (growth) is smaller than R (the required return).
- Firms typically go through life cycles and so exhibit different dividend patterns over time. This requires the use of a mixed-growth dividend model.
- Under the dividend growth assumption, cash dividends are not constant but grow at an average rate from one period to another indefinitely.
- For many firms, it is not appropriate to assume that dividends will grow at a constant rate, but, rather, they have two stages of growth: (1) rapid growth early in their lives and (2) lower rates of growth at maturity.
- To value a share of stock for a firm with a mixed (supernormal) dividend growth pattern, it is possible to apply the general dividend model and the equation for the price of a share of stock with constant dividend growth at any point in time.

- One type of dividend is the scrip or stock dividend. A stock split involves the distribution of a larger multiple of the outstanding shares.
- The regular cash dividend is the most common dividend. This is a dividend paid in cash on a regular basis. Extra cash dividends are generally paid at the same time. Some companies use extra dividends to make sure they distribute a minimum portion of their earnings every year. Special cash dividends and extra cash dividends are one-time payments.
- The timeline for the payment of dividends is fairly standard and starts with a board vote, followed by a declaration date, then the ex-dividend date, the record date and finally the payable date.
- The three dates that are most important with dividends are: the declaration date, the record date and the payment date.

MIND MAP



LIST OF FORMULAS

Present value of a stock

$$PV(\text{stock}) = PV(\text{dividend}) + PV(\text{sale price})$$

Single-period valuation

$$P_0 = \frac{D_1 + P_1}{1 + R}$$

where:

- P_0 = Current price of a stock,
- D_1 = Dividend paid at the end of the period,
- P_1 = Stock price at the end of the period,
- R = Required return on common stock in a particular risk class.

Zero-growth valuation

$$P_0 = D / R$$

Where:

- P_0 = Current price of the share,
- D = Constant cash dividend in each period,
- R = Required return.

Constant-growth valuation

$$P_0 = \frac{D_1}{R - g}$$

Where:

- P_0 = Current price of share,
- D_1 = Dividend in the next period ($t = 1$),
- g_0 = Constant dividend growth rate,
- R = Required return.

Mixed-growth valuation

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \dots + \frac{D_t}{(1+R)^t} + \frac{P_t}{(1+R)^t}$$

where:

- P_0 = Current price of share,
- D_t = Dividend in period t ,
- R = Required return.

Preference share with fixed maturity

$$P_0 = \frac{D/m}{1+i/m} + \frac{D/m}{(1+i/m)^2} + \frac{D/m}{(1+i/m)^3} + \dots + \frac{D/m + P_{mn}}{(1+i/m)^{mn}}$$

where:

- P_0 = Current price of the stock,
- D = Annual preference share dividend payment,
- P = Stated value of preference share,
- i = Yield to maturity,
- m = Number of times dividend payments are made per year,
- n = Number of years to maturity.

Preference share with no maturity

$$P_0 = \frac{D}{R}$$

where:

- P_0 = Current price of share,
- D = constant cash dividend, and
- R = interest rate or required rate of return.

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. Firms can pay or not pay dividends to shareholders.

Answer B is incorrect. Stocks are not liabilities, no interest payments involved.

Answer C is correct. Common have limited liability and their losses are limited to the original amount of the investment in their investment in the firm.

Answer D is incorrect. Stocks are not liabilities, no credit rating involved.

Question 2

Answer A is incorrect. By definition, firms are to pay fixed dividends to preference shareholders.

Answer B is correct. Preference shareholders have no voting right and no control of the company.

Answer C is incorrect. By definition, preference shares are part of the equity.

Answer D is incorrect. By definition, preference shareholders can claim their dividends earlier than common shareholders.

Question 3

Answer A is correct. The dividend Enrique will receive is proportional to the percentage of share that he owns, the dividend which he will receive is HK\$60,000 × 30% = HK\$18,000.

Question 4

Answer A is incorrect. Cash is not the only form of dividend.

Answer B is incorrect. Scrip is not the only form of dividend.

Answer C is incorrect. Property is not the only form of dividend.

Answer D is correct. By definition, dividends can take four forms: Cash, property, scrip or stock.

Question 5

Answer B is correct. $P_3 = D_4 \div (R - g) = 3 \times (1.10)^4 \div (0.14 - 0.1) = \text{HK\$}109.81$

Question 6

Answer A is incorrect. Dividends may or may not vary each year.

Answer B is incorrect. Dividends are not required to be constant.

Answer C is correct. The simplifying assumptions of the dividend discount model include dividends are constant, dividends have a constant-growth pattern and dividend growth rate is not equal to the required rate of return.

Answer D is incorrect. Dividends may grow positively.

Question 7

Answer C is correct. Using a financial calculator, the price of the stock can be calculated as HK\$101.34, by entering the following data.

N	I	PMT	FV
8	7	6.25	110

Question 8

Answer B is correct. Annual dividend payments = HK\$3 × 4 = HK\$12; Value of preference share = HK\$12 ÷ 0.10 = HK\$120.

Question 9

Answer A is correct. Companies split their stock so as to bring its price down to an appropriate trading range.

Answer B is incorrect. The value change will take place when the stock split is exercised.

Answer C is incorrect. The value change will take place when the stock split is exercised.

Answer D is incorrect. It will not change the value of total value of stock.

Question 10

Answer C is correct. Total value of 1,000 shares = 1,000 × HK\$24 = HK\$24,000. Total number of shares = 3 × 1,000 = 3,000

Question 11

Answer A is correct. By definition, regular cash dividends are frequently used ways for companies to return some of their profits to shareholders.

Answer B is incorrect. Even with higher-than-expected earnings regular cash dividends may not necessarily go up.

Answer C is incorrect. These are liquidating dividends.

Answer D is incorrect. Regular cash dividends are set by the board, and are not predetermined fixed payments.

Question 12

Answer A is incorrect. A regular cash dividend is a cash dividend that is paid on a regular basis.

Answer B is incorrect. A firm pays extra cash dividends if its earnings are higher than expected.

Answer C is correct. A special dividend is a one-time payment to stockholders that is normally used to distribute a large amount of value.

Answer D is incorrect. Liquidating dividends are usually paid when the company is going to be liquidated.

Question 13

Answer A is incorrect. Declaration date is when the board announced its decision to pay dividends.

Answer B is correct. By definition, share prices change on the ex-dividend date even without any new information.

Answer C is incorrect. Record date is the date of which the investors must be shareholders of record to be entitled to dividends.

Answer D is incorrect. Dividends are paid on payable date.

Question 14

Answer A is incorrect. This is the date the board declared the dividends.

Answer B is correct. By definition, the ex-dividend date is the latest date before which an investor has to purchase stock to receive a declared dividend.

Answer C is incorrect. Dividends are paid on the payable date.

Answer D is incorrect. The record date is the date of which the investors must be shareholders to receive dividends.

 EXAM PRACTICE**QUESTION 1**

Identify which of the following is NOT true about common stock.

- A.** Common shareholders have the right to vote for company directors.
- B.** Common stock is considered to have no fixed maturity.
- C.** Holders of common stock are guaranteed dividend payments.
- D.** Common shareholders have limited liability in regard to the obligations of a company.

QUESTION 2

Identify which of the following is true about common stock.

- A.** Common stock has a fixed maturity.
- B.** Common shareholders are guaranteed dividend payments.
- C.** Common shareholders have the lowest priority claim on the assets of a company after a bankruptcy.
- D.** Common shareholders have unlimited liability towards the obligations of the corporation.

QUESTION 3

Identify which of the following explains why a preference share is sometimes treated like a debt security.

- A.** Legally preference share is a debt security.
- B.** Preference share dividend payments are similar to bond interest payments and are fixed in nature regardless of the firm's earnings.
- C.** Preference share dividends are deductible from taxable income just like interest payments on bonds.
- D.** Preference shareholders receive a residual value and not a stated value.

QUESTION 4

You are considering purchasing stock that pays dividends of HK\$4.50. You will be able to sell the stock for HK\$85.00 in a year. Your required rate of return is 15%. Calculate how much you would be willing to pay for the stock.

- A.** HK\$89.50
- B.** HK\$65.37
- C.** HK\$94.10
- D.** HK\$77.82

QUESTION 5

Identify which of the following statements is NOT true about the general dividend valuation model.

- A.** The model assumes no specific pattern for future dividends, such as a constant-growth rate.
- B.** It makes a specific assumption about when the share will be sold.
- C.** The model calls for forecasting an infinite number of dividends for a stock.
- D.** The price of a share of stock is the present value of all expected future dividends.

QUESTION 6

A toy company introduces a new line through which it expects to rapidly grow its earnings over the next three years. After paying a HK\$2.00 dividend last year, it does not plan to pay dividends over the next three years. After that time, the company plans to pay dividends of HK\$4.00 starting in year 4 and increase those dividends by 10% in years 5 and 6. Calculate the present value of dividends to be paid out over the next six years if the required rate of return is 15%. (Do not round intermediate calculations. Round final answer to two decimal places.)

- A.** HK\$13.24
- B.** HK\$12.00
- C.** HK\$6.57
- D.** HK\$10.24

QUESTION 7

Identify which of the following is the most typical example of a zero-growth dividend stock.

- A.** The common stock of a firm in the biotechnology industry.
- B.** The preference shares of a utility company.
- C.** The common stock of a firm in the healthcare industry.
- D.** The common stock of a firm in the information technology industry.

QUESTION 8

A company is growing at a constant rate of 8%. The company paid a dividend of HK\$3.00 a week earlier. If the required rate of return is 15%, calculate the price of the stock three years from now. (Do not round intermediate calculations. Round final answer to two decimal places.)

- A.** HK\$58.31
- B.** HK\$46.29
- C.** HK\$51.02
- D.** HK\$42.83

QUESTION 9

An investor is interested in buying stock in a bank whose management expects no growth in the near future. The investor is attracted by the fact that last year the bank paid a dividend of HK\$5.65. If the investor requires a return of 14%, compute the maximum price he should be willing to pay for a share of the bank's stock.

QUESTION 10

Investors expect Company J to grow at a constant rate of 7%. The company's next dividend is expected to be paid in one year and will be HK\$1.15. The stock is currently priced at HK\$22.35. Calculate the required rate of return on Company J's stock.

QUESTION 11

Company K issued perpetual preference shares that have a par value of HK\$100. These shares pay a dividend of 4.5%. The required rate of return is 8.25%. Calculate the market price of the stock.

QUESTION 12

Ms. Tan wants to purchase common stock of Company L. The stock is priced at HK\$37.45 and the company is expected to pay a dividend of HK\$2.58 next year. The dividends are expected to increase at a constant rate of 7%.

- (a) Calculate the market value of the stock if the required rate of return is 14%.
- (b) Evaluate and explain if this is a good buy.

QUESTION 13

Management of Company M forecast the following growth rates for the next three years: 35%, 28% and 22%. Management then expects the company to grow at a constant rate of 9% indefinitely. The company paid a dividend of HK\$1.75 a week earlier and the required rate of return is 20%. Calculate the value of this stock.

QUESTION 14

Company N is expected to grow at a rate of 22% over the next five years. After that, growth is expected to settle down to 6%. The company paid a dividend of HK\$2.35 and the required rate of return is 15%.

Required

- (a) Calculate the present value of dividends during the rapid-growth period, assuming they grow at the same rate as the company.
- (b) Calculate the value of the stock after year 5.
- (c) Calculate the value of the stock today.

QUESTION 15

Stock in Company P is worth HK\$10.00 per share the day before the ex-dividend date. The dividend is HK\$0.25. Calculate the price of the shares on the ex-dividend date assuming there are no taxes.

QUESTION 16

The outstanding shares of Company Q were worth HK\$120 before a 2-for-1 stock split. After the split, the shares were valued at HK\$62 each. Explain why the market might be increasing the total value of the equity of the company.

QUESTION 17

You work for Big Rich Asset Management. You have a portfolio of stocks under management. They are as follows:

- Stock 1 has forecasted a growth rate of 40% for two years and growth rates of 25% and 20% for the next two years. Growth is then expected to stabilise at 7.5% forever. The firm paid a dividend of HK\$3.50 recently and the required rate of return is 18%.
- Stock 2 issued perpetual preference shares some years ago. The stock, a bank, pays an annual dividend of HK\$4.27. The required rate of return is 12.2%.

- Stock 3 has a market price of HK\$36.34. Dividends are expected to grow at a constant rate of 6%. The last dividend was HK\$3.25. The required rate of return is 16%.
- Stock 4 paid a HK\$2.50 dividend a day before. The company has forecasted a constant-growth rate of 7% forever. The required rate of return is 18%.
- Stock 5 is a fast-growing drug company. Management forecasts that in the next three years its growth rates will be 30%, 28% and 24%, respectively. A week earlier the company paid a dividend of HK\$1.67. Management expects growth to stabilise after three years at 8% and the required rate of return is 14%.

Required

- Calculate the current value of Stock 1.
- Calculate the value of Stock 2 given your required rate of return. Explain whether you should buy this stock if its current market price is HK\$34.41.
- Calculate the value of Stock 3 stock. Based on your answer, explain whether you should buy additional shares in Big Rich Asset Management.
- Compute the expected dividends D_1 , D_2 , D_3 and D_4 for Stock 4.

Compute the present value of the four dividends.

Calculate the expected value of Stock 4 in four years (i.e. P_4).

Calculate the value of the stock today.

Use the equation for constant growth and compute the price of Big Rich Asset Management today.

- Calculate the dividends for Stock 5 for each of the next three years as well as their present value.

Calculate the price of Stock 5 after year 3 when the firm settles to a constant-growth rate.

Calculate the current price of Stock 5.

ANSWERS TO EXAM PRACTICE

Question 1

Answer A is incorrect. The statement is actually correct.

Answer B is incorrect. The statement is actually correct.

Answer C is correct. The statement is wrong because common shareholders are not guaranteed dividends.

Answer D is incorrect. The statement is actually correct.

Question 2

Answer A is incorrect. Common stock has no fixed maturity.

Answer B is incorrect. Common stock is not guaranteed dividends.

Answer C is correct. Owners of common stock will receive value only after all creditors have been satisfied.

Answer D is incorrect. Common shareholders have limited liabilities.

Question 3

Answer A is incorrect. Preference share is treated like equity.

Answer B is correct. Both dividend and interest are fixed in nature.

Answer C is incorrect. Preference share dividends are not deductible from taxable income.

Answer D is incorrect. Preference shareholders do not receive a residual value, only a stated value.

Question 4

Answer D is correct.

$$P_0 = \frac{D_1 + P_1}{1 + R} = \frac{4.50 + 85.00}{1 + 0.15} = \frac{89.50}{1.15} = \text{HK\$}77.82$$

Question 5

Answer A is incorrect. The statement is actually true.

Answer B is correct. The general dividend valuation model makes no specific assumption about when the share of stock is going to be sold in the future.

Answer C is incorrect. The statement is actually true.

Answer D is incorrect. The statement is actually true.

Question 6

Answer C is correct.

$$D_0 = 2.00; \quad D_1 = D_2 = D_3 = 0$$

$$D_4 = 4.00$$

$$D_5 = D_4(1 + g) = 4.00(1.10) = 4.40$$

$$D_6 = D_5(1 + g) = 4.40(1.10) = 4.84$$

$$\text{Present value of the dividends} = \text{PV}(D_1) + \text{PV}(D_2) + \dots + \text{PV}(D_6)$$

$$\begin{aligned} &= 0 + 0 + 0 + \frac{4.00}{(1.15)^4} + \frac{4.40}{(1.15)^5} + \frac{4.84}{(1.15)^6} \\ &= 0 + 0 + 0 + 2.29 + 2.19 + 2.09 \\ &= \text{HK\$}6.57 \end{aligned}$$

Question 7

Answer A is incorrect. The common stock of a firm in the biotechnology industry should have high growth.

Answer B is correct. The preference share dividend is fixed and does not grow.

Answer C is incorrect. The common stock of a firm in the health care industry should have high growth.

Answer D is incorrect. The common stock of a firm in the information technology industry should have high growth.

Question 8

Answer A is correct.

$$R = 15\%; \quad D_0 = \text{HK\$}3.00; \quad g = 8\%$$

$$\begin{aligned} P_3 &= \frac{D_4}{R - g} = \frac{D_0(1+g)^4}{R - g} \\ &= \frac{3.00(1.08)^4}{0.15 - 0.08} = \text{HK\$}58.31 \end{aligned}$$

Question 9

$$D_0 = \text{HK\$}5.65; \quad g = 0; \quad R = 14\%$$

$$P_0 = \frac{D}{R} = \frac{5.65}{0.14} = \text{HK\$}40.36$$

Question 10

$$D_1 = \text{HK\$}1.15; \quad P_0 = \$23.00; \quad g = 7\%$$

$$\begin{aligned} P_0 &= \frac{D_1}{R - g} \\ R &= \frac{D_1}{P_0} + g \\ &= \frac{1.15}{22.35} + 0.07 \\ &= 0.0515 + 0.07 = \text{HK\$}12.15\% \end{aligned}$$

Question 11

$$D = 4.5\% (\text{HK\$}100) = \text{HK\$}4.50; \quad R = 8.25\%$$

$$P_0 = \frac{D}{R} = \frac{4.50}{0.0825} = \text{HK\$}54.55$$

Question 12

$$(a) \quad P_0 = \frac{D_1}{R - g} = \frac{2.58}{0.14 - 0.07} = \text{HK\$}36.86$$

(b) The stock is overpriced and not a good buy.

Question 13

$$g_1 = 35\%; \quad g_2 = 28\%; \quad g_3 = 22\%; \quad g_4 = 9\%; \quad D_0 = \text{HK\$}1.75; \quad R = 20\%$$

$$D_1 = D_0(1 + g_1) = 1.75 \times 1.35 = 2.36$$

$$D_2 = D_1(1 + g_2) = 2.36 \times 1.28 = 3.02$$

$$D_3 = D_2(1 + g_3) = 3.02 \times 1.22 = 3.69$$

$$D_4 = D_3(1 + g_4) = 3.69 \times 1.09 = 4.02$$

$$P_3 = \frac{D_4}{R - g} = \frac{4.02}{0.20 - 0.09} = 36.57$$

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{P_3}{(1+R)^3}$$

$$P_0 = \frac{2.36}{1.20} + \frac{3.02}{(1.20)^2} + \frac{3.69}{(1.20)^3} + \frac{36.57}{(1.20)^3}$$

$$P_0 = 1.97 + 2.10 + 2.14 + 21.16$$

$$P_0 = \text{HK\$}27.37$$

Question 14

(a) $g_{1-5} = 22\%$; $g = 6\%$; $D_0 = \text{HK\$}2.35$; $R = 15\%$

$$D_1 = D_0(1 + g_1) = 2.35 \times (1.22) = 2.867$$

$$D_2 = D_1(1 + g_2) = 2.867 \times (1.22) = 3.498$$

$$D_3 = D_2(1 + g_3) = 3.498 \times (1.22) = 4.267$$

$$D_4 = D_3(1 + g_4) = 4.267 \times (1.22) = 5.206$$

$$D_5 = D_4(1 + g) = 5.206 \times (1.22) = 6.351$$

$$\begin{aligned} \text{PV (Dividends)} &= \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{D_4}{(1+R)^4} + \frac{D_5}{(1+R)^5} \\ &= \frac{2.867}{1.15} + \frac{3.498}{(1.15)^2} + \frac{4.267}{(1.15)^3} + \frac{5.206}{(1.15)^4} + \frac{6.351}{(1.15)^5} \\ &= 2.49 + 2.65 + 2.81 + 2.98 + 3.16 \\ &= \text{HK\$}14.09 \end{aligned}$$

(b) $D_6 = D_5(1 + g) = 6.351 \times (1.06) = 6.732$

$$P_5 = \frac{D_6}{R - g} = \frac{6.732}{0.15 - 0.06} = \text{HK\$}74.80$$

(c) $\text{PV}(P_5) = \frac{74.80}{(1.15)^5} = 37.19$

$$\begin{aligned} P_0 &= \text{PV (Dividends)} + \text{PV}(P_5) \\ &= 14.09 + 37.19 \\ &= \text{HK\$}51.28 \end{aligned}$$

Question 15

Since there are no taxes, the value of the shares should drop by the amount of the dividend. Therefore, the shares should trade for HK\$9.75 on the ex-dividend date.

Question 16

Since the stock split did not create value for investors by itself, there must be information in the split that accounts for the HK\$4 increase in value to shareholders. Firms have a tendency to increase their dividend rate after a stock split. So, if a stock split is generally followed by dividend rate increases, the increase tends to generate information for investors that there is a potential increase in cash flow to investors of the firm's equity.

Question 17

(a) $g_{1-2} = 40\%$; $g_3 = 25\%$; $g_4 = 20\%$; $g = 7.5\%$; $D_0 = \text{HK\$}3.50$; $R = 18\%$

$$D_1 = D_0 \times (1 + g_1) = 3.50 \times 1.40 = 4.90$$

$$D_2 = D_1 \times (1 + g_2) = 4.90 \times 1.40 = 6.86$$

$$D_3 = D_2 \times (1 + g_3) = 6.86 \times 1.25 = 8.575$$

$$D_4 = D_3 \times (1 + g_4) = 8.575 \times 1.20 = 10.29$$

$$D_5 = D_4 \times (1 + g) = 10.29 \times 1.075 = 11.06$$

$$P_4 = \frac{D_5}{R - g} = \frac{11.06}{0.18 - 0.075} = 105.35$$

$$P_0 = \frac{D_1}{1+R} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{D_4}{(1+R)^4} + \frac{P_4}{(1+R)^4}$$

$$P_0 = \frac{4.90}{1.18} + \frac{6.86}{(1.18)^2} + \frac{8.575}{(1.18)^3} + \frac{10.29}{(1.18)^4} + \frac{105.35}{(1.18)^4}$$

$$P_0 = 4.15 + 4.93 + 5.22 + 5.31 + 54.34$$

$$P_0 = \text{HK\$}73.94$$

(b) $D = \text{HK\$}4.27$; $R = 12.2\%$

$$P_0 = \frac{D}{R} = \frac{4.27}{0.122} = \text{HK\$}35.00$$

Since the stock is worth HK\$35.00 but can be purchased for HK\$34.41, you should buy this stock.

$$(c) P_0 = \frac{D_1}{R - g} = \frac{3.25 \times (1.06)}{0.16 - 0.06} = \text{HK\$}34.45$$

No more shares should be bought. The stock is overpriced and the shares should be sold.

(d) $D_0 = \text{HK\$}2.50$; $g = 7\%$; $R = 18\%$

$$D_1 = 2.50 \times (1.07) = 2.675$$

$$D_2 = 2.50 \times (1.07)^2 = 2.86$$

$$D_3 = 2.50 \times (1.07)^3 = 3.063$$

$$D_4 = 2.50 \times (1.07)^4 = 3.277$$

$$\begin{aligned}\text{PV (Dividends)} &= \frac{2.675}{(1.18)^1} + \frac{2.86}{(1.18)^2} + \frac{3.063}{(1.18)^3} + \frac{3.277}{(1.18)^4} \\ &= 2.27 + 2.05 + 1.86 + 1.69 \\ &= \text{HK\$}7.87\end{aligned}$$

$$D_5 = D_4(1 + g) = 3.277 \times (1.07) = 3.506$$

$$P_4 = \frac{D_5}{R - g} = \frac{3.506}{0.18 - 0.07} = \text{HK\$}31.88$$

$$\text{PV}(P_4) = \frac{31.88}{(1.18)^4} = 16.44$$

$$\begin{aligned}P_0 &= \text{PV (Dividends)} + \text{PV}(P_4) \\ &= 7.87 + 16.44 \\ &= \text{HK\$}24.31\end{aligned}$$

For a constant-growth stock:

$$P_0 = \frac{D_1}{R - g} = \frac{2.675}{0.18 - 0.07} = \text{HK\$}24.32$$

(e) $g_1 = 30\%$; $g_2 = 28\%$; $g_3 = 24\%$; $g = 8\%$; $D_0 = \text{HK\$}1.67$; $R = 14\%$

$$D_1 = D_0(1 + g_1) = 1.67 \times (1.30) = 2.171$$

$$D_2 = D_1(1 + g_2) = 2.171 \times (1.28) = 2.779$$

$$D_3 = D_2(1 + g_3) = 2.779 \times (1.24) = 3.446$$

$$\begin{aligned}\text{PV (Dividends)} &= \frac{D_1}{1 + R} + \frac{D_2}{(1 + R)^2} + \frac{D_3}{(1 + R)^3} \\ &= \frac{2.171}{1.14} + \frac{2.779}{(1.14)^2} + \frac{3.446}{(1.14)^3} \\ &= 1.90 + 2.14 + 2.33 = \text{HK\$}6.37\end{aligned}$$

$$D_4 = D_3(1 + g) = 3.446 \times (1.08) = 3.722$$

$$P_3 = \frac{D_4}{R - g} = \frac{3.722}{0.14 - 0.08} = \text{HK\$}62.03$$

$$\text{PV}(P_3) = \frac{62.03}{(1.14)^3} = 41.87$$

$$\begin{aligned}P_0 &= \text{PV (Dividends)} + \text{PV}(P_3) \\ &= 6.37 + 41.87 \\ &= \text{HK\$}48.24\end{aligned}$$



Part C

Capital Market Models and Theories



Chapter 6 Capital Asset Pricing Model

Chapter 7 Cost of Capital

Chapter 8 Capital Structure Theories

6

Capital Asset Pricing Model

CHAPTER TOPIC LIST

6.1 Risk and Return

- 6.1.1 Measures of Return
- 6.1.2 Measures of Risk

6.2 Diversification and Portfolio

- 6.2.1 Diversification
- 6.2.2 Portfolio
- 6.2.3 Limitations of Diversification

6.3 Covariance and Correlation Coefficient of a Portfolio

- 6.3.1 Covariance
- 6.3.2 Correlation Coefficient

6.4 Systematic Risk

- 6.4.1 Total Risk versus Systematic Risk

6.4.2 Importance of Systematic Risk

6.4.3 Measuring Systematic Risk

6.5 Capital Asset Pricing Model

6.5.1 Assumptions

6.5.2 Risk-free Interest Rate

6.5.3 Compensation for Bearing Systematic Risk

6.5.4 Risk Premium

6.5.5 Calculating the Expected Return by Capital Asset Pricing Model

6.5.6 Efficient Frontier and Capital Allocation Line

6.6 Security Market Line



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE**LO7.01: Explain and analyse the theories of major capital market models**

7.01.07 Analyse the risk and return relationship for individual securities and a portfolio of securities, and apply the capital asset pricing model (CAPM) to the pricing of securities



OPENING CASE

SYSTEMATIC RISK AND THE GLOBAL FINANCIAL CRISIS

Effective financial management is grounded in minimising risk and maximising returns, but firms and investors may find themselves blinded to risk when presented with attractive returns based on unsound fundamentals. Such is the lesson of the 2007–2008 global financial crisis, a classic case of a global market slump caused by ‘systematic risk’.

Total risk can be broken down into ‘systematic risk’ and ‘unsystematic risk’. The distinction is crucial.

In the late 2000s, the US Federal Reserve was keeping interest rates low, while a government initiative to make home mortgage loans available to lower-income Americans led to a surge in lending to this demographic, a small portion of whom did not have the economic wherewithal to pay back the loans. These loans were euphemistically called ‘subprime’, or high-risk, loans.

Growing numbers of subprime loans subsequently led to a boom in the US housing market, with banks leveraging heavily, making loans to a growing pool of what ended up being an even higher-risk group of homebuyers – new entrants from the middle and upper-income levels with higher credit scores than those receiving subprime loans.

These individuals tended to already own homes and were focused on making money by what was called ‘flipping’ homes, in which little time passed between the purchase of a home and its resale, with perhaps some renovation work done in the interim to boost resale value and maximise profits months later. The returns were simply too attractive for many investors, who neglected to use diversification to minimise or eliminate risk.

This risk was further amplified by the use of financial products that can be used for corporate financing or risk management. Amongst these instruments were bundled mortgages, derivatives and a new range of products known as mortgage-backed securities. The market for such products swelled, and many financial institutions prioritising returns over risk assessment went all in.

Irresponsible lending practices, lax government regulation and negligent risk assessment by credit rating agencies also contributed to a growing financial bubble that many observers preferred to think of as a new economy.

That bubble quickly began to deflate in the wake of large-scale defaults, especially amongst the speculative home ‘flippers’. This triggered a swift and substantial devaluation of the financial products, which in turn affected the financial institutions who had bet heavily on them. Fannie Mae and Freddie Mac, the quasi-governmental entities that sat at the heart of

the American home financing system, had to be taken over by the US federal government on 7 September 2008.

Shocks continued with the bankruptcy of Lehman Brothers one week later, which at the time was the biggest corporate collapse in US history. Investor confidence evaporated, as did stock values in the United States and other major markets. Liquidity of the already hard-hit banks continued to dry up.

A subsequent US congressional investigation determined that the crisis was avoidable, it was 'not a natural disaster, but the result of high-risk, complex financial products; undisclosed conflicts of interest; and the failure of regulators, the credit rating agencies, and the market itself to rein in the excesses of Wall Street'.



OVERVIEW

Understanding the trade-off between risk and return makes it possible to make investment decisions that best meet the financial needs and goals of a corporate entity. When evaluating potential investments in financial assets, expected return and risk are two dimensions of the decision-making process. The concepts presented in this chapter include the development of measures of expected return and risk on an individual financial asset and on a portfolio of financial assets, the principle of diversification, and the capital asset pricing model (CAPM).

Diversification is a useful method for lessening, or ideally eliminating, business risk within a portfolio or grouping of investible assets. This entails spreading investment funds within a portfolio across different assets. Investing in assets whose returns are less than perfectly positively correlated is key to this strategy, but as the number of assets increases up to a certain point, the diversification benefit of each individual asset decreases. Total risk for well-diversified portfolios is basically equal to systematic risk. Asset markets, therefore, only reward systematic risk.

The risk-free interest rate embodies the amount of interest that would make an investment without risk over a defined period of time. The beta coefficient in CAPM represents systematic risk and volatility for assets, whereas the market risk premium in CAPM defines the gap between the risk-free interest rate and a market portfolio's expected returns.

The CAPM comprises these concepts to calculate the required rate of return for an individual investment or portfolio. By employing CAPM, one can ideally optimise the risk and return relationship when investing in stock by calculating the cost of equity – either in the form of common stock or preference share – or the cost of debt.



6.1 RISK AND RETURN

Risk and return are directly related concepts, in which the degree of risk and the expected level of compensation are positively correlated. The greater the perceived risk to an investment, the greater the return investors will require in order to compensate for undertaking that risk and vice versa.

6.1.1 Measures of Return

There are three main methods that can be used to calculate the return of investment. These methods are:

1. holding period returns,
2. average returns, and
3. expected returns.

6.1.1.1 Holding Period Returns

The total return over a specific holding period includes both capital appreciation and income. Capital appreciation (R_{CA}) for a return is calculated using the following equation:

$$R_{CA} = \frac{\text{Capital Appreciation}}{\text{Initial Price}} = \frac{P_1 - P_0}{P_0} = \frac{\Delta P}{P_0}$$

Income for a return (R_I) is calculated using the following equation:

$$R_I = \frac{\text{Cash Flow}}{\text{Initial Price}} = \frac{CF_1}{P_0}$$

The total holding period return (R_T) is calculated thus:

$$R_T = R_{CA} + R_I = \frac{\Delta P}{P_0} + \frac{CF_1}{P_0} = \frac{\Delta P + CF_1}{P_0}$$

where:

- P_1 = Price at period 1
- P_0 = Price at period 0
- ΔP = Change in price
- CF_1 = Cash flow at period 1

6.1.1.2 Average Returns

There are two main metrics that are used to calculate average returns.

The first is the arithmetic mean (AM), which is most often used in statistics and is suitable for investment horizons of about 12 months.

Arithmetic Mean

The arithmetic mean can be determined using the following equation:

$$\text{Arithmetic mean (AM)} = \frac{\sum_{i=1}^n R_i}{n}$$

Geometric Mean

The second metric is the geometric mean (GM), which is used to determine the compound growth rate over multiple periods and longer time horizons.

In turn, the geometric mean can be calculated using the following equation:

$$\text{Geometric mean (GM)} = [(1 + R_1)(1 + R_2)(1 + R_3) \dots (1 + R_n)]^{\frac{1}{n}} - 1$$

where,

- R_i = Interest rate of i th period of time
- n = Number of periods

Calculating returns using these two approaches can lead to different results.

The arithmetic mean simply adds all the return together and averages them out. This approach is simple but not always accurate. The arithmetic mean always overstates returns of an investment.

Investment annual returns are not independent of one another, so using the geometric mean may lead to more accurate results. If an investment suffers a loss, it will have less capital to generate returns in the coming years, and vice versa.

With higher volatility, the difference between arithmetic mean and the geometric mean will usually be higher, which means the arithmetic return has a more exaggerated overstatement in the returns.

6.1.1.3 Expected Returns

Expected return is the anticipated profit or loss on an investment. It is also known as the expected rate of return.

Apart from the expected returns, there are always unexpected returns which together make up total returns. Total returns, also called actual returns and realised returns, are the actual returns received by the investors on an investment, such as dividends, coupon payments, capital gain and so on.

Total returns are equal to the sum of the expected return and unexpected return, as calculated using the following equation:

$$\text{Total return (R)} = \text{expected return (E(R))} + \text{unexpected return (U)}$$

The unexpected returns can be further broken down into systematic and unsystematic as shown in the following:

$$\text{Unexpected return (U)} = \text{systematic portion (m)} + \text{unsystematic portion (}\epsilon\text{)}$$

By substituting the part of unexpected return into the total return equation, it is possible to arrive at the following equation:

Total return = expected return + systematic portion + unsystematic portion

$$R = E(R) + m + \varepsilon$$

The expected value (EV) refers to the sum of the products of the various possible outcomes as well as the probability that those outcomes will actually happen. The expected return of asset, $E(R_{Asset})$, in turn, refers to the average of possible returns that an investment might generate. Each return is assigned a weight corresponding to the probability that it will occur.

This is represented by the following equation:

$$E(R_{Asset}) = \sum_{i=1}^n (p_i \times R_i) = (p_1 \times R_1) + (p_2 \times R_2) + \dots + (p_n \times R_n)$$

If the likelihood of each of the possible outcomes is equal, that is,

$p_1 = p_2 = p_3 = \dots = p_n = p = 1/n$, the formula subsequently reduces to:

$$E(R_{Asset}) = \frac{\sum_{i=1}^n (R_i)}{n} = \frac{R_1 + R_2 + \dots + R_n}{n}$$

where:

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R_{Asset})$ = Expected return of an asset.

Illustrative Example 1

Mr. Leung invested in China Mobile stock. His investment will generate returns that will be influenced by the condition of the economy during the year. Based on the following information, calculate the return for Mr. Leung's investment in China Mobile.

State	Return %	Probability
Weak	13	0.30
OK	20	0.40
Strong	25	0.30

Illustrative Example 1 (continued)

Using the equation for expected returns it is possible to calculate the following:

$$E(R_{\text{CMobile}}) = p_{\text{weak}} * R_{\text{weak}} + p_{\text{ok}} * R_{\text{ok}} + p_{\text{strong}} * R_{\text{strong}}$$

$$E(R_{\text{CMobile}}) = (0.30 \times 0.13) + (0.40 \times 0.20) + (0.30 \times 0.25) = 0.194$$

Mr. Leung's return on his investment should be 19.4%.

6.1.2 Measures of Risk

Risk refers to the possibility that the expected return on an investment will differ from the actual return.

Downside risk refers to the chance that returns will fall short of expectations, while **upside risk** refers to the possibility that returns will exceed expectations. Generally speaking, risk is a measure of uncertainty, typically in the context of actual returns failing to meet expected returns.

Risk can be measured by calculating variance and standard deviation, which is the square root of the variance. As standard deviation from the mean increases, so does risk, and therefore expected return.

6.1.2.1 Calculating the Variance and Standard Deviation

The **variance** (σ^2) squares the difference between each possible instance and the mean (μ) while multiplying each difference by its corresponding probability before summing everything.

In finance, the mean and the variance (or standard deviation) can be interpreted as the return and risk of an asset. See the following equation:

$$\text{Var}(R) = \sigma_R^2 = \sum_{i=1}^n \left\{ p_i \times [R_i - E(R)]^2 \right\}$$

where:

$\text{Var}(R)$ = variance of return of an asset,

σ_R^2 = variance of return of an asset,

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R)$ = Expected return of an asset.

Should all the possible outcomes share the same likelihood, the formula then simplifies to:

$$\sigma_R^2 = \frac{\sum_{i=1}^n [R_i - E(R)]^2}{n}$$

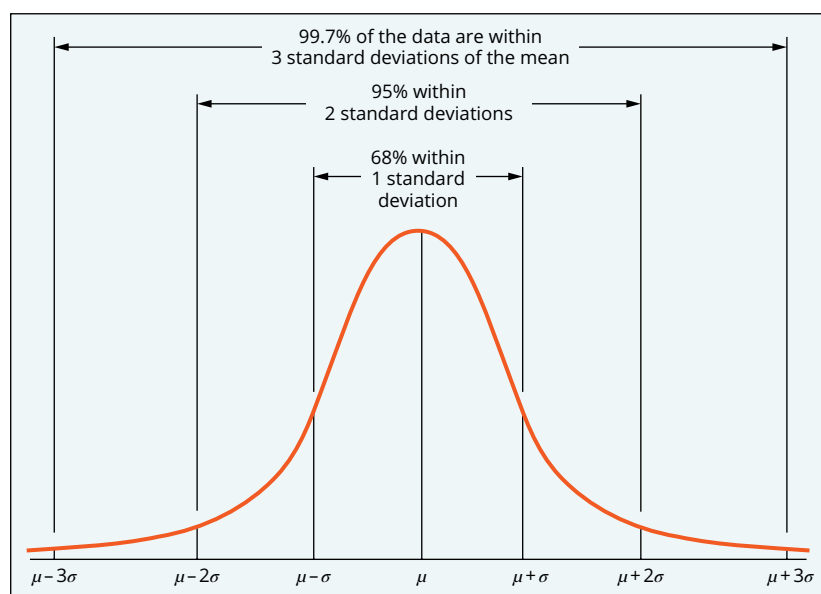
The standard deviation can be calculated by taking the square root of the variance:

$$\sigma = \sqrt{\sigma^2}$$

6.1.2.2 Interpreting the Variance and Standard Deviation

Consider Exhibit 6.1 as an example of how to interpret variance and standard deviation. Using the diagram in Exhibit 6.1 it is possible to see that:

- μ and σ are the mean and the standard deviation of the distribution respectively. The mean and the standard deviation can be interpreted as the return and risk of an investment or a portfolio respectively. A higher standard deviation implies a higher volatility or fluctuation in stock price, and vice versa, which means the risk is higher.
- A **normal distribution** is a symmetric frequency distribution totally described by its mean and standard deviation.
- The left and right sides of a graph of a normal distribution are mirror images. The mean is in the centre of the distribution. The probability of a particular outcome being a particular distance from the mean is the same in both the left or right side of the distribution.
- Standard deviation shows the probability of an outcome falling within a certain distance from the mean, or within a certain range.



► **EXHIBIT 6.1** Variance and standard deviation (Source: data from Dan Kernier [CC BY-SA 4.0].)

The graph shows the following:

Number of standard deviations from the mean	Fraction of total observations
1	68%
2	95%
3	99.7%

If the return of an asset follows the normal distribution with a mean of 5% and standard deviation of 15%, it is possible to say that the probability that the return of asset falls between $(5\% - 15\%) = -10\%$ and $(5\% + 15\%) = 20\%$ is 68% (= 0.68).

6.1.2.3 Difference Between Value at Risk and Expected Shortfall

There are some key differences between value at risk (VaR) and expected shortfall. While VaR considers the proportion of an investment or asset that faces a particular risk, the expected shortfall considers the possibility that the risk will actually impact the returns.

Value at Risk

When using value at risk (VaR), the goal is to make a statement such as: 'There is a 99% possibility that losses will be no more than HK\$V in 10 days'.

The variable HK\$V is the portfolio's VaR (10 days, 99%), which is a function of two parameters:

- The time horizon of 10 days, and
- The confidence level of 99%.

Value at risk, $VaR > 0$, is defined as:

$$P(L > VaR) = \alpha$$

where:

- P = the probability,
- L = the loss, and
- $1 - \alpha$ is the confidence level.

VaR is an attractive measure because it is easy to understand. However, when VaR is used in an attempt to limit the risks taken by traders, banks and the other financial market participants, it can lead to undesirable results.

Expected Shortfall

Expected shortfall (ES) is defined as:

$$ES = E(L | L > VaR), VaR > 0$$

where:

- E = expected value,
- L = the loss, and
- VaR = the value at risk.

ES is a measure that can produce better incentives for traders than VaR.

VaR looks at how bad things could get while ES considers the actual loss in the event of trouble.

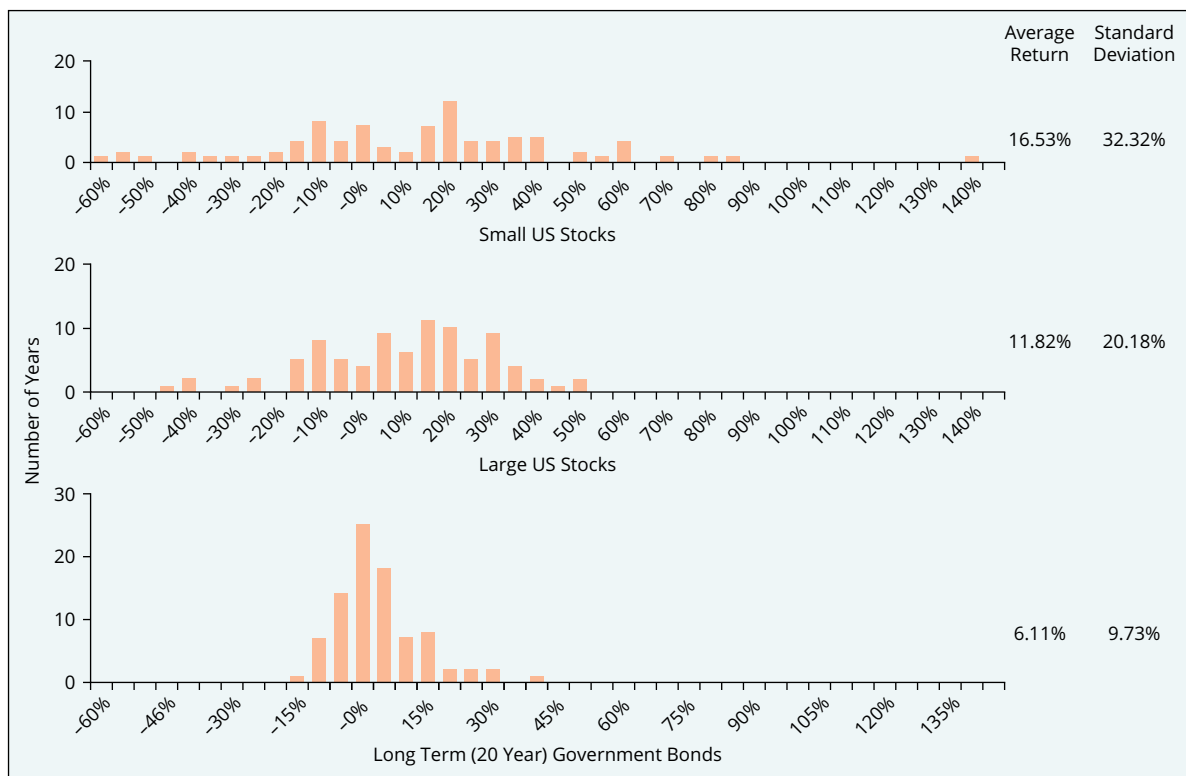
ES has better properties than VaR in that it always recognises the benefits of diversification. After the 2007–2008 global financial crisis, ES has become a more commonly used measurement of risk as it emphasises the risk of suffering the maximum loss which is exactly the risk that arises during a period of financial crisis.

The key for ES is that it also provides a signal of extreme severe loss which can be higher than the loss at threshold expected by VaR.

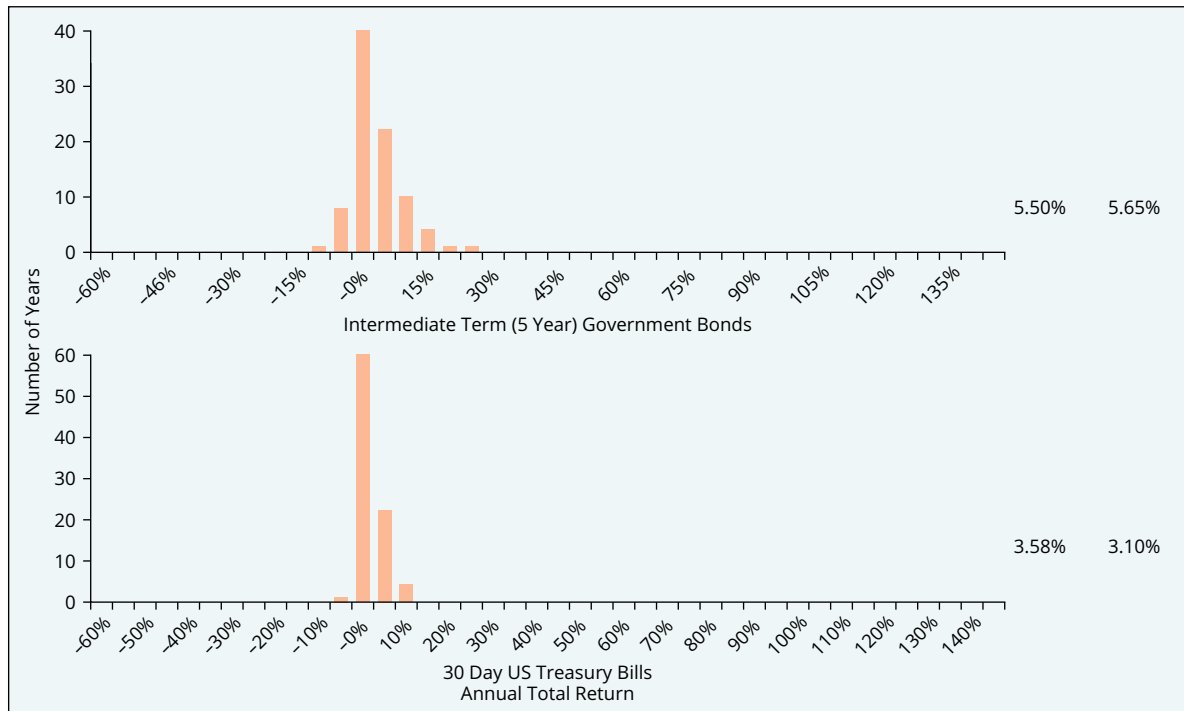
6.1.2.4 Historical Market Performance

Annual returns have tended to be higher for securities with higher risk levels. For example, small stocks that feature the largest standard deviation of total returns feature the highest average return. Likewise, treasury bills possess the smallest standard deviation, and therefore the lowest annual return.

Exhibit 6.2 shows the distributions of annual total returns for US stocks and bonds from 1926 through 2012. Note the correlation between higher/lower standard deviations of returns and higher/lower average returns.



► **EXHIBIT 6.2** Return on investment of US securities (Source: data from U.S. Department of the Treasury.)



► **EXHIBIT 6.2 (Continued)**

Key Learning Point

There are three main methods that can be used to calculate the return of investment: holding period returns, average returns and expected returns.

► Knowledge Check Questions

Question 1

Identify which of the following statements is correct.

- A** The greater the risk associated with an investment, the lower the return that investors expect from it.
- B** When making a choice between two investments with similar levels of risk, investors tend to choose the investment with the higher return.
- C** When making a choice between two investments with the same expected return, investors will choose the riskier one.
- D** When making a choice between two investments with the same level of risk, investors prefer the investment with the lower return.

Question 2

Last year, Ms. Tan bought a small piece of real estate for HK\$85,000 that is now valued at HK\$102,000. She needs a total return of 25%. Calculate how much income she needs to reach the objective.

► Knowledge Check Questions (*continued*)

Question 3

Baxter Inc's annual returns for the past four years were 2.75%, -1.8%, 7.2% and 6.5%. Calculate the average annual arithmetic and geometric returns for the company over the four-year period.

6.2 DIVERSIFICATION AND PORTFOLIO

A key concept in investing is portfolio **diversification**. This concept is easy to understand but can be difficult to implement and it is where many investors make big mistakes, either concentrating too much or settling for average performance. Diversification is achieving variety. When it comes to a portfolio, diversification is a risk management strategy to combine various assets to reduce overall risk.

The purpose of portfolio diversification is to manage risk. And risk management plans should include diversification rules. These rules should always be followed. Diversification lowers volatility by taking advantage of the fact that asset categories, different industries and different stocks do not move together. Diversification can lower unsystematic risk that is specific to each asset.

By owning investments in different industries and companies, investors minimise risk. And, at the same time, by diversifying, investors can decrease volatility in their portfolios.

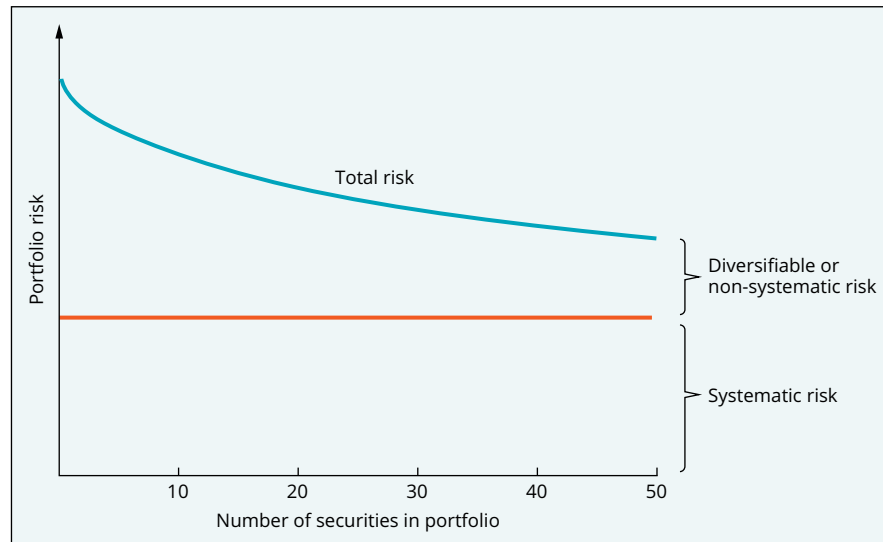
6.2.1 Diversification

All portfolios' risks can be reduced by spreading investment funds across several assets. The key to this strategy lies in the investor's ability to select assets whose returns are less than perfectly positively correlated. Even with random selection or naive diversification, the risk of a portfolio can be reduced.

Exhibit 6.3 shows that as the number of securities increases, the diversification benefit decreases. In other words, additional securities eventually become superfluous.

It is impossible to eliminate total risk through diversification because diversification cannot eliminate market risk. It is possible to eliminate unique or non-systematic risk, however, through effective diversification.

As a result, diversification adds value to a portfolio by reducing risk without sacrificing returns. The majority of the benefits derived from diversification can typically be achieved by investing in 40 to 50 different securities. However, this may become a form of over-diversification which leads to an outcome below expected return. The appropriate number of securities may vary based on markets, portfolio policies and defined risk profile.



► EXHIBIT 6.3 Portfolio risk

In theory, international diversification can reduce more risk than investing solely in domestic capital markets, as international economies may not be well correlated. However, evidence suggests the benefits of international diversification are declining as global securities markets become increasingly integrated.

6.2.2 Portfolio

The term **portfolio** refers to a grouping of securities, such as stocks and bonds, which are combined and dealt with as a single investible asset.

Investors can reduce the risk to their portfolio of investments by investing in at least two assets whose values do not in all cases move in the same direction simultaneously. This notion is the core idea behind diversification.

There are two main types of portfolios, single-asset portfolios and multi-asset portfolios.

In single-asset portfolios, returns for individual stocks from day to day tend to be independent and more or less normally distributed. The coefficient of variation (CV) is useful in making an initial assessment of risk and return for individual stocks as shown in the following equation:

$$CV_{\text{Asset}} = \frac{\sigma_{R_{\text{Asset}}}}{E(R_{\text{Asset}})}$$

An investor seeking to minimise risk will look for a lower CV. If an investor can earn a positive risk-free rate which is the interest of risk-free asset, for example, US Treasury bills while not bearing any risk, then it makes sense to compare the investment's risk with the return that the investor expects to earn above the risk-free rate. This concept is referred to as the modified coefficient of variation and can be expressed as follows:

$$CV_{\text{Asset}}^* = \frac{\sigma_{R_{\text{Asset}}}}{E(R_{\text{Asset}}) - R_{\text{rf}}}$$

In multi-asset portfolios, the expected return on a portfolio containing multiple assets is derived by taking the sum of the product of the expected returns on each individual security, all of which are weighted.

The weights, or portfolio weights, represent the ratio of the portfolio that is invested in the respective stocks. A multi-asset portfolio's expected return can be calculated by the following equation:

$$E(R_p) = \sum_{i=1}^m w_i E(R_i)$$

where:

- $E(R_p)$ = Expected return on the portfolio,
- m = Number of assets in the portfolio,
- w_i = Proportion of the portfolio invested in asset i ,
- $E(R_i)$ = Expected return on asset i .

Illustrative Example 2

Suppose that Mr. Leung is holding a HK\$100,000 portfolio comprising two stocks: HSBC and HKEx. The expected returns on HSBC and HKEx are 12% and 10%, respectively. Mr. Leung invests HK\$60,000 on HSBC and the remainder on HKEx. Calculate the expected return of Mr. Leung's portfolio.

To calculate the portfolio's expected return, use the equation to calculate the return as follows:

$$\begin{aligned} E(R_{\text{Portfolio}}) &= W_{\text{HSBC}} * E(R_{\text{HSBC}}) + W_{\text{HKEx}} * E(R_{\text{HKEx}}) \\ &= \text{HK}\$(60,000 / 100,000) * 12\% + \text{HK}\$(40,000 / 100,000) * 10\% \\ &= 0.112 \text{ or } 11.2\% \end{aligned}$$

The total return Mr. Leung can expect is 11.2%.

6.2.3 Limitations of Diversification

If returns on individual stocks do not change in the same way, increasing the number of stocks can lower the standard deviation further.

However, as more assets are added to a portfolio, the decrease in the overall standard deviation gets increasingly smaller. As the number of assets gets larger and larger, the portfolio standard deviation only decreases up to a point. It does not approach zero. That is because

investors can diversify away risks associated with individual assets. However, diversification cannot be used to do away with risks that are common to all the assets. This describes why over-diversification is counter-productive to the purpose behind diversifying a portfolio.

Risks that can be dealt with through diversification are 'diversifiable risks' or 'unsystematic risks', as opposed to 'systematic risk' or 'non-diversifiable risk'.

Key Learning Point

Portfolios' risks can be reduced by spreading investment funds across several assets. The key to this strategy lies in the investor's ability to select assets whose returns are less than perfectly positively correlated.

► Knowledge Check Questions

Question 4

Identify which of the following two statements is correct.

- (i) The expected return on a portfolio is the equally weighted average of the expected returns on the individual securities in the portfolio.
- (ii) The standard deviation of a portfolio reflects the weighted impact of the individual securities' standard deviations and the relationship amongst the co-movements of the returns on those individual securities.

- A** (i) is incorrect, (ii) is correct.
- B** (i) is correct, (ii) is incorrect.
- C** (i) and (ii) are incorrect.
- D** (i) and (ii) are correct.

Question 5

Identify which of the following investors would pay most for a particular asset.

- A** An investor holding a single-asset portfolio.
- B** An investor holding a diversified portfolio.
- C** An investor who does not have a completely diversified portfolio.
- D** A risk-averse investor who does not recognise the benefits of diversification.

Question 6

Calculate the expected return for a portfolio that has HK\$800 invested in Stock O and HK\$1,200 invested in Stock K, if the expected returns on Stock O and Stock K are 10% and 18%, respectively.

6.3 COVARIANCE AND CORRELATION COEFFICIENT OF A PORTFOLIO

The previous section discussed how to calculate the expected return of a portfolio. The variance or standard deviation of a portfolio represents not only the variance or standard deviation of that portfolio's assets, it also represents how the returns on the assets within the portfolio vary together.

Covariance and the **correlation coefficient** are two instruments used for measuring how the returns on a pair of assets vary together.

6.3.1 Covariance

It is very rare for the prices of two assets in a portfolio to change in exactly the same manner.

When two assets in a portfolio move in opposing directions, the gains made by one asset's price change will offset the losses in the price of the other, at least to some degree.

Consequently, the risk level of a two-asset portfolio is less than that of the average of the risks attributed to the individual assets themselves.

Variance of a two-asset portfolio is calculated according to the following formula:

$$\sigma_{R_2 \text{ Asset portfolio}}^2 = w_A^2 \sigma_{R_A}^2 + w_B^2 \sigma_{R_B}^2 + 2w_A w_B \text{COV}_{R_{A,B}}$$

where:

- $\sigma_{R_2 \text{ Asset portfolio}}^2$ = Variance of return of the portfolio,
- w_A = Weight of return of asset A in the portfolio, and
- w_B = Weight of return of asset B in the portfolio.

$\text{COV}_{R_{A,B}}$ represents the covariance between asset A and asset B. This is a measure of how the returns on the assets move together, or covary:

$$\text{COV}_{R_{AB}} = \sum_{i=1}^n (p_i \times [R_{Ai} - E(R_A)] \times [R_{Bi} - E(R_B)])$$

where:

- p_i = Probability of event i,
- R_{Ai} = Return of asset A under event i,
- R_{Bi} = Return of asset B under event i,
- $E(R_A)$ = Expected return of asset A, and
- $E(R_B)$ = Expected return of asset B.

The standard deviation of a two-asset portfolio can be calculated by taking the square root of the variance of a two-asset portfolio. This is represented in the following formula:

$$\sigma_{\text{portfolio}} = \sqrt{w_{R_A}^2 \sigma_{R_A}^2 + w_{R_B}^2 \sigma_{R_B}^2 + 2w_{R_A} w_{R_B} \text{COV}_{R_{AB}}}$$

where,

$\sigma_{\text{portfolio}}$ = Standard deviation of return of the portfolio,

w_{R_A} = Weight of return of asset A in the portfolio,

w_{R_B} = Weight of return of asset B in the portfolio.

6.3.2 Correlation Coefficient

In order to facilitate interpretation of covariance, it is possible to divide the covariance by the product of the two assets' standard deviations.

The statistical measure that identifies how asset returns move in relation to each other is known as the correlation coefficient (ρ):

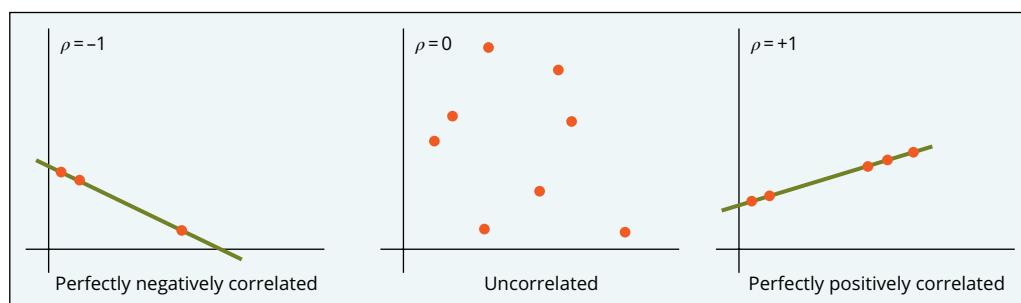
$$\rho_{R_{AB}} = \frac{\text{COV}_{R_{AB}}}{\sigma_{R_A} \sigma_{R_B}}$$

A negative correlation indicates that the returns are moving in opposite directions. A positive correlation indicates movement in the same direction.

A correlation of 0 indicates that the returns on the assets in question are not correlated. If the correlation between assets is imperfect, meaning that the correlation coefficient is less than +1, there is a benefit from diversifying by holding multiple assets with different risk characteristics.

When there are more stocks in a portfolio, it becomes more complex to calculate variance due to the need to account for the covariance between each pair of assets.

Perfectly negatively correlation, uncorrelation, perfectly positively correlation are shown in Exhibit 6.4 and explained below:



► **EXHIBIT 6.4** Correlation coefficient graph

- Value ranges between -1 and 1
- -1 = Perfectly negatively correlated (i.e. as one goes up, the other goes down);
- 0 = uncorrelated, not related;
- +1 = Perfectly positively correlated, i.e. both go up/down together.

In a two-stock portfolio, if $\rho = -1$, the two stocks should be combined so that the relative weight and mix of the stocks will eliminate all portfolio risks.

On the other hand, if $\rho = +1$, there is no diversification potential when combining the two stocks. In general, portfolios have $\rho \approx 0.65$ which means risk is lowered but not eliminated.

Illustrative Example 3

Consider the returns from two stocks with the following data. Calculate the covariance and correlation coefficient of the returns for the stocks. Stock A is expected to return 10.8% and Stock B 9.7%.

Prob	Stock A	Stock B
0.4	0.09	0.11
0.5	0.11	0.08
0.1	0.17	0.13

$$\text{COV}_{R_{AB}} = 0.4 \times (0.09 - 0.108) \times (0.11 - 0.097) + 0.5 \times (0.11 - 0.108) \times (0.08 - 0.097) + 0.1 \times (0.17 - 0.108) \times (0.13 - 0.097) = 0.000094$$

$$\sigma_{R_A}^2 = 0.4 (0.09 - 0.108)^2 + 0.5 (0.11 - 0.108)^2 + 0.1 (0.17 - 0.108)^2 = 0.000516$$

$$\sigma_{R_A} = 0.02271563$$

$$\sigma_{R_B}^2 = 0.4 (0.11 - 0.097)^2 + 0.5 (0.08 - 0.097)^2 + 0.1 (0.13 - 0.097)^2 = 0.000321$$

$$\sigma_{R_B} = 0.01791647$$

$$\rho_{R_{AB}} = \frac{0.000094}{(0.02271563)(0.01791647)} = 0.230967$$

Key Learning Point

Covariance and the correlation coefficient are two instruments used for measuring how the returns on a pair of assets vary together.

► Knowledge Check Questions

Question 7

If the covariance between the returns on two assets is equal to zero, explain why the correlation coefficient must be zero as well.

Question 8

Two stocks have the information included below. With this information, calculate the covariance of the returns for the stocks. In the calculation, assume expected return for Stock 1 of 14.4% and for Stock 2 of 15.9%. Round the final answer to five decimal places.

Prob	Stock 1	Stock 2
0.5	0.11	0.18
0.3	0.17	0.15
0.2	0.19	0.12

Question 9

Using the information from Question 8, calculate the correlation coefficient of the returns for the two stocks.

Question 10

The covariance of the returns between Stock A and Stock B is 0.09875. The variance is 0.2116 for Stock A, and the variance is 0.1369 for Stock B. Compute the correlation coefficient between the returns of the two.

6.4 SYSTEMATIC RISK

Complete diversification can eliminate unique risk from a portfolio, however, the investor still faces **systematic risk**.

Systematic risk, sometimes called market risk, is the risk of being in the market rather than in any specific investment. For example, changes in gross domestic product (GDP), interest rates and inflation will affect a large number of assets, although at varying degrees in the market.

6.4.1 Total Risk versus Systematic Risk

The expected return on an asset depends on systematic risk or market risk, because the unsystematic risk can be diversified away. The implication is that regardless of total risk, expected returns depend on systematic risk.

Total risk assesses all the risk factors associated with a particular investment. It takes into account systematic factors but also factors that are unique to the particular investment. Investors do not like risk, which means that they are likely to hold a portfolio of securities rather than just one and benefit from diversification.

As a result, when making a decision about whether to invest in a stock, they want to know how the stock will affect the risk and expected returns of their portfolio. The standard deviation, which measures the risk of a stock, does not show how that stock contributes to the risks and returns of a portfolio. Another measure of risk is known as systematic risk.

Unsystematic risk is risk that affects a small number of assets or even just one. It is also known as unique risk, asset-specific risk or diversifiable risk. Strikes, parts shortage, accidents and takeovers are examples of unsystematic risk.

The systematic risk principle states that there will be reward for bearing risk but not reward for bearing risk unnecessarily.

6.4.2 Importance of Systematic Risk

Systematic risk is important because it can lead to losses caused by factors that are external (or exogenous) to the specific investment.

Investors with portfolios that are not well diversified face systematic risk in addition to unsystematic risk. Diversified investors face only systematic risk, which typically cannot be diversified away.

Investors expect returns on individual assets that are higher than the total risk, which include the sum of both systematic and unsystematic risk.

However, it is worth noting that asset markets only reward systematic risk, and this is why finance is particularly concerned about systematic risk when thinking about the relationship between risk and return.

6.4.3 Measuring Systematic Risk

If systematic risk is the sole consideration in measuring expected returns, using standard deviation as a measure of risk is impossible. Standard deviation measures total risk.

Given that systematic risk is non-diversifiable risk, the systematic risk or market risk for an individual asset is simply a representation of the relation between individual asset returns and market returns.

Finding the slope of the line of best fit between the returns of the stock and the general market is the method by which to quantify the relation between returns from a stock and the market.

The slope represents the best fit **beta** (β).

The beta of an asset is defined as follows:

$$\beta_i = \frac{\rho_{R_{Mi}} \times \sigma_i}{\sigma_M} = \frac{\text{COV}_{R_{Mi}}}{\sigma_M^2}$$

where:

- $\rho_{R_{Mi}}$ = Correlation coefficient of the asset i 's returns with the market,
- σ_i = Standard deviation of returns of asset i ,
- σ_M = Standard deviation of return of market,
- $\text{COV}_{R_{Mi}}$ = Covariance between returns of asset i and the market.

If the beta is:

- =1: The asset's systematic risk is the same as that of the market.
- >1: The asset's systematic risk is higher than the market.
- <1: The asset's systematic risk is lower than the market.
- =0: The asset has zero systematic risk.

For example, consider a stock with a covariance with the market of 0.2, and the variance of the market is 12%, the beta of the stock will be $\frac{0.2}{0.12} = 1.667$.

The beta of the stock is greater than 1, which means that the stock has more systematic risk than the market.

Ethics in Practice

Keeping meticulous records and knowing your clients are not just about meeting legal requirements and observing best practices, they can also be life-saving for a business facing unfair accusations when, in the aftermath of a crisis triggered by systematic risk, scapegoats are sought. They are also helpful for weeding out unethical staff at lower levels whose activities are not known by upper management.

The 2008 global financial crisis was caused by systematic risk, but when it came to determining accountability, only one bank was singled out for punishment.

Abacus Federal Savings Bank is a bank founded in New York City in 1984 by Chinese Americans. In the wake of the Great Financial Crisis, while large Wall Street investment banks were deemed 'too big to fail' by the US government, Abacus was the only bank indicted, accused by the Manhattan District Attorney of misleading the Federal National Mortgage Association, better known as Fannie Mae, regarding the incomes and other qualifications of their loan applicants.

While many observers were happy to see a bank – any bank – face punishment for the devastation wrought by the crisis, others wondered why a small Asian-American financial institution with no previous reputation for mismanagement or ethics violations was the only bank standing accused of wrongdoing.

Abacus, to the surprise of many, fought the government charges in a costly trial that lasted three years and cost it HK\$10 million in legal fees. The bank's extensive documentation showed that its loan applicants defaulted at a small fraction of the national average, ultimately saving it from legal repercussions and damage to its reputation.

Ethics in Practice (*continued*)

Although the bank's name was ultimately cleared, it was discovered that some low-level loan officers had encouraged loan applicants to inflate their stated incomes and their job titles. These officers had a financial incentive to secure loans from Fannie Mae, as they received commissions on all approved loans. No proof was found that bank management had any knowledge of these activities.

In 2015, the US legal system acquitted Abacus, along with its chief credit officer and loan supervisor. In 2017, it was awarded a 5-star rating, the highest possible rating, from bank-rating firm Bauer Financial.

Key Learning Point

The expected return (or market required return) on an asset depends only on that asset's systematic risk or market risk, because the unsystematic risk can be diversified away.

► Knowledge Check Questions

Question 11

Explain why beta is the appropriate measure of risk for a diversified portfolio.

Question 12

Identify which of the following will be beta of the portfolio if it has the same risk with the market.

- A More than one
- B Equal to one
- C Less than one
- D Equal to zero

➤ 6.5 CAPITAL ASSET PRICING MODEL

The **capital asset pricing model (CAPM)** is an economic model used to determine the correlation between risk and expected return, which it employs to value capital assets such as securities, portfolios and capital projects.

CAPM comprises both the time value of money and the element of risk – the time value of money is incorporated as the risk-free rate of return and the element of risk is incorporated in the form of the beta coefficient. CAPM acknowledges that an investment's expected rate of return must provide for, and be equal to or greater than, the rate on a risk-free investment plus a premium for the risk inherent in the investment.

Therefore, CAPM is used to determine the economic feasibility of an investment. If the expected rate of return is equal to or greater than the computed rate obtained using CAPM, then the investment is considered economically feasible and vice versa.

The formula for CAPM is:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

where:

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

6.5.1 Assumptions

CAPM relies on a number of assumptions, which vary in their ability to influence outcomes.

The following are amongst the most influential assumptions, and thus significant limitations, on CAPM:

- That all investors have equal access to all available investments and that all investors are using a one-period time horizon.
- That variance from the asset class benchmark is the sole measure of asset risk.
- That external costs such as commissions, taxes, etc. are non-existent.
- That restrictions on borrowing or lending at the risk-free rate do not exist, i.e. all parties are permitted to do so.
- That all asset classes have a market, and, by extension, a market benchmark – CAPM cannot be applied without a market or benchmark for a particular asset class.
- The use of historical data which may not be relevant in calculating future expected returns.

6.5.2 Risk-free Interest Rate

Every investment comes with an expected **risk-free interest rate**. This rate is a reflection of the reward anticipated for the deferred current consumption that is incurred when an investment is made.

In the United States, the risk-free rate of return is typically measured by the rates on US Treasury securities (bills and notes), while in Hong Kong it is the rate of Hong Kong Exchange Fund Bills and Notes issued by the Hong Kong Monetary Authority.

Investors or firms can earn the rates paid on these securities without undertaking the risk attached to commercial securities or project undertakings.

6.5.3 Compensation for Bearing Systematic Risk

The compensation investors require for bearing risk is represented by the differences between risk free rate of return and required returns for risky investments, or:

$$E(R_i) = R_{rf} + \text{Compensation for taking risk}_i$$

If the compensation for taking risk varies with asset risk, and that systematic risk is the determining factor, then:

$$E(R_i) = R_{rf} + (\text{Units of systematic risk}_i \times \text{Compensation per unit of systematic risk})$$

If beta, β , is the correct measure for the number of units of systematic risk, then:

$$\text{Compensation for taking risk} = \beta \times \text{Compensation per unit of systematic risk}$$

The rate of return on the market above the risk-free return, represents the compensation investors require for assuming market or systematic risk. Therefore:

$$\text{Compensation per unit of systematic risk} = E(R_m) - R_{rf}$$

And by extension, the equation for expected return is therefore:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

6.5.4 Risk Premium

Under the systematic risk principle, the higher the beta, the higher the **risk premium**, as beta is a measure of systematic risk.

When considered within a capital budgeting context, a project's beta coefficient is a quantification of the relative riskiness of that project.

From the perspective of a firm the weighted average beta of all its project investments is a measure of that firm's total project risk. Given that the return to investors in the firm is dependent on the returns earned by the firm's projects, the weighted average beta of the firm's projects is effectively a measure of investor risk.

6.5.5 Calculating the Expected Return by Capital Asset Pricing Model

The capital asset pricing model (CAPM) describes the relationship between risk and expected return:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

The equation shows how to calculate expected return for a single security using CAPM.

The expected return of a portfolio of CAPM can be found using the following equation:

$$E(R_{n \text{ Asset Portfolio}}) = R_{rf} + \beta_{n \text{ Asset Portfolio}} [E(R_m) - R_{rf}]$$

The above can be found by applying the expected return and the beta of a portfolio:

$$\begin{aligned} E(R_{n \text{ Asset Portfolio}}) &= \sum_{i=1}^n (x_i \times E(R_i)) \\ &= (x_1 \times E(R_1)) + (x_2 \times E(R_2)) + \dots + (x_n \times E(R_n)) \end{aligned}$$

$$\beta_{n \text{ Asset Portfolio}} = \sum_{i=1}^n x_i \beta_i = x_1 \beta_1 + x_2 \beta_2 + x_3 \beta_3 + \dots + x_n \beta_n$$

where:

- x_i = Weighting of asset i ,
- $E(R_i)$ = Expected return of asset i ,
- β_i = Beta of asset i .

Illustrative Example 4

Suppose that Ms. Lau is planning to invest in Tencent. The beta of Tencent is 1.2, the expected market return is 10% and the rate of Hong Kong Exchange Fund Bills and Notes is 3%.

Calculate Ms. Lau's expected return on Tencent.

To determine the expected return of Tencent, we can make use of the equation for expected return as follows:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

where:

- $R_{rf} = 3\%$
- $E(R_m) = 10\%$
- $\beta_{\text{Tencent}} = 1.2$
- $E(R_{\text{Tencent}}) = 3\% + 1.2 \times (10\% - 3\%) = 0.114$ or 11.4%

With this data, Ms. Lau's expected return on Tencent is 11.4%

6.5.6 Efficient Frontier and Capital Allocation Line

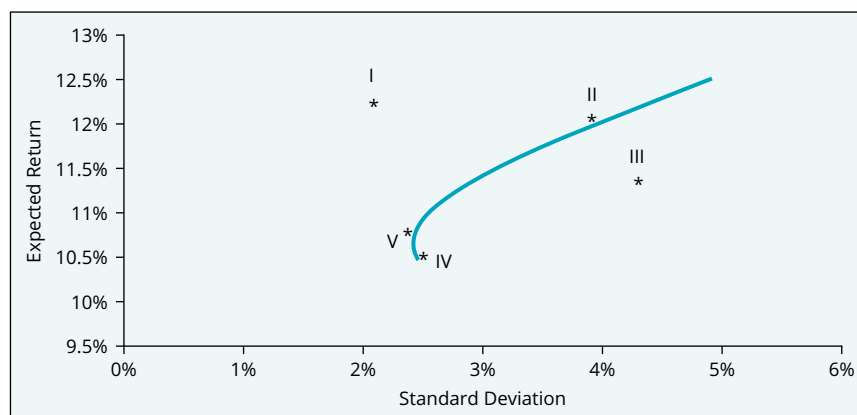
Economist Harry Markowitz won a Nobel prize for his contributions to the field of finance. He showed investors the optimal way to diversify portfolios. He made arguments based on several assumptions, including that investors are rational decision makers, that they are risk averse, that their preferences are based on the expected returns on their portfolios and risks measured by variance or standard deviation.

Markowitz introduced the idea of an **efficient portfolio**. This theoretical portfolio can be constructed from a set of assets available on the market. Efficient portfolios provide the highest expected return for a particular level of risk or the lowest risk for a particular return.

Efficient portfolios can be identified by specifying expected return and minimising risk at that particular level of return or specifying a portfolio risk and maximising expected return with that particular level of risk.

The efficient frontier considers the optimal portfolio to arrive at the highest expected return for a certain market risk or the lowest risk for a certain expected return.

The first step in the analysis is to determine the expected return-risk combinations available to investors from a given set of assets, by allowing the portfolio weights to vary. The entire curve is referred to as the minimum variance frontier or efficient frontier, which is shown in Exhibit 6.5.



► **EXHIBIT 6.5 The efficient frontier** (Source: data from Booth 2014. Reproduced with permission of John Wiley & Sons.)

The variance is simply the standard deviation squared. The shape is virtually identical to the frontier that would be constructed using just two assets. By using all available assets, however, it is possible to generate a set of more efficient portfolios, in the sense of offering higher expected return for a given risk level, or lower risk for a given expected return level.

The capital allocation line (or capital market line) represents the boundary separating risky assets and risk-free assets in a portfolio.

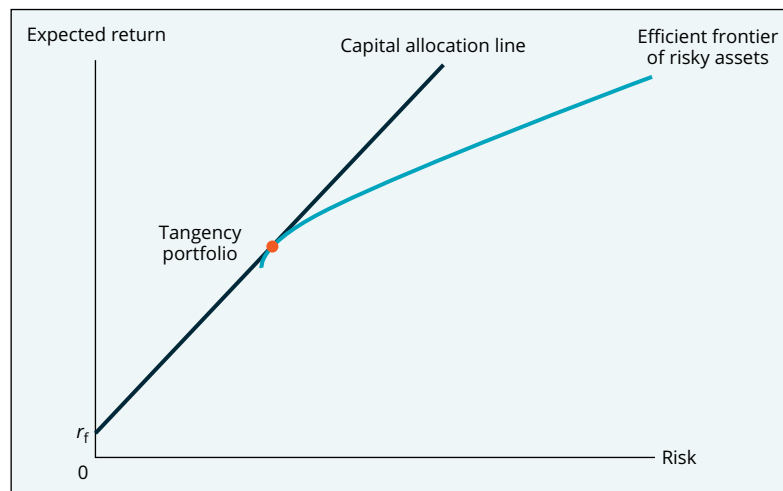
To determine the capital allocation line (CAL) of a portfolio, the first step is to determine the expected market return, which can be arrived at using the formula:

$$E(r_p) = E(r_r) \times w_r + (1 - w_r) \times E(r_f)$$

The next step is to determine how risky a portfolio is, this time using the formula discussed earlier:

$$\sigma_p = W_r \times \sigma_r$$

With these two variables it is possible to determine all the different combinations of both assets in a graph, Exhibit 6.6, that includes expected returns on one axis and risk on the other. The line is the capital allocation line and shows the possible asset allocations.



► **EXHIBIT 6.6 Capital allocation line** (Source: data from Manusca (CC BY-SA 4.0.))

While both these concepts are useful to allocate portfolios and portfolio weightings it is also important to account for the fact that they are based on sweeping assumptions, assumptions that are rarely true in real life. The assumption that investors are rational actors, for example, has to be considered in light of the goals of each investor along with the risk tolerance of each one, a fact that goes to another assumption that investors are risk averse. Yet another assumption, that asset returns follow a normal distribution pattern, is also rarely true.

Key Learning Point

The capital asset pricing model (CAPM) is an economic model used to determine the correlation between risk and expected return, which it employs to value capital assets such as securities, portfolios and capital projects.

► Knowledge Check Questions

Question 13

Stock C has a beta of 1.8. The risk-free rate of return is 3%, while the market risk premium is 6%. Calculate the expected return.

Question 14

Stock D has an expected return of 17.9% with an expected return on the market of 13%, and the beta is 1.7. Calculate the risk-free rate.

► **Knowledge Check Questions (continued)**

Question 15

Stock E has an expected return of 16.5% that has the risk-free rate at 5% and the beta at 2.3. Calculate the risk premium on the market.

Question 16

Stock F has an expected return of 16.6% that has the risk-free rate at 4% and the expected return on the market at 10%. Calculate the beta of Stock F.

6.6 SECURITY MARKET LINE

The **security market line (SML)** shows the expected total return predicted by the CAPM for various values of beta.

The price of the asset drives the actual expected total return as per the following equation:

$$R_T = \frac{\Delta P + CF_1}{P_0}$$

The SML represents the trade-off between market risk and the required rate of return for any investment with significant risk, be it an individual security or a portfolio.

It is calculated with the following equation:

$$k_i = R_{rf} + [E(R_m) - R_{rf}] \beta_i$$

where:

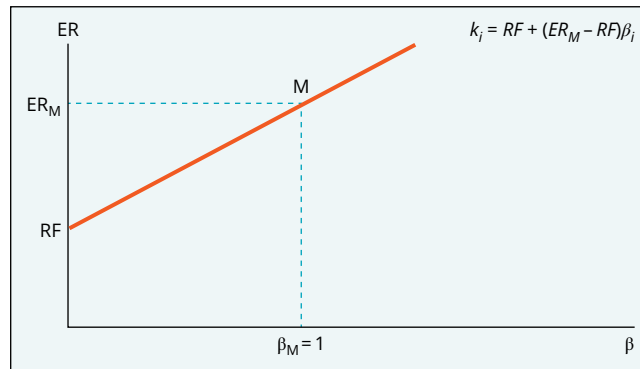
- k_i = required return of an asset i ,
- R_{rf} = Risk-free rate,
- $E(R_m)$ = Expected return of the market,
- β_i = Beta of an asset i .

The equation used to determine the required return on security notes that i is the risk-free rate plus a market risk premium. This is shown in Exhibit 6.7, which shows the SML at equilibrium.

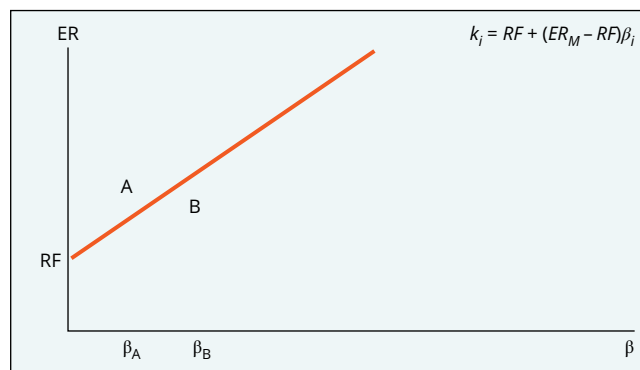
In equilibrium, the expected return on all accurately priced securities will lie on the SML.

Securities can be mispriced according to the CAPM. Securities or portfolios with expected returns that exceed their required returns are considered undervalued and will lie above the SML, as shown in Security A in Exhibit 6.8.

Conversely, securities or portfolios with expected returns lower than their required returns are considered to be overvalued and will be beneath the SML, as shown by Security B in Exhibit 6.8.



► **EXHIBIT 6.7 Security market line at equilibrium** (Source: data from Booth 2016. Reproduced with permission of John Wiley & Sons.)

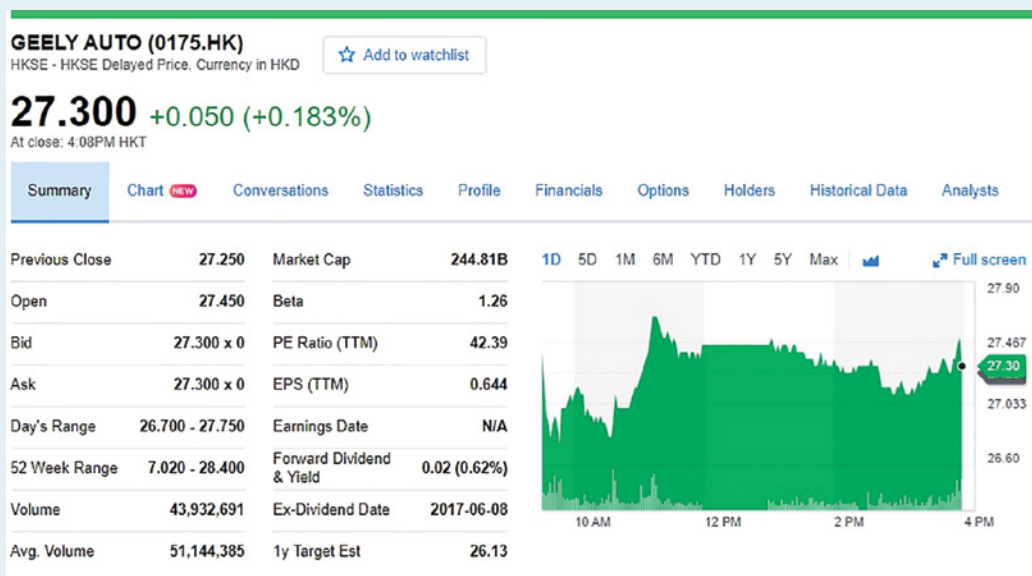


► **EXHIBIT 6.8 Under and overvalued securities on security market line** (Source: data from Booth 2016. Reproduced with permission of John Wiley & Sons.)

Illustrative Example 5

To illustrate the use of the security market line (SML) to identify any undervalued/overvalued stocks listed on the Hong Kong Stock Exchange, let us consider the case of Geely Auto (SEHK: 0,175).

Assume that the required return is 7%.



Source: Stock quote of Geely Auto from HK Yahoo Finance at 18 November 2018.

Illustrative Example 5 (continued)

According to the stock quote of Geely Auto, the company's beta is 1.26. The risk-free rate, which is the most-recent three months' 91-day rate of Hong Kong Exchange Fund Bills, is 0.85% (period beginning with 1 August 2017).

Performance of the Tracker Fund of Hong Kong

As of 31 October 2017	
<i>Cumulative return</i>	
1 month	2.5
3 months	3.4
6 months	14.7
<i>Annualised return</i>	
YTD	28.3
1 year	23.0
3 years	5.4
5 years	5.4
10 years	-1.1
Since inception (11 November 1999)	3.9

Source: Data from Tracker Fund of Hong Kong

The return of the Tracker Fund of Hong Kong (SEHK: 2,800), which tracks the performance of the Hang Seng Index, will be used as the market return, which is the annualised return since inception, or 3.9%, as shown in the above table.

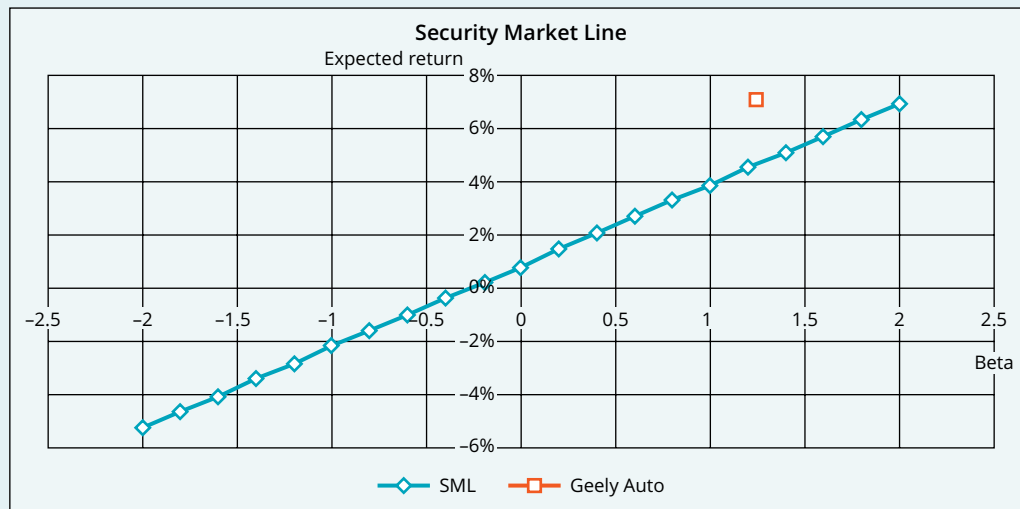
With this information, it is possible to determine the expected return $E(R)$ by employing the CAPM:

$$E(R) = R_f + \beta \times [E(R_m) - R_f]$$

$$E(R) = 0.85\% + 1.26 \times (3.9\% - 0.85\%) = 4.693\%$$

Illustrative Example 5 (continued)

Subsequently it is possible to chart the security market line and compare it with Geely Auto.



The graph shows that Geely Auto lies above the security market line, indicating that the stock is undervalued.

Apply and Analyse 1**Apply**

ABC Semiconductors Ltd is listed on the stock exchange of Hong Kong. It has beta of 2. An equity benchmark index has produced 10% return per annum in the past 20 years while the government 10-year bond yield is 4%.

Using this information, construct and evaluate the security market line for ABC Semiconductors Ltd.

Analysis

In order to construct the SML, it is necessary to calculate market risk premium, which turns out to be 6% which is the difference between benchmark index return and risk-free rate ($10\% - 4\%$).

In the SML graph, a beta of zero corresponds to expected return of 4% (risk-free rate). A beta of 1 corresponds to expected return of 10% ($4\% + 1 \times 6\%$). A beta of 2 corresponds to expected return of 16% ($4\% + 2 \times 6\%$). Connecting the three dots would create SML that represents what the market expects in return for any given amount of risk (beta).

Because ABC Semiconductors has a beta of 2, its expected return predicted by SML is 16% ($4\% + 2 \times 6\%$). If its share price had been priced accurately by the investors, the company's beta and expected return would place it exactly on SML (X coordinate matches beta of 2 and Y coordinate matches expected return of 16%).

Actual expected return of ABC Semiconductors Ltd based on market consensus estimate and current share price is 20%, hence the company is sitting above SML.

Apply and Analyse 1 (continued)

SML expects a return of 16% for beta of 2. The actual expectation of return for ABC Semiconductors is higher than what SML predicts.

SML graph would suggest ABC Semiconductor's share price is potentially undervalued. If rational investors agreed with this conclusion, they would bid up the share price until actual expectation of return is brought down to 16%, the level predicted by SML under the market equilibrium state. When that happens, the company would sit on SML, no longer above it.

Key Learning Point

The security market line (SML) indicates what the CAPM predicts the expected total return should be for various values of beta.

► Knowledge Check Questions

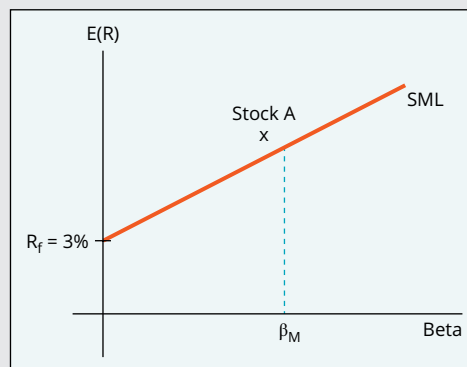
Question 17

Identify which of the following represents a plot of the relation between expected return and systematic risk.

- A** The beta coefficient
- B** The covariance of returns line
- C** The security market line
- D** The variance

Question 18

Identify which of the following is most likely to be the $E(R)$ of stock A.



- A** $E(R) = 3\%$, Beta = 0.8
- B** $E(R) = 4\%$, Beta = 1
- C** $E(R) = 5\%$, Beta = 0.9
- D** $E(R) = 2\%$, Beta = 0.6

► Knowledge Check Questions (*continued*)

Question 19

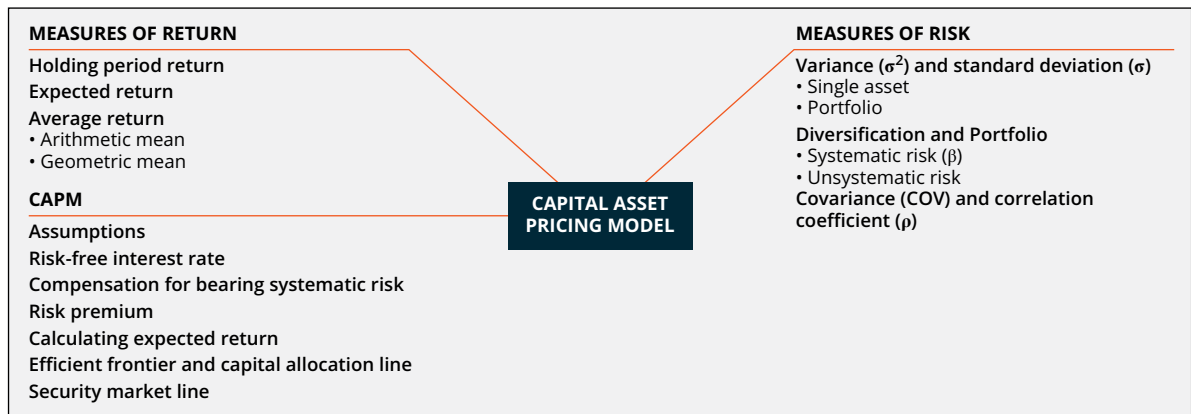
Explain which stock(s) in the following graph is/are undervalued.





SUMMARY

- Risk and return are core concepts to financial management, representing the potential positive and negative consequences of an investment, respectively.
- Financial managers should seek to minimise risk and maximise returns.
- Risks can be classified as diversifiable or non-diversifiable. Diversifiable risks can be dealt with partially or fully through effective diversification within a portfolio, in which asset returns are not perfectly positively correlated, although the diversification benefits of each individual asset in the portfolio decrease as the number of assets increases beyond fifty different securities.
- The volatility of returns of assets within a given portfolio are measured by variance and standard deviation.
- Covariance and the correlation coefficient measure how the returns for a pairing of assets vary together.
- Although useful for minimising risk, diversification has its limits. Non-diversifiable, or systematic risks, cannot be managed via diversification.
- Systematic risk represents the responsiveness of individual asset returns to the market returns. For investors with well-diversified portfolios, systematic risk is of paramount importance. Systematic risk is the fundamental consideration when considering the trade-off between risk and return in finance.
- By using the beta coefficient, which measures assets' systematic risk and volatility, a market risk premium can be established. If an asset's beta is zero, it has zero systematic risk, whereas if it is 1, it has the same systematic risk as the market. If an asset's beta is higher than 1, then it has more systematic risk than the market. Conversely, a beta lower than one indicates less systematic risk than that of the market.
- Based upon the systematic risk principle and diversification principle, the market will only reward for bearing systematic risk that cannot be diversified by investing into a well-diversified portfolio themselves (whereas the market will not reward for bearing unsystematic risks).
- The capital asset pricing model (CAPM) takes the diversifiable systematic risk as measured by the beta coefficient one step further, by determining the required rate of return for an asset or portfolio given existing systematic risk.
- Operating on a number of assumptions, CAPM uses the risk-free rate of return and the beta coefficient to achieve this end. CAPM's primary value to a firm or investor is that it can be used to ascertain whether or not an investment is economically feasible.
- The security market line (SML) indicates what the CAPM predicts the expected total return should be for various values of beta.

 MIND MAP **LIST OF FORMULAS**

Capital appreciation component of a return:

$$R_{CA} = \frac{\text{Capital Appreciation}}{\text{Initial Price}} = \frac{P_1 - P_0}{P_0} = \frac{\Delta P}{P_0}$$

Income component of a return:

$$R_I = \frac{\text{Cash Flow}}{\text{Initial Price}} = \frac{CF_1}{P_0}$$

Total holding period return:

$$R_T = R_{CA} + R_I = \frac{\Delta P}{P_0} + \frac{CF_1}{P_0} = \frac{\Delta P + CF_1}{P_0}$$

where:

- P_1 = Price at period 1
- P_0 = Price at period 0
- ΔP = Change in price
- CF_1 = Cash flow at period 1

Arithmetic mean (AM):

$$AM = \frac{\sum_{i=1}^n R_i}{n}$$

Geometric mean (GM):

$$GM = [(1+R_1)(1+R_2)(1+R_3)\dots(1+R_n)]^{\frac{1}{n}} - 1$$

where:

- R_i = Interest rate of i th period of time
- n = Number of periods

Total return:

Total return = expected return + systematic portion + unsystematic portion

$$R = E(R) + m + \varepsilon$$

Expected return with different probability:

$$E(R_{Asset}) = \sum_{i=1}^n (p_i \times R_i) = (p_1 \times R_1) + (p_2 \times R_2) + \dots + (p_n \times R_n)$$

Expected return with same probability:

$$E(R_{Asset}) = \frac{\sum_{i=1}^n (R_i)}{n} = \frac{R_1 + R_2 + \dots + R_n}{n}$$

where:

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R_{Asset})$ = Expected return of an asset.

Variance with different probability:

$$\text{Var}(R) = \sigma_R^2 = \sum_{i=1}^n \left\{ p_i \times [R_i - E(R)]^2 \right\}$$

where:

$\text{Var}(R)$ = variance of return of an asset,

σ_R^2 = variance of return of an asset,

R_i = Return of an asset in occurring situation i ,

n = Number of occurring situation,

p_i = probability of occurring situation i ,

$E(R)$ = Expected return of an asset.

Variance with same probability:

$$\sigma_R^2 = \frac{\sum_{i=1}^n [R_i - E(R)]^2}{n}$$

Standard deviation:

$$\sigma = \sqrt{\sigma^2}$$

Coefficient of variation:

$$CV_{Asset} = \frac{\sigma_{R_{Asset}}}{E(R_{Asset})}$$

Modified coefficient of variation:

$$CV_{Asset}^* = \frac{\sigma_{R_{Asset}}}{E(R_{Asset}) - R_{rf}}$$

Portfolio's expected return:

$$E(R_p) = \sum_{i=1}^m w_i E(R_i)$$

where:

- $E(R_p)$ = Expected return on the portfolio,
- m = Number of assets in the portfolio,
- w_i = Proportion of the portfolio invested in asset i ,
- $E(R_i)$ = Expected return on asset i .

Variance of a two-asset portfolio:

$$\sigma_{R_{2 \text{ Asset portfolio}}}^2 = w_A^2 \sigma_{R_A}^2 + w_B^2 \sigma_{R_B}^2 + 2w_A w_B \text{COV}_{R_{A,B}}$$

where,

$\sigma_{\text{portfolio}}^2$ = Variance of return of the portfolio,

w_{R_A} = Weight of return of asset A in the portfolio,

w_{R_B} = Weight of return of asset B in the portfolio.

Covariance of two securities:

$$\text{COV}_{R_{AB}} = \sum_{i=1}^n (p_i \times [(R_{Ai} - E(R_A)) \times (R_{Bi} - E(R_B))])$$

where:

- p_i = Probability of event i ,
- R_{Ai} = Return of asset A under event i ,
- R_{Bi} = Return of asset B under event i ,
- $E(R_A)$ = Expected return of asset A, and
- $E(R_B)$ = Expected return of asset B.

Standard deviation of a two-asset portfolio:

$$\sigma_{\text{portfolio}} = \sqrt{w_{R_A}^2 \sigma_{R_A}^2 + w_{R_B}^2 \sigma_{R_B}^2 + 2w_{R_A} w_{R_B} \text{COV}_{R_{AB}}}$$

where:

$\sigma_{\text{portfolio}}$ = Standard deviation of return of the portfolio

w_{R_A} = Weight of return of asset A in the portfolio

w_{R_B} = Weight of return of asset B in the portfolio

Correlation coefficient of two securities:

$$\rho_{R_{AB}} = \frac{\text{COV}_{R_{AB}}}{\sigma_{R_A} \sigma_{R_B}}$$

Beta of asset:

$$\beta_i = \frac{\rho_{R_{Mi}} \times \sigma_i}{\sigma_M} = \frac{\text{COV}_{R_{Mi}}}{\sigma_M^2}$$

Capital asset pricing model:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

where:

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

Portfolio's expected return under CAPM:

$$E(R_{n \text{ Asset Portfolio}}) = R_{rf} + \beta_{n \text{ Asset Portfolio}} [E(R_m) - R_{rf}]$$

while:

$$\begin{aligned} E(R_{n \text{ Asset Portfolio}}) &= \sum_{i=1}^n (x_i \times E(R_i)) \\ &= (x_1 \times E(R_1)) + (x_2 \times E(R_2)) + \dots + (x_n \times E(R_n)) \end{aligned}$$

$$\beta_{n \text{ Asset Portfolio}} = \sum_{i=1}^n x_i \beta_i = x_1 \beta_1 + x_2 \beta_2 + x_3 \beta_3 + \dots + x_n \beta_n$$

where:

- x_i = Weighting of asset i ,
- $E(R_i)$ = Expected return of asset i ,
- β_i = Beta of asset i .

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. When the risk is greater, investors will expect higher return as a reward of bearing more risk.

Answer B is correct. We assume that people are rational.

Answer C is incorrect. Economists assume that people are rational and risk adverse, if the expected return of the two investments are same, investors will prefer the one with lower risk.

Answer D is incorrect. Similar to Answer C, the investor will prefer the one will higher return.

Question 2

$$\frac{\text{HK\$}102,000 - \text{HK\$}85,000 + \text{HK\$}X}{\text{HK\$}85,000} = 0.25, \text{HK\$}X = \text{HK\$}4,250$$

Question 3

Arithmetic mean, $AM = (2.75\% - 1.8\% + 7.2\% + 6.5\%) / 4 = 0.036625 \rightarrow 3.66\%$

Geometric mean, $GM = [(1 + 0.0275)(1 - 0.018)(1 + 0.072)(1 + 0.065)]^{(1/4)} - 1 = 0.036 \rightarrow 3.60\%$

Question 4

Answer C is correct.

(i) is incorrect because the expected return is weighted according to the proportion of each stock in the portfolio.

(ii) is incorrect because the standard deviation of the portfolio is the square root of variance of the portfolio, while the variance measures the differences between expected return of an individual security and the portfolio in a weighted average.

Question 5

Answer A is incorrect. A single-asset portfolio may not offer the highest return.

Answer B is correct. Diversification can substantially reduce the variability of returns (i.e. risk) without an equivalent reduction in expected returns.

Answer C is incorrect. An incompletely diversified portfolio would suffer from higher risk and lower expected return.

Answer D is incorrect. A non-diversified portfolio would suffer from higher risk and lower expected return.

Question 6

Portfolio weight in O = HK\$800/(HK\$800 + HK\$1,200), portfolio weight in K = 1 – 40% = 60%

$E(R) = 10\% \times 40\% + 18\% \times 60\% = 14.80\%$

Question 7

From the equation, $\rho_{R_{AB}} = \frac{COV_{R_{AB}}}{\sigma_{R_A} \sigma_{R_B}}$, $COV_{R_{AB}}$ is the numerator, if it is zero, then the $\rho_{R_{AB}}$ will be 0.

Question 8

$Cov(R_1, R_2) = 0.5 \times (0.11 - 0.144) \times (0.18 - 0.159) + 0.3 \times (0.17 - 0.144) \times (0.15 - 0.159) + 0.2 \times (0.19 - 0.144) \times (0.12 - 0.159) = -0.00079$

Question 9

$$\sigma_1^2 = 0.5 (0.11 - 0.144)^2 + 0.3 (0.17 - 0.144)^2 + 0.2 (0.19 - 0.144)^2 = 0.001204$$

$$\sigma_1 = 0.0346987$$

$$\sigma_2^2 = 0.5 (0.18 - 0.159)^2 + 0.3 (0.15 - 0.159)^2 + 0.2 (0.12 - 0.159)^2 = 0.000549,$$

$$\sigma_2 = 0.02343075$$

$$\rho_{R_{1,2}} = \frac{-0.00079}{(0.0346987)(0.02343075)} = -0.96677$$

Question 10

The correlation coefficient between the returns of the two stocks:

$$\rho = \frac{cov(R_1, R_2)}{\sigma_1 \sigma_2} = \frac{0.09875}{\sqrt{0.2116 \times 0.1369}} = 0.580199$$

Question 11

A diversified portfolio eliminates the unsystematic risk, consider only the systematic risk, which is the risk that beta measures.

Question 12

Answer A is incorrect. More than one beta represents higher risk than the market portfolio.

Answer B is correct. The market portfolio is recognised as a diversified portfolio with no unsystematic risk. For a portfolio that does not contain unsystematic risk, the beta is 1.

Answer C is incorrect. Less than one beta represents lower risk than the market portfolio.

Answer D is incorrect. A zero beta means the asset has zero systematic risk.

Question 13

$$E(R_C) = R_{rf} + \beta_C (E(R_M) - R_{rf}) = 0.03 + (1.8 \times 0.06) = 0.138 = 13.8\%$$

Question 14

$$\begin{aligned} E(R_D) = 0.179 &= R_{rf} + \beta_D (E(R_M) - R_{rf}) = R_{rf} + 1.7 (0.13 - R_{rf}) \Rightarrow R_{rf} = 0.06 \\ &= 6\% \end{aligned}$$

Question 15

$$\begin{aligned} E(R_E) = 0.165 &= R_{rf} + \beta_E (E(R_M) - R_{rf}) = 0.05 + 2.3 (\text{Risk Premium}) \Rightarrow \text{Risk Premium} = 0.05 \\ 0.05 + 2.3 (\text{Risk premium}) &= 0.165. \text{ Therefore, risk premium} = 0.05 = 5\% \end{aligned}$$

Question 16

$$E(R_F) = 0.166 = R_{rf} + \beta_F (E(R_M) - R_{rf}) = 0.04 + \beta_F (0.10 - 0.04) \Rightarrow \beta_F = 2.1$$

Question 17

Answer A is incorrect. A beta coefficient shows how an individual security reacts to the change in market.

Answer B is incorrect. A market premium is the excess return for bearing systematic risk.

Answer C is correct. By definition, security market line represents a plot of the relation between expected return and systematic risk.

Answer D is incorrect. Variance measures the total risk.

Question 18

Answer A is incorrect. The expected return is higher than risk-free rate.

Answer B is incorrect. The beta of stock A is less than the beta of market (beta = 1).

Answer C is correct. The beta of stock is less than the beta of market and stock A has an expected return higher than risk-free rate.

Answer D is incorrect. The expected return is higher than risk-free rate.

Question 19

Stock A and D lie above the SML, they are undervalued. Stock B lies on the SML which means it is correctly priced. Stock C lies below SML which is overvalued.

EXAM PRACTICE

QUESTION 1

Identify which one of the following statements is the movement of Company A's stock price if it has a beta of 1 and the stock market dropped by 200 points today.

- A. Company A's stock price decreased by HK\$200.
- B. Company A's stock price increased by HK\$200.
- C. Company A's stock price decreased by the same percentage decrease as the stock market.
- D. Company A's stock price increased by the same percentage decrease as the stock market.

QUESTION 2

Identify which one of the following statements is not the assumption of CAPM.

- A. Assuming no transactional costs.
- B. Assuming no personal income taxes.
- C. Assuming all investors have different expectations about expected returns, and standard deviations for all traded securities.
- D. Assuming all investors can borrow/lend at the risk-free rate.

QUESTION 3

Compute the expected return and standard deviation of Min's portfolio if she has HK\$5,000 to invest. The expected return on the market portfolio is 11% with a standard deviation of 15% and she borrowed HK\$2,000 at the risk-free rate of 4% to invest in the market portfolio.

- A. Expected return = 19.40%; standard deviation = 15.40%
- B. Expected return = 15.40%; standard deviation = 19.40%
- C. Expected return = 13.80%; standard deviation = 21.00%
- D. Expected return = 21.00%; standard deviation = 13.80%

QUESTION 4

Identify the most accurate description(s) from the statements below:

- (i) The CAPM points out that rational investors should be compensated for unique risk.
 - (ii) The CAPM implies that unsystematic risk is the appropriate measure of risk to determine the risk premium required by investors for holding a risky security.
 - (iii) The expected return from unsystematic risk is zero.
- A. (i), (ii) and (iii) are correct.
 - B. (i) and (ii) are incorrect, (iii) is correct.
 - C. (i), (ii) are correct, (iii) is incorrect.
 - D. (i), (iii) are incorrect, (ii) is correct.

QUESTION 5

Identify which of the following is *NOT* a correct statement of beta.

- A. It is a measure of market risk.
- B. It measures the risk of an individual stock or portfolio relative to the market portfolio.
- C. It is the slope of the security market line.
- D. It changes with time as the risk of the underlying security or portfolio changes.

QUESTION 6

Compare the difference between expected returns using subjective probabilities and expected returns based on historical values.

QUESTION 7

Compare the difference between systematic and unsystematic risk.

QUESTION 8

Describe the components of total return and total risk.

QUESTION 9

Explain the expected return with regard to individual securities and a portfolio as a whole.

QUESTION 10

Justify why diversification always reduces the overall risk.

QUESTION 11

You have been provided with a HK\$20,000 fund to invest in two securities: Spot and Dot. After an extensive analysis of the economy and the two securities, you have compiled the following forecasts:

State of the economy	Probability of occurrence	Spot expected return	Dot expected return
Boom	15%	−6%	35%
Normal	60%	12%	20%
Bust	25%	18%	−10%

- Calculate the expected returns on Spot and Dot.
- Calculate the standard deviations of the returns on Spot and Dot.
- Calculate the covariance of the returns on Spot and Dot.
- Calculate the correlation between Spot and Dot.
- Calculate the composition of the portfolio if you wish to have an expected return of 12% on the portfolio.
- Calculate the standard deviation of the portfolio.

QUESTION 12

Risk-return in a portfolio. The following table presents some statistics about the returns of three assets, Assets A, B and C, respectively, under three possible scenarios (boom, normal, and recession). The expected probabilities of each state are also specified in the table.

Scenario	Probability	Asset A	Asset B	Asset C
<i>Boom</i>	10%	8%	5%	6%
<i>Normal</i>	60%	6%	(C)	5%
<i>Recession</i>	30%	4%	5%	−1%
Expected return		(A)	5.00%	3.30%
Standard deviation		(B)	(D)	2.83%

- (a) Solve the blanks in the above table. Show your calculations.
- (b) Suppose you wish to combine Assets A and B in order to create a portfolio that has the same total risk as Asset C. Compute the weight that you should invest in Asset A and Asset B respectively. Compute the expected return of the portfolio made of A and B.

QUESTION 13

Mr. Lee owns 1,000 shares of Swire Pacific which are listed on the Hong Kong Stock Exchange and valued at HK\$100,000 (HK\$100 per share). The expected annual return on Swire Pacific's shares is 15% with a standard deviation of 35%. Suppose Mr. Lee has an additional HK\$100,000 to invest. The money could be used to buy another 1,000 shares in Swire Pacific or to buy 800 shares in Hutchinson, whose shares are trading at HK\$125 each. The expected annual return on Hutchinson is also 15% with a standard deviation of 35%. The correlation coefficient between the returns on Swire and Hutchison shares is 0.

Required:

- (a) Advise Mr. Lee whether to invest the additional HK\$100,000 in 1,000 Swire Pacific shares or 800 Hutchison shares.
- (b) Analyse under what circumstances would Mr. Lee be indifferent between the two choices, considering the requirements for portfolio diversification.

QUESTION 14

- (a) Great Stationary Limited is a small company and is considering the following projects:

Project	A	B
Projected return	12%	10%
Correlation of returns with the market	0.8	0.6
Covariance of returns with the market	16	14
Standard deviation of returns	8%	9%
Expected market return is 10%.		
Standard deviation of market return is 5%.		
Risk-free return is 3%.		

Required:

- (i) Identify which project has higher total risk. Explain total risk first and then the answer.
- (ii) Identify which project has higher systematic risk. Explain systematic risk and the answer.
- (iii) Use the capital asset pricing model (CAPM) to determine the required return for project A and project B. Explain which project Great Stationary Limited should undertake.
- (b) The owners of Great Stationary Limited would like to know why investment decisions are so complex, given the assumption that people are assumed to be risk averse (i.e. afraid of risk) in finance. If the assumption was correct, the company would not need any financial consultants and should simply invest in risk-free asset and earn risk-free rate returns.

Required:

- (i) Explain the concept of a risk-free asset and give one example.
- (ii) Explain whether the company should just invest in risk-free assets.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. The stock price would change by the same percentage increase or decrease as the stock market with a beta of one, not a fixed price decrease.

Answer B is incorrect. The stock price would change by the same percentage increase or decrease as the stock market with a beta of one, not a fixed price increase.

Answer C is correct. Beta equal to one means market risk the same as the average security in the market. A beta of one implies that if the market return increased (or decreased) by 1%, the return on the security (or portfolio) would, on average, increase (decrease) by 1%.

Answer D is incorrect as the decrease in the stock market would not result in an increase in stock price with a beta of one.

QUESTION 2

Answer A is incorrect. Transactional costs are stated in the assumption of CAPM.

Answer B is incorrect. Personal income tax is stated in the assumption of CAPM.

Answer C is correct. Expected returns of CAPM are calculated by risk-free rate, market return and beta which should be the same for all investors while the standard deviation is not considered in CAPM.

Answer D is incorrect. Borrowing/lending at a risk-free rate is stated in the assumption of CAPM.

QUESTION 3

Answer C is correct.

Portfolio expected return, $E(R_p) = 4\% \times -0.4 + 11\% \times 1.40 = 13.80\%$

Portfolio standard deviation, $\sigma_p = 15\% \times 1.40 = 21\%$

QUESTION 4

Answer B is correct.

- (i) Rational investors will only be compensated for bearing systematic risk, which is faced by all investors but not unique for rational investors.
- (ii) The CAPM implies that systematic risk (beta) is the appropriate measure of risk instead of unsystematic risk.
- (iii) According to the systematic principle, bearing unnecessary risk will not be rewarded, therefore the return of bearing unsystematic risk will be zero.

QUESTION 5

Answer A is incorrect. Beta is a measure of market risk.

Answer B is incorrect. Beta measures the risk of an individual stock or portfolio relative to the market portfolio.

Answer C is correct. The slope of the security market line is risk premium instead of beta.

Answer D is incorrect. Beta changes with time as the risk of the underlying security or portfolio changes.

QUESTION 6

Historical expected returns assume that, on average, the return of the next period will be similar to the past. On the other hand, subjective probabilities incorporate actual information about the state of the stock to predict the future returns.

QUESTION 7

Systematic risk, also called market risk, refers to how all securities in a portfolio tend to be directly related to overall movements in the general market or economy, so diversification cannot eliminate this risk.

QUESTION 8

Total return equals the income yield plus the capital gain (or loss) yield on the security.

Total risk can be viewed as the sum of market (systematic) risk and unique (non-systematic) risk.

QUESTION 9

The expected return of an individual security is the weighted average of the most likely returns under various future scenarios, where the weights correspond to the probabilities of each scenario actually occurring.

QUESTION 10

As long as the individual securities are not perfectly positively correlated, portfolio risk is always less than a weighted average of the risk of the individual securities. Therefore, there are benefits of diversifying. The only case in which there are no risk-reduction benefits to be obtained from diversification is when the correlation coefficient is +1.0; in this case the portfolio standard deviation is simply the weighted average of the standard deviations of the individual securities.

QUESTION 11

- (a) Expected return on Spot = $-6\% \times 15\% + 12\% \times 60\% + 18\% \times 25\% = 10.8\%$
 Expected return on Dot = $35\% \times 15\% + 20\% \times 60\% + (-10\%) \times 25\% = 14.75\%$
- (b) Spot's standard deviation,
 $\sigma_S = \sqrt{0.004234 + 0.000086 + 0.001296} = 7.494\%$
 Dot's standard deviation,
 $\sigma_D = \sqrt{0.006151 + 0.001654 + 0.015314} = 15.2049\%$
- (c) The covariance of the returns on Spot and Dot,
 $COV_{SD} = \sum_{i=1}^3 \Pr_i(r_{S,i} - 10.8\%)(r_{D,i} - 14.75\%) = -.0051 + .000378 - .00446 = -0.00918$
- (d) The correlation between Spot and Dot, $\rho_{SD} = COV_{SD} / \sigma_S \sigma_D = -0.00918 / (0.07494 \times 0.152049) = -0.80565$
- (e) Portfolio weight in Spot,
 $w_S = (ER_p - ER_D) / (ER_S - ER_D) = (12\% - 14.75\%) / (10.8\% - 14.75\%) = 69.62\%$
 Portfolio weight in Dot, $w_D = 1 - 69.62\% = 30.38\%$
- (f) The portfolio standard deviation,
 $\sigma_D = \sqrt{(w_S^2 \sigma_S^2 + w_D^2 \sigma_D^2 + 2w_S w_D COV_{SD})} = \sqrt{(0.002722 + 0.002134 - 0.003883)} = 3.1186\%$

QUESTION 12

(a) A: $5.60\% = 10\% \times 8\% + 60\% \times 6\% + 30\% \times 4\%$

B: $1.20\% = [10\% \times (8\% - 5.60\%)^2 + 60\% \times (6\% - 5.60\%)^2 + 30\% \times (4\% - 5.60\%)^2]^{0.5}$

C: $5\% = 5\% \times 3 - 5\% - 5\%$

D: $0\% = [10\% \times (5\% - 5\%)^2 + 60\% \times (5\% - 5\%)^2 + 30\% \times (5\% - 5\%)^2]^{0.5}$

(b) Since risk of Asset B is 0, the total risk of the portfolio can be written as:

$$\sigma^2 = W_A^2 \sigma_A^2$$

Hence: $2.83\%^2 = W_A^2 \times (1.20\%)^2$

$$W_A = 2.358$$

$$W_B = -1.358 = 1 - 2.358$$

Expected return of the portfolio is:

$$E(R_{A,B}) = W_A \times E(R_A) + W_B \times E(R_B)$$

$$= 2.358 \times 5.60\% + (-1.358) \times 1.20\%$$

$$= 11.58\%$$

Remark: The portfolio is invested 236% in Stock A and -136% in the risk-free asset, representing borrowing at the risk-free rate to buy more of Stock A.

QUESTION 13

Since both stocks individually offer the same return and standard deviation, it seems to suggest it is irrelevant which stock is bought. However, that would be wrong, because it fails to consider the benefit of diversification, which exists as long as the correlation coefficient between the two stocks is less than 1.

If Mr. Lee buys 1,000 Swire Pacific shares, the expected return and standard deviation of his HK\$200,000 investment would simply be 15% and 35% respectively.

If, however, he buys 800 Hutchinson shares, his portfolio's expected return and standard deviation will be:

$$R_p = 0.5 \times 15\% + 0.5 \times 15\% = 15\%$$

$$\tilde{\sigma}_p = \sqrt{(0.5^2 \times 0.35^2) + (0.5^2 \times 0.35^2)} = 0.2475 = 24.75\%$$

Therefore, Mr. Lee would still have the expected return of 15%, but a lower standard deviation. Thus, he should buy 800 Hutchinson shares.

QUESTION 14

- (a) (i) Total risk refers to the uncertainty or variability of return. It can be measured by variance or standard deviation. It consists of systematic risk and unsystematic risk. Based on the standard deviations given, project B is riskier.

- (ii) Systematic risk refers to the variability of return of an investment that relates to or can be explained by the variability of return of the general market. It can be measured by the beta coefficient in the capital asset pricing model.

$$\beta_A = (0.8 \times 0.08) / 0.05 = 1.28$$

$$\beta_B = (0.6 \times 0.09) / 0.05 = 1.08$$

Based on systematic risk, project A is riskier.

- (iii) Required return of X = $3\% + (1.28)(10\% - 3\%) = 11.96\%$

$$\text{Required return of Y} = 3\% + (1.08)(10\% - 3\%) = 10.56\%$$

Since the estimated return of B is lower than the required return, B should not be accepted.

Since the estimated return of A is higher than the required return, A should be accepted.

- (b) (i) Risk-free asset refers to an asset which has no uncertainty or variability of return (i.e. no systematic risk and no unsystematic risk). In reality, it is hard to find a truly risk-free asset. In practice, we often use some short-term government debt as an example or proxy of risk-free asset.
- (ii) No definite answer as it all depends on the company's risk preference. In the context of the capital asset pricing model, all investments lying on the capital market line (including the risk-free asset) are efficient. Investors should choose a point on the capital market line which represents all combinations (different weightings) between the risk-free asset and the market portfolio depending on their individual risk preference.

7

Cost of Capital



CHAPTER TOPIC LIST

7.1 Cost of Equity (Common Stock)

7.1.1 Dividend Growth Model

7.1.2 Capital Asset Pricing Model

7.1.3 Comparison of Models to
Compute the Cost of Equity

7.2 Cost of Equity (Preference Share)

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7.4 Weighted Average Cost of Capital

7.5 Limitations of Weighted Average Cost of Capital – Risk-adjusted Weighted Average Cost of Capital



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE
--

LO7.01: Explain and analyse the theories of major capital market models
--

7.01.08 Explain and calculate the cost of equity, cost of debt and the weighted average cost of capital (WACC)



OPENING CASE

ABC INVESTMENT LTD

Accounting staff at ABC Investment Ltd are becoming increasingly involved in coordinating the corporate finance planning of its various subsidiaries.

Staff are already quite familiar with various concepts including discounted cash flow, the evaluation of equity value and how to calculate bond values. Staff are now shifting their focus to apply their existing skills and knowledge to utilise the capital asset pricing model (CAPM) and weighted average cost of capital (WACC).

Staff review the corporate finance plans of various subsidiaries to determine their cost of common shares, cost of preferred shares, weighted average cost of debts and WACC for the entire group. These calculations are complicated by the fact that various companies use a great variety of funding instruments.

However, as staff's experience with corporate finance grows, it becomes clear that WACC calculations have their shortcomings. One lesson that emerges is to avoid mechanically applying the formula without considering practical difficulties in the real market of corporate finance. Sometimes, the mechanical application of WACC may lead to the wrong investment decisions.

As staff work their way through the particular situation of each subsidiary, ABC Investment Ltd's staff become more familiar with the theory and practice of cost of capital. 

OVERVIEW

A very important and frequently applied consideration for financial managers is determining the cost of equity and debt that a firm holds. This is what this chapter explores.

A firm's capital (or capital structure) is made up of the long-term sources of funds used in financing the firm's assets. The major categories of capital elements typically include long-term debt, preference shares and common stock, though a variety of other instruments with various characteristics are also used. There is a cost associated with using each element of capital structure. The cost for each element identified above is defined in a similar way.

Determining the cost of equity and debt is done through a number of models that require an understanding of the target company's capital structure and the market in which it operates.

Determining the cost of equity, both in regard to common stock and preference shares, is a key priority. There are some similarities between these two classes but there are also some important differences. Determining the cost of equity relies on the use of the dividend growth model (DGM) as well as calculations of the capital asset pricing model (CAPM).

Unsystematic risk may be eliminated through the use of a diversified portfolio. This means that investors only require compensation for bearing systematic risk. This chapter addresses the practical concerns that can make it more difficult to determine a price for systematic risk. Not all the companies issue publicly traded shares for specific projects. This means companies have no direct way to estimate a discount rate that may reflect the risk of incremental cash flows from a specific project.

Financial managers tackle this problem by estimating the cost of capital as a whole and then using this cost of capital to discount the cash flows for all projects.

 7.1 COST OF EQUITY (COMMON STOCK)

Finding the actual market price of a share of publicly traded stock is easy, it is basically a matter of looking it up online. But the market price fluctuates, sometimes dramatically.

When stock prices rise or fall, investors or financial managers have to make decisions about whether to sell or buy. To do this, they have to determine whether the market price of a stock actually reflects its true value.

An approach to do this is to develop a stock-valuation model and then compare the value generated from the model with the price of the stock. A market price below the estimate would suggest the stock may be undervalued, so investors might be encouraged to buy it.

There are different types of equity securities. The best known is the **common stock** (or **ordinary shares**), which represents a basic ownership claim in a corporation.

A basic right of owners of common stock is to vote on all important matters with impact on the company, such as on the choice of members of the board of directors or mergers and acquisitions. Dividend payments to holders of common stock are not guaranteed and these shareholders have the lowest-priority claim on the assets of a company in the event of bankruptcy.

However, legally, common shareholders have limited liability. Their losses are limited to the original amount of their investment and their personal assets are not tied to the obligations of the company.

At the same time, common stocks are perpetuities because they do not have a maturity date. Common stock is only retired if management repurchases it in the open market or if the firm is liquidated.

As discussed in earlier chapters, the value of an asset is the present value of its future cash flows. There are three steps that go into valuing an asset:

1. Estimate future cash flows.
2. Determine the required rate of return, or discount rate. This rate reflects how risky future cash flows are.
3. Compute the present value of the future cash flows.

Applying these steps is fairly easy when determining the value of a bond because cash flows are stated in the bond contract and the rate of return required is generally the yield to maturity on similar bonds with comparable risk.

Valuing common stock is more difficult.

One problem is that cash flows from stock are less certain than with bonds. A company may or may not pay dividends during a particular period.

A second problem is that common stocks are true perpetuities without a maturity date. Companies never have to redeem them, unlike most bonds that have a finite maturity. A final problem is that the rate of return on common stock cannot be observed directly, which makes it more difficult to group stocks into risk classes than it is to group bonds.

The cost of equity for a company is a weighted average of the costs of different types of stock that the company may have outstanding. Information about market rates of return can be used to determine the cost of debt, similarly market information can be used to estimate the cost of equity. There are several ways to do this. The most appropriate approach will depend on the information that is available and its measure of reliability to the analyst.

7.1.1 Dividend Growth Model

The **dividend growth model (DGM)** – or constant dividend growth model as it is sometimes called – is a valuation model used to calculate the fair value of a stock and makes an assumption that dividends grow at either a stable rate forever or at a different rate in the current period.

This means that the value of a stock today is equal to the value of next year's dividends divided by the difference between the required rate of return and the dividend's assumed constant growth rate.

DGM, based on a constant dividend growth model, is expressed as the following formula:

$$P_0 = \frac{D_1}{R - g}$$

We can rearrange the formula to solve for R or k_{cs} as follows:

$$k_{cs} = \frac{D_1}{P_0} + g$$

where:

- k_{cs} = Required returns on common stock,
- D_1 = Dividend paid in the next period ($t = 1$),
- P_0 = current price per share,
- g = Expected growth rate of dividends.

This is an approach for a firm that pays dividends that will grow at a constant rate, such as an electric utility, will find useful but not as much for fast growing stocks that do not pay dividends, like high-tech firms.

For fast-growing firms, it may be more useful to use a DGM for a **multistage growth dividend model**.

The multistage-growth dividend model allows for faster dividend growth rates in the near term, followed by a constant long-term growth rate. The approach is based on the mixed (supernormal) DGM.

There is a level of complexity in choosing the correct number of stages of forecasted growth as well as how long each stage will last. Since it is complex to solve algebraically for the required rate of return, the value is generally solved using a trial-and-error method, after forecasting the different stages of dividend growth.

7.1.2 Capital Asset Pricing Model

Another important model in determining the cost of common stock is the **capital asset pricing model (CAPM)**, which describes the relationship between systematic risk and expected return for an asset.

The CAPM equation is as follows:

$$E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}]$$

where:

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

The equation for CAPM suggests that the cost of stock is equal to the risk-free rate of return plus compensation for systematic risk.

For instance, to estimate the expected return for a stock with a beta of 1.5, an expected return on the market of 10% and a risk-free rate of 4%, the equation for this calculation would be:

$$\begin{aligned} E(R_i) &= R_{rf} + \beta_i [E(R_m) - R_{rf}] \\ &= 0.04 + [1.5 \times (0.10 - 0.04)] = 0.13, \text{ or } 13\% \end{aligned}$$

Note that this requires three pieces of information:

1. Risk-free rate,
2. Beta,
3. Market risk premium or expected return on the market. Remember that the market risk premium is the difference between the expected return on the market and the risk-free rate $[E(R_m) - R_{rf}]$.

In the example, the expected return on the market is known but this cannot generally be observed in real life. Financial analysts base their estimate of the market risk premium using historical data.

There are some important practical considerations when choosing the appropriate risk-free rate, beta and market risk premium for this calculation.

First, examine the **risk-free rate**. The CAPM equation should always use the current effective annual yield on a risk-free asset because the risk-free rate at a specific point in time reflects market expectations for the rate of inflation. Expected rates of inflation change over time, so an old risk-free rate may not reflect current circumstances.

Analysts must choose between short-term rates or long-term rates when determining a risk-free rate.

The most appropriate choice has long been the subject of heated debate in finance. A useful approach is to use the risk-free rate on long-term treasuries to determine the cost of equity capital. This is because an equity claim is a long-term claim on the cash flows of a company.

Shareholders have a perpetual claim on the cash flows of a company. It is possible to match long-term risk-free rates with a long-term claim by using a long-term treasury security.

A long-term risk-free rate is more effective than a short-term rate to reflect long-term inflation expectations as well as the cost of getting investors to part with their money for a long period of time.

Identifying the appropriate **beta** is much more complicated if the common stock is not publicly traded. The beta of a stock is a measure of its volatility or systematic risk in comparison to the overall market.

Beta is calculated using a regression analysis. The process is more complicated if the common stock is not publicly traded but this problem may be overcome by using a 'comparable' company that also has stock in a public exchange and operates in the same

business and carries similar levels of debt. Another approach is to use an average of the betas for the public firms in the same industry.

Finally, it is not possible to directly observe the market risk premium because the rate of return investors expect for the market portfolio, $E(R_m)$, at a particular point in time is simply not known.

Therefore, it is not possible to calculate the market risk premium as the difference between the expected return on the market and the risk-free rate, $[E(R_m) - R_{rf}]$. Financial analysts tend to use a measure of average risk premiums that investors have earned in the past to calculate the risk premiums they may expect in the present. In the United States, for instance, stock market returns have been higher than returns on government bonds by more than 5.5% since 1926.

A financial analyst might use a value of 5.71% (or a value close to it) as the market risk premium for the future, based on a reasonable belief that the market risk premium in the past is a reasonable estimate of the risk premium today.

Illustrative Example 1

An investor is considering buying 100 shares of Sun Hung Kai Properties (SHKP) stock.

The beta for SHKP is 1.59. The risk-free rate is 4%. The market risk premium is 6%. Determine the expected rate of return on SHKP stock using CAPM.

Calculate the expected return:

$$E(R) = R_{rf} + \beta \times [E(R_m) - R_{rf}] = 0.04 + (1.59 \times 0.06) = 0.1354, \text{ or } 13.54\%$$

7.1.3 Comparison of Models to Compute the Cost of Equity

7.1.3.1 Advantages and Disadvantages of Dividend Growth Model

The dividend growth model (DGM) is relatively easy to implement, which is an obvious advantage, but this model also has a number of drawbacks. The first of these drawbacks is that it only applies to companies that pay dividends. Added to this, the model does not work when dividends don't grow at a constant rate. Dividend growth is very sensitive to the estimated growth rate of the company. A 1% increase in growth rate basically increases the cost of equity by 1%. The dividend growth model does not take these risk factors into account.

7.1.3.2 Advantages and Disadvantages of Capital Asset Pricing Model

In turn, CAPM also has its pros and cons.

In terms of advantages, CAPM is particularly useful in its ability to make adjustments for systematic risk. CAPM is also applicable to all companies for which the beta can be estimated.

But this approach also has disadvantages. For starters, CAPM requires an estimate of the expected market risk premium, a premium that changes over time. Another disadvantage is that it estimates the beta, which also changes. Finally, CAPM relies on past experience to predict future behaviour, an approach that is not always reliable.

Ultimately, the biggest problem with CAPM is that it often uses historical averages to estimate the expected market risk premium. The approach has the same problem with its estimation of beta. In essence, it can be unreliable on both.

7.1.3.3 Comparing Dividend Growth Model and Capital Asset Pricing Model

Companies may ultimately find that the two models do not arrive at the same cost of equity, even for the same company, because the underlying assumptions are different.

The constant growth model is similar to a growing perpetuity model. It requires that dividends grow at a reasonably constant rate forever, which does not always happen, and applies a discount rate that should be greater than the growth rate.

The CAPM approach, however, assumes that returns are normal. This approach does not account for taxes, transaction costs or other market imperfections.

Given the above, it often makes sense to compute the cost of equity using both methods and then average them out.

Illustrative Example 2

Consider the following data related to the ordinary shares of ABC Company Ltd and calculate the cost of equity:

- Beta of ABC = 1.8
- Average market return = 10.4%
- Risk-free rate of return = 4%
- Analysts' estimates of growth in dividends and earnings = 6% per year
- Last dividend per shares = HK\$2.2
- Current market price of outstanding shares = HK\$18.65

Answer

Capital asset pricing model (CAPM) formula in Section 7.1.2:

$$E(R_i) = R_{rf} + \beta_i (R_m - R_{rf})$$

Dividend growth model (DGM) formula in Section 7.1.1:

$$k_{cs} = \frac{D_1}{P_0} + g$$

- Using CAPM: $R_E = 4\% + 1.8(10.4\% - 4\%) = 15.52\%$
- Using DGM: $R_E = [2.2(1.06) / 18.65] + 0.06 = 18.50\%$
- Average: $(15.52\% + 18.50\%) / 2 = 17.01\%$

Key Learning Point

There are two models to compute the cost of equity: DGM and CAPM.

► Knowledge Check Questions

Question 1

Identify which of the following is equal to the cost of equity.

- A Expected market return.
- B Rate of return sought by shareholders.
- C Cost of retained earnings plus dividends.
- D Risk of financing.

Question 2

If an all-equity company with cost of equity of 15% is expected to earn and pay out a HK\$5.50 dividend forever (in perpetuity), calculate the value of the company's stock.

- A HK\$36
- B HK\$37
- C HK\$38
- D HK\$40

Question 3

If the market risk premium is currently 6% and the risk-free rate of return is 4%, calculate the expected return on a common shares with a beta equal to 2.

- A 8%
- B 10%
- C 12%
- D 16%

➤ 7.2 COST OF EQUITY (PREFERENCE SHARE)

Preferred stock (or **preference shares**), like common stock, represent an ownership stake in a company. However, preference shares receive preferential treatment. Preference shareholders have a higher precedence when it comes to the payment of dividends and in the distribution of assets in the case of a liquidation or bankruptcy.

Preference shares are a type of equity, legally speaking. Preference shares dividends are paid by using after-tax dollars but preference shareholders have no voting privileges, unless they convert the stock into common stock.

Preference shares have no maturity and are generally seen as perpetuities. Still, preference shares are not true perpetuities because shares contracts may contain various provisions such as call provisions and sinking fund provisions that may eventually force management to buy a percentage of stock every year until all the preference shares are retired.

One question among finance professionals is whether preference shares should be considered debt or equity. There is a strong argument that preference shares are a special type of bond, as opposed to equity, because they do not come with voting rights; receive a fixed dividend regardless of earnings as well as the stated value in the case of a liquidation; tend to have credit ratings that are similar to those of bond issues; may be converted into common stock and have a retirement date (which make them not true perpetuities). Ultimately, preference shares are hybrid securities with characteristics that place them somewhere between bonds and common stock.

It is important, however, when calculating the value of preference shares, to consider whether the issue has a fixed maturity. Preference shares may be treated as a bond with a fixed maturity if they are attached to a sinking fund clause that calls for the mandatory retirement of the issue over a period of time.

The characteristics of preference shares (without fixed maturity) make it possible to use the perpetuity model to estimate the cost of preference shares.

Just as with common stock, the cost of preference shares can be determined by rearranging the pricing equation for preference shares as follows:

$$k_{ps} = \frac{D_{ps}}{P_{ps}}$$

D = dividend; P = current price; k = cost, or discount rate.

It is important to note that CAPM can also be used to estimate the cost of preference shares.

Illustrative Example 3

An accountant is asked to calculate the annual discount rate used by the market for a preferred stock of Dragon Utilities Ltd. Dragon has preferred stock outstanding with a stated value of HK\$100. It pays HK\$2 in dividend every quarter. The current price is HK\$95 per share.

The calculation of the preferred stock discount rate is based on the equation presented in section 7.2.

$$k_{ps} = \frac{D_{ps}}{P_{ps}} = 2 / 95 = 2.11\%$$

Since the yield on the preferred stock is 2.11% per quarter, the annual discount rate is 8.44%.

Key Learning Point

When calculating the value of preference shares, a key issue is whether the preference shares have fixed maturity. Preference shares linked to a sinking fund clause that calls for mandatory retirement over a set period of time may be treated as bonds with a fixed maturity.

► Knowledge Check Questions

Question 4

Tang Insurance Ltd issued fixed-rate perpetual preference shares three years ago to institutional investors for HK\$25 per share. The stock pays a dividend of HK\$1.75. If Tang Insurance were to issue the preference shares today the yield would be 6.5%. Calculate the current value of the stock.

- A HK\$25.00
- B HK\$26.92
- C HK\$37.31
- D HK\$40.18

Question 5

Billy's Goat Coats has a preference shares issue outstanding with a current price of HK\$38.89. The firm last paid a dividend on the issue of HK\$3.50 per share. Compute the firm's cost of preference shares.

- A 7%
- B 8%
- C 9%
- D 10%

Question 6

Melba's Toast has a preference shares issue outstanding with a current price of HK\$19.50. The firm is expected to pay a dividend of HK\$2.34 per shares a year from today. Compute the firm's cost of preference shares.

- A 11.50%
- B 11.75%
- C 12.00%
- D 12.25%



7.3 COST OF DEBT

Almost all firms use debt financing of some kind. Financial managers generally arrange for revolving lines of credit to ensure they have enough working capital for inventories or accounts receivable or other operating requirements. Companies sometimes also use fixed-term loans, such as bank loans, or sell bonds to finance their operations or acquire long-term assets.

There are costs associated with each type of debt that a firm uses. However, when estimating the cost of capital for a firm, what is of particular interest is the cost of the long-term debt of a company.

The focus, when considering the value of the assets of a company, is on long-term assets. Long-term debt matures in more than one year and typically includes fixed-term bank loans as well as bonds sold on public debt markets. The one-year horizon is used because firms often borrow money to pay off the debt at maturity.

Lines of credit generally do not enter into calculations about the cost of debt because these tend to be temporary and banks require that the outstanding balances be paid off from time to time.

When estimating the cost of long-term debt, analysts usually estimate the cost at a specific date. In practical terms, this means that the interest rate that a company pays on outstanding debt may not reflect its current cost of debt.

Interest rates change over time, so the cost of debt also changes. The rate a company paid three years ago for a five-year loan is not likely to be the same as the rate for a new loan.

For instance, suppose ABC Entertainment Ltd issued bonds for 7% five years ago. Since then, interest rates have fallen, so the same bonds could be sold at par value today for 6%.

Today's cost of debt today is 6%, not 7%. So, 6% is the cost of debt management will use to calculate the **weighted average cost of capital (WACC)**. However, financial statements will show that the company is paying 7%, which is what the company agreed to pay five years earlier and not the rate at which it will sell the same bonds today. The financial statements reflect the cost of debt sold in the past.

The current cost of long-term debt is the appropriate cost of debt for calculations of the WACC. This is the relevant cost because the WACC is the opportunity cost of capital for the firm's investors as of today.

The current cost of debt for a publicly traded bond is the yield-to-maturity calculation.

The first step to estimate this cost is to convert the bond data to reflect the compounding frequency as well as account for the effective annual interest rate (EAR) to account for the actual current annual cost of the debt.

It is also necessary to account for the cost of issuing the bond – issuance costs using the net proceeds that the company receives for the bond, and not the price that is paid by the investor.

For the current cost of long-term bank debt or other private debt, it would be simple for the firm to contact its banker to find out the rate the bank it would charge if it decided to refinance the debt today.

Companies can deduct interest payments for tax purposes. The after-tax cost of interest payments is equal to the pre-tax cost times 1 minus the tax rate:

$$k_{\text{Debt after-tax}} = k_{\text{Debt pretax}} \times (1 - t)$$

To estimate the cost of debt for a company with several debt issues, it is necessary to first estimate the cost of the individual debt issues and then calculate the weighted average of costs.

Apply and Analyse 1

Apply

An executive completed a leveraged buyout of the company with HK\$35 million. To finance the transaction, the executive and three partners put up a HK\$10 million in equity capital and borrowed HK\$25 million.

The bank debt includes HK\$10 million of secured debt borrowed at 6% from Bank 1 and HK\$7 million of senior unsecured debt borrowed at 7% from Bank 2. The remaining HK\$8 million was borrowed from an investment group. The rate on this subordinated (or junior) unsecured debt is 9.5%. What is the overall after-tax cost of the debt financing if the company's average and marginal tax rates are 25%?

Analysis

The overall after-tax cost of debt can be calculated through the following three steps:

1. Calculate the portion of debt for each debt issue.
2. Calculate the weighted average pre-tax cost of debt.
3. Calculate the after-tax average cost of debt.

The weights for the three types of debt are as follows:

$$\text{Secured debt} = \text{HK\$10,000,000} / \text{HK\$25,000,000} = 0.40$$

$$\text{Senior unsecured debt} = \text{HK\$7,000,000} / \text{HK\$25,000,000} = 0.28$$

$$\text{Subordinated unsecured debt} = \text{HK\$8,000,000} / \text{HK\$25,000,000} = 0.32$$

The weighted average pre-tax cost of debt is:

$$\begin{aligned} k_{\text{Debt pretax}} &= X_{\text{Secured debt}} k_{\text{Secured debt pretax}} + X_{\text{Senior unsecured debt}} k_{\text{Senior unsecured debt pretax}} \\ &\quad + X_{\text{Subordinated unsecured debt}} k_{\text{Subordinated unsecured debt pretax}} \\ &= (0.40)(0.06) + (0.28)(0.07) + (0.32)(0.095) = 0.0740, \text{ or } 7.40\% \end{aligned}$$

The after-tax cost of debt is therefore:

$$k_{\text{Debt after-tax}} = k_{\text{Debt pretax}} \times (1 - t) = 7.40\% \times (1 - 0.25) = 5.55\%$$

Key Learning Point

The current cost of long-term debt is the appropriate cost of debt for calculations of the weighted average cost of capital (WACC). This is the relevant cost because the WACC is the opportunity cost of capital for the firm's investors as of today.

► Knowledge Check Questions

Question 7

Identify what values a company would use to calculate the current cost of debt for outstanding bonds.

- A The coupon rate of the bonds.
- B The current yield to maturity of the bonds.
- C The current coupon yield of the bonds.
- D None of the above.

Question 8

Identify the effect of income taxes.

- A Increase the cost of debt.
- B Decrease the cost of debt.
- C Decrease the cost of equity.
- D They decrease both the cost of debt and cost of equity for a firm.

Question 9

Bellamee Inc. has semi-annual bonds outstanding with five years to maturity, and the bonds are priced at HK\$920.87. If the bonds have a coupon rate of 7%, calculate the yield to maturity (YTM) for the bonds.

- A 4.5%
- B 7.0%
- C 9.0%
- D 9.2%

Question 10

Lewis Corporation has semi-annual bonds outstanding with 13 years to maturity and the bonds are currently priced at HK\$746.16. If the bonds have a coupon rate of 8.5%, calculate the after-tax cost of debt for Lewis that has a marginal tax rate of 35%.

- A 6.25%
- B 8.125%
- C 12.5%
- D 12.80%

7.4 WEIGHTED AVERAGE COST OF CAPITAL

In any discussion of how to calculate the weighted average cost of capital (WACC), the assumption is that costs of the different types of financing are known.

This assumption makes it possible to plug those costs into the equation below:

$$k_{\text{Firm}} = \sum_{i=1}^n x_i k_i = x_1 k_1 + x_2 k_2 + x_3 k_3 + \cdots + x_n k_n$$

where:

- k = The cost of funds;
- x = The weight of the amount within the total amount.

With the information and the equation, it is possible to calculate the WACC. Unfortunately, it is not that simple, because in the real world analysts have to estimate each of the individual costs.

It is important, however, to stress one point: All of these calculations depend in some part on financial markets being efficient. The reason is that analysts may not be able to observe the rate of return investors required. Instead, they must rely on the security prices that can be easily seen in financial markets to estimate the required rate.

When working under the assumption that financial markets are reasonably efficient at taking in new information into these prices, it makes sense to rely on security prices.

WACC refers to the average combined cost of the different sources of finance, with each source weighted according to its respective proportion of funds. WACC makes it possible to apply the overall cost of capital or the discount rate into the evaluation of discounted cash flows by tapping into information from individual sources of finance. The cost of capital used to make decisions on investments is, ultimately, the marginal cost of funds raised to finance investments.

The WACC may be the most reliable guide to determine the marginal cost of capital, assuming that the company continues to invest in projects with standard levels of business risk by continuing to raise funds in the same proportions as the company's existing capital structure.

It is possible to estimate the beta for the entire firm as a weighted average of the betas for the individual project, if the beta for individual projects can be estimated. Since it is not always possible for outside stakeholders to estimate betas for individual projects, this approach is generally not used.

Analysts must instead use their knowledge of the statement of financial position and the concept of market efficiency to arrive at an estimate for the cost of capital for a company.

An alternative approach, rather than calculating the cost of capital for individual projects, analysts may instead focus on the different types of financing, such as debt and equity.

If it is possible to estimate the cost of each type of financing from the cost in the capital markets. It is possible to calculate the cost of capital for a company by using the following equation:

$$k_{\text{Firm}} = \sum_{i=1}^n x_i k_i = x_1 k_1 + x_2 k_2 + x_3 k_3 + \dots + x_n k_n$$

Dividing the costs of capital into debt and equity portions of the firm, it is possible to use the above equation to arrive at the WACC for a firm through the following equation:

$$k_{\text{Firm}} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{Equity}} k_{\text{Equity}}$$

Apply and Analyse 2

Apply

A subsidiary, Happy Products Ltd, has outstanding debt with a market value of HK\$300 million. The debt consists of bonds with a coupon of 9% with payments made semi-annually that mature in 15 years. The bonds are currently priced at HK\$1,440.03 each.

The company also has 2 million preferred shares outstanding priced at HK\$12.00. The preferred shares pay an annual dividend of HK\$1.20.

The company also has 14 million shares of common stock outstanding with a price of HK\$20.00 per share. It is expected to pay a HK\$2.20 common dividend a year from now. The dividend is expected to increase by 5% per year indefinitely.

If the company is subject to a 40% marginal tax rate, calculate the WACC.

Analysis

Step 1: Determine total debt, common equity and preferred equity:

Debt = HK\$300,000,000 (given)

Preferred equity = $12 \times 2,000,000 = \text{HK\$}24,000,000$

Common equity = $20 \times 14,000,000 = \text{HK\$}280,000,000$

Total capital = HK\$604,000,000

$X_{\text{Debt}} = 300/604 = 0.4967$

$X_{\text{ps}} = 24/604 = 0.0397$

$X_{\text{cs}} = 280/604 = 0.4636$

Step 2: Calculate cost of capital components:

Cost of debt:

Solve using,

$N = 30$

$PV = -1,440.03$

$PMT = 45$

$FV = 1,000$

Apply and Analyse 2 (continued)

$$1440.03 = \frac{90}{2} \left[\frac{1 - \left(1 + \frac{\text{YTM}}{2}\right)^{-15 \times 2}}{\frac{\text{YTM}}{2}} \right] + \frac{1000}{\left(1 + \frac{\text{YTM}}{2}\right)^{2 \times 15}}$$

We find that YTM = 0.0484 (this is a pre-tax number).

Cost of preferred equity:

$$K_{ps} = D / P_{ps} = \text{HK\$}1.20 / \text{HK\$}12.00 = 0.10$$

Cost of common equity:

$$K_{cs} = (D_1 / P_{cs}) + g = (\text{HK\$}2.20 / \text{HK\$}20.00) + 0.05 = 0.16$$

Step 3: Combine using the WACC formula.

$$\text{WACC} = (0.4967 \times 0.0484 \times (1 - 0.4)) + (0.0397 \times 0.10) + (0.4636 \times 0.16) = 0.0926, \text{ or } 9.26\%$$

Key Learning Point

WACC is the combined average cost of a variety of sources of finance (e.g. bonds, common stock and preference shares) weighted according to the proportion each component bears to the total pool of funds.

► Knowledge Check Questions

Question 11

Company One Inc has a cost of capital of 10%, a capital structure that is 50% debt and 50% equity, before-tax cost of debt is 5% and its marginal tax rate is 20%. Calculate the cost of equity capital (correct the answer to integer).

- A 10%
- B 12%
- C 14%
- D 16%

Question 12

Droz's Hiking Gear Inc has found that its common equity capital shares have a beta equal to 1.5 while the risk-free return is 8% and the expected return on the market is 14%. It has 7-year semi-annual maturity bonds outstanding with a price of HK\$767.03 that have a coupon rate of 7%. The firm is financed with HK\$120,000,000 of common shares (market value) and HK\$80,000,000 of debt. Calculate the after-tax weighted average cost of capital for Droz's, if it is subject to a 35% marginal tax rate.

► Knowledge Check Questions (continued)

- A 10.20%
- B 11.76%
- C 11.88%
- D 13.32%

Question 13

Poly's Parrot Shops has found that its cost of common equity capital is 17%. It has 7-year maturity semi-annual bonds outstanding with a price of HK\$767.03 that have a coupon rate of 7%. The firm is financed with HK\$120,000,000 of common shares (market value) and HK\$80,000,000 of debt. Calculate the after-tax weighted average cost of capital for Poly's, if it is subject to a 35% marginal tax rate.

- A 10.20%
- B 11.76%
- C 11.88%
- D 13.32%

 7.5 LIMITATIONS OF WEIGHTED AVERAGE COST OF CAPITAL – RISK-ADJUSTED WEIGHTED AVERAGE COST OF CAPITAL

It is very difficult to directly estimate the discount rate for individual projects. Financial managers will often require analysts to use the current cost of capital to discount cash flows for individual projects.

The WACC is the discount rate that analysts are often required to use. Using the WACC to discount the cash flows for a project can make sense under certain circumstances but can be very dangerous in other circumstances.

The goal of forecasting project cash flows is to forecast incremental after-tax free cash flows at the company level, cash flows that represent the difference between the cash flows that a company will generate if it undertakes or does not undertake the project.

The rate used to discount these incremental cash flows, according to financial theory, should be the rate that reflects the systematic risk. The WACC is the appropriate discount rate to evaluate a project only when the systematic risk of the cash flows from the project are the same as those for the entire company.

This is rarely true. Most companies have multiple projects with various degrees of risk.

If a single rate is used to discount cash flows for projects with different levels of risk, the rate may prove to be too low or too high in specific cases. If the discount rate is too low, a

company may move forward on projects with a negative net present value (NPV) project. The estimated NPV will be positive even though the true NPV is negative. When the discount rate is too high, there is a risk of the firm rejecting a positive NPV project. The estimated NPV will be negative even though the true NPV is positive.

At the same time, there is a risk of bias towards riskier project when a company uses just one rate to evaluate all projects. Over time, average risk associated with company assets increases over time. What's more, because some positive NPV projects may be accepted and negative NPV projects may move forward, there is a possibility that less value will be created for shareholders than if the correct discount rate is used on all projects and this can put the firm at a disadvantage.

Ultimately, it is only appropriate to use the company's WACC to discount project cash flows if the expected cash flows have the same systematic risk as the expected cash flows from the entire company.

There are a few points to take into consideration when determining whether the systematic risk for a project matches that of the company as a whole.

Systematic risk of project cash flows depends on the nature of the business. Revenues and expenses may be affected to different degrees by changes in economic conditions in different industries or even among different companies in similar industries. A company that makes pastry and a company that makes machinery, for example, are likely to be affected by different factors. Demand for pastry is likely to be constant in all economic conditions. Alternatively, demand for machinery will fluctuate depending on the economic conditions.

Total volatility is not the same as systematic volatility. Companies with more total volatility tend to have more systematic volatility.

Beta is a measure of systematic risk, which in turn is a key factor in determining WACC. This means that WACC should be used only for projects with business risks similar to those for the entire company.

It is ultimately correct to use WACC to discount project cash flows as long as two conditions hold. First, the WACC of a company should be used to evaluate project cash flows only if systematic risk for the project matches the systematic risk for the company's portfolio of projects. Second, a company's WACC should be used to evaluate a project only if the mix of financing matches the mix of the company as a whole.

Different divisions or investment projects of a company may have different risk characteristics and should not be using the overall average cost of capital to evaluate projects. In such cases, the division's or project's WACC should be adjusted to reflect the appropriate risk and capital structure. This could be achieved by identifying other public listed companies with similar risk profiles as the project under consideration. The WACC of these comparable companies would be relevant benchmarks.

Illustrative Example 4

Consider a project that requires an investment of HK\$100 and should produce inflows of HK\$40 per year for three years. The correct discount rate is 8%. The approach to calculating the NPV is:

Illustrative Example 4 (continued)

$$\begin{aligned}\text{NPV} &= \text{FCF}_0 + (\text{FCF}_1 / (1+k)) + (\text{FCF}_2 / (1+k)^2) + (\text{FCF}_3 / (1+k)^3) \\ &= -100 + 40 / (1+0.08) + 40 / (1+0.08)^2 + 40 / (1+0.08)^3 = \text{HK\$}3.08\end{aligned}$$

Based on the correct discount rate, the project is attractive because it returns more than the cost of capital and produces positive NPV.

Suppose, however, that you are required to use your firm's WACC of 11% to evaluate this project. The calculation then is as follows:

$$\text{NPV} = -100 + 40 / (1+0.11) + 40 / (1+0.11)^2 + 40 / (1+0.11)^3 = -\text{HK\$}2.25$$

The NPV is now negative. When the WACC is used incorrectly to discount cash flows, the company may end up rejecting a positive NPV project. This is the danger of using firm-wide WACC to discount new investment.

Key Learning Point

A project's WACC should be adjusted to reflect the appropriate risk and capital structure.

► Knowledge Check Questions**Question 14**

Identify which of the following must hold to use a company's WACC to evaluate future cash flows from a project.

- I. Project financing is done in the same proportion of debt and equity as the company as a whole.
- II. Systematic risk for a project is the same as overall systematic risk of the company.
- III. The project should have conventional cash flows.

- A** I only.
- B** II only.
- C** I and II.
- D** I, II and III.

Question 15

Identify which of the following risks firms have no way of directly estimating the discount rate.

- A** A security that is publicly traded.
- B** Its debt securities.
- C** Incremental cash flows from a project.
- D** None of the above.

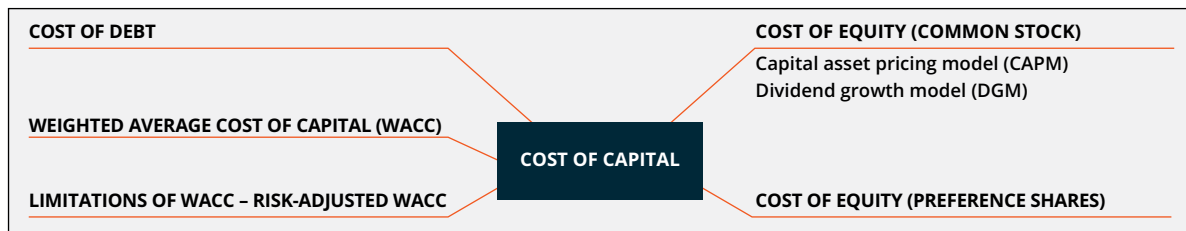


SUMMARY

- There are several types of equity securities, and the one most prevalent is common stock, which represents a basic ownership claim in a company.
- Three steps go into valuing an asset: First, estimate expected future cash flows. Second, determine the required rate of return, which reflects how risky future cash flows are. Third, compute the present value of the future cash flows.
- It is easy to apply these steps to determine the value of a bond but common stock valuation is more difficult because expected cash flows are less certain. Common stocks are true perpetuities, because they do not have a final maturity date but the rate of return is not directly observable; and the cost of equity is a weighted average of the costs of different types of stock that the company may have at a specific point in time.
- The dividend growth model (DGM) – or constant dividend growth model as it is sometimes called – is a valuation model used to calculate the fair value of a stock and makes an assumption that dividends grow at either a stable rate forever or at a different rate in the current period.
- The DGM approach is useful for a firm that pays dividends that will grow at a constant rate, such as an electric utility, but not be as useful for fast growing stocks that don't pay dividends, such as high-tech firm.
- The capital asset pricing model (CAPM) describes the relationship between systematic risk and expected return for an asset.
- The current effective annual yield on a risk-free asset should always be used in the CAPM equation.
- A long-term risk-free rate is better than a short-term rate to reflect the expectations for long-term inflation and the cost of investment over the long term.
- The beta of a stock is a measure of its volatility or systematic risk in comparison to the overall market.
- It is not possible to directly observe the market risk premium because the rate of return that investors expect for the market portfolio is simply not known.
- Preferred stock represents an ownership interest in a company. Preferred stock receives preferential treatment. Preferred stocks are hybrid securities, positioned between bonds and common stock.
- In computing the value of preferred stock, the important issue is whether the preferred stock has a fixed maturity.
- Long-term debt usually refers to debt that, when it was borrowed, was set to mature in more than one year.

- The current cost of long-term debt is the appropriate cost of debt for calculations of the weighted average cost of capital (WACC).
- In discussion of how the WACC for a firm is calculated, it is generally assumed that the costs of the different types of financing are known.
- The WACC is computed by weighing the costs of individual sources of finance based on their relative importance.
- The goal of forecasting cash flows for a project is to forecast incremental after-tax free cash flows at the company level. These cash flows represent the difference between the cash flows if the project is adopted and if it is not adopted.
- Total volatility is not the same as systematic volatility. Companies with more total volatility have more systematic volatility.
- Different divisions or investment projects of a company may have different risk characteristics and should not be using the overall average cost of capital to evaluate projects.

MIND MAP



LIST OF FORMULAS

Dividend growth model (DGM)

$$K_{cs} = \frac{D_1}{P_0} + g$$

Where

- k_{cs} = Required returns on common stock,
- D_1 = Dividend paid in the next period ($t = 1$),
- P_0 = current price per share,
- g = Expected growth rate of dividends.

Capital asset pricing model (CAPM)

$$E(R_i) = R_{rf} + \beta_i (R_m - R_{rf})$$

Where

- $E(R_m)$ = Expected return of market portfolio m ,
- R_{rf} = Risk-free rate of return,
- β_i = Beta coefficient of asset i ,
- $E(R_i)$ = Expected rate of return of asset i .

Pricing of preference shares

$$k_{ps} = \frac{D_{ps}}{P_{ps}}$$

Where

- D is dividend,
- P is current price,
- k is cost, or discount rate.

Weighted average cost of capital (WACC)

$$k_{\text{Firm}} = \sum_{i=1}^n x_i k_i = x_1 k_1 + x_2 k_2 + x_3 k_3 + \cdots + x_n k_n$$

$$k_{\text{Firm}} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{Equity}} k_{\text{Equity}}$$

Where

- k = The cost of funds;
- x = The weight of the amount within the total amount.

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. The expected market return is used to calculate the cost of equity by CAPM.

Answer B is correct. The cost of equity is defined as the rate of return required by shareholders.

Answer C is incorrect. The cost of keeping retained earnings is not related to the cost of equity.

Answer D is incorrect. The risk the company incurs when financing is defined as financial risk.

Question 2

Answer B is correct. Value of stock = HK\$5.50 ÷ 0.15 = HK\$36.67 per share

Question 3

Answer D is correct. $E(R_i) = R_{rf} + \beta_i [E(R_m) - R_{rf}] = 0.04 + 2.0 \times 0.06 = 0.04 + 0.12 = 0.16$

Question 4

Answer B is correct. The company can issue preferred stock at 6.5%. $P_p = \text{HK\$}1.75 / 0.065 = \text{HK\$}26.92$

Question 5

Answer C is correct. The cost of preferred equity. $k_{ps} = D_{ps} / P_{ps} = \text{HK\$}3.50 / \text{HK\$}38.89 = 9\%$.

Question 6

Answer C is correct. The cost of preferred equity. $k_{ps} = D_{ps} / P_{ps} = \text{HK\$}2.34 / \text{HK\$}19.50 = 12\%$.

Question 7

Answer A is incorrect. A bond can be sold at premium or discount and it may not be able to correctly reflect the cost of debts.

Answer B is correct. Current yield to maturity can discount the value of a company's bond into present value.

Answer C is incorrect. As the unpaid coupon payment should also be included in calculating in cost of debts.

Answer D is incorrect because B is correct.

Question 8

Answer A is incorrect. Debt is tax deductible and decreases the cost of debt.

Answer B is correct. Debt is tax deductible.

Answer C is incorrect. Equity is not tax deductible.

Answer D is incorrect. Although cost of debt is decreased, income taxes do not increase the cost of equity of a firm.

Question 9

Answer C is correct. Semi-annual coupon payment = $1,000 \times 7\% \div 2 = \text{HK\$}35$

The semi-annual YTM can be calculated as 4.5% by entering the following data in a scientific calculator.

PMT = 35, PV = -920.87, FV = 1,000, N = 10

Hence, the YTM = $2 \times 4.5\% = 9\%$

Question 10

Answer B is correct. Using the formula for pricing bonds, we have

$$\text{HK\$}746.16 = 42.50 \times \text{PVIFA}(\text{YTM} / 2, 26) + \text{HK\$}1,000 \times \text{FVIF}(\text{YTM} / 2, 26)$$

After solving for YTM / 2 = 6.25%, we find that YTM = 12.5%

Note that we assume semi-annual coupons for the bond.

Now, we adjust taxes: $12.5\% \times (1 - 0.35) = 8.125\%$

Question 11

Answer C is correct. $r_e = r_a + (r_a - r_d)(D / E)$

If $D / (D + E) = 0.50$, then $D / E = 1.0$

$$r_e = 0.10 + (0.10 - 0.05)(1.0)(1 - 0.2)$$

$$r_e = 0.10 + [0.05(0.90)] = 0.10 + 0.04 = 0.14, \text{ or } 14\%$$

Question 12

Answer D is correct. Noting that the proportion of debt and equity is:

$$X_{\text{debt}} = \text{HK\$}80,000,000 / (\text{HK\$}120,000,000 + \text{HK\$}80,000,000) = 0.4,$$

$$X_{\text{cs}} = \text{HK\$}120,000,000 / (\text{HK\$}120,000,000 + \text{HK\$}80,000,000) = 0.6$$

The return on common shares can be found with the CAPM:

The equation for the CAPM is:

$$E(R_i) = R_{\text{rf}} + \beta_i [E(R_m) - R_{\text{rf}}] \Rightarrow$$

$$E(R_i) = 0.08 + 1.5 \times [0.14 - 0.08] = 0.17$$

The price of the bonds can be used to find the YTM:

$$\text{HK\$}767.63 \times \text{PVIFA}(\text{YTM} / 2, 14) + \text{HK\$}1,000 \times \text{PVIF}(\text{YTM} / 2, 14) \Rightarrow$$

$$\text{YTM} / 2 = 6\% \Rightarrow \text{YTM} = 12\% = k_{\text{Debt pretax}}$$

The formula for WACC is:

$$\text{WACC} = X_{\text{Debt}} k_{\text{Debt pretax}} (1 - t) + X_{\text{ps}} k_{\text{cs}} = 0.4 \times 0.12 \times (1 - 0.35) + 0.6 \times 0.17 = 0.1332$$

Question 13

Answer D is correct. $x_{\text{Debt}} = \text{HK\$}80,000,000 / (\text{HK\$}120,000,000 + \text{HK\$}80,000,000) = 0.4$

$$x_{\text{CS}} = 120,000,000 / (120,000,000 + 80,000,000) = 0.6$$

$$\text{Coupon payment} = 1,000 \times 7\% / 2 = \text{HK\$}35$$

By entering the following data in a scientific calculator, the YTM, which is the cost of debt, of the bond can be calculated as 12%.

$$\text{PMT} = 35, \text{PV} = -767.03, \text{FV} = 1,000, \text{N} = 14.$$

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt pretax}} \times (1 - t) + x_{\text{ps}} k_{\text{CS}} = 0.4 \times 0.12 \times (1 - 0.35) + 0.6 \times 0.17 = 0.1332 = 13.32\%$$

Question 14

Answer C is correct. (Both I and II hold.)

I holds because a firm's WACC may be used to evaluate a project only if the financing mix of the project matches the proportions of debt, preferred shares and common shares across the company.

II holds because a firm's WACC should also be used to evaluate cash flows for new projects only if systematic risk for that project matches systematic risk of the company's project portfolio.

III does not hold because the frequency of cash flows cannot make the result more accurate when using WACC.

Question 15

Answer A is incorrect. Firm can use DGM or CAPM to estimate the discount rate of the publicly traded security.

Answer B is incorrect. A firm estimates its yield to maturity to estimate the debt securities.

Answer C is correct. The discount rate has to be adjusted in order to make a more accurate reflection on the risk and capital structure.

Answer D is incorrect. C is correct.

 EXAM PRACTICE**QUESTION 1**

Identify what the beta is of a portfolio with a level of systematic risk that is the same as that of the market.

- A. Equal to zero.
- B. Equal to one.
- C. Less than the beta of the risk-free asset.
- D. Less than zero.

QUESTION 2

Identify which of the following is not an assumption of the capital asset pricing model (CAPM).

- A. Capital markets are in equilibrium.
- B. Transaction costs are the same for all investors.

- C. All investors can borrow or lend money at the risk-free rate of return.
- D. All investors have identical expectations about expected returns, standard deviations, and correlation coefficients for all securities.

QUESTION 3

You have invested all of your retirement funds in your company's stock, which has a beta of 1.0. Identify which of the following you would expect from your company stock in your retirement plan when you hear on the radio that the stock market is down 150 points.

- A. Should be a great day, your stock should be up significantly.
- B. Rats! Your company stock is likely down more than the market today.
- C. Your company stock is likely down with the market today in similar proportion.
- D. It's anyone's guess as the market does not move at all in relation to your company stock.

QUESTION 4

Identify which of the following statements is incorrect.

- A. Betas are constant.
- B. High growth companies tend to have betas much larger than one.
- C. Portfolio betas are the weighted average of the betas of the stocks in the portfolio.
- D. Betas for companies in the same industry are generally similar, but will exhibit differences based on financial risk, size and other factors.

QUESTION 5

You and a group of nine other investors decide to buy three small properties. If all 10 of you pool your money, there will be enough to buy all three. Each investment requires the same amount of money and each investor will have a 10% interest in each property. Each investor will, in effect, have one-third of their portfolio in each property.

Each property costs the same but they are in different locations and neighbourhoods, which might give each one different levels of systematic risk. The betas for the properties are 1.2, 1.3 and 1.5. If the risk-free rate is 4% and the market risk premium is 6%, calculate the expected return on your real estate portfolio after making all three investments.

QUESTION 6

Hong Ltd is financed with debt, preferred equity and common equity with values of HK\$20 million, HK\$10 million and HK\$30 million. The betas for the debt, preferred stock and common stock are 0.2, 0.5 and 1.1. If the risk-free rate is 2.55%, the market risk premium is 5.63%, and the average and marginal tax rates are both 30%, state the company's weighted average cost of capital.

QUESTION 7

Management of NT Ltd believes that the company's WACC is too high because the firm has been too secretive in its operations. Evaluate this statement.

QUESTION 8

RT Ltd has a preferred shares issue outstanding that pays an annual dividend of HK\$1.30 per year. The current cost of preferred equity is 9%. If the company issues more preferred shares that pay exactly the same dividend and the investment banker retains 8% of the sale price proceeds, state the cost of new preferred shares for the company.

QUESTION 9

The cost of capital for a common stock, when calculated using a dividend discount model, is much lower than the calculation for the cost of capital using the CAPM. Explain this discrepancy.

QUESTION 10

The return of common shares of Tang Cyclic Inc is 1.6 times as sensitive to macroeconomic information as the market returns. When the risk-free rate of return is 4% and the market risk premium is 5.63%, state the company's cost of common equity capital.

QUESTION 11

The managers of a firm financed through common equity are considering two different projects. The first project has a large amount of unsystematic risk and a small amount of systematic risk. The second project has a small amount of unsystematic risk and a large amount of systematic risk. Identify which project, if taken, is more likely to increase the firm's cost of capital.

 ANSWERS TO EXAM PRACTICE**QUESTION 1**

Answer A is incorrect. Market beta is equal to one.

Answer B is correct. Market beta is equal to one.

Answer C is incorrect. Risk-free asset has zero systematic risk.

Answer D is incorrect. Market beta is equal to one.

QUESTION 2

Answer A is incorrect. It is an assumption of CAPM. Capital markets are in equilibrium.

Answer B is correct. It is not an assumption of CAPM. CAPM includes the assumption that there are no transaction costs. CAPM does not include an assumption that transaction costs are the same for all investors.

Answer C is incorrect. It is an assumption of CAPM. All investors can borrow or lend money at the risk-free rate of return.

Answer D is incorrect. It is an assumption of CAPM. All investors have identical expectations about expected returns, standard deviations and correlation coefficients for all securities.

QUESTION 3

Answer A is incorrect. Your stock should be down similar to the market.

Answer B is incorrect. Your company stock is likely down in similar magnitude as the market today.

Answer C is correct. Beta equal to one means market risk is the same as the average security in the market. A beta of 1 implies that if the market return increased (or decreased) by 1 percent, the return on the security (or portfolio) would, on average, increase (decrease) by 1 percent. Your company stock is likely down with the market today in similar proportion.

Answer D is incorrect. Your stock should be down in similar magnitude as the market.

QUESTION 4

Answer A is correct. Betas also may change over time; for example, high-growth companies tend to have betas much larger than 1.0, but over time, as the company matures and growth slows, the beta approaches 1.0.

Answer B is incorrect. It is true that high growth companies tend to have betas much larger than one.

Answer C is incorrect. It is true that portfolio betas are the weighted average of the betas of the stocks in the portfolio.

Answer D is incorrect. It is true betas for companies in the same industry are generally similar, but will exhibit differences based on financial risk, size and other factors.

QUESTION 5

$$\begin{aligned}
 E(R_i) &= R_{rf} + \beta_i [E(R_m) - R_{rf}] \\
 &= 0.04 + (1.2 \times 0.06) = 0.112, \text{ or } 11.2\%, \text{ for cottage 1} \\
 &= 0.04 + (1.3 \times 0.06) = 0.118, \text{ or } 11.8\%, \text{ for cottage 2} \\
 &= 0.04 + (1.5 \times 0.06) = 0.130, \text{ or } 13.0\%, \text{ for cottage 3} \\
 E(R_{\text{Portfolio}}) &= [x_1 \times E(R_1)] + [x_2 \times E(R_2)] + [x_3 \times E(R_3)] \\
 &= (1/3 \times 0.112) + (1/3 \times 0.118) + (1/3 \times 0.13) = 0.12, \text{ or } 12.0\%
 \end{aligned}$$

QUESTION 6

The portions of total financing that debt, common equity and preferred equity represent are as follows:

Debt = HK\$20 million

Preferred equity = HK\$10 million

Common equity = HK\$30 million

Total capital = HK\$60 million

$$x_{\text{debt}} = 20 / 60 = \text{HK\$}0.3333$$

$$x_{\text{ps}} = 10 / 60 = \text{HK\$}0.1667$$

$$x_{\text{cs}} = 30 / 60 = \text{HK\$}0.5000$$

The costs of debt, common equity and preferred equity are as follows:

$$E(R_{\text{Debt}}) = \text{Return}_{rf} + \beta_{\text{Debt}} (E(\text{Return}_m) - \text{Return}_{rf}) = 0.0255 + (0.2 \times 0.0563) = 0.03676, \text{ or } 3.676\%$$

$$E(R_{\text{ps}}) = \text{Return}_{rf} + \beta_{\text{ps}} (E(\text{Return}_m) - \text{Return}_{rf}) = 0.0255 + (0.5 \times 0.0563) = 0.05365, \text{ or } 5.365\%$$

$$E(R_{\text{cs}}) = \text{Return}_{rf} + \beta_{\text{cs}} (E(\text{Return}_m) - \text{Return}_{rf}) = 0.0255 + (1.1 \times 0.0563) = 0.08743, \text{ or } 8.743\%$$

$$\text{WACC} = (0.3333 \times 0.03676 \times (1 - 0.3)) + (0.1667 \times 0.0563) + (0.5 \times 0.08743)$$

$$= 0.061677, \text{ or } 6.1677\%$$

QUESTION 7

WACC is a function of perceived risk involved in cash flows of projects. When risk perceptions are higher than actual risk due to a lack of information available to the market in regard to those projects, the company may be able to lower perceived risk by sharing more information. This approach could actually lower WACC.

QUESTION 8

The current cost of preferred shares for RT Ltd is:

$$P_{ps} = \frac{D}{k_{ps}} = \frac{1.30}{0.09} = \text{HK\$}14.44$$

so RT Ltd would receive 92% of the proceeds. It is possible, then, to adapt the cost of preferred equity to:

$$k_{ps} = \frac{D}{P_{ps}(1-F)} = \frac{1.30}{14.44(1-0.08)} = \frac{1.30}{13.29} = 0.0978, \text{ or } 9.78\%$$

QUESTION 9

Comparing the formulas for both methods, we arrive at:

$$E(R_{cs}) = R_{rf} + \beta(E(R_m) - R_{rf}) \text{ and } k_{cs} = \frac{D_1}{P_{cs}} + g$$

With these two sources of information, it is possible to see that the only variable it is not possible to get from the market is the growth rate in dividends. Future dividends are also a function of this estimated growth rate. Since the dividend discount method provides a lower cost of capital than CAPM. It is possible that the estimated growth rate is lower than what the aggregate market has assumed. This assumes the market is pricing the stock efficiently. An incorrect market price might cause the difference.

QUESTION 10

The beta for Tang Cyclic Inc is 1.6. It is possible to use the remaining information in the CAPM as:

$$\begin{aligned} E(R_{cs}) &= R_{rf} + \beta(E(R_m) - R_{rf}) \\ &= 0.0255 + 1.6(0.0563) = 0.11558 = 11.56\% \end{aligned}$$

QUESTION 11

The level of systematic risk associated with a particular project helps inform the cost of capital. As a result, a project with the greatest level of systematic risk may have the greatest impact on the cost of capital for a company, even when it has the lowest level of unsystematic risk.

8

Capital Structure Theories

CHAPTER TOPIC LIST

8.1 Capital Structure Theories

- 8.1.1 Modigliani–Miller Propositions and Their Implications
- 8.1.2 Agency Theory
- 8.1.3 Trade-off Theory
- 8.1.4 Signalling Theory
- 8.1.5 Pecking Order Theory

8.2 Key Considerations for Establishing a Corporate Capital Structure

8.3 Financial Products for Financing and Risk Management

- 8.3.1 Hedging
- 8.3.2 Speculating
- 8.3.3 Arbitraging

8.4 Use of Financial Experts for Fund Raising and Optimising Capital Structure

- 8.4.1 Financing Plan
- 8.4.2 Making Capital Structure Decisions



LEARNING OUTCOMES

PRINCIPAL LO7: EXPLAIN AND ANALYSE DIFFERENT SOURCES OF FINANCE AND CAPITAL STRUCTURE

LO7.01: Explain and analyse the theories of major capital market models

- 7.01.10** Explain and illustrate through calculations the theory of capital markets including: portfolio theory; capital asset pricing model; cost of capital; Modigliani Miller theories; and efficient market hypothesis

- 7.01.11** Explain, other than Modigliani Miller theories, various types of capital structure theory, including agency theory, trade-off theory, signalling theory and pecking order theory

- 7.01.12** Explain the key considerations for establishing a corporate capital structure

- 7.01.13** Describe the use of financial products, including derivatives, in relation to corporate financing and risk management

- 7.01.14** Explain the use of external experts for advisory on transaction services for fund raising



OPENING CASE

ABC INVESTMENT LTD

As the experience and skills of the financial management staff at ABC Investment Ltd continue to grow, it becomes increasingly clear that corporate finance management is a key practice to ensure both the returns of investments and effective fundraising.

Important considerations to this work are the various approaches to management that can help prioritise the importance of factors and decision making. Successful finance professionals should be familiar with and regularly apply these ideas to their work. Financial management staff also have to consider the implications of more qualitative theories in corporate finance and come to understand that corporate finance decision-making involves a lot more than straightforward equations.

All these considerations come into play in the financial management work that ABC Investment Ltd undertakes for itself and other companies in its group.

An important part of this work is calculating the costs of funds for the companies' various ventures. This includes considering factors that are important to designing an optimal capital structure for a company.

In the end, finance professionals have to match theory to practice combining the two and applying them to the management of the company and the various subsidiaries within the corporate group. 



OVERVIEW

Companies have different needs and use those needs to determine the capital structure that best suits them. As this chapter explores, the capital structure policies of companies balance out the benefits companies can get against the costs of capital and debt.

There are a number of different approaches to determining the optimal capital structure for a company. The Modigliani and Miller (M&M) propositions provide some insights into the optimal capital structure of a firm with a pair of propositions that are widely considered. There are other considerations, including agency theory, trade-off theory, signalling theory and pecking order theory, all of which should be taken into account when considering the optimal capital structure.

There are a number of key considerations for managers as they determine the optimal capital structure for the companies they manage. These include financial flexibility, access to opportunities, financial leverage and income volatility, to name a few. The impact of any move on earnings is also a key consideration.

There are different approaches to security capital and funding operations, all the while managing risk. From a financial point of view, companies that frequently engage in hedging, speculation and arbitrage do not only tackle any potential risk, but also profit from opportunities that may arise.

Finally, this chapter explores how experts may use all these considerations in developing a financial plan that can help companies determine how much external funding they have access to and need and work to maximise the value of the firm.



8.1 CAPITAL STRUCTURE THEORIES

The benefits and costs of debt are reflected in the **capital structure** policies of companies.

For instance, interest payments may be tax deductible so companies with a low level of debt pay higher taxes than they would if they had more debt. However, these companies have more flexibility to react to market changes or borrow more. Flexibility is useful in times of recession or allow companies to take advantage of opportunities.

The appropriate capital structure for a particular firm depends on that firm's characteristics. When it comes to capital structure policy, one size does not fit all. The appropriate mix of debt and equity financing differs across firms and, even within the same firm, it can change over time.

This mix of debt and equity may include common stock, debt and preference shares. Some companies have different types of common stock with different voting rights and claims on cash flows.

There are different types of debt: long term or short term, secured or unsecured, convertible or non-convertible. Preference shares may be cumulative or non-cumulative and convertible or non-convertible into common stock. The portion of debt included in the total financing of a company represents the financial leverage in the capital structure of a company, with more debt suggesting higher degrees of financial leverage.

The amount of financial leverage a company carries is important as it impacts the value of the firm.

8.1.1 Modigliani–Miller Propositions and Their Implications

Understanding how financing decisions affect the costs of finance is necessary to better understand how the optimal capital structure for a particular firm is determined.

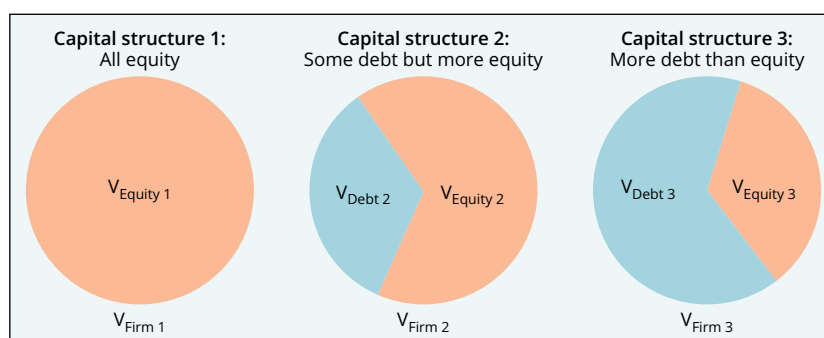
Key insights into these mechanisms can be found in the **Modigliani–Miller (M&M) propositions** developed by Franco Modigliani and Merton Miller. These propositions were developed more than half a century ago but are still relevant.

Modigliani and Miller's Proposition 1 says capital structure decisions will not impact the value of a firm as long as six conditions are met:

1. There are no taxes.
2. There are no costs from information or transactions.
3. There are no bankruptcy costs.
4. Companies and investors have equivalent borrowing costs.
5. Market information is symmetrical.
6. Capital structure decisions do not affect the real investment policies of the company.

Companies choose what real assets projects to invest by developing a real investment policy.

For instance, without taxes, the value of free cash flows from assets may be represented in pie graphs such as the ones shown in Exhibit 8.1. The slice labelled V_{Equity} represents cash flows to be received by the shareholders, and the slice labelled V_{Debt} represents the value to be received by the debt holders.



► **EXHIBIT 8.1 Equity and debt pie** (Source: data from Parrino 2016. Reproduced with permission of John Wiley & Sons.)

The market value of the debt added to the market value of the equity should equal the market value of cash flows that assets generate (V_{Assets}). In practice, V_{Assets} also refers to the firm's enterprise value (V_{Firm}), which means that it is possible to write the equation as:

$$V_{\text{Firm}} = V_{\text{Assets}} = V_{\text{Debt}} + V_{\text{Equity}}$$

M&M Proposition 1 says that if the size of the pie is fixed and only shareholders and debt holders are getting a slice, then the total value of equity and debt claims does not change when the capital structure changes. Exhibit 8.1 shows three pies, each of which represents a different capital structure. Regardless of how the pies are sliced, the total value of debt and equity remains the same.

If the conditions specified by M&M hold, the capital structure of the firm specifies how that pie is to be sliced, but it does not change the overall size of the pie or the combined size of the debt and equity slices.

For example, ABC Entertainment Ltd is totally financed using equity. ABC Entertainment Ltd produces annual cash flows of HK\$100M and these cash flows are expected to continue indefinitely.

If the discount rate for ABC Entertainment Ltd's cash flows is 10%, it is possible to use a perpetuity model to calculate the value of the firm using the following equation:

$$V_{\text{Firm}} = \text{PVP} = \text{CF} / i = 100\text{M} / 0.1 = \text{HK\$1,000M}$$

Since the firm is financed entirely with equity, the equity is also worth HK\$1,000M.

Suppose that management is considering changing its capital structure from HK\$1,000M (100%) equity to HK\$800M (80%) equity and HK\$200M (20%) debt. The company could accomplish this by selling perpetual bonds worth HK\$200M and paying HK\$200M to shareholders through a one-time special dividend.

What ABC Entertainment Ltd is considering is a financial restructuring, a mix of financial transactions such as issuing debt or repurchasing stock that ultimately changes the capital structure of the company but has no impact on the real assets of the company. A financial restructure does not lead to new investment.

Now suppose that the company has only one investor who owns 100% of the outstanding stock of the company and the company has to pay 5% of the interest on the debt after the restructuring. If the restructuring took place, the investor would immediately receive a HK\$200M special dividend. After that, each year the investor would be entitled to the HK\$90M that is left over after the HK\$10M interest payment on the bonds.

M&M showed that if an investor does not like the effect of the restructuring, management could go ahead with the restructuring and the investor could undo its effect on the cash flows received by making off-setting trades in an investment account.

To undo the effects of the proposed financial restructuring, the investor would simply use the entire HK\$200M special dividend to redeem all of the perpetual bonds the company issued. From there, an investor may receive the first HK\$10M the company earns every year as interest payments on the bonds. At the same time, an investor who owns 100% of the stock might receive remaining cash flows from that ownership stake.

As happened before the restructuring, the investor should receive all cash flows generated by the firm but instead of receiving those cash flows as dividends, some of them might be in the form of interest payments.

If the company has more than one stockholder the result is the same.

Perfect financial markets would meet all the conditions outlined in M&M Proposition 1. Investors can make changes in their investment accounts that will match the cash flows for any capital structure that the company might carry.

Investors may not be willing to pay more for the stock of a company that does this for them because they can do this on their own. As a result, the value of the company will be the same regardless of the capital structure. This happens because changes in capital structure will not change the total value of any claims debt holders and shareholders have.

The conditions outlined in the first proposition suggest that the capital structure of a company does not impact the value of real assets because the capital structure decisions do not have an impact on the level, timing or risk associated with the cash flows that real assets generate.

Changes to the capital structure of the firm do not affect the risk of the cash flows that assets generate but affect the risk of the equity claims on those cash flows, so the required return on equity would be affected.

M&M's Proposition 2 says that the cost of (i.e. required return on) the common stock of a company is directly related to the debt-to-equity ratio. The formula for the weighted average cost of capital (WACC), first discussed in Chapter 7, helps illustrate this point:

$$WACC = x_{Debt} k_{Debt \text{ pretax}} (1-t) + x_{ps} k_{ps} + x_{cs} k_{cs}$$

If there are no taxes, as M&M Proposition 1 assumes, then $t = 0$ and the equation is:

$$WACC = x_{Debt} k_{Debt} + x_{ps} k_{ps} + x_{cs} k_{cs}$$

Furthermore, assuming (for simplicity) that the firm has no preference shares, then this equation can be simplified further:

$$WACC = x_{Debt} k_{Debt} + x_{cs} k_{cs}$$

where: $x_{Debt} + x_{cs} = 1$

Under M&M Proposition 1 conditions, changes to the capital structure do not affect the risk of the cash flows that assets generate, so the WACC does not change with the capital structure.

This is because, as a weighted average of the cost of debt and equity, WACC reflects the riskiness of the cash flows that assets generate (k_{Assets}). Recognising that the proportions of debt and equity in the firm's capital structure are calculated as:

$$x_{Debt} = \frac{V_{Debt}}{V_{Debt} + V_{cs}}$$

and

$$x_{cs} = \frac{V_{cs}}{V_{Debt} + V_{cs}}$$

where:

- V_{Debt} = the dollar value of the debt, and
- V_{cs} = the dollar value of the common stock.

So the equation can be written as:

$$WACC = k_{Assets} = \frac{V_{Debt}}{V_{Debt} + V_{cs}} k_{Debt} + \frac{V_{cs}}{V_{Debt} + V_{cs}} k_{cs}$$

Finally, using basic algebra, it is possible to rearrange this equation to solve for k_{cs} in terms of k_{Assets} and k_{Debt} :

$$k_{cs} = k_{Assets} + \left(\frac{V_{Debt}}{V_{cs}} \right) (k_{Assets} - k_{Debt})$$

This equation is M&M's Proposition 2, which shows that the required return on the common stock of a company is directly linked to the debt-to-equity ratio.

This is visible in the equation by noting that as the ratio V_{Debt}/V_{cs} increases on the right-hand side, k_{cs} will increase on the left-hand side.

This holds not only for firms that have only common stock outstanding but also for firms that have preference shares outstanding as well.

The two terms on the right side of the equation can be considered to be sources of risk in the cash flows.

First is the underlying risk of the assets, which is identified in the required return on assets (k_{Assets}). This is generally called **business risk**, and represents the risk associated with the specific characteristics of particular projects or assets.

A second source of risk is **financial risk**, which is also reflected in the second term and comes from the capital structure of a company. Financial risk reflects the impact of financing decisions on the riskiness of cash flows. Companies that use more debt financing have greater financial risk.

Debt holders have the first claim on cash flows from company assets. Any payments of interest and principal should be made before payments are made to shareholders. As a result, when the proportion of debt in the capital structure of a firm is larger so are the interest and principal payments and, in turn, the uncertainty associated with cash flows to be claimed by stockholders is greater (Exhibit 8.2).

Revenue	}	Business risk
- Cost of goods sold		
Gross profit	}	×
- Selling, general and admin. expenses		
Operating profit	}	Financial risk
- Interest expense		
Earnings before tax	}	=
- Income tax		
Net income		Total equity risk

► **EXHIBIT 8.2 Risk relationships** (Source: data from Parrino 2016. Reproduced with permission of John Wiley & Sons.)

By assuming that a firm's net income is a reasonable measure of the cash flows to which shareholders have a claim, it is possible to use a simple statement of profit or loss and other comprehensive income (like the one in Exhibit 8.3) to illustrate the distinction between business and financial risk.

Column	No Financial Risk				Financial Risk			
	Low Operating Leverage		High Operating Leverage		Low Operating Leverage		High Operating Leverage	
	1	2	3	4	5	6	7	8
Variable costs as a percent of total costs	80%		40%		80%		40%	
Interest expense	HK\$ 0.00		HK\$ 0.00		HK\$ 15.00		HK\$ 15.00	
	Before	After	Before	After	Before	After	Before	After
Revenue	100.00	80.00	100.00	80.00	100.00	80.00	100.00	80.00
– Cost of goods sold (VC)	60.00	48.00	30.00	24.00	60.00	48.00	30.00	24.00
Gross profit	40.00	32.00	70.00	56.00	40.00	32.00	70.00	56.00
– Selling, general, and admin. (FC)	15.00	15.00	45.00	45.00	15.00	15.00	45.00	45.00
Operating profits	25.00	17.00	25.00	11.00	25.00	17.00	25.00	11.00
– Interest expense	0.00	0.00	0.00	0.00	15.00	15.00	15.00	15.00
Earnings before tax	25.00	17.00	25.00	11.00	10.00	2.00	10.00	–4.00
– Income taxes (35%)	8.75	5.95	8.75	3.85	3.50	0.70	3.50	–1.40
Net income	HK\$ 16.25	HK\$11.05	HK\$ 16.25	HK\$ 7.15	HK\$ 6.50	HK\$ 1.30	HK\$ 6.50	HK\$–2.60
Percent change in net income		–32%		–56%		–80%		–140%

► **EXHIBIT 8.3 Financial statement** (Source: data from Parrino 2016. Reproduced with permission of John Wiley & Sons.)

The example of Exhibit 8.3 shows that business risk is associated with the operations of the business. Thinking of business risk as the systematic risk related to operating profits, it is possible to see that a firm's business risk reflects the systematic variation in:

1. unit sales,
2. unit prices,
3. the costs of producing and selling the firm's products, and
4. the degree of operating leverage in the production process.

In contrast, financial risk is associated with the required payments to a firm's lenders. The total risk of the cash flows that the shareholders have is dependent on both the business risk and the financial risk.

The numerical example in Exhibit 8.3 also shows the distinction between business risk and financial risk and how they combine to determine total equity risk.

Elegant analyses of how capital structure choices are related to firm value and how financial leverage affects the cost of equity were provided by M&M. However, they recognised that the conditions underlying their analyses are unrealistic. Firms do pay taxes, there are information and transaction costs and financing decisions have an impact on investment policies.

M&M analysis helps identify what factors to consider to understand better how the capital structure affects the value of a company and the cost of equity. Financial policies matter because taxes, information and transaction costs and choices of the capital structure are all important.

Illustrative Example 1

ABC Investment Ltd has a subsidiary called Keyboard Chiropractic Clinics Ltd that produces HK\$300,000 of cash flow each year. The firm does not have any debt outstanding and its cost of equity capital is 25%. The firm's management would like to repurchase HK\$600,000 of its equity by borrowing a similar amount at a rate of 8% per year.

Assuming that the debt will be perpetual, what is the cost of equity capital for Keyboard Chiropractic Clinics Ltd after it changes its capital structure. Assuming that Modigliani–Miller Proposition 1 holds and using the 25% cost of equity capital, the value of the firm today can be calculated:

$$300,000 / 0.25 = \text{HK\$}1,200,000$$

HK\$1,200,000 must be the value of the firm after the repurchase.

At the time of borrowing, the value of the debt claim must be HK\$600,000. Therefore, the value of the outstanding equity after the repurchase must be:

$$1,200,000 - 600,000 = \text{HK\$}600,000$$

Since it is known that HK\$48,000 ($\text{HK\$}600,000 \times 0.08$) in cash flow must be paid to the debt holders each year, it is therefore also known that $\text{HK\$}252,000 = (\text{HK\$}300,000 - \text{HK\$}48,000)$ will be available to the shareholders on an annual basis.

This means that the new cost of equity capital for the firm must be:

$$k_{\text{Equity}} = \text{HK\$}252,000 / \text{HK\$}600,000 = 42.00\%$$

8.1.2 Agency Theory

When one party, known as the principal, delegates decision-making authority to another party, known as the agent, it is referred to as an **agency relationship**.

In agency relationships, the agent is expected to act in the interest of the principal. However, agents' interests can sometimes be in conflict with the interests of the principal.

Agency theory attempts to explain the dynamic between principals and agents in business transactions.

The aim of the theory is to avoid and solve problems that may arise in any such agency relationship. These problems can emerge when the goals of the agent and the principal (a company) are different and so are the risk levels with which each side is comfortable.

When it comes to the capital structure of companies, the most frequent agency relationship is between shareholders (the principal) and company executives (the agents). For instance, the executives of a company might be primarily interested in their salaries and bonuses as opposed to share value, which is the primary concern of shareholders.

In effect, agency theory suggests that both sides may have goals that are not in line.

This problem is exacerbated when one side has limited information or ability to control what the other side is doing.



Company executives seeking to expand into a new product or service line, for example, will risk the company's short-term profitability for future growth and higher earnings. Shareholders who prefer high current capital growth might not be aware of those plans.

Appetite for risk is also a focus of agency theory. Should the agent be charged with the resources of the principal, they incur low risk or none at all, because the losses will be incurred by the principal.

The managers and shareholders of a firm often behave in ways that reduce a firm's value when the firm becomes financially distressed. The resulting costs are a type of agency cost.

Agency costs result from conflicts of interest between principals and agents.

To illustrate, let us use a pizza parlour example. As the owner of a pizza parlour, you have decided your goal is to maximise the value of the business and thereby your ownership interest. There is no conflict of interest in your dual roles as owner and manager because your personal and economic self-interest is tied to the success of the pizza parlour. The restaurant has succeeded because you have worked hard and have focused on customer satisfaction.

Now suppose you decide to hire a college student to manage the restaurant. Will the new manager always act in your interest? Or could the new manager be tempted to give free pizza to friends now and then or, after an exhausting day, leave early rather than spend time cleaning and preparing for the next day? From this example, you can see that once ownership and management are separated, managers may be tempted to pursue goals that are in their own self-interest rather than the interests of the owners.

The board of directors is responsible for approving major decisions and monitoring the performance of the managers, who make most decisions. The board is rarely privy to most of these management decisions. A concern here is that the incentives of managers and shareholders are not always aligned, so managers may make decisions that benefit them rather than shareholders.

Debt financing can help align the interests of both managers and shareholders. Debt financing creates incentives for managers to focus on maximising cash flows while also limiting the ability of bad managers to waste money on projects with negative net present value (NPV), which leads to lower agency costs that may arise from the principal-agent relationship between shareholders and managers. However, debt financing can also increase agency costs by changing how managers behave. A large portion of a manager's remuneration may depend on the success of the firm, which could mean that they prefer a lower amount of risk.

The use of debt increases the volatility related to the earnings of a company and the chances that the company gets into financial difficulty. This higher level of risk could also lead to management taking a more conservative approach. For instance, when a company has more financial leverage, managers may have greater incentives to turn down risky projects with a positive NPV than managers at companies with less leverage. At the same time, managers at companies with high levels of leverage may tend to distribute fewer profits to shareholders to create a bigger cushion against possible bankruptcy. Both these efforts may actually reduce the value of a company.

The sixth M&M condition is that using debt financing does not affect the real investment policy of a company but if the use of debt financing leads managers to turn down projects with a positive NPV, then the use of financing does have an impact on real investment policies. The fact that managers may act in this way is another reason that debt financing affects the value of the firm.

Investors lend money to firms and leave it up to the shareholders to decide how to use the money. This leads to the emergence of a principal–agent relationship. In turn, shareholders appoint managers to invest the money and make all of interest and principal payments. Shareholders may not always use the money in a way that is of the best interest of lenders.

For instance, shareholders may want to distribute dividends rather than reinvest in a company. Loans are contracts between a lender and the company, so any dividends paid out may impact the ability of a company to repay the loan and lenders may not get that money back. This is known as a wealth transfer from lenders to the shareholders. To deal with this, lenders often include clauses in their contracts to limit dividends, but these provisions are hardly foolproof.

Notice that any discussion of stockholder–lender agency costs assumes managers do what shareholders would like them to do, but this is not always the case. As a result, there are different possibilities of how financial leverage affects the decisions of managers. For instance, managers of a company that uses debt financing may want to invest in projects with low-risk projects but shareholders may prefer high-risk projects. The question of whether shareholders get their wishes depends on how strong corporate governance is.

For example, imagine a company has invested HK\$50 million in 10% risk-free bonds that will pay HK\$55 million in one year. The company also has one-year debt for which HK\$50 million of interest and principal will be due in a year. The company is solvent and can repay the debt but the equity is just HK\$5 million, since this is all that is left over after the payments.

Now, consider that shareholders desire to sell the risk-free bonds and invest in a project that has a 50–50 chance of returning HK\$95 or HK\$15 million in one year. Instead of receiving just HK\$50 million with no risk, the lenders have a 50% chance of receiving the HK\$50 million they are owed and a 50% chance of receiving only HK\$15 million. This means that the lenders ultimately expect to receive HK\$32.5 million:

$$E(V_{\text{Bonds}}) = (0.50 \times 50) + (0.50 \times 15) = \text{HK\$}32.5 \text{ million}$$

The amount of HK\$32.5 million is HK\$17.5 million less than the HK\$50 million that the lenders expected to receive when the firm held the risk-free bonds. The value that the shareholders expect to receive, however, has increased by HK\$17.5 million, from HK\$5 million to HK\$22.5 million:

$$E(V_{\text{Stock}}) = [0.50 \times (95 - 50)] + (0.50 \times 0) = \text{HK\$}22.5 \text{ million}$$

By changing to riskier assets, a HK\$17.5 million transfer of wealth has resulted. This is part of what is known as the asset substitution problem. After a company receives a loan, the shareholders in a company have an incentive to substitute more risky assets for less risky ones.

Sometimes, shareholders have incentives to invest in risky negative NPV projects. For example, assume shareholders sell the HK\$50 million in risk-free bonds and invest in a project with a 50% chance of returning HK\$70 million and a 50% chance of returning HK\$10 million. This scenario means that the expected return on the investment is HK\$40 million $[(0.50 \times 70) + (0.50 \times 10) = \text{HK\$}40 \text{ million}]$.

Despite this being a negative NPV project, shareholders can expect to receive a value of HK\$10 million, which is twice as much as the HK\$5 million they could expect to receive when the firm owned the risk-free bonds:

$$E(V_{\text{Stock}}) = [0.50 \times (70 - 50)] + (0.50 \times 0) = \text{HK\$10 million}$$

The lenders bear the HK\$15 million loss in the value of the firm (HK\$55 million – HK\$40 million = HK\$15 million), and they pay for the HK\$5 million gain to the shareholders. By selling the risk-free bonds, the lenders now expect to receive HK\$20 million less than the HK\$50 million they would have received:

$$E(V_{\text{Bonds}}) = (0.50 \times 50) + (0.50 \times 10) = \text{HK\$30 million}$$

In a situation known as the underinvestment problem, shareholders of financially distressed firms can also have incentives to turn down positive NPV projects. This happens when the value that is created by investing in a positive NPV project is likely to go to the lenders of a financially distressed firm, instead of the shareholders.

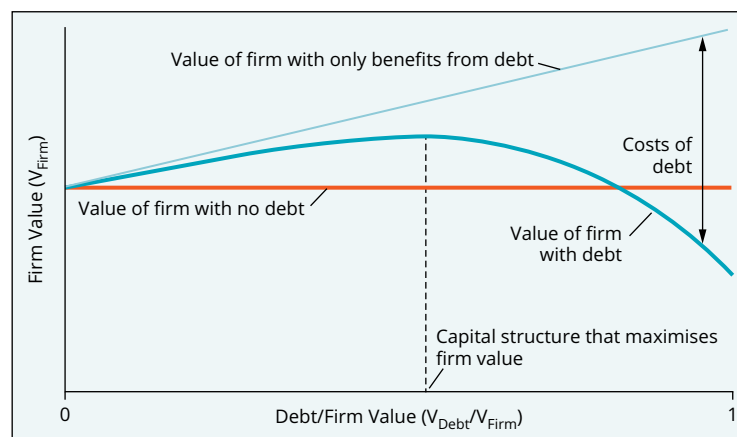
For example, imagine a financially distressed firm that has debt with a face value of HK\$50 million outstanding and the value of its assets is HK\$32.5 million. If the assets of the company were sold today, the lenders would receive the whole HK\$32.5 million. The shareholders would receive nothing.

Companies in financial distress often have a difficult time raising capital. For instance, consider the possibility that managers identify a HK\$5 million investment project that will return HK\$17.5 million in one day without risk. But, because the firm is distressed, management has to sell stock to raise the HK\$5 million. This investment does not make sense for the company because if shareholders invest the HK\$5 million, they will get nothing back if the company is sold. Both the HK\$5 million that shareholders invest and the HK\$12.5 million NPV (HK\$17.5 million – HK\$5 million = HK\$12.5 million) will ultimately go to the lenders. Instead of receiving HK\$32.5 million, lenders will get HK\$50 million and shareholders will lose HK\$5 million.

Still, without leverage, there would be no problems related to asset substitution or underinvestment. Regardless of risk, shareholders would always want to invest in positive NPV projects and reject projects with negative NPV projects. Debt provides shareholders with incentives to change investment policies, engage in asset substitution and turn down positive NPV projects. Writing contracts that protect lenders is no easy task, so lenders compensate by increasing the interest rate that they charge on funds which, in turn, adds more debt to the capital structure of a company.

8.1.3 Trade-off Theory

The **trade-off theory** of capital structure says managers choose the capital structure of their companies based on the trade-offs between benefits and costs. The aim of this target capital structure is to maximise the value of the company. This is shown in Exhibit 8.4.



► **EXHIBIT 8.4 Trade-off theory** (Source: data from Parrino 2016. Reproduced with permission of John Wiley & Sons.)

The idea that a small amount of debt financing may allow for a tax shield helps explain the trade-off theory. Since leverage is low, the chances that the company will get into trouble are low but the benefits are more significant and this helps raise the value of the firm. The challenge is to make sure that the company does not keep adding debt to the point where the cost associated with the next dollar that is borrowed equals the benefit, the point when the costs start to exceed the benefits, and more debt simply lowers firm value. The trade-off theory of capital structure says managers will increase debt until the costs and benefits are equal because this is the capital structure that maximises firm value.

8.1.4 Signalling Theory

Signalling theory, sometimes also called debt signalling theory, is a theory that links the future performance of a company's stock with announcements that are made with regard to its debt.

In general, announcements in which a company is taking on debt are viewed positively. This is because investors assume that the company is capable of paying interest of the debt.

Alternatively, investors are likely to view any announcement of a debt reduction negatively, seeing it as a sign that management lacks confidence in its ability to service its debt.

8.1.5 Pecking Order Theory

Another popular theory of how companies determine their capital structures is the **pecking order theory**, which recognises that different types of capital have different costs. The result is a pecking order in financing choices. Managers first choose the least expensive capital then move on to capital that costs more, and so on.

The cheapest funds are generated internally. Debt costs more to obtain but is still relatively inexpensive. Raising money by selling stock can be expensive. The out-of-pocket costs of selling equity are much higher than the comparable costs for bonds.

In addition, the filings required by regulatory agencies, are greater. Stock markets often react badly to announcements that companies are selling stock, because these sales can be seen as evidence that the company is not profitable enough to fund investments internally. In turn, lower stock prices reduce the value of shares and makes future stock issues more expensive.

Key Learning Point

The amount of financial leverage in the capital structure of a company affects the value of the firm.

► Knowledge Check Questions

Question 1

Identify which of the following is a part of the capital structure of a company.

- A Accounts receivable
- B Bonds
- C Prepaid expenses
- D Notes receivable

Question 2

Identify which is a benefit of using debt.

- A Lower cost
- B Availability of funds
- C Attracting top talent
- D Low cost of subsequent offerings

Question 3

Identify which of the following disadvantages emerge for a company from the inclusion of debt in the capital structure.

- A Lower transactions costs
- B Higher agency costs
- C Maximising cash flows
- D Lower taxes

Question 4

A company has a WACC of 11%, a cost of debt of 8% and a 12% cost of equity. Calculate the debt-to-equity ratio.

- A $\frac{1}{2}$
- B $\frac{1}{4}$
- C $\frac{1}{6}$
- D $\frac{1}{3}$

Question 5

Ming Motors has current pre-tax annual cash flows of HK\$1,000 and taxes of 35%. The discount rate for its cash flows is 12%. The company issues a HK\$1,500 bond and uses the proceeds to pay a one-time special dividend. Calculate Ming Motors' value after the debt issuance. Assume that the pre-tax annual cash flows are perpetual.

- A HK\$5,417
- B HK\$5,942
- C HK\$6,392
- D None of the above

► Knowledge Check Questions (*continued*)

Question 6

A company issues HK\$1 million in debt at a YTM of 9%. The debt trades at par. The company's marginal corporate tax rate is 35%. Calculate the present value of the tax savings in perpetuity.

- A HK\$11,025
- B HK\$20,475
- C HK\$350,000
- D HK\$227,500

Question 7

Identify which sources of financing are managers less likely to use according to the pecking order theory.

- A Retained earnings
- B New debt
- C New preferred stock
- D New common stock

Question 8

Identify the agency costs for a company.

- A Costs from conflicts of interest between shareholders and managers.
- B Costs that cannot be recovered.
- C Out-of-pocket costs that the company faces from being in financial distress.
- D Costs from passing on an alternative option.

8.2 KEY CONSIDERATIONS FOR ESTABLISHING A CORPORATE CAPITAL STRUCTURE

When making choices about the best capital structure for their companies, managers consider factors such as trade-off or a pecking order but also how their financing decisions will affect company operations. For instance, another important consideration is financial flexibility, which requires managers to keep sufficient funds on hand to take advantage of opportunities that arise or deal with unexpected challenges.

When a **net present value (NPV)** investment becomes possible, managers should be able to obtain financing but such financing might not be available at a reasonable cost at all times. It might, for instance, be difficult to convince investors to invest and a company may be forced to offer a higher return, making the project's NPV negative and causing the firm to pass up a good opportunity. Similarly, a company might not have the financial capabilities to deal with unexpected problems and this might result in higher cost.

Suppose some kind of natural catastrophe damages company facilities. Those facilities may be insured but by the time insurance payments are received, the company may be out of business. Companies should have access to enough financial flexibility to deal with problems that may arise.



Company managers are also concerned with the impact of **financial leverage** on the net income of the company. Most companies have to deal with fluctuations in profits and fixed-interest payments that magnify these fluctuations. Company managers dislike **volatility** in reported net income because it may hurt relationships with outside investors and increase the risk of defaulting on debt. Managers refer to these potential fluctuations as 'risk' and often work to manage the capital structure of the companies that they manage to minimise risk.

Another consideration is the impact that financial leverage has on earnings per share. Interest expenses associated with debt financing reduce the value of net income, but, depending on the value of the stock of a company, using debt instead of equity financing might increase the value of earnings per share. Managers worry about earnings per share because this affects the price of a company's stock.

Financial theory holds that managers should not be concerned with accounting earnings and focus on cash flows.

Managers also take into account the impact of their choices on their control over company operations. The choice between equity and debt financing affects how much control managers have. For instance, if the founders of a company own a majority of the stock, but resort to using equity financing to fund a new project, the capital raised may lead to growth in the value of the company but it may also require the company to issue more shares which could, in turn, reduce the shareholding and the amount of control that the founders have. To deal with this, the founders might prefer to tap into more expensive debt financing.

These are some examples of practical considerations. Ultimately, there is no formula to follow, but the key consideration for managers to choose a capital structure is to maximise the value of the company.

Illustrative Example 2

There are three practical considerations that concern accountants and managers when they make capital structure decisions:

1. *Financial flexibility*: Managers should minimise the cost of capital and, at the same time, ensure companies have flexibility to raise new capital quickly to deal with problems or tap into opportunities.
2. *Net income risk*: Increasing leverage also increases risk to net income and the risk of default.
3. *Earnings impact*: When companies use debt to finance projects, the interest payments reduce the accounting dollar value of net income. But, when using debt, no new shares are issued and this means earnings per share should increase when undertaking positive NPV projects. Financial theory suggests these effects should not matter but managers still consider them when making financing decisions.

Key Learning Point

In theory, if a positive net present value (NPV) investment becomes available, managers should be able to obtain financing for it. Managers are also concerned about the impact of financial leverage on the volatility of the firm's net income and do not like volatility in net reported income.

► Knowledge Check Questions

Question 9

State which of the following is not the concern of manager when they decide the company's capital structure.

- A Financial flexibility
- B Earnings per share
- C Accounting earnings
- D Net present value

Question 10

Two companies are in the same line of business and have similar amounts of operating leverage. Tang Ltd has no debt in its capital structure. Central Company Ltd has a capital structure that consists of 50% debt. Identify which of the following is true.

- A Central Company Ltd's total leverage exceeds that of Tang Ltd by 50%.
- B Central Company Ltd's sensitivity of net earnings to changes in earnings before interest and taxes is the same as Tang Ltd.
- C The sensitivity of earnings before interest and taxes to changes in sales of Central Company Ltd is the same as Tang Ltd.
- D Central Company Ltd has the same sensitivity of net earnings to changes in sales as Tang Ltd.



8.3 FINANCIAL PRODUCTS FOR FINANCING AND RISK MANAGEMENT

Companies have to both finance their operations (and fuel growth) while protecting themselves from risks. Over time, companies have found ways to finance their operations that do both.

In terms of finance, the three main approaches companies use to do both are hedging, speculation and arbitrage. The effective use of these tools can help companies protect their bottom lines and grow.

8.3.1 Hedging

Hedging is the act of entering into a contract with the goal of lowering existing exposure to risk, such as price fluctuations.

Most corporations and financial institutions become involved in forward markets for hedging purposes, rather than for speculation. In fact, most businesses and financial institutions are prohibited from engaging in speculation. Hedgers enter into derivative contracts to protect a short or long position in the underlying asset.

The most common approach to speculate as a form of risk control and to finance operations may be engaging in **future contracts** or **forwards**.



A future contract is a legal agreement to buy or sell a particular asset (typically stocks, commodities or foreign currency) at a predetermined price at some point in the future. Futures contracts are typically standardised and sold through futures exchanges.

A forward is similar to a future contract but it is standardised and is done between two parties to buy or sell a particular asset in the future. Forwards differ from future contracts in that they can be customised to any commodity, total amount and date of delivery. Forwards can be based on a cash basis or delivery basis.

Suppose that a Hong Kong company, Wing Ming Furniture Company Ltd, is owed EUR1 million and expects to receive this in a year. This means that Wing Ming has a naked position in euros. More specifically, it has euro foreign exchange exposure as it is long in the underlying asset, which is euros. While this position is the result of a commercial contract, the payoff is identical to a speculative purchase of a euro forward contract.

If the Hong Kong dollar depreciates then the company will receive more Hong Kong dollars when the debt is paid and it exchanges the proceeds at the future spot rate.

Alternatively, if the Hong Kong dollar appreciates, Wing Ming will receive less Hong Kong dollars.

This scenario of long euro exposure is common for Hong Kong companies that do business with Europe and invoice in euros. If a Hong Kong company, such as Wing Ming, exports to Europe and waits for its European customer to pay, it has long euro exposure. Similarly, short euro exposure arises when a Hong Kong company imports goods from Europe and then has to pay the euro invoice at some future time, such as in 90 days.

Most companies in this situation are not willing to speculate, because they are in business to manufacture and sell things, not to speculate on future exchange rates. In this regard, however, note that doing nothing and exchanging the euros when they are received is equivalent to engaging in currency speculation – regardless of whether it is to euros, Chinese yuan renminbi, US dollars, Japanese yen, British pounds or other currencies.

Some companies with extensive foreign currency exposure do find that it makes sense to hold foreign currency accounts and actively manage their foreign currency (forex) exposure, but most simply want to remove this exposure.

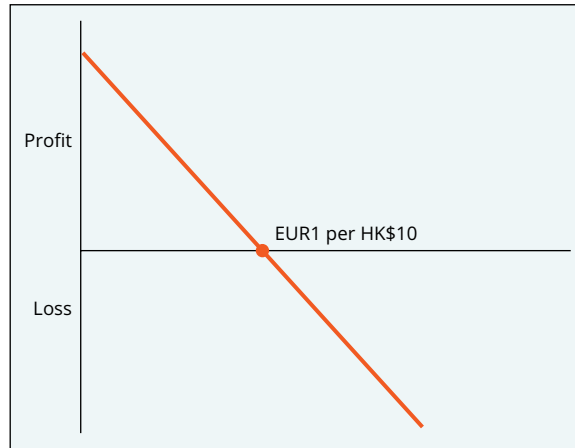
Removing a naked position is called **covering** (or hedging). In fact, the dictionary definition of hedging describes it as betting on both sides in order to remove risk.

The opposite of a firm's long position in a foreign currency, such as euros, is a short position in euros forward contracts, so the company needs to sell euros forward for Hong Kong dollars.

Before discussing the 'hedged' position, consider Exhibit 8.5, which shows a payoff diagram for a naked sale of foreign currency short position.

It is worth noting that the payoff for the naked short position is the complete opposite of the naked long position. If EUR1 million is sold forward for HK\$10 million and the Hong Kong dollar depreciates to EUR1.20, the forward contract loses money. In this case, the forward contract requires that EUR1 million be exchanged for HK\$10 million. On the other hand, to get the EUR1 million now requires HK\$12 million in the spot market which leads to an HK\$2 million loss.

Alternatively, if the Hong Kong dollar appreciates to EUR8, then buying EUR1 million to fulfil the forward contract requires only HK\$8 million, meaning a HK\$2 million profit.



► **EXHIBIT 8.5 Foreign currency short** (Source: data from Booth 2016.)

Thus, the gains and losses from the naked short position are the exact opposite of a naked long position and can be determined by using the following equation:

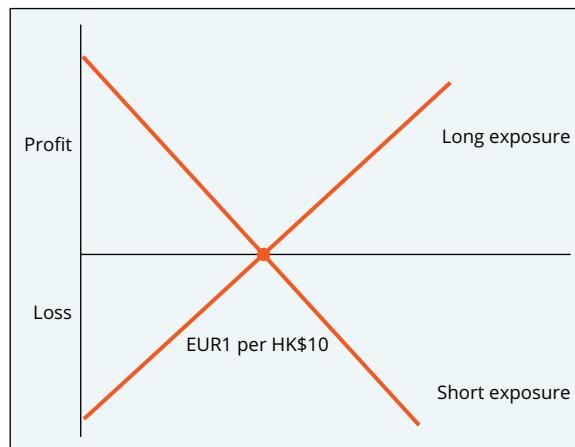
$$\text{Profit (loss) from short position} = (F - S_T) \times n$$

Where

- F = Forward price
- S_T = Spot price
- n = number of contracts

Now consider the company with a long euro exposure of EUR1 million that buys a forward contract to sell euros for Hong Kong dollars, thereby creating a EUR1 million short position.

That company's combined position is shown in Exhibit 8.6.



► **EXHIBIT 8.6 Foreign currency hedge** (Source: data from Booth 2016.)

If the Hong Kong dollar depreciates to EUR12, the company earns a profit of HK\$2 million on its underlying euro exposure but then loses HK\$2 million on the forward sale of euros.



Similarly, if the HKD appreciates to HK\$8, it loses HK\$2 million on the underlying euro exposure but gains HK\$2 million on the forward sale of euros.

Regardless of what happens to the value of the exchange rate, it is locked in (or covered or hedged) against any future fluctuations in the exchange rate, so the company is no longer exposed to exchange rate changes.

Financial futures work in similar fashion, except that companies hedge movements in interest rates rather than fluctuations in the value of a currency. For companies holding bond portfolios, the goal of hedging is to minimise exposure to **basis risk**.

The risk associated with a hedged position and linked to the fact that the hedged asset is not identical to the asset used as the hedge is basis risk. It may be impossible to create a perfectly hedged position because any changes to the price of the underlying asset may not be predictable, particularly not in relation to the asset being hedged. Forward contracts can be structured to minimise or eliminate basis risk because they are not standardised, unlike futures contracts. As a result, they can be tailored to specific circumstances, assets and maturity dates.

A final type of linear hedging contract is the swap, an agreement between two counterparties to exchange cash flows. This is a direct agreement between two parties and there is no formal exchange to guarantee performance. Such contracts require a dealer or an over-the-counter (OTC) market. They also involve credit risk.

Swaps have evolved, with the banks or swap dealers acting as intermediaries between the two counterparties to the swap. Now we use an **interest rate swap** as an example. Such a swap can be initiated by considering two hypothetical companies, A and B. Company A is a top-rated AAA company, whereas Company B is a lower-rated BBB company. Remember that AAA is the highest possible rating. BBB is a good bond rating but it is the lowest 'investment grade' rating. Company A wants to raise debt and pay a floating interest rate to finance short-term receivables and credit which earn short-term interest rates. Company B, on the other hand, wants long-term fixed rate financing. After approaching the capital market, Company A is quoted a rate of 10.8% for fixed rate financing or 0.25% over HIBOR (Hong Kong Interbank Overnight Rate). In contrast, Company B is quoted fixed rate financing at 12% and floating rate financing at a 0.75% spread over HIBOR.

The quotes look reasonable; the BBB-rated borrower is quoted a higher yield in both the fixed rate and the floating rate markets and would have borrowed at 12%. The AAA company would have borrowed at HIBOR plus 0.25%.

Why is the BBB company quoted a spread of 1.2% more than the AAA in the fixed rate market and only 0.5% more in the floating rate market?

Although Company A had an absolute financing advantage in fixed and the floating rate markets, it also had a comparative advantage in the fixed rate market, where its financing cost is 1.2% lower than Company B. Alternatively, Company B had a comparative advantage in borrowing in the floating rate market, paying just 0.5% more than A. To benefit from the difference in spreads, the companies can borrow where they have an advantage and swap payments.

This means Company A should use fixed rate financing to borrow HK\$50 million at 10.8%, which it does not need, and Company B should borrow another HK\$50 million through a floating rate note at HIBOR 0.75%. Company A signs a five-year swap contract and makes payments to Company B at HIBOR. Company A, in turn, gets a fixed 10.9% rate from Company B. The overall cost to Company A is HIBOR –0.1% because it pays HIBOR to

Company B and 10.8% on its own financing but gets 10.9% from Company B. As a result, Company A saves 0.35% over the quoted cost of capital of HIBOR + 0.25%.

For its part, Company B borrows funds at HIBOR + 0.75% and pays Company A based on a 10.9% fixed rate and receives payments based on HIBOR. The net effect is an 11.65% fixed rate, the 10.9% plus 0.75% for financing. Company B also saves 0.35% from its own quoted capital costs.

A 'fixed for floating' interest rate swap denominated in one currency is generally known as a plain vanilla interest rate swap. This name reflects the fact that it is the most common type.

In the example, both Company A and Company B gain 0.35% but in real life the gain for each company depends on negotiations. The principle is the same, however, even if the total spread of 0.7% can be divided in different ways.

This comparative advantage argument is basic in finance. Any company that is offered a good deal in floating rate funds should borrow them, regardless of whether they are needed, and then use swaps to exchange them for what is actually needed. The result is to lock in the advantage in financing. Many companies enter into swap arrangements.

The swap arrangement above is not without its problems. One main concern is that Company A, which is a highly rated AAA company, is relying on the payments from B, a BBB-rated company, to make its fixed rate payments. In the example, Company A takes on extra credit risk because if Company B defaults then Company A is stuck with its own 10.8% fixed rate financing and could also have to continue paying HIBOR to B. This risk is why early swaps were negotiated directly between counterparties and included 'set-off' rights, which meant that if either party defaulted, the other party would stop making payments. This practice eventually changed, with required payments becoming net payments. Instead of exchanging the total interest, parties exchange payments that account for the difference between the different rates and payments are typically made every six months.

8.3.2 Speculating

Companies occasionally speculate that the price of an asset or interest rates will move in one direction or another. In other words, they speculate with the expectation of generating returns or as a means to control risk.

Speculation, then, refers to the practice of trading a financial instrument involving high risk in the expectation of generating returns. The goal is to take advantage from fluctuations in the market.

Speculative opportunities are one reason why firms can hold cash. In broad strokes, there are four main reasons why companies hold cash:

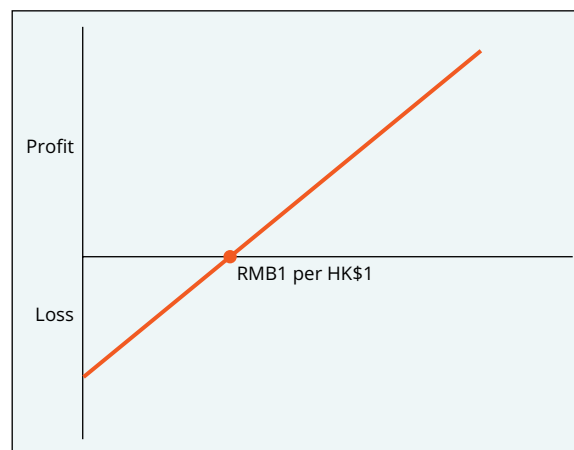
1. *Transactions motive*: Cash that is required for normal operations. For some companies, such as manufacturing firms, the actual 'cash' portion required will be minimal, but they will require money in their chequing account (called a 'current account' for companies) to handle bill payments and so on. Other firms, such as retail stores or restaurants, will need to have actual cash on hand, as well as money in their current account to cover cheques.
2. *Precautionary motive*: Cash that firms keep on hand to take care of unanticipated required outlays of cash, such as unexpected repairs on equipment.
3. *Finance motive*: This refers to the cash that firms will accumulate in anticipation of any major outlays, such as lump-sum loan repayments, dividend payments and so on.

4. *Speculative motive:* Firms may keep extra cash available to take advantage of unexpected 'bargains,' such as the opportunity to purchase raw materials very cheaply or to speculate in the price of future assets or protect themselves if they foresee rapid changes in the price of a particular asset.

Companies and investors can use future contracts and forwards to speculate.

For example, suppose the Hong Kong dollar is trading at par with the RMB (it is not but let's assume so to make the arithmetic easier) for much of 2018. This would mean that the exchange rate is HK\$1.00 for RMB1.00. Making it simpler, suppose both the spot and one-year forward rates are 1.00. A company may decide to speculate on the forward value of the RMB and can buy a RMB1 million forward that obligates the company to pay HK\$1 million and receive RMB1 million in a year. The company is speculating, which means it does not have any underlying RMB. This is a **naked position**, which means the company is exposed to changes in the value of the underlying asset, the RMB.

The company has agreed to buy RMB and sell Hong Kong dollars forward. This means it is *long* RMB and *short* HKD. 'Long' means the investor owns something. 'Short' means the investor owes something. The possible payoffs depending on various exchange rates are shown in Exhibit 8.7, with the horizontal axis showing the value of the Hong Kong dollar in terms of the RMB.



► **EXHIBIT 8.7 Speculative position** (Source: data from Booth 2016.)

Consider that the spot exchange rate for the RMB in one year depreciates to HK\$1.20 or appreciates to HK\$0.80. If the HK\$ appreciates, then the RMB depreciates relative to the Hong Kong dollar and vice versa, because the exchange rate is the ratio of HK\$ to RMB. If the HK dollar goes to 1.2, each RMB would be worth HK\$1.20, so the RMB has appreciated and the investor fulfils the forward foreign exchange contract. The company does this by exchanging HK\$1 million for RMB1 million and selling the RMB in the spot market to get HK\$1.2 million and earn HK\$200,000 in profit.

If the opposite happens, the company would incur a HK\$200,000 loss.

The key point is that the investor has to fulfil the contract regardless of the movement in the value of the underlying asset. A forward contract can generate both gains and losses and the payoff is linear in regard to the value of the underlying asset.

Exhibit 8.7 shows the possible payoff from the forward contract as a 45-degree line going through the forward rate. If the spot exchange rate in the future exceeds the forward rate by HK\$0.01, then the speculator earns HK\$0.01 profit for every HK dollar sold forward for RMB. The opposite is also true.

A forward contract is a legal contract between a speculator and a bank to exchange funds in the future at a price that is set today. However, it is also necessary to consider the cost of the contract. The forward rate represents the price for future delivery but, while the bank makes money from buying and selling forward contracts there is no immediate cash outlay; rather, the contract stipulates terms for delivery in the future.

For banks, this creates a problem. If the HK dollar depreciates to 1.2, the investor will execute the forward contract, buy more RMB and make HK\$200,000 in profits. On the other hand, if the RMB appreciates to 0.8, the bank faces credit risk and the investor may lose HK\$200,000 and, possibly, renege on the contract. This risk is also called counterparty risk.

Banks typically only sell forward contracts for legitimate business purposes and to companies with approved credit limits. Banks generally avoid selling forward contracts for speculative purposes.

As the example illustrates, speculation refers to the use of trading in assets or carrying out a financial transaction that carries significant risk but with the expectation of significant gain. The danger of speculating is that the risk of loss may be greater than the possibility of gain.

8.3.3 Arbitraging

Arbitrage refers to the concurrent purchase and sale of an asset in order to profit from an imbalance in its price in different markets or forms. Arbitrage exploits existing market inefficiencies, without which it would be impossible to engage in arbitrage.

Academics who study finance and markets have rediscovered the idea that there are limits to arbitrage and that speculation can be destabilising. The result is a field of financial thought known as behavioural finance, which suggests that investors make systematic mistakes in both their processing of information and their investing activities.

Another well-known multifactor asset pricing model is the arbitrage pricing theory (APT). The APT holds under very few assumptions, unlike the CAPM. In particular, the APT does not depend on the existence of an underlying market portfolio, and it allows for the possibility that several types of risk may affect security returns.

In fact, APT is based on the no-arbitrage principle, which states that two otherwise identical assets cannot sell at different prices.

Development of the formal APT assumes that asset returns are linearly related to a set of indexes, which proxy risk factors that influence security returns. The underlying risk factors in the APT represent broad economic forces, which are unpredictable. The APT can be expressed as in the equation below:

$$ER_i = a_0 + b_{i1}F_1 + b_{i2}F_2 + \dots + b_{in}F_n$$

where

- ER_i = expected return on security i ,
- a_0 = expected return on a security with zero systematic risk,

- b_i = sensitivity of security i to risk factor 1 to n ,
- F = risk premium for risk factor 1 to n .

The equation shows that the risk associated with a security is based on its sensitivity to basic economic factors, while expected return increases proportionately to this risk. The sensitivity measures (b_i) play a role similar to beta in the CAPM, because they measure the relative sensitivity of a security's return to a particular risk premium.

In fact, the APT would 'collapse' into the CAPM if there were only one risk factor (market returns) influencing security returns. The main problem with the APT is that the factors are not specified ahead of time. In fact, APT does not even specify the number of risk factors that exist, or state which factors will be the most important.

As a result, these factors, as well as their relative importance, must be identified empirically.

Most empirical evidence suggests that three to five factors influence security returns and are priced in the market. These five factors are:

1. changes in expected inflation;
2. unanticipated changes in inflation;
3. unanticipated changes in industrial production;
4. unanticipated changes in the default-risk premium; and
5. unanticipated changes in the term structure of interest rates.

Another approach to arbitrage, which can be used in securities, is the use of options.

Financial options are derivative securities whose value is derived from the value of another asset, called the underlying asset. Owners of options have the right, but not the obligation, to buy or sell another asset at or before a specific date known as the exercise date and for a specified price called the exercise price or strike price.

The value of an option is derived from the value of an underlying asset. Assume an investor owns an option to buy one share of a stock for US\$150 per share and the exercise date is today. If the investor does not exercise the option today, it expires and become worthless. If the price of the stock is less than US\$150 per share, there is no need to exercise the option. This is because exercising the option would mean the investor is paying US\$150 for something that can be bought for a cheaper price on the market. If the price of the stock is exactly US\$150, there is also no benefit from exercising the option.

Alternatively, if the price of the stock goes above US\$150, the investor would benefit from exercising the option. The investor may not necessarily want to own the stock but it could buy it for US\$150 and sell it at a profit almost immediately. This means that the value of the option is, in effect, the difference between the market price of the stock and the exercise price of the option. For instance, if the stock is trading for US\$160 per share, then the option is worth US\$10 (US\$160 – US\$150 = US\$10). And if the stock is trading for US\$170, then the option is worth US\$20 and so on. The relation between the value of an option and the value of the underlying asset is known as the **option payoff function**.

The option in the example above is a **call option**. Call options give owners the right to 'call' or buy the underlying asset.

When the exercise price is US\$150 and the value of the underlying the stock is also US\$150, then the call option is worth US\$0 and the owner of the option can let the option expire, which limits the downside to US\$0. If the price of the stock goes higher than the strike price, then the value of the call option goes up just as much.

At the instant before it expires, the value of a call option equals US\$0 if the underlying asset is worth less than or equal to the exercise price. At the same time, the value of the seller's position equals US\$0 if the value of the underlying asset is less than or equal to the exercise price.

Any gains for the investor, or the owner or buyer of the option, represent a loss for the seller. In the example above, if the stock is trading at US\$160 when the option expires then the owner of the option earns a profit of US\$10 but the seller is now obligated to sell shares that are worth US\$160 for US\$150, a US\$10 loss per share.

At first glance, it does not seem to make sense for anyone to sell call options but sellers collect a fee known as a call premium on the options they sell and these fees make selling options a profitable venture if the price of the underlying asset is close to or below the strike price at the strike date. A **call premium** is a lot like an insurance premium, except that instead of insurance the seller pays out in case a specific event happens, in this case the value of the underlying asset going above the exercise price.

The owner of a **put option**, in turn, has the right to sell an underlying asset at a prespecified price. The payoff in a put option is similar to that of a call option but in reverse. Owners of put options benefit if the price of the underlying asset is below the exercise price.

The value of a put option to the owner equals US\$0, if the value of the underlying asset is greater than or equal to the exercise price at the time of expiry. On the other hand, the value to the seller of a put option equals US\$0 if the value of the underlying asset is greater than or equal to the exercise price.

There are two main types of options. Some, known as European options, can be exercised only on the expiration date. American options, however, can be exercised at any point before the expiration date and are the most commonly traded on stock exchanges. These are the most common types of options but there are other, more exotic, kinds. Options known as Bermudan options can be exercised only on specific dates during the life of option.

Illustrative Example 3

Another subsidiary, Lantau Ltd has just made a sale to a British customer. The sale was for a total value of GBP135,000 and is to be paid 60 days from now.

Lantau Ltd is concerned that the British pound will depreciate against the US dollar (and in turn against the HK dollar) and has management plans to hedge.

The company's bank informs management that the spot rate is US\$1.8133/GBP and the 60-day forward rate is US\$1.7864/GBP.

If Lantau Ltd sells its pounds receivable at the forward rate, what is the dollar value of its receivables? If it did not enter into a forward contract and the spot rate 60 days later was US\$1.7635/GBP, how much did the company lose by not hedging?

Illustrative Example 3 (continued)

Calculate the impact on Lantau Ltd's profitability.

Amount received by Lantau Ltd by selling at the forward rate:

$$= \text{GBP}135,000 \times \text{US}\$1.7864 / \text{GBP} = \text{US}\$241,164$$

Amount received by Lantau Ltd by selling at the spot rate 60 days later:

$$= \text{GBP}135,000 \times \text{US}\$1.7635 / \text{GBP} = \text{US}\$238,072.50$$

$$\text{Loss incurred by not hedging} = \text{US}\$241,164 - \text{US}\$238,072.50 = \text{US}\$3,091.50$$

Key Learning Point

Hedging is the act of entering into a contract with the goal of lowering existing exposure to risk, such as price fluctuations. Speculation, then, refers to the practice of trading a financial instrument involving high risk in the expectation of generating returns. Arbitrage refers to the concurrent purchase and sale of an asset in order to profit from an imbalance in its price in different markets or forms.

► Knowledge Check Questions**Question 11**

Identify which of the following statements about the mechanics of futures contracts is false.

- A** All futures contracts are made with a futures exchange, not with an individual.
- B** Oil can be a kind of underlying asset of a future contract.
- C** Future contracts can be used to speculate.
- D** All futures contracts are marked to market each month, as the value of the contract changes.

Question 12

Suppose ABC Entertainment Ltd pays a fixed rate of 9.5% while DEF Ltd pays a floating rate of HIBOR + 5%. The two companies enter into a swap. ABC Entertainment Ltd agrees to pay HIBOR to DEF Ltd and DEF Ltd agrees to pay 9.5% to ABC Entertainment Ltd. Calculate the net paying interest rate for ABC Entertainment Ltd.

- A** HIBOR + 1%
- B** HIBOR
- C** 12%
- D** HIBOR + 0.5%

► **Knowledge Check Questions (continued)**

Question 13

ABC Entertainment Ltd borrows at a fixed rate of 9.5% or a floating rate of HIBOR + 1%. DEF Ltd borrows at a fixed rate of 12% or a floating rate of HIBOR + 5%. ABC Entertainment Ltd is currently borrowing at its fixed rate. DEF Ltd is borrowing at its floating rate. So, the two companies enter into a swap. ABC Entertainment Ltd agrees to pay HIBOR to DEF Ltd and DEF Ltd agrees to pay 9.5% to ABC Entertainment Ltd. Calculate how much ABC Entertainment Ltd and DEF Ltd could save through the swap in percentages.

- A** 1% and 1%
- B** 0.5% and 0.5%
- C** 1.5% and 0.5%
- D** 2% and 0%

8.4 USE OF FINANCIAL EXPERTS FOR FUND RAISING AND OPTIMISING CAPITAL STRUCTURE

After setting the capital budget, management has to decide how to finance assets. The easiest approach is to use internally generated funds, meaning earnings minus any capital used on dividends or to repurchase stock. In reality, few firms finance their projects in this way because most firms have more projects than they can fund internally. This means management must look for other sources of funding such as loans, long-term debt or issuing additional equity.

Companies often rely on financial experts, both internal and external, to determine the ideal mix of internal and external funds to use, to raise funds for external funds and to optimise the capital structure of the company. Financial experts can help put together a financing plan, working in conjunction with the board and top management.

External experts are particularly helpful in providing advisory services on transactions that companies undertake to raise funds.

Experts can help companies determine the best approach to fundraising, from tapping venture capital investors to issuing stock in public markets.

Two important participants in the direct financial markets are investment banks and money centre banks. Investment banks help companies sell debt or equity and provide broker and dealer services as well as other, more traditional, banking services. Money centre banks are large commercial banks that offer both traditional and investment banking services throughout the world.

Other than investment banks and money centre banks, companies could also hire fundraising consultants whose end-goal is to raise money for a company. The scope of work

typically includes the production of marketing materials. These include documents such as a teaser or executive summary, a business plan and/or placement memorandum, and management presentation. Other deliverables include financial projections and models. Organisation and management of a data room for due diligence are often included in the scope of work. It is not unusual for a fundraising consultant to help management with business development initiatives to make the company more attractive to investors. Consultants could also define the prospective investor universe, initiate investor outreach, review term sheet and documentation.

These financial institutions and financial experts typically participate in the origination, underwriting and distribution of funds.

The process of preparing a security issue for sale is called origination. Investment bankers work with companies to determine the feasibility of specific fundraising projects and amount and type of capital that needs to be raised during the origination phase. The investment banker helps secure a credit rating, if it is needed. The investment banker also determines the sale date, obtains legal clearances to sell the securities and gets the securities printed. If securities are to be sold in the public markets, the issuer must also file a registration statement with the appropriate regulator.

The process by which the investment banker helps the company sell a new security issue is called underwriting. The most common type of underwriting arrangement is called firm commitment underwriting. This is where the investment banker buys the new securities from the company that issues them and sells them to investors. The investment banker buys the entire security issue at a fixed price, which means the issuing company is guaranteed that price. By doing this, the investment banker is taking the risk that the actual price at which the securities are sold is less than the price that is paid to the company. Since issuing companies typically need a certain amount of money to fund specific projects or for operations, and receiving less than this amount would be a serious problem, it is typically preferred by financial managers to have their new security issues underwritten on a firm-commitment basis.

Once the securities have been bought from the issuer, the investment bankers immediately offer to resell individual securities to institutional investors and the public at a specified offering price. The underwriters hope to be able to sell the offering at the market-clearing price, which is the price that will allow the entire security issue to be sold during the first day of sale. There is considerable risk involved with underwriting because it is difficult to estimate the price that will clear the market. The investment bank will suffer a financial loss if it has to sell the securities at a price below the price that it paid to the issuing company.

The process of marketing and reselling securities to investors is called distribution. Security prices can take large, unexpected swings, so a quick resale of all the securities is important to accomplish. To achieve this aim, the underwriters sell the securities, often by forming sales syndicates, which consist of a number of different investment banking firms. The syndicate is disbanded if the securities are not sold in a few days, and individual syndicate members may sell the unsold securities at whatever price they can.

8.4.1 Financing Plan

The goal of the **financing plan** is to determine the amount of external funding a company needs. There are three parts to a financing plan. The first is the amount of external funding needed and any sources of funds available. The second part is the capital structure for

the company that management desires. The third part sets out the company's payout policy, which determines the amount of capital the company may be able to tap for new investment projects.

An important point to note here is that the investment (capital budgeting) and financing decisions cannot be made independently – they must be considered together. The reason is that when management makes an investment decision, it must already have identified a source of funds to pay for the investment.

8.4.2 Making Capital Structure Decisions

Managers make financing decisions in light of the practical considerations of running the operations of the business. Managers have to make sure they have enough financial resources to tap into opportunities or overcome challenges.

Without enough financial flexibility, an unforeseen problem might end up being more costly than it needs to be or even put the company out of business. As discussed in section 8.2, financial leverage can have an impact on the volatility of the net income of a company and managers do not like volatility because it may impact the relationship with investors. Financial leverage also impacts earnings per share as well as the amount of control that shareholders have.

There is no one way to make financing decisions. Ultimately, these will be based on the needs of specific companies at a particular point in time and plans for projects going forward. Nevertheless, it is safe to say that the ultimate objective of shareholders and good managers is to choose a capital structure that maximises the value of the firm.

Illustrative Example 4

The optimal capital structure is influenced by many factors but companies in the same industry tend to adopt similar capital structure as they react to similar competitive dynamics.

According to a study by credit rating agency Moody's, global companies in non-financial and non-utility sectors with Moody's Baa (lowest investment grade band) and Ba (highest non-investment grade band) ratings had the following median debt/book capitalisation ratio as of December 2016.

	Baa	Ba
Global aggregate	46.4%	55.7%
Aerospace and defence	60.9%	62.0%
Automotive	45.0%	55.1%
Chemical	46.2%	57.5%
Consumer products	45.6%	55.0%
Energy	44.2%	45.5%
Healthcare	43.4%	53.9%
Manufacturing	43.1%	47.3%
Media	52.2%	73.5%



Illustrative Example 4 (continued)

	Baa	Ba
Metal and mining	35.9%	41.9%
Retail and distribution	55.1%	64.2%
Services	49.3%	57.5%
Telecommunications	54.7%	56.4%
Transport	41.9%	61.6%

For each of the two rating categories shown here, it is clear that some industries carry higher leverage, such as aerospace and defence, retail and distribution, telecommunications and media. Some industries carry lighter leverage, such as metal and mining, manufacturing, healthcare and energy.

Therefore, as each company's management tries to determine the optimal capital structure in financial planning, they should keep in mind the benchmark the rest of the companies in their global industry may be following.

Key Learning Point

The goal of the financing plan is to determine how much external funding the firm needs. The financing plan has three components: the amount of external funding and the sources of funds, the desired capital structure of the firm and the company's payout policy.

► Knowledge Check Questions

Question 14

Identify which of the following is a characteristic of the optimal capital structure of a company.

- A** Minimises the cost of financing projects.
- B** Minimises interest payments.
- C** Maximises value of the company.
- D** A and C.

Question 15

Identify which of the following is true of debt financing.

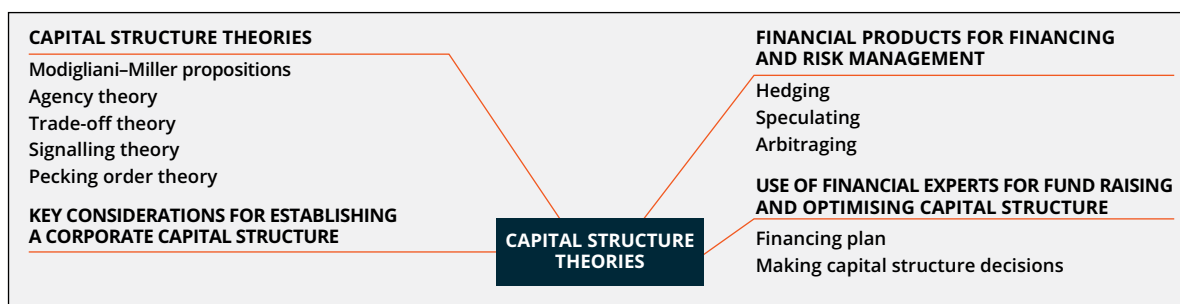
- A** Causes managers to take on riskier projects to make interest payments.
- B** Is more expensive than issuing equity.
- C** Allows companies to make discretionary interest payments.
- D** Limits the ability of managers to waste money.



SUMMARY

- The capital structure policies at firms reflect trade-offs between the benefits and the costs of capital and debt.
- The appropriate capital structure for a particular firm depends on that firm's characteristics.
- The Modigliani and Miller (M&M) propositions provide essential insights into the mechanisms through which financing decisions affect financing costs.
- M&M's Proposition 1 says that capital structure decisions in a company will not affect the value of the company if some conditions are met.
- M&M's Proposition 2 says that the cost of the common stock of a company is related to the debt-to-equity ratio.
- Agency theory attempts to explain the dynamic between principals and agents in business transactions.
- The trade-off theory of capital structure says managers choose a specific target capital structure by balancing the benefits and costs.
- Signalling theory, sometimes also called debt signalling theory, is a theory that links the future performance of a company's stock with announcements that are made with regard to its debt.
- The pecking order theory points out that different types of capital have different costs. This naturally leads to a pecking order in financing choices.
- Managers worry about how financing decisions impact the practical issues of managing a business.
- Managers are also concerned about the impact of financial leverage on the volatility of the firm's net income.
- Managers dislike volatility in reported net income as it can cause conflict with outside investors, who are adverse towards unpredictable earnings.
- Hedging is the act of entering into a contract with the goal of lowering existing exposure to risk, such as price fluctuations.
- Speculation refers to the practice of trading a financial instrument involving high risk in the expectation of generating returns. The goal is to take advantage from fluctuations in the market.
- There are four main reasons for companies to hold cash: transactions, precaution, financing and speculation.
- Arbitrage refers to the concurrent purchase and sale of an asset in order to profit from an imbalance in its price in different markets or forms. Arbitrage exploits existing market inefficiencies, without which it would be impossible to engage in arbitrage.
- The goal of the financing plan is to determine how much external funding the firm needs. The financial plan has three components: the amount of external funding and the sources of funds, the desired capital structure and the payout policy.

MIND MAP



LIST OF FORMULAS

Firm's enterprise value

$$V_{\text{Firm}} = V_{\text{Assets}} = V_{\text{Debt}} + V_{\text{Equity}}$$

Weighted average cost of capital (WACC)

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt pretax}} (1 - t) + x_{\text{ps}} k_{\text{ps}} + x_{\text{cs}} k_{\text{cs}}$$

Weighted average cost of capital (WACC) – simplified (assuming no taxes and preference shares)

$$\text{WACC} = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{cs}} k_{\text{cs}}$$

Proportion of debt and equity in a firm's capital structure

$$x_{\text{Debt}} = \frac{V_{\text{Debt}}}{V_{\text{Debt}} + V_{\text{cs}}}$$

$$x_{\text{cs}} = \frac{V_{\text{cs}}}{V_{\text{Debt}} + V_{\text{cs}}}$$

Required return on common stock

$$k_{\text{cs}} = k_{\text{Assets}} + \left(\frac{V_{\text{Debt}}}{V_{\text{cs}}} \right) (k_{\text{Assets}} - k_{\text{Debt}})$$

Gains and losses from naked short position

$$\text{Profit (loss) from short position} = (F - S_T) \times n$$

Where

- F = Forward price
- S_T = Spot price
- n = number of contracts

Arbitrage pricing theory

$$ER_i = a_0 + b_{i1}F_1 + b_{i2}F_2 + \dots + b_{in}F_n$$

Where

- ER_i = expected return on security i ,
- a_0 = expected return on a security with zero systematic risk,
- b_i = sensitivity of security i to risk factor 1 to n ,
- F = risk premium for risk factor 1 to n .

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. The capital structure of a company is the mix of debt and equity used to finance activities and accounts receivable is an asset.

Answer B is correct. The capital structure of a company is the mix of debt and equity used to finance activities. The mix typically includes common stock and, frequently, debt and preferred stock.

Answer C is incorrect. The capital structure of a company is the mix of debt and equity it uses to finance activities and prepaid expenses are assets.

Answer D is incorrect. A company's capital structure is the mix of debt and equity used to finance activities and notes receivable is an asset.

Question 2

Answer A is correct. The most important benefit from including debt in a firm's capital structure is that firms would be able to deduct interest payments for tax purposes but would not be able to deduct dividend payments for tax purposes. It is therefore less costly to distribute cash to investors through interest payments than through dividends.

Answer B is incorrect. Increased availability of funds can also come from other sources such as equity or working capital.

Answer C is incorrect. There is no direct relationship between debt burden and attraction to management talent.

Answer D is incorrect. The cost of subsequent security issuances depends on many market and company factors at that time.

Question 3

Answer A is incorrect. Lower transactions costs from use of debt are advantages.

Answer B is correct. The costs of debt include agency costs. Agency costs are associated with actions of management and shareholders that may act in their own interests as opposed to what is best for the company.

Answer C is incorrect. Debts actually force management to focus better on maximising firm's cash flows to service debts. This increased focus on cash flow is an advantage.

Answer D is incorrect. Lower taxes associated with debt shield are advantages to the company.

Question 4

Answer D is correct.

$$WACC = x_{\text{Debt}} k_{\text{Debt}} + x_{\text{Equity}} k_{\text{Equity}} = x_{\text{Debt}} \times 0.08 + x_{\text{Equity}} \times 0.12 = 11\%$$

Also, $D + E = 1$.

Therefore, substituting gives:

$$x_{\text{Debt}} \times 0.08 + (1 - x_{\text{Debt}}) \times 0.12 = 0.11$$

Solving for x_{Debt} gives you $\frac{1}{4}$, which means that x_{Equity} must be $\frac{3}{4}$. Therefore, the debt-equity ratio is $\frac{1}{3}$.

Question 5

Answer B is correct.

$$V_{\text{Firm}} = [CF \times (1 - t)] / i = [1,000 \times (1 - 0.35)] / 0.12 = \text{HK\$}5,417$$

$$\text{PV of tax shield} = 1,500 \times 0.35 = \text{HK\$}525$$

$$V_{\text{Firm}} = V_{\text{Unleveraged}} + \text{PV of the tax shield} = 5,417 + 525 = \text{HK\$}5,942$$

Question 6

Answer C is correct.

$$V_{\text{Tax-savings debt}} = D \times t = 1,000,000 \times 0.35 = \text{HK\$}350,000$$

Question 7

Answer A is incorrect. Retained earnings are first to be used within the pecking order.

Answer B is incorrect. New debt is second to be used within the pecking order.

Answer C is incorrect. New preferred stock is more likely to be used before new common stock.

Answer D is correct. Pecking order theory states managers first use retained earnings, seen as the least expensive form of capital, to finance projects. They might then move on to debt then to externally raised equity, which is the most expensive.

Question 8

Answer A is correct. Managers and shareholders of a firm often behave in ways that reduce a firm's value when the firm becomes financially distressed. The resulting costs are a type of agency cost.

Answer B is incorrect. This is sunk cost.

Answer C is incorrect. This is financial distress cost.

Answer D is incorrect. This is opportunity cost.

Question 9

Answer A is incorrect. Managers have to make sure sufficient financial resources are retained to tap unexpected opportunities and to overcome unexpected problems.

Answer B is incorrect. Earnings per share will affect the stock price, that is, the firm's value.

Answer C is correct. Cash flows are what really matter rather than accounting earnings.

Answer D is incorrect. NPV will determine whether managers should finance for that project.

Question 10

Answer A is incorrect. The degree of total leverage of Central Company Ltd exceeds that of Tang Ltd, but the extent of the difference is not dependent on the amount of debt. It depends on the amount of interest expense. In the case of financial leverage, it is the interest that acts as a fulcrum.

Answer B is incorrect. Central Company Ltd's net earnings are more sensitive to changes in earnings before interest and taxes than Tang Ltd.

Answer C is correct. Central Company Ltd's sensitivity to earnings before interest and taxes from changes in sales is the same as Tang Ltd.

Answer D is incorrect. Central Company Ltd has greater sensitivity of net earnings to changes in sales than Tang Ltd.

Question 11

Answer A is incorrect. Futures contracts are typically standardised and sold through a futures exchange.

Answer B is incorrect. Oil is a commodity that usually trades with future contract.

Answer C is incorrect. Both forward and future contracts can be used to speculate.

Answer D is correct. All future contracts are marked to market each day, as the value of the contract changes.

Question 12

Answer B is correct.

$$\text{Net interest rate (ABC)} = \text{HIBOR} + 9.5\% - 9.5\% = \text{HIBOR}$$

Question 13

Answer A is correct.

Net interest rate (ABC) = HIBOR

ABC saving = (HIBOR + 1%) – HIBOR = 1%

Net interest rate (DEF) = (HIBOR + 1.5%) + 9.5% – HIBOR = 11%

DEF saving = 12% – 11% = 1%

Question 14

Answer A is incorrect. Minimising the cost of financing the firm's projects is only one of the objectives.

Answer B is incorrect. Minimising interest payments to creditors is not the objective of optimal capital structure.

Answer C is incorrect. Maximising overall value of the firm is only one of the objectives.

Answer D is correct.

Question 15

Answer A is incorrect. A manager is less likely to take on riskier projects in order to reduce credit risk.

Answer B is incorrect. Debt is less expensive than issuing equity.

Answer C is incorrect. Managers have no discretionary power over making interest payments.

Answer D is correct. Debt limits the ability of managers to waste shareholders' money.

 EXAM PRACTICE**QUESTION 1**

Identify which of the following is true according to trade-off theory.

- A.** How much debt a company has is irrelevant.
- B.** Debt is used only as a last resort.
- C.** Debt is not used when tax rates are high.
- D.** The level of debt of companies is optimal.

QUESTION 2

Identify which of the following leads to the asset substitution problem.

- A.** To the detriment of bondholders, managers substitute more risky assets for less risky assets.
- B.** To the detriment of bondholders, managers substitute less risky assets for more risky assets.
- C.** To the detriment of equity holders, managers substitute more risky assets for less risky assets.
- D.** To the detriment of equity holders, managers substitute less risky assets for riskier assets.

QUESTION 3

Identify which of the following creates a problem of underinvestment in a company that is financially distressed.

- A. The value from an investment in a positive NPV project could go to debt holders rather than equity holders.
- B. The value from an investment in a positive NPV project is likely to go to equity holders rather than debt holders.
- C. Managers choose to invest in negative NPV projects to reduce their own risk.
- D. Higher risk makes issuing equity difficult.

QUESTION 4

Assuming all Modigliani and Miller assumptions apply, the weighted average cost of capital for a company is 15%. Calculate the current cost of equity capital for the company if the proportion of debt to the value of the firm is 0.5 and the cost of debt is 10%.

QUESTION 5

Ming Processed Foods Ltd is valued at HK\$500 million. Ming plans to repurchase HK\$100 million of its by issuing a non-maturing debt at an annual interest rate of 10%. The company also faces a 30% marginal tax rate. Calculate the value of the company after recapitalisation assuming most M&M assumptions hold except for the taxes.

QUESTION 6

A company has a WACC of 12.6% and faces a 40% marginal tax rate. The company also has HK\$250 million of outstanding debt at 9% and HK\$750 million of equity outstanding. Calculate the expected return on the equity.

QUESTION 7

Use the information in the following table to identify the proportion of debt a company should use.

Benefit or (cost)	No debt	25% debt	50% debt	75% debt
Tax shield	HK\$0	HK\$10	HK\$20	HK\$30
Agency cost	– HK\$10	– HK\$5	– HK\$5	– HK\$20
Financial distress cost	– HK\$1	– HK\$3	– HK\$10	– HK\$10

QUESTION 8

There is an order of financial sources for managers who subscribe to the pecking order theory of financing. Describe the order and evaluate it by observing the costs of each source relative to the costs of other sources.

QUESTION 9

A computer manufacturer in Canada sells a large system to a German firm, which will be delivered in six months. The seller expects a payment of US\$1.5 million upon delivery. The spot exchange rate is CAD1.15 per US\$ while the six-month forward rate is CAD1.25 per USD.

- (a) Identify the risks the seller faces.
- (b) Describe how the seller can hedge its currency risk.

- (c) Determine the profit or loss on the hedge if the spot rate in six months is:
 - (i) CAD0.75 per USD
 - (ii) CAD1.50 per USD
- (d) Given your answers to (c)(i) and (ii), explain why or why not the seller should hedge.

QUESTION 10

Central Grocers Inc produces annual cash flows of HK\$175,000, which are expected to continue indefinitely. The company is financed entirely with equity capital at an annual cost of 12%. Management is considering borrowing HK\$400,000 at an annual interest rate of 6% to repurchase HK\$400,000 of the company's outstanding stock. (You can assume that the debt will be outstanding into the indefinite future.)

- (a) Calculate the total value of Central Grocers' stock before the stock repurchase.
- (b) Under Modigliani and Miller's Proposition 1, calculate the value of the total claims on the company's assets after the stock repurchase.
- (c) Calculate the rate of return on common stock required by investors after the repurchase.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. The amount of debt a company has represents costs. It is relevant.

Answer B is incorrect. Debt should be used only if its benefits outweigh the costs.

Answer C is incorrect. Debt will be used if a company's after tax cost is less than the benefit of carrying debt.

Answer D is correct. Companies have an optimal level of debt. The trade-off theory suggests there is a trade-off between a tax shield from interest on debt and the costs of financial distress. This trade-off leads to an optimal range of debt.

QUESTION 2

Answer A is correct. Managers substitute more risky assets for less risky ones to the detriment of bondholders.

Answer B is incorrect. Managers substitute more risky assets.

Answer C is incorrect. It is done to the detriment of bond holders.

Answer D is incorrect. Managers substitute more risky assets for less risky ones to the detriment of bondholders.

QUESTION 3

Answer A is correct. The value of investing in a positive NPV project is likely to go to debt holders, not equity holders.

Answer B is incorrect. The value from investing in a positive NPV project may go to debt holders.

Answer C is incorrect. Management would not invest in negative NPV projects.

Answer D is incorrect. It has nothing to do with issuing equity.

QUESTION 4

Since $V_{\text{Debt}} / V_{\text{Firm}} = 0.5$, $V_{\text{Debt}} / V_{\text{cs}} = 0.5 / 0.5 = 1$

Using the formula:

$$k_{\text{cs}} = k_{\text{Assets}} + (V_{\text{Debt}} / V_{\text{cs}}) \times (k_{\text{Assets}} - k_{\text{Debt}})$$

$$= 0.15 + (1) \times (0.15 - 0.10) = 0.2, \text{ or } 20\%$$

QUESTION 5

Ming Processed Foods Ltd will be worth HK\$500 million plus the present value of the tax shield, which is $\text{HK\$}100,000,000 \times 0.3 = \text{HK\$}30,000,000$. In turn, Ming Processed Foods Ltd will be worth HK\$530,000,000 after recapitalisation.

QUESTION 6

Using the WACC formula when taxes are included:

$$\text{WACC} = (1 - t) \times k_{\text{Debt}} \times (V_{\text{Debt}} / V_{\text{Firm}}) + k_{\text{cs}} \times (V_{\text{cs}} / V_{\text{Firm}})$$

It is possible to solve for the missing variable:

$$0.126 = (1 - 0.4) \times 0.09 \times [250 \text{ million} / (250 \text{ million} + 750 \text{ million})] + k_{\text{cs}} \times [750 \text{ million} / (250 \text{ million} + 750 \text{ million})]$$

$$k_{\text{cs}} = 0.15, \text{ or } 15\%$$

QUESTION 7

Adding the cost and benefits for each proportion of debt it is possible to see:

Benefit or (cost)	No debt	25% debt	50% debt	75% debt
Total cost/benefit	-HK\$11	HK\$2	HK\$5	HK\$0

Therefore, this firm can maximise firm value by choosing a 50% debt capital structure.

QUESTION 8

According to the pecking order theory, the costs, from lowest to highest, are:

- Internally generated funds (or retained earnings) – This will actually be the cheapest source in this list.
- New issue debt – This will be more costly than internally generated funds, which are the cheapest option in the pecking order, but relatively inexpensive compared to other sources in this list.
- New issue equity – This will be the most expensive source in this list.

Managers who subscribe to the pecking order theory appear to choose the cheapest sources of financing before moving on to more expensive sources.

QUESTION 9

- The seller faces two main risks:

- (1) The German firm may not pay: credit risk.
- (2) The Canadian dollar may have appreciated relative to the US dollar (for the same number of US\$, the seller will receive fewer Canadian dollars): currency risk.

- (b) The seller will be long US dollar in 6 months when the German firm pays. Therefore, the seller could sell US dollar (or conversely buy Canadian dollar) using a 6-month forward contract to counter its long US dollar and short Canadian position. This will fix the CAD/USD exchange rate and eliminate the currency risk.
- (c) (i) In six months, the seller will convert its US\$1.5 million into Canadian dollar using the forward rate and receive $1.5\text{m} \times 1.25 = \text{CAD}\1.875 million. If they had not hedged, they could have converted at the spot and obtained $1.5\text{m} \times 0.75 = \text{CAD}\1.125 million. Therefore, the profit on the hedge is: $(1.25 - .75) \times 1.5 \text{ million} = \text{CAD}\$750,000$.
- (ii) In six months, the seller will convert its US\$1.5 million into Canadian dollar using the forward rate and receive $1.5\text{m} \times 1.25 = \text{CAD}\1.875 million. If they had not hedged, they could have converted at the spot and obtained $1.5\text{m} \times 1.5 = \text{CAD}\2.25 million. Therefore, the loss on the hedge is: $(1.25 - 1.5) \times 1.5\text{million} = -\text{CAD}\$375,000$.
- (d) Yes, it should hedge. At the time the seller enters the forward contract (today) they have no idea what the exchange rate will be in the future. The forward contract is the market's best estimate of exchange rate in the future. By not hedging, the seller management would be betting that they can forecast exchange rate better than market forces. This would be difficult to justify if their bet turned out to be horribly wrong in six months' time.

QUESTION 10

- (a) Using the perpetuity formula, we can compute the value of Central Grocers' stock before the repurchase to be $\text{HK}\$175,000 / 0.12 = \text{HK}\$1,458,333$ if the cash flow is expected to be produced into the infinite future.
- (b) Under M&M Proposition 1, the value of the firm will be the same before and after the repurchase. Therefore, since the value of the debt claim will be $\text{HK}\$400,000$ at the time of the transaction, the value of the outstanding equity after the repurchase must be $1,458,333 - 400,000 = 1,058,333$.
- (c) Since interest of $\text{HK}\$400,000 \times 0.06 = \text{HK}\$24,000$ must be paid to the debt holders each year, the amount that will be available for the shareholders each year will equal $175,000 - 24,000 = \text{HK}\$151,000$. This implies that the new cost of equity capital for the firm will be $\text{HK}\$1,058,333 = \text{HK}\$151,000 / k_{cs}$, $k_{cs} = 0.1427$, or 14.27%.



Part D

Financial Analysis and Forecasts



Chapter 9 Financial Analysis and Profitability Projections for
Financial Strategy

Chapter 10 Financial Forecasts and Business Planning

9

Financial Analysis and Profitability Projections for Financial Strategy

CHAPTER TOPIC LIST

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- 9.1.1 Price-to-Book Ratio
- 9.1.2 Earnings per Share Ratio
- 9.1.3 Price-Earnings Ratio

9.2 Accounting and Finance Ratios

- 9.2.1 Short-Term Solvency/Liquidity Ratios
- 9.2.2 Financial Leverage/Long-Term Solvency Ratios
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9.10 Symptoms of and Remedies for Overtrading and Excessive Leverage

9.10.1 Overtrading

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9.11 Financial Strategy Evaluation and Recommendation



LEARNING OUTCOMES

PRINCIPAL LO5: PRODUCE FINANCIAL ANALYSIS

LO5.01: Apply techniques of financial analysis

5.01.01 Identify and calculate main stock exchange indicators

5.01.02 Analyse financial performance using accounting and finance ratios, and compare the result with other market players/competitors

LO5.02: Evaluate a business entity's financial strategy

5.02.01 Evaluate the financial strategy of an entity and recommend financial actions that will add value to the entity

5.02.02 Evaluate the credit policy of company's suppliers and credit terms given to the company's customers to identify any substantial gap in between

LO5.03: Produce a forecast cash flow statement for a business entity

5.03.01 Analyse the role of cash flow analysis in the evaluation of an entity's strategic and operational plans

5.03.02 Produce cash flow statements to analyse the financial strategy of an entity and its short- and medium-term financial requirements

5.03.03 Explain how an entity can use cash flow reporting systems to monitor and manage its financial strategy

LO5.04: Prepare profitability projections

5.04.01 Explain how profitability projections can be used to examine the impact of an entity's financial strategy

5.04.02 Prepare projections of future profitability to evaluate an entity's financial strategy

LO5.05: Analyse the liquidity and solvency positions for a business entity

5.05.01 Analyse the impact of an entity's financial strategy on its liquidity and solvency through the use of ratios and other techniques

5.05.02 Describe the symptoms of and remedies for overtrading and excessive leverage



OPENING CASE

ABC INVESTMENT LTD

As management accountants grow in their scope of jobs and level of responsibility, their roles typically become more demanding. They are expected to tackle more complicated financial and strategic issues.

As the management accountant at ABC Investment Ltd, you might delve deeper into the financial management of various subsidiaries and have to consider more strategic issues that include public market valuation, leverage, efficiency, liquidity, working capital, capital budgeting, credit policy, cash flow forecast, business risk, degree of operating leverage and degree of combined leverage.

There are no easy answers to many issues that arise out of these different factors, even if they seem to have straightforward formulas. These issues are frequently interrelated.

Actions that make one set of financial ratios better may result in trade-offs that make other sets of ratios worse. It is important to be thoughtful in the various considerations and in any recommendations to management. 



OVERVIEW

Financial analysis, projections and a clear financial strategy are important considerations for companies – considerations that could mean the difference between success and failure.

This chapter examines various tools and avenues open to companies in determining their financial strategy.

First, a successful financial strategy depends on clear-eyed analysis of the operations of a company. This analysis begins with careful consideration of some of indicators and ratios that, while not always informative on their own, can combine to paint a very accurate picture of the current state and future of a company.

These financial ratios help inform the financial strategy of a company, but they are not enough. Financial managers should also take into consideration cash flows, credit policies and terms and cash flow reporting. This chapter considers all of these, as well as the importance of cash flow reporting systems, the impact of profitability projections on financial strategy and how future profitability may impact any evaluation of a company's financial strategy.

Finally, the chapter considers a couple of red flags for finance professionals setting a financial strategy for a company, including symptoms of overtrading and excessive leverage, before briefly considering some of the finer points of evaluating and recommending a financial strategy.



9.1 STOCK EXCHANGE INDICATORS

Investors that acquire shares in a company through a stock market use various approaches to evaluate the value of a company.

Stock markets typically evaluate a company's liquidity, efficiency, leverage and profitability using some ratios often known as market-value ratios, which combine market-value data with data from the financial statements of a firm.

The most frequently used market-value ratios are the price-to-book ratio, the earnings per share ratio and the price-earnings ratio.

9.1.1 Price-to-Book Ratio

The **price-to-book ratio (P/B)**, sometimes called the market-to-book ratio, is a comparison of the market value of a company's investments to its book value. The formula for this ratio is:

$$\text{Price to book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

Book value per share represents the amount of investment accumulated in the equity account of a firm for every share of stock. The market value of equity per share is the market share price.

When the price-to-book ratio is higher, the company has likely been more effective at creating value for shareholders. When a price-to-book ratio is less than one, it means the company has not created reasonable value for shareholders.

9.1.2 Earnings per Share Ratio

The net income of a company represents its revenues relative to expenses for a particular time period. A company generates net income for a period when revenues exceed expenses. When the opposite happens, the company incurs a net loss.

Net income reported on a per-share basis is called **earnings per share (EPS)**. EPS is calculated by dividing the net income by the number of ordinary shares currently outstanding. EPS can help shareholders or investors determine whether the company has earned or lost money for each share of stock.

The formula for earnings per share (EPS) is as follows:

$$\text{Earning per share} = \frac{\text{Net income}}{\text{Shares outstanding}}$$

9.1.3 Price-Earnings Ratio

The price-earnings (P/E) ratio relates earnings per share to price per share. The formula is as follows:

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

Price per share on a given date is easily found through listings, apps, stock markets or any other online source.

What does it signify for a firm to have a price-earnings ratio of, for example, 25.4?

It means that the stock market places a value of HK\$25.40 on every HK\$1 of net income. Investors are willing to pay HK\$25.40 for a claim on HK\$1 of earnings because stock prices reflect earnings in the current year and also future cash flows.

A high P/E ratio may suggest investors expect earnings to grow rapidly, but may also be caused by low earnings in a particular year when investors might expect earnings to recover soon. As with other measures, to understand whether the P/E ratio is too high or too low, it is necessary to compare the P/E ratio of a company with those of its competitors and market trends.

Illustrative Example 1

One potential investment target is Cora's Candies Limited.

The shares of Cora's Candies Limited are currently trading at HK\$18.20. There are 4 million shares outstanding. The company's 2018 net income was HK\$5.2 million. What is the market value for the equity of the company and what is the P/E ratio of the shares?

The market value of equity is simply the market price of a share times the number of shares outstanding. This is calculated as:

$$\text{MVE} = 18.20 / \text{share} \times 4 \text{ million shares} = \text{HK\$}72.8 \text{ million}$$

The P/E ratio can be found with the net income and market value figures, but it is usually calculated using per share amounts. The earnings per share calculation is:

$$\begin{aligned} \text{EPS} &= \text{HK\$}5.2 \text{ million} / 4 \text{ million shares} = \text{HK\$}1.30 \text{ per share. Therefore, } \text{P/E} = \text{Price/EPS} \\ &= \text{HK\$}18.20 / \text{HK\$}1.30 = 14.0. \end{aligned}$$

Key Learning Point

The price-to-book ratio, earnings per share and price-earnings ratio are three frequently used measures of the value of a company.

► Knowledge Check Questions

Question 1

Remar Electronics Corp reported that its net income for a particular period was HK\$1,276,351. The firm has 420,000 shares outstanding and a PE ratio of 11.2. Calculate the firm's share price.

- A HK\$34.05
- B HK\$3.68
- C HK\$11.20
- D HK\$36.80

Question 2

RTR Corp reported net income of HK\$812,425 for the year. The company's share price is HK\$13.45, with 312,490 shares outstanding. Compute the firm's price-earnings ratio.

- A 4.87 times
- B 8.12 times
- C 5.17 times
- D None of the above

9.2 ACCOUNTING AND FINANCE RATIOS

Some financial statement users have a different interest when examining financial statements. Banks, for example, and suppliers shipping to customers on credit, want to know whether a firm has the means to pay off its debts. This leads to a focus on the liquidity of the company.

Banks, investors and other stakeholders are also interested in the levels of financial leverage at a company, the quality of asset management, profitability, the value of shares as well as how the performance of a company compares to others in the same industry.

9.2.1 Short-term Solvency/Liquidity Ratios

Liquidity refers to how easily something can be converted into cash.

The focus in analysing the liquidity of the company is on the overall liquidity of a firm's assets and on the available assets to meet current liabilities. In terms of overall liquidity, there are a number of ratios that investors and other stakeholders typically consider.

The first is the **current ratio**, which is estimated as current assets (CA) divided by current liabilities (CL):

$$\text{Current ratio} = \frac{\text{Current assets (CA)}}{\text{Current liabilities (CL)}}$$

Bankers consider this ratio important because bank loans are normally short term and part of a company's current liabilities. A current ratio of 0.9, could mean that a company does not have enough current assets available to pay off a loan.

If a bank has to force a company to liquidate assets, a current ratio less than 1 could suggest that some assets are probably not worth much.

The inventory of a failing company may not be worth its book value, making it difficult for creditors to recover funds. Therefore, banks also look at the **quick ratio** or acid test ratio. The quick ratio is the sum of cash (C), marketable securities (MS) and accounts receivable (AR) divided by current liabilities:

$$\text{Quick ratio} = \frac{\text{Cash (C) + Marketable securities (MS) + Accounts receivable (AR)}}{\text{Current liabilities (CL)}}$$

A quick ratio of less than 1, for example, indicates that a company may not be able to pay off all its current liabilities from its 'quick' (i.e. most liquid) assets.

9.2.2 Financial Leverage/Long-term Solvency Ratios

Leverage ratios, also known as long-term solvency ratios, measure how much debt a firm uses as opposed to equity financing. These ratios also point to whether a company is able to meet its long-term obligations such as interest and lease payments. Creditors, shareholders and managers care about these ratios.



Financial leverage refers to the use of debt in the capital structure of a firm. The use of debt financing, instead of equity financing, can magnify shareholders' returns because interest payments on most long-term debt and some short-term debt are fixed, regardless of operating profits.

If operating profits increase, fixed-interest payments do not change and the increases go to shareholders.

However, when profits fall, fixed-interest payments also stay the same (assuming the company continues to meet its obligations) and the loss is charged against the shareholders' equity. So, debt boosts returns to shareholders during good periods and reduces them during bad periods.

At the same time, using debt in the capital structure increases the risk of default (insolvency risk), i.e. the risk that the company will not be able to pay debts. Regardless of the company's performance, debtholder must be paid the interest and principal payments owed to them on time.

If a company does not make an interest payment on the prescribed date, it defaults on its debt and could be pushed into bankruptcy.

There are three leverage ratios which focus on how much debt a company uses in its capital structure. The more debt, the more volatile the earnings, and the greater its risk of default.

The **total debt ratio** measures how much the finances of the company derive from sources other than shareholders. The higher the total debt ratio, the more debt the company has. The total debt ratio is:

$$\text{Total debt ratio} = \frac{\text{Total liabilities (TL)}}{\text{Total assets (TA)}}$$

The easiest way to determine the total debt ratio is to subtract total equity from total assets, which also means total debt is equal to total liabilities. An alternative approach is to add all interest-bearing liabilities. To determine whether a high or low total debt ratio is good or bad, it is necessary to determine how the capital structure affects the value of a company.

Two common variations of this ratio are the debt-to-equity ratio and the equity multiplier.

The **debt-to-equity ratio** makes it possible to determine the amount of debt for each dollar of equity. The formula for this ratio is:

$$\text{Debt-to-equity ratio} = \frac{\text{Total debt}}{\text{Total equity}}$$

The **equity multiplier** makes it possible to determine the amount of assets that a company has for every dollar of equity. It is calculated using the following formula:

$$\text{Equity multiplier} = \frac{\text{Total assets}}{\text{Total equity}}$$

It is worth noting that the equity multiplier is related to the debt-to-equity ratio:

$$\text{Equity multiplier} = 1 + \text{Debt-to-equity ratio}$$

Recall the statement of financial position identity:

$$\text{Total assets} = \text{Total liabilities (debt)} + \text{Total shareholders equity}$$

All three leverage ratios are related by the statement of financial position identity, and once one is known it is possible to compute the other two.

Apart from these three leverage ratios, times-interest-earned ratio and cash coverage ratio are also important ratios which measure a company's ability to pay interest. The **times-interest-earned ratio** can be calculated using the following formula to arrive at how much interest was earned:

$$\text{Times-interest-earned} = \text{Earnings before interest and tax (EBIT)} / \text{Interest}$$

In turn, the **cash coverage ratio**, which shows the amount of cash a company has after adding back the non-cash deduction from 'Earnings before interest and tax (EBIT)' to get 'cash flow before taxes' for interest payment, is arrived at with the following formula:

$$\begin{aligned} \text{Cash coverage ratio} \\ = \frac{\text{Earnings before interest and tax (EBIT)} + \text{Depreciation and/or Amortisation}}{\text{Interest}} \end{aligned}$$

9.2.3 Asset Management Efficiency Ratios

Another form of financial ratio is measurement of the efficiency in assets and sales generation. The asset management efficiency ratios reveal internal policies regarding management of credit and inventory and are also called termed activity ratios or turnover ratios.

Amongst the ratios typical to asset management efficiency is the receivables turnover ratio. This reports the time required to generate sales from accounts receivable, those sales made on credit. It reflects the company's credit policies. For example, if a company lets too many collections expand for too long, it becomes less efficient.

The formula to calculate the receivables turnover is:

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable average / ending balance}}$$

For example, a company has a current accounts receivable balance of HK\$416,800 and total credit sales for the last quarter were HK\$3,404,298. The receivables turnover is: $\text{HK\$3,404,298} \div \text{HK\$416,800} = 8.17$. This reveals that the turnover rate is 8.17 times receivables.



A higher turnover rate reflects faster collections, so the higher the turnover, the more positive the ratio. A related calculation is the days' receivables. This tells how long, on average, it took for the company to collect outstanding receivables. It is calculated by dividing 365 (days per year) by receivables turnover:

$$\text{Days' receivables} = \frac{365}{\text{Receivables turnover}}$$

For example, consider that receivables turnover was reported as 8.17. The calculation for days' receivables is: $365 \div 8.17 = 45$. This reflects that the average collection period is 45 days.

On the other hand, payables turnover measures how fast a company pays its suppliers. If the ratio falls from one period to the next, this indicates that the company is paying its suppliers more slowly. This could be a signal of a liquidity problem or it could be a financially strong company that has negotiating power with its suppliers and uses them as an inexpensive form of financing. If a company is paying its suppliers in a quick manner, it may mean that the suppliers require fast payment terms or that the company is able to pay earlier for early payment discounts.

The payables turnover ratio is calculated by dividing credit purchases from suppliers by the amount of accounts payable. The formula is:

$$\text{Payables turnover} = \frac{\text{Net credit purchases}}{\text{Account payable (average/ending balance)}}$$

For example, a company has an accounts payable balance of HK\$842,000 and total purchases for the last quarter were HK\$6,450,780. Based on this information, the payables turnover is: $\text{HK\$6,450,780} \div \text{HK\$842,000} = 7.66$.

Alternatively, the cost of goods sold can also be used as the numerator in the payables turnover formula when the net credit purchases cannot be estimated. The alternative formula is:

$$\text{Payables turnover} = \frac{\text{Cost of goods sold}}{\text{Account payable (average/ending balance)}}$$

A related calculation is the days' payables. This tells how long, on average, it took for the company to pay outstanding payables. It is calculated by dividing 365 (days per year) by payables turnover:

$$\text{Days' payables} = \frac{365}{\text{Payables turnover}}$$

For example, consider that the payables turnover was reported as 7.66. The calculation for the days' payables is $365 \div 7.66 = 48$. This reflects that the average payment period is 48 days.

Another form of turnover is the **inventory turnover ratio**. This shows how many times the typical inventory level is replaced. In practice, inventory is not completely used up and then replaced, so this represents an average. To calculate, divide the cost of goods sold (costs required to generate sales) by the inventory balance:

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory average / ending balance}}$$

For example, the current year's cost of goods sold was HK\$8,524,009 and closing inventory balance was HK\$1,225,241. Inventory turnover was: HK\$8,524,009 ÷ HK\$1,225,241 = 7.0.

On average, inventory was replaced seven times during the year. This can also be reflected in the days' inventory ratio, showing how long items remain in inventory (on average) before being sold. To calculate:

$$\text{Days' inventory ratio} = \frac{365}{\text{Inventory turnover}}$$

For example, the inventory turnover ratio was reported as 7.0. The days' inventory ratio is: 365 ÷ 7.0 = 52. The average inventory item is held in the inventory for 52 days before being replaced.

The fixed asset turnover ratio is a different form of test. It measures a company's management of capital assets in the generation of sales. For example, a construction company investing in heavy earth-moving equipment would hope for a return on its investment by use of that equipment. The ratio is calculated by dividing sales by net fixed assets (capital assets minus accumulated depreciation). The calculation:

$$\text{Fixed asset turnover} = \frac{\text{Sales}}{\text{Net fixed assets}}$$

For example, a company's annual sales were HK\$18,667,012. As of the end of the year net fixed assets were reported as HK\$8,400,450. The fixed asset turnover ratio was: HK\$18,667,012 ÷ HK\$8,400,450 = 2.2. The net fixed assets were turned 2.2 times during the year in terms of total sales.

Another variation tests sales in comparison to total assets, the total asset turnover ratio. The formula is:

$$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$$

For example, total sales were reported as HK\$18,667,012 and total assets were HK\$12,999,023. The total asset turnover was: HK\$18,667,012 ÷ HK\$12,999,023 = 1.4.



9.2.4 Profitability Ratios

Profitability ratios consider how efficiently management uses assets to generate sales and manage operations. Along with managers, shareholders and creditors care about these measures because they provide insight on earnings.

In general, the companies that perform better have higher profitability ratios.

Gross profit margin measures the percentage of net sales remaining after the cost of goods sold is paid. The gross profit margin considers how companies manage the expenses of producing products or services.

The gross profit margin formula is:

$$\text{Gross profit margin} = \frac{\text{Net sales} - \text{Cost of goods sold}}{\text{Net sales}}$$

Moving further down the statement of profit or loss and other comprehensive income, it is possible to measure the percentage of sales that remains after all expenses, which are paid except for interest and taxes.

Operating profit is generally measured as earnings before interest and taxes (EBIT). The operating profit margin provides an indication of profitability regardless of financing policies or tax strategies. The formula for operating profit margin is:

$$\text{Operating profit margin} = \frac{\text{Earnings before interest and taxes (EBIT)}}{\text{Net sales}}$$

Sometimes, this ratio is calculated using earnings before interest, taxes, depreciation and amortisation (EBITDA) instead of EBIT, which is known as the **EBITDA margin**.

Net profit margin is the percentage of sales that remains after all expenses, including interest and taxes, have been paid. The formula is:

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Net sales}}$$

Different industries have different levels of profit. A rule of thumb is that the higher the profit margin, the better a company is performing.

It is also essential to analyse profitability as a percentage of investment in assets or equity.

The first of these measures is return on assets (ROA), which can be calculated in two ways.

The first way to calculate ROA looks at EBIT for every dollar of assets. This measure indicates how efficiently management uses its assets. ROA can be used to measure pre-tax returns on total net investment from operations. The formula **EBIT return on assets (EROA)** is:

$$\text{EBIT return on assets} = \frac{\text{EBIT}}{\text{Total assets}}$$

Return on assets (ROA) can also be calculated as follows:

$$\text{Return on assets} = \frac{\text{Net income}}{\text{Total assets}}$$

Although it is a common calculation, it is not always the best approach. The ROA calculation divides net income by total assets (or debt plus equity). Debt plus equity measures how much investment a company has received from shareholders and creditors. This ratio provides information how efficiently a company uses its assets but is obscured by financing decisions and taxes.

EROA is more efficient than ROA to measure how assets are used in operations. EROA divides a measure of earnings to debtholders and shareholders by how much both groups have invested.

In financial ratios, if the numerator is a measure of income to shareholders, only investments by shareholders should be used in the denominator. If the numerator is a measure of total profits from operations, the denominator should be a measure of total investments by both debtholders and shareholders.

Return on equity (ROE) measures net income as a percentage of the shareholders investment in the company. The return on equity formula is as follows:

$$\text{Return on equity} = \frac{\text{Net income}}{\text{Total equity}}$$

As with efficiency ratios, the calculation of ROA and ROE involves dividing a value from the statement of profit or loss and other comprehensive income from a particular period of time by a value from the statement of financial position from the end of the period.

Calculating ROA and ROE using the average asset value or equity value is another approach that is often preferred, where the average value is determined as follows:

$$\text{Average asset or equity value} = \frac{\text{Beginning value} + \text{Ending value}}{2}$$

It makes sense to use average asset or equity values because the average value of assets or equity help generate earnings over a period.

9.2.5 Market-based Ratio

There are also a number of standard ratios used to understand the relative valuation of a company.

Some ratios that are useful in assessing a company's relative value include earnings per share (EPS) and the equity **book value per share (BVPS)**, which is the ending shareholders' equity (SE) divided by the ending number of shares outstanding:

$$\text{Book value per share} = \frac{\text{Shareholders' equity (SE)}}{\text{Number of shares}}$$

Usually, reporters of financial information will adjust the previous per-share values to reflect any changes in the number of shares outstanding.

In terms of valuation, one of the most basic ratios an analyst can look at is a company's **dividend yield**, which is the current dividend per share (DPS) divided by the current share price (P):

$$\text{Dividend yield} = \frac{\text{Dividend per share (DPS)}}{\text{Current share price (P)}}$$

For example, a hypothetical dividend yield of 0.63% is quite low. A low dividend yield can be caused by low dividend payouts or high share prices or both. Because investors are concerned about the company's ability to sustain its dividend payments, one ratio that is useful to assess this ability is the **dividend payout ratio**, which is the DPS divided by the EPS:

$$\text{Dividend payout} = \frac{\text{Dividend per share (DPS)}}{\text{Earnings per share (EPS)}}$$

One of the most important and widely followed value ratios is the **price-earnings (P/E) ratio** (discussed earlier), which is estimated as the share price (P) divided by the EPS:

$$\text{P/E ratio} = \frac{\text{Price per share (P)}}{\text{Earnings per share (EPS)}}$$

Whenever share prices increase or EPS declines, P/E ratios increase. Some analysts tend to focus on the prices reflected in the numerator of the P/E. Others say that when the P/E ratio gets too high, shares become too expensive. Yet others argue that a high P/E ratio may point to expectations of substantial growth in future earnings or low risk.

For companies, higher P/E ratios are generally good because they suggest that markets have confidence in them.

When the P/E ratio is estimated based on the current share price and a forecast or estimated EPS (EEPS) figure, it is known as a forward P/E ratio:

$$\text{Forward P/E ratio} = \frac{\text{Price per share (P)}}{\text{Estimated earnings per share (EEPS)}}$$

Security analysts who recommend companies tend to use forward P/E ratios because they normally make shares look cheaper (i.e. as long as EPS continues to increase).

The higher the P/E ratio, the longer the period of time until shareholders recover their investments in the form of profits. Shares with low P/E ratios are considered value stocks, since they have shorter payback periods. By comparison, shares with high P/E ratios are generally regarded as growth stocks that rely on future growth.

P/E ratios are meaningless when EPS is negative or very low and, for firms that do not pay dividends, dividend yields equal zero. In these cases, analysts look to other ratios to understand the relative value of a company.

One commonly used ratio is the price-to-book ratio (also discussed earlier), sometimes known as the market-to-book (M/B) ratio, which is defined as the market price per share (P) divided by the book value per share (BVPS):

$$\text{Price-to-book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

Another valuation ratio that is often considered is the EBITDA multiple, which is calculated as the total enterprise value (TEV) divided by earnings before interest, taxes, depreciation and amortisation (EBITDA):

$$\text{EBITDA multiple} = \frac{\text{TEV}}{\text{EBITDA}}$$

The total enterprise value is an estimate of the total market value of the firm, which equals the equity market value plus the market value of the firm's debt. This is the market value of what is called 'invested capital' in the company's financial statements.

This ratio is of particular interest when a firm is considering the takeover or purchase of another firm, since it values the whole firm and not just the common equity. It also recognises that the firm can always issue or buy back common equity and debt.

9.2.6 Comparison with Other Market Players/Competitors

Is it possible to ascertain whether a ratio value is too high or too low? Is the value good or bad?

The starting point for ascertaining whether a ratio value is too high or too low or whether it represents good value is to select the right benchmark that will become a basis for comparisons.

There are three main ways to gather financial data. The first is trend analysis, the second is industry group analysis and the third is peer group analysis.

Trend analysis considers the performance of a company over time using historical information. Management uses trend analysis to determine the movement of a ratio over time (i.e. refer to Module 2, Chapter 11).

An increase or decrease in a ratio value is neither good nor bad in a vacuum. However, a changing ratio value can prompt financial managers to investigate and determine if any action is necessary.

Industry group analysis is another way to establish a benchmark, and this requires that a group of companies with similar size and product lines who compete in the same market be considered together.

The average ratio values for these companies become the benchmarks.

Since no two companies are exactly the same, deciding which companies to include in the analysis is always a judgement call but if it is possible to put together a reasonably sized sample the average values provide benchmarks.



There are multiple sources of information about financial ratios and financial data, including, for example, companies such as Dun & Bradstreet and Standard & Poor's (S&P).

There are a number of industry databases but none of them are perfect. When trying to find a sample of companies that are similar to the company in focus, it may be hard to find classifications that are not too broad.

Another problem is that different databases may calculate ratios differently.

A third way to establish benchmark information is to identify a peer group of competing firms that are, ideally, in similar lines of business, of similar size and direct competitors. A peer group can help management compute average ratio values to compare the performance of companies.

The senior management team in a company may identify its competitors to determine which companies to include in the peer group. It is also possible to look at financial statements and analysts reports to determine peer groups.

The peer group methodology may be the best way to establish a benchmark if financial data for peer companies is publicly available. It is important to note that a comparison against a single company may be appropriate when there is a market leader and the goal is to compare the company in question against the best.

9.2.7 DuPont System

The ROA equation, as discussed earlier, links the firm's return on assets with its total asset turnover and net profit margin.

This can be derived from the ROA equation as follows:

$$\begin{aligned}\text{ROA} &= \text{Net income} / \text{Total assets} \\ &= (\text{Net income} / \text{Net sales}) \times (\text{Net sales} / \text{Total assets}) \\ &= \text{Net profit margin} \times \text{Total asset turnover}\end{aligned}$$

This calculation starts with the ROA formula. Then it can be broken down into $\text{ROA} = (\text{Net income} / \text{Net sales}) \times (\text{Net sales} / \text{Total assets})$. The first ratio is the firm's net profit margin and the second ratio as the firm's total asset turnover.

This equation shows that a firm's ROA is determined by the firm's net profit margin and the firm's total asset turnover.

There are some managerial implications of each of these terms.

The net profit margin ratio can be written as:

$$\text{Net profit margin} = \text{Net income} / \text{Net sales} = (\text{EBIT} / \text{Net sales}) \times (\text{EBT} / \text{EBIT}) \times (\text{Net income} / \text{EBT})$$

The net profit margin may be considered the product of three ratios:

1. the operating profit margin (EBIT/Net sales);
2. a ratio for the impact of interest expenses on profits (EBT/EBIT), and
3. a ratio for the impact of taxes on profits (Net income/EBT).

The net profit margin focuses on how management generates profits from sales by managing operating expenses, interest and tax expenses.

Total asset turnover (Net sales/Total assets) measures how efficiently management uses company assets. (i.e. the output management generates with a given asset base).

Some interesting managerial insights can be provided by the ROA equation. To increase ROA, management can increase net profit margin, total asset turnover or both. Every firm wants to make both as large as possible but competition, marketing realities, technology and manufacturing capabilities are amongst the factors that limit both asset turnover and net profit margins and, in turn, ROA.

There are two possible approaches to manage ROA. The first puts emphasis on high profit margin and low asset turnover. High-end department stores use this approach. The second relies on low profit margins and high turnover, an approach favoured by discount stores or low-end supermarkets.

To obtain the ROE equation, we start with the formula discussed earlier:

$$\begin{aligned}\text{ROE} &= \text{Net income/Total equity} \\ &= (\text{Net income/Total assets}) \times (\text{Total assets/Total equity}) \\ &= \text{ROA} \times \text{Equity multiplier}\end{aligned}$$

The ROE equation can be broken down into (Net income / Total assets) × (Total assets / Total equity). Based on this, ROE is the product of two ratios already discussed: ROA and the equity multiplier.

An interesting fact is that ROE is determined by the firm's ROA and its use of leverage.

ROE increases as the use of debt in the firm's capital structure increases. Thus, increasing the use of leverage is one way that management can increase the firm's ROE. However, this comes at a price; the greater the use of financial leverage, the more risky it is for the firm, as discussed in Chapter 8. Management's preference for risk and the willingness of creditors to lend money determines how a company can use this strategy.

It is now possible to combine the two equations into a single equation. $\text{ROE} = \text{ROA} \times \text{Equity multiplier}$ and $\text{ROA} = \text{Net profit margin} \times \text{Total asset turnover}$. With some substitutions, this can be morphed into an expression formally called the **DuPont equation**, as follows:

$$\text{ROE} = \text{Net profit margin} \times \text{Total asset turnover} \times \text{Equity multiplier}$$

The DuPont equation can also be expressed as a ratio:

$$\text{ROE} = (\text{Net income/Net sales}) \times (\text{Net sales/Total assets}) \times (\text{Total assets/Total equity})$$

The DuPont system uses financial ratios to determine the financial health of a company by taking three steps. First, management uses DuPont ratios to assess the financial health of a company. Second, management corrects any identified problems. Third, management monitors the financial performance over time.

The DuPont system puts the onus on management to maximise ROE, not just maximise the value of shares. The system was originally designed for management to diagnose and correct problems, but investors also find the diagnostic powers of interest.

The DuPont system is derived from two equations that link ROA and ROE and identifies three areas to maximise ROE: (1) profit on sales, (2) how efficiently company assets are used and (3) the amount of financial leverage. A single ratio monitors each of these areas and together these ratios make up the DuPont equation.

Illustrative Example 2

Your boss is interested in Excelsior Limited. Analyse the following available information to answer the questions presented to you and calculate the required ratios.

Excelsior Limited's statement of profit or loss for Year 2 (HK\$ million)

Net sales	1,850
Less: Cost of goods sold	605
Less: Depreciation	180
Earnings before interest and taxes	1,065
Less: Interest paid	80
Taxable income	985
Less: Taxes	156
Net income	829
Dividends paid	585
Addition to retained earnings	244

Excelsior Limited's statement of financial position as at 31 December Year 2 (HK\$ million)

	Year 1	Year 2		Year 1	Year 2
Cash	100	112	Accounts payable	400	350
Accounts receivable	330	234	Notes payable	390	370
Inventory	410	435	Subtotal	790	720
Subtotal	840	781	Long-term debt	500	550
Net fixed assets	1,556	1,888	Common shares	550	599
			Retained earnings	556	800
			Subtotal	1,106	1,399
Total assets	2,396	2,669	Total liability and equity	2,396	2,669

Calculate leverage ratios, including debt ratio and debt-equity ratio, in both years. Has Excelsior Limited improved on its leverage ratios in Year 2?

Illustrative Example 2 (continued)

In Year 1,

$$\text{Debt ratio} = \text{TL} / \text{TA} = (2,396 - 1,106) / 2,396 = 0.54$$

$$\text{Debt equity ratio} = \text{D} / \text{SE} = 500 / 1,106 = 0.45$$

In Year 2,

$$\text{Debt ratio} = \text{TL} / \text{TA} = (2,669 - 1,399) / 2,669 = 0.48$$

$$\text{Debt equity ratio} = \text{D} / \text{SE} = 550 / 1,399 = 0.39$$

The company improved on both leverage ratios in Year 2.

Calculate the company's efficiency ratios including gross profit margin and operating margin in Year 2. Explain the differences.

$$\text{Gross profit margin} = (\text{S} - \text{CGS}) / \text{S} = (1,850 - 605) / 1,850 = 0.67$$

$$\text{EBITDA} = \text{net sales} - \text{cost of goods sold} = 1,850 - 605 = 1,245$$

$$\text{Operating margin} = \text{NOI} / \text{S} = (\text{EBITDA} - \text{depreciation}) / \text{sales} = (1,245 - 180) / 1,850 = 57.6\%$$

Gross profit margin excludes the impact of depreciation, while operating margin includes it.

Calculate liquidity ratios including working capital ratio, current ratio and quick ratio for Year 2. Explain the differences amongst the ratios.

$$\text{Working capital ratio} = \text{CA} / \text{TA} = 781 / 2,669 = 0.29$$

$$\text{Current ratio} = \text{CA} / \text{CL} = 781 / 720 = 1.08$$

$$\text{Quick ratio} = (\text{CA} - \text{inventory}) / \text{CL} = (781 - 435) / 720 = 0.48$$

These ratios all indicate whether the corporation is capable of paying its current liabilities. Working capital ratio measures the percentage of liquid assets amongst all assets. The current ratio indicates whether it can pay off current liabilities using the current assets. Since inventory has the least liquidity of the current assets, the quick ratio removes it from the calculation.

Key Learning Point

Accounting and finance ratios may be used to analyse short-term solvency and liquidity, financial leverage and long-term solvency, asset management efficiency, profitability and share prices.

9.2.8 Short-Term Financial Planning

Cash is as important to a company as blood is to human beings. A firm can make use of the operating cycle and the cash conversion cycle to project the amount of the working capital it needs for its daily operation.

9.2.8.1 Operating Cycle

The **operating cycle** measures the average period of time a firm takes from purchasing and selling inventory to collecting cash from its customers. It is calculated using the following formula:

$$\text{Operating cycle} = \text{Days' inventory} + \text{Days' receivables}$$

As explained in Section 9.2.3, the days' inventory is the amount of time inventory is stored before it is sold. The days' receivables is the amount of time it takes to receive payment from the sale of the inventory. Note that the units (e.g. days, weeks) must be the same for both ratios in the equation.

9.2.8.2 Cash Conversion Cycle

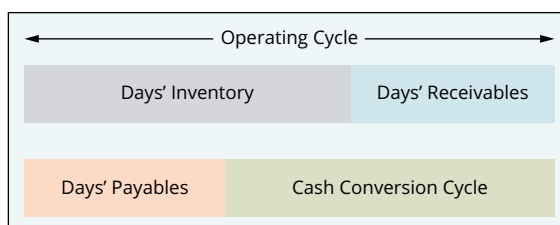
The **cash conversion cycle (CCC)** (also called the cash operating cycle) is a type of liquidity analysis. The cash conversion cycle indicates the number of days it takes for the cash invested to be converted back to cash. In other words, it measures the time used to sell the inventory, collect the receivables, and pay the creditors.

The Cash conversion cycle is measured as follows:

$$\text{Cash conversion cycle (CCC)} = \text{Days' inventory} + \text{Days' receivables} - \text{Days' payables}$$

If CCC is negative, it indicates that the company has excess cash to invest. A CCC of -10 indicates that the company has excess cash to invest for 10 days.

A graphical representation of the operating cycle and cash conversion cycle is shown in Exhibit 9.1.



► **EXHIBIT 9.1** Operating Cycle and Cash Conversion Cycle

There are mainly three factors that will affect the length of the cash conversion cycle:

- When the suppliers extend the payment terms of the company, the cash conversion cycle will be shortened, as the company can repay the overdue trade debts later.
- During the peak seasons of high demand, the company has to increase the amount of inventory on hand to satisfy the customer purchase orders, which will lengthen the cash conversion cycle.
- A lenient credit policy and related payment terms allow the customers to repay cash later which will increase the cash conversion cycle.

The managerial decisions and business terms with trading parties can influence the operating and cash conversion cycles of a firm. In most situations, the shorter the cycle, the more beneficial to the company's short-term financial planning and the cash requirements of maintaining the operation.

Illustrative Example 3

	2019	2018
	HK\$ (in thousand)	HK\$ (in thousand)
Sales	1,641,171	3,590,109
Cost of goods sold	795,314	921,550
Inventory	258,736	202,781
Accounts receivable	57,157	57,664
Accounts payable	170,293	193,760
Average Inventory	$(258,736 + 202,781)/2 = 230,759$	
Average A/R	$(57,157 + 57,664)/2 = 57,411$	
Average A/P	$(170,293 + 193,760)/2 = 182,027$	

(Figures extracted from Bossini International Holding Limited Annual report 2018/19
<https://doc.irasia.com/listco/hk/bossini/annual/2019/car2019.pdf>)

Calculate CCC for 2019.

Days' inventory = $\text{HK\$}230,759 / (\text{HK\$}795,314 / 365 \text{ days}) = 105.90 \text{ days}$

Days' receivables = $\text{HK\$}57,411 / (\text{HK\$}1,641,171 / 365 \text{ days}) = 12.77 \text{ days}$

Days' payables (using 'purchases') = $\text{HK\$}182,027 / [(\text{HK\$}795,314 + \text{HK\$}258,736 - \text{HK\$}202,781) / 365 \text{ days}]$
 $= 78.05 \text{ days}$

*Or Days' payables (using 'COGS') = $\text{HK\$}182,027 / (\text{HK\$}795,314 / 365 \text{ days}) = 83.54 \text{ days}$

CCC = $105.90 + 12.77 - 78.05 = 40.62 \text{ days}$

*Or CCC = $105.90 + 12.77 - 83.54 = 35.13 \text{ days}$

The cash conversion cycle is a good measurement tool for analysing a company's liquidity.

A low CCC implies a high liquidity and that the company will have to fund its inventory and receivables for a shorter period. The CCC should be used in comparison with different time periods and the company's competitors.

For example, if Company A has CCC of 85 days in 2019 and 80 days in 2020, then Company A has shown an improvement in liquidity during the year. Taking a deeper look into days' inventory, days' receivables or days' payables can indicate which part contributes most to the improvement.

If Company B, which is the competitor of Company A, has CCC of 75 days in 2020. Company A should then focus on what it should improve as compared to its competitor.

► Knowledge Check Questions

Question 3

Identify what a debt ratio of 0.9 means.

- A The firm has HK\$0.90 of short-term debt per HK\$1.00 of assets.
- B The firm has HK\$0.90 of short-term debt per HK\$1.00 of long-term debt.
- C The firm has HK\$0.90 of total debt per HK\$1.00 of assets.
- D The firm has HK\$0.90 of total debt per HK\$1.00 of equity.

Question 4

Identify which of the following investors are likely to be most interested in.

- A The current ratio.
- B The price-earnings ratio.
- C The times-interest-earned ratio.
- D The total asset turnover ratio.



9.3 IMPACT OF FINANCIAL STRATEGY ON CORPORATES' LIQUIDITY AND SOLVENCY

Company management has to make acquisition and investment decisions, which can be generally divided into real asset decisions and financial asset acquisition decisions. Investment decisions are considered in terms of financial management and financial strategy.

Financial management refers to the management of the financial resources of a company, which includes assets and the financing of those assets.

In turn, strategic financial management refers to the planning and use of financial resources to accomplish the objectives of a company while returning the maximum value to shareholders.

A **financial strategy** outlines the short- and long-term goals of an organisation and can have a real impact on the liquidity and solvency of a company.

In broad strokes, a financial strategy has three main components:

- financing,
- investment, and
- dividends.

The framework for analysing long-term investment or asset decisions is known as **capital budgeting** or capital expenditure analysis, and these decisions are some of the most important that a company can make. They are based on the financial strategy of the company.

These long-term investments include such activities as building a new plant, introducing a new product and acquiring another company.

To evaluate capital budgets, management has to consider the future incremental cash flows that a project can generate and consider the **cost of capital** associated with the project. Most capital investments require a substantial outlay up front but also the expectation that the project will generate incremental cash flows for multiple accounting periods. Incremental cash flows refer to the change in the total cashflows of the company after the investment is made.

The financing part of the financial strategy focuses on the cost of capital, which is the return that providers of capital – creditors and owners – require with their investment in the company.

A project's cost of capital reflects the cost of funds, considering the risk of the project.

An investment in plant, property and equipment represents an investment in a long-lived asset that will generate benefits in the long term. To support these long-term investments, companies invest in operating assets that are short-lived or current assets. These operating assets support day-to-day operations.

These operating assets are the company's working capital because they are put to work to support the long-term investments of the company and consist of cash, inventory and accounts receivable. The amount of working capital a company needs is determined, in large part, by the company's line of business.

To manage short-term obligations management needs to understand what the company will need in regards to short-term borrowing and trade credit. There are different types of short-term financing. Financial manager must consider the cost of each type.

One important consideration for every company is its capital structure, as discussed in Chapter 8. A company's capital structure is the mix of debt and equity that the company uses to finance its business. Debt capital consists of interest-bearing debt obligations, which may be notes or bonds, whereas equity capital consists of stock issues and retained earnings and is the ownership interest in a business enterprise.

Companies have to pay interest on loans as well as the principal but they do not have similar obligations with stock. As a result, there is some risk when utilising debt. Ultimately, financial risk is affected by how management designs the capital structure of a company. The more debt capital a company has relative to equity capital, the greater the financial risk.



Different industries balance short-term liabilities, long-term liabilities and equity differently, due to customary practices, credit practices or business risk.

The relative amounts of liabilities and equity raise a number of questions including: How do companies choose between debt and equity? What is the best avenue to raise equity, retained earnings or new equity? Why does a company choose to go public? How do companies choose between bank debt and public debt? What are the factors that determine whether a company accesses short-term money market versus bank borrowings? These basic questions represent the core of liability management for corporations, which we refer to as their corporate financing decisions and are at the heart of a financial strategy.

Illustrative Example 4

A chief information officer (CIO) would like to discuss with you the key elements of capital budgeting. What are the basic areas of capital budgeting?

Capital budgeting considers some basic questions, including:

1. How does a firm decide to expand its existing buildings or to construct or buy another building?
2. How does a firm decide to replace machinery and equipment? Should companies continue to use machinery just because it is in working order?
3. How do firms choose between buying or leasing machinery and equipment?

Key Learning Point

A financial strategy outlines the short- and long-term goals of an organisation and can have a real impact on the liquidity and solvency of a company.

► Knowledge Check Questions

Question 5

Identify which of the following is not one of the three main components of financial strategy.

- A** Financing
- B** Investment
- C** Dividend
- D** Operations

Question 6

Describe the kind of cash flow that is used in evaluating a company's capital budget.



9.4 GAP BETWEEN CREDIT POLICY AND CREDIT TERMS

A basic problem companies often face is that profits are not cash and a firm can only pay its bills with cash.

Without considering other investments, companies receive cash from sales, or more precisely, accounts receivable. It is reasonable to assume that the percentage of sales will remain constant as long that credit policies remain constant, the mix between credit sales and cash sales does not change and the economy does not crash. More lenient credit terms could lead to an increase in receivables as well as a higher proportion of credit to cash. These same things might happen when the economy goes through a downturn. It is important to understand the macroeconomic environment that generates the sales forecast to refine the percentage of sales tied in receivables.

9.4.1 Cash Flow and Credit

Cash flow from operations is the correct measure of the cash generating ability of a company. Depreciation and amortisation are non-cash charges and need to be added back to net income to get cash flow, but this is only part of the story; other cash items need to be considered in determining cash flow.

When firms sell on credit, they have to make sure that they are actually collecting the cash from those sales. Similarly, the statement of profit or loss and other comprehensive income includes only the cost of producing items that are sold. If production is unsold, it goes into inventory, but the firm has still paid to produce it, so there is still a cash outflow.

It is possible to define a company's credit policy as the percentage of sales collected this month and the balance collected with a one-month delay.

In practice, sales can be collected after two or more months, but the important thing is that by varying, the company's credit policy changes and this affects its cash flow from operations.

Rapid growth may lead to problems with cash flow, because investments may require more cash than sales can generate. This use of cash is called the **burn rate**. This is also why slow-growing profitable companies are known as 'cash cows'.

Companies that are growing quickly need more liquidity or external financing to manage a higher burn rate. Companies generally want to lower their burn rate.

The burn rate points to how long a company can continue operating before it has to raise more funds.

One way to lower the burn rate is to, essentially, collect receivables more quickly. This changes a company's **credit policy**.

For most companies, the decision of whether or not to extend credit – the 'trade credit decision' – depends on the business they are in and the policies of their competitors. If a firm extends credit, for example, it has to decide which of its customers qualify and this decision is based a formal credit analysis designed to assess the risk of non-payment.



This process involves collecting information about their credit history, financial strength, cash flows and overall financial stability.

9.4.2 Extending Credit

Before considering how firms make these decisions, it is important to note that a company's decision to extend trade credit is subtly different from a bank's decision to extend a loan.

Suppose, for example, a firm is considering a HK\$100,000 order for widgets. The customer promises to pay in 60 days. If the cost of trade credit is 12% per year, the firm might sell the widgets for HK\$102,000 (i.e. $\text{HK\$}100,000 + (\text{HK\$}100,000 \times 12\% \times 2/12)$), due in 60 days.

This is similar to a bank lending the customer HK\$100,000 to buy the widgets for cash, then charging 2% interest on the loan so that the customer pays HK\$102,000 in 60 days.

In both cases, the purchaser owes HK\$102,000, either to the company or to the bank. The bank's profit from the loan is limited to the interest rate and if the customer defaults, the bank could be out of the full HK\$100,000 loan.

However, the firm is in a slightly different position.

First, the firm's cost is not the HK\$100,000 it charges for the widgets, because it has to factor a profit margin into the calculation. If the firm's profit margin is 10%, and if it cannot sell the widgets to other customers, then its decision involves a potential loss of HK\$90,000 if the customer does not pay, rather than the HK\$100,000 the bank would lose.

Further, if a customer pays, it may make more purchases in future and generate more profits. By offering credit to customers that would not otherwise get similar terms from a bank, a firm might be putting more emphasis on developing a good customer at the risk of just losing its production costs.

This balance may also help explain why some firms grant credit and others do not. Companies with short collection periods, such as restaurants, may not need to give credit because the benefits are generally limited. Alternatively, a company in a high-margin industry such as car sales may benefit by offering to finance products.

Terms of credit can vary from one customer to another and the cost of credit is often based on the traditional terms in the form of a discount, discount period and net days. For example, terms of 2/10, net 30 indicate that a 2% discount is paid within 10 days or the full amount would be due at the end of 30 days.

What's more, this means the credit period is the time period after 10 days and the cash price is 100% minus 2%, or 98%. In turn, the rate of interest is 2% on the cash price.

When a company extends more lenient credit terms, it expects revenues to increase as the number of units sold also increases. The company may be able to charge higher prices. The company may also have higher costs, including higher financing costs and higher risk of non-payment.

On the other hand, more restrictive credit policies may result in faster collections but also lower sales. Companies can consider how best to make these decisions by using the net present value (NPV) framework, which is the present value of future cash flows minus initial cash outlays.

Future cash flows will, in broad strokes, be positive when companies loosen credit terms and negative when they tighten them. The loosening of credit terms might lead to cash outlays

from the required additional investment in receivables, while there is cash inflow when credit terms are tightened as the level of receivables goes down.

The after-tax cost of short-term debt is typically the appropriate discount rate because receivables are low risk and may be financed over the short-term debt.

Apply and Analyse 1

Apply

ABC Entertainment Ltd has a subsidiary that is considering a credit policy to allow customers to pay full price for purchases without penalties within 40 days of purchase, credit terms known as net 40. Currently, the company does not grant credit.

The company expects to be able to increase prices by HK\$1 per unit by adopting the new policy, leading to a new price per unit of HK\$111. Further, the company also expects sales to increase by 1,000 units per year (to 11,000 units) and to keep variable costs at HK\$99 per unit.

The company anticipates that bad debt losses will add up to HK\$6,000 per year.

The company also plans to use its line of credit to finance any required additional investment in receivables. It pays 6% interest on the line of credit. The tax rate of the firm is 40%. The question is whether the firm should extend credit under these terms?

Analysis

In the past, the firm did not have any receivables, so the initial investment may be considered as the additional funds that will be tied up in receivables. This amount equals the number of days that sales must be financed multiplied by estimated sales per day:

$$\begin{aligned} 40 \times \text{Sales per day} &= 40 \times [(111 \times 11,000 \text{ units}) / 365 \text{ days}] \\ &= 40 \text{ days} \times 3,345.21 / \text{day} = \text{HK\$}133,808 \end{aligned}$$

It is now necessary to estimate the PV (Future CFs), which equals the present value of the incremental after-tax cash flows generated by extending credit.

Incremental before-tax annual CFs = (1,000 extra units sold) $\times$ ((HK\$111 – HK\$99) profit per unit) + (10,000 units sold) $\times$ (HK\$1 increase in price per unit sold) – (HK\$6,000 in bad debt losses) = (1,000) (HK\$12) + (10,000) (HK\$1) – HK\$6,000 = 12,000 + 10,000 – 6,000 = HK\$16,000

The appropriate after-tax discount rate is $6\% \times (1 - 0.4) = 3.6\%$

If we assume the firm reaps the benefits of this change in policy indefinitely, we can find the present value of the future benefits by viewing the incremental after-tax annual CFs as a perpetuity:

$$\begin{aligned} \text{PV (Future CFs)} &= \text{Annual after-tax incremental CFs} / k \\ &= [16,000 \times (1 - 0.4)] / 0.036 = \text{HK\$}266,667 \\ \text{NPV} &= \text{PV (Future CFs)} - \text{CF}_0 = 266,667 - 133,808 = \text{HK\$}132,859 \end{aligned}$$

Thus, it might be best for the subsidiary to begin extending credit under the terms described above.

Key Learning Point

One way to lower the burn rate is to collect receivables more quickly. This changes a company's credit policy.

► Knowledge Check Questions**Question 7**

Identify which of the following best describes the period of time that a company can operate before it has to raise funds, either by issuing stock or borrowing.

- A** Cash burn
- B** Burn rate
- C** Net working capital
- D** Working capital

Question 8

Identify which of the following companies will have a low collection period.

- A** Computer manufacturer
- B** Fast food shop
- C** Electronic appliance producer
- D** Mining company



9.5 ROLES OF CASH FLOW ANALYSIS ON STRATEGIC AND OPERATIONAL PLANS

Working capital is the sum of a company's current assets and **net working capital** (NWC) is the difference between current assets and current liabilities. Working capital management refers to how the company manages its current assets such as cash and marketable securities, accounts receivable and inventories and its current liabilities like accounts payable, notes payables and short-term borrowings.

It is critical to manage working capital well for the viability of a company, both short- and long-term. The mismanagement of working capital can lead to liquidity problems and may even force a company into bankruptcy.

Companies have to manage working capital on a day-to-day basis but decisions regarding the capital structure, dividend policy and capital budgeting typically are made quite infrequently.

Defining what constitutes good working capital management can be complicated but it does have a number of characteristics:

1. Maintaining optimal cash balances.
2. Investing excess funds in marketable securities with the best possible returns while considering risks.
3. Managing accounts receivables properly.
4. Managing inventory efficiently.
5. Choosing the right level of short-term financing in the cheapest and most flexible manner.

Profits are not cash and a company can only pay its bills with cash. A business with profits, but no cash, cannot survive. Therefore, it is important for financial managers to be aware of the cash flow consequences of decisions.

Consider the example of Well View Company Ltd, which produces touch-screen display screens for e-readers for HK\$75 each and sells them for HK\$100 each. Well View has a policy of keeping a 30-day supply of inventory on hand and requires customers to pay within 30 days.

For the months January through May, Well View Company Ltd has the following activity:

Each month, Well View Company Ltd produced enough screens for 30 days.

For example, it sold 1 million screens in January, but produced 1.5 million to meet demand in February. And in February it sold 1.5 million units but produced 2 million to meet demand in March.

Therefore, its inventory activity was:

Number of screens (in thousands)				
Month	Beginning balance	Add: Produced	Subtract: Sold	Ending balance
January	1,000	1,500	1,000	500
February	500	2,000	1,500	1,000
March	1,000	2,500	2,000	1,500
April	1,500	3,000	2,500	2,000
May	2,000	3,500	3,000	2,500

The cost of goods sold and ending balance of inventory, with each screen with a unit cost of HK\$75, are:

(in HK\$ millions)		
Month	Cost of goods sold	Ending balance in inventory
January	75	37.5
February	112.5	75
March	150	112.5
April	187.5	150
May	225	187.5
Total	750	

Well View Company Ltd's customers pay in 30 days, so the company collects January's sales in February; February sales in March, and so on:

Accounts Receivable (in HK\$ millions)				
Month	Beginning balance	Add: sales on credit	Subtract: collections	Ending balance
January	100	100	100	100
February	100	150	100	150
March	150	200	150	200
April	200	250	200	250
May	250	300	250	300

Companies produce four statements:

1. the statement of profit or loss and other comprehensive income;
2. the statement of financial position;
3. the statement of cash flows; and
4. the statement of changes in equity.

Of the four statements, financial analysts tend to focus on the statement of cash flows because the other statements usually have accounting adjustments that make it difficult to find the financial issues.

Exhibit 9.2 shows a simplified statement of profit or loss and simplified statement of cash flows for Well View Company Ltd as of 1 June. The focus of this simplified example is on the importance of working capital.

January through 1 June (in HK\$ millions)	
Statement of profit or loss	
Sales	1,000
Cost of goods sold	750
Net income	250
Statement of cash flows	
Net income	250
Increase in accounts receivable	200
Increase in inventory	150
Less: Change in working capital	350
Cash flow from operations	–HK\$100

► **EXHIBIT 9.2** Statement of profit or loss and statement of cash flows

Keeping things simple, Well View Company Ltd does not have any investment or financing cash flows, so therefore the change in cash is the same as the cash flow from operations.

The company is profitable. Adding up sales from January through May, it is possible to see that the company had sales of HK\$1 billion. The screens had a HK\$75 cost, so Well View had HK\$750 million in cost of goods sold and therefore an operating profit was HK\$250 million.

So, if Well View Company Ltd has a profit of HK\$250 million, is the company doing OK?

The problem is the change in working capital. Well View uses credit to sell screens to customers. Customers pay with a 30-day lag. In typical trade fashion, customer order screens, the company ships them along with an invoice and the terms of payment, which are 30 days.

By 1 June, Well View's accounts receivable has increased from HK\$100 to HK\$300 million, an increased investment in receivables of HK\$200 million. The reason for the increase is the increased sales level from HK\$100 million in January to HK\$300 million in May, meaning that Well View granted more credit to its customers, even though its credit policy had not changed.

Similarly, Well View Company Ltd has an inventory policy of keeping a 30-day supply on hand so that when customers place orders, Well View can box the order and send it out immediately.

In January, Well View Company Ltd had 1 million screens on hand worth HK\$75 million. On 1 June the company had 3 million screens worth HK\$75 each or HK\$225 million, 2 million more in June than in January, or HK\$150 million.

The company increased its inventory because of an increase in sales from HK\$100 to HK\$300 million. As a result, Well View has to increase inventory by HK\$200 million worth of screens to keep a month worth of sales at hand.

Well View has a higher investment in inventory of HK\$150 million. The company pays its bills right away so it has no accounts payable. The change in net working capital is the change in receivables plus inventory minus any change in payables, i.e. –HK\$350 million.

When the change in net working capital is added to operating cash flows, the cash flow from operations ends up at –HK\$100 million. Because Well View started with HK\$100 million cash in the bank, by the end of May the cash has been depleted and the company has run out of cash to finance its operations. At this point, the company's cheques would bounce.

It is possible to confirm this by tracing Well View Company Ltd's cash flows through this period:

Month	Cash Flows (in HK\$ millions)			
	Beginning cash balance	Collections from customers	Payments to suppliers	Ending cash balance
January	100.0	100	112.5	125.0
February	125.0	100	150.0	75.0
March	75.0	150	187.5	37.5
April	37.5	200	225.0	12.5
May	12.5	250	262.5	0.0



It is important to note that cash flow from operations is the correct measure of the cash generating ability of the company.

If Well View Company Ltd had a cash-only sales policy and did not extend credit, then with the same level of sales, the company would have HK\$1,000 million in cash. But, because the company grants 30-day credit, it did not get any money from the HK\$300 million of sales in May.

Companies that sell on credit have to make sure to collect on those sales but they also have to plan for a lag in receiving cash. Similarly, the statement of profit or loss and other comprehensive income includes only the cost of producing items that the company sells.

When production is not sold, it goes into inventory but the company still had cash outlays to produce the inventory. The most important net working capital items are the increase in receivables and inventory, net of payables.

Illustrative Example 5

Another subsidiary of ABC Entertainment Ltd, DEF Ltd, has the following financial information.

Net operating income	HK\$40,000
Depreciation on equipment and building	10,000
Sale of land	25,000
Purchase of equipment	15,000
Retirement of long-term debt	30,000
Dividends declared	15,000
Long-term investments	10,000

Which of these are sources of funds? Which are uses of funds?

The sources of funds are the net operating income; depreciation and sale of land.

$$40,000 + 10,000 + 25,000 = \text{HK\$}75,000$$

The uses are purchase of equipment; retirement of long-term debt; dividends and long-term investments.

$$15,000 + 30,000 + 15,000 + 10,000 = \text{HK\$}70,000$$

Key Learning Point

Managing working capital effectively is critical to both the short-term and the long-term viability of any company.

► Knowledge Check Questions

Question 9

Describe how net working capital is calculated.

Question 10

Identify which of the following actions will increase the net working capital, if other factors remain unchanged.

- A** Increase in inventory
- B** Decrease in cash
- C** Increase in accounts payable
- D** Increase in long-term debt

9.6 STATEMENT OF CASH FLOW PREPARATION AND ANALYSIS

The statement of cash flows provides financial information about cash receipts and payments for a specific period of time.

To help stakeholders such as investors, creditors and others analyse the cash position of a company, the statement of cash flows reports the effects on cash of any activities related to operating, investing and financing. The statement also shows any net increases or decreases in cash and how much cash the company has at the end of the period.

Reporting any sources, uses and changes in the cash position of a company is useful because it helps investors, creditors and others understand what is happening with the company's cash. The statement of cash flows provides answers to questions like: Where did cash come from during the period? What was cash used for? What was the change in the cash balance during the period?

Other financial statements such as the statement of financial position or the statement of profit or loss and other comprehensive income provide only limited information about a company's cash flows.

For example, statements of financial position show the increase in property, plant and equipment during the year. However, they do not show how the additions were financed or paid. The statement of profit or loss and other comprehensive income shows net income. However, it does not indicate the amount of cash generated by operating activities.

The statement of changes in equity shows cash dividends declared but not payments of cash dividends for the year. None of these statements shows a detailed summary of the source of cash and how it was being used.

The statement of cash flows reports cash receipts, payments and the net change in cash resulting from operating, investing and financing activities during a specific period. The information in this statement helps stakeholders assess:

- The company's ability to generate future cash flows. Investors may be able to foresee the amounts, timing and uncertainty of future cash flows by analysing the relationships between the various items in the statement.
- The company's ability to pay dividends and meet obligations, which is particularly important for employees, creditors and shareholders.
- Any reasons for differences between net income and net cash that may result from operating activities. Net income helps explain the success or failure of a company. Some users of these statements are judgemental of accrual-basis net income because it may require some estimates, which may not always be reliable. This is not the case when it comes to cash.
- The cash investing and financing transactions during the period. By examining these transactions of a company, the reader of a financial statement may be able to understand why assets and liabilities changed during the period in question.

The statement of cash flows classifies receipts and payments as operating, investing and financing activities:

- Operating activities, including the cash effects of transactions that create revenue and expenses.
- Investing activities, such as acquiring and disposing of investments and property, plant and equipment and lending money and collecting the loans.
- Financing activities, such as obtaining cash from debt issuance and repaying borrowings as well as obtaining cash from shareholders, repurchasing shares and paying dividends.

The operating activities is important as it shows the cash generated by company operations, which may be the most accurate measure of the ability of the company to generate cash to continue operating as a going concern.

Exhibit 9.3 lists standard cash receipts and cash payments within each of the three classifications.

Types of Cash Inflows and Outflows

Operating activities – statement of profit or loss items

Cash inflows:

From sale of goods or services.

From interest received and dividends received.

Cash outflows:

To suppliers for inventory.

To employees for wages.

To government for taxes.

To lenders for interest.

To others for expenses.

► **EXHIBIT 9.3 Typical cash receipts and payments** (Source: Weygandt 2016. Reproduced with permission of John Wiley & Sons.)

Types of Cash Inflows and Outflows

Investing activities – changes in investments and long-term assets

Cash inflows:

- From sale of property, plant and equipment.
- From sale of investments in debt or equity securities of other entities.
- From collection of principal on loans to other entities.

Cash outflows:

- To purchase property, plant and equipment.
- To purchase investments in debt or equity securities of other entities.
- To make loans to other entities.

Financing activities – changes in long-term liabilities and shareholders' equity

Cash inflows:

- From sale of common stock.
- From issuance of debt (bonds and notes).

Cash outflows:

- To shareholders as dividends.
- To redeem long-term debt or reacquire capital stock (treasury stock).

► EXHIBIT 9.3 (continued)

The general guidelines are:

- Operating activities include statement of profit or loss items.
- Investing activities include cash flows resulting from changes in investments and long-term asset items.
- Financing activities include cash flows resulting from changes in long-term liabilities and shareholders' equity items.

Certain cash flows related to investing or financing activities can be classified as operating activities. For example, receipts of investment revenue (interests and dividends) can be classified as operating activities; so are payments of interest to lenders. These are considered as operating activities because companies report these items in the statement of profit or loss and other comprehensive income, where results of operations are shown.

Cash may not be involved in all of a company's significant activities. The list shows examples of significant non-cash activities:

- Direct issuance of common stock to purchase assets.
- Conversion of bonds into common stock.
- Direct issuance of debt to purchase assets.
- Exchanges of plant assets.



Significant financing and investing activities that are unaffected by cash are not reported in the statement of cash flows. These activities are, however, reported in either a separate schedule at the bottom of the statement of cash flows or in a separate note or supplementary schedule to the financial statements.

Investors and creditors typically use ratios based on accrual accounting but cash ratios are gaining acceptance.

In the statement of cash flows, net cash from operating activities is supposed to point to the ability of a company to generate cash. However, net cash generated by operations does not consider what a company must invest in new fixed assets to maintain its operations at current levels.

Companies are generally also expected to maintain dividends at current levels to keep investors happy. The measurement of free cash flow provides some insight on the ability of a company to generate cash.

9.6.1 Free Cash Flow

Free cash flow is an important measurement since it shows how efficient a company is at generating cash. Investors use free cash flow to measure whether a company might have enough cash, after funding operations and capital expenditures, to pay investors through dividends and share buy-backs.

Free cash flow defines the net cash provided by operating activities after adjusting capital expenditures and dividends.

In this following example, suppose MPC Ltd produced and sold 10,000 keyboards this year. It reported HK\$1,000,000 net cash provided by operating activities. In order to maintain production at 10,000 keyboards, MPC Ltd invested HK\$150,000 in equipment. It chose to pay HK\$50,000 in dividends. Its free cash flow was HK\$800,000 ($\text{HK\$1,000,000} - \text{HK\$150,000} - \text{HK\$50,000}$).

The company could use this HK\$800,000 either to purchase new assets to expand the business or to pay an HK\$800,000 dividend and continue to produce 10,000 keyboards.

Exhibit 9.4 provides another example. It provides basic information (in millions of US dollars) taken from the 2015 statement of cash flows of Microsoft Corporation.

Microsoft generated approximately US\$13.25 billion of free cash flow. This incredible amount of cash generated in a single year can be used for the acquisition of new assets, the retirement of stock or debt or the payment of dividends.

Also note that Microsoft's cash from operations of US\$29.1 billion is greater than its 2015 net income of US\$12.2 billion. This lends additional credibility to Microsoft's income number as an indicator of potential future performance.

Microsoft Corporation Statement of Cash Flows (partial) 2015		
Cash provided by operating activities		HK\$29,080
Cash flows from investing activities		
Additions to property and equipment	HK\$(5,944)	
Purchases of investments	(98,729)	
Sales of investments	70,848	
Acquisitions of companies	(3,723)	
Maturities of investments	15,013	
Other	(466)	
Cash used by investing activities		(23,001)
Cash paid for dividends		(9,882)

Microsoft's free cash flow is calculated as below:

Cash provided by operating activities	HK\$29,080
Less: Expenditures on property, plant and equipment	5,944
Dividends paid	9,882
Free cash flow	HK\$13,254

► **EXHIBIT 9.4 Microsoft cash flows in 2015** (Source: Weygandt 2016. Reproduced with permission of John Wiley & Sons.)

Illustrative Example 6

The subsidiary New Territories Ltd of ABC Investment Ltd has produced the following statement of cash flows. The group's CIO needs to know the free cash flow generated.

Figures in HK\$ million

Cash flows from operating activities		
Net income		19,000
Adjustments to reconcile net income to net cash provided by operating activities:		
Depreciation expense	8,100	
Loss on disposal of plant assets	1,300	
Decrease in accounts receivable	6,900	
Increase in inventory	(4,000)	
Decrease in accounts payable	(2,000)	10,300
Net cash provided by operating activities		29,300
Cash flows from investing activities		
Sale of investments	1,100	
Purchase of equipment	(19,000)	

Illustrative Example 6 (continued)

Net cash used by investing activities		(17,900)
Cash flows from financing activities		
Issuance of stock	10,000	
Payment on long-term note payable	(5,000)	
Payment for dividends	(9,000)	
Net cash used by financing activities		(4,000)
Net increase in cash		7,400
Cash at beginning of year		10,000
Cash at end of year		17,400

Computing the free cash flow for New Territories Ltd makes it possible to generate better information than 'net cash provided by operating activities'.

$$\text{Free cash flow} = 29.3 \text{ billion} - 19 \text{ billion} - 9 \text{ billion} = \text{HK\$1.3 billion}$$

Net cash provided by operating activities does not account for the fact that the maintenance of the current level of operations requires a company to invest in new plant assets. Dividends must be maintained at current levels to satisfy investors. The measurement of free cash flow provides additional insight about the ability of a company to generate cash.

9.6.2 Two Methods of Preparing a Statement of Cash Flows

There are two methods of preparing statement of cash flows. The direct method seeks to systematically track all transactions and events that involve cash movements. This can be achieved by appropriate coding of input data to distinguish cash transactions from accrual entries, or it might be worked out from cash-based accounting before it is amended by adjusting entries.

The indirect method is used to calculate the cash flows by working from accrual-based balances. The prime example of this is to calculate cash flow from operating activities by adjusting profit for the year for non-cash transactions (e.g. depreciation).

The indirect method is illustrated in Exhibit 9.5 below. The direct method is then shown in Exhibit 9.6.

In theory these two methods should come to the same cash flows, though the former is capable of being set up to achieve the required classifications more accurately and in more detail and once automated can be produced without extra effort.

Indirect method statement of cash flows	
Cash flows from operating activities	
Profit before taxation	3,350
Adjustments for:	
Depreciation	450
Foreign exchange loss	40
Investment income	(500)
Interest expense	<u>400</u>
	3,740
Increase in trade and other receivables	(500)
Decrease in inventories	1,050
Decrease in trade payables	<u>(1,740)</u>
Cash generated from operations	2,550
Interest paid	(270)
Income taxes paid	<u>(900)</u>
<i>Net cash from operating activities</i>	1,380
Cash flows from investing activities	
Acquisition of subsidiary × net of cash acquired	(550)
Purchase of property, plant and equipment	(350)
Proceeds from sale of equipment	20
Interest received	200
Dividends received	<u>200</u>
<i>Net cash used in investing activities</i>	(480)
Cash flows from financing activities	
Proceeds from issue of share capital	250
Proceeds from long-term borrowings	250
Payment of finance lease liabilities	(90)
Dividends paid	<u>(1,200)</u>
<i>Net cash used in financing activities</i>	(790)
Net increase in cash and cash equivalents	110
Cash and cash equivalents at beginning of period	120
Cash and cash equivalents at end of period	<u>230</u>

► **EXHIBIT 9.5** Statement of cash flows – indirect method (HK\$ million)

Direct method statement of cash flows		
Cash flows from operating activities		
Cash receipts from customers	30,150	
Cash paid to suppliers and employees	(27,600)	
Cash generated from operations	2,550	
Interest paid	(270)	
Income taxes paid	(900)	
<i>Net cash from operating activities</i>		1,380
Cash flows from investing activities		
Acquisition of subsidiary X, net of cash acquired	(550)	
Purchase of property, plant and equipment	(350)	
Proceeds from sale of equipment	20	
Interest received	200	
Dividends received	200	
<i>Net cash used in investing activities</i>		(480)
Cash flows from financing activities		
Proceeds from issuance of share capital	250	
Proceeds from long-term borrowings	250	
Payment of finance lease liabilities	(90)	
Dividends paid	(1,200)	
<i>Net cash used in financing activities</i>		(790)
Net increase in cash and cash equivalents		110
Cash and cash equivalents at beginning of period		120
Cash and cash equivalents at end of period		230

► **EXHIBIT 9.6** Statement of cash flows – direct method (HK\$ million)

Key Learning Point

The primary purpose of a statement of cash flows is to provide financial information about the cash receipts and cash payments of a company for a specific period of time.

► Knowledge Check Questions

Question 11

Identify which of the following best describe the activities of lending money and collecting loans.

- A** Operating activities.
- B** Investing activities.
- C** Financing activities.
- D** Non-cash investing and financing activities.

► Knowledge Check Questions (*continued*)

Question 12

Identify which of the following is not reported in the statement of cash flows.

- A** Cash receipts from operating activities.
- B** Cash payments from investing activities.
- C** The net changes in cash.
- D** Credit sales.

9.7 FUNCTION OF CASH FLOW REPORTING SYSTEMS ON FINANCIAL STRATEGY

For many companies, cash flow problems are not always apparent. What is needed to help predict a cash flow problem is the statement of cash flows that reports cash every month.

To this end, companies create a cash budget, which is essentially a cash flow statement projected for each month (i.e. refer to Module 2, Chapter 9).

The important feature about the cash budget is that it allows the financial manager to forecast cash inflows and outflows over a forecast horizon and to see the cumulative effect on the company's cash balances, and in turn makes it easier to develop and stick to a financial strategy.

Typically, companies prepare a cash budget for at least the upcoming year on at least a monthly (and sometimes even a weekly or daily) basis. These cash budgets are important planning tools for the company and help shape their financial strategy.

For example, a cash budget can indicate when and for how long a company can expect to have excess cash balances that it can invest in marketable securities. Cash budgets also show any additional borrowings that a company may need to cover cash shortfalls. The cash budget may help a company determine when to arrange for short-term borrowing.

Banks often ask for cash budgets when processing a loan application because they need to determine whether the loan is needed for a short period or is permanent financing.

Companies do not enhance their reputation with the people at the bank by asking for help at the last minute. At the very least, such an approach indicates poor cash management, and the lending officer would pay close attention to any loans made to such a company.

Instead, effective cash reporting can help a company predict cash needs and perhaps set up a facility whereby it could invest surplus funds at the start and end of the year. A bank could, in turn, design a borrowing and lending facility.

It is important for small companies to build a good relationship with the bank and establish a good reputation. The cash budget is a key tool towards these ends as it allows for better understanding of cash inflows and outflows of the company.



Sales forecasts, estimated production schedules and estimates of other major inflows or outflows are the major components of the cash budget.

The basic tool for forecasting the cash inflows and outflows through a company is the cash budget. Although it is helpful in identifying key items, it does not explain the cause of the problem. Instead, to identify problems, it is necessary to look at the drivers of cash flow, which include the credit policy, payment policy and inventory policy.

Most companies have a cash and cash equivalents line on their statement of financial position. The cash is in the form of currency and deposits at financial institutions, whereas cash equivalents are marketable securities with maturities less than three months. Cash and cash equivalents are the most liquid assets of a business entity and serve to meet the day-to-day operations of the business.

As discussed in chapter 8, companies hold cash for:

- Transactions motive: this refers to the cash that is required for a company's normal operations.
- Precautionary motive: the cash that companies keep on hand in the event of unanticipated outlays of cash, such as repairs.
- Finance motive: cash accumulated in anticipation of major outlays, like lump-sum loan repayments or dividend payments.
- Speculative motive: extra cash that companies keep to deal with any unexpected opportunities that may arise, including an opportunity to get a reduced price on raw materials.

Most analysts refer to 'cash on hand' when speaking of 'cash', that is, money in a current account and short-term marketable securities – essentially cash available almost instantaneously.

Investments in short-term marketable securities are often called 'near-cash' or cash equivalents. They can be quickly and inexpensively converted into cash. Examples are treasury bills, bankers' acceptances or commercial paper. They are all part of the financial strategy of a company.

The company's ability to borrow on a line of credit or to have a bank overdraft also helps determine how much cash and cash equivalents a company holds. Most companies hold more liquidity in borrowing facilities than in cash and equivalents, because the cost of these facilities is relatively cheap compared to the cost of raising capital to invest in marketable securities.

In particular, precaution and speculation can all be satisfied by using near-cash items, such as investments in marketable securities, and by maintaining additional short-term borrowing capacity.

What is the optimal amount of cash companies should maintain on hand? Like most decisions in finance, it comes down to a classic trade-off between risk and expected return. Effective cash reporting can assist in this decision.

Cash and near-cash generally provide low returns compared to other investments but they provide liquidity and have minimal additional risks. A good rule of thumb is to assume that cash is low-risk, low-expected return.

A company can be conservative and maintain a large portion of assets in cash and near-cash, which minimises the risk of not being able to satisfy liquidity requirements. On the other hand, the approach sacrifices potential returns.

Other companies may choose a more aggressive approach to cash management, keeping minimal balances in hand. This approach may lead to higher returns but also increases risks of becoming illiquid. This is a problem because it is often expensive to create liquidity quickly through facilities such as short-term loans with higher borrowing rates. At other times, companies may be forced to sell assets, such as inventory or even fixed assets, at discounted prices, which may also be expensive.

The optimal amount of cash balances the risks of illiquidity with any sacrificed returns to maintain cash. The optimal cash balance, as a percentage of total assets, varies from one company to another. Some companies have more predictable cash requirements, and will require less cash compared to others with less predictable cash flows.

Near-cash items may help tackle the issues associated with the low returns and holding cash without sacrificing too much liquidity. A company may, for instance, set up a line of credit or keep more of their liquidity in money market instruments, which provide higher returns than cash.

The general approach to good cash management is to speed up inflows and slow down outflows as much as possible. The qualifier 'as much as possible' is important because companies always face constraints that will delay inflows and prevent them from delaying outflows too much.

For example, a company may speed up inflows from sales by giving out less credit, but this may also impact sales. At the same time, a company may delay payments to suppliers, but this risks hurting those relationships.

A key factor in speeding up cash inflows is putting in place an efficient credit policy, which specifies the policies for collecting receivables.

Illustrative Example 7

Your subsidiary Caine Ltd has submitted the following monthly cash flow items for the current month. It is necessary to calculate the ending cash balance of the current month.

Beginning cash	HK\$10,000
Prior month sales	50,000
Current month sales	80,000
Current purchase	35,000
Prior month purchase	20,000
Sale of old equipment	70,000
Dividend payment	60,000
Wages and others	35,000
Inflows = 50,000 + 80,000 + 70,000 = HK\$200,000	
Outflows = 35,000 + 20,000 + 60,000 + 35,000 = HK\$150,000	
Ending balance = 10,000 + 200,000 – 150,000 = HK\$60,000	

Key Learning Point

The important feature about the cash budget is that it allows the financial manager to forecast cash inflows and outflows over a forecast horizon and to see the cumulative effect on the company's cash balances, and in turn makes it easier to develop and stick to a financial strategy.

► Knowledge Check Questions**Question 13**

Identify which of the following is not a motive for holding cash.

- A** Transactions motive
- B** Precautionary motive
- C** Finance motive
- D** Investing motive

Question 14

Identify which of the following is the correct formula to determine ending cash balance.

- A** Beginning cash balance + collections from customers – payments to suppliers = ending cash balance.
- B** Beginning cash balance – collections from customers + payments to suppliers = ending cash balance.
- C** Beginning cash balance + collections from customers + payments to suppliers = ending cash balance.
- D** Beginning cash balance – collections from customers – payments to suppliers = ending cash balance.

9.8 IMPACT OF PROFITABILITY PROJECTIONS ON FINANCIAL STRATEGY

To do a reasonable job of financial forecasting, it is necessary to completely understand the relationships in the company's financial statements through a comprehensive financial analysis.

Financial forecasting is important to make finance decisions. Without knowing its financing needs, companies cannot know if it needs short-term or long-term funds and whether its financing needs will grow over time.

Financial advisers and bankers must understand the needs of the company to recommend a strategy or structure loans or credit facilities. At the same time, equity investors have to understand financing needs when investing in a company to avoid investing in a company that suddenly will need to raise more money in the future.

An accurate sales forecast is the most important input in financial forecasting because sales determine the financing needs of a company. So, all managers have input into the forecasting process but the input from marketing managers is particularly important. The **percentage of sales method** is the most basic forecasting technique.

The percentage of sales method involves the following steps:

1. Determine which financial policy variables you are interested in.
2. Set all the non-financial policy variables as a percentage of sales.
3. Use percentage of sales to extrapolate the statement of financial position.
4. Estimate of future retained earnings.
5. Re-examine and refine the forecast until it makes sense.

Treasurers are concerned with the financial policy variables, such as common equity, long-term funds or total external financing requirements. For instance, assume the treasurer is interested in total **external financing requirements (EFR)**.

Consider a simple statement of financial position for a hypothetical firm, shown in Exhibit 9.7 to see what this means. This firm has total assets of HK\$150 million that have been financed with HK\$80 million of common equity, HK\$40 million of long-term debt, and HK\$20 million of short-term debt. Taken together, these are referred to as invested capital, because both the common shareholders and the bank and long-term debt investors have made a decision to invest in the firm.

Statement of financial position (HK\$ million)			
Cash	5	Accruals	5
Securities	10	Payables	5
Receivables	10	Bank debt	20
Inventory	25	Current liabilities	30
Current assets	50	Long-term debt	40
Net fixed assets	100	Common equity	80
Total assets	150	Total liabilities and equity	150

► **EXHIBIT 9.7 Hypothetical statement of financial position (HK\$ million)**
 (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

On the other hand, accruals and payables are referred to as **spontaneous liabilities**. Companies normally generate payables and accruals as long as they are in business: they have credit with suppliers and employees are paid wages at the end of every month.

A five-year financial forecast assumes accounts automatically increase with sales and only financial policy variables stay constant.

The next step is to convert non-policy variables to a percentage of sales. For example, assume sales are HK\$120 million and that this sales level generates reasonable future percentages.

Analysts usually look at ratios over the previous five years or so and factor in current conditions to determine reasonable values going forward.

The next step is to extrapolate the statement of financial position based on the sales forecast by using these percentages of sales. For this example, assume that sales are forecast to increase by 10% a year. The result is a basic initial forecast shown in Exhibit 9.8.

	Base case	% of total asset	10% sales increase in year 1	10% sales increase in year 2	10% sales increase in year 3
Sales	120		132	145	160
Cash	5	4.2	5.5	6.1	6.7
Marketable securities	10	8.3	11	12.1	13.3
Accounts receivable	10	8.3	11	12.1	13.3
Inventory	25	20.8	27.5	30.3	33.3
Net fixed assets	100	83.3	110	121	133
Total assets	150	125	165	181.6	199.6
Accruals	5	4.2	5.5	6.1	6.7
Accounts payable	5	4.2	5.5	6.1	6.7
Short-term debt	20	16.7	20	20	20
Long-term debt	40	33.3	40	40	40
Equity	80	66.7	80	80	80
Total liabilities and equity	150	125	151	152.2	153.4
Cumulative EFR			14	29.4	46.2

► **EXHIBIT 9.8 Initial forecast (HK\$ million)** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

The top line is the sales forecast, with sales going from the current level of HK\$120 million to HK\$132 million one year out, and then to HK\$145 million, and finally to HK\$160 million.

An analyst would then express asset accounts and spontaneous liabilities as a percentage of sales. These are non-policy variables. They are extrapolated based on the sales forecast.

For example, consider cash, which is HK\$5 million or 4.2% of the sales level of HK\$120. Based on a forecast of 10% sales growth, cash would increase by 10% to HK\$5.5 million, HK\$6.1 million and HK\$6.7 million.

The values of the other assets are also increased by 10% per year, resulting in a forecast value of HK\$199.6 million for total assets.

On the liability side, as discussed, it is possible to assume that spontaneous liabilities increase at the rate of sales, so it is possible to arrive at a forecast value of HK\$153.4 million.

Subtracting this figure from the total assets forecast of HK\$199.6 million, arrives at the required invested capital figure of HK\$186.2 million.

However, recall no change in the amount of invested capital was assumed, leaving only HK\$140 million in invested capital, which suggests a shortfall of HK\$46.2 million.

This shortfall is the statement of financial position plug, labelled as the EFR.

Notice that this is a cumulative not an annual requirement – that is, the firm needs to raise HK\$46.2 million in total financing over the next three years.

The forecast above is very simple. It ignores any new equity that the firm could generate through retained earnings and that existing debt will still be there three years from now, which is unlikely to be the case – some of it may be refinanced or banks may refuse to renew the short-term loans.

For the sake of the example, assume all debt can be renewed and focus on retained earnings.

To do this, it is necessary to look at the company's statement of profit or loss and other comprehensive income. A simple example is shown in Exhibit 9.9.

Statement of profit or loss (HK\$ million)	
Sales	120
Gross operating profit	48
Fixed costs	31
EBIT	17
Interest	5
Taxes (50%)	6
Net income	6
Dividends	3

► **EXHIBIT 9.9 Sample statement of profit or loss and other comprehensive income (HK\$ million)**

(Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

At the current level of sales of HK\$120 million, EBIT is HK\$17 million. From that, it subtracts HK\$5 million in interest on HK\$60 million of debt and then pays income taxes of HK\$6 million. As a result, the company has net income of HK\$6 million and a 5% net profit margin.

However, the company pays out 50% of earnings as dividends, so only HK\$3 million is retained within the firm. What is important is that the firm's retained earnings as a percentage of sales are its net profit margin, multiplied by one minus the dividend payout, which equals 2.5% in this example.

This information can be adapted to revise the initial forecast, adding 2.5% of sales each year to retained earnings, which leads to a first revision to the basic forecast, as shown in Exhibit 9.10.

	Base case	% of total asset	10% sales increase in year 1	10% sales increase in year 2	10% sales increase in year 3
Sales	120		132	145	160
Cash	5	4.2	5.5	6.1	6.7
Marketable securities	10	8.3	11	12.1	13.3
Accounts receivable	10	8.3	11	12.1	13.3
Inventory	25	20.8	27.5	30.3	33.3
Net fixed assets	100	83.3	110	121	133
Total assets	150	125	165	181.6	199.6
Accruals	5	4.2	5.5	6.1	6.7
Accounts payable	5	4.2	5.5	6.1	6.7
Short-term debt	20	16.7	20	20	20
Long-term debt	40	33.3	40	40	40
Equity	80	66.7	83.3	86.9	90.9
Total liabilities and equity	150	125	154.3	159.1	164.3
Cumulative EFR			10.7	22.5	35.3

► **EXHIBIT 9.10 Revised forecast (HK\$ million)** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

The only difference between this revision and the initial forecast is that it takes into account future retained earnings of 2.5% per Hong Kong dollar of sales, which reduces the EFR by HK\$10.7 million.

However, this forecast is still relatively basic and does not make use of skills in financial analysis.

It is possible to improve on this basic forecast. To do this, an analyst would go through all the non-financial policy assumptions to see whether they can be improved.

The percentage of sales method creates a strict relationship between assets and sales. For instance, the assumption that cash is 4.2% of sales assumes it will remain the same regardless of the level of sales, HK\$10 million or HK\$1 billion.

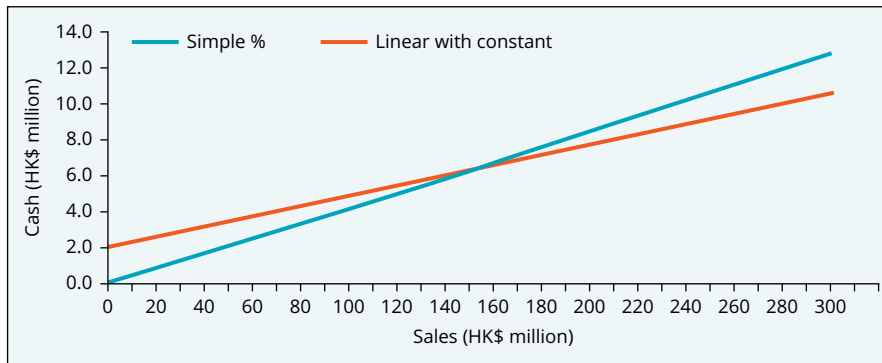
Graphically, this forces the relationship between cash and sales into a straight line that passes through the origin.

It would be more reasonable to assume that there are economies of scale associated with the management of cash and that the marginal impact of the growth in sales is less than the 4.2% of the average percentage of sales.

For instance, the relationship at a sales level of HK\$120 million might be HK\$2 million plus 2.5% of sales so that the marginal impact of sales growth is only 2.5% of sales.

Exhibit 9.11 depicts the differences in the cash forecasts that arise by changing this assumption.

How cash varies with sales can be determined by statistically analysing previous sales levels and cash balances. It is possible to use an ordinary least squares (OLS) linear regression to estimate a linear relationship with or without a constant. Software such as Microsoft Excel



► **EXHIBIT 9.11 Revised forecast** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

or statistical packages can be used for this regression. Assume that the treasurer is satisfied, for this forecast, that operations can be managed over a three-year horizon from existing cash balances without any more investment. As a result, cash will be constant at HK\$5 million.

Let's now consider accounts receivable. In this case, it is assumed to be a strict percentage of sales.

As long as conditions (e.g. credit policy and the mix between credit and cash sales) do not change, it is reasonable to assume that the percentage of sales does not change. However, changes to credit policy or the proportion of credit to cash sales could impact the percentage of sales tied up in receivables. An economic downturn could also have an impact on the percentage.

Assume that for the period of the forecast, it is reasonable to estimate receivables at 8.3% of sales in this case.

Inventory and receivables are treated in similar fashion. If the inventory turnover ratio is likely to be constant, inventory will increase along with sales.

However, if there are changes in the inventory management system and in the economic environment, the percentage of sales for inventory will change as well.

Net fixed assets is the next item to consider. In this case, assume that the constant percentage of sales for inventory is reasonable, consistent with the treatment of receivables. Whether a constant percentage of sales is reasonable depends on whether the firm has one plant or more. For single-plant companies, assuming a constant percentage of sales is unreasonable as investment in plant and equipment often lasts many years and the companies do not need to replace plant and equipment every year. For multi-plant companies, replacement of plant and equipment occurs over large numbers of plants.

These considerations circle back to the question of why a company might hold marketable securities. Companies do not create value by investing in marketable securities but hold them as a temporary place for cash until it is needed.

In this case, suppose the treasurer realises that the company needs HK\$10 million in marketable securities for a plant expansion in the next year but has no further capital expenditures in the forecast. This means that it might not make sense to extrapolate marketable securities into the future and, additionally, the value of plant and equipment will decline in future through depreciation.

Assume that the company depreciates the plant and equipment by HK\$10 million per year but spends HK\$10 million in marketable securities in the first year to maintain the plant and equipment at HK\$100 million.

The last item to consider are spontaneous liabilities, which act like accounts receivable because accounts payable is the major item. If the company does not expect to change its payment policy, payables will rise in line with sales. In this case, it is reasonable to assume payables will remain a constant percentage of sales.

Making the adjustments discussed above, it is possible to arrive at the second revision to the forecast, which is shown in Exhibit 9.12.

	Base case	% of total asset	10% sales increase in year 1	10% sales increase in year 2	10% sales increase in year 3
Sales	120		132	145	160
Cash	5	4.2	5	5	5
Marketable securities	10	8.3	0	0	0
Accounts receivable	10	8.3	11	12.1	13.3
Inventory	25	20.8	27.5	30.3	33.3
Net fixed assets	100	83.3	100	90	80
Total assets	150	125	143.5	137.4	131.6
Accruals	5	4.2	5.5	6.1	6.7
Accounts payable	5	4.2	5.5	6.1	6.7
Short-term debt	20	16.7	20	20	20
Long-term debt	40	33.3	40	40	40
Equity	80	66.7	83.3	86.9	90.9
Total liabilities and equity	150	125	154.3	159.1	164.3
Cumulative EFR			-10.8	-21.7	-32.7

► **EXHIBIT 9.12 Second forecast revision (HK\$ million)** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

This revision has different implications for the treasurer. The company had an external financing requirement in the past but it can now repay some financing obligations because EFR is negative. This is because the company does not need to increase cash or marketable securities and to spend more money on plant and equipment.

The **net profit margin** is the next step to consider. The previous assumption was that net profit margin remained the same as sales changed but if sales grow by 10% as forecasted due to a recovery from recession, the growth may happen without any new fixed costs. In this case, if the company's costs are fixed, net profit margin may increase.

Exhibit 9.13 shows the statement of profit or loss and other comprehensive income that arise if gross profits are kept at 40% of sales but maintain fixed costs at HK\$31 million.

Statement of profit or loss (HK\$ million)				
	Base case	Year 1	Year 2	Year 3
Sales	120	132	145	160
Gross margin (40%)	48	53	58	64
Fixed costs	31	31	31	31
Interest	5	5	5	5
Tax	6	8.5	11	14
Net income	6	8.5	11	14
Net profit margin	5%	6.4%	7.6%	8.8%

► **EXHIBIT 9.13 Statement of profit or loss (HK\$ million)** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

Notice that the profit margin increases dramatically from 5% to 8.8% over the three-year period. This is the typical 'recovery from recession' pattern, in which the firm lowers fixed costs during a recession, through restructuring and cost-cutting, so that the immediate impact of a recovery causes greatly improved profit margins.

Finally, companies do not generally follow a constant dividend payout policy but tend to increase dividends slowly over time in line with sustainable earnings. In this case, if management expects the dividend to stay at HK\$3 million for the next three years, retained earnings may increase faster than net income. If this higher level of retained earnings is added to the forecast, then a final forecast can be derived as shown in Exhibit 9.14.

	Base case	% of total asset	10% sales increase in year 1	10% sales increase in year 2	10% sales increase in year 3
Sales	120	%	132	145	160
Cash	5	4.2	5	5	5
Marketable securities	10	8.3	0	0	0
Accounts receivable	10	8.3	11	12.1	13.3
Inventory	25	20.8	27.5	30.3	33.3
Net fixed assets	100	83.3	100	90	80
Total assets	150	125	143.5	137.4	131.6
Accruals	5	4.2	5.5	6.1	6.7
Accounts payable	5	4.2	5.5	6.1	6.7
Short-term debt	20	16.7	20	20	20
Long-term debt	40	33.3	40	40	40
Equity	80	66.7	85.5	93.5	104.5
Total liabilities and equity	150	125	156.5	165.7	177.9
Cumulative EFR			-13	-28.3	-46.3

► **EXHIBIT 9.14 Final forecast (HK\$ million)** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)



The upshot of this final revision is that the surplus is even greater, and the treasurer has to develop a plan for managing these funds, rather than arranging for the firm to raise money.

In considering how the simple percentage of sales forecast changed, it is obvious that some effects were exaggerated. For example, fixed costs are only fixed in the sense that they do not vary with sales. However, they still tend to increase over time as a result of wage increases and the general tendency to hire more staff as the company's profits increase.

Likewise, it would be difficult for the firm to maintain the dividend at HK\$3 million as profits and cash start to pile up.

Different assumptions about the economic outlook could turn a surplus into a deficit.

For instance, if a slowdown is forecast, then a stable or declining sales forecast could lead to lower profit margins and require higher receivables and inventory as a percentage of sales.

There are multiple possible scenarios. The case here is to show how the simple percentage of sales forecasting method, when used in combination with basic financial analysis skills, could lead to an effective forecast.

On the other hand, if used blindly, the method may generate misleading results.

9.8.1 Internal Growth Rate

Management often has an interest in knowing the rate at which the firm can grow employing just internally generated funds of accumulated profits, i.e. addition to retained earnings. This rate is referred to the **internal growth rate (IGR)**. The IGR represents the maximum growth rate that a firm can achieve without external funding.

The application of what formula to use to calculate IGR depends on the how return on asset (net income divided by total asset) is calculated.

If the opening asset balance is used for calculating ROA, the following equation should be applied:

$$\text{Internal growth rate} = \text{Return on asset (ROA)} \times \text{Retention rate (} b \text{)}$$

where

ROA = Return on assets = Net income/Total assets

b = Plowback (retention) ratio

= Addition to retained earnings/Net income

= 1 – Dividend payout ratio

= 1 – Dividend/Net income

If the closing asset balance is used for calculating ROA, the sustainable growth rate can be calculated as:

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}$$

Illustrative Example 8

The return of assets (ROA) of Company C is 10.15% (closing asset balance is used), calculate the internal growth rate using the closing asset balance where the retention ratio of 65% is:

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}$$

$$\text{Internal growth rate} = \frac{10.15\% \times 65\%}{1 - 10.15\% \times 65\%} = 7.06\%$$

9.8.2 Sustainable Growth Rate

The **sustainable growth rate (SGR)** is the maximum rate of growth that a company can sustain without changing its capital structure.

As long as the calculation of ROA is matched up with the corresponding IGR formula, theoretically the IGR calculated using the two formulas either using the opening and closing assets balance will be exactly the same. The situation will also be applicable to the calculation of SGR.

The SGR formula using the opening equity balance is:

$$\text{Sustainable growth rate} = \text{Return on equity (ROE)} \times \text{Retention rate (} b \text{)}$$

where

ROE = Return on equity = Net income/Total equity

b = Plowback (retention) ratio

The SGR formula employing the closing equity balance is:

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - (\text{ROE} \times b)}$$

Illustrative Example 9

A company has the following information in its financial statements for the year ended 31 December 2020:

Interest-bearing debt	HK\$900,000
Shareholders' equity	HK\$2,500,000
Sales	HK\$1,050,000
Net income	HK\$670,000
Dividends	HK\$200,000
Sales growth	5%

Calculate both internal growth rate and sustainable growth rate either using 'opening' or 'closing' balance of assets and equity.

Internal growth rate (using 'opening' total assets balance in 2020):

$$\begin{aligned}\text{Opening shareholders' equity} &= \text{HK\$2,500,000} - (\text{HK\$670,000} - \text{HK\$200,000}) \\ &= \text{HK\$2,030,000}\end{aligned}$$

$$\begin{aligned}\text{Opening total asset} &= \text{total debt} + \text{total equity} = \text{HK\$900,000} + \text{HK\$2,030,000} \\ &= \text{HK\$2,930,000}\end{aligned}$$

$$\begin{aligned}\text{Return on asset (ROA)} &= \text{net income} / \text{total asset} = \text{HK\$670,000} / \text{HK\$2,930,000} \\ &= 22.87\%\end{aligned}$$

$$\begin{aligned}\text{Retention rate} &= 1 - (\text{dividend} / \text{net income}) = 1 - (\text{HK\$200,000} / \text{HK\$670,000}) \\ &= 0.7015\end{aligned}$$

$$\begin{aligned}\text{Internal growth rate} &= \text{Return on asset (ROA)} \times \text{Retention rate (} b \text{)} \\ &= 22.87\% \times 0.7015 = 16.04\%\end{aligned}$$

Internal growth rate (using 'closing' total assets balance in 2020):

$$\begin{aligned}\text{Closing total asset} &= \text{total debt} + \text{total equity} = \text{HK\$900,000} + \text{HK\$2,500,000} \\ &= \text{HK\$3,400,000}\end{aligned}$$

$$\text{ROA} = \text{net income} / \text{total asset} = \text{HK\$670,000} / \text{HK\$3,400,000} = 19.71\%$$

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - (\text{ROA} \times b)} = \frac{19.71\% \times 0.7015}{1 - (19.71\% \times 0.7015)} = 16.04\%$$

Sustainable growth rate (using 'opening' shareholders' equity balance in 2020):

$$\begin{aligned}\text{Return on equity (ROE)} &= \text{net income} / \text{shareholders' equity} \\ &= \text{HK\$670,000} / \text{HK\$2,030,000} \\ &= 33\%\end{aligned}$$

$$\text{Sustainable growth rate} = \text{Return on equity} \times \text{Retention rate} = 33\% \times 0.7015 = 23.15\%$$

Illustrative Example 9 (continued)

Sustainable growth rate (using 'closing' shareholders' equity balance in 2020):

ROE = net income/shareholders' equity = HK\$670,000/HK\$2,500,000 = 26.8%

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - (\text{ROE} \times b)} = \frac{26.8\% \times 0.7015}{1 - (26.8\% \times 0.7015)} = 23.15\%$$

Key Learning Point

Financial forecasting is critical for companies, its advisers and external analysts. Before making decisions about how to finance itself, the company has to assess its financing needs to determine if it needs funds for the short-term or long-term and whether its financing needs will grow over time.

► Knowledge Check Questions

Question 15

Explain the difference between invested capital and spontaneous liabilities.

- A** The firm knows who invests capital in the firm, but not the identity of who the firm owes its spontaneous liabilities.
- B** Invested capital is randomly determined, so the corresponding liabilities must be spontaneous and therefore also random.
- C** Invested capital is the amount of equity invested in the firm, while spontaneous liabilities are the sum of long-term and short-term debt.
- D** Invested capital is the result of investor decisions while spontaneous liabilities arise from the firm's business operations.

Question 16

Determine the three key steps to conduct a financial assessment of a business strategic plan.

- (i) Estimate revenue and expenses.
- (ii) Conduct market research of the products/services.
- (iii) Conduct a contribution analysis to determine if the business strategies positively contribute to the bottom line.
- (iv) Analyse the source of funding (i.e. equity or debt).
- (v) Combine all the numbers in one-year and three-year financial projections.

- A** (i), (ii) and (iii)
- B** (i), (iii) and (v)
- C** (ii), (iii) and (iv)
- D** (iii), (iv) and (v)



9.9 FUTURE PROFITABILITY PROJECTIONS FOR EVALUATING FINANCIAL STRATEGY

Expectation of future profits is important in determining the financial strategy of a company. A company that expects significant profits is likely to take a different approach to investing and raising capital than one that is not expecting to be profitable for some time.

Although the analytical methods discussed earlier in this chapter are critical components of the capital budgeting process, they are not always very useful in determining a course of action when, as is often the case, a company does not have sufficient money to invest in all available positive NPV projects.

In other words, they are not helpful in identifying the bundle or combination of positive NPV projects that creates the greatest total value for shareholders when there are capital constraints or **capital rationing**.

In an ideal world, companies would be able to accept all positive NPV projects because they would have the ability to finance them. The managers and investors would just need to agree on the projects with positive NPVs, and investors would provide capital to those projects because returns from them would be greater than the returns the investors could earn elsewhere in the capital markets.

However, companies often do not have the ability to invest in all of the available projects with positive NPVs. It can be difficult for outside investors to accurately assess the risks and returns associated with the company's projects.

As a result, investors may want higher returns for their capital and the company may face capital constraints. Managers might be forced to forego some positive NPV projects since investors are not providing enough capital to fund those projects at reasonable rates or because the company is not generating enough profits to invest in them.

One goal for financial decision-making is profit maximisation.

However, although profit maximisation may seem a logical goal, it has serious drawbacks. One problem is that it may be difficult to determine what 'profit' is. To most businesspeople, profits may be just revenues minus expenses but accountants may consider that the same decisions could increase profit under one set of rules and reduce it under another – the origin of the term creative accounting.

Another problem is that accounting profits are not the same as cash flows. For instance, companies recognise revenues when a sale is made, generally before payment is received for a sale. Management wants cash because it can be used for investments or to buy goods and services.

Yet another problem with profit maximisation as a goal is that it does not distinguish between getting a dollar today and getting a dollar sometime in the future. In finance, the timing of cash flows is extremely important (i.e. refer to Chapter 3).

For example, the longer a company goes without paying its bills, the more interest is accrued. The interest accrues because of the time value of money; the longer a company has access to money, the more it has to pay for it. The time value of money is one of the most important concepts in finance.

Finally, profit maximisation also overlooks the uncertainty and risk that may stem from cash flows. It is a basic principle of finance that there is a trade-off between risk and return. When given a choice, most companies tend to pick less risky investments.

The goal of profit maximisation ignores differences in value caused by different levels of risk. The trade-off between risk and return have to be considered.

One discounted cash flow technique that companies and analysts use to evaluate capital budgeting projects is the **profitability index** (PI), also called the benefit/cost ratio. The PI method computes the ratio between the present values of the inflows and outflows:

$$\text{Profitability index} = \frac{\text{Present value of future cash flows}}{\text{Initial cost}}$$

The PI measures the relative benefits of a project; specifically, the present value (PV) of those benefits. For example, a PI of 2 means the project returns HK\$2 for every HK\$1 invested. It would be foolish to invest in a project that returns less than the value of the investment. The PI includes a natural decision rule: Companies accept projects with a PI greater than 1 and reject those with PIs of less than 1.0.

Ultimately, it is not just individual projects but a company's growth prospects that affect financing decisions.

Apply and Analyse 2

Apply

Consider a company that is evaluating the following independent investments, A through F:

Project	Initial cash outlay	NPV	Profitability index
A	HK\$100,000	HK\$13,646	1.136
B	HK\$50,000	– HK\$3,342	0.934
C	HK\$80,000	HK\$10,558	1.132
D	HK\$60,000	HK\$4,320	1.072
E	HK\$75,000	HK\$10,825	1.144
F	HK\$90,000	HK\$7,225	1.080

Which projects make the best selections and are worth recommending to the company's CIO for investment?

Apply and Analyse 2 (continued)

Analysis

In the absence of capital rationing, which projects should be selected? If you look at projects ranked by NPV and PI, you see the following rankings:

Rank	Rank based on NPV	Rank based on PI
1	A	E
2	E	A
3	C	C
4	F	F
5	D	D
6	B	B

Why do they differ? Because the NPV is in monetary terms, and PI is in relative terms (benefit/cost ratio).

The company should select projects A, C, D, E, and F because they all have positive NPVs and are assumed to be independent. Therefore, the capital spending is HK\$100,000 + 80,000 + 60,000 + 75,000 + 90,000 = HK\$405,000, and the expected value added from this set of projects is HK\$13,646 + 10,558 + 4,320 + 10,825 + 7,225 = HK\$46,574.

Now suppose a capital budget constraint of HK\$250,000 is placed on new investments. Which projects should be selected? The company should take on the combination of positive NPV projects that maximises total NPV, within the given budget constraints. All possible combinations of projects are the following:

Project selected	Total NPV	Capital spending required
A, C, D	HK\$28,524	HK\$240,000
A, D, E	HK\$28,791	HK\$235,000
A, D, F	HK\$25,191	HK\$250,000
C, D, E	HK\$25,703	HK\$215,000
C, D, F	HK\$22,103	HK\$230,000
C, E, F	HK\$28,608	HK\$245,000
D, E, F	HK\$22,370	HK\$235,000

If the capital budget constraint is fixed on HK\$250,000, the company should select only projects A, D and E. The reason for this is because the spending fits in the budget, and this is the set of projects within the budget that has the highest total NPV.

Key Learning Point

One discounted cash flow technique that companies and analysts use to evaluate capital budgeting projects is the profitability index (PI), also called the benefit/cost ratio. The PI method computes the ratio between the present values of the inflows and outflows.

► Knowledge Check Questions

Question 17

If a project has a positive net present value (NPV), identify which of the following is the profitability index.

- A Greater than one
- B Less than one
- C Equal to one
- D Negative

Question 18

Identify which of the following are the limitations of considering only profit maximisation when establishing a financial strategy.

- (i) Limited capital
- (ii) Delaying cash flow
- (iii) Ignoring risk

- A (i) and (ii)
- B (i) and (iii)
- C (ii) and (iii)
- D (i), (ii) and (iii)



9.10 SYMPTOMS OF AND REMEDIES FOR OVERTRADING AND EXCESSIVE LEVERAGE

Companies rely on financial strategies to achieve their goals and remain profitable. At times, however, the same strategies that can help companies can turn against them.

Trading in stocks, bonds and foreign currency, for example, can help a company speculate profitably or hedge against risks but it might also turn against the company if the trading becomes so frequent that it leads to losses. On the other hand, leverage or debt can be a powerful financial tool but excessive leverage could be dangerous, even fatal, for a company.

9.10.1 Overtrading

Companies engage in **overtrading** when they buy or sell stocks or bonds excessively, to the point that the trade becomes counter-productive to the financial strategy and investment objectives of a company.

There are a few remedies that financial managers can put in place to tackle overtrading. These remedies revolve around three key considerations:

1. Self-awareness: Companies, like individual investors, should make frequent assessments of their trading activities looking for any telltale signs of overtrading. One such sign is a progressive increase in the number of trades per month.

2. Pause: Sometimes, companies make trades that result in a loss, and then ramp up their trading activity to make up the lost ground. Pausing trading activities and reassessing the trading strategy could ensure companies return to a profitable strategy.
3. Rules: Developing and adhering to rules for trading can help companies deal with overtrading. Rules can be linked, for example, with the moving average price of a stock or the dividend yield.

9.10.2 Excessive Leverage

Leverage, as discussed, is the use of debt as a source of funding when investing or expanding the asset base of a company to generate returns. While leverage can be a useful and important part of a company's financial strategy it is also possible to over-leverage.

An important point is that leverage magnifies both gains and losses.

If a company uses leverage to make an investment and the investment does not work out, the losses may be much greater than they would have been without the use of debt financing. In this case, both the interest expense and credit risk work against the company and result in a greater loss of shareholder value.

A company's financial risk reflects its interest expenses or financial leverage.

Computing the degree of financial leverage a company has, makes it possible to determine its exposure to financial risk. The **degree of financial leverage (DFL)** measures how sensitive earnings per share are to changes in EBIT:

The equation is as follows:

$$\text{Degree of financial leverage (DFL)} = \frac{\text{Percentage change in earnings per share (EPS)}}{\text{Percentage change in EBIT}}$$

DFL represents the percentage change in earnings per share from a 1% change in EBIT. For instance, a DFL of 1.25 means EPS will rise or fall by 1.25% for every 1% increase or decrease in EBIT.

Total earnings risk, or total variability in earnings per share, is the result of combining the effects of business risk and financial risk. When combined, **operating leverage** and financial leverage magnify changes in sales much more than changes in earnings.

When considered together, operating and financial leverage generate **combined leverage**.

The **degree of combined leverage (DCL)** is the percentage change in EPS from a 1% change in sales volume.

The equation is as follows:

$$\text{Degree of combined leverage (DCL)} = \frac{\text{Percentage change in EPS}}{\text{Percentage change in sales}}$$

The **degree of operating leverage (DOL)** measures are used to evaluate how company income changes with respect to a percentage change in sales. Operating leverage includes both fixed and variable costs. Companies with a high DOL have high fixed costs relative to its variable costs and if

the DOL is high, EBIT is volatile compared to a percentage change in sales, all else being equal. The opposite is also true. There are different formulas to calculate DOL. One such formula is:

$$\text{Degree of operating leverage (DOL)} = \frac{\text{Percentage change in EBIT}}{\text{Percentage change in sales}}$$

The main approach to calculate DOL is to divide the percent change in EBIT by the percent change in sales.

A straightforward relationship exists amongst the degree of operating leverage, the degree of financial leverage and the degree of combined leverage. A firm's DCL is the product of its degree of operating leverage and its degree of financial leverage.

The use of debt increases the risk of default because debt payments are a fixed obligation and debtholders must be paid both interest and principal regardless of how the company performs.

If a company has excessive leverage and does not make an interest payment on the prescribed date, the company defaults on its debt and might be forced into bankruptcy by creditors.

Illustrative Example 10

The statement of profit or loss of a subsidiary, Queen's Road Manufacturing Ltd, for 2019 and 2020 are as follows:

Statement of profit or loss (HK\$)	2019	2020
Net sales	1,300,000	1,500,000
Cost of goods sold	780,000	900,000
Gross profit	520,000	600,000
General and administrative	150,000	150,000
Marketing expenses	130,000	150,000
Depreciation	40,000	53,000
Interest	45,000	57,000
Earnings before taxes	155,000	190,000
Income taxes	62,000	76,000
Net income	93,000	114,000

Assuming one-half of the general and administrative expenses are fixed costs, calculate Queen's Road Manufacturing Ltd's DOL, DFL and DCL in 2019 and 2020.

	2019 (HK\$)	2020 (HK\$)
Sales	1,300,000	1,500,000
Variable costs ^a	985,000	1,125,000
Fixed costs ^b	115,000	128,000
EBIT	200,000	247,000
Interest	45,000	57,000

Illustrative Example 10 (continued)

	2019 (HK\$)	2020 (HK\$)
EBT	155,000	190,000
Tax	62,000	76,000
NI	93,000	114,000

^a COGS, marketing expenses; one-half of G&A expense

^b Depreciation and one-half of G&A expense

2019: $DOL = (1,300,000 - 985,000) / 200,000 = 1.575$

$DFL = 200,000 / (200,000 - 45,000) = 1.29$

$DCL = 1.575 \times 1.29 = 2.03$

2020: $DOL = (1,500,000 - 1,125,000) / 247,000 = 1.52$

$DFL = 247,000 / (247,000 - 57,000) = 1.30$

$DCL = 1.52 \times 1.30 = 1.98$

Illustrative Example 11

Management is grappling with the issues of business risk and degree of operating leverage for a number of subsidiaries. The table below shows how certain changes would affect the business risk of the company.

Changes	Effect of changes
Imports lead to more competition.	Higher levels of competition will lead to pressure to lower prices, thus lowering EBIT and increasing DOL.
Labour costs go down.	Lower costs will lead to wider price-variable cost margins and higher levels of EBIT and lowering DOL.
Healthcare costs, provided for employees, increase.	Higher costs will lead to narrower price-variable cost margins, lower levels of EBIT and higher levels of DOL.
Social security and unemployment insurance taxes go up.	As a component of labour costs, these tax increases will increase labour costs, narrow the price-variable cost difference, decrease EBIT, and raise DOL.
New technology makes it possible for the firm to generate the same output with fewer employees.	Variable costs, in the form of labour costs, will fall. But the net effect on business risk depends on the technology's impact on fixed costs.

Key Learning Point

Companies engage in overtrading when they buy or sell stocks or bonds excessively, to the point that the trade becomes counter-productive to the financial strategy and investment objectives of a company. While leverage can be a useful and important part of a company's financial strategy it is also possible to over-leverage.

► Knowledge Check Questions

Question 19

A company experienced a 12% increase in earnings before interest and taxes and a 3% increase in sales. Calculate the degree of operating leverage for this company.

- A 36
- B 6
- C 4
- D 9

Question 20

If a firm has current earnings per share of HK\$2 and a degree of financial leverage of 4, calculate what earnings per share if there is a 10% increase in sales.

- A HK\$2.80
- B HK\$8
- C HK\$2.20
- D Cannot tell from this information

Question 21

Explain the degree of operating leverage (DOL) and its relationship with the cost structure of a company.

9.11 FINANCIAL STRATEGY EVALUATION AND RECOMMENDATION

Ultimately, the decisions that a company makes regarding financing and capital structure choices are based on a number of theoretical and real-world considerations.

Companies in the same industry generally face similar business risks. Many financial managers examine their competitors' capital structures to determine if their own financial strategies are appropriate.



9.11.1 Considerations for Leverage Decision

Companies with highly variable EBIT cannot afford to issue large amounts of debt, as the combined effects of high business and financial risk may imperil the company's future. In general, greater EBIT variability reduces the company's reliance on debt.

Tax regulations are another consideration. Debt interest deductions are a strong motivator of debt. The tax incentive for debt financing may fall as a company accumulates non-debt tax shields like depreciation expense, research and development (R&D) and advertising.

Another consideration is the mix of tangible and intangible assets. Agency costs and **bankruptcy costs** can make debt less attractive, particularly for companies with intangible assets like goodwill, customer loyalty, R&D and growth opportunities.

Financial managers try to maintain access to capital. Without financing, a company may have to pass up investment opportunities or face a cash crunch.

A loss of financial flexibility can disrupt company operations. Some companies seek flexibility by maintaining financial slack or unused debt capacity. This can be done by maintaining an investment-grade bond rating or large lines of credit.

A company can arrange for debt financing with maturity that matches the time that the company needs the money. For maturity matching, debt holds an important advantage over stock financing because equity securities do not have a stated maturity.

Leases may also be useful because they may be set for terms that coincide with the period of time that the company needs the assets.

Additional equity issuance reduces per-share dividends and dilutes control. Shareholders who worry about this may force a company to use more debt financing and less external equity financing.

Companies with above-average profitability may lower flotation costs and restrictive debt covenants by using internal equity to finance capital projects. The more profitable companies tend to use less debt financing.

Companies can minimise financing costs by issuing debt when interest rates are low, particularly if companies expect rates to go up.

Similarly, issuing stock is advantageous when stock prices are high.

Ultimately it is up to the management team to determine financing policy. Their judgment and expectations for the financial future determine the capital structure of the company.

9.11.2 Key Stakeholders

The discussion above leads to the role of a number of stakeholders in evaluating financial strategies and making decisions.

The capital structure of a company is affected by the cost of raising capital. Depository institutions and investment banks channel savings to companies that want to raise funds and do so in ways that lower flotations and the transactions costs of raising funds, making it easier for companies to raise funds.

Investors consider the prospects for a company and trade-off between risk and return. The market prices of a company's securities and the ratings determine the cost of financing, which affects the company's long-term capital budgeting decisions.

Financial capital is expensive, therefore the cost of financing assets is a key part of the investment decisions of a company.

Key Learning Point

The decisions that a company makes regarding financing and capital structure choices are based on a number of theoretical and real-world considerations.

► Knowledge Check Questions

Question 22

Explain the key internal and external stakeholders of an organisation who are interested in its financial aspects, such as financial strategy and performance.



SUMMARY

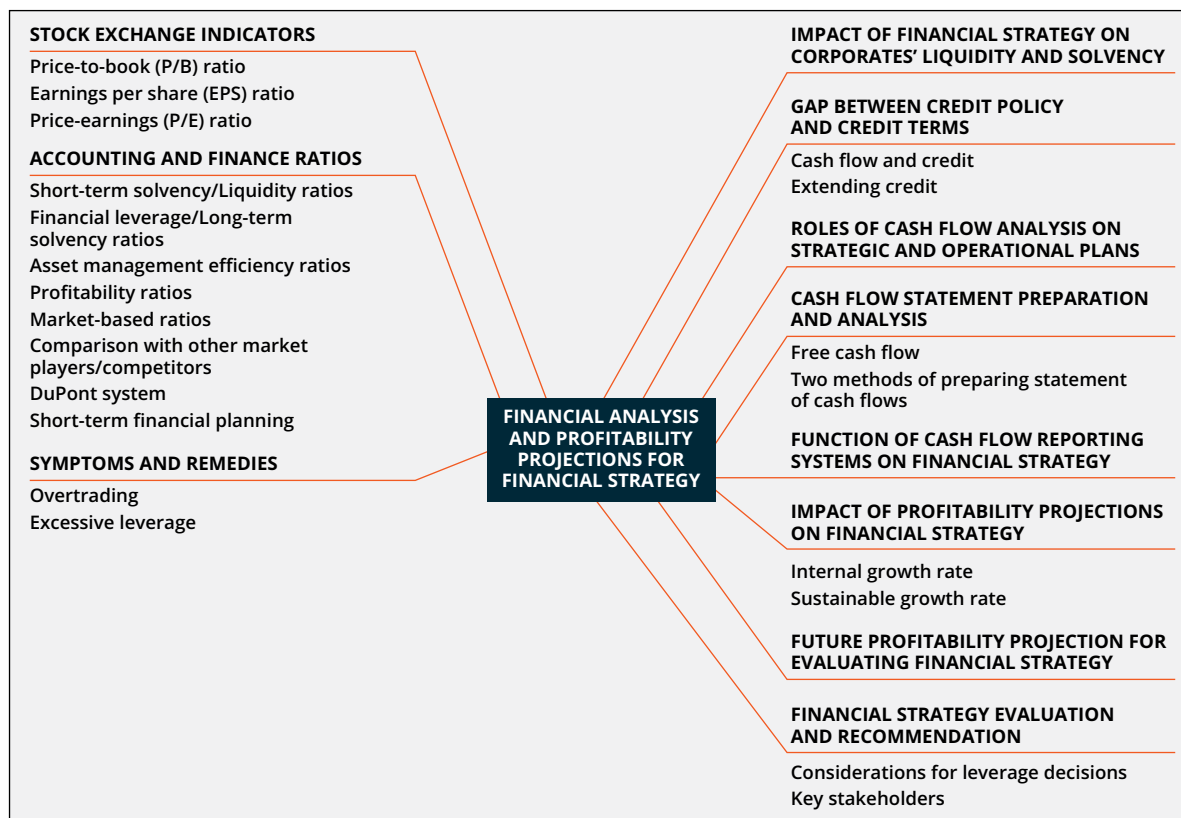
- Stock markets typically evaluate a company's liquidity, efficiency, leverage and profitability by using a number of ratios often known as market-value ratios, which combine market-value data with data from a firm's financial statements.
- The price-to-book ratio, sometimes called the market-to-book ratio, compares the market value of a company's investments to its book value.
- A company's net income shows its accomplishments (revenues) relative to its efforts (expenses) within a time period. Net income is usually reported on a per-share basis called earnings per share (EPS), where EPS equals net income divided by the number of common shares outstanding.
- The price-earnings (P/E) ratio relates earnings per share to price per share. A high P/E ratio can suggest that investors expect earnings to grow quickly but may also be caused by unusually low earnings in any one year and investors might anticipate earnings to recover to a normal level soon.
- The focus in analysing the liquidity of the company is on the overall liquidity of a firm's assets and on the available assets to meet current liabilities.
- The current ratio is estimated as current assets divided by current liabilities (CL).
- The quick ratio or acid test ratio, which is cash (C) plus marketable securities (MS) plus accounts receivable (AR) divided by current liabilities.
- Leverage ratios measure how much debt rather than equity financing a company uses and points to the company's ability to meet long-term financial obligations, such as interest payments on debt and lease payments. These ratios are also known as long-term solvency ratios.
- The total debt ratio measures how many of its assets the company finances from sources other than shareholders.
- The debt-to-equity ratio makes it possible to determine the amount of debt for each dollar of equity.
- The equity multiplier makes it possible to determine the amount of assets that a company has for every dollar of equity.
- Asset management efficiency ratios measure how efficiently a dollar of revenue is turned into profits.
- Profitability ratios measure the ability of management to use company assets to generate sales and manage operations. Shareholders, creditors and managers are interested in these numbers because they focus on earnings.
- The gross profit margin measures the percentage of net sales remaining after the cost of goods sold is paid.

- Operating profit is generally measured as EBIT and the operating profit margin gives an indication of how profitable operations are regardless of financing policies or tax management strategies.
- The net profit margin is the percentage of sales that remain after expenses, including interest and taxes, are paid.
- Return on equity (ROE) measures net income as a percentage of the shareholders' investment in the company.
- There are a number of standard ratios used to understand the relative valuation of a company.
- Some ratios that are useful in assessing a company's relative value include earnings per share (EPS) and the equity book value per share (BVPS).
- In terms of valuation, one of the most basic ratios an analyst can look at is a company's dividend yield, which is the current dividend per share (DPS) divided by the current share price (P).
- One of the most important and widely followed value ratios is the price-earnings (P/E) ratio, which is estimated as the share price (P) divided by the EPS.
- When the P/E ratio is estimated based on the current share price and a forecast or expected EPS (EEPS) figure, it is known as a forward P/E ratio.
- Another commonly used ratio is the price-to-book ratio, which is defined as the market price per share (P) divided by the book value per share (BVPS).
- The EBITDA multiple is calculated as the total enterprise value (TEV) divided by earnings before interest, taxes, depreciation and amortisation (EBITDA).
- Financial managers can collect appropriate benchmark data in three ways: trend analysis, industry analysis and peer group analysis.
- Determining ROA requires knowing the net profit margin and total asset turnover.
- The DuPont system uses financial ratios to determine the financial health of a company. The first step is for management to assess the financial health of the company. The second is to correct any problems. The third is to monitor financial performance over time.
- Financial management is the management of the financial resources of a company, which includes assets and the financing of those assets.
- A financial strategy outlines the short- and long-term goals of an organisation and can have a real impact on the liquidity and solvency of a company.
- The framework for analysing long-term investment or asset decisions is known as capital budgeting or capital expenditure analysis, and these decisions are some of the most important that a company can make.
- Cash flow from operations is the correct measure of the cash generating ability of a company.
- Rapid growth normally causes cash flow problems as investment soaks up cash more quickly than incremental sales generate cash: this is generally referred to as a company's burn rate.
- One way to lower the burn rate is to, essentially, collect receivables more quickly. This changes a company's credit policy.



- Managing working capital effectively is critical to both the short-term and the long-term viability of any company. In fact, working capital mismanagement can often cause liquidity problems, which, in an extreme case, can force a company into bankruptcy.
- The primary purpose of a statement of cash flows is to provide financial information about the cash receipts and cash payments of a company for a specific period of time.
- Companies create a cash budget, which is essentially a cash flow statement projected for each month.
- Effective cash reporting can help a company predict cash needs and perhaps set up a facility whereby it could invest surplus funds at the start and end of the year.
- How much a company holds in cash and cash equivalents is also partly determined by its ability to borrow on an operating line of credit or to run an overdraft.
- Financial forecasting is critical for a company, financial advisers and external analysts. Without knowing what its financing needs are, a company cannot decide what funds to tap, short or long term, and how the financing needs will evolve.
- The most basic forecasting technique is the percentage of sales method.
- Expectations of future profits are another important consideration in determining the financial strategy of a company. A company that expects significant profits is likely to take a different approach to investing and raising capital than one that is not expecting to be profitable for some time.
- Internal growth rate (IGR) is the maximum growth rate that a firm can achieve without external trading.
- Sustainable growth rate (SGR) is the maximum rate of growth a company can sustain without changing its capital structure.
- One discounted cash flow technique that companies and analysts use to evaluate capital budgeting projects is the profitability index (PI), also called the benefit/cost ratio.
- Companies engage in overtrading when they buy or sell stocks or bonds excessively, to the point that the trade becomes counter-productive to the financial strategy and investment objectives of a company.
- While leverage can be a useful and important part of a company's financial strategy it is also possible to over-leverage.

MIND MAP



LIST OF FORMULAS

Price-to-book ratio

$$\text{Price-to-book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

Earnings per share ratio

$$\text{Earnings per share} = \frac{\text{Net income}}{\text{Shares outstanding}}$$

Price-earnings ratio

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

Current ratio

$$\text{Current ratio} = \frac{\text{Current assets (CA)}}{\text{Current liabilities (CL)}}$$



Quick ratio

$$\text{Quick ratio} = \frac{\text{Cash (C)} + \text{Marketable securities (MS)} + \text{Accounts receivable (AR)}}{\text{Current liabilities (CL)}}$$

Total debt ratio

$$\text{Total debt ratio} = \frac{\text{Total liabilities (TL)}}{\text{Total assets (TA)}}$$

Debt-to-equity ratio

$$\text{Debt-to-equity ratio} = \frac{\text{Total debt}}{\text{Total equity}}$$

Equity multiplier

$$\text{Equity multiplier} = \frac{\text{Total assets}}{\text{Total equity}}$$

Equity multiplier related to debt-to-equity ratio

$$\text{Equity multiplier} = 1 + \text{Debt-to-equity ratio}$$

Financial position identity

$$\text{Total assets} = \text{Total liabilities (debt)} + \text{Total shareholders equity}$$

Times-interest-earned ratio

$$\text{Times-interest-earned} = \text{Earnings before interest and tax (EBIT)} / \text{Interest}$$

Cash coverage ratio

$$\text{Cash coverage ratio} = \frac{\text{Earnings before interest and tax (EBIT)} + \text{Depreciation and/or Amortisation}}{\text{Interest}}$$

Receivables turnover

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable average / ending balance}}$$

Days' receivables

$$\text{Days' receivables} = \frac{365}{\text{Receivables turnover}}$$

Payables turnover ratio

$$\text{Payables turnover} = \frac{\text{Net credit purchases}}{\text{Account payable (average/ending balance)}}$$

Alternative formula of Payables turnover ratio

$$\text{Payables turnover} = \frac{\text{Cost of goods sold}}{\text{Account payable (average/ending balance)}}$$

Days' payables

$$\text{Days' payables} = \frac{365}{\text{Payables turnover}}$$

Inventory turnover ratio

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory average / ending balance}}$$

Days' inventory ratio

$$\text{Days' inventory ratio} = \frac{365}{\text{Inventory turnover}}$$

Fixed asset turnover ratio

$$\text{Fixed asset turnover} = \frac{\text{Sales}}{\text{Net fixed assets}}$$

Total asset turnover ratio

$$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$$

Gross profit margin

$$\text{Gross profit margin} = \frac{\text{Net sales} - \text{Cost of goods sold}}{\text{Net sales}}$$



Operating profit margin

$$\text{Operating profit margin} = \frac{\text{Earnings before interest and taxes (EBIT)}}{\text{Net sales}}$$

Net profit margin

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Net sales}}$$

EBIT return on assets

$$\text{EROA} = \frac{\text{EBIT}}{\text{Total assets}}$$

Return on assets

$$\text{ROA} = \frac{\text{Net income}}{\text{Total assets}}$$

Return on equity

$$\text{ROE} = \frac{\text{Net income}}{\text{Total equity}}$$

Average value

$$\text{Average asset or equity value} = \frac{\text{Beginning value} + \text{Ending value}}{2}$$

Book value per share

$$\text{Book value per share} = \frac{\text{Shareholders' equity (SE)}}{\text{Number of shares}}$$

Dividend yield

$$\text{Dividend yield} = \frac{\text{Dividend per share (DPS)}}{\text{Current share price (P)}}$$

Dividend payout ratio

$$\text{Dividend payout} = \frac{\text{Dividend per share (DPS)}}{\text{Earnings per share (EPS)}}$$

Price-earnings (P/E) ratio

$$\text{P/E ratio} = \frac{\text{Price per share (P)}}{\text{Earnings per share (EPS)}}$$

Forward P/E ratio

$$\text{Forward P/E ratio} = \frac{\text{Price per share (P)}}{\text{Estimated earnings per share (EEPS)}}$$

Price-to-book ratio

$$\text{Price-to-book ratio} = \frac{\text{Market value of equity per share}}{\text{Book value of equity per share}}$$

EBITDA multiple

$$\text{EBITDA multiple} = \frac{\text{Total enterprise value (TEV)}}{\text{Earnings before interest, taxes, depreciation and amortisation (EBITDA)}}$$

Derived from the ROA equation

$$\begin{aligned} \text{ROA} &= \text{Net income/Total assets} \\ &= (\text{Net income/Net sales}) \times (\text{Net sales/Total assets}) \\ &= \text{Net profit margin} \times \text{Total asset turnover} \end{aligned}$$

Net profit margin ratio can be written as:

$$\text{Net profit margin} = \text{Net income/Net sales} = (\text{EBIT/Net sales}) \times (\text{EBT/EBIT}) \times (\text{Net income/EBT})$$

ROE equation, we start with the formula discussed earlier:

$$\begin{aligned} \text{ROE} &= \text{Net income/Total equity} \\ &= (\text{Net income/Total assets}) \times (\text{Total assets/Total equity}) \\ &= \text{ROA} \times \text{Equity multiplier} \end{aligned}$$

DuPont equation

$$\text{ROE} = \text{Net profit margin} \times \text{Total asset turnover} \times \text{Equity multiplier}$$

DuPont equation ratio

$$\text{ROE} = (\text{Net income/Net sales}) \times (\text{Net sales/Total assets}) \times (\text{Total assets/Total equity})$$

Operating cycle

$$\text{Operating cycle} = \text{Days' inventory} + \text{Days' receivables}$$



Cash conversion cycle (CCC)

$$\text{CCC} = \text{Days' inventory} + \text{Days' receivables} - \text{Days' payables}$$

Internal growth rate

With opening asset balance

$$\text{Internal growth rate} = \text{Return on asset (ROA)} \times \text{Retention rate (} b \text{)}$$

With closing asset balance

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - (\text{ROA} \times b)}$$

Sustainable growth rate

With opening equity balance

$$\text{Sustainable growth rate} = \text{Return on equity (ROE)} \times \text{Retention rate (} b \text{)}$$

With closing equity balance

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - (\text{ROE} \times b)}$$

Profitability index

$$\text{Profitability index} = \frac{\text{Present value of future cash flows}}{\text{Initial cost}}$$

Degree of financial leverage

$$\text{DFL} = \frac{\text{Percentage change in earnings per share (EPS)}}{\text{Percentage change in EBIT}}$$

Degree of combined leverage

$$\text{DCL} = \frac{\text{Percentage change in EPS}}{\text{Percentage change in sales}}$$

Degree of operating leverage

$$\text{DOL} = \frac{\text{Percentage change in EBIT}}{\text{Percentage change in sales}}$$

► Answers to Knowledge Check Questions

Question 1

Answer A is correct.

Net income = HK\$1,276,351

Shares outstanding = 420,000

EPS = HK\$1,276,351 ÷ 420,000 = HK\$3.04 per share

PE ratio = 11.2 times

Price per share = 11.2 × HK\$3.04 = HK\$34.05

Question 2

Answer C is correct.

Net income = HK\$812,425

Share price = HK\$13.45

Shares outstanding = 312,490

EPS = HK\$812,425 ÷ 312,490 = HK\$2.60 per share

PE Ratio = HK\$13.45 ÷ HK\$2.60 = 5.17

Question 3

Answer C is correct. The total debt ratio measures the extent to which the firm finances its assets from sources other than the shareholders. Total debt ratio = Total debt / Total assets, a debt ratio of 0.9 means the firm has HK\$0.90 of total debt per HK\$1.00 of assets.

Question 4

Answer A is incorrect. The current ratio is a liquidity ratio. Short-term creditors are more concerned with this. Investors are likely to be more interested in market-value ratios and price-earnings ratio is one of them.

Answer B is correct. Investors are likely to be more interested in market-value ratios and price-earnings ratio is one of them.

Answer C is incorrect. Times-interest-earned ratio is a leverage ratio, and creditors are more concerned with this. Investors are likely to be more interested in market-value ratios like price-earnings ratio.

Answer D is incorrect. Total asset turnover ratio is an efficiency ratio. Investors are likely to be more interested in market-value ratios like price-earnings ratio.

Question 5

Answer A is incorrect. Generally, a financial strategy has three main components: financing, investment, and dividends.

Answer B is incorrect. Generally, a financial strategy has three main components: financing, investment, and dividends.

Answer C is incorrect. Generally, a financial strategy has three main components: financing, investment, and dividends.

Answer D is correct. It is not one of the three main components of financial strategy.

Question 6

Evaluating capital budgets requires analysing the future incremental cash flows that a project is expected to generate, considering the project's cost of capital.

**Question 7**

Answer A is incorrect. Cash burn refers to a company's cash on hand and the rate that cash is used in operations due to negative operating cash flows.

Answer B is correct. The burn rate is how long a company can operate before it has to raise funds, either by issuing stock or borrowing.

Answer C is incorrect. Net working capital is current assets minus current liabilities.

Answer D is incorrect. Working capital is the sum of a company's current assets.

Question 8

Answer A is incorrect. A computer manufacturer grants credit to customers which make a longer collection than a fast food shop.

Answer B is correct. Fast food shops generally have very low collection periods; that is, they generally do not give credit.

Answer C is incorrect. Electronic appliance producers grant credit to customers which make a longer collection than a fast food shop.

Answer D is incorrect. A mining company grants credit to customers which make a longer collection than a fast food shop.

Question 9

Net working capital is current assets minus current liabilities.

Question 10

Answer A is correct. Increase in inventory will increase the current assets and thus the net working capital increases.

Answer B is incorrect. Decrease in cash will decrease the current assets and thus the net working capital decreases.

Answer C is incorrect. Increase in accounts payable will increase the current liabilities and the net working capital decrease.

Answer D is incorrect. The changes in long-term debt have no relationship with either the current assets or current liabilities.

Question 11

Answer A is incorrect. Operating activities involve statement of profit or loss items.

Answer B is correct. Investing activities involve cash flows resulting from changes in investments and long-term asset items.

Answer C is incorrect. Financing activities involve cash flows resulting from changes in long-term liability and shareholders' equity items.

Answer D is incorrect. There are no non-cash investing and financing activities in the statement of cash flows.

Question 12

Answer A is incorrect. Cash receipt from operating activities is recorded in the statement of cash flow

Answer B is incorrect. The cash payment from investing activities is recorded in the statement of cash flow.

Answer C is incorrect. The net change in cash is calculated in the statement of cash flow.

Answer D is correct. Credit sales is recorded in the statement of profit or loss.

Question 13

Answer A is incorrect. Transaction motive refers to the cash that is required for a company's normal operations.

Answer B is incorrect. Precautionary motive refers to the cash that companies keep on hand to take care of unanticipated required outlays of cash, such as unexpected repairs on equipment.

Answer C is incorrect. Finance motive refers to the cash that companies will accumulate in anticipation of any major outlays, such as lump-sum loan repayments, dividend payments, and so on.

Answer D is correct. Companies hold cash for transactions, precaution, finance and strategy or speculation.

Question 14

Answer A is correct. Beginning cash balance + collections from customers – payments to suppliers = ending cash balance.

Question 15

Answer A is incorrect. The firm knows who invests capital in the firm and who the firm owes its spontaneous liabilities.

Answer B is incorrect. Both invested capital and spontaneous liabilities are not randomly determined.

Answer C is incorrect. Invested capital is the amount of equity invested in a firm but spontaneous liabilities are the accruals and payables of firms.

Answer D is correct. Invested capital is the investors' investment in the firm and spontaneous liabilities generally arise when the firm keeps on running its business.

Question 16

Answer A is incorrect. Of which the item (ii) market research is not part of financial assessment.

Answer B is correct. They are key steps in sequence, whereas item (v) is assessing the short- and medium-term financial viability of the strategic plan.

Answer C is incorrect. Of which items (ii) and (iv) are not the key steps.

Answer D is incorrect. Of which item (iv) is not a key step.

Question 17

Answer A is correct. PI greater than 1 indicates a positive return.

Answer B is incorrect. PI less than 1 indicates a negative return.

Answer C is incorrect. PI equal to 1 indicates no returns.

Answer D is incorrect. PI will not be negative under normal circumstances.

Question 18

(i) is correct because with limited capital, a company cannot invest in all the positive NPV projects that can maximise its profit.

(ii) is correct because the profits recognised when sales occurred instead of cash received, however, for a company to make investment or buy goods and services, cash is needed.

(iii) is correct because the profit maximisation goal ignores differences in value caused by differences in risk.

Therefore, A, B and C are incorrect, and D is correct.

Question 19

Answer C is correct.

$DCL = \text{Percentage change in EPS} \div \text{Percentage change in sales} = 12\% \div 3\% = 4$

Question 20

Answer D is correct. $DFL = \text{Percentage change in EPS} \div \text{Percentage change in EBIT}$. The percentage change in EBIT is missing.

Question 21

The degree of operating leverage (DOL) is a measure used to evaluate how a company's operating income changes after a percentage change in its sales. A company's operating leverage involves fixed costs and variable costs. A company with a high degree of operating leverage has high fixed costs relative to its variable costs. If the degree of operating leverage is high, the earnings before interest and taxes (EBIT) experiences volatility with respect to a percentage change in sales, all else remaining the same, and vice versa.

Question 22

Stakeholders may be interested in the financial nature of the management and performance of a company and how that could impact them. Any perceptions of what stakeholders want to see impacts how financial information is used in financial statements.

Internal stakeholders include management, employees and the board of directors. External stakeholders include customers or clients, owners or shareholders, lenders, financiers and guarantors, suppliers of goods and services, donors, government agencies and local communities. Competitors, pressure groups, the media and the general public may also be interested.

Different stakeholders may have different perspectives. Employees may be interested in the stability of the company while suppliers may want to determine its creditworthiness. Customers may want to know how profitable a company is to determine whether it will be reliable in future. A shareholder may want to consider returns and risks.

EXAM PRACTICE

QUESTION 1

Identify which of the following ratios is an indicator of market value.

- A. Total asset turnover ratio
- B. Price-earnings ratio
- C. Quick ratio
- D. Return on equity

QUESTION 2

Calculate a firm's equity multiplier given a debt ratio of 0.65.

- A. 2.86
- B. 2.28
- C. 1.53
- D. 3.35

QUESTION 3

Jet Inc has net sales of HK\$326,352, accounts receivable of HK\$98,765 and inventory of HK\$72,989. Calculate the firm's accounts receivable turnover.

- A. 4.4 times
- B. 3.3 times
- C. 0.3 times
- D. 0.2 times

QUESTION 4

For the year ended 30 April 2018, Clotheswear Company has total assets of HK\$87,631,181, ROA of 11.67%, ROE of 21.19% and a net profit margin of 11.59%. What are the company's net income and net sales? Calculate the firm's debt-to-equity ratio.

QUESTION 5

Starock Jewellers management announced that the company had net earnings of HK\$6,481,778 for this year. The company has 2,543,800 shares outstanding, and the year-end stock price is HK\$54.21. Calculate Starock's earnings per share and P/E ratio.

QUESTION 6

Shiny Automotive's statement of financial position at the end of its most recent fiscal year shows the following information:

Shiny Automotive Statement of financial position as at 30 April 2019			
Assets		Liabilities and equity	
Cash and marketable sec.	HK\$23,015	Accounts payable and accruals	HK\$163,257
Accounts receivable	141,258	Notes payable	21,115
Inventory	212,444	Total current liabilities	HK\$184,372
Total current assets	HK\$376,717		
		Long-term debt	168,022
Net plant and equipment	711,256	Total liabilities	HK\$352,394
Goodwill and other assets	89,879	Common stock	313,299
		Retained earnings	512,159
Total assets	HK\$1,177,852	Total liabilities and equity	HK\$1,177,852

It was also reported that the firm had a net income of HK\$156,042 on net sales of HK\$4,063,589.

- (a) Calculate the current ratio and quick ratio.
- (b) Calculate the days' receivables, total asset turnover ratio and the fixed asset turnover ratio.

QUESTION 7

The Sanders Electric Company is evaluating two projects for possible inclusion in the firm's capital budget. Project M will require a HK\$37,000 investment while project O's investment will be HK\$46,000. After-tax cash inflows are estimated as follows for the two projects:

Year	Project M	Project O
1	HK \$12,000	HK \$10,000
2	12,000	10,000
3	12,000	15,000
4	12,000	15,000
5		15,000

Calculate the NPV and PI for each project based on a 10% cost of capital. Explain which, if either, of the projects is acceptable.

QUESTION 8

The statement of financial position and statement of profit or loss and other comprehensive income of Cookie & Coffee Creations Company for its first year of operations, the year ended 31 October 2020, are as follows.

COOKIE & COFFEE CREATIONS COMPANY
Statement of Financial Position
31 October 2020
(all amounts in HK\$)

Current Assets		
Cash	72,732	
Account receivable	3,900	
Inventory	21,476	
Prepaid expenses	7,560	
		105,668
Property, plant, and equipment		
Furniture and fixtures	15,000	
Accumulated depreciation – furniture and fixtures	(1,500)	13,500
Computer equipment	5,040	
Accumulated depreciation – computer equipment	(720)	4,320
Kitchen equipment	34,800	
Accumulated depreciation – kitchen equipment	(2,460)	32,340
Total assets		155,828
Current liabilities		
Accounts payable	7,018	
Income tax payable	23,628	
Dividends payable	840	
Salaries and wages payable	4,700	

COOKIE & COFFEE CREATIONS COMPANY
Statement of Financial Position
31 October 2020
(all amounts in HK\$)

Interest payable	225	
Note payable – current portion	4,800	
		41,211
Long-term liabilities		
Note payable – long-term portion		9,200
Total liabilities		50,411
Stockholders' equity		
Paid in capital		
Preferred stock, 2800 share issued and outstanding	16,800	
Common stock, 25,930 shares issued and 25,180 outstanding	31,116	47,916
Retained earnings		58,101
Total paid-in capital and retained earnings		106,017
Less: Treasury stock (750 common shares)		(600)
Total stockholders' equity		105,417
Total liabilities and stockholders' equity		155,828

COOKIE & COFFEE CREATIONS COMPANY
Statement of Profit or Loss and Other Comprehensive Income
Year Ended 31 October 2020
(all amounts in HK\$)

Sales		555,000
Cost of goods sold		277,500
Gross profit		277,500
Operating expenses		
Salaries and wages expense	111,000	
Depreciation expense	4,680	
Other operating expenses	43,184	158,864
Income from operations		118,636
Interest expense		496
Income before income tax		118,140
Income tax expense		23,628
Net income		94,512

**Additional information:**

The company is thinking about borrowing an additional HK\$50,000 to buy more kitchen equipment. The loan would be repaid over a 2-year period. The terms of the loan provide for equal semi-annual installment payments of HK\$5,000 on May 1 and November 1 of each year, plus interest of 5% on the outstanding balance.

Required:

- (a) Calculate the following ratios:
- Current ratio
 - Receivables turnover
 - Inventory turnover
 - Total debt ratio
 - Times-interest-earned ratio
 - Gross profit margin
 - Net profit margin
 - Total asset turnover
 - Return on assets
- (b) Average industry financial ratios are listed as follows; appraise your findings from part (a).

Current ratio	2.11 times
Receivables turnover	145.1 times
Inventory turnover	14.4 times
Total debt ratio	20.2%
Times-interest-earned ratio	206 times
Gross profit margin	36.7%
Net profit margin	15%
Total asset turnover	4.02 times
Return on assets	36.7%

- (c) Based on your analysis in parts (a) and (b), explain whether a bank would lend Cookie & Coffee Creations Company HK\$50,000 to buy the additional equipment.
- (d) Identify alternatives Cookie & Coffee Creations Company could consider instead of bank financing.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. The total asset turnover ratio is an efficiency ratio. Price-earnings ratio is a market-value ratio.

Answer B is correct. Price-earnings ratio is a market-value ratio.

Answer C is incorrect. Quick ratio is a liquidity ratio. Price-earnings ratio is a market-value ratio.

Answer D is incorrect. Return on equity is a profitability ratio. Price-earnings ratio is a market-value ratio.

QUESTION 2

Answer A is correct.

Debt ratio = Total debt/Total assets

Total debt = 0.65 × Total assets

Total assets = Total debt + Total equity = 0.65 × Total assets + Total equity

Total equity/Total assets = 0.35

Equity multiplier = Total assets / Total equity = 1/0.35 = 2.86.

QUESTION 3

Answer B is correct. Accounts receivable turnover = Net sales/Accounts receivable = HK\$326,352/HK\$98,765 = 3.3 times.

QUESTION 4

Total assets = HK\$87,631,181

ROA = 11.67%, ROE = 21.19%, PM = 11.59%

Net income = Total assets × ROA = HK\$87,631,181 × 11.67% = HK\$10,226,559

Net sales = net income/profit margin = HK\$10,226,559/11.59% = HK\$88,236,055

ROE = net income/equity.

Equity = net income/ROE = HK\$10,226,559/21.19% = HK\$48,261,251

Debt = Total assets – equity = HK\$87,631,181 – HK\$48,261,251 = HK\$39,369,930

Debt to equity ratio = debt/equity = HK\$39,369,930/HK\$48,261,251 = 0.8158

QUESTION 5

Net earnings = HK\$6,481,778

Shares outstanding = 2,543,800

Year-end stock price = HK\$54.21

EPS = net earnings/shares outstanding = HK\$6,481,778/2,543,800 = HK\$2.548

PE ratio = share price / EPS = HK\$54.21/2.548 = 21.28

QUESTION 6

$$(a) \text{ Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}} = \frac{\text{HK\$376,717}}{\text{HK\$184,372}}$$

$$= 2.04 \text{ times}$$



$$(b) \text{ Quick ratio} = \frac{\text{Current asset} - \text{Inventory}}{\text{Current liabilities}} = \frac{\text{HK\$}376,717 - \text{HK\$}212,444}{\text{HK\$}184,372}$$

$$= 0.89 \text{ times}$$

$$\text{Days' receivables} = \frac{365}{\text{Accounts receivables turnover}} = \frac{365}{\text{Net sales} / \text{Accounts receivables}}$$

$$= \frac{365 \times \text{Accounts receivables}}{\text{Net sales}} = \frac{365 \times \text{HK\$}141,258}{\text{HK\$}4,063,589}$$

$$= 12.69 \text{ days}$$

$$\text{Total asset turnover} = \frac{\text{Net sales}}{\text{Total assets}} = \frac{\text{HK\$}4,063,589}{\text{HK\$}1,177,852}$$

$$= 3.45 \text{ times}$$

$$\text{Fixed asset turnover} = \frac{\text{Net sales}}{\text{Net fixed assets}} = \frac{\text{HK\$}4,063,589}{\text{HK\$}711,256}$$

$$= 5.71 \text{ times}$$

QUESTION 7

$$\text{NPV (M)} = \text{HK\$}12,000 \times \text{PVIFA} (10\%, 4) - 37,000 = \text{HK\$}12,000 \times 3.170 - 37,000 = \text{HK\$}1,040$$

$$\text{PI(M)} = \text{HK\$}38,040 / 37,000 = 1.028$$

$$\text{NPV(O)} = \text{HK\$}10,000 \times \text{PVIFA} (10\%, 5) + 5,000 (\text{PVIF}(10\%, 3) + \text{PVIF}(10\%, 4) + \text{PVIF}(10\%, 5)) - 46,000$$

$$= 10,000 \times 3.791 + 5,000(.751 + .683 + .621) - 46,000$$

$$= \text{HK\$}48,185 - 46,000 = \text{HK\$}2,185$$

$$\text{PI(O)} = 48,185 / 46,000 = 1.048$$

Both projects have positive NPVs, so both are acceptable.

QUESTION 8

(a) i. Current ratio

$$\frac{\text{HK\$}105,668}{\text{HK\$}41,211} = 2.56 \text{ times}$$

ii. Receivables turnover

$$\frac{\text{HK\$}555,000}{\text{HK\$}3,900} = 142.31 \text{ times}$$

iii. Inventory turnover

$$\frac{\text{HK\$}277,500}{\text{HK\$}21,476} = 12.92 \text{ times}$$

iv. Total debt ratio

$$\frac{\text{HK\$}50,411}{\text{HK\$}155,828} = 32.35\%$$

- v. Times-interest-earned ratio

$$\frac{\text{HK\$118,636}}{\text{HK\$496}} = 239.19 \text{ times}$$

- vi. Gross profit margin

$$\frac{\text{HK\$277,500}}{\text{HK\$555,000}} = 50.0\%$$

- vii. Net profit margin

$$\frac{\text{HK\$94,512}}{\text{HK\$555,000}} = 17.03\%$$

- viii. Total asset turnover

$$\frac{\text{HK\$555,000}}{\text{HK\$155,828}} = 3.56 \text{ times}$$

- ix. Return on assets

$$\frac{\text{HK\$94,512}}{\text{HK\$155,828}} = 60.65\%$$

- (b) For the profitability performance, the company has higher gross and net profit margins than the industry average, which means it is more profitable than its competitors. However, the difference above the industry average for the net profit margin is much less than that for the gross profit margin, which signifies the company has more operating expenses or poorer cost control than its peers. On the other hand, the return on assets remarkably outperformed the industry average. This implies that the company can generate profit by managing its assets effectively.

For the company's solvency, the company relies on more debt for financing than the other market participants as it has a higher total debt ratio but the times-interest-earned ratio is higher than the industry average, showing that the company has high earning ability to fulfil its debt obligation.

However, the company has a slightly unsatisfactory performance in the area of asset management efficiency, as all the turnovers are lower than the industry average.

- (c) The bank should have no doubt about lending money to the company. The new total debt ratio would still be acceptable $[(\text{HK\$50,411} + \text{HK\$50,000}) \div (\text{HK\$155,828} + \text{HK\$50,000}) = 48.78\%]$. Even if there is no increase in sales, operating income should still be more than enough to cover the additional interest expense. The company is profitable and is within an acceptable credit risk range for the bank.
- (d) Instead of bank financing, Cookies & Coffee Creations Company could also consider leasing the equipment instead of purchasing it.

10

Financial Forecasts and Business Planning



CHAPTER TOPIC LIST

10.1 Cash Flow Formulation and Evaluation

10.1.1 Free Cash Flow Generated from Operations

10.2 Profit Forecasts and Projection – Pro Forma Financial Statements

10.3 Business Planning Formulation and Evaluation

10.3.1 Cash Flow Break-even

10.3.2 Cash Budget

10.4 Roles and Composition for a Business Plan



LEARNING OUTCOMES

PRINCIPAL LO9: PRODUCE FINANCIAL FORECASTS AND EVALUATE BUSINESS PLANS
LO9.01: Formulate plans and forecasts for a business entity
9.01.01 Formulate and evaluate cash flow and profit projections for a business
9.01.02 Formulate and evaluate business plans for an entity
9.01.03 Explain the role and composition of business plans in helping an entity achieve its objectives



OPENING CASE

ABC INVESTMENT LTD

As the CFO of ABC Investment Limited, your job has become even more interesting as you delve into new areas of financial management of your subsidiaries, as required by your CEO and Board of Directors.

In order to formulate more precise business planning and minimise uncertainty, you work through financial management issues of different subsidiaries, by using incremental cash flow, pro forma financial statement, break-even analysis and forecasting techniques. You realise that financial planning and management really involves both uncertainty and precision. Uncertainty must be reduced to sufficiently low levels to make effective and efficient business decisions. However, any seemingly precise calculation must be tempered with realism, which forecasting is at best, but as accurate as possible by employing various techniques.

This tug and pull of financial management is what makes financial management challenging yet interesting. As you struggle with these issues, the investors and the financial market also struggle with the same issues. Somehow out of this uncertain yet precise business planning process, investments are made or rejected, and money is made or lost. 

OVERVIEW

Cash flow is the life blood of any company. Cash helps companies acquire productive assets which in turn generate more cash flow in future to support operations. The job of ensuring enough financing and cash from operating activities is the focus of this chapter.

The job is an important one for both new and existing businesses. Developing financial forecasts and projections requires a particular set of skills that are essential to ensuring sufficient cash flow. For established companies, the process typically includes also developing **pro forma** – or projected – financial statements.

The process of ensuring that a company – or even a project within a company – has enough cash flow starts earlier, in the planning stages. Business planning and evaluation is key, as is the creation of a well thought-out business plan that can act as a road map for the business.

This chapter covers all these subjects, with a particular focus on cash flows and how companies can give themselves an edge by carefully planning and forecasting their financing needs before embarking on any new ventures.

10.1 CASH FLOW FORMULATION AND EVALUATION

To produce its products or services, a company needs to purchase a range of assets. Most will be long-term assets, which are also known as **productive assets**.

Productive assets can be tangible assets, such as equipment, machinery or a manufacturing facility, or intangible assets, such as patents, trademarks, technical expertise or other types of intellectual capital. Regardless of the type of asset, the company tries to select assets that will generate the greatest **cash flows** for the owners.

The decision-making process through which the company purchases productive assets is called **capital budgeting**, and it is one of the most important decision processes in a company.

After selecting their productive assets, managers must find the money to pay for them. The approach of finding and managing long-term funding to acquire productive assets is based on financing decisions. Debt and equity are the two basic sources of funds.

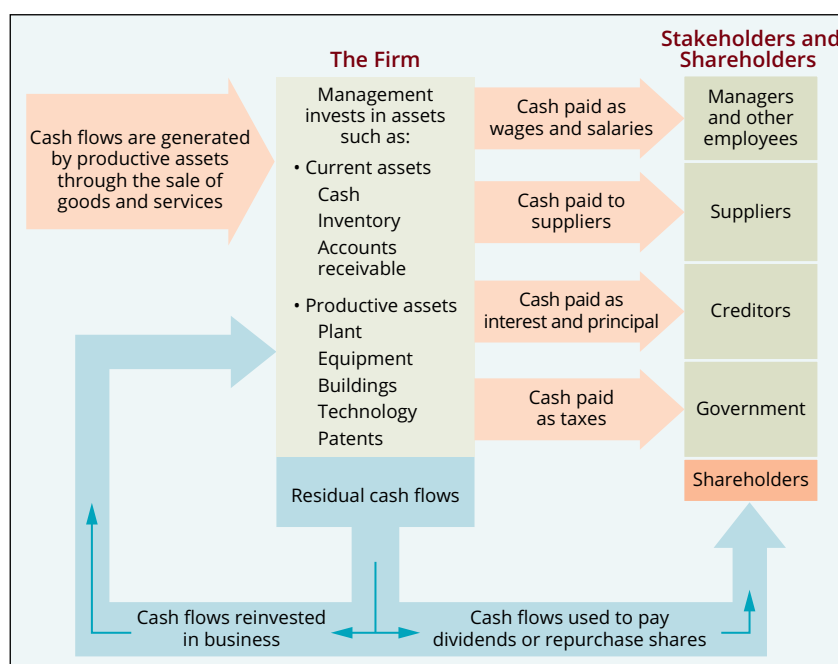
Equity represents ownership in a company and is made up of contributions by owners as well as any cash flows reinvested in the company. Most companies borrow from banks or issue long-term debt. After purchasing productive assets and the business is operating, managers want to make the products that the company offers or provide services at the lowest possible cost while maintaining a certain level of quality. In practical terms, this means buying raw materials at the lowest cost possible, keeping production and labour costs down, minimising management and administrative costs and ensuring efficient shipping and delivery to ensure competitive costs.

Companies also need to manage their daily finances to make sure they have enough cash for salaries, to purchase supplies, ensure inventories, taxes and pay for any other expenses that have to be paid. Working capital management focuses on managing current assets and current liabilities.

A company generates cash flows by selling the goods and services it produces. Companies whose inflows of cash exceed outflows are successful. Any cash left over after paying for operating expenses, creditors and taxes, the **residual cash flows**, can be distributed as cash dividends or by repurchasing some shares, or reinvest the cash in the business.

The reinvestment of residual cash flows in the business to buy more productive assets is a very important concept. If these funds are invested well they can make it possible for the company to grow and generate larger residual cash flows over time.

This reinvestment of cash flows is the most fundamental way in which businesses grow in size, as shown in Exhibit 10.1, which illustrates how the revenue generated by productive assets ultimately becomes residual cash flows.



► **EXHIBIT 10.1** Cashflows from company to owners (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

Companies that do not generate enough cash to pay operating expenses, creditors and taxes fail. Over time, unprofitable companies are forced into **bankruptcy** by creditors if the owners do not shut them down first. In bankruptcy, companies are reorganised or their assets sold off or liquidated. If the company is liquidated, creditors are paid in a priority order according to the structure of the firm's financial contracts and prevailing bankruptcy law. Anything left after creditor claims and tax claims have been paid is distributed to owners, but this rarely happens.

10.1.1 Free Cash Flow Generated from Operations

A good way to begin considering cash flows in capital budgeting is by describing the mechanics of cash flow calculations and the rules for estimating cash flows for individual projects.

The approach used to calculate project cash flows is similar to that used to calculate the cash flow to investors with the following two key aspects:

1. In capital budgeting calculations, the cash flows used are based on forecasts of future cash revenues, expenses, taxes and investment outlays.
2. In capital budgeting the focus is on estimating the cash flows expected from individual project to be produced in the future, known as incremental after-tax free cash flows.

The cash flows discounted in a net present value (NPV) analysis are incremental after-tax free cash flows that a project will generate. Incremental after-tax free cash flows represent the change in the company's total after-tax free cash flows if the project is undertaken.

Thus, the incremental after-tax free cash flows (FCF) for a project is calculated as the total after-tax free cash flows that the company would produce with the project, less the total after-tax free cash flows that the company would produce without the project:

$$FCF_{\text{Project}} = FCF_{\text{Firm with project}} - FCF_{\text{Firm without project}}$$

FCF_{Project} equals the net impact on the cash revenues, costs, taxes and investment outlays of the company.

The term free cash flows is commonly used and refers to the fact that companies are free to distribute them to creditors or shareholders. These are cash flows left over after the company has made any necessary investments. Cash flows associated with financing specific projects are not included in the calculation of FCF because these are accounted for in the discount rate that is used in an NPV analysis.

The FCF calculation is illustrated in Exhibit 10.2.

When calculating the FCFs for a project, the first step is to compute the incremental cash flow from operations (CF_{Ops}) for each year during the project's life. This is the cash flow that a project should generate after paying for operating expenses and taxes.

To obtain the FCF, it is necessary to subtract the incremental capital expenditures (Cap Exp) and the incremental additions to working capital (Add WC) essential for the project.

Explanation	Calculation	Formula
The change in the firm's cash income, excluding interest expense, resulting from the project	Revenue	Revenue
	– Cash operating expenses	– Op Ex
	Earnings before interest, taxes, depreciation and amortisation	EBITDA
	– Depreciation and amortisation	– D&A
	Operating profit	EBIT
	× (1 – Firm's marginal tax rate)	× (1 – t)
	Net operating profit after tax	NOPAT
Adjustments for the impact of depreciation and amortisation and investments on FCF	+ Depreciation and amortisation	+ D&A
	Cash flow from operations	CF Opns
	– Capital expenditures	– Cap Exp
	– Additions to working capital	– Add WC
	= Free cash flow	= FCF

► **EXHIBIT 10.2 FCF calculation** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

Cap Exp and Add WC represent the investments in long-term assets (i.e. property, plant and equipment) and in working capital items (i.e. accounts receivable, inventory and accounts payable) which must be made if the project is pursued.

The calculation of free cash flows for capital budgeting is very similar to the calculation of the cash flows to investors. This is expected as managers evaluate projects based on the present value of the cash flows produced for investors in their company.

Despite that, there is a key difference in the computation. The taxes for a project analysis are computed by multiplying the project's operating profit (EBIT) by the company's marginal tax rate, giving us the amount of taxes that would be owed by the firm on a project if no debt is used to finance that project.

In contrast, the calculation of cash flows to investors uses the actual taxes paid by the company. This means that the effect that interest deductions have on the taxes owed is included. Although interest payments will reduce the taxable income from a project if any debt financing is used, it is possible to ignore this reduction when evaluating a project for two reasons:

1. Excluding the effects associated with how the project is financed to isolate the cash flows from the project itself is desirable.
2. The cost of both debt and equity financing for a project, including the impact of using debt financing on the firm's tax obligations, are reflected in the discount rate.

Doing the calculation in this way makes it easier to estimate the NPV of a project under alternative financial structures. Since the FCF calculation gives us the after-tax cash flows from

operations in addition to what is necessary to make any required investments, the FCFs for a project are the cash flows that the firm's investors can expect to receive from the project. This is why it is necessary to discount the FCFs when computing the NPV.

The formula for the FCF calculation can also be written as:

$$FCF = [(Revenue - Op\ Ex - D\&A) \times (1 - t)] + D\&A - Cap\ Exp - Add\ WC$$

Where:

- Revenue is the incremental revenue (net sales) associated with the project,
- D&A is the incremental depreciation and amortisation associated with the project, and
- t is the company's marginal tax rate.

The FCF calculated with the above equation equals the total annual cash flow with a project minus the total cash flow without the project. Still, to put a value on a project, it is not necessary to estimate the total cash flows of the company in an NPV analysis but only to estimate cash outflows and inflows directly related to the project.

This idea that it is possible to evaluate cash flows from a project independently is called the stand-alone principle.

In the **stand-alone principle**, it is possible to treat the project as if it were a stand-alone company that has its own revenue, expenses and investment requirements.

NPV analysis compares the present value of the FCF from this stand-alone company with the cost of the project.

Exhibit 10.2 makes it easier to understand why FCF is calculated as it is.

First, note that the incremental cash flow from operations, CF_{Opns} , equals the incremental **net operating profits after tax (NOPAT)** plus D&A. NOPAT is a measure of incremental net income from a project without accounting for interest expenses. It represents how a project will impact the cash flows of a company, excluding interest expenses. Interest expenses are excluded because the cost of financing is considered in the discount rate.

The company marginal tax rate, t , is used to calculate NOPAT because project profits are assumed to be incremental. Since companies already pay taxes, the tax rate to be used for FCF is the tax rate on additional profits generated from the project. This rate is the marginal tax rate.

The calculation then adds incremental depreciation and amortisation to NOPAT to calculate CF_{Opns} . D&A represents non-cash charges that may reduce tax obligations. It is possible to subtract D&A before calculating taxes because this accounts for the fact that a company can deduct D&A to calculate taxes but, because D&A is a non-cash charge, it has to be added back to NOPAT to get cash flow from operations.

Once CF_{Opns} has been calculated, it is a matter of simply subtracting cash flows related to the required investments to get the FCF for a project in a particular period.

Investments are used to purchase long-term tangible assets like real property or intangible assets like patents or mailing lists as well as to fund current assets like inventories. All investments incremental to a project have to be accounted for. Investments in land, buildings,

machinery and equipment are the most obvious in any project but investments in intangible assets may also be needed. Incremental capital expenditures are incremental investments in long-term tangible assets and intangible assets (Cap Exp).

Beyond tangible and intangible assets, it is also needed to account for incremental additions to working capital such as credit for sales that generate additional accounts receivable.

If it will be necessary to hold inventory of a product, financing costs for that inventory should be considered. Also, incremental changes in current liabilities linked to the project will have to be considered.

Illustrative Example 1

Panda Guitar Shop is a music retail store, owned by Mr. Ho, that specialises in guitars and other musical instruments. Mr. Ho's main competition is Dragon Guitar Centre. He is exploring ways to improve his business. Panda's year-end financial statements present the following numbers:

- Net income: HK\$100,000
- Depreciation: HK\$10,000
- Change in accounts receivable: +HK\$50,000
- Change in inventory: -HK\$20,000
- Change in accounts payable: -HK\$25,000

The indirect method is used to convert accrual net income into cash basis net income. Increases in assets must be subtracted and decreases in assets must be added. Decreases in liabilities must be subtracted and increases in liabilities must be added. For example, a drop in inventory during the year means that cash was received and the decreases have to be added to income.

Calculating operating cash flow (OCF) for Mr. Ho's store using the indirect method requires the following process:

Operating cash flow

$$100,000 - 50,000 + 20,000 - 25,000 + 10,000 = \text{HK\$55,000}$$

Mr. Ho was able to generate HK\$55,000 of cash flow from operations, after paying all bills. This money left over could be reinvested into the business through purchases of inventory, for instance, or Mr. Ho could pay himself a dividend. In any case, the operation was successful.

Many investors prefer analysing cash flow numbers and compare them to ratios that are difficult to alter. Management can manipulate performance ratios with ease but cash flow are harder to manipulate because companies earn and spend cash. The company earns cash and spends cash. At the same time, cash flows present a clear picture of where the company stands and any problems with operations. For example, a company may have

Illustrative Example 1 (continued)

high net income but low OCF due to problems collecting from customers or a low net profit and a high cash flow, as profits are how the company uses accelerated depreciation on the statement of profit or loss and other comprehensive income to lower profit levels. The cash flow trail cannot be manipulated with such ease and can better indicate the health and future success of a company.

The operating section of the statement of cash flows of Mr Ho's business (i.e. using the indirect method) in the above illustrative example can be shown through the direct or indirect methods. In either case, the difference is in the operating section with the investing and financing sections identical. The direct method shows the major classes of gross cash receipts and payments. However, detailed cash receipts and payments are not given in the above illustrative example, thus the direct method in compiling the statement of cash flows cannot be demonstrated.

Ethics in Practice

A company should consider the inherent conflict that can exist between the planning and control phases of budgeting and cash forecasting, which is the life blood of a business concern.

During planning, companies care about accurate estimates that could lead to good results. The control phase requires evaluating the performance of CEOs, managers and staff by comparing results to the budget. CEOs, managers and employees often have to balance their interests and those of the company's, which is the shared goal between the company and individual employees.

Apply and Analyse 1

Apply

An automotive company, Wong Motor Ltd, is launching a new product line.

Prices of fuel are spiking, so Wong Motor is debating the introduction of a new production line of hybrid cars. The expected annual unit sales of the hybrid cars are 30,000 units for HK\$22,000 per car. The variable costs of production of each car is HK\$10,000 and fixed overhead is HK\$80 million per year.

The effect of introducing the new hybrid model is that it will lower sales of regular cars by 10,000 units every year. The unit price of regular cars is HK\$20,000 with a unit variable cost of HK\$12,000. Depreciation costs of the production plant are HK\$50,000 per year. The marginal tax rate is 40%. Calculate the incremental annual cash flow from operations.

Analysis

Step One. Revenue: $22,000 \times 30,000$ cars = HK\$660,000,000



Apply and Analyse 1 (continued)

Step Two. Op Exp: $10,000 \times 30,000$ cars = HK\$300,000,000, plus lost net revenue from regular sedans = $(20,000 - 12,000) \times 10,000$ cars = HK\$80,000,000; total Op Exp = HK\$380,000,000

Step Three. D&A: HK\$50,000

Step Four. Calculate using the template in Exhibit 10.2.

ΔNR	660,000,000
- $\Delta Op\ Ex$	<u>-380,000,000</u>
= $\Delta EBITDA$	280,000,000
- $\Delta D\&A$	<u>-50,000</u>
= $\Delta EBIT$	279,950,000
$\times (1 - t)$	<u>0.60</u>
= $\Delta NOPAT$	167,970,000
+ $\Delta D\&A$	<u>50,000</u>
= ΔCFO	168,020,000
- $\Delta CapEx$	0
- ΔAWC	<u>0</u>
= ΔFCF	168,020,000

Another way to calculate the incremental annual cash flow from operations is:

$$((22,000 - 10,000) \times 30,000 - (20,000 - 12,000) \times 10,000) \times (1 - 0.4) + 50,000 \times 0.4 = \text{HK\$}168,020,000$$

Fixed costs are not included in the calculations for incremental cash flows because they have to be paid regardless of whether the company invests in developing hybrid cars.

Key Learning Point

The reinvestment of residual cash flows is the most fundamental way that businesses grow in size.

► Knowledge Check Questions

Question 1

Identify what is involved in developing a capital budget.

- A Determining how to finance assets.
- B Determining how much to pay in dividends.
- C Determining what productive assets to purchase.
- D Deciding how much debt to carry in the capital structure.

Question 2

Explain the difference between profit and cash flows.

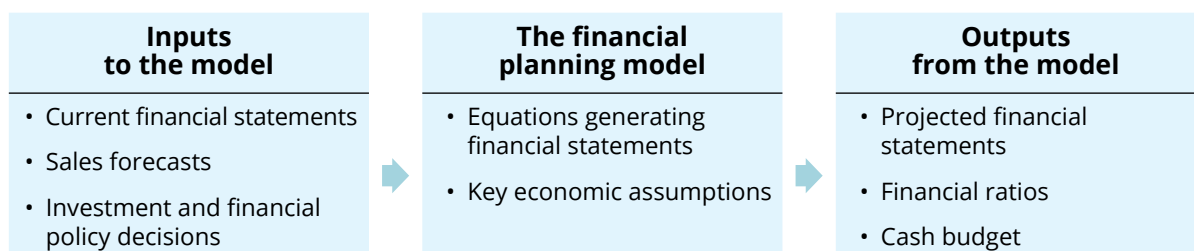
- A Profit is positive, but cash flow is a negative in every case.
- B Cash flows are the amount of cash left over after profits have been committed to expenditures.
- C Profit may involve non-cash transactions and accruals; cash flows are cash-based profits only, plus or minus transactions in capital assets.
- D Cash flows are strictly related to acquisition of capital assets and how they are financed.

➤ 10.2 PROFIT FORECASTS AND PROJECTION – PRO FORMA FINANCIAL STATEMENTS

A **financial planning model** is no more than a series of equations that is used to generate projected financial statements for a company, such as a statement of profit or loss and other comprehensive income or a statement of financial position.

As demonstrated in Exhibit 10.3, the three basic components of a financial planning model are:

1. Inputs to the model,
2. The model itself, and
3. Outputs from the model – the projected financial statements.



► **EXHIBIT 10.3 Components of a financial planning model** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

Important inputs to the financial planning model include current financial statements, sales forecasts and investment and financial policy decisions.

The starting point for constructing a financial planning model are a company's current statement of profit or loss and other comprehensive income and the statement of financial position.

These statements serve as a baseline.

For most financial planning models, the principal input variable is a forecast of sales or the sales growth rate. The sales forecast is the key driver in financial models because so many items on the statement of profit or loss and other comprehensive income and the statement of financial position changes depending on sales.

For instance, if sales increase, the company will likely need to use more labour and raw materials, and possibly more investments in capital assets and working capital.

Sales forecasts are developed for specific periods and expressed as percent change in sales:

$$\% \Delta S = \frac{(S_{t+1} - S_t)}{S_t}$$

where:

- $\% \Delta S$ = percent change in net sales,
- S_t = level of net sales in period t ,
- S_{t+1} = level of net sales in period $t + 1$.

Sales are calculated as the number of units sold multiplied by the price at which they are sold. For example, if the current year's sales (S_t) are HK\$100 million and the forecasted sales for next year (S_{t+1}) are HK\$120 million, applying the equation above yields the percent growth in sales over the coming year:

$$\% \Delta S = \frac{(S_{t+1} - S_t)}{S_t} = \frac{(\text{HK\$}120 - 100)}{100} = 0.20, \text{ or } 20\%$$

Putting together a financial planning model requires a number of investment and financial policy decisions which, in turn, impose constraints on outputs. These constraints also have to be recognised in preparing the model.

These key investment and financial policy decisions include:

- Investment policy decisions: Investment decisions typically involve large capital expenditures, such as new factories, new lines of business or a business acquisition.
- Financial policy decisions: Capital structure decisions, such as determining a targeted capital structure. Financing decisions, such as determining acceptable types of financing like retained earnings, equity, preferred stock or debt. And, payout decisions, such as identifying the company's dividend and stock repurchase policies for the sales period.

A financial planning model is a set of equations that generates projected financial statements. Along with the sales forecast and the investment and financial policy decisions, management must specify key assumptions regarding how the statement of profit or loss and other comprehensive income items and the statement of financial position accounts vary with sales.

For instance, imagine that, based on historical data, the cost of goods sold is 80% of sales and inventory and accounts receivable are 15% of sales. It makes sense that these will hold for the projected statement of profit or loss and other comprehensive income and statement of financial position.

Thus, if sales are projected to be HK\$100 million next year, the projected cost of goods sold will be HK\$80 million ($\text{HK\$100 million} \times 0.80 = \text{HK\$80 million}$), and inventory and accounts receivable will both be HK\$15 million ($\text{HK\$100 million} \times 0.15 = \text{HK\$15 million}$).

The financial planning model generates projected financial statements or **pro forma financial statements**. 'Pro forma' refers to forecasts or projections. These projected statements are based on inputs and assumptions. Apart from the pro forma financial statements, computer-based planning models usually generate a set of financial ratios and include features that enable management to prepare a cash budget.

Let's consider the example of Wong Sales Company Ltd, a small, privately owned company in Wan Chai, Hong Kong. Management owns 75% of the company stock, with the balance owned by friends and outside investors.

Wong Sales is projecting a banner year, as sales are expected to increase 30%. Management is conservative in regard to risk. Part of the management team is worried about the risk of increasing sales by 30% in a one-year period. The following information is available in their financial statements:

- Net sales during the fiscal year were HK\$2 million.
- The cost of production of goods and services averages 85% of sales.
- The average tax rate is 34.1%.
- The company pays 33.5% of earnings as cash dividends.

Exhibit 10.4 shows the firm's current (most recent) and pro forma statement of profit or loss and other comprehensive income.

The pro forma statement of profit or loss and other comprehensive income for Wong Sales assumes that the items vary directly with sales.

	Current	Pro forma	Assumptions
Net sales	2,000	2,600	Sales increase: 30%
Costs	<u>1,700</u>	<u>2,210</u>	Total costs = 85% of sales
Taxable income	300	390	
Taxes (34.1%)	<u>102</u>	<u>133</u>	
Net income	198	257	
Dividends	66	86	Dividend policy: 33.5% of net income
Addition to retained earnings	132	171	

► **EXHIBIT 10.4 Wong Sales current and pro forma statement of profit or loss and other comprehensive income (HK\$'000)** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

The calculations used to arrive at this pro forma statement of profit or loss and other comprehensive income take into account management's expected sales growth of 30%, so projected sales are $\text{HK\$}2 \text{ million} \times 1.30 = \text{HK\$}2.6 \text{ million}$.

Since total costs have averaged 85% of sales, projected total costs are $\text{HK\$}2.6 \text{ million} \times 0.85 = \text{HK\$}2.21 \text{ million}$.

Projected taxes, which are 34.1% of taxable income, are $0.341 \times \text{HK\$}390,000 = \text{HK\$}132,990$, which can be rounded to $\text{HK\$}133,000$ for simplicity.

Subtracting taxes from taxable income, makes it possible to calculate the company's projected net income of $\text{HK\$}257,000$.

Wong Sales cash dividend is $\text{HK\$}86,000$ ($0.335 \times \text{HK\$}257,000 = \text{HK\$}86,095$, which can be rounded to $\text{HK\$}86,000$) and the remaining $\text{HK\$}171,000$ of net income ($0.665 \times \text{HK\$}257,000 = \text{HK\$}170,905$) is retained in the company as an addition to retained earnings.

These amounts relate to two important ratios:

- The dividend payout ratio, and
- The retention ratio, or plowback ratio.

The formulas of the ratios and calculations for Wong Sales Company Ltd are as follows:

$$\text{Dividend payout ratio} = \text{Cash dividends} / \text{Net income}$$

$$= \text{HK\$}86,000 / \text{HK\$}257,000 = 0.335, \text{ or } 33.5\%$$

$$\text{Retention (plowback) ratio} = \text{Addition to retained earnings} / \text{Net income}$$

$$= \text{HK\$}171,000 / \text{HK\$}257,000 = 0.665, \text{ or } 66.5\%$$

The dividend payout ratio represents the portion of earnings paid as cash dividends while the retention ratio refers to the percentage of earnings retained in the company. Smaller, fast-growing companies retain more earnings (or most earnings) while more established companies that are growing more slowly and have larger cash flows distribute more dividends.

It is worth noting that the sum of the retention ratio and the dividend payout ratio equals 1 ($0.335 + 0.665 = 1$) since every dollar of net income is paid out or retained.

To generate a pro forma statement of financial position, a good place to start is the current statement of financial position, as shown in Exhibit 10.5. For each account that varies directly with sales, a relation is given as a percent of sales for the current year. Note that these percentages differ among the accounts.

The process of determining which accounts vary with sales, and how to calculate the relevant percentages is straightforward.

The process begins by looking at statement of financial position accounts that might vary with sales. This is done by expressing accounting data from four or five years as a percentage of sales. Trends may be easy to spot or trend lines fit into the data. This process makes it possible for management to decide which financial accounts may be estimated as a percentage of sales and which ones must be forecast from other data.

In this statement of financial position, many accounts vary directly with sales. The projected percent of sales is shown for each of these accounts. The accounts labelled 'n/a' do not change proportionately with sales.

Assets			Liabilities and shareholders' equity		
	Current	Projected % of sales		Current	Projected % of sales
Current assets			Current liabilities		
Cash	100	5%	Accounts payable	60	4%
Accounts receivable	120	6	Notes payable	<u>140</u>	<u>n/a</u>
Inventory	<u>140</u>	<u>7</u>	Total	200	n/a
Total	360	18%	Long-term debt	200	n/a
			Owner's equity		
			Common stock	10	n/a
			Retained earnings	<u>590</u>	<u>n/a</u>
			Total equity	<u>600</u>	<u>n/a</u>
Net fixed assets	<u>640</u>	<u>32</u>	Total liabilities and shareholders' equity	<u>HK\$1,000</u>	50%
Total assets	<u>HK\$1,000</u>	50%			

► **EXHIBIT 10.5 Statement of financial position (HK\$'000)** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

Exhibit 10.6 shows several years of historical data from accounts in the statement of financial position. The far-right column includes forecast values for 2018.

	Percent of sales				
	2014	2015	2016	2017	Forecast 2018
Cash	5%	5%	4%	5%	5%
Accounts receivable	10	9	9	9	6
Inventory	7	8	7	6	7
Accounts payable	4	4	4	3	4
Net fixed assets	30	32	34	32	32

► **EXHIBIT 10.6 Historical data** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

There is a rationale for assigning a percent of sales figure to these statement of financial position accounts.

The key working capital accounts generally vary with sales. For example, inventory has to rise along with any increases in sales. The historical data show how this works. Inventory levels stay at a constant percentage of sales ranging 6% to 8%. Management has to determine the optimal level of inventory as it undertakes plans ahead. Because it can be expensive to finance inventory, management often works to keep levels down. At the same time, if inventory becomes too low, companies may lose out on sales.



Consider Wong Sales Company Ltd, which has determined that the right inventory-to-sales ratio is 7%. The ratio of accounts receivable to sales is 9% but other companies with similar credit policies have receivables-to-sales ratios of 6%. As sales increase, Wong Sales has given more credit to customers. Still, Wong Sales wants to move its ratios to industry standard, which means a more aggressive approach to collections. At the same time, management wants cash accounts to total 5% of sales, a ratio that may provide a better level of liquidity to fund operations and pay for unexpected emergencies without tying up cash into low-yielding assets. In regard to liabilities, there is historical information to show that accounts payable shift alongside sales. This is intuitive because, as it sells more product, the company has to order more materials from suppliers. Management expects accounts payable to be 4% of sales.

Another variable that moves along with sales is net fixed assets. Here again, historical data confirms this assumption. Management at Wong Sales plans to use a four-year historical average of 32% as the target for the fixed assets to sales ratio. This means that for every HK\$100 in sales, the company needs HK\$32 in fixed assets.

Such a relation between fixed assets and sales may not always hold. The reason is that fixed assets may vary directly with sales only when a firm is operating at full capacity and fixed assets can be added in small increments.

As a final comment, notice that on Exhibit 10.5 on the asset side of the statement of financial position the total percent of sales for asset items adds up to 50%. This means that total assets are 50% of sales.

The ratio of total assets to sales is called the capital intensity ratio and is calculated for Wong Sales Company Ltd as follows:

$$\text{Capital intensity ratio} = \text{Total assets} / \text{Net sales}$$

$$= \text{HK\$1 million} / \text{HK\$2 million} = 0.5, \text{ or } 50\%$$

The **capital intensity ratio**, which is the inverse of the total asset turnover ratio, suggests how much in assets the company needs for every HK\$1 in sales. The higher the ratio, the more capital the company needs to generate sales. In other words, the higher the ratio, the more capital intensive the company is.

Companies that are more capital intensive are thought to have more risk than other companies that use fewer fixed assets. A high capital intensity is tied to high fixed assets and high fixed costs, including depreciation. In the event of a drop in sales, profits for firms with high fixed costs are likely to drop because these costs cannot be quickly reduced, at least not in the short term.

With a 50% capital intensity ratio, Wong Sales Company is not a highly capital-intensive firm.

For most companies, the remaining liability accounts on the statement of financial position do not change with sales. Rather, these values are more affected by management decisions. This means that every single liability and equity account has to be evaluated separately.

In the case of individual accounts, short-term borrowing is represented by notes payable. The value of this account changes only in response to management decisions like making

payments or borrowing. As a result, the value of this particular account does not change with sales.

Similarly, the account value for long-term debt changes only when management decides to issue or retire debt. The same argument holds for the common stock account, which changes only when management decides to sell or repurchase common shares. The last account is retained earnings, which may or may not change in response to movements in sales totals. The total funds held in retained earnings depends on both earnings and the company's dividend payout policy that management sets.

For the time being, both common stock and retained earnings accounts are entered as n/a. The retained earnings accounts can also be derived from the pro forma statement of profit or loss and other comprehensive income as demonstrated above.

It is now possible to construct a preliminary pro forma statement of financial position, as shown in Exhibit 10.7. The preliminary pro forma statement of financial position is a first approximation in deciding how the company should fund the assets it needs to support an increase in sales of 30%.

Assets			Liabilities and shareholders' equity		
	Projected	Changed		Projected	Changed
Current assets			Current liabilities		
Cash	130	30	Accounts payable	104	44
Accounts receivable	156	36	Notes payable	140	0
Inventory	182	42	Total	244	44
Total	468	108	Long-term debt	200	0
			Owner's equity		
			Common stock	10	0
			Retained earnings	761	171
Net fixed assets	832	192	Total equity	771	171
Total assets	1,300	300	Total liabilities and shareholders' equity	1,215	215
			External funding needed (EFN)	85	

► **EXHIBIT 10.7 Preliminary pro forma statement of financial position (HK\$'000)** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

This preliminary pro forma statement of financial position is a first approximation in deciding how to fund anticipated growth. At this stage of the analysis, the statement of financial position will not balance (total assets will not equal liabilities and shareholders' equity) and the difference will be the plug value, which is usually the amount of external funding the firm will need in order to fund investments and operations.

Once it is constructed, management can develop a suitable financing plan.

To construct the preliminary pro forma statement of financial position it is necessary to follow the following steps:

1. Calculate the projected values for all the accounts that vary with sales and enter these values into the preliminary pro forma statement of financial position.
2. Compute and enter the projected value of any other statement of financial position accounts for which an end-of-period value can be forecast or otherwise determined.
3. For all the accounts for which end-of-period values could not be forecast or otherwise determined, enter the current year's value.
4. Typically, the statement of financial position will not balance at this point. Thus, it is necessary to compute the plug value, which balances it. The plug value will involve the accounts marked "n/a" in the initial statement of financial position. These accounts must be analysed in light of the company's capital structure and dividend policies. The plug value is usually the amount of external funding needed (EFN), because it usually requires adding new assets to the statement of financial position to support growth; thus, total assets exceed total liabilities plus equity.

What does all the information tell management of Wong Sales Company Ltd?

First, if sales increase as projected, the company's total assets will expand by HK\$300,000. Of that HK\$300,000 increase, HK\$108,000 will go to increase current assets and HK\$192,000 will go to increase the firm's fixed assets.

Second, the HK\$300,000 in additional assets could be financed as follows:

- HK\$171,000 from internally generated funds (the addition to retained earnings),
- HK\$44,000 from expanded trade credit (the increase in accounts payable), and
- HK\$85,000 of external funding from the sale of debt or equity or both.

How should Wong Sales Company Ltd fund the HK\$300,000 to support the 30% increase in sales?

The company could issue debt, equity or reduce dividends. Alternatively, it could also rethink its strategy and scale back the 30% targeted growth figure.

In the end, the decision is made to pay no cash dividend to shareholders for the coming year. As a result, the HK\$300,000 increase in assets is funded entirely from earnings. Management has decided to avoid the risks associated with additional debt and the dilution of earnings and/or voting right that would result from issuing additional common stock.

Exhibit 10.8 shows the final pro forma statement of financial position reflecting the decision to temporarily suspend dividends and fund the expansion with internal funds (retained earnings). Although financial models can determine the amount of EFN, management must make the final decision about how to fund the firm's capital requirements.

Assets			Liabilities and shareholders' equity		
	Projected	Changed		Projected	Changed
Current assets			Current liabilities		
Cash	130	30	Accounts payable	104	44
Accounts receivable	156	36	Notes payable	139	-1
Inventory	182	42	Total	243	43
Total	468	108	Long-term debt	200	0
			Owner's equity		
			Common stock	10	0
			Retained earnings	847	257
Net fixed assets	832	192	Total equity	857	257
Total assets	1,300	300	Total liabilities and shareholders' equity	1,300	300
			External funding needed (EFN)	0	

► **EXHIBIT 10.8 Pro forma statement of financial position (HK\$'000)** (Source: Parrino 2017. Reproduced with permission of John Wiley & Sons.)

Lastly, only management makes decisions, not financial models. Financial models provide numbers, but these numbers are the result of the inputs and assumptions that go into the models. Management uses financial models to consider its strategic alternatives, assess financial impact of various decisions and determine whether they are consistent with financial policies.

Illustrative Example 2

Pro forma means 'for the sake of form'. Pro forma financial statements are modified actual financial statements to show what these documents would look like in some theoretical scenarios.

For instance, entrepreneurs that want to pitch an idea to investors prepare pro forma financial statements to show what the company could look like after it is in operation. The entrepreneur is also likely to prepare pro forma financial statements when the company undertakes a one-time expenditure to show the impact of that expenditure. In this case, the pro forma financial statements would be included in annual reports.

Below is a very simple example of a pro forma statement of profit or loss and other comprehensive income. Assume a company just had a restructure. Accounting regulations require that company to include any restructuring charge on the statement of profit or loss and other comprehensive income. Since the restructuring charge was huge, the company's income was wiped out and the company showed a loss for that period.

However, this restructuring charge is not part of normal business operations, so the company included a pro forma version of the statement of profit or loss and other comprehensive income in its annual report.



Illustrative Example 2 (*continued*)

This is to show investors and other interested parties what the company's statement of profit or loss and other comprehensive income would have looked like without that one-time restructuring charge.

The difference between the original statement of profit or loss and other comprehensive income and the pro forma statement of profit or loss and other comprehensive income are shown below. Removing the one-time restructuring charge turns the loss into profit. Management may want to show why the company recorded a loss during a particular period so that investors and other stakeholders understand.

Original statement of profit or loss and other comprehensive income

Revenue	HK\$500,000
Cost of goods sold	(250,000)
Restructuring charge	(300,000)
Interest income	50,000
Interest expense	(25,000)
Net income (loss)	<u>(HK\$25,000)</u>

Pro forma statement of profit or loss and other comprehensive income

Revenue	HK\$500,000
Cost of goods sold	(250,000)
Interest income	50,000
Interest expense	(25,000)
Net income (loss)	<u>HK\$275,000</u>

Ethics in Practice

Forecasting is often a politically charged activity. Strategic business decisions, resource allocations in terms of budgeting and human resource careers depend on predictions and not actual events. Forecasters are often in a difficult position of having to bear bad news or contradict company officials.

Additionally, as you are well aware by the above profit forecasts and projection – pro forma financial statements, it is always possible for a forecast to be biased by restrictions to the models or by the explanatory variables that are considered, by the length of the sample, by deciding what observations to include or by focusing on short-term or long-term trends, etc. These are just some examples of variables that can impact a forecast. A clever, unscrupulous or careless analyst could steer a forecast in a particular direction.

Key Learning Point

The three basic components of a financial planning model are: inputs to the model, the model itself, and outputs from the model.

► Knowledge Check Questions

Question 3

Excellent Traders has total assets of HK\$513,480 and sales of HK\$723,062. Calculate the firm's capital intensity ratio.

- A 1.41
- B 0.71
- C 1.23
- D 0.45

Question 4

A company paid out HK\$163,961.60 as dividends on net income of HK\$298,112. Calculate the firm's retention ratio.

- A 55%
- B 45%
- C 50%
- D 60%

10.3 BUSINESS PLANNING FORMULATION AND EVALUATION

Businesses are launched for different reasons. For example, some companies start with an idea for a new product while others deal with a market need. An important decision is whether to start a new business or purchase one that already exists. Starting a business may provide more potential rewards but come with more risk while growing an existing business may come with less risk but more limited growth.

Company founders have to choose products or services to sell and the markets in which to sell them as well as the strategy for the company. Founders have to raise money, acquire assets and hire people. Founders have to manage day-to-day operations and make sure they implement plans efficiently.

There are multiple reasons behind business failures. Consumers may reject the products or services of the company or management may follow a weak strategy or lack the management



skills to carry out the right strategy. Some companies fail to properly estimate how much capital they will need or the amount and timing of cash inflows and outflows, which means that the company does not have enough money to operate. Many new businesses fail, which means it is important to carefully plan any new venture.

From a financial perspective, there are a number of factors that affect the choice of the appropriate organisational form of a new business venture.

Making a company independent can make ownership interests more liquid and facilitate access to capital or investors. For both partnerships and corporations, raising capital can be expensive.

Agency problems may arise when owners pass on the process of decision-making to professional managers but the benefits might outweigh the costs. Separating ownership from management control gives the firm the ability to raise more capital from investors that are not interested in being directly involved. This can drastically increase the number of investors to which the company has access.

Also, using this approach, an entrepreneur may be able to turn control of the company over to a more effective manager, become less involved but still reap the benefits of success.

It is important for companies to obtain funding, so understanding cash requirements is key. A miscalculation may spell ruination for the company because raising capital can take time and becomes even more difficult when the cash position of a company become tighter.

Investors are reluctant to invest in companies that are short of cash because a business that is in this position may not be viable or the owner may not be the right person to run it.

To understand the cash requirements of a company, there are two particularly useful tools. The first is the cash flow break-even analysis. The second is the cash budget.

10.3.1 Cash Flow Break-even

A pre-tax operating cash flow (EBITDA) break-even analysis can be used to determine how much sales the company needs to break even on operations from a pre-tax operating cash flow perspective.

It is calculated using the following equation:

$$\text{EBITDA Break-even} = \text{FC} / (\text{Unit Price} - \text{Unit VC})$$

Where:

- FC = the fixed costs associated with the business,
- Unit VC = the variable cost per unit,
- Unit Price – Unit VC = the contribution margin per unit.

The concept of EBITDA break-even is important to understand. The focus of this calculation is to maximise the per-unit contribution of each product and minimise overheads. At the same time, this calculation may offer a way to estimate how long it takes a product to break even and how much capital will be needed to launch a new product.

Entrepreneurs should want to maximise per-unit contributions and minimise costs but they often lose sight of this simple need by getting caught up in developing the best possible product and not considering how much customers are willing to pay. For instance, adding new features can be expensive, more expensive than what customers are willing to pay.

Businesses should always look for ways to maximise the per-unit contribution of products while keeping a competitive position.

10.3.2 Cash Budget

The cash budget is very useful for management planning efforts. The budget summarises cash flows in and out of the company over a period of time. The cash budget presents inflows and outflows on a monthly basis and can be prepared for any period of time.

Preparing a cash budget gives a clear picture where capital comes from and where it goes and how much cash and investment a company needs (see Module 2, Chapter 9).

Understanding what happens to the money, where it comes from and where it goes, makes it possible to better control finances and undertake fundraising in time to make sure the company succeeds.

Illustrative Example 3

You are considering investing in a small retail business that has monthly fixed costs of HK\$5,500 and sells only one product. To produce this product, the variable cost per unit is HK\$35, and can sell for HK\$90 per unit. What is the annual pre-tax operating cash flow break-even point for this business?

You can solve the monthly pre-tax operating cash flow break-even point:

$$\begin{aligned}\text{EBITDA Break-even} &= \text{FC} / (\text{Unit Price} - \text{Unit VC}) = 5,500 / (90 - 35) \\ &= 100 \text{ units per month}\end{aligned}$$

Therefore, the annual EBITDA break-even point is 100 units per month × 12 months per year = 1,200 units.

Illustrative Example 4

HKQP Furniture Company currently manufactures 1,000 units of a plastic chair P369.

The per unit selling price and cost information are as follows.

	HK\$ per unit
Selling price (SP)	150
Variable costs (VC):	
Direct materials	40
Direct labour	20
Manufacturing overhead costs	15
Selling and distribution costs	6
Total VC per unit	81
Fixed costs (FC):	
Manufacturing overhead costs	10
Selling and distribution costs	4
Finance and administrative costs	5
Total FC per unit	19

- Compute the break-even quantity in units.
- Compute break-even in sales dollars.
- Compute the break-even sales quantity if variable cost is increased by 5%.
- Compute the break-even sales dollars if fixed cost is increased by HK\$1,000.

Answer

(a) Total fixed cost (FC) = 10,000 + 4,000 + 5,000 = HK\$19,000

Unit selling price (SP) = HK\$150

Unit variable cost (VC) = HK\$81

Contribution margin per unit (CM) = (SP – VC) = (150 – 81) = HK\$69

Therefore, break-even quantity in units (Q) = 19,000 / 69 = 276 units

For break-even units, round up to the next whole unit.

Illustrative Example 4 (*continued*)

(b) Contribution margin ratio = $CM / SP = 69 / 150 = 46\%$

Therefore, break-even in sales dollars = $19,000 / 46\% = \text{HK\$}41,305^*$

or

$Q \times SP = 276 \text{ units} \times 150 = \text{HK\$}41,400^*$

*Difference due to rounding up number of units.

(c) Contribution margin per unit (CM) = $(SP - VC) = (150 - 81 \times 105\%) = \text{HK\$}65$

Therefore, break-even quantity in units (Q) = $19,000 / 65 = 292 \text{ units}$

For break-even units, round up to the next whole unit.

(d) Contribution margin ratio = $CM / SP = 69 / 150 = 46\%$

Therefore, break-even in sales dollars = $\text{HK\$} (19,000 + 1,000) / 46\% = \text{HK\$}43,478$

Key Learning Point

Two tools are especially useful in understanding the cash requirements: the cash flow break-even analysis and the cash budget.

► Knowledge Check Questions

Question 5

Jimmy's Products Inc is looking to invest in a product line that will increase its fixed costs to HK\$35,000. If each unit of the product sells for HK\$600 and costs HK\$390 to produce, calculate the projects' pre-tax operating cash flow break-even.

- A 191 units
- B 149 units
- C 167 units
- D 145 units



10.4 ROLES AND COMPOSITION FOR A BUSINESS PLAN

Financing a business is rarely simple. An important tool in financing a business – as well as in managing it – is the **business plan**.

Some of the sources of the equity capital used by entrepreneurs are: their own money, investments from friends and family, investments by venture capitalists and equity raised by selling shares in the stock market. Debt financing can also come from a wide variety of sources. Aside from the entrepreneur, banks, local individual investors, other businesses and the sale of debt in the public debt markets are among the sources that debt financing can come from.

For a young, rapidly growing business, having enough cash is a simple matter if the money comes only from the entrepreneur as the entrepreneur only needs to make the decision on their own. With other sources of financing, things can be more complicated.

Potential investors have to be convinced that purchasing debt or equity in the company will yield attractive returns, that the entrepreneur will have to be persuasive enough and make investors feel that they will be adequately compensated for bearing the risk.

Even if the business has a well-established track record, it is still difficult enough to convince people to invest.

To overcome the scepticism of outside investors, businesses must prepare a business plan, especially when the business is in the very early stages of development.

In addition to its usefulness in raising capital, a business plan can help a company set the goals and objectives, serve as a benchmark for evaluating and controlling the company's performance, and communicate ideas to managers, outside directors, customers, suppliers and other stakeholders.

A thoroughly thought-out business plan can help a company avoid problems and position them to be able to deal with problems that do occur. In short, business planning is extremely important to the set-up, survival and expansion of a company.

The depth and breadth of business plans vary widely. Most well thought-out business plans include:

- An executive summary.
- An overview of the company that includes any comparative advantages and a description of what it does.
- A detailed description of the products and services, the state of development, competitive advantages and the product life cycle.
- A market analysis that looks into the company's products and highlights the characteristics of the markets in which the company operates.
- Details of marketing and sales activities.
- A discussion of how the company will operate and distribute products and services.

- A list of the management team, the organisational structure of the company and the talents and skills of managers.
- The ownership structure, including any securities issued by the company and the value of those securities.
- Capital requirements.
- Historical financial results, if available, financial forecasts and pro forma financial statements. Analysis of results based on financial statement analysis tools also goes in this section along with a month-by-month cash budget for two or three years and yearly forecasts of operating results.
- Various appendices may include detailed supporting information.

A business plan is a road map that outlines the results of careful strategic planning and paints a picture of how a business will be run and grow over time. A strong business plan makes it easier to communicate to investors what the business can become in the future and how it will get there.

Illustrative Example 5

Business plans are critical to the success of any new venture. They serve as the framework for a company and provide benchmarks for future goals. They are key to helping entrepreneurs think through their business and keep them on track. They should outline possible business strategies and consider all variables.

The size and scope of a business plan depends on the specific goals. A business plan drafted for investors should be detailed and clearly explain the industry and where the concept fits. A business plan developed for the entrepreneurs themselves or their partners would not be as detailed but should still outline goals and proposals on how to attain them.

If the product or service is not excessively complex, the business plan does not need to be very lengthy.

Although the exact structure of business plans varies, the following 10 components can be presented in a business plan or proposal.

1. Mission or vision statement;
2. Description of the company or its offerings;
3. Description of its products or services;
4. Analysis of the market, competitors, where the company fits and potential market share;
5. Description of the management team;
6. A marketing plan;
7. Analysis of strengths, weaknesses, opportunities and threats;
8. Cash flow statement;
9. Revenue projections; and
10. Summary and conclusion to bring it all together.

Key Learning Point

A business plan is helpful to raise capital, set the goals for a company and objectives for growth as well as setting a benchmark to evaluate and control company performance while communicating ideas to all stakeholders – managers, directors, customers, suppliers and others.

► Knowledge Check Questions

Question 6

Determine which of the elements will not be included in a well-developed business plan.

- A** Executive summary
- B** Company overview
- C** Market analysis
- D** Employees' personal information

Question 7

Writing a business plan is important to help a company launch, grow and succeed. Drafting a business plan is best done step by step. Describe briefly the following three main sections of a business plan and what each should include.

- (a) Products and services
- (b) Market analysis
- (c) Strategy and implementation

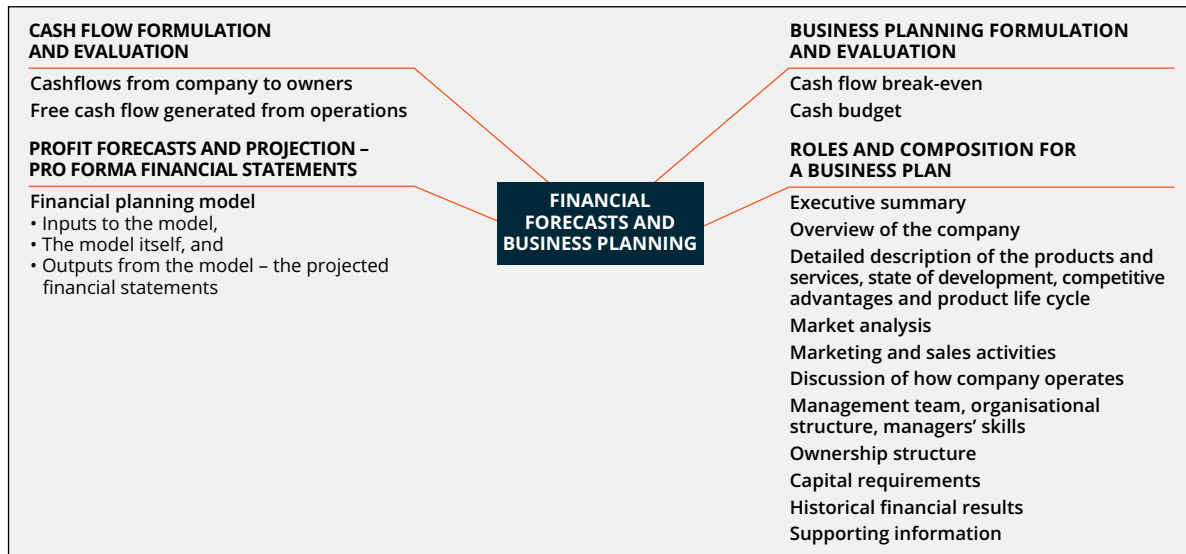


SUMMARY

- To produce its products or services, a company needs to purchase a range of assets. Most will be long-term assets, which are also known as productive assets.
- Regardless of the type of asset, the company tries to select assets that will generate the greatest cash flows for the owners.
- The decision-making process through which the company purchases productive assets is called capital budgeting, and it is one of the most important decision processes in a company.
- There are two basic sources of funds: debt and equity.
- Companies generate cash flows by selling the goods or services. Successful companies have cash inflows that are larger than cash outflows after paying operating expenses, creditors and taxes.
- Managers can distribute any cash remaining after settling obligations. The leftover cash is called residual cash flows. These residual cash flows can be distributed to owners as a cash dividend, by repurchasing some shares or reinvesting in the business.
- Companies that are unprofitable over time will be forced into bankruptcy by their creditors if the owners do not shut them down first.
- The approach used to calculate project cash flows is similar to that used to calculate the cash flow to investors with two key aspects: Any cash flows used in capital budgeting are based on forecasts of future revenues, expenses, taxes and investment outlays. In capital budgeting, the focus is on estimating the cash flows expected from individual project to be produced in the future, known as incremental after-tax free cash flows.
- The incremental after-tax free cash flows (FCF) for a project is calculated as the total after-tax free cash flows the company would produce with the project, less the total after-tax free cash flows the company would produce without the project.
- Free cash flows refer to cash flows that companies can distribute to creditors and shareholders because they are left in the company after any necessary investments in working capital and long-term assets are made.
- The incremental cash flow from operations equals the incremental net operating profits after tax (NOPAT) plus D&A. NOPAT is essentially a cash flow measure of the incremental net income from the project without interest expenses.
- A financial planning model is no more than a series of equations that is used to generate projected financial statements for a company, such as a statement of profit or loss and other comprehensive income or a statement of financial position.
- The three basic components of a financial planning model are: Inputs to the model, the model itself and outputs from the model.

- Preparing a financial planning model requires top management to make a number of investment and financial policy decisions.
- The outputs from the financial planning model are projected financial statements called pro forma financial statements.
- To generate a pro forma statement of financial position, a good place to start is the current statement of financial position.
- The capital intensity ratio, which is the inverse of the total asset turnover ratio, suggests the amount of assets that the company needs to generate for a dollar in sales.
- Financial models can only generate numbers based on the inputs and assumptions made when the model was constructed. Once financial models have been constructed, they can help management evaluate strategic alternatives, assess their financial impact on the company and whether they are consistent with the company's financial policies.
- The cash flow break-even analysis and the cash budget are useful to understand the cash requirements of a company.
- An important tool in financing a business – as well as in managing it – is the business plan.
- A business plan provides a road map for a business and outlines the results from a strategic planning process, describes where the company is going and how it will get there.

MIND MAP



LIST OF FORMULAS

Free cash flows (FCF)

$$FCF = [(Revenue - Op\ Ex - D\ \&\ A) \times (1 - t)] + D\ \&\ A - Cap\ Exp - Add\ WC$$

Dividend payout ratio

$$Dividend\ payout\ ratio = Cash\ dividends / Net\ income$$

Retention ratio

$$Retention\ ratio = Addition\ to\ retained\ earnings / Net\ income$$

Capital intensity ratio

$$Capital\ intensity\ ratio = Total\ assets / Net\ sales$$

Pre-tax operating cash flow

$$EBITDA\ Break-even = \frac{FC}{Price - Unit\ VC}$$

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. Capital budgeting does not help how a firm's assets should be financed. Instead, it helps in determining which asset should be purchased. Capital budgeting is a decision-making process through which the firm purchases long-term productive assets.

Answer B is incorrect. Dividend payments are not part of the capital budgeting decision. Capital budgeting is a decision-making process through which the firm purchases long-term productive assets.

Answer C is correct. Capital budgeting is a decision-making process through which the firm purchases long-term productive assets.

Answer D is incorrect. Capital budgeting does not involve determining a firm's capital structure. Capital budgeting is a decision-making process through which the firm purchases long-term productive assets.

Question 2

Answer A is incorrect. The positive or negative aspects of the two are not the defining attributes.

Answer B is incorrect. Cash flows are not an aspect of how profits are committed to expenditures.

Answer C is correct. The distinction is that profits are not all cash-based and exclude non-profit expenditures such as capital asset purchases.

Answer D is incorrect. Cash flows involve much more than acquisition of capital assets.

Question 3

Answer B is correct. Capital intensity ratio = Total assets / Net sales = 0.71

Question 4

Answer B is correct. Dividend payout ratio = Dividends paid / Net income = HK\$163,961.60 / HK\$298,112 = 55%. Retention (plowback) ratio = (1 – Dividend payout ratio) = (1 – 0.55) = 45%

Question 5

Answer C is correct. Pre-tax operating cash flow (EBITDA) break-even point = FC / (Price – Unit VC) = HK\$35,000 / (HK\$600 – HK\$390) = 167 units

Question 6

Answer A is incorrect. It summarises the key points made in the plan.

Answer B is incorrect. It describes what the company does and what its comparative advantages are.

Answer C is incorrect. It considers the markets for products and the characteristics of these markets in relation to the company.

Answer D is correct. It is part of employee's privacy and should not be disclosed in the business plan.

Question 7

(a) Products and services

Describe in detail what is being sold and how it will benefit customers with details about suppliers, product or service costs along with net revenue from sales that can be expected. Pictures or diagrams may be useful.

This section should include:

- A description of the product or service and how it helps customers;
- An explanation of the role that the product or service will play in the market and any pros and cons over competitors;
- Details of the product or service life cycle;
- Copyright, patent or intellectual property data;
- Research and development activities.

(b) Market analysis

Show industry knowledge and include conclusions from market research. Detailed findings should be included in an appendix. The market analysis should include:

- Targeted customer segments;
- Industry description and outlook;
- Historical, current and projected marketing data;
- An evaluation of competitors.

(c) Strategy and implementation

Summarise your sales and marketing strategy, and how you will implement them with an operating plan. This section should include:

- An explanation of how the business will be promoted and how it will enter the market;
- Information on costs, pricing, promotions and distribution;
- Details of operations, from acquisition of supplies to product delivery;
- Details about labour and number of employees;
- Operating hours and facilities.



EXAM PRACTICE

QUESTION 1

Identify which of the following best describes the cash flow statement of a company.

- A.** A listing of the flows of cash into and out of the business or project.
- B.** Cash deposits and withdrawals in a company's checking account at the bank.
- C.** The balance in a company's bank at a specific point in time.
- D.** A projection of future flows of cash.

QUESTION 2

Identify which of the following describes a pro forma forecast.

- A.** The initial groundwork for forecasting incomes and expenses.
- B.** A pro forma forecast is a financial forecast based on pro forma financial statements.
- C.** Pro forma financials reflecting the expected state of a business after a significant transaction.
- D.** Inclusion of future events in pro forma financial statements

QUESTION 3

As business planning formulation and evaluation is crucial, identify which of the following best describes what a business plan is.

- A.** A complete story about the business.
- B.** A description of various details that impact and contribute to the success of the business.
- C.** A document that provides a guide to a company and outlines its prospects and growth potential with the goal of obtaining financing.
- D.** A clear and concise model about the business project.

QUESTION 4

Identify which of the following explains what can be done through the process of creating a business plan.

- A.** Secure funding for a start-up or expanding operations.
- B.** Force managers to look at their business and evaluate what is working and what is not.
- C.** Help managers focus on the right things and give them a roadmap to future success.
- D.** All of the above.

QUESTION 5

Explain the difference between profitable and unprofitable firms.

QUESTION 6

List the appropriate decision rule for a firm considering undertaking a capital project with a real-life example.

QUESTION 7

Bubbles Inc had revenues of HK\$12 million, cash operating expenses of HK\$8 million, and depreciation and amortisation of HK\$1.5 million during 2018. HK\$700,000 worth of equipment was purchased during the year while the firm increased its inventory by

HK\$500,000 (with no corresponding increase in current liabilities). The marginal tax rate for Bubbles is 30%.

- Compute Bubbles's cash flow from operations for 2018.
- Compute Bubbles's free cash flow for 2018.
- Determine Bubbles's cash flows associated with investments for 2018.

QUESTION 8

McDonald Metal Works has been able to generate net sales of HK\$13,445,196 on assets of HK\$9,145,633. Compute the firm's capital intensity ratio.

QUESTION 9

For McDonald Metal Works in Question 8, calculate how much must net sales grow if the capital intensity ratio has to drop to 60%. State your answer as both a percent of sales and a dollar sales increase.

QUESTION 10

For each of the three production choices, calculate the accounting operating profit break-even point and pre-tax operating cash flow break-even point.

Choice	Price	Unit VC	FC	D&A
A	HK\$250	HK\$160	HK\$15,000	HK\$3,000
B	HK\$55	HK\$10	HK\$1,100	HK\$200
C	HK\$10	HK\$1.50	HK\$100	HK\$100

QUESTION 11

State what a business plan is. Explain how a business plan can help an entrepreneur to build a successful business.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is correct. As a comprehensive description and definition.

Answer B is incorrect. They represent the daily transactions of a company's bank account.

Answer C is incorrect. It represents the ending cash balance of a company's bank account of a certain accounting period.

Answer D is incorrect. A projection of future flows of cash is a cashflow forecast or budget for planning purpose.

QUESTION 2

Answer A is incorrect. They are only the basic ground work.

Answer B is correct. A comprehensive description and definition.

Answer C is incorrect. They should be done before the event or business decision was made.

Answer D is incorrect. Those future events should be quantified in monetary terms as far as possible.

QUESTION 3

Answer A is incorrect. It is just the historical background of a company.

Answer B is incorrect. Both strengths, weaknesses, opportunities and threats should be described in a business plan.

Answer C is correct. A comprehensive description and definition.

Answer D is incorrect. It is only a component in a business plan.

QUESTION 4

Answer A is incorrect. It is just the initial purpose of a business plan in its infant stage.

Answer B is incorrect. It is just one of the operational guides of a business plan.

Answer C is incorrect. It is just a forward-looking aspect of a business plan.

Answer D is correct. A, B and C encompass the process of business planning.

QUESTION 5

A profitable firm generates enough cash flows to cover operating expenses, taxes and payments to creditors. Unprofitable firms do not and may be forced into bankruptcy.

QUESTION 6

A firm should undertake a capital project only if the value of the project's future cash flows exceeds the cost of the project. For example, a financial manager would not invest HK\$10,000,000 in a new production line if the future cash flows from that line are expected to produce only HK\$9,000,000 in future cash flows. That would be like throwing HK\$1,000,000 away.

QUESTION 7

(a) Bubbles Inc		
	Revenue	HK\$12,000,000
–	Operating Ex	8,000,000
	EBITDA	HK\$4,000,000
–	D&A	1,500,000
	EBIT	HK\$2,500,000
×	(1 – <i>t</i>)	70%
	NOPAT	HK\$1,750,000
+	D&A	1,500,000
	CF Opns	HK\$3,250,000

(b)

Bubbles Inc		
Revenue		HK\$12,000,000
- Operating Ex		8,000,000
EBITDA		HK\$4,000,000
- D&A		1,500,000
EBIT		HK\$2,500,000
× (1 - t)		70%
NOPAT		HK\$1,750,000
+ D&A		1,500,000
CF Opns		HK\$3,250,000
- Cap Exp		HK\$700,000
- Add WC		500,000
FCF		HK\$2,050,000

- (c) Cash flows associated with investments equal the purchase of tangible and intangible assets as well as increases in working capital. Therefore, the cash flows associated with investments equal 700,000 + 500,000 = HK\$1,200,000.

QUESTION 8

Sales = HK\$13,445,196

Total assets = HK\$9,145,633

$$\begin{aligned}\text{Capital intensity ratio} &= \frac{\text{Total assets}}{\text{Net sales}} \\ &= \frac{\text{HK\$9,145,633}}{\text{HK\$13,445,196}} = 0.6802 \text{ or } 68.02\%\end{aligned}$$

QUESTION 9

Target capital intensity ratio = 60%

$$\begin{aligned}\text{Capital intensity ratio} &= \frac{\text{Total assets}}{\text{Net sales}} \\ 0.60 &= \frac{9,145,633}{\text{Net sales}} \\ \text{Net sales} &= \frac{9,145,633}{0.60} = \text{HK\$15,242,721.67}\end{aligned}$$

Growth in net sales needed = New sales - Old sales = 15,242,722 - 13,445,196 = HK\$1,797,526

Percentage change in net sales needed = 1,797,526 / 13,445,196 = 13.37%

QUESTION 10

Choice	Unit price	Unit VC	Fixed costs	D&A	EBIT break-even ^a	Pre-tax operating cash flow break-even ^b
A	HK\$250	HK\$160	HK\$15,000	HK\$3,000	200	167
B	HK\$55	HK\$10	HK\$1,100	HK\$200	29	24
C	HK\$10	HK\$1.5	HK\$100	HK\$100	24	12

^a EBIT break-even = (Fixed Costs + D&A) / (Price – Unit VC)

^b Pre-tax operating cash flow break-even = Fixed costs / (Price – Unit VC)

QUESTION 11

A business plan describes in detail how a business will evolve. A well thought-out business plan can help an entrepreneur or company raise capital and communicate any expectations for what the business will look like and how it will get there as well as the returns investors can expect. At the same time, a business plan can help entrepreneurs set goals and objectives, provide benchmarks to evaluating company performance and communicate ideas to managers and stakeholders.



Part E

Strategic Management Accounting



Chapter 11 Strategic Management Accounting Framework

Chapter 12 Pricing Strategies and Decisions

Chapter 13 Investment Project Appraisal and Evaluation

Chapter 14 Post-Appraisal Audit of Projects

Chapter 15 Effective Performance Measurement

11

Strategic Management Accounting Framework

CHAPTER TOPIC LIST

11.1 Current Issues in Strategic Management Accounting

11.2 Linkages Between Management Accounting and Strategic Management of an Organisation

11.3 Business Planning Processes for Corporate Strategic Plans

11.4 Strategic Cost Management Techniques

11.4.1 Just-in-Time

11.4.2 Total Quality Management Concepts

11.4.3 Product Life Cycle Costing

11.4.4 Target Costing

11.4.5 Quality Costing

11.4.6 Theory of Constraints

11.4.7 Throughput Analysis



LEARNING OUTCOMES

PRINCIPAL LO4: ANALYSE THE STRATEGIC MANAGEMENT ACCOUNTING FRAMEWORK

LO4.01: Describe the key elements of a strategic management accounting framework

4.01.01 Explain current issues in strategic management accounting

4.01.02 Explain the linkages between management accounting and the strategic management of an organisation

4.01.03 Describe the business planning processes including the development of corporate strategic plans

4.01.04 Describe strategic cost management techniques, including just-in-time (JIT) and total quality management (TQM) concepts, product life cycle costing, target costing, quality costing, theory of constraints and throughput analysis



OPENING CASE

ABC INVESTMENT LTD

Your job at ABC Investment Ltd has pushed you deeper into the strategic management accounting of the company's subsidiaries. Key decisions must be made based on the skilful use of accounting information in order to maximise investment returns and minimise the use of capital.

Working to help the company's subsidiaries, your job requires you to work through strategic management accounting, business planning processes, strategic planning and inventory costing. This increasingly leads to an appreciation for the qualitative factors that go into strategic financial management.

With greater experience and competence increase, your job increasingly requires you to handle some of these qualitative factors effectively to enhance strategic financial management. 

OVERVIEW

The Chartered Institute of Management Accountants (CIMA) defines management accounting as 'the process of identification, measurement, accumulation, analysis, preparation, interpretation and communication of information used by management to plan, evaluate and control within an entity and to assure appropriate use of and accountability for its resources. Management accounting also comprises the preparation of financial reports for non-management groups such as shareholders, creditors, regulatory agencies and tax authorities' (CIMA Official Terminology).

The Institute of Management Accountants (IMA) made a recent update to its definition: 'management accounting is a profession that involves partnering in management decision making, devising planning and performance management systems, and providing expertise in financial reporting and control to assist management in the formulation and implementation of an organisation's strategy'.

Strategic management accounting helps companies aim for and achieve success. This is a form of management emphasising factors outside of the organisation, as well as internally based non-financial information.

A strategic management accounting framework is designed to take into consideration a number of forward-looking concepts to help build a forward-looking plan of action that not only takes into consideration factors internal to the company but also external factors that other types of accounting do not consider.

This chapter provides an overview of the main issues associated with strategic management accounting and links management accounting and strategic management. This includes considerations of business planning and several concepts that can aid in the planning process, as discussed in Chapter 10.

The concepts for planning and managing costs include just-in-time, total quality management, product life cycle costing, target costing, quality costing, theory of constraints and throughput analysis.



11.1 CURRENT ISSUES IN STRATEGIC MANAGEMENT ACCOUNTING

Strategic management accounting (SMA) is a specific field of accounting that brings together the business objectives with the information generated by **management accounting**. The combination of strategic objectives and management accounting information generates forward-looking projections and models that help management in its decision-making process.



While management accounting focuses on internal metrics, SMA considers external information such as cost trends, prices, market share of the company and competitors, cash flows, access to resources and more. Strategic management accounting also is based on analysis of significant information about competitors, suppliers and technologies.

Strategic management accounting has a triangular structure with three primary considerations:

1. Technical analysis focused on information about specific events.
2. Behavioural considerations about the best path to achieve the company's objectives.
3. Cultural considerations based on shared beliefs within the organisation.

Combined the technical, behavioural and cultural considerations create an effective SMA programme.

These broad considerations are examined in light of three important factors for any company:

1. Quality,
2. Cost, and
3. Time.

These three factors, known as QCT, are used by companies to set themselves apart within the market in which they operate. Companies regularly consider QCT factors and contrast them with the preferences of customers or the demands of the market. For example, since the 1970s, McDonald's restaurants have been promulgating the QCT factors as their business and operational strategies – high quality of all the foods and drinks produced; continuous cost control in order to offer affordable prices for all customers; and fast delivery time of foods and drinks ordered by customers. Of course, clean and comfortable dining environment is also a key success factor in McDonald's operation.

At times, companies may be more focused on one factor or another. Sometimes, getting a product to market first (time) is more important than its price (cost), for example, the iPhone products. For other companies, demands for quality from customers may be the primary consideration, such as the car industry.

The key difference between strategic management accounting and traditional management accounting is this focus on competitive conditions. Ultimately, SMA looks at the entirety of a company's operations and how it gets its products or services to customers to develop forward-looking strategies for the success of the company.

Another distinguishing feature is that SMA may consider a sequence of decisions and patterns, compared to management accounting and its tendency to focus on a single issue (i.e. cost, volume or profit). In other words, management accounting is introspective in nature, and SMA is designed to look beyond the company to understand markets and competitive challenges.

Management accounting most often is an expansion of **cost accounting** (processes designed to analyse cost inputs and to assign them to various production processes or segments). Consequently, management accounting tends to focus on manufacturing. SMA is quite different in the sense that it focuses more on the nature of competition and a broader analysis of relevant factors (such as competitive profitability or market share).

Current issues involved with SMA include adjustments made to accounting frameworks due to the globalisation of companies. Today, a growing portion of companies operates in several countries and regions; this was not the case a few decades ago. This raises many management-based issues that must be addressed to remain competitive.

For example, supply chain risk management in the past focused on domestic processes with a very small number of vendors or suppliers located overseas. Today, many organisations rely on widespread resources, which increases risks on many levels and demands a strategic approach to management accounting. This was the genesis of SMA in the 1980s out of concern that in the advancing global economy, management accounting was losing its value as a business practice.

The traditional branches of accounting were management and financial. In other words, the internal versus external focus of effort was considered separately and not brought together. Management accounting was focused current and future events and trends, and financial accounting was based primarily on the past. With expanding markets and global emphasis of those markets, it became essential to develop a strategic combination of these disciplines.

In today's global market and supply chain, SMA adopts a different point of view to previous accounting sciences. The modern accounting department must be able to evaluate not only its own competitive stance and product pricing, but also its position related to competitors. This 'position' may relate to market share, cost structure, company size, financial resources and product features.

A modern challenge in SMA is that it is a relatively new approach. Therefore, many companies apply its principles only partially. Management in the accounting industry faces the modern challenge of not only identifying what has happened in the past, but also what has not been done.

Another modern trend, and one meeting resistance amongst some in the accounting profession, is the idea that 'value' is based on more than just measurements in a financial manner (profitability, cash flows or capitalisation, for example). Marketing and sales strategies, increasing market share and firms environmental, social and governance (ESG) impacts are examples of non-financial issues. As we will discuss in Section 13.8.2, around the world there is an increasing need for firms to report their ESG impacts as part of their Director's report within their annual report or as a separate ESG Report.

Why do these issues trouble traditional accounting thought? The historical approach to accounting has been based on the assumption that everything can be measured and quantified. Under SMA and its external factors beyond monetary methodology, the non-financial issues cannot be exactly quantified; SMA requires an understanding of impacts in other ways. These may include social perception, corporate governance, brand loyalty and consumer opinion. None of these can be assigned a monetary value.

This is a new field of study. The current issues are focused on definitions as well as roles SMA can and should play in the broader management accounting sphere. But SMA is not intended as a replacement for traditional management accounting. SMA is meant to formulate forward-looking strategies; traditional management accounting is well suited for implementing those strategies. However, this demands coordination between the two sides. Perhaps the most severe issue and challenge is to make the transition so that both points of view can be coordinated to work together.



Illustrative Example 1

Staff at ABC Investment Ltd have questions about behavioural finance, an arm of behavioural economics which uses psychology as a tool to explore the ways in which investors make financial decisions and thus can help explain otherwise irrational-appearing movements in share prices.

Is loss aversion the same thing as risk aversion? If a financial manager is interested in maximising owners' wealth, how could the manager be overconfident and still be honest? If a person gives too much weight to recent information, rather than long-term averages, what type of behaviour does this exhibit?

Risk aversion is the preference not to have volatility, either up or down. Loss aversion is preference not to have the value decrease.

It is possible that someone who works for a company may have a more hopeful view of the company's prospects than someone who is independent of the company.

Giving too much weight to recent information is an example of anchoring behaviour.

A distinction is made between management accounting and strategic management accounting, in numerous ways. Whereas management accounting tends to be based on historical analysis for single entities and single fiscal periods, the strategic approach takes a view of current and future analyses, and even more so, for several periods.

Apply and Analyse 1

Apply

The CFO of your company comes from the 'old school' in which management accounting focuses internally and is based on the importance of quantifying all information. Identify what the CFO is overlooking in considering current issues and their practical challenges in relation to strategic management.

Analysis

The traditional management accounting role is based on policy implementation and tends to contain two characteristics. First, it is internally focused. Second, it grows from a discipline demanding that all issues be quantified.

Strategic management accounting, or SMA, is more future-oriented and externally focused. Its purpose is to develop business plans based on analysis of the competition, market trends (especially related to the expanding global market environment), and to develop ideas for how the company may position itself based on where these matters appear to be going.

Both sides offer great value. The challenge and greatest current issue is to get these two sides to work together. SMA is best suited to analyse trends and develop strategic methods to prepare for future changes. Management accounting is best suited to quantify and implement those ideas.

The great challenge is getting both sides to recognise the overall value of both efforts, and to devise ways to work together.

Apply and Analyse 1 (continued)

Strategic management accounting that has been developed for the last thirty years varies in terms of comprehension of the concepts and the techniques. Strategic management accounting techniques are adapted depending on viewpoints towards the concepts of strategic management itself. In order to achieve competitive advantage, various concepts and applications of accurate strategic management accounting have been developed and studied. The challenge is for the practice of accounting to provide relevant information in making strategic decisions for the company.

Key Learning Point

Strategic management accounting (SMA) is a specific field of accounting that brings together the strategic business objectives of a business with the information generated by management accounting.

► Knowledge Check Questions

Question 1

Describe the attributes of strategic management accounting.

- A** SMA functions as an internal system to coordinate all aspects of accounting-based planning and forecasting.
- B** This is a system designed to work with automated transaction entry processes.
- C** SMA is meant to combine internal and external attributes and to realise the objectives of the organisation through projections and analysis.
- D** SMA originates from the budgeting and budgetary control process for planning, control, performance measurement and decision making.

Question 2

Identify how management accounting originated.

- A** It was developed as an expansion of the cost accounting system.
- B** Management accounting is practised exclusively by the CFO.
- C** This is the operational segment of the treasury function.
- D** Management accounting is automated order entry software.



11.2 LINKAGES BETWEEN MANAGEMENT ACCOUNTING AND STRATEGIC MANAGEMENT OF AN ORGANISATION

Strategic management consists of top-level management activities usually undertaken by the board of directors and senior management of a company. The board of directors of a company sets the mission, vision and goals of a company to be achieved in the long term, sometimes beyond 5 or 10 years. Based on these directions, the senior management of the company will formulate the company's strategic plans (e.g. marketing, sales, financial, etc.), which are implemented through tactics and quantified by annual budgets, used as financial targets for planning, control and performance measurement; as part of the management accounting processes.

Both management accounting and strategic management accounting (SMA) rely on information generated by virtually all fields of accounting. **Financial accounting**, however, is focused on production of financial statements and relies on accounting principles used in creating those statements.

Management accounting, with its focus on internal information, relies heavily on financial accounting but also relies on information that financial accountants do not often require or consider – such as cost of producing products at the various stages of production, for example, which is the basis of cost accounting.

When combined, management accounting and cost accounting help managers make better decisions for the company by providing them with complete information for exercising their management functions, including planning, control and performance evaluation.

Management accounting goes further, however, by considering a range of factors such as cost systems, cost behaviour, profit planning, operational budgeting, capital budgeting and costing information.

There is much overlap between management accounting and cost accounting. And the links between costs and strategic management are also significant.

Management accounting includes analysis of product costs, both internally and for competitors. Strategic management accounting in comparison, studies costs that the market may be willing to absorb and whether production of a product makes sense.

A statement of profit or loss and other comprehensive income reports the cost of goods sold and the statement of financial position reports the value of ending inventories. Management accounting extends that information to focus on product costs, period costs, job order costing, process costing, how overhead is allocated and more.

Strategic management accounting takes it a step further by drawing conclusions from all these information and data and using those conclusions to make strategic decisions for the company.

Strategic management can also be defined as ‘the identification of the purpose of the organisation and the plans and actions to achieve the purpose. It is that set of managerial decisions and actions that determine the long-term performance of a business enterprise. It involves formulating and implementing strategies that will help in aligning the organisation and its environment to achieve organisational goals’. Strategic management accounting (SMA) is an accounting tool or system helping the senior management of a company to formulate, implement, monitor and attain its business strategies and strategic planning.

Illustrative Example 2

Part of the job of the CFO of a subsidiary of ABC Investment Ltd is to make the final decision on large capital expenditures.

One recent proposal is the purchase of a new computer system at a cost of HK\$5,000,000. The proposal is based on expectations that the new system will deliver savings of HK\$600,000 in the accounting department and could be useful elsewhere as well.

The company's treasurer takes a different view of the proposal, claiming the company will have to borrow money to buy the computer system and this will cost HK\$1,000,000 in interest. She is also concerned that the promised savings will not happen.

It is up to the CFO to determine what factors should be considered in the decision and, based on these facts, to decide whether to make the purchase.

The factors to consider include:

- The number of years the new system will benefit the company.
- The risk that there will be less than expected savings.
- The cost of capital for the company, including the cost of equity, the amount of debt and equity the company uses and the tax rate (because the HK\$1,000,000 in interest may be tax deductible).

Key Learning Point

At its very core, both management accounting and strategic management accounting rely on information generated by all fields of accounting.

► Knowledge Check Questions

Question 3

Explain the primary function of financial accounting.

- A** This is an alternate name for management accounting.
- B** It is the branch of accounting responsible for development of information needed to publish and analyse financial statements.
- C** This branch of accounting is responsible for obtaining and repaying loans.
- D** Financial accounting oversees cash flows and working capital control.

Question 4

Identify the two primary types of statements generated by financial accounting.

- A** Budgets and forecasts.
- B** Analysis of cash flows and summaries of net income.
- C** Statement of goals and objectives, and budget variance reports.
- D** Statement of financial position and statement of profit or loss.

11.3 BUSINESS PLANNING PROCESSES FOR CORPORATE STRATEGIC PLANS

Financial planning is a dynamic system that includes forecasting revenues, budgeting expenses, planning capital expenditures, managing cash flows and managing risks. The corporate planning process that is used to develop **corporate strategic plans** goes further by considering all these internal factors as well as external considerations that could impact a company.

Planning not only incorporates predictions, but requires examining possible outcomes ranging from a worst-case to a best-case scenario.

Planning is an ongoing process, with projections adjusted with additional information from time to time as situations evolve.

Business planning begins with the company's strategy, as developed by the company's board of directors or senior management. A company's strategy is the direction that the entity intends on moving in the long term and is expressed financially (profitability) and in terms of markets (market share). This strategy is the primary focus of the planning process and is used as a benchmark for evaluating performance.

Once the company's strategy is clarified, management solicits information from across the operations of a company about a wide range of factors, including the quantity of goods likely to be sold, sales price and state of the market. This data may include expectations about the prospects for different products, customer decisions or variations between markets in different regions and it may attempt to take into account possible economic changes, including forecasts or estimates of market opportunities and what might endanger them.

These findings are then used to prepare a formal operating plan which may be in the form of budgets or pro forma statements.

While financial planning is principally aimed at allowing a company to estimate future cash flows, strategic management plans are aimed at forecasting company performance for a period from one year to five years or more.

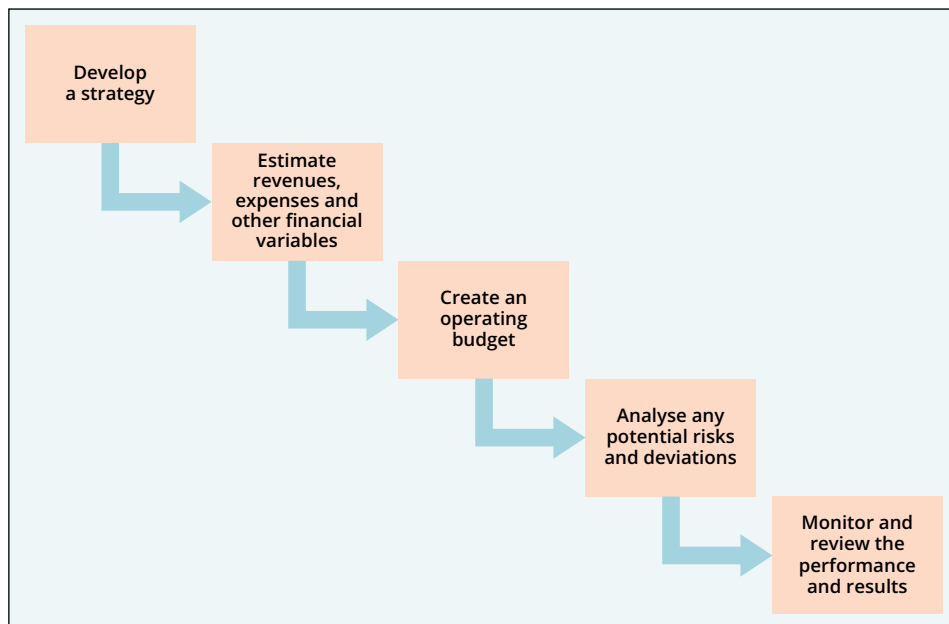
While financial managers rely on budgets, often accompanied by pro forma financial statements to estimate short-term and long-term financing needs (see Chapter 10), strategic managers need to also be aware of a wider set of variables.

Nevertheless, the information generated by financial accounting and cost accounting is key to the business planning process.

Estimates indicating a surplus of funds may help managers decide what to do with these funds. Invest them? Pay down debt? Repurchase stock? Pay a dividend? Launch a new product? Build a new plant? Tap into a new market?

The planning process for businesses ultimately follows five steps, as shown in Exhibit 11.1:

1. Develop a strategy;
2. Estimate revenues, expenses and other financial variables;
3. Create an operating budget;
4. Analyse any potential risks and deviations; and
5. Monitor and review the performance and results.



► **EXHIBIT 11.1 The planning process** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)



Illustrative Example 3

Your company's annual report suggests advancing on four fronts:

1. Ensuring that the cost of reaching market share targets is balanced with expected revenue growth;
2. Expediting the development of new products;
3. Having the finance needed for our plan in place and rebuilding the statement of financial position; and
4. Working together to leverage our resources around the world.

What are the implications of this strategy for financial management of your company?

These implications include:

- Although market share and revenues are important, financial managers must consider the impact on the company's costs from efforts to grow market share and revenues.
- More investment in research and development.
- Reduce reliance on debt financing and focusing on net income that is reinvested in the firm.
- Being more efficient in the use of its assets.

Key Learning Point

Planning not only incorporates predictions, but also requires examining possible outcomes ranging from a worst-case to a best-case scenario.

► Knowledge Check Questions

Question 5

Describe the functions of corporate strategic plans.

- A** This includes forecasts, budgets and projections, planning for capital expenditures, managing cash flows and managing risks.
- B** These are limited to the budgeting process.
- C** This refers to strategic management accounting alone and is not of concern on the operational level.
- D** As a strategically-based function, this involves identification of sources for long-term capitalisation.

Question 6

Explain what strategic planning is in business.

11.4 STRATEGIC COST MANAGEMENT TECHNIQUES

Cost considerations are key to the practice of strategic management accounting. For executives making decisions for their companies, it is essential to understand where costs come from and what costs will be in the future. This necessitates understanding the factors driving costs and how those factors will change in future.

Supply gluts, for example, that drive up the cost of raw materials may be predictable. Minimising the cost of storage and ensuring supply are available in the right quantity at the right time are also strategic considerations. So is producing enough goods and services to supply clients and support expansion but not so much that there are large overhangs of inventory. These are just a few of the considerations for strategic management accountants.

11.4.1 Just-in-Time

Many companies have significantly reduced their inventory levels and costs by using **just-in time (JIT) inventory** methods, an innovation resulting from focusing on supply chain, notably as this has become global and more complex than in the past.

Under a JIT method, goods are manufactured, purchased or delivered at the time they are needed, but not kept ready for an extended period of time.

Most inventory control systems used by large companies employ automated computerised systems to track all kinds of inventory items as they flow into and out of the company (e.g. DHL, Amazon, Toyota). These systems control inventory levels by relating them to the production process, which in turn is related to the company's projected sales levels.

Any approach to inventory management must balance the benefits of holding inventory. Three of the most common approaches are:

- **Economic order quantity (EOQ) model:** The EOQ model defines (and determines) the optimal inventory level as the one that minimises both shortage and carrying costs.

It is aimed at allowing a business to figure out what quantity of inventory will best allow it to offset the costs of holding a good, such as storage, financing, insurance and spoilage, against the cost of having a shortage of that good. The model is aimed at establishing the point at which carrying costs and shortage costs are the same (see Module 2, Chapter 3). As the model assumes that items are sold at the same rate throughout the year, it is better suited to products with this kind of sales-profile than those which have large seasonal or monthly variations.

- **Materials requirement planning (MRP):** MRP is a computerised system aimed at ensuring inventory holdings are aligned with production schedules. MRP works by calculating what volume of raw materials and in-process inventory are needed to be held in order to meet any specific volume of orders. Combined with accurate sales forecasts, MRP allows businesses to maintain low levels of inventory.

- **Just-in-time (JIT) inventory systems:** JIT systems aim at ensuring that components and raw materials reach a factory at precisely the time they are needed. This allows companies to maintain very low inventory levels. To work smoothly, they also call for close relationships with suppliers and finely tuned logistics operations. Another version of JIT refers to delivery of finished goods to the market. Just as producers do not want to invest in large levels of inventory for future use, the end user does not want to devote warehouse space, insurance and security. The JIT on the end user side is aimed at identifying product requirements and having them transported on that basis. The ideal JIT system takes goods directly from delivery to the shelf with little or no time needed for storage.

11.4.2 Total Quality Management Concepts

To work effectively, JIT systems place a high demand on quality throughout the production process. Any interruption to production incurs costs for a company, therefore it would want to ensure that both its inputs, production processes and final products have as few defects as possible – with the ideal rate being zero.

To realise such targets, many companies use **total quality management (TQM)** systems. Such systems can collect information across a company's production processes, including the number and type of product and input defects, any rework costs and costs arising from meeting service or warranty agreements. This information can also be used to redesign products and their production processes to lower defect rates and to shorten set-up and production times.

TQM systems can also be used to collect information about other areas of a company's business, such as customer support needs, product satisfaction levels and the cost of internal procedures. Such data can then be used to gauge and raise efficiency levels and profitability across the entire length of the company's supply and value chains.

Four steps are involved in a TQM process. These are continually repeated to locate where improvements are needed:

1. *Identify and plan.* Determine a problem, collect information and identify the root cause.
2. *Take action.* Develop and implement a fix and launch a measurement system to track the results.
3. *Check the results.* Analyse the measurement using testing before and after to determine whether the root of the problem has been eliminated or reduced.
4. *Disseminate the information.* Document the results of the process and issue recommendations to avoid similar problems in the next production cycle.

These four steps can be summarised with an acronym, PDCA, for 'plan, do, check and act'.

11.4.3 Product Life Cycle Costing

The product life cycle is a step-by-step description that a product goes through before it reaches customers.

The product life cycle has four phases:

- *Introduction:* Building market acceptance for a new product. This phase is marked by marketing expenditures and chasing after early adopters. Profits may be higher before competitors enter the market or pricing may be set in such a way that it deters others (i.e. Apple's iPhone). Success is uncertain at this stage, so many companies look to avoid large investments in production.
- *Growth:* During this phase, companies work on building market share. They may release new versions of the product or spin-offs and expand distribution networks.
- *Maturity:* With many competitors in the market, the primary task for companies at this state of the product life cycle is to defend market share. With constant downward pressure on prices, companies may introduce target costing. Cash flows may be positive at this stage.
- *Decline:* Sales drop and eventually the product is terminated. Costs drop to preserve cash flows. Distribution channels are reduced. Excess inventory is sold off at large discounts and production lines shut down.

The duration of the product life cycle depends on the type of product and the market. Some last for decades (e.g. automobile) and others for less than a year (e.g. fashion).

The use of the product life cycle to guide costs is known as life cycle costing. A life cycle costing system tracks and accumulates actual costs and revenues from invention until termination.

Product life cycle costing represents the total cost of ownership from the moment a product is conceived until it is terminated. By adding up all the costs, it makes it possible to evaluate the success of a product by considering the costs of procuring, operating, maintaining and, ultimately, disposing of a product.

Life cycle costing (LCC) is different from traditional cost accounting that focuses on immediate accounting periods. Rather, LCC provides a long-term picture of a product's profitability and the success of the planning around the product. It can also be used to improve controls of manufacturing costs.

While the product life cycle can be broken down into the four stages described above, product life cycle costing involves a few more steps:

- Market research to determine what customers want and what they are willing to pay.
- Details of the product, costs, delivery dates, performance and so on.
- Design of a product.
- Prototype manufacture of small batches for testing.
- Development of a product based on testing and altering the design as needed.
- Tooling of production facilities.
- Manufacture of the product in larger batches.
- Sale.
- Distribution.

- Support after the product hits the market.
- Decommissioning of the product at the end of its life.

Product life cycle costing can lead to lower costs, more realistic assessments of cash flows, facilitation of long-term planning and better understanding of incremental costs.

11.4.4 Target Costing

As the prices of all goods are principally set by the laws of supply and demand, over which no single company has any significant say, the main way in which a business can pursue profitability is by controlling its costs.

This requires setting a **target cost** that generates a desired profit. The relationship between price and profit is shown in Exhibit 11.2.



► EXHIBIT 11.2 Price, profit and cost

For the company to meet its profit goal, it must produce its products for the target cost (or less). If target cost is not achieved, it will fail to produce the desired profit. A company chooses the market segment it wants to compete in, that will become its market niche. For example, the company will have to make a choice between selling luxury goods or economy goods to focus its efforts on one of these segments.

Once the company has decided which part of the market it will target, it then needs to conduct research to establish what price potential customers will pay for its good or service, set a desired level of profit and then, with these two figures in place, set a target cost.

After the company has determined the target cost, employees with expertise in production and operations, marketing and finance are assembled. The task of the team is to design and develop a product that meets quality specifications without exceeding the target cost. The target cost must include all production costs over the entire expected life cycle of a product or service.

Many industries use price points that customers expect to pay – setting the price of their products at certain sums, such as HK\$199, HK\$249 and so on. Under the target pricing concept, companies then set out to design goods that can be manufactured and sold at these prices, taking into account factors such as what consumers expect to pay and the prices of goods made by other businesses. In doing so, they reverse the normal procedure of designing a product, figuring out how much it will cost to make it and then setting a price that reflects this cost.

To succeed, companies must of course be able to produce their goods at a cost that allows them to realise their desired profit. If they cannot manage this, they will end up with goods that are priced far higher than those of competitors or have very high manufacturing costs.

Illustrative Example 4

Cute Toys Limited is a large seller of low-cost stuffed toys. They are restricted to selling their products at HK\$50 per unit because they have so many competitors. If the company set its profit margin at 10% on the selling price, calculate the target cost per unit.

Answer

Target profit margin = 10% of HK\$50 = HK\$5 per unit

Target cost = Selling price – Profit margin (HK\$50 – HK\$5)

Target cost = HK\$45 per unit

11.4.5 Quality Costing

Quality costing is used to show management the importance of quality-related activities.

Quality costing demonstrates the ultimate cost of poor quality, it tracks the causes and effects of problems and the ultimate impact and looks for ways to find both problems and solutions. Ultimately, quality costing can be used to boost profits.

Quality costing can serve as a gateway to introduce a TQM system, as discussed in Section 11.4.2 above, and can be a powerful tool to boost the efficiency of a company.

The cost of quality may have different interpretations. 'Quality costs' can refer to costs of poor quality due to identifying and correcting defective products. Quality costs can also mean costs to attain good quality. They may also mean the costs of running the quality assurance department. In cost of quality (COQ) theory, quality cost refers to the cost of poor quality.

It is not surprising that quality costs may account for 10% to 30% of sales of a company and may occupy 25% to 40% of its operating expenses. However, many companies do not assign clear responsibility for action to reduce them. Nor do they create a structured approach for doing so.

Quality cost assessments should include hidden or hard-to-measure components such as sales lost because of poor product or service quality.

Since 2016, some companies have reported their quality cost information as part of their sustainability reporting, including it as part of their Director's report within their Annual Report or as a separate ESG Report as we will discuss in Section 18.3.2.

There are four categories of COQ:

1. Prevention costs. Costs aimed at preventing quality-related problems incurred during the design, implementation and running of a quality management system. These can include:



- (a) Drawing up specifications for all inputs, processes, final products and supporting services.
 - (b) Drafting quality, operational and inspection procedures.
 - (c) Establishing and maintaining a quality assurance system and department.
 - (d) Training for the implementation and upkeep of total quality management (TQM) schemes.
- 2. Appraisal costs. Costs associated with all quality-related data and monitoring processes, which are associated with the suppliers' and customers' evaluation of purchased materials, processes, products and services to ensure that they conform to specifications. These can include:
 - (a) Pre-approvals assessments of suppliers and their products and services.
 - (b) Audits of all quality systems to ensure they meet previously agreed specifications.
 - (c) Verification that all inputs, processes and products meet previously agreed specifications.
- 3. Internal failure costs. Costs incurred to remedy all defects or other quality issues identified before a product or service is delivered to the customer. These can include:
 - (a) Expenses incurred in determining the reason for defects appearing in a product or service.
 - (b) Costs incurred by having to rework defective materials or processes.
 - (c) Losses incurred from holding stocks of products or materials that cannot be repaired, used or sold.
- 4. External failure costs. Costs incurred in remedying defects discovered by customers after delivery. These can include:
 - (a) Handling customer complaints.
 - (b) Repairing products either at customers premises or returned to the company.
 - (c) Meeting warranty undertakings for goods or services covered by guarantees.
 - (d) Transport and other handling costs arising from the return or recall of products.

The total cost of quality is the sum of the above four categories. **Activity-based costing (ABC)** is a proven management accounting tool in identifying and quantifying the above four categories of TQM costs.

11.4.6 Theory of Constraints

Theory of constraints (TOC) is a methodology that allows companies to identify the major factors limiting their growth. The theory was first outlined by Dr. Eliyahu Goldratt in his book, *The Goal* (1984). Since then, it has been applied by companies in both Europe and North America.

The starting point for the theory is that all companies face constraints that hold back their potential and, therefore, that fixing such constraints by identifying and managing them should result in a rise in profits. The theory also proposes that companies are more profitable when they direct their energies at fixing constraints rather than addressing other aspects of their business, such as whether they are operating at full capacity. Fixing such bottlenecks improve a company's ability to effectively use or limit overtime labour while meeting customer demands.

Since Goldratt first expounded his theory, TOC has evolved to become one of the most widely used management practices worldwide.

At TOC's core lies a series of processes aimed at improving business effectiveness, efficiency and productivity by identifying and removing the main constraints holding a company back from reaching its various production, sales and profitability goals.

A core insight of TOC is that all companies are complex systems, each comprising multiple linked activities. Often, however, their businesses are held back because one or more of their activities acts as a 'bottleneck' hindering the operation of their overall businesses.

TOC offers a powerful set of tools that allow a business to identify which of its processes are acting as constraints in realising its long-term profitability. These tools are:

The five focusing steps – which allow a business to identify and eliminate constraints.

The thinking processes – which allow a business to analyse and solve problems.

Throughput accounting – which allows a business to measure its performance and come up with managerial decisions.

TOC can also help a company prioritise the order in which it addresses any constraints that are found, allowing it rapidly to put together a programme of change. This can be particularly useful when a company faces a changing or deteriorating business environment that is harming its performance.

The successful implementation of TOC programme will generate at least one of several performance improvements:

- Improved operational effectiveness
- Raised output efficiencies
- Shorter lead times
- Reduced inventory
- Enhanced profitability

1. Five focusing steps

The five focusing steps are:

- (a) Identify the constraint or bottleneck which is holding back the overall effectiveness of the system.
- (b) Determine how this constraint can be reworked to allow the system to work at the highest possible rate of effectiveness.
- (c) Make eliminating this constraint the highest priority, including, where necessary aligning other processes to ensure maximum benefit is secured from removing the constraint.



- (d) Identify and make any other changes needed to eliminate the constraint.
- (e) Once completed, repeat the cycle from step (a) for the system's next biggest constraint.

2. The thinking processes

At the heart of TOC analysis lies what are known as the thinking processes – a problem-solving methodology that has been specifically developed to handle the complex, interdependent systems of many manufacturing processes. These processes use a series of diagnostic tools that can trace 'undesirable effects' (UDEs) back to their underlying causes and then allowed these causes to be addressed without generating any other UDEs.

The procedures of the thinking processes centre on three crucial questions:

- (a) What needs changing?
- (b) What should it be changed to?
- (c) What actions will bring about this change?

3. Throughput accounting

Throughput accounting is a form of accounting aimed at removing the practices used in many traditional accounting systems that inadvertently encourage practices that run counter to long-term increases in profitability.

One example of such a distortion is the way in which traditional accounting treats inventory as an asset with a cash value. Doing so may encourage a company to increase its inventory holdings, for example, by manufacturing excess products in order to inflate its apparent assets and so produce an accounting profit. TOC, in contrast, treats inventory as a liability because it both ties up cash and incurs storage costs.

Traditional accounting also has a bias towards reducing expenses in order to improve profitability, whereas TOC emphasises the benefits of raising throughput. The theoretical issue with cutting expenses is that its ultimate target is zero, even though this would likely result in production also falling to zero.

As for TOC, a constraint is anything that prevents the system from achieving its goal, any complex system may have multiple constraints. A key task therefore is figuring out which one or few constraints should be investigated to result in the maximum benefit.

It should also be noted that constraints can be either internal or external. An important priority for all companies should be identifying internal constraints and removing them using the five focusing steps.

An example of an internal constraint would be if a company had far greater demand for products than capacity to make them. Such a constraint could in turn arise from various other constraints, such as having equipment incapable of producing the volume of output needed, lacking staff with the right skills or mindset necessary to make more goods or having policies (written or unwritten) which hinder the manufacture of more goods.

An external constraint happens when the system produces more products than the market will require. If this happens, the focus for the organisation should be on mechanisms to create more demand for its products or services.

11.4.7 Throughput Analysis

Throughput analysis is looking at investment decisions in terms of their impact on the entire company and not just on the specific area of investment.

The approach is based on the idea that production costs do not change at the level of the individual unit produced. Manufacturing a single unit requires companies to incur costs of the relevant materials only, all other costs happen regardless of whether a unit is produced. For instance, before any product can be manufactured, a company must build a factory.

The throughput analysis system approach uses a different terminology to explore cost concepts. These terms include:

- *Throughput*. Sales minus totally variable expenses. Few costs are totally variable, so throughput as a percentage of sales should be quite high.
- *Operating expenses*. All expenses except the variable expenses used to calculate throughput.
- *Investment*. The cash invested to increase production capacity.

These concepts can be summarised in three formulas:

$$\text{Revenue} - \text{Totally variable expenses} = \text{Throughput}$$

$$\text{Throughput} - \text{Operating expenses} = \text{Net profit}$$

$$\text{Net profit} \div \text{Investment} = \text{Return on investment (ROI)}$$

These formulas can be used to make decisions on whether to change production systems. The best improvements increase throughput, for which there is no theoretical upper limit. Alternatively, reducing operating expenses are less important since expenses can only be reduced to zero.

Key Learning Point

Strategic management accountants have a range of tools they use to help companies control costs. The most important of these are just-in-time (JIT) systems, total quality management (TQM), product life cycle costing, target costing, quality costing, the theory of constraints and throughput analysis.



► Knowledge Check Questions

Question 7

Define the term JIT.

- A** This is just-in-time making of needed materials in a way that minimises the need to store as inventory.
- B** The term describes the 'January inventory timing' attribute, a tendency to reduce inventories to avoid paying taxes.
- C** It means 'juxtaposed interest trend', or the cost of borrowing money to buy inventory.
- D** JIT means 'just in time', or acquisition of financing immediately before payment for goods is due.

Question 8

Identify the steps in a product life cycle.

- A** Production, marketing, expansion and profits.
- B** Patent, production, marketing and competition.
- C** Expansion, peak, contraction, trough.
- D** Introduction, growth, maturity, decline.

Question 9

Explain the theory of constraints.

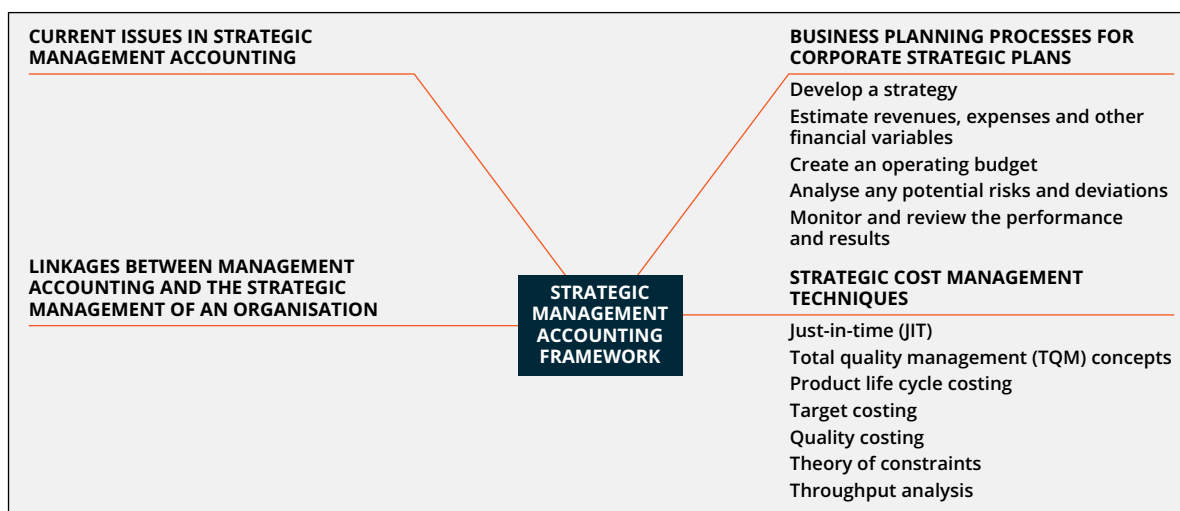
- A** Competition makes it impossible for any one producer to dominate the market.
- B** Customers saturate the market until a product is no longer desired.
- C** Bottlenecks in production create defects and delays.
- D** Generating ever-higher profits eventually becomes impossible due to resistance in the market.



SUMMARY

- Strategic management accounting (SMA) is a field of accounting that brings together the strategic business objectives of a business with the information generated by management accounting.
- Strategic management accounting has a triangular structure with three primary considerations: technical analysis, behavioural considerations and cultural considerations.
- SMA considerations are examined in light of quality, cost and time (QCT).
- Both management accounting and strategic management accounting rely on information generated by all fields of accounting.
- Management accounting, with its focus on internal information, relies heavily on financial accounting but also studies the cost of producing products at the various stages of production, which is the basis of cost accounting.
- There is much overlap between management accounting and cost accounting. And the links between costs and strategic management are also significant.
- Strategic management takes it a step further by drawing conclusions from all this information and data and using those conclusions to make strategic decisions for the company.
- The corporate planning process that is used to develop corporate strategic plans goes further by considering all these internal factors but also the many external considerations that could impact a company.
- Planning is an ongoing process, with projections adjusted with additional information.
- The planning process follows five steps: Develop a strategy; estimate revenues and expenses; create a budget; analyse any potential risks and deviations; and monitor and review the performance and results.
- Under a just-in-time method, goods are manufactured or purchased just in time for use. JIT systems fine-tune the receipt of raw materials so that they arrive exactly when they are required in the production process. The objective is to reduce inventory to its lowest possible level.
- The aim of total quality management (TQM) systems is to aim for zero defects.
- The use of the product life cycle to guide costs is known as life cycle costing. A life cycle costing system tracks and accumulates actual costs and revenues from invention until termination.
- Target costing examines the relationship between price and profit.
- Quality costing is used to show management the importance of quality-related activities.
- The theory of constraints is a specific approach used to identify and manage constraints to achieve the company's goals.
- Throughput analysis is the study of investment decisions in terms of their impact on the entire company and not just on the specific area of investment.

MIND MAP



► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. SMA is not solely based internally.
 Answer B is incorrect. This is not about automation of accounting functions.
 Answer C is correct. This is a combination of internal emphasis and external factors, intended to realise the objectives of the company.
 Answer D is incorrect. This is not restricted to the budgeting process.

Question 2

Answer A is correct. Management accounting originated as an expansion of cost accounting.
 Answer B is incorrect. This is not a practice limited to the CFO.
 Answer C is incorrect. This system is not part of the treasury function.
 Answer D is incorrect. It is not a form of software.

Question 3

Answer A is incorrect. Financial accounting is entirely separate from management accounting.
 Answer B is correct. Financial accounting is focused on development of financial reports and analysis of the results.
 Answer C is incorrect. This branch is not involved in debt or its repayment.
 Answer D is incorrect. This branch of accounting is not restricted to cash flows or working capital controls.

Question 4

Answer A is incorrect. These would not be considered as forms of statements.

Answer B is incorrect. Financial accounting reports are not identified as cash flows or income summaries.

Answer C is incorrect. These are not standard forms of financial statements.

Answer D is correct. Under these or other names, the statement of financial position and statement of profit or loss are the primary forms of published statements from financial accounting.

Question 5

Answer A is correct. The range of functions is broad and includes forecasting, budgeting, capital expenditure planning and risk management.

Answer B is incorrect. Strategic planning encompasses more than just budgeting.

Answer C is incorrect. This is not part of strategic management accounting functions.

Answer D is incorrect. It is not limited to capitalisation of the operation.

Question 6

Strategic planning is the means by which a company sets and realises its priorities. This calls for the company to direct its resources and improve its operational ability to meet these priorities, in turn requiring it to ensure its employees and other stakeholders are clear and agree on the intended results and making any changes in all the above in the event of any major change in the environment.

Question 7

Answer A is correct. JIT means delivering required goods when they are needed and not before.

Answer B is incorrect. This is not the meaning of JIT.

Answer C is incorrect. This is not the meaning of JIT.

Answer D is incorrect. This is not related to obtaining of financing.

Question 8

Answer A is incorrect. The product life cycle is not defined in these ways.

Answer B is incorrect. The product life cycle is not defined in these ways.

Answer C is incorrect. The product life cycle is not defined in these ways.

Answer D is correct. The four steps in the product life cycle are introduction, growth, maturity, decline.

Question 9

Answer A is incorrect. This is not a theory related to competition.

Answer B is incorrect. Constraints are not an attribute of customer behaviour.

Answer C is correct. Bottlenecks cost time and money and need to be reduced, with the goal of zero defects.

Answer D is incorrect. This is not related to market resistance to products.

EXAM PRACTICE

QUESTION 1

Describe the current issues regarding strategic management accounting.

- A.** Internet-based information is highly unreliable and cannot be used in SMA.
- B.** Of the many issues, amongst the most critical are the continuing globalisation of supply chain and value chain participation; coordination with traditional management accounting; and recognition that value is not necessarily financial only.
- C.** The problems include unfair trade treaties favouring one country over another.
- D.** Cyberattacks are on the rise and this threatens all SMA initiatives.

QUESTION 2

Explain what is included in business planning processes.

- A.** This must include production analysis, outcome planning, internal and external influences and conditions and anything else that might impact the company.
- B.** The only items included are budgets and forecasts.
- C.** An analysis of current and future markets.
- D.** A study of cash flows and planning for future capitalisation.

QUESTION 3

Describe the applications of JIT.

- A.** This is limited to 'just-in-time' production of finished goods.
- B.** It extends only as far as warehouse space limitations mandate.
- C.** JIT refers to providing of materials required for production when needed and not before; and to delivery of finished goods to end users to avoid needing to build a large inventory.
- D.** The term applies only when available capitalisation is limited.

QUESTION 4

Explain the most relevant current issues involved with strategic management accounting.

QUESTION 5

List the key attributes of total quality management.

QUESTION 6

Define how target costing functions.

QUESTION 7

List the elements of the quality costing system.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. This is not determined by Internet information.

Answer B is correct. These are the current issues of greatest importance to the SMA process.

Answer C is incorrect. Unequal trade treaties are not amongst the most prominent SMA issues.

Answer D is incorrect. Cyberattacks are a serious issue, but not exclusively related to SMA.

QUESTION 2

Answer A is correct. The business planning process is described by the listed influences.

Answer B is incorrect. Budgets and forecasts are not the only portions of business planning.

Answer C is incorrect. Current and future market conditions are not exclusively relevant to business planning.

Answer D is incorrect. Cash flows and working capital are not the only items of concern in business planning.

QUESTION 3

Answer A is incorrect. It is not limited to production levels.

Answer B is incorrect. This is not determined by limited storage space.

Answer C is correct. The concept refers to provision of resources in production, or to timely delivery of finished goods to end users.

Answer D is incorrect. This is not a term used about working capital.

QUESTION 4

The current issues concerning SMA are numerous, as this is a developing concept in the accounting industry. Amongst the issues are continuing globalisation and how this affects risk management in the value chain and supply chain. Another challenge is the lack of understanding and cooperation between internally based management accounting and more externally based and forward-looking strategic management accounting. A third issue is the need to identify 'value' beyond the financial realm. Traditional management accounting tends to see value only in terms of profit or loss.

QUESTION 5

TQM is designed to reduce defects and has the goal of attaining a zero-defect environment. This is accomplished through four stages: Identify and plan, act, check the result and disseminate the information. Any one of these four not placed into action will nullify the entire effort, so the success of TQM relies on thorough application of these ideas.

**QUESTION 6**

Target costing is a method for establishing the most desirable cost of a product, even before production is begun. The idea is to address the specific price point and quality requirements of a target population. Considerations in setting costs include the pricing by competitors of the same or similar products; demographics of the target population, including age, gender, disposable income and preferences.

QUESTION 7

Quality costing is an essential element in SMA. The accounting industry has come to the realisation that trying to save production costs by producing goods of low quality, ultimately leads to loss of market share and loss of profits. The cause and effect of quality-based decisions is part of the broader vision that SMA strives to apply in decisions concerning price and quality. A well-priced and high-quality product not only improves efficiency; it also boosts long-term profitability.

12

Pricing Strategies and Decisions

CHAPTER TOPIC LIST

12.1 Factors Affecting Pricing Policies

- 12.1.1** Cost Data
- 12.1.2** Demand Factor
- 12.1.3** Consumer Psychology
- 12.1.4** Competition Profit
- 12.1.5** Government Policy

12.2 Types of Pricing Strategies

- 12.2.1** Cost-Plus
- 12.2.2** Skimming
- 12.2.3** Penetration
- 12.2.4** Complementary Product
- 12.2.5** Product Line
- 12.2.6** Volume Discounting
- 12.2.7** Discrimination
- 12.2.8** Relevant Cost

12.3 Impact of Corporate Social Responsibility on Pricing Strategies

- 12.3.1** Cost of Quality
- 12.3.2** Environmental Costs

12.4 External Market Factors and Internal Cost Structures on Pricing Strategies

- 12.4.1** External Market Factors
- 12.4.2** Internal Cost Structures

12.5 Pricing Strategies for Non-Profit making Entities

- 12.5.1** Voluntary Pricing
- 12.5.2** Ability to Pay
- 12.5.3** Versioning
- 12.5.4** Bundle Services/Memberships



LEARNING OUTCOMES

PRINCIPAL LO3: APPLY PRICING STRATEGIES AND DECISIONS

LO3.01: Calculate prices for products and services

3.01.01 Explain the factors affecting pricing policies

3.01.02 Describe different pricing strategies:

- All forms of cost-plus
- Skimming
- Penetration
- Complementary product
- Product-line
- Volume discounting
- Discrimination
- Relevant cost

3.01.03 Explain the concept of cost of quality and environmental costs

3.01.04 Apply different approaches of pricing products and services considering external market factors and internal cost structures

3.01.05 Describe and explain possible pricing strategies for non-profit making entities



OPENING CASE

ABC INVESTMENT LTD

As your job evolves at ABC Investment Ltd, you become deeply involved in pricing strategies within the organisation. There are many theories and practices associated with setting prices. It is important to consider all external and internal factors in recommending the best pricing strategy.

You are also increasingly called upon to consider issues of corporate social responsibility that impact the company's pricing decisions. As a responsible corporate citizen, ABC Investment Ltd must be mindful of the social consequences of its pricing actions in the marketplace. Legal and reputational damages could be great if serious missteps in pricing are taken.

It is also necessary to consider some of the pricing strategies undertaken by not-for-profit organisations, some of which may be in ways associated with ABC Investment Ltd, who may be involved in social enterprises in the near future. Perhaps some of their pricing strategies merit serious consideration by your own profit making operations. 

OVERVIEW

Pricing policies at entities of all types – for profit or otherwise – are amongst the most important and difficult decisions that management has to make. In making pricing decisions, management consider multiple factors and use different types of data depending on the type of organisation and product or service being priced.

One consideration when determining prices is cost data, including both direct and indirect costs. Other considerations include demand for a particular product and service, consumer psychology, competition and government policy.

This chapter also examines pricing strategies that companies can apply, including total cost-plus pricing, skimming, penetration, complementary product, product line pricing strategy, volume discounting, price discrimination and relevant cost.

The chapter also explores the impact of corporate social responsibility (CSR) on pricing, including variables such as the cost of quality and environmental costs.

Pricing is also driven by considerations of external market factors and internal cost structures. This chapter also examines pricing strategies that non-profitmaking entities can apply, including voluntary pricing, ability to pay, versioning and bundle services.

 12.1 FACTORS AFFECTING PRICING POLICIES

Establishing the price for any good or service is affected by many factors.

A good example is the pharmaceutical industry. Companies spend heavily on research and development searching for new drugs that they can patent, then try to recoup their investment and make a profit by pricing new products high and marketing them aggressively.

Various health crises have pressured the drug industry to lower the prices of drugs used to treat diseases such as HIV/AIDS or cancer. Individuals and health care providers around the world have long questioned whether prices are set exorbitantly high. The pharmaceutical companies' response is that high prices are needed to offset the expense and the risk of researching and developing new products, many of which are never actually brought to the marketplace.

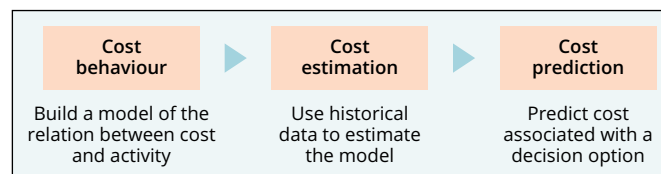
In the long run, drug companies are not unlike companies in other industries in that the prices they set must cover the product's costs and eventually earn a reasonable profit. But to price products appropriately, it is necessary to understand the financial and economic market forces at work.

12.1.1 Cost Data

There are a number of costs and benefits linked to every decision and cost and benefit have a different level of priority in the decision-making process. A first step in this process is knowing what to measure. The next step is how to estimate costs and benefits.

The core concept behind these estimates is that costs and benefits are the result of business activities. Advertising, for example, is an activity that has costs but also generates benefits, such as revenues. So, costs and benefits can be determined by considering the changes in activity and the financial impact of the changes in activity.

Exhibit 12.1 shows the rationale behind estimating costs, a process that starts with specifying how a cost is linked to an activity. A couple of principles underpin this process, specifically variability and traceability. Variability deals with how activities influence costs and benefits. Traceability refers to how a cost or benefit can be directly associated with a particular choice.



► **EXHIBIT 12.1 Estimating costs** (Source: Balakrishnan 2012.
Reproduced with permission of John Wiley & Sons.)

As the volume of activity changes, some costs change but others remain the same. This is linked to variability, the relation between a cost or a benefit and a particular activity. Three ideas are important to consider (see also Module 2, Chapter 2):

1. **Variable costs** are proportional to the volume of activity;
2. **Fixed costs** do not change along with changes in the level of activity;
3. A **mixed cost** contains both fixed and variable components.

The same concepts apply to variable, fixed and mixed benefits.

The other important concept is traceability, the degree to which a cost or revenue can be directly correlated to a decision. A cost or revenue that can be directly traced to a particular decision is a direct cost or a direct benefit. Alternatively, if only a part of the cost or revenue is linked to a decision then it is indirect cost or benefit. Indirect costs are often called common costs. Indirect manufacturing costs are known as manufacturing overhead.

It can be easy to confuse direct and common costs with variable and fixed costs, but they are not the same. Both direct and indirect costs can be fixed or variable. Raw materials or parts used to make a good are variable direct costs; the cost of a piece of machine used to make a product only is a fixed direct cost. The regular salary paid to the manager of a plant is a fixed and indirect cost. The cost of oils and lubricants used in the running of machines is indirect and variable because the amount depends on the volume of production (how much the machines are used) and indirect because it is not possible to identify the exact amount of oil that the particular product used in production of individual units.

There are direct and indirect costs in all functional areas of a company. Direct marketing expenses include sales commissions. Alternatively, the cost of a sales office is an indirect cost. Similarly, royalty payments for a patent is a direct cost, while the cost of research staff is indirect.

Direct costs and benefits are easier to estimate than indirect costs and benefits because direct costs and benefits are directly related to a particular decision. Indirect costs or benefits require assumptions to be attributed and lead to a more imprecise calculation.

These considerations of costs are related to setting prices. The first consideration in setting prices is the **cost of production**.

Prices need to be sufficiently high for a company to recoup its costs of research, development, production, sales and distribution and allow for profit margins, as in the pharmaceutical industry example above.

Prices should cover both direct costs such as materials and labour as well as indirect costs such as costs not directly connected with revenue for an identified product.

Direct costs make up a consistent percentage of the cost of producing a good or service.

Indirect costs, however, change as a percentage of total costs as the volume of production grows.

In other words, the portion of the price that goes towards indirect costs falls as production volumes increases. This is because indirect costs do not change drastically with increases in production. For instance, the lease costs of a factory do not change if the factory produces 1,000 widgets or 1,000,000 widgets.

12.1.2 Demand Factor

In many industries, businesses have little say in the general prices of a good or service. Instead they have to accept whatever price is determined by the competition between **supply and demand** in the marketplace. Such businesses are called **price takers**.

Petrol retailers are a good example of price takers. The price of petrol is set according to the volume of oil supplied by oil producers such as the Organization of the Petroleum Exporting Countries (OPEC) and the demand for oil from end users. Most commodities, such as corn, wheat, coal or copper, are products whose sellers have little say over prices beyond those set by the wider forces of supply and demand.

Certain kinds of companies have greater freedom to set their own prices. Amongst them are companies which make goods specifically tailored to the needs or wishes of individual customers, such as custom-designed clothing or specialised equipment.

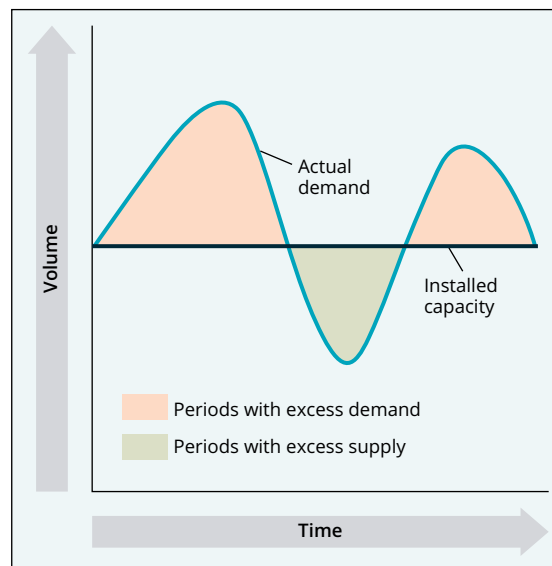
Companies with products that other companies cannot make also have greater freedom to set their own prices. Makers of patent-protected goods are one example of such companies. Intel, for example, can always charge a high price for its most recently developed chips until other companies figure out a way of manufacturing their own version. Demand for such unique products is likely to be high and the company can be a price setter. However, as demand tends to fall when prices rise, even with an exclusive right to market a product, a producer is likely to find itself subject to changing demand.

A company will also be able to set the price when it has been successful at distinguishing its product or service from others. Even in a competitive market like coffee, some companies have been able to differentiate their products and charge more (e.g. Starbucks Coffee).

Another important consideration is capacity, or the maximum volume of production or services a business can generate given the resources available to it.

Deciding how much capacity to install requires long-term forecasting and planning. In order to figure out what facilities, equipment and staff will be needed, companies need to estimate the likely demand for their products and services up to several years ahead. Once capacity is in place, it is often expensive either to reduce or to change capacity levels. As a result, many businesses struggle to cope with major fluctuations in demand levels.

Exhibit 12.2 shows the temporary imbalances between the demand and supply of organisational resources. It shows that it is difficult to change the supply of capacity in the short term and that demand varies. There will be times where demand exceeds available supply, while at other times available supply exceeds demand.



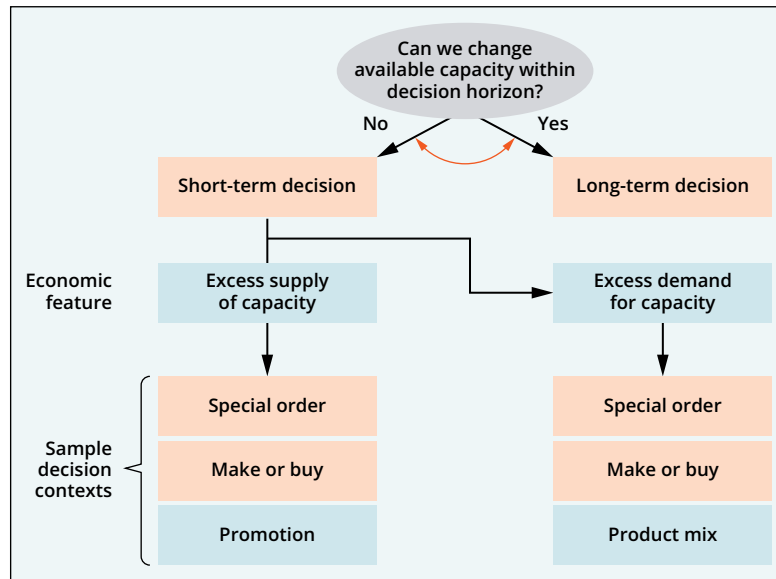
► **EXHIBIT 12.2 Capacity and demand** (Source: Balakrishnan 2012. Reproduced with permission of John Wiley & Sons.)

When planning for capacity, effective managers anticipate the likelihood of such short-term gaps between supply and demand.

Most short-term decisions fall into one of two broad categories:

- Those for handling excess supply, which can include cutting prices or offering special promotions in order to boost demand.
- Those for handling excess demand, which can include raising prices or altering the product mix so as to sell more of a company's goods.

Exhibit 12.3 shows the context in which these decisions have to be made. The exhibit shows how short-term decisions in most business contexts involve dealing with temporary gaps between capacity and demand.



► **EXHIBIT 12.3 Decision context** (Source: Balakrishnan 2012.
Reproduced with permission of John Wiley & Sons.)

Short-term decisions often have a long-term impact. Rushing to complete a job, for example, might help a company establish a reputation for being a good business to turn to in an emergency, helping it acquire new customers. As it is usually hard to gauge the longer-term impact of such behaviour, companies tend to restrict their business analysis of specific projects or processes to looking at the short-term costs and benefits. As a result, the possible longer-term implications of most short-term decisions tend either to be overlooked or looked at in more qualitative terms as part of a separate process.

12.1.3 Consumer Psychology

Psychological pricing refers to the practice of setting prices slightly lower than rounded numbers. It is based on the idea that customers do not round up and will treat them as lower prices than they actually are.

Customers look at tagged prices from left to right and ignore that last few digits. With this idea in mind, a car may be priced at HK\$149,999 rather than HK\$150,000. While the difference is minimal, customers might identify the price with HK\$140,000 rather than HK\$150,000.

Psychological pricing is most commonly used with consumer goods.

Another approach is to set higher prices based on the belief that customers will pay more attention to a more expensive product and consider it to be premium, such as jewellery.

There are a number of advantages to psychological pricing.

One is that it may place a product within a lower or higher price band, depending on the approach. A customer who wants to buy a car worth less than HK\$150,000 may consider one worth HK\$149,999 but is not likely to pay much attention to one priced at the upper ceiling of her budget.

Another advantage, one that is hard to verify, is that customers may be induced to buy products based on the price reductions that psychological pricing advocates as most supermarkets do.



A third advantage is inventory control. It is harder for staff to 'steal' funds if numbers are not round.

A fourth advantage is that psychological prices make it easier to offer discounts by changing a single digit in the pricing. A retailer may offer a 10% discount on any item whose price ends in '98' for instance. This makes it much easier to implement discounts.

Still, there are some difficulties associated with psychological pricing. At the top of the list is that customers may be more rational than this type of pricing assumes and ultimately base their purchases on value rather than perceived price.

In spite of this, psychological pricing is widely used and businesses regularly use this approach to compete. Setting prices higher than the competition might lead to drops in sales so, even if only following herd mentality, companies regularly use psychological pricing and so do their competitors.

12.1.4 Competition Profit

The price setting process is directly affected by pricing policies of the **competition**.

This is particularly true for price takers. For example, a fuel station that consistently charges HK\$5 more than others for a litre of fuel is likely to go out of business. The same is true for any products that are not differentiated with benefits beyond price.

Before setting prices, companies research their markets and discover what their competitors are charging. They then consider any competitive advantages and how much more or less customers may be willing to pay for their products. This involves comparisons of quality in materials, reputation and customer preferences for one product over another, even when they offer approximately the same attributes.

12.1.5 Government Policy

Government price controls and government policy also affect pricing policies for producers.

Governments may decide to set minimum or maximum prices for goods or services. Drug prices, for example, are often capped by governments. So are utility prices. Governments may also impose taxes that must be included in the price of products such as fuel for cars or cigarettes.

Price ceilings set maximum prices while price floors set minimum prices.

To avoid a decrease in profits, companies need to formulate pricing strategies that consider regulations affecting their products.

Beyond setting price floors or price ceilings, regulations may restrict how businesses approach pricing or determine their pricing strategies. For instance, predatory pricing, in which a company sets a very low market price to attract customers, may not be allowed. Governments may impose penalties for violating these regulations, as the Hong Kong Competition Ordinance (Cap.619) enacted on 13 January 2013 does.

Companies operating in multiple jurisdictions may have to contend with varying levels of government regulation.

Regulations may also have indirect impact on prices, particularly when they affect the supply or demand of products or raw materials. Throughout 2018, for example, there was a

move to increase import tariffs of some raw materials between China and the United States, such as steel and aluminium, and those tariffs would have a definite impact on the prices set by, for example, the property companies that use those raw materials in their products.

The impact can be even more nuanced. For example, tariffs may be set on imports of a raw material for a particular country and after a certain quantity of imports has been reached. Such nuanced regulations require companies to carefully consider their pricing strategies.

Illustrative Example 1

What is the best way to use gross profit to monitor the effectiveness of a company's overall pricing policies?

Gross profit margin (i.e. sales minus variable cost and then divided by sales) is a common method in setting product price. Accountants and business analysts regard the gross profit rate of a company as holding far more information than the gross profit amount.

For example, a gross profit of HK\$100 million might appear to be a large sum of money to an uninformed observer. However, if this sum was the outcome of a gross profit rate of just 3%, it would be unimpressive. The gross profit rate shows what percentage of each dollar in sales goes to the gross profit.

Gross profit, calculated as the difference between the cost of producing the good or service and the revenue it earns, shows the merchandising profit of a company. As it excludes operating expenses, it is not a measure of overall profitability. Much useful information about a company can be gained by studying the amount and rate of its gross profit, comparing those figures with the rates and totals for previous years and with those of competitors. Comparison of gross profit rates between similar companies can offer a broad indication of the quality of their respective purchasing and pricing policies. For example, analysts can usefully compare the gross profit margins of Hong Kong's three largest fast-food chain restaurants, Café De Coral, Fairwood and Maxim's.

Key Learning Point

Establishing the price for any good or service is affected by many factors such as cost data, demand, competition, consumer psychology and government regulation.

► Knowledge Check Questions

Question 1

Identify which of the following describes the difference between direct and indirect costs.

- A** Direct costs are paid by the company and indirect costs are assigned to vendors or suppliers for payment.
- B** Direct costs can be directly traced to a product while indirect costs cannot.

► Knowledge Check Questions (*continued*)

- C** Whereas direct costs are specific in amount, indirect costs identify payments of different price levels.
- D** Direct costs are true costs, but indirect costs are classified as operating expenses.

Question 2

Identify which of the following best describe attributes and reasoning behind psychological pricing.

- A** Consumers will have natural resistance to prices above a known limit. With this in mind, producers have to always keep prices below that level.
- B** The price assigned to a product should always be slightly lower than the competitor's price for the same product.
- C** Wealthier consumers always are willing to pay more for the same product, and poorer consumers will only be willing to pay the minimum.
- D** Prices are set to appeal to consumers seeking a bargain. For example, asking HK\$39.99 instead of HK\$40 is associated more with a price in the 30s than in the 40s of Hong Kong dollars.

Question 3

Identify the ways in which government policy may restrict or encourage producers to set prices.

- A** Governments may set minimum or maximum prices, impose taxes or regulations or place tariffs on imports, making them less competitive with domestically produced goods.
- B** Governments may pass laws requiring consumers to buy particular products and restricting them from buying others.
- C** By lowering taxes on goods and services, governments increase sales but add to the problem of deflation.
- D** When a government wants a product off the market, it appeals to producers to cooperate under threat of higher taxes.

➤ 12.2 TYPES OF PRICING STRATEGIES

Pricing decisions are amongst the most difficult decisions that managers face and the pricing strategies they adopt may determine the success or failure of a company. The end goal of a pricing strategy is to maximise profit. This may not necessarily mean setting the highest possible price, not if a lower price can lead to increased sales. Determining the optimal price or mix of pricing strategies is, for most companies, an ongoing consideration.

According to economic theory, the size of demand for any product or service is a function of the price charged for it – the higher the price that is charged for a good or service, the lower will be the quantity demanded. Companies that are able accurately to estimate the quantity

demand at different prices for a good, find themselves in the fortunate position of being able to discover an optimal price for it. All they have to do is calculate the contribution margin by deducting their variable costs for each price, multiplying the contribution margin by the quantity demanded, then subtracting their fixed costs.

Determining the demand function, the relation between price and the quantity demanded, is the most difficult part of determining the profit-maximising price. There are many different approaches that can be used.

By asking sales managers in different regions to estimate the quantities they can sell at different prices, the sum of the total quantity at different prices would provide an estimate of total quantity demanded for each price level. Another method would be to extrapolate the entire market based on test-marketing of the product with a number of different customers.

Estimating demand function can be challenging, which is why some companies turn to the simpler approach of using cost-plus pricing.

12.2.1 Cost-Plus

In a competitive environment, product prices are set according to market demand. Companies which want to meet a set profit must therefore focus on meeting their target costs. The less competitive a market is, the more freedom a company has to set its own prices.

When companies set their product prices, they usually do so in terms of the cost of their product or service – a process known as **total cost-plus pricing**.

Under total cost-plus pricing, a company arrives at its **target selling price** by adding a **mark-up** to the cost base. Providing that the cost base includes a correct calculation of the costs of making and selling a product, the mark-up should be the company's desired profit. Exhibit 12.4 illustrates the steps involved in this process, with the mark-up representing the difference between the selling price and cost – the profit on the product.

Selling price	–	Cost	=	Mark-up (Profit)
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► EXHIBIT 12.4 Mark-up

Ideally, the amount of the mark-up is set according to the return on its investment that the company wants to receive. In reality, however, its mark-up will be set by a host of factors, including how much competition it faces; what demand there will be for its good; whether it faces any political or legal issues and so on.

Once the company knows its cost base and desired mark-up, it can add these two figures together to set a target selling price, as shown in Exhibit 12.5.

Cost	+	Mark-up percentage × Cost	=	Target selling price
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► EXHIBIT 12.5 Cost-plus formula

The ultimate measure used to evaluate the performance of all companies is their return on investment (ROI). ROI is calculated by dividing the income earned from a business by the average amount invested in that business. The higher the rate, the better.

To achieve a desired return on investment for a product or service requires setting its mark-up according to the desired ROI per unit of that product or service. This figure is calculated by multiplying the desired ROI percentage by the total investment costs of making the product, then dividing this resultant sum by the total number of units made.

Total cost-plus pricing has the advantage of being simple to calculate. Its shortcoming is that it ignores demand, or whether customers will pay the price set by the company. Further complicating matters is the role sales volume plays in determining per-unit costs. The lower the sales volume, the higher the price a company must charge to meet its desired ROI.

Illustrative Example 2

You own and manage a printer company. You have determined that the total cost of development, production and marketing of 3XA model printer is HK\$780. By using the cost-plus pricing method and assuming that the company does not pay any interest and income taxes, what is the unit selling price of a 3XA printer if making a:

- A. net profit margin of 25%; or
- B. gross profit margin of 30%

- A. $780 / (1 - 25\%) = \text{HK\$}1,040$
- B. $780 \times (1 + 30\%) = \text{HK\$}1,014$

Using cost-plus pricing, the estimated price of the printer is either HK\$1,040 or HK\$1,014. The company will be able to recoup the cost of producing the printer, while earning a net profit margin of 25% or 30% gross profit margin on each unit sold.

Illustrative Example 3

Cost-plus pricing can be used for goods as well as services. Many services companies, such as management consultants or law firms, use cost-plus pricing to set their fees.

Take the case of an IT maintenance company with repair staff it pays HK\$100 an hour and other expenses such as telephone support, transport and material costs that average HK\$40 an hour. For the company to earn a 30% profit for its repair services, and assuming its staff work continuously, how much should it charge for each hour of work completed?

$$\begin{aligned} \text{Break-even price} &= 100 \text{ for repair personnel cost} + 40 \text{ in additional expenses} \\ &= \text{HK\$}140 \end{aligned}$$

$$\text{Cost-plus pricing} = \text{break-even price} \times \text{profit margin goal}$$

$$\text{Cost-plus pricing} = \text{HK\$}140 \times 1.3$$

$$\text{Cost-plus pricing} = \text{HK\$}182$$

To achieve the 30% profit margin goal, the company must bill clients at HK\$182 per hour of customer service.

Illustrative Example 4

HKQP Engineers & Contractors (HKQP) has a five-year contract to supply electricity to I&S Manufacturing Ltd (I&S) using a diesel-fuelled power plant that it has installed to help the company cope with the frequent power cuts it has experienced.

Under the terms of the contract, HKQP will be reimbursed every month for the cost of any fuel it uses plus a 20% profit on its total costs.

Staff working at the power plant earn a total of HK\$15,000 per month. Head office expenses allocated to the power plant in the form of a management fee total HK\$50,000. The plant is depreciated at the rate of HK\$20,000 per month over the five-year contract period.

During the first month, HKQP used 15,000 litres of diesel priced at HK\$10 per litre.

Calculate the amount you will invoice the company for the first month.

Answer

Total cost for the first month
 = diesel cost + labour cost + overheads cost (including depreciation)
 = 150,000 (diesel cost @ 10 per litre for 15,000 litres)
 + 15,000 (direct labour) + 50,000 (management fee)
 + 20,000 (depreciation)
 = HK\$235,000

Invoice amount for first month
 = 235,000 + 20% of total cost
 = HK\$282,000

Illustrative Example 5

ABC Investment Ltd's subsidiary Intelligent Products Ltd is in the process of setting a sales price for a new slim pen. The per-unit variable cost estimates for the pen is HK\$60.

To produce and sell its pen, Intelligent Products Ltd incurs fixed manufacturing overhead of HK\$280,000 and fixed selling and administrative expenses of HK\$240,000. The company is aiming for a 20% ROI and has invested HK\$1 million.

Assuming production of 10,000 units, calculate the mark-up and target selling price.

To arrive at the cost per unit, divide total fixed costs by the number of units the company expects to produce.

Fixed manufacturing overhead = $280,000 / 10,000 = \text{HK\$}28$

Fixed selling and administrative expenses = $240,000 / 10,000 = \text{HK\$}24$

Total fixed cost per unit = HK\$52

(Desired ROI percentage $\times$ Amount invested) / Units produced = Mark-up

$(20\% \times 1,000,000) / 10,000 \text{ units} = \text{HK\$}20$

Illustrative Example 5 (continued)

Intelligent Products Ltd expects to receive income of HK\$200,000 ($20\% \times \text{HK\$1,000,000}$) on its HK\$1 million investment. On a per-unit basis, the mark-up based on the desired ROI per unit is HK\$20 ($\text{HK\$200,000}/10,000$ units). Given the per-unit costs shown above, the sales price should be HK\$132.

Variable cost	HK\$60
Fixed cost	HK\$52
Total cost	HK\$112
Desired ROI	HK\$20
Selling price per unit	HK\$132

12.2.2 Skimming

While economic theory plays an important role in pricing decisions, so too does business strategy.

One strategic approach to pricing is called **skimming**.

With a skimming strategy, the initial price of a product is set higher than competing products. With skimming, a company can take advantage of consumers willing to pay a high price to be first to have an exciting new product, such as Apple's iPhones.

This segment of buyers is often referred to as early adopters. After introduction, the price is lowered to capture additional customers at a lower price point.

12.2.3 Penetration

Another strategic approach to pricing is the **market penetration strategy**. Under this strategy, the price is set lower than competing products. The idea is to rapidly achieve high volume and economies of scale while gaining market share when entering the market, as the fast-moving consumable goods (FMCG) do in supermarkets (e.g. 'No Frill' private brand offered by Wellcome Supermarket).

When applied well, market penetration pricing can help companies boost their market share and volume of sales. In turn, higher sales volume can lead to lower production costs (fixed costs per unit) and higher profits. A low price set as part of an introductory campaign may entice customers to switch over to a new product.

In many ways, a penetration strategy is the opposite of a skimming strategy, that relies on setting prices high initially to boost market share and profits early in the product life cycle (e.g. pharmaceutical products).

12.2.4 Complementary Product

A **complementary product pricing strategy** is, in some ways, a combination of skimming and penetration pricing. The approach is most often used by retailers (e.g. the few supermarket chains in Hong Kong).

In this approach, a company may set the price of one product to achieve the largest possible volume of sales regardless of cost or profits, in order to increase demand for another product that may have higher profit margins.

This is an approach that may be taken by companies that sell shaving kits, for example. A razor may be priced low enough to attract new customers but the razor blade refills represent the profit margin for the company. Complementary product pricing often works with products with accessories, that can range from computers to cars.

The idea is that the loss associated with selling one product too cheaply is more than offset by the gains of selling the complementary product at a significant margin.

12.2.5 Product Line

A **product line pricing strategy** is used by companies that sell multiple related products and set prices in combination rather than setting prices for each product on its own.

The idea is to maximise sales by offering complementary – as opposed to competitive – products. This strategy requires companies to consider the impact that the price of one product or service may have on another.

A product line pricing strategy may attract different kinds of customers, from those looking to buy premium products and are willing to pay a premium price to those looking for a basic product based on low price, and every customer in between.

Some variations to this approach include:

- **Captive pricing:** Companies sell some products at a low price to get consumers to use the base product but then encourage them to buy add-ons (e.g. side dishes to the KFC meal sets).
- **Loss leaders:** Selling products at or below cost to attract customers and drive sales of other products. This approach is often used by supermarkets.

12.2.6 Volume Discounting

Volume discounting involves incentives for purchasing large volumes of a product or service. The concept is that a customer purchases a lot of a particular product to get a lower price per unit.

This is an approach made famous by discount retailers such as Costco Wholesale megamarkets in the United States (not available in Hong Kong) who sell products in huge quantities.

There are three key reasons that companies may choose a volume discount strategy.

The first is to attract new or first-time customers. Selling a new customer a large volume of a product gives a company time to sell them other products with higher margins.

The second reason is to be more competitive. A volume discount may attract both customers and larger orders at the cost of the profit margin.



A third reason is to facilitate distribution, which lowers costs. By selling large quantities of a single product in one transaction, companies avoid having to make many different sales, which add to the costs of distribution.

There are some drawbacks, however. One is that margins become smaller. Another is that a volume discount approach may devalue a product. A third consideration is that it may be impossible to raise prices later without losing customers.

12.2.7 Discrimination

Price discrimination is another strategy based on charging different customers different prices for the same product or service.

One approach to price discrimination is to charge customers the maximum amount they are willing to pay. Another approach is to group customers based on their characteristics. Banks, for example, charge customers with more money in deposits lower fees for their services.

Price discrimination can be valuable if the profits that can be generated by separating the markets are greater. And this depends on the ultimate elasticity of demand of the market segment. Customers that have more inelastic demand (meaning that they need the good or service more) are charged more (e.g. funeral services).

It is important, however, that different types of customers do not overlap.

There are various types of price discrimination. First-degree discrimination happens when companies charge the maximum possible price for every unit sold. Second-degree price discrimination refers to the practice of setting different prices for different quantities consumed, such as mobile data or hours with a personal trainer. Third-degree price discrimination refers to companies that charge different prices to different consumers groups, such as transport fares or movie tickets for senior citizens.

12.2.8 Relevant Cost

Relevant cost is cost associated with a management decision that is likely to change in the future. This eliminates costs not relevant to a product and helps determine price.

Relevant costs can be useful in setting prices by stripping away irrelevant costs, such as costs that may have already been incurred. For example, an airline or train company may sell last minute tickets at a cheaper price because the only relevant cost associated with the sale of those tickets is the cost of loading the passenger into the plane or train. All other costs have already been incurred.

The same may be said for orders for products that come in after all fixed costs have been covered. A price quote for a special order at the last minute may only take into account relevant costs, such as incremental material and labour costs and not indirect costs such as management salaries that have already been covered in the pricing of goods or services.

Incremental costs, sometimes known as **differential costs**, are the higher costs that arise from a short-term change in business activity. For example, if a company found its total monthly costs rising to HK\$360,00 from HK\$320,000 after it increased its machine hours from

8,000 to 10,000 for accepting a special order, the incremental cost of the 2,000 machine hours would be HK\$40,000.

The principle of relevance is the term used to describe the method that a company uses to compare options. It allows managers to decide which costs and benefits are significant and should be included in their thinking, and which are irrelevant and can be ignored.

The principle of relevance can simplify the analysis of determining the best option. According to this principle, it is only necessary to consider the benefits and costs that differ from one option to another, like differential costing.

When identifying which costs and benefits should be borne in mind, only their impact on future revenues and future costs need be considered. Sunk costs should not influence decisions because the past cannot be changed. Sunk costs have already been incurred and will not change regardless of what choices might be made going forward. As a result, sunk costs are not relevant because they do not differ from one option to another.

Apply and Analyse 1

Apply

Ten years ago, HKQP Company Limited manufactured 500 air conditioners for a customer with the following per unit selling price and cost information.

	HK\$ per unit
Selling price	15,000
Variable costs:	
Direct materials	4,000
Direct labour	2,000
Manufacturing overhead costs	1,500
Selling and distribution costs	600
Total variable cost per unit	8,100
Fixed costs:	
Manufacturing overhead costs	1,000
Selling and distribution costs	400
Finance and administrative costs	500
Total fixed cost per unit	1,900
Net profit per unit	5,000

However, the designated customer was bankrupt just before delivery of the air conditioners. Therefore, the order was cancelled at the last minute and the 500 air conditioners were stored in the warehouse for 10 years.

Apply and Analyse 1 (continued)

Further information:

Now, HKQP Company Limited finds a new customer who is interested in the batch of air conditioners. In selling this old stock of air conditioners, the company needs to spend:

- refurbishment cost of HK\$500 for each air conditioner, and
- incur HK\$100,000 transportation expenses for delivering the air conditioners to the new customer.

Required:

1. Calculate the minimum selling price that HKQP Company Limited should set for the air conditioners; and
2. Explain why such a pricing strategy is adopted.

Analysis

1. Minimum selling price per air conditioner

Variable/incremental cost:

Refurbishment cost	HK\$500
Transportation cost (HK\$100,000 / 500)	200
Total variable/incremental cost	<u>HK\$700</u>

2. All the variable and fixed costs incurred in manufacturing these 500 air conditioners 10 years ago are irrelevant since they are 'sunk cost' and are irrelevant costs for pricing consideration. Hence, the only relevant and variable/incremental costs are the refurbishment cost and transportation cost at a total of HK\$700 per unit (i.e. HK\$500 + HK\$100,000/500). The company can quote a selling price of anything above HK\$700 to make a profit. This is the relevant or incremental pricing strategy that HKQP Company Limited should adopt.

Another similar analogy is selling equity shares in the stock market. Irrespective of the purchase price that we have paid for a certain ordinary share in the Hong Kong Stock Exchange (HKSE) 10 years ago, our cash outflow is sunk cost or irrelevant cost for making the decision today of whether selling the shares at today's quoted price in the HKSE, and the only consideration is the transaction cost (i.e. stamp duty, agent's fee) to be paid. However, there is always a psychological pricing factor where the original purchase price, paid 10 years ago, must always be considered.

Key Learning Point

Cost-plus, skimming, penetration, complementary product, product line, volume discounting, discrimination and relevant costs are pricing strategies that companies can apply when setting selling prices.

► Knowledge Check Questions

Question 4

Identify which of the following explain how cost-plus pricing operates.

- A** This is a method of setting price based on the known cost and mark-up needed to generate a profit level.
- B** The original purchase price is increased gradually over time.
- C** The cost of a product is increased by selling warranty contracts.
- D** Groups of purchasers band together to buy products at cost plus a small profit for the producer.

Question 5

Describe the attributes of product-line pricing.

- A** Every product is priced according to the geographic market where it is sold.
- B** Products in different lines are priced the same to encourage cross-line purchases.
- C** The entire product line is given identical prices, collectively covering the mark-up and net profit goals.
- D** A product line is sold for one price that is lower than it would be to sell each product individually.

Question 6

Explain price discrimination and how it works.

- A** This is illegal price setting by producers.
- B** Prices are established based on socioeconomic characteristics of customers.
- C** The price level is increased for exports but kept lower for domestic markets.
- D** Different prices are set for the same product, based on customers' willingness to pay, characteristics (such as low fees charged to bank depositors with large balances), or the quantity sold (buy two, get one free, for example).

➤ 12.3 IMPACT OF CORPORATE SOCIAL RESPONSIBILITY ON PRICING STRATEGIES

Corporate social responsibility (CSR) refers to the impact that any company has on society and the environment. CSR takes into account outcomes for all stakeholders in a business, outcomes based on long-term sustainability and not just short-term returns to shareholders.

Strong CSR practices require companies to take actions beyond the narrow interests of the company and to go beyond legal or regulatory requirements to consider the responsibility that companies have to society. These responsibilities include efforts to reduce carbon footprints or deal with employees ethically, or engage in philanthropy or volunteer events.



CSR can help improve the environment or tackle societal issues but also build the reputation of an organisation.

12.3.1 Cost of Quality

Quality, or the lack of it, comes at a cost for companies as explained in Chapter 11.

The **cost of quality** is associated with the cost of producing products of inferior quality. The cost of quality includes the cost of reworking a product, testing it, providing after sales service or replacing faulty products.

Customers associate quality with their expectations. This means that high quality is not necessarily associated with the highest standards but, rather, with products that best meet expectations.

Creating products that go beyond what customers want may create considerable expense and waste and not be considered high quality.

Companies may reduce costs by eliminating expenses on products or parts of products over which customers do not have an obvious opinion or perception of quality (e.g. packaging of chocolates or biscuits).

These cost reductions can include lower quality materials in areas that are out of sight or allow lower quality production where customers are not discerning or perceptive.

Companies are most often concerned with a number of quality considerations:

- Quality of design – the ability of a company to meet the expectations of a customer. Good CSR requires that product design be of quality high enough to meet the expectations of customers.
- Quality of conformance – a company's ability to produce products that match the specifications of the original design (e.g. car industry). This may have particular costs, including the purchase of the proper materials (which may be more expensive) as well as quality and undamaged materials.

In turn, quality expectations impact different categories of costs for companies including:

- Preventive costs – incurred to avoid product failures and include the development of procedures, training, testing, preventive maintenance on production machinery and supplier qualification assessments.
- Appraisal costs to reduce risks associated with shipping defective products to customers – such costs include testing both components from suppliers and final products before shipping, plus the cost of any testing equipment.
- Internal failure costs associated with defective products uncovered prior to delivery – these costs include reworking the defective products, additional testing, scrap, purchasing replacement parts and sales of seconds.
- External failure costs linked to defective products, including lost revenue and lost customers, processing of returned goods, administering warranty claims, field service costs, lawsuits or product recalls.

It is often cheaper to pay for improvements internally rather than waiting for customers to point out defects. This is because a customer who buys a product of low quality may not

buy from the company again and may publicly announce their dissatisfaction of the product. However, it is important to keep in mind that the cost of quality may be significant.

12.3.2 Environmental Costs

The management of **environmental costs** is increasingly important to companies for three reasons.

First, environmental awareness is more widespread with customers concerned about their carbon footprint and recycling of products. Companies can attract customers by shoring up their environmental credentials.

Second, environmental costs in some industries are high, particularly in areas related to oil production where they can add up to 20% of operating costs that have to be carefully managed.

Third, increasing environmental regulations and higher penalties add to company costs for environmental mismanagement.

Environmental regulations are sometimes difficult to define and costs are hard to identify. And when they can be identified, they may be hard to control. Environmental costs are broken down into four categories:

- Conventional costs of raw material and energy.
- Hidden costs mixed in with overhead.
- Contingent costs incurred in the future, such as the cost of an environmental clean-up.
- Image and relationship costs such as those associated with negative reports.

Some definitions describe environmental costs as those incurred to protect the environment and dealing with waste materials.

Illustrative Example 6

It has become increasingly important to measure how organisations perform in terms of environmental sustainability.

An example of how a company can improve is Canada's Aéroports de Montréal (ADM), the non-profit entity that operates Pierre Elliott Trudeau and Dorval airports in Montreal. The company has received certification under ISO 14001, the global standard on environmental initiatives from the International Organization for Standardization. In so doing, ADM had to commit to meeting environmental goals every year, through continuous improvement of its operations.

The measures ADM took included building a high-performance thermal plant that raised energy efficiency by 70%, installing computer-controlled blind systems that maximised the use of natural light and replacing sliding doors with revolving ones that cut the amount of energy used for heating and air conditioning.

The above environmental facilities and measures implemented by ADM might have increased its investment in non-current assets in the short term but reduced the operating costs in the long term, which would reduce or maintain its product and service prices.



Illustrative Example 7

Industries need to comply with environmental regulatory laws and manage their waste to protect the environment. Some industries rarely report the costs incurred in complying with environmental law and waste management. Overlooking these costs may mislead the company's management in the setting of their product prices.

A study focused on the brewing industry located in Lagos State, examined the effect of environmental protection costs on product price in Nigeria. The management accountants who took part in the survey were from various companies: Nigeria Brewing Plc, Guinness Plc, Coca-Cola Plc and Seven-Up Plc.

The survey design was used for the study and a questionnaire was administered to generate data. Researchers employed purposive sampling technique in selecting the sample frame.

Based on the data collected, the study discovered that there was a negative relationship between environmental regulatory costs and product pricing decisions. The researchers recommended that companies should design accounting systems that captured expenses incurred on environmental matters. This would enable the firm to appreciate the amount that they have invested in managing their waste and the cost incurred to comply with environmental protection rules and regulations.

Illustrative Example 8

'Turning into the 21st century, it is a general belief that customers' perceptions of firms' corporate social responsibility (CSR) and fairness in their pricing strategy determine customers' behaviour as users of financial services, such as the banking industry'. Briefly comment on this statement.

Answer

Research in the banking industry suggests that companies that pay attention to CSR can benefit in multiple ways. Notably, undertaking actions that demonstrate social responsibility can have a positive influence on perceptions of whether a company is charging a fair price.

The research, which was based on a survey of 300 customers of banks and saving banks, found that CSR activities appear to result in customers seeing socially oriented businesses as having fairer pricing strategies. It also found that CSR activities led to a rise in customer loyalty levels. It appears that roles of satisfaction and commitment play a role as mediating variables in the formation of such positive options.

Key Learning Point

Strong CSR practices require companies to take actions extending beyond the narrow interests of the company, beyond legal or regulatory requirements, and to consider the responsibility companies have to society.

► Knowledge Check Questions

Question 7

Identify which of the following defines 'cost of quality' and provides examples.

- A** This means that setting standards for high-quality products cost money. For example, responding to customers places a demand for added labour.
- B** When competitors offer a higher-quality product, it is easier to undercut their price.
- C** There are costs associated with poor quality. These include having to rework products, testing and replacing faulty items.
- D** Higher-quality products can be priced above competitors' products, but that involves added costs.

Question 8

Identify why environmental costs need to be considered in a full-cost pricing strategy.

- A** Environmental costs include direct and indirect costs.
- B** Environment costs have been increasing.
- C** Environmental costs involve hidden costs to be incurred after selling products or services.
- D** All of the above.

12.4 EXTERNAL MARKET FACTORS AND INTERNAL COST STRUCTURES ON PRICING STRATEGIES

Pricing decisions are affected by both external and internal factors. External factors, such as demand for a product, competition or regulation, impact pricing as much as internal factors, such as the mix of variable and fixed costs.

12.4.1 External Market Factors

The key external factors impacting pricing strategies are:

- Demand,
- Competition, and
- Government regulation.

Different groups of customers have different levels of demand for a product or service. In turn, the level of demand amongst different customers impacts pricing.

A first consideration is the user group to which a good or service is aimed. Companies may charge different prices to different types of customers.

A second consideration is the location or point of consumption where the good or service is offered. Prices may vary from one location to another. Points of production and points of service impact pricing. For example, hotels in tourist spots may charge more than those in locations with less traffic.

The time of production also impacts pricing. Some goods or services are perishable and pricing may drop as time passes (e.g. air or train tickets). High-tech goods may become less valuable as new products and technologies are introduced.

Competitive considerations also affect pricing decisions. The more competition in a market or industry, the more careful managers have to be with their pricing decisions.

A final consideration is government regulation. Different industries, sectors or locations may operate under different levels of regulations, which affect pricing; and the level of government regulations (including taxes) affects pricing decisions.

12.4.2 Internal Cost Structures

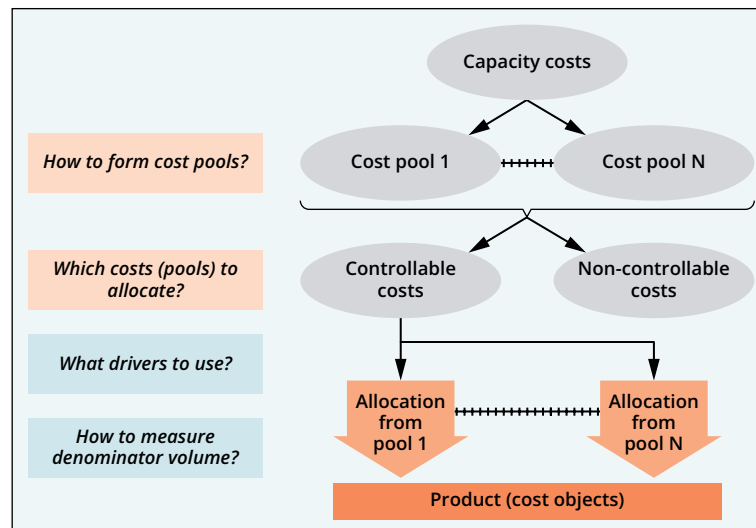
Pricing is also affected by internal cost factors.

Pricing structures are affected by the proportion of fixed costs, such as cost of machine, and variable costs, such as direct materials or supplies, that businesses incur.

Companies use their cost structure to determine prices.

A **cost-based pricing strategy** relies on costs to determine the right price. Costs may be difficult to determine, so companies may then choose **activity-based costing**, which aligns pricing with the cost structure of specific products or services.

A business could become more or less competitive by changing its cost structure in both the totality of its components or for specific parts.



► **EXHIBIT 12.6 Four steps of a product costing system** (Source: Balakrishnan 2012. Reproduced with permission of John Wiley & Sons.)

Companies use cost allocations to determine the capacity resources needed for a product. Exhibit 12.6 shows the four stages of a product costing system:

1. Decide what cost pools to form;

2. Decide which costs to allocate to which pool;
3. Identify the cost drivers to use in each pool; and
4. Determine the appropriate denominator volume measure for each pool.

The first, third and fourth steps relate to defining elements that are common to all cost allocation procedures: cost pools, cost drivers and the denominator volume. The second step represents a refinement to allocation procedures. This refinement separates the costs of capacity resources into controllable and non-controllable costs. Doing this allows a business to evaluate how different long-term decisions can affect profitability.

Activity-based costing (ABC) is an approach to determining product costs that assigns the controllable costs of capacity resources to products more reliably than traditional costing systems. (Activity-based costing differs from traditional systems across all four of the steps listed above; see Module 2, Chapter 6 for more information.)

Underlying activity-based costing is the idea that all companies operate around a hierarchy of coordinated business activities. For example, a company may have sales activities, manufacturing activities, inventory activities and so on, each in turn comprising further activities. Purchase order processing, for example, incorporates activities such as soliciting, approving, executing and paying purchase orders. Each of these processes and activities takes organisational inputs and transforms them into measurable outputs.

These activities can be divided into four types:

1. Unit-level activities, such as assembly and inspection, which are proportional to the volume of production undertaken.
2. Batch-level activities, which pertain to a group of units, such as the setting up machine when starting production. These activities relate to executing production orders and customer orders that may include large units.
3. Product- or customer-level activities relating to the needs of a specific product or specific customer, such as specific product design and engineering or advertising.
4. Facility-level activities that ensure the smooth running of the business as a whole, such as building maintenance or company administration.

The cost pools of activity-based costing systems correspond to these four categories of activities.

The first step in designing a product costing system is forming cost pools. A cost pool is a collection of the costs of similar resources and can be formed by functions such as departments or on any other logical basis. ABC systems use activities as the basis for forming cost pools.

Accounting records are used to decide which costs should be placed in which pool. Companies typically tend to organise their accounts by type, i.e. in terms of salaries, equipment, supplies, rent and so on, rather than by the activity supported by that expense. This means that any single account type – departmental salaries, for instance – might contain costs relating to several different activities, such as customer support, inventory management, account administration and so on. The first step of establishing an ABC system is to analyse each type of account and determine the specific activities that it supports.



The second step is deciding which costs to allocate. To obtain an accurate estimate of product costs and profitability requires including all capacity cost that would change if a decision was taken to produce a product. It excludes all capacity costs that would remain unaltered. Because activity-based costing focuses on controllable costs, it differs from traditional systems which allocate all manufacturing and sales costs regardless of their controllability.

The third step is identifying cost drivers. A suitable cost driver is one that has the strongest causal relation with the costs in the cost pool. In the long term, any change in driver volume should trigger a proportionate change in the costs contained in the pool.

The final step is to allocate costs to products by measuring denominator volume. ABC systems use practical production capacity as the denominator volume when computing allocation rates.

In traditional systems, allocation costs are usually calculated by dividing the costs in a cost pool by the actual/budgeted production volume for a given period. Doing this, however, has the weakness of spreading all capacity costs over the actual/budgeted volume and so paying no attention to capacity is used efficiently or not. As a result, when production volumes are low and a company has idle capacity, allocation costs will rise and products will appear more expensive.

Because ABC systems use practical capacity as the denominator volume when allocating capacity costs, if a product consumes the same amount of resources, the cost allocated to it will remain the same. As this method assigns capacity costs to products strictly in proportion to use, unallocated activity costs – the difference between allocated activity costs and actual spending on capacity resources – can be more readily uncovered than in traditional costing systems.

Being able to identify unused capacity can be of great assistance in managing costs by giving companies a tool to track down and eliminate unused capacity.

Illustrative Example 9

Framing is the manner in which something is presented. The manner in which something is presented may influence how others feel about it.

A case in point is special offers by stores and other companies. Bars and restaurants, for example, often offer 'happy hour' specials with some type of discount for drinks, such as selling two for the price of one during certain hours that may match hours when traffic is slowest.

By making purchases during happy hour, buyers believe they are getting a deal.

But turn this around and offer the same products for the same price but with a different advertising or framing. For example, what if the advertisement pointed not to the prices during happy hour but to the higher prices at other times? Framing the same situation in this way is only likely to leave customers feeling taken advantage of and negative about the business.

In effect, customers' attitudes are shaped by this framing.

Illustrative Example 10

Explain how activity-based costing facilitates product pricing in a business organisation.

Answer

Activity-based costing can provide relatively accurate cost information because ABC systems analyse cost behaviour on activities. More accurate cost information can better serve the enterprise's product pricing decisions. Besides, ABC helps companies to manage cost by identifying unused capacity.

Key Learning Point

The key external factors impacting pricing strategies include demand, competition and government regulation. Internally, pricing decisions are impacted by the cost structure of a company.

► Knowledge Check Questions

Question 9

Explain what is meant by cost-based pricing.

- A** The price of products is set based on the known costs to produce.
- B** Prices are set based on desired net profits as a percentage of revenues.
- C** Costs are increased to covered low sales volume.
- D** As general expenses rise, costs have to be reduced to maintain profits.

Question 10

Describe, showing examples, a few external factors that can affect setting product or service prices.



12.5 PRICING STRATEGIES FOR NON-PROFIT MAKING ENTITIES

Non-profit entities have to make pricing decisions like every company; but for them, these decisions may be based not on the goal of maximising profits but on achieving a specific



mission or outcomes. With this in mind, non-profit entities adopt different pricing strategies that include voluntary pricing, ability to pay, versioning or bundling to achieve their goals.

12.5.1 Voluntary Pricing

Voluntary pricing or pay what you want (PWYW) is an approach to pricing giving customers the choice on what to pay for goods or services. Their choice can be as low as zero, although a floor price or suggested price is sometimes set. Buyers can also choose to pay more than the floor price or suggested prices.

Voluntary pricing eliminates the disadvantages of traditional approaches to pricing and may attract buyers with its flexibility, eliminating fear that the product or service is not ultimately worth the price.

For the seller, voluntary pricing eliminates the need to set the right price and changes the pricing game into a relatively friendly win-win exchange.

Voluntary pricing has been traditionally used by non-profits and some for-profit companies, such as social enterprises, on the margin of the economy but the approach is gaining traction in some sectors, particularly as a way to gain market share.

12.5.2 Ability to Pay

The **ability to pay** approach is based on the idea that buyers should not pay more than they are able or may face an unnecessary burden.

This approach makes it possible to introduce progressive pricing schemes, mixed with some buyers who can afford to pay more for a particular product or service.

Ability to pay is not the same as straight income brackets often used to determine taxation levels.

Advocates of this approach states that it makes it possible for buyers with more resources to pay more and, perhaps, provide a level of subsidy to those with fewer resources who may also need access to the particular good or service.

Critics of the approach argue that the system makes the playing field uneven and penalises those with more resources.

In some industries such as banking, ability to pay is called 'capacity'. For example, it is used by some companies to determine a borrower's ability to make interest and principal repayments on loans.

Illustrative Example 11

The Saint Anthony's Church near the University of Hong Kong's main campus operates a café on Sundays inside its community centre staffed by many volunteers, which offer various kinds of food, drink and refreshment. Visitors can order whatever types and quantities of food and drink, as well as pay whatever amount of cash into a collection box, which is made of transparent plastic so that people can see through all the money inside. By adopting these practices, the café does not have to set the selling prices of its foods, drinks and refreshments, as well as saving cashier manpower.

12.5.3 Versioning

Another pricing approach frequently used by non-profit entities is called **versioning** (sometimes also known as 'quality discrimination').

This approach to pricing, which is at times used by for-profit companies, relies on businesses using different models of the same product and charging different prices for each model. Versioning gives consumers or users of a non-profit product or service the option to buy or tap into a higher-valued model, like elderly home services.

Versioning is typically used when products have large fixed costs of production but relatively small variable costs. Software companies, for example, often used versioning to attract more customers with different needs.

For-profit companies and non-profit entities in various sectors use versioning when they can provide different levels of quality or depth of service.

12.5.4 Bundle Services/Memberships

Bundle pricing allows entities to sell packages or sets of goods or services for a lower price than they would charge if each product or service was bought separately. Examples include option packages in cars or value meals at fast-food restaurants.

Memberships serve a similar purpose, as they allow members to pay a price for the ability to access a good or service at special prices or included in the membership, such as social clubs (i.e. the supermarkets operated by the Hong Kong Teachers' Association).

Some memberships require fees to join or participate. Members are entitled to special privileges or benefits, often preferential pricing on goods and services, as are members of the Foreign Correspondents Club in Hong Kong.

Bundle pricing is based on the idea that every customer is willing to pay a price for a particular good or service. If the price is equal to or lower than what the customer is willing to pay, the customer will buy. The difference between what a customer pays and what a customer is willing to pay is known as **consumer surplus**. Bundle pricing aims to capture consumer surplus.

Bundling is useful to target products to particular groups or services better and perhaps to differentiate between groups of customers with greater ease.

Offering products may also make it easier to get more revenue from each customer by selling more products in a bundle rather than just a single product or service.

Illustrative Example 12

Voluntary pricing or 'pay what you want' (PWYW) models must work or would not be used. The catch is that they do not always work.

PWYW is more likely to work for a small and local business like a café than a large chain coffee shop, because it is much more difficult to make customers care about large businesses. In essence, the system relies on people wanting to do the right thing. This is motivated by self-image.



Illustrative Example 12 (continued)

It may not be intuitive but when people are allowed to pay anonymously, they are likely to pay more, suggesting that they are generous.

In one PWYW experiment in Germany, no customers chose not to pay at a PWYW restaurant. Most people paid about 19% less than the restaurant's fixed prices. However, the experiment attracted more customers and overall sales increased.

In another experiment in which photographs were sold to customers, the average payment was US\$0.92 but when buyers were told the money would go to charity, the average donation increased to US\$5.33, but there were fewer sales.

Illustrative Example 13

Non-profit organisations face many of the same challenges as commercial companies when it comes to setting prices. Whether you are running a sports club or a charity for children, determining what prices to charge for membership fees or seats to a charity dinner, you are still trying to find a strategy that will allow you to have the optimum number of members or the highest possible donations. However, as with businesses in general, focusing on value is the best way of setting an effective pricing strategy that will maximise your revenues, profits and supporter numbers.

As with commercial businesses, a common pricing mistake is to adopt a cost mark-up pricing strategy using a set percentage above cost to set prices. If you price too high, then you will find fewer people willing to become supporters or buy your goods. If you price too low, then you will unnecessarily lower your revenues and profit.

A better approach is to look at the value of your services or products have for your supporters. In many instances, there will be no connection between cost and value, so what you must do is establish the value that customers or donors place on what you offer. Even if it costs little to produce, that good or service may have great value for someone who supports what you are doing because of the value your users see in the charitable activities that your organisation runs. In turn, this gives potential customers a reason to pay the prices that you ask.

Sponsorships or donations, for example, may be secured for nothing more than putting a person's name in a prominent position on a programme or website. Or, if you are running an art gallery or museum, you may like to reference the recommended admission donations of other similar institutions and keep your entry price in line with theirs.

As a result, your time may be well spent trying to figure out what value supporters or others aligned with your cause place on the various activities or goods that you offer. Surveys are one possible source of such information, as are comparisons with the prices offered by organisations similar to yours and questions put discretely to knowledgeable sources.

Illustrative Example 13 (*continued*)

You may also want to investigate the potential benefits of having a value statement that explains why supporters or customers should donate money or otherwise support your organisation. Such statements typically should highlight the value of the work your organisation does and how effective it is at doing that work. Ideally, it should also make the prospective donor or supporter feel that their involvement would really count.

Key Learning Point

Non-profit entities adopt different pricing strategies that include voluntary pricing, ability to pay, versioning or bundling to achieve their goals.

► Knowledge Check Questions

Question 11

Identify which of the following best describes voluntary pricing and how it is used.

- A** This means that prices are set not by the laws of supply and demand, but by voluntary cooperation between producers.
- B** This system, often used by not-for-profit companies, does not set a fixed price but invites buyers to determine price (pay what you want).
- C** Producers voluntarily reduce prices to fight inflation.
- D** Consumers agree to voluntarily pay a higher price if the producer pledges to make contributions to charities.

Question 12

Identify the characteristics of bundle pricing.

- A** A set of identified goods are sold for a single price, used as incentive for membership in a group.
- B** Goods are bundled together after purchase to facilitate delivery.
- C** A group of customers bands together to make a single purchase that no one of them could afford on their own.
- D** A producer sets identical prices to a bundle of similar products.

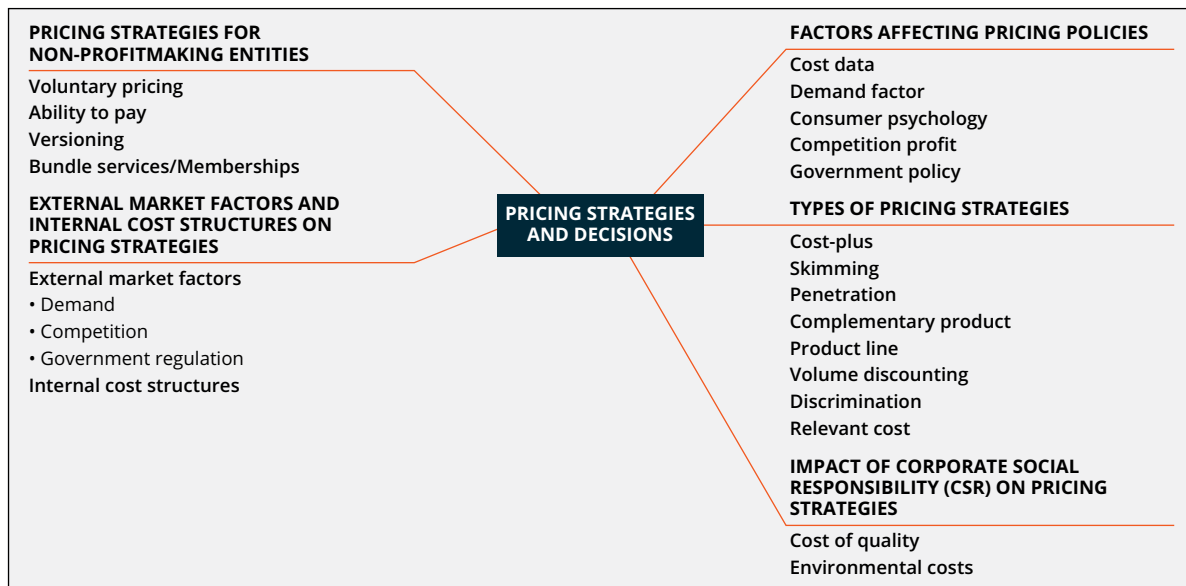


SUMMARY

- Establishing the price for any good or service is affected by many factors.
- One consideration in setting prices is the cost of production.
- Direct costs make up a consistent percentage of the cost of producing a good or service.
- Indirect costs change as a percentage of total costs as the volume of production grows.
- Usually, companies do not set prices. Instead, prices are set by market competition according to the laws of supply and demand.
- Psychological pricing refers to the practice of setting prices slightly lower than rounded numbers. It is based on the idea that customers do not round up and will treat them as lower prices than they actually are.
- Another consideration in price setting is competition.
- Government price controls and regulations also affect pricing policies.
- The end goal of a pricing strategy is to maximise profit.
- When a company sets the price of a product, it usually does so as a function of the cost of that product or service. This practice is known as total cost-plus pricing.
- Under total cost-plus pricing, a company determines its target selling price by first calculating its cost base and then adding a mark-up.
- Management is ultimately evaluated based on its ability to generate a high return on the company's investment.
- With a skimming strategy, the initial price of a product is set higher than competing products. With skimming, a company can take advantage of consumers willing to pay a high price to be first to have an exciting new product.
- The market penetration strategy is a strategy in which the price is set lower than competing products.
- In a complementary product pricing strategy a company sets the price of one product to achieve the largest possible volume of sales regardless of cost or profits, which increases demand for another product with a higher profit margin.
- A product line pricing strategy is used by companies that sell multiple related products and set prices in combination rather than looking at each product on its own to maximise sales by offering complementary – as opposed to competitive – products.
- A volume discount strategy involves incentives for purchasing large volumes of a product or service.
- Price discrimination is charging different customers different prices for the same product or service.

- Relevant cost is associated with a particular management decision that is likely to change in the future.
- Corporate social responsibility (CSR) is the impact that any company has on society and the environment.
- Cost of quality is associated with the cost of not producing quality products. The cost of quality includes the cost of reworking a product, testing it, providing aftermarket service or replacing faulty products.
- The management of environmental costs is increasingly important to companies.
- The key external factors impacting pricing strategies are demand, competition and government regulation.
- Pricing structures are affected by the proportion of fixed costs, such as cost of machine, and variable costs, such as direct materials or supplies.
- Voluntary pricing or pay what you want (PWYW) is an approach to pricing giving customers the choice of how much to pay for a good or service.
- The ability to pay approach is based on the idea that buyers should not pay more than they are able, or would face an unnecessary burden.
- Versioning (sometimes also known as 'quality discrimination') is another pricing approach.
- Bundle pricing allows entities to sell packages or sets of goods or services for a lower price than they would charge if each product or service was sold separately.
- Memberships serve a similar purpose, allowing members to pay a price for the right to access a good or service at special prices or included in the membership.

MIND MAP



LIST OF FORMULAS

Cost-plus formula

$$\text{Cost} + \text{Mark-up percentage} \times \text{Cost} = \text{Target selling price}$$

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. It is not the definition of direct and indirect costs.

Answer B is correct. Direct costs are directly traceable to a product while indirect costs are not.

Answer C is incorrect. The level of dollars spent is not the distinguishing feature between these costs.

Answer D is incorrect. Indirect costs are not the same as operating expenses.

Question 2

Answer A is incorrect. It is not the definition of psychological pricing.

Answer B is incorrect. Price comparisons to competitors are not psychological pricing.

Answer C is incorrect. The term is not defined by income levels of consumers.

Answer D is correct. Reducing a cost by a few cents may encourage more purchases by consumers, who may be less resistant to the price even though the approximate level is known.

Question 3

Answer A is correct. These methods are examples of how government policies are used to influence the economy.

Answer B is incorrect. Laws have an effect on spending patterns, but not by legislating what consumers must or must not buy.

Answer C is incorrect. Lowering taxes is one form of government policy, but there is much more.

Answer D is incorrect. It is not an example of commonly practised steps by governments.

Question 4

Answer A is correct. The process of setting price based on known costs and desired markup level.

Answer B is incorrect. It does not refer to changes made to the price.

Answer C is incorrect. This is not associated with offering warranties.

Answer D is incorrect. Cost-plus is not driven by consumer cooperation to gain discounts.

Question 5

Answer A is incorrect. This is not determined by geography of the market.

Answer B is incorrect. Pricing between product lines is not what this refers to.

Answer C is incorrect. Uniform pricing between product lines is not the correct definition.

Answer D is correct. By combining an entire line into one product, the price is lower than it would be to sell each component.

Question 6

Answer A is incorrect. It is not the same as an illegal version of discrimination.

Answer B is incorrect. It does not refer solely to price adjustments based on consumer income levels.

Answer C is incorrect. Discrimination is not distinguished between exports and domestic sales.

Answer D is correct. It describes price differences based on type of customer, characteristics or the quantity sold.

Question 7

Answer A is incorrect. It does not refer to added costs to generate higher quality.

Answer B is incorrect. The term does not mean prices can be more competitive with higher-quality competitor products.

Answer C is correct. There are costs associated with poor quality, and that should be a consideration when setting quality standards.

Answer D is incorrect. It is not a reference to the cost of creating higher quality.

Question 8

Answer A is incorrect. Other two choices are also correct.

Answer B is incorrect. Other two choices are also correct.

Answer C is incorrect. Other two choices are also correct.

Answer D is correct. All the choices A, B and C above are correct. Environment costs may include both direct and indirect costs. Environment costs have been increasing due to more stringent legal requirements. There may be hidden or future costs to be incurred after sales, such as legal lawsuits or penalties claimed by government or regulators due to environmental pollution.

Question 9

Answer A is correct. It refers to pricing of products based on identified direct costs.

Answer B is incorrect. It does not mean that prices are set based on desired net profit levels.

Answer C is incorrect. It is not an adjustment to costs based on sales volume.

Answer D is incorrect. It does not identify cost reductions to offset expense increases.

Question 10

Economic factors such as inflation, deflation and interest rate affect selling price as they directly affect the cost of production and consumer perceptions related to the product price. Government policies is another factor that also affect the pricing decision, such as sales tax.

Question 11

Answer A is incorrect. It does not refer to producer price-setting.

Answer B is correct. This is a system asking customers to determine the price of products.

Answer C is incorrect. This is not an action by producers to reduce inflation.

Answer C is incorrect. It does not refer to collective purchasing actions.

Question 12

Answer A is correct. Bundle pricing is a combination of goods into a single price, often available only to members.

Answer B is incorrect. It is not a method for delivery of purchased goods.

Answer C is incorrect. It does not refer to customers buying products together.

Answer D is incorrect. It is not a reference to how producers set prices for similar products.

EXAM PRACTICE

QUESTION 1

Identify which attributes explain the 'demand factor' in pricing.

- A. Producers set prices based on demand and will raise prices as high as possible.
- B. Producers create artificial demand by limiting quantity.
- C. Prices are set by market demand, not by producers.
- D. Demand is a determining factor that dictates levels of supply.

QUESTION 2

Identify the attributes of skimming as a price practice.

- A. The producer sets prices above competitor prices to appeal to early adopters, those willing to pay more to be the first.
- B. Employees of producers take some profits without reporting them.
- C. Producers offer little or no after-sale support to customers.
- D. Product quantities are limited so that prices can be raised.

QUESTION 3

Identify the external market factors that impact price.

- A. Supply, demand, inflation and taxes.
- B. Supply, demand, competition and customer income levels.
- C. Demand, competition and government regulation.
- D. Business cycle, competition and taxes.

QUESTION 4

Define 'ability to pay' as applied to pricing policies.

- A. The producer sets prices based on income, so higher income consumers pay more.
- B. Buyers should not be expected to pay a price that creates a burden.
- C. This refers to choices made by consumers, who may not have the ability to buy two products and must choose one or the other.
- D. Producers cannot manufacture goods when raw material costs are too high.

QUESTION 5

Explain the market penetration strategy and why it is used.

QUESTION 6

Define corporate social responsibility (CSR) and its scope of activities.

QUESTION 7

Describe the attributes of the internal cost structure.

QUESTION 8

Explain how versioning works and why it is practised.

QUESTION 9

Describe the pricing strategy for market penetration and provide one example.

QUESTION 10

Hong Kah Song and Dance Club was registered under the Societies Ordinance with the Hong Kong Police in 2016 by over fifty working professionals who were interested in singing and dancing after work. Since establishment, Hong Kah has been offering a performance every night in the year. They used to rent the auditoriums managed by the Hong Kong government, and pay an average rental charge of HK\$150,000 per performance. Each auditorium can accommodate 2,000 people, even though Hong Kah has recorded that the average number of people who have purchased tickets is 1,200 per performance. The variable costs per performing hour are HK\$75,000, and each performance lasts three hours. The marketing and customer service costs per performance night are HK\$150,000 and HK\$135,000 respectively.

**Required:**

- (a) In order to accumulate a critical mass of audiences, Hong Kah has been charging an across-the-board single ticket price of HK\$400. Calculate the surplus or deficit per performance.
- (b) Calculate the break-even ticket price that Hong Kah should charge for the rest of their performances in 2019, assuming that a single ticket price is still adopted as their pricing strategy.
- (c) Calculate the break-even number of people for each performance, assuming the same cost structure remains in 2019.
- (d) Explain the changes in pricing strategies that you would suggest to Hong Kah and their possible implications.

QUESTION 11

Without parental financial support, it has been almost impossible for young university graduates in Hong Kong to purchase their own flat, even after working for 10 years. To alleviate this acute social problem to some extent, the Hong Kong government has been implementing a low-rental scheme by offering youth hostels to young people, mainly university graduates aged between 21 and 30, at below market rental prices for a maximum period of five years. A few of these hostel projects have been assigned to local NGOs by the government for implementation as from 2019.

HKQP Society is being contracted by the government to construct and manage a 20-storey youth hostel in Shatin, New Territories. There will be 800 apartments in total, of which the average size of each apartment is 250 square feet. By cooperating with a local construction company to build the hostel, the land and building costs are HK\$4,000 and HK\$3,000 per square foot. HKQP needs to pay administration, interest and general overheads which are expected to be 30% of land and construction costs. For each apartment, initial expenditure for internal furnishing and construction costs will be HK\$80,000 and for utility supplies will be HK\$40,000. Furthermore, an average marketing expense of HK\$5,000 will be incurred for each apartment. Since this is HKQP's first experience in managing this huge social project, a buffer of 10% of the total construction cost of the hostel is being expected.

The hostel will be completed and rented out in 2021 and the average monthly rental for each apartment will be HK\$4,000, which is about 35% of the market rental price of similar apartment size.

Required:

- (a) Calculate the total cost of a hostel apartment, and the number of years to recover the total cost of provision if the monthly rental of HK\$4,000 will not be adjusted indefinitely.
- (b) Compared with the market rental charges, explain whether the planned hostel apartment rate is reasonable.
- (c) Explain the rationale behind this pricing strategy.
- (d) Besides costs, list the other considerations or qualitative factors that HKQP should take into account throughout the project.

QUESTION 12

The New Territories Small House Policy in Hong Kong is a policy approved by the executive council in November 1972 and has been implemented since December 1972. The purpose of the policy created was to allow one indigenous villager to apply for permission to erect a small house on a suitable site within his own village. The small house is restricted to a maximum of three floors of 700 square feet in size for each floor.

ABC Construction Limited specialises in constructing such New Territories small houses, and in 2009 it had built such a small house in Fanling, New Territories. However, one month before handing over the finished small house to the owner, Mr. Chan, he declined to take the legal title of the house and only paid the agreed penalty to ABC Construction Limited.

Now, ABC Construction Limited is discussing with a potential New Territories indigenous villager, Mr. Lee, who has indicated his interest in purchasing the already built-up small house in Fanling, which is still owned by ABC Construction Limited as a property asset ready for sales, although Mr. Lee requested for some conversion work to be performed by ABC Construction Limited before his outright purchase. The costs of conversion work are estimated as follows:

	HK\$
Material A (200 units at HK\$400 per unit)	80,000
Material B (500 units at HK\$200 per unit)	100,000
Technical labour (100 hours at HK\$200 per hour)	20,000
Non-technical labour (150 hours at HK\$100 per hour)	15,000
Variable overheads (HK\$50 per labour hour)	12,500
Fixed overheads (HK\$20 per labour hour)	5,000
	<u>232,500</u>

The following information is also relevant:

1. The cost incurred on the original small house built in 2009 was HK\$2,000,000.
2. The small house can be sold without conversion now at HK\$3,000,000.
3. Material A is in continuous use by the company. The company has 300 units of material A in inventory which was purchased at a unit cost of HK\$400. New supplies of material A would cost HK\$450 per unit.
4. The company has 400 units of material B in inventory which was purchased at a cost of HK\$200 per unit some time ago. This material is no longer in use by the company and if not used in the conversion, it would be sold at HK\$250 per unit. New supplies of material B would cost HK\$350 per unit.
5. Technical labour are currently fully occupied. The company could get the technical labour to work overtime on the conversion by paying 50% overtime allowance, or they could divert labour hours from other jobs that currently earn a contribution of HK\$400 per labour hour.
6. Non-technical labour are paid an hourly rate and are currently working below full capacity.
7. Variable overheads for the conversion at HK\$50 per labour hour are accurately estimated.

8. Fixed overheads are absorbed at HK\$20 per labour hour according to the company's costing records.
9. ABC Construction Limited used to adopt a cost-plus pricing policy of adding 20% on full cost.

Required:

- (a) Calculate the minimum price that should be quoted to Mr. Lee for the existing small house with conversion work done.
- (b) State, by way of notes, your explanations on the relevance of items 1 to 8 above.
- (c) Explain what pricing strategy is adopted by ABC Construction Limited in this potential deal.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. The supply side of the producer has to be considered as well.

Answer B is incorrect. Limiting supply quantity may not create demand.

Answer C is correct. The prices are set by markets, not by producers.

Answer D is incorrect. Other factors such as costing have to be considered as well.

QUESTION 2

Answer A is correct. Skimming the setting prices to take advantage of early adopters.

Answer B is incorrect. This is an unethical and even illegal practice.

Answer C is incorrect. This is only a minor consideration although has become more important irrespective of whatever pricing strategy is undertaking.

Answer D is incorrect. Other financial and non-financial factors have to be considered.

QUESTION 3

Answer A is incorrect. Supply is mainly an internal operating factor.

Answer B is incorrect. Supply is mainly an internal operating factor.

Answer C is correct. External market factors are demand, competition and government regulation.

Answer D is incorrect. Business cycle involves internal operating factors.

QUESTION 4

Answer A is incorrect. This is a factor for discrimination pricing.

Answer B is correct. Recognition that buyers should not be unnecessarily burdened due to levels of price.

Answer C is incorrect. This is not a decision to be made by the buyers in relation to budget or opportunity cost.

Answer D is incorrect. The producers may have to increase selling price or cut costs.

QUESTION 5

This strategy involves setting a product's price below the price charged by competitors for the same product or similar products. The rationale is to gain sales volume as rapidly as possible and to seize market share. It may be necessary to later raise prices, especially if the initial price is below the marginal cost to produce the product.

QUESTION 6

CSR is a range of initiatives practised by companies to reduce negative social or environmental impacts of their products and services. For example, an organisation takes steps to reduce its carbon footprint, to give money to worthy causes or to encourage employees to take part in volunteering.

QUESTION 7

The internal cost structure describes factors within the organisation, versus those on the outside (external cost structure). The structure determines prices for products based on the attributes of the organisation, such as its known direct and indirect costs, general expense levels and scope of its markets.

QUESTION 8

Versioning is a system for creating different models of the same product and marketing them at different prices. This may be done based on the characteristics of a market or on the advertised purpose of a product. For example, a plastic box with many movable compartments may be sold in sporting goods stores as a fishing tackle box for HK\$14.95. The same box, packaged differently, is sold in art supply stores as a box for storing paints, and is priced at HK\$24.95. Although this is the same product, the producer has concluded that some markets will willingly pay more for it than others, based on how the product is presented.

QUESTION 9

Penetration strategies offer lower prices on goods and services as a way of attracting new customers. Many new companies use penetration strategies to establish a foothold in a market where they face competitors. It often leads to a short-term fall in income for the business.

If a company can use a penetration strategy to establish a reputation and market share for its products, it may be able eventually to start raising its prices, thus becoming profitable and better reflecting its market position.

A typical company adopting this pricing strategy is the smart phones produced by Xiao Mi Company Limited in China, who sold their smart phones at very low prices, compared with the national (e.g. Huawei) and overseas (e.g. Apple) competitors in the early 2010s. In order to capture a sizeable market Xiao Mi raised the selling prices of its smart phone after listing in the Hong Kong Stock Exchange in 2018.

QUESTION 10

(a) Surplus/deficit per performance

	HK\$	HK\$
Ticket sales (HK\$400 × 1,200)		480,000
Less expenses:		
Rental charges	150,000	
Variable costs (3 hours × HK\$75,000)	225,000	
Marketing costs	150,000	
Customer service costs	135,000	510,000
Total deficit		30,000

(b) Break-even ticket price

$$\begin{aligned}\text{Break-even price} &= \text{Total operating cost} / \text{average number of people} \\ &= 510,000 / 120,000 = \text{HK\$}425\end{aligned}$$

(c) Break-even number of people

$$\begin{aligned}\text{Break-even number of people} &= \text{Total operating cost} / \text{Unit ticket price} \\ &= 510,000 / 400 = 1,275 \text{ audiences}\end{aligned}$$

(d) Changes in pricing strategies

Having had three years of open-stage performance experiences, Hong Kah should have gained some word of mouth and sense of belonging from some audiences. It can consider increasing the unit ticket price above the break-even price of HK\$425 in order to generate reasonable surpluses for sustaining the singing and dancing performances. However, the increase should not be substantial, otherwise some low-income audiences might be lost.

Alternatively, Hong Kah can consider differentiating the ticket prices, for example, in accordance with the distance between the audience and the performers (i.e. the audience sitting in the front rows will be charged with a higher ticket price). Nevertheless, this differential pricing needs to balance amongst the numbers of tickets of various sittings.

Furthermore, Hong Kah can offer discounts (e.g. 10% to 20%) for early-bird clients who purchase the tickets before a certain timeline. Yet, it may be difficult to determine how much early-bird discount would be appropriate and effective.

QUESTION 11

(a)	(HK\$)	Area	Total (HK\$'000)
Land cost	4,000	250	1,000
Building cost	3,000	250	750
Admin, interest, overhead			525
Initial room expenditure			80
Utility supplies			40
Marketing			5
Total			2,400
Buffer			240
Total cost per apartment			2,640

If the monthly rental of HK\$4,000 will not be adjusted indefinitely, it takes 55 years to recover the total cost in providing one such hostel apartment (i.e. HK\$2,640,000 / HK\$4,000/12).

- (b) Paying a monthly rental of HK\$4,000 for such a hostel apartment, which is about 35% of the prevailing market rental price, would be very attractive to the young university graduates, who would like to start-up their independent life and be away from parents,

who may not be wealthy enough to subsidise their children to purchase their own flat from the private property market.

- (c) HKQP is adopting a social pricing strategy of which the primary objective is helping the young people, especially the young university graduates to start out on their own and live apart from their parents, which is also the aim of the Hong Kong government as a social policy. The system should be a success, although the payback period will be as long as 55 years.
- (d) As explained in (c) above, social consideration is of paramount importance to be duly considered by all stakeholders, including HKQP. Although charging at a far-below market rental charge, the quality of hardware and services provided by the hostel apartment should be good enough for the young tenants to experience their independent life. In addition, HKQP in collaboration with the Hong Kong government and a local constructor should tightly monitor and control the building costs of the hostel in order to spend wisely the subsidies invested by the government. Furthermore, HKQP should constantly review, and adjust, if necessary, the monthly rental charges in order to combat inflation and provide quality services along the years in operation.

QUESTION 12

- (a) Minimum price to be quoted to the potential customer with conversion work:

	Note	HK\$
Disposal value of the small house	2	3,000,000
Relevant costs of conversion:		
Material A ($200 \times \text{HK\$}450$)	3	90,000
Material B ($100 \times \text{HK\$}350 + 400 \times \text{HK\$}250$)	4	135,000
Technical labour ($100 \times (\text{HK\$}200 + \text{HK\$}100)$)	5	30,000
Non-technical labour ($150 \times \text{HK\$}100$)	6	15,000
Variable overheads ($250 \times \text{HK\$}50$)	7	12,500
		<u>3,282,500</u>

- (b) Notes:

- (i) Cost of production for the set of kitchen equipment is a sunk cost and is thus irrelevant.
- (ii) Disposal value of HK\$3,000,000 for the small house is the opportunity cost and should be included.
- (iii) Since material A will be used regularly by the company in other production, the company needs to buy the required material A if it proceeds with the conversion. The historical cost is irrelevant and the relevant cost is the replacement cost of HK\$450 per unit and of HK\$90,000 in total ($\text{HK\$}450 \times 200$).
- (iv) Material B is no longer used in production. If it is used for the conversion, it could not be sold. The historical cost is also irrelevant. The relevant costs are the net realisable value of the 400 units in inventory and the replacement cost for purchasing 100 units: $250 \times 400 + 350 \times 100 = \text{HK\$}13,500$.



- (v) The company will rationally take the lower of the:
 - basic pay plus overtime allowance: $100 \times \text{HK}\$(200 + 100) = \text{HK}\$30,000$, and
 - basic pay plus contribution forgone for other jobs: $100 \times \text{HK}\$(200 + 400) = \text{HK}\$60,000$.Thus the relevant cost of skilled labour is HK\$2,400.

- (vi) Since the unskilled labour is currently working below full capacity, the hourly rate will be paid for the conversion cost.

- (vii) Variable overheads are incremental costs and are thus relevant.

- (viii) Fixed overheads are absorbed and do not change as a result of the conversion and are thus unavoidable fixed costs. It is thus irrelevant for the conversion.

Therefore, the minimum selling price to be quoted by ABC Construction Limited on the existing small house after conversion work should be $(3,282,500 \times 110\%)$ or HK\$3,610,750.

- (c) The pricing strategy adopted by ABC Construction Limited in this potential deal is 'relevant cost-plus method', of which the total relevant cost is ascertained first (i.e. HK\$3,282,500) before adding up a 10% profit mark-up used to be practised by the company, in order to arrive at the selling price to be quoted (i.e. HK\$3,610,750).

13

Investment Project Appraisal and Evaluation

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LEARNING OUTCOMES

PRINCIPAL LO4: ANALYSE THE STRATEGIC MANAGEMENT ACCOUNTING FRAMEWORK

LO4.02: Analyse proposed investment projects

- 4.02.01** Calculate relevant cash flows and discount rates for investment appraisal purposes, taking into account inflation and tax
- 4.02.02** Apply various models to establish the value of a business
- 4.02.03** Analyse a proposed investment project by applying appropriate investment appraisal techniques given the objectives and circumstances of an organisation
- 4.02.04** Analyse an appropriate investment strategy when capital is rationed
- 4.02.05** Explain other non-financial considerations for investment appraisal
- 4.02.06** Identify social responsibility, sustainability and environmental consequences of investment decisions and explain the impact on investment appraisal

PRINCIPAL LO8: APPLY APPRAISAL TECHNIQUES FOR CAPITAL INVESTMENT PROJECTS

LO8.01: Apply appropriate appraisal techniques to capital investment projects

- 8.01.01** Apply different appraisal techniques to evaluate capital investment projects including payback period, discounted payback, accounting rate of return, net present value and internal rate of return
- 8.01.02** Describe the impact of taxation, inflation, risk and uncertainty in capital investment appraisal
- 8.01.03** Explain and compare the strengths and weaknesses of different project appraisal techniques
- 8.01.04** Apply the concept of incremental cash flows in capital investment appraisal
- 8.01.05** Calculate relevant discount rates for each capital investment projects, taking into account inflation and tax



OPENING CASE

ABC INVESTMENT LTD

As the CFO, your work at ABC Investment Ltd exposes you to a variety of ways to value companies to be invested in, merged with or acquired. You must apply dividends, earnings, enterprise value and free cash flow measures as you assess corporate valuation. None of the valuation techniques is perfect by itself, so it is important to understand the strengths and weaknesses of each one.

You are also learning capital budgeting principles and grappling with incremental cash flow calculations. Not every cost item is relevant to investment decisions. You must be discerning in assessing incremental cash flow impacts. Tax is another major cost consideration in investment.

You are using several techniques in investment appraisal work. Again, no single appraisal technique is perfect. You must use several of them in combination to reach the best conclusion. You are well versed in the pros and cons of each technique. Capital rationing is a reality because ABC Investment Ltd does not have unlimited funds. You must recommend to your boss how to ration the funds for the most worthwhile investments.

Your job is becoming more challenging. But you are also having a lot more fun grappling with all these interesting issues and considerations. 



OVERVIEW

Companies undertake investments to expand and grow their market share. The process of choosing how and where to invest can be complex and make the difference between success and failure.

This chapter explores the approaches that companies can use to evaluate business projects and establish standards for evaluation. This process determines relevant and incremental cash flows in specific businesses. These approaches, including investment strategies, must consider the impact of inflation and taxes; social responsibility; and non-financial considerations.

The chapter also explores the various capital investment appraisal techniques and considers how to build an investment strategy when capital is limited. Beyond these considerations, the chapter also explores and compares the various investment appraisal methods including the payback period method and the discounted payback period approach, the accounting rate of return approach, the two related approaches of the internal rate of return and the net present value method as well as the profitability index.

This chapter discusses other considerations that companies undertake when making investment decisions, including the impact of taxes, inflation, risk, corporate social responsibility, sustainability, the environment, organisation strategy, legal issues, cultural impact and operational challenges. These issues, as forms of risk, anticipation of consequences and other challenges, will affect profits but more significantly, will also affect cash flows for the long term.



13.1 BUSINESS VALUATION MODELS

Business valuation models are used to determine the economic value of a business and arrive at a **fair value** for the business. Determining a fair value for the business is the first step in deciding whether an investment in a company is worthwhile.

In corporate finance and management accounting, business valuation is a frequent theme. It comes up whenever a company is considering investment in a new venture or in the shares or debt of another company. Determining the value of a company is also a key first step in determining the value of the shares of a company – with the addition of other factors such as dividends, cash flow and future growth projections.

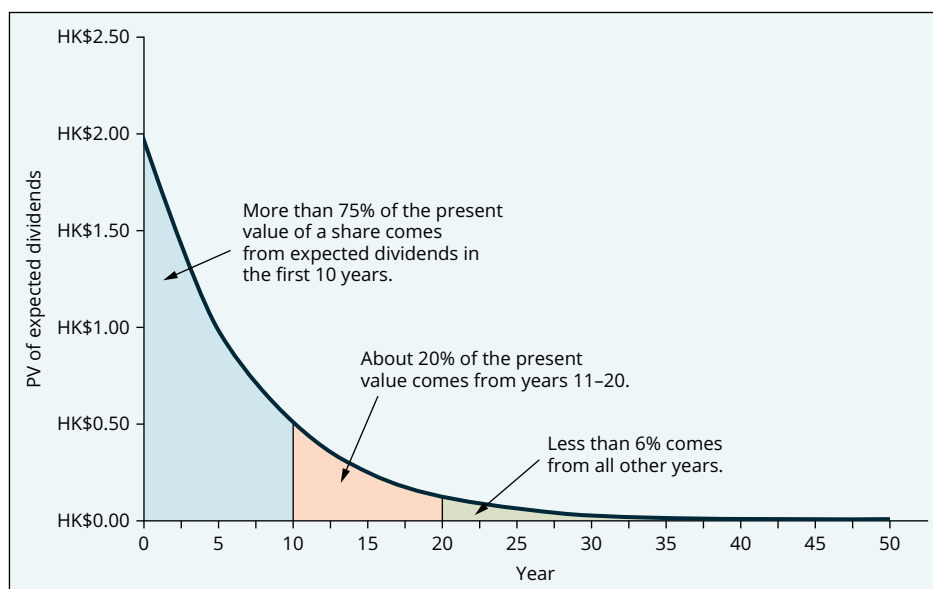
Various tools that analysts can use to determine the value of a business include reviewing financial statements and using models that are based on the information available.

For most day-to-day work of management accountants, business valuation models are used to determine share values and make decisions as to whether an investment is worthwhile. The most commonly used models used to determine share values are the **dividend growth model**, the **free cash flow model**, the **price-to-earnings ratio** (P/E ratio) and the enterprise value to EBITA (earnings before interest, taxes and amortisation) ratio.

13.1.1 Dividend Growth Model

As discussed in chapter 5, the dividend growth model is most commonly used to determine the value of the shares of a company by considering the growth of dividends over time. In a sense, this approach is like the approach used to value a perpetual annuity, a bond that pays a coupon rate in perpetuity, but with some wrinkles. An important difference is that cash dividends grow from one period to the next rather than staying constant.

The assumption of an infinite time horizon is not a problem, even though it is true that most companies do not continue dividend growth forever. The further in the future a cash flow occurs, the smaller its present value, as shown in Exhibit 13.1. Therefore, distant dividends have a small present value and contribute very little to the share price.



► **EXHIBIT 13.1 Impact on share price of dividends** (Source: Kidwell 2018. Reproduced with permission of John Wiley & Sons.)

For example, assume a company pays constant dividends and has a 15% discount rate. In this case, dividends paid over the first 10 years will make up more than three-quarters of the value of a share, whereas those paid after the 20th year will account for less than 6%.

Under the dividend growth model, the value of a share is calculated using the following formula:

$$P_0 = \frac{D_1}{R - g}$$

Where:

- P_0 = current value, or price per share,
- D_1 = dividend paid in the next period ($t = 1$),
- g = constant growth rate for dividends,
- R = required return on ordinary shares or discount rate.

The constant growth dividend model thus tells us that the current price of a share is the next period dividend divided by the difference between the discount rate and the dividend growth rate.

The dividend growth model holds true only when the growth rate is less than the discount rate or required rate of return. Using this equation when R is equal to or less than g ($R \leq g$) will only produce a meaningless figure.

Also note that if $g = 0$ there is no dividend growth. The equation becomes $P_0 = D / R$ and the dividend payment pattern offers a no-growth dividend stream. However, there are limitations to this model. The assumption of stability in the growth rate is unrealistic at some time hence a weakness of the model. Owing to the changes in the earnings of the company the assumption of stability is violated.

Illustrative Example 1

The current cash dividend on the ordinary shares of Ming Wong Investments Ltd's ordinary shares is HK\$0.27. Financial analysts expect the dividends to grow at a constant rate of 6% per year and investors require a 12% return on this class of shares. Calculate the current share price of Ming Wong Investments Ltd.

In this scenario, $D_0 = \text{HK\$}0.27$, $R = 0.12$ and $g = 0.06$. We first find D_1 and then calculate the value of a share.

$$\text{Dividend: } D_1 = D_0 \times (1 + g) = 0.27 \times 1.06 = 0.2862$$

$$\text{Value of a share: } P_0 = D_1 / (R - g) = 0.2862 / (0.12 - 0.06) = 0.2862 / 0.06 = \text{HK\$}4.77$$

The current share price for Ming Wong Investments Ltd should be HK\$4.77.

13.1.2 Free Cash Flow Model

The **free cash flow (FCF) model**, as the name suggests, considers the cash flow that a company will generate to determine the value of shares. This approach is also useful in capital budgeting to determine the viability of a specific project based on the cash flows that the project will generate over time.

As explained in Chapter 10, the cash flows discounted in an NPV analysis are incremental after-tax free cash flows (FCF). Thus, **incremental after-tax free cash flows (FCF)** are the total after-tax FCF the company would produce. The term **free cash flows** is commonly used in practice, due to it being the cash flows that remain after a company has made all its necessary

investments in working capital and non-current assets and so has the freedom to distribute to debtholders and shareholders.

The amount of a company's FCF can be seen in two ways. First, it is equal to any change in the company's cash income, excluding interest expenses, plus any depreciation and amortisation expenses, minus all capital expenditures and investments in working capital. And second, it is also equal to the incremental after-tax cash flows from operations, minus capital expenditures and investments in working capital.

Capital expenditure and working capital are respectively investments in fixed assets, such as buildings and equipment, and working capital items, such as accounts receivable or payable and inventory.

The formula for the FCF is relatively simple (even if the calculation can be complex depending on the size of the company). The complications can be seen below in Exhibit 13.2. FCF are calculated as:

$$FCF = [(Revenue - Op\ Ex - D\ \&\ A) \times (1 - t)] + D\ \&\ A - Cap\ Exp - Add\ WC$$

Explanation	Calculation	Formula
The change in the company's cash income, excluding interest expense, resulting from the project.	Revenue	Revenue
	– Cash operating expenses	– Op Ex
	Earnings before interest, tax, depreciation and amortisation	EBITDA
	– Depreciation and amortisation	– D&A
	Earnings before interest and tax	EBIT
	× (1 – Company's marginal tax rate)	× (1 – t_c)
	Net operating profit after tax	NOPAT
	+ Depreciation and amortisation	+ D&A
Adjustments for the impact of depreciation and amortisation and investments on FCF.	Cash flow from operations	CF Opns
	– Capital expenditures	– Cap Ex
	– Additions to working capital	– Add WC
	Free cash flow	FCF

► **EXHIBIT 13.2 FCF calculation** (Source: Kidwell 2018. Reproduced with permission of John Wiley & Sons.)

13.1.3 Price-to-Earnings Ratio

Relative valuation can be used to estimate the value of a company's common shares in comparison to those of other similar companies. It uses one or more of several shared variables, such as income, sales, cash flow or book value. As multiples are used a lot in relative valuation, it is also referred to the **method of multiples**.



Relative valuation is simple to apply, as the relevant financial and market data is readily available for most companies. What can make it hard, however, is finding a group of comparable companies.

Even after finding a group of comparable companies, it is still necessary to estimate the appropriate multiple because values differ even for comparable companies, so the exercise involves substantial analysis and judgement.

The most commonly used relative valuation multiple is the **price-to-earnings (P/E) ratio**.

The P/E ratio represents the number of times investors are willing to pay for a company's earnings, as expressed in the share price, or the share price divided by the earnings per share. The resulting **multiple** represents the numbers of years of earnings in current price.

The P/E approach calls for estimating a company's earnings per share (EPS) and then multiplying this figure by a credible P/E multiple. Such a multiple is one that appears sustainable for a certain foreseeable period. The typical P/E formulation used for valuation purposes takes an estimated EPS figure for the next 12 months. This basic valuation is expressed as:

$$P_0 = \text{Estimated EPS}_1 \times \text{Justified P/E}$$

The P/E ratio used in this equation is based on expected future earnings (Estimated EPS_1) and is therefore a leading P/E ratio, also called the forward P/E.

By comparison, analysing financial ratios, the reported P/E ratios are often based on earnings over the previous 12 months (EPS_0), resulting in lagging P/E ratios. Because the two sides of this equation (price and earnings) are based on different periods (current price versus latest reported earnings), a single P/E value is not entirely accurate. In applying the lagging P/E, it is more accurate to compare the P/E range from high to low for several past fiscal years.

For valuation purposes, it is more common to focus on the leading P/E ratio because market values are based on expectations about the future. However, because this requires estimation, leading P/E is subjective. This does not reduce its value; however, being aware of its limitations as a cautionary point of view makes sense.

Using the P/E ratio is relatively easy. If a company's forecast earnings per share are HK\$2 and the P/E ratio for comparable companies is 20, that leads to a share value of HK\$40 using the P/E approach.

In this case, the multiple of earnings is 20 times. In this sense the multiple is an example of a payback period. The higher the multiple, the longer the payback period and the more the investor expects earnings to increase. Alternatively, it could be that the shares are simply expensive. P/E analysis and the resulting multiple are often used to identify whether shares are bargain-priced or overpriced.

Relative valuation values the company relative to comparable companies. This means that if the comparable companies are all overvalued, using the P/E approach will overvalue the company in question. As a result, if the market is in a 'speculative bubble' the P/E approach will not detect it directly.

The P/E ratio is an attractive (and commonly reported) statistic for investors and analysts for several reasons. First, it relates the price to the earnings owned by shareholders. Second, it is easy to compute and as a result is commonly available. This makes comparisons relatively straightforward. Third, it is intuitive, as it indicates the payback period and can be related to several other company characteristics, such as growth opportunities and risk.

Unfortunately, obtaining reasonable estimates requires a substantial amount of analysis and judgement.

13.1.4 Enterprise Value to EBITA

It is possible to form multiples using earnings before interest and taxes (**EBIT**), or earnings before interest, taxes and amortisation (**EBITA**), or earnings before interest, taxes, depreciation and amortisation (**EBITDA**) instead of net income to eliminate a significant proportion of volatility caused in EPS figures using debt, and by depreciation and amortisation (which are non-cash expenses).

EBIT, EBITA and EBITDA are earnings to all suppliers of capital (that is, bondholders and shareholders), so it is a common practice to measure these earnings against the value of an entity's capital.

The market value of an entity's capital, less cash and cash equivalents, is known as that entity's **enterprise value**. As this value is also the value that anyone wanting to buy the company would have to pay, it is also often known as the takeover value.

Enterprise value can be compared to the various earning measures in the following ways: enterprise value to EBIT, enterprise value to EBITA and the enterprise value to EBITDA. Enterprise value is used in the numerator to reflect the fact that EBIT, EBITA and EBITDA represent income available to both debt and equity holders. EBITA may be used in place of EBITDA for companies heavily invested in capital assets. In such instances, EBITDA may distort the outcome due to exceptionally high annual depreciation.

Key Learning Point

The most commonly used business valuation models are the dividend growth model, the free cash flows models, the price-to-earnings (P/E) ratio and the enterprise value to EBITA ratio.

► Knowledge Check Questions

Question 1

Identify which of the following is not a workable assumption in the constant growth dividend discount model.

- A Dividends do not grow.
- B Dividends are expected to decline at a constant rate.
- C The required rate of return is less than the growth rate.
- D The required rate of return is more than the growth rate.

► **Knowledge Check Questions (*continued*)**

Question 2

Using the method of multiples to value the equity of a private company, you estimate that the private company and its peers all have a price-earnings ratio of around 15. If the private company's earnings are HK\$3 million, which of the following values is its equity is closest to.

- A HK\$0.2 million
- B HK\$5 million
- C HK\$15 million
- D HK\$45 million

13.2 RELEVANT/INCREMENTAL CASH FLOW CALCULATION IN MAKING INVESTMENT DECISIONS

Management's decision-making process does not always follow the same pattern, because decisions vary significantly in their scope, urgency and importance. It is possible, though, to identify some steps that management uses in the process. Management accounting's contribution, for example, tends to occur mostly in the process of evaluating choices and reviewing results. This includes providing relevant revenue and cost data, the expected overall effect on net income, and providing reports on the impact of previously taken decisions.

When making business decisions, company managers take into account both financial and non-financial factors. Financial factors include data about revenues, costs and profitability. Non-financial factors include changes in staff morale and the company's image; such non-financial factors are important. This section considers only the financial factors that shape business decision-making.

In business, the process of identifying a change in financial expenses and its effect on future earnings arising from deciding to pursue an alternative course of action is called **incremental analysis**.

The data for incremental analysis can come from a range of sources, including market analysts, designers and engineers, as well as from a company's financial department. Of course, the more reliable the data, the better the likely findings. But incremental analysis draws on estimates of future earnings, its findings necessarily contain an element of uncertainty.

Exhibit 13.3 shows an example of a basic approach in incremental analysis.

	Alternative A	Alternative B	Net income increase (decrease)
Revenues	HK\$125,000	HK\$110,000	HK\$(15,000)
Costs	100,000	80,000	20,000
Net income	HK\$ 25,000	HK\$ 30,000	HK\$ 5,000

► **EXHIBIT 13.3 Incremental analysis** (Source: Weygandt 2015. Reproduced with permission of John Wiley & Sons.)

This example compares the change in net income between Alternative A and Alternative B. In this case, incremental revenue will be HK\$15,000 less under alternative B than under alternative A. However, the company will realise a HK\$20,000 incremental cost savings under alternative B. Thus, alternative B will produce HK\$5,000 more net income than alternative A.

Incremental analysis can be a useful tool for uncovering non-intuitive findings. Sometimes, what is normally regarded as a variable cost – staff salaries, for example – may turn out not to be an incremental cost when considering whether to buy some new equipment if that equipment uses the same volume of staff time. Likewise, rent – usually treated as a fixed cost – may become an incremental cost if a company is considering whether to continue leasing premises or buy them instead.

Several types of decisions involve incremental analysis. The more common ones are to decide whether to do the following (see Module 2, Chapter 8 for more information):

- Make or buy components or other inputs.
- Sell products as they are or add value by processing them further.
- Take an order for goods at a non-standard price.
- Retain or replace machinery or equipment.
- Retain or close down an unprofitable business unit.
- Decide how to allocate limited resources.

Each of these may, at some point, be part of a capital budgeting process for a company.

13.2.1 Capital Budgeting Principles

Many companies follow carefully set processes in capital budgeting. Proposals for projects are requested, a capital budgeting committee examines the proposals and submits its findings. Management choose the projects that they believe are worthy of funding and submit this list of projects to the board of directors. The directors make the final decision to approve or reject the capital expenditure budget.

Capital budgeting decisions have a significant impact on a company's future profitability and cash flows. Making a poor capital budgeting decision can cost the company a lot of money.

Ultimately, the value of all financial investments is determined by the status of cash flows. Cash flow information may not always be available. In this situation, to estimate cash flows, adjustments can be made to accrual accounting numbers. Often, net annual cash flows are calculated by adding depreciation expenses back into net income. This is done because such expenses involve no cash outflow.



Cash flows are the inputs that are considered relevant in capital budgeting decisions. Such decisions typically involve a large number of considerations, including the availability of funds, the company's tolerance for risk, the nature of its various activities and the relations between them, and its overall approach to decision-making.

13.2.1.1 Sunk Costs

Sunk costs are costs that have already been incurred and will not be changed or avoided by any future decision. For example, if shortly after a company buys a piece of equipment, a new, more productive version of that equipment is released, then the book value of the original machine, as a sunk cost, should have no effect on any decision to buy the new machine.

In sum, sunk costs are not relevant costs. And all that matters to evaluate a project is how much must be invested in the future and what can be expected to be received in return for that investment. Past investments are irrelevant.

For example, Company X has invested HK\$10 million in a project that had been expected to generate cash inflows with a present value of HK\$20 million. Before starting operation, however, a change in the business environment means that the project will now only generate cash inflows with a value of HK\$2 million – and even to earn this HK\$2 million will call for the investment of another HK\$1 million. Should Company X invest this money?

Of course. If the company is to stop investing in the project, it will lose its HK\$10 million investment. If the company makes the investment, the total loss will be HK\$9 million. Although neither is an attractive alternative, it is obviously better to lose HK\$9 million than to lose HK\$10 million. The conclusion is the same if the previous investment is ignored and the decisions is between never receiving anything and receiving an NPV of HK\$1 million (HK\$1 million investment and HK\$2 million return).

The point is that, while it is often painful to do, sunk costs should be ignored when calculating FCF.

13.2.1.2 Opportunity Costs

Opportunity cost refers to the fact that when they take one action, companies often find themselves having to forgo the benefit from an alternative opportunity. For instance, using a machine to make chairs might mean not being able to use it to make tables – which could be a lost opportunity if the market was found to be flooded with chairs but short of tables. Such a lost benefit is called an opportunity cost.

The concept of opportunity cost here is like the opportunity cost of capital. Capital used in one investment cannot be used for a different investment.

Opportunity costs occur in many forms. A business might have some spare space – leasing it out could bring in some revenues, but at the expense of expanding its own production facilities. To the extent that an action was ruled out by another represents the opportunity cost of that action.

To make this clearer, a company is considering a project that will need some equipment that it already has but which could be sold for HK\$500,000 on the used-equipment market. If the project is accepted, the company will lose the opportunity to sell the piece of equipment for HK\$500,000. This HK\$500,000 cost must be included in the project analysis because accepting the project will reduce the amount of money that the company can earn from selling excess equipment by this amount.

13.2.1.3 Financing Costs Are Ignored

The cash flows associated with financing a project (cash outflows or inflows to or from debtholders or shareholders) are not included in the calculation of FCFs because they are accounted for in the discount rate used in an NPV analysis.

The incremental cash flows from operations equals the incremental net operating profits after tax (NOPAT) plus D&A. **NOPAT** is a cash measure of the incremental profit from the project that excludes interest expenses. Interest expenses are omitted because the cost of financing a project is reflected in the discount rate. Consequently, NOPAT shows the impact of the project on the company's cash profit apart from the effects of any interest expenses from its financing.

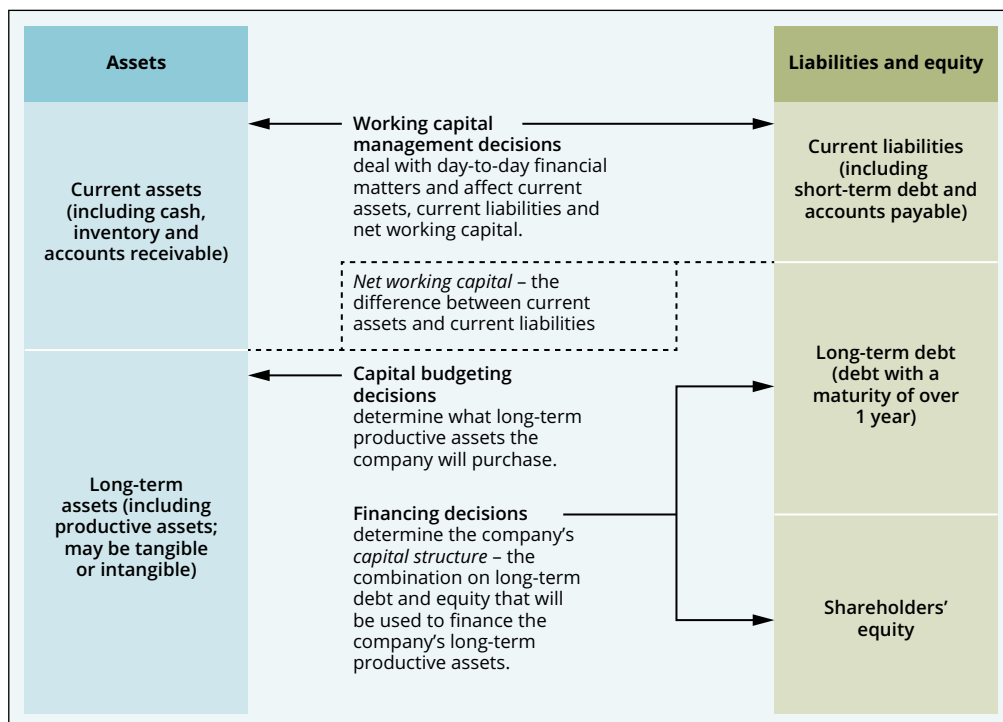
NOPAT is calculated using a company's marginal tax rate. This is because any profits arising from a new project are assumed to be incremental for the company. Assuming the company already pays taxes, that means that the appropriate tax rate for FCF calculations is the tax rate that the company would pay on any additional profits earned from the new project, which is the marginal tax rate.

D&A is subtracted before calculating the taxes the company would pay on the incremental earnings for a project. This accounts for the ability of a company to deduct D&A when calculating tax. However, since D&A is a non-cash charge, it must be added back to NOPAT to derive the cash flows from operations accurately.

13.2.1.4 Net Working Capital

Management must also decide how to manage the company's current assets, such as cash, inventory and accounts receivable, and its current liabilities, such as trade credit and accounts payable when making investment decisions.

The dollar difference between current assets and current liabilities is called **net working capital**, as shown in Exhibit 13.4.



► **EXHIBIT 13.4 Net working capital** (Source: Kidwell 2018. Reproduced with permission of John Wiley & Sons.)



Working capital management refers to the day-to-day management of the company's current assets and liabilities. The goals of managing working capital are making sure enough money is available to pay bills and earning interest on any spare cash by investing it as profitably as possible.

Even though the company may be profitable, mismanagement of working capital can cause it to default on its debt. For example, if a company is not diligent in collecting the account receivable on sales that are made to customers on credit, it can find itself without the ability to pay its bills.

Trade creditors can then force the company into insolvency if it ends up not being able to obtain alternative financing.

Profitability can be affected at the inventory level as well. If the company has more inventory than it needs to meet customer demands, too much money is tied up in non-earning assets. If the company holds too little inventory, it can lose sales because it does not have products to sell on hand when customers want them. The optimal inventory level is something the company has to work out.

Similar considerations involving net working capital can be extended to investment decisions. Investment decisions are also affected by considerations of net working capital. The incremental additions to working capital from an investment are calculated by adding the change in cash and cash equivalents to the change in accounts receivable and to the change in inventories minus changes in accounts payable.

For any project, capital has to be invested in cash and cash equivalents, accounts receivable and inventory. Suppliers also provide capital through accounts payable. When a project ends, inventories are sold, accounts receivable are collected and accounts payable are paid. Cash and cash equivalents are thus no longer needed, and so companies recover the net working capital invested in that project.

To reflect this, the cash flows in the last year of a project typically includes a negative investment in working capital that equals the cumulative investment in working capital over the life of the project. Management has to make sure the recovery of working capital is reflected in the cash flows in the last year of a project. Working capital can account for 20% or more of revenue in some businesses and excluding recovery of working capital from the calculations can lead to a substantial underestimation of the NPV of a project.

13.2.1.5 Externalities

Externalities are the side effects or consequences of an economic activity, such as an investment, that affects other parties in positive or negative ways, without being reflected in the cost of the goods or services involved.

Externalities impact investment decisions in myriad ways and can add to the cost of investment, often by their relationship to government regulations. For example, an investment in a new factory may create pollution. That pollution is an externality. Government regulations may require companies pay carbon tax to make up for the additional pollution. The cost of that carbon tax must be considered in investment decisions.

Externalities are impacts that are not directly related to an investment. While there are both positive and negative externalities, most of them are negative. At times, investments may result in externalities but the cost of dealing with those externalities may be lower than the returns generated by the investment of project.

There are some positive externalities. Research and development activities, for example, may be considered a positive externality if it increases both the profits of a company and the sum of knowledge in society.

13.2.2 Discount Rate for Capital Budgeting

A question related to the application of **net present value (NPV)** is to consider how a **discount rate** (or the required rate of return) is chosen in real capital budgeting decisions.

The discount rate, as has been discussed previously, is the interest rate that can be used to determine the actual value of future cash flows. The use of the discount rate is standard in all analysis of cash flows. This standard discounted cash flow analysis is key for companies to gauge how much finance they will need in future to meet their obligations and carry on operations.

In most instances, companies can use a discount rate that is equal to its cost of capital. The cost of capital is the rate it pays for funding from creditors and shareholders. It is calculated as a weighted average of the rates paid for loans or other borrowed funds and for funds from investors in the company's common and preferred shares. This figure is the weighted average cost of capital (WACC) which has also been discussed and explained. In other words, companies use the WACC as the discount rate when budgeting future operations or specific projects.

Another approach that investors and companies sometimes use to determine a discount rate is to set a required return. Because capital comes at a cost, companies have to include that cost in their budgeting. This cost is the required rate of return of lenders and investors and also represents how much companies would have to pay to access capital from either source.

The discount rate can also be used to assess the viability of a specific project. If a project is believed to be of higher risk than the company's usual line of business, the discount rate should be increased. That is, the discount rate has two elements: a cost of capital element and a risk element. Using an incorrect discount rate can lead to incorrect capital budgeting decisions.

The discount rate is often given other names, including the hurdle rate, the required rate of return and the cut-off rate. The determination of the cost of capital varies somewhat, depending on whether the entity is a for-profit or not-for-profit enterprise.

Illustrative Example 2

ABC Investment Ltd's subsidiary Golden Company Ltd is a maker of golf balls. It sells these balls to retailers at HK\$7 each.

Its costs for making 20,000 balls are:

Materials	HK\$10,000
Labour	30,000
Variable overhead	20,000
Fixed overhead	40,000
Total	<u>HK\$100,000</u>

Golden also incurs 5% sales commission (HK\$0.35) on each ball sold.

Illustrative Example 2 (continued)

Ball Corporation offers Golden HK\$4.80 per ball for 5,000 balls that it will sell under its own brand name in markets where Golden has no presence. If Golden accepts the offer, its fixed overhead will increase from HK\$40,000 to HK\$46,000 as it needs to purchase a new imprinting machine. No sales commission will result from the special order.

Your CIO requests the following:

1. Calculate the incremental analysis for the special order.
2. Explain whether Golden should accept the special order.
3. Describe the assumptions that underlie the decision made in part (2).

Answer

1. Variable cost per unit = $(10,000 + 30,000 + 20,000) \div 20,000$ units = HK\$3

$$\text{Contribution margin per unit} = 4.80 - 3 = \text{HK\$}1.80$$

Incremental contribution margin (5,000 units $\times$ HK\$1.80)	HK\$9,000
Less incremental cost:	
Fixed cost	6,000
Incremental income (loss)	<u>HK\$3,000</u>

2. The company should accept the special order, as it will increase net income by HK\$3,000.
3. The assumptions underlying the decision is that current sales will not be affected if Golden accepts the offer, and they have the capacity to produce the 5,000 additional balls.

Key Learning Point

Decisions require choosing one of two or more possible courses of action. The process of identifying the financial expenses associated with these alternative possible courses of action is incremental analysis.

► Knowledge Check Questions**Question 3**

Identify which of the following is correct when you undertake incremental analysis of the financial data arising from alternative courses of action.

- A** Financial data does not change.
- B** Financial data changes.
- C** Financial data is mixed.
- D** Financial data would change more than 20%.

► **Knowledge Check Questions (continued)**

Question 4

Identify what the book value of the old equipment is in a decision to retain or replace equipment.

- A Opportunity cost
- B Sunk cost
- C Incremental cost
- D Marginal cost

Question 5

Identify what happens if an unprofitable segment is eliminated.

- A Net income increases.
- B Other segments absorb the variable expenses of the eliminated segment.
- C Other segments absorb the fixed expenses allocated to the eliminated segment.
- D Net income decreases.

13.3 CASH FLOW ANALYSIS OF CAPITAL PROJECT INVESTMENT WITH IMPACT OF INFLATION, TAX AND DEPRECIATION

Several issues arise when undertaking an analysis of the cash flows of a company or a project.

The first of these issues is determining which cash flows are incremental to a project and which are not, and there are various rules to do this. Just as importantly, however, are the effects of inflation, taxation and depreciation on cash flows.

Because capital expenditures typically involve the estimation of cash flows several years in the future, **inflation** can play an important role in determining these estimates because it will affect future levels of sales and expenses. In addition, inflation alters the level of interest rates and therefore affects the company's cost of capital that is to be used in discounting these future cash flows.

It is important to treat inflation consistently. One way to do this is to estimate future cash flows on a nominal (or actual) basis, that is, estimating the actual cash flow that will result in year two, and so on, without adjusting for inflation, and then discount the cash flows using nominal (or actual) discount rate.

Another approach is to estimate future real or inflation-adjusted cash flows then discount them using real discount rates. A problem with this approach is that estimates of depreciation tax savings represent the actual amount of depreciation charged. Usually, the simplest way to get around this problem is to estimate the nominal cash flows then discount them using nominal discount rates.

Companies must always bear in mind the effect that inflation will have on sales and expenses. While it can often be assumed that this effect will be the same for both sales and expenses, sometime this will not be the case.

For example, a company is considering a three-year project with the following conditions:

- Unit sales of 80,000 a year for the entire three years, with a real sales price forecast at HK\$21 per unit, but with forecast price inflation of 4% a year.
- Real production and sales expenses forecast at HK\$15 per unit, but with expense inflation forecast of 4% a year.
- An initial cost of assets of HK\$1 million with no salvage value expected.
- Depreciation for Years 1, 2 and 3 are HK\$333,300, HK\$444,500 and HK\$148,100 respectively. There will be a tax benefit from loss on disposal of HK\$74,100.
- A marginal tax rate is 35% for the company.
- A nominal cost of capital for the project of 8%.

The net present value of this project is HK\$58,684, as detailed in Exhibit 13.5.

	Year 0 (HK\$)	Year 1 (HK\$)	Year 2 (HK\$)	Year 3 (HK\$)
Unit sales		80,000	80,000	80,000
Sale price per unit		21.00	21.84	22.71
Operating expense per unit		15.00	15.60	16.22
Investment cash flow	-1,000,000			
Salvage value				0
Tax benefit from loss on disposal				-74,100
Terminal cash flow		0	0	-74,100
Change in revenues		1,680,000	1,747,200	1,817,088
<i>Less</i> Change in expenses		1,200,000	1,248,000	1,297,920
<i>Less</i> Change in depreciation		333,300	444,500	148,100
<i>Equals</i> Change in taxable income		146,700	54,700	371,068
<i>Less</i> Change in taxes		51,345	19,145	129,874
<i>Equals</i> Change in income after tax		95,355	35,555	241,194
<i>Plus</i> Change in depreciation		333,300	444,500	148,100
Operating cash flow		428,655	480,055	389,294
Net cash flow	-1,000,000	428,655	480,055	315,194
Net present value	58,584			

► **EXHIBIT 13.5** Changes in cash flow and NPV at 4% inflation (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

If the inflation expectations change, so that the rate of inflation of revenues differs from that of expenses, the outcome changes. For example, if the rate of inflation on units sold results in an increase of the sale price by 6% per year, and if the rate of inflation on operating expenses per unit is 8% each year, the net present value changes to HK\$34,560, as shown in Exhibit 13.6.

	Year 0 (HK\$)	Year 1 (HK\$)	Year 2 (HK\$)	Year 3 (HK\$)
Unit sales		80,000	80,000	80,000
Sales price per unit		21.00	22.26	23.60
Operating expense per unit		15.00	16.20	17.50
Investment cash flow	-1,000,000			
Salvage value				0
Tax benefit from loss on disposal				-74,100
Terminal cash flow				-74,100
Change in revenues		1,680,000	1,780,000	1,887,648
Less Change in expenses		1,200,000	\$1,296,000	1,399,680
Less Change in depreciation		333,300	444,500	148,100
Equals Change in taxable income		146,700	40,300	339,868
Less Change in taxes		51,345	14,105	118,954
Equals Change in income after tax		95,355	26,195	220,914
Plus Change in depreciation		333,300	444,500	148,100
Operating cash flow		428,655	470,695	369,014
Net cash flow	-1,000,000	428,655	470,695	294,914
Net present value	34,560			

► **EXHIBIT 13.6 Cash flows and NPV at different inflation rates** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

Under both analyses, depreciation is the same, but the operating cash flows are different due to the different inflation rates used to calculate revenues and expenses.

By changing the rates of inflation used, a project can change from viable to non-viable or vice versa. In the example above, if the inflation rate for expenses were raised from 5% to 10%, the net present value of the project would fall to HK\$5,811, making it unviable.

Such disparate inflation rates are a definite possibility. In the United States, for example, the producer price index (PPI) tracked an increase in the price of sugar during one period in 2010 of 21.1% but confectionary producers were only able to raise their prices by just about 0.5%.

Taxation and depreciation also have important implications in capital budgeting decisions, as Exhibits 13.5 and 13.6 show.

The impact of taxation on cash flows is generally predictable. Although there are changes to tax rates, these changes are infrequent and driven by regulatory requirements. It is important, however, to consider different tax levels in different jurisdictions when analysis cash flows.

In Hong Kong, the profits tax rate averaged 16.5% for two decades. It peaked at 17.5% in 2003/2004–2007/2008. In January 2018, Hong Kong introduced a two-tiered profits tax regime for all types of business. For corporations, the tax rate on the first HK\$2 million of assessable profits is now 8.25% and the tax rate on assessable profits over HK\$2 million is 16.5%. For unincorporated businesses, the tax rate on the first HK\$2 million of assessable profits is 7.5% and it is 15% on assessable profits over HK\$2 million.

When analysing inflation and taxes together, a break-even can be calculated, revealing the required rate of return after both inflation and taxes. This is a significant calculation because the required break-even rate often is much greater than might otherwise be assumed.

To calculate the break-even, divide the assumed inflation rate by the after-tax rate (100 minus effective tax rate).

After-inflation, after-tax break-even rate:

$$I \div (100 - E) = B$$

I equals the inflation rate; E is the effective tax rate; and B is the required break-even rate of return. For example, the effective tax rate is 16.5% and current inflation is 5%. The break-even rate is:

$$5\% \div (100 - 16.5) = 6.0\%$$

To generate a break-even return, the company needs to generate 6% net return under these conditions, just to break even after inflation and taxes.

Depreciation is another important consideration from the capital budgeting perspective when it comes to analysis of cash flows.

Generally, assets that a company invests in will lose value (or depreciate) over time. Depreciation charges effectively write off, or 'expense' the cost of assets over a recovery period of years. Despite depreciation not being a cash item, depreciation on assets is treated as a deduction in determining a company's taxable income. As depreciation can lead to tax savings in this way, it is sometimes referred to as the depreciation tax shield.

The written-down value of an asset is calculated as the total cost of investment minus accumulated depreciation. For capital budgeting, depreciation charges are usually calculated using either the straight-line method or the diminishing-balance method. Under the straight-line method, the total cost of an investment is divided by its estimated useful life. Under the diminishing-balance method, the cost is obtained by multiplying a fixed percentage of the asset's written-down value.

The main difference between these two methods is that the diminishing-balance method has a higher depreciation expense in the first years of an asset's usage and a lower one in its later years. As a consequence of the higher expenses in the first period, a company also has a higher tax saving and so greater after-tax net cash flows. This can be of considerable benefit to a company during the first years of a new project.

Recall that the FCF calculation includes incremental depreciation along with incremental amortisation (D&A). Depreciation and amortisation go together in the calculation because amortisation is a non-cash charge (deduction) like depreciation.

It is important to know that amortisation, like depreciation, is a deduction that is allowed under tax law to compensate for the decline in value of certain, mainly intangible, assets used by a business. It is also used to write off some tangible assets such as organisational costs (costs incurred before operations begin, set up as an asset and then amortised over several years).

Illustrative Example 3

The Morgantown Company Ltd (MC) has a project that requires the purchase of production equipment that costs HK\$800,000 initially. MC's management estimated that the project would generate revenues of HK\$440,000 and operating expenses of HK\$250,000 each year.

At the end of five years, the equipment will be sold for HK\$300,000. The equipment is depreciated annually at 20%, 32%, 19.20%, 11.52%, 11.52% and 5.76%. The tax rate = 35%, and the project's cost of capital is 10%.

If the revenues and operating expenses increase at a rate of 3% each year, calculate the net present value of this project.

Initial cost: HK\$800,000

Salvage: HK\$300,000

Useful life: 5

Tax rate: 35%

Year	0	1	2	3	4	5
Initial outlay and asset sale	–HK\$800,000					HK\$300,000
Tax on sale of asset						–HK\$88,872
Change in working capital						
Investment cash flow	–HK\$800,000	HK\$0	HK\$0	HK\$0	HK\$0	HK\$211,128
Change in revenues		HK\$440,000	HK\$453,200	HK\$466,796	HK\$480,800	HK\$495,224
Change in operating expenses		HK\$250,000	HK\$257,500	HK\$265,225	HK\$273,182	HK\$281,377
Change in depreciation		HK\$160,000	HK\$256,000	HK\$153,600	HK\$92,160	HK\$92,160
Change in taxable income		HK\$30,000	HK\$60,300	HK\$47,971	HK\$115,458	HK\$121,687
Change in taxes		HK\$10,500	HK\$21,105	HK\$16,790	HK\$40,410	HK\$42,590
Change in income after tax		HK\$19,500	HK\$39,195	HK\$31,181	HK\$75,048	HK\$79,096
Add: depreciation		HK\$160,000	HK\$256,000	HK\$153,600	HK\$92,160	HK\$92,160
Operating cash flow		HK\$179,500	HK\$216,805	HK\$184,781	HK\$167,208	HK\$171,256
Net cash flow	–HK\$800,000	HK\$179,500	HK\$216,805	HK\$184,781	HK\$167,208	HK\$382,384

**Illustrative Example 3 (continued)**

Net present value: HK\$32,824.07

Tax on asset sale is computed as follows:

$$\begin{aligned} (\text{Sales proceeds} - \text{remaining book value}) \times \text{tax rate} &= (300,000 - (800,000 \times 5.76\%)) \times 35\% \\ &= \text{HK\$88,872} \end{aligned}$$

Key Learning Point

Because capital expenditures typically involve the estimation of cash flows several years in the future, inflation plays a role in determining these estimates because it will affect future levels of sales and expenses. Taxation and depreciation are related considerations when analysing cash flows.

► Knowledge Check Questions**Question 6**

Identify which element of a project's cash flow inflation does not affect.

- A** Sales
- B** Expenses
- C** Depreciation
- D** Cost of capital

Question 7

Identify the impact of increasing inflation.

- A** It reduces the present value of depreciation tax shields.
- B** It increases the present value of depreciation tax shields.
- C** It does not affect the present value of depreciation tax shields.
- D** It would not affect the present value of depreciation tax shields if inflation remains below 5%.

13.4 INVESTMENT APPRAISAL TECHNIQUES

The planning process used to determine whether an organisation's long-term investments are worth funding is called capital budgeting or **investment appraisal**.

Companies have multiple avenues to appraise their investments. Some of these approaches, such as the payback period or the discounted payback period approaches consider how long it will take for a project to become cash positive. Other approaches, such as the accounting rate of return (ARR), the internal rate of return (IRR), the net present value (NPV) and the profitability index (PI) take more variables into account.

13.4.1 Payback Period

Companies often use a measure of time to screen or evaluate projects. Conceptually and mathematically easier than the net present value and the internal rate of return, the **payback period** is the number of years required to fully recover the initial cash outlay associated with the capital expenditure.

Shorter payback periods are preferable, and usually this decision criterion is implemented by choosing a cut-off date and rejecting projects whose payback period is longer than the cut-off period.

The payback period is determined by calculating cumulative cash flows over time.

For example, an investment requires an outlay of HK\$5 million initially, with expected cash flows of HK\$1 million, HK\$0 and HK\$5 million for the next three years. The payback period is easy to calculate in this case by simply adding up the inflows as follows:

Year	Cash flows (in millions)	Cumulative cash flows recovered (in millions)
1	HK\$1	HK\$1
2	HK\$0	HK\$1
3	HK\$5	HK\$6

In this example, if the cash flows are received evenly throughout the year, the payback period would be about three years ($2 + 4/5 = 2.8$ years).

13.4.2 Discounted Payback

The **discounted payback period** is defined as the number of years required to fully recover the initial cash outlay in terms of discounted cash flows. Shorter periods are preferable, and projects with discounted payback periods before the cut-off date are typically accepted.

For example, a project expects outlays of HK\$12 million and subsequent end-of-year cash inflows of HK\$5, HK\$5 and HK\$8 million, respectively, if the discount rate is 15%. The

discounted payback period is defined by calculating the cumulative discounted cash flows over time as follows:

Year	End-of-year cash flows (in millions)	Calculation	Accumulated discounted cash flows (in millions)
1	5	$5 / 1.15$	HK\$4.34783
2	5	$4.3473 + [5 / (1.15)^2]$	HK\$8.12854
3	8	$8.12854 + [8 / (1.15)^3]$	HK\$13.38867

The discounted payback period in this case is about three years. It is in the third year that the HK\$12 million initial outlay is recovered.

13.4.3 Accounting Rate of Return

The **accounting rate of return (ARR)** (also known as average rate of return) approach to appraisals of investments or capital projects calculates the return on a capital project using accounting data gathered from the project's net income (NI) and book value (BV). It is an alternative to approaches that use cash flow information.

The accounting rate of return (ARR) can be calculated in numerous ways. The most common definition is:

$$\text{ARR} = \text{Average net income} / \text{Average book value}$$

Where:

- Average net income = $(\text{NI}_1 + \text{NI}_2 + \dots + \text{NI}_n) / n$
- Average book value = $(\text{BV}_1 + \text{BV}_2 + \dots + \text{BV}_n) / n$
- n = the project's estimated life

Another approach is to determine average book value is simply to arrive at the average by dividing opening book value and closing book value by two, as follows:

$$(\text{Opening book value} + \text{Closing book value}) / 2$$

This method does not consider the time value of money or risk factors, but rather is based solely on the rate of return, usually calculated per year. When ARR is greater than the required return, the project is in positive territory. The higher a calculated ARR, the more attractive a potential project will be. A drawback to this method is that it can be manipulated by adjusting the method used to calculate depreciation (and as a result, calculated net income). For this reason, ARR may yield misleading estimates of investment evaluation. The solution to this problem is to ensure that the same net income estimates are used for comparative analysis.

13.4.4 Internal Rate of Return (IRR)

The **internal rate of return** (IRR), is a legitimate alternative to the net present value (NPV) method.

These two approaches have several similarities. For example, unlike ARR, they both take the time value of money into account by discounting a project's cash flows.

The IRR method uses the rate of return of a project to decide whether it offers a better return than the project's opportunity cost of capital. In contrast, the NPV method values a capital project by using the discount rate as the rate of return required by investors for investments with similar risk as its opportunity cost of capital.

The IRR is defined as the discount rate that matches the present value of a project's expected cash inflows with the present value of its outflows:

$$PV(\text{Project's future cash flows}) = PV(\text{Cost of the project})$$

This means that the IRR can be seen as the discount rate for which NPV equals zero. This relationship is summarised as:

$$\begin{aligned} NPV &= NCF_0 + \frac{NCF_1}{1+IRR} + \frac{NCF_2}{(1+IRR)^2} + \dots + \frac{NCF_n}{(1+IRR)^n} \\ &= \sum_{t=0}^n \frac{NCF_t}{(1+IRR)^t} = 0 \end{aligned}$$

If the required return of a project is smaller than the IRR, the project should be accepted. Since both the IRR and the NPV methods are based on whether the project's return exceeds the cost of capital and, hence, whether the project will add value to the company, it may seem that they are interchangeable, and that either should accept or reject the same capital projects.

The IRR and NPV methods give the same answer in many situations. However, some of the mathematical properties of the IRR equation can lead to incorrect decisions concerning whether to accept or reject a capital project.

13.4.5 Modified Internal Rate of Return

Another way to consider whether an investment is worthwhile is to use the modified internal rate of return (MIRR) approach. MIRR is similar to IRR, it is based on the rate of return that a project promises. MIRR considers that a project has positive cash flows and that those cash flows are reinvested at the same cost of capital of the firm.

The formula for MIRR is:

$$MIRR = \sqrt[n]{\frac{FV(\text{Positive cash flows} \times \text{Cost of capital})}{PV(\text{Initial outlays} \times \text{Financing cost})}} - 1$$

There are several advantages to MIRR. It considers the time value of money as well as the cash flows from a project. This approach also indicates the returns that a project might generate based on a realistic reinvestment of future inflows. MIRR also has objective criteria.

On the other hand, MIRR is not useful to decide between mutually exclusive projects. Likewise, it is also not useful when a company has a limited capital budget.

13.4.6 Net Present Value

Recognising the time value of money can significantly alter the long-term impact of capital budgeting decisions. For example, cash flows early in the life of an investment will be worth more than those occurring later because of the time value of money. It is therefore useful to recognise the timing of cash flows when evaluating projects.

Capital budgeting techniques that consider both the time value of money and the estimated net cash flow from an investment are called discounted cash flow techniques. They are largely recognised as the most informative and best-conceived approaches to making capital budgeting decisions.

The expected net cash flows used in discounting cash flows are calculated using annual net cash flows plus the estimated liquidation earnings arising from the sale of the asset is sold at the end of its life (its salvage value).

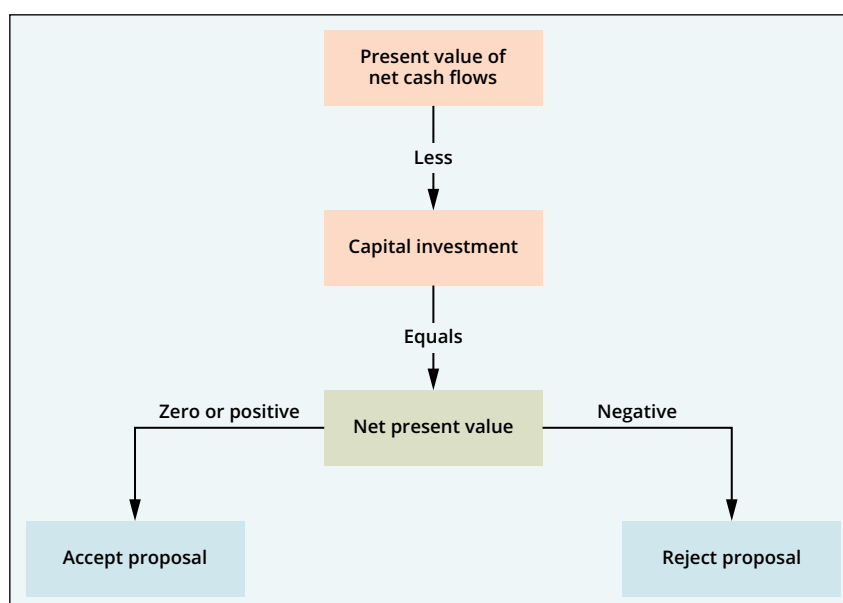
The NPV method is the primary discounted cash flow technique used for such calculations (though IRR as an alternative possibility). This method calls for discounting net cash flows to their present value, after that comparing that value with the capital investment needed for the project.

The interest rate used to discount future net cash flows is often called the discount rate or required rate of return. Its value is set by a company's management.

Under the NPV decision rule, a proposal is:

- Acceptable if the NPV zero or positive. At such a figure, the rate of return on an investment will equal or be greater than the required rate of return.
- Unacceptable if NPV is negative.

Exhibit 13.7 shows the net present value decision criteria.



► **EXHIBIT 13.7 NPV decisions** (Source: Weygandt 2015. Reproduced with permission of John Wiley & Sons.)

When selecting between acceptable proposals, the higher the positive net present value, the more attractive the investment.

13.4.7 Profitability Index

The **profitability index (PI)** is another discounted cash flow (DCF) approach used to evaluate capital expenditure decisions.

Like the internal rate of return (IRR), the PI measures project attractiveness. A PI is calculated by dividing the ratio of a project's discounted net incremental after-tax cash inflows by its discounted cash outflows. Usually, the cash outflows are the project's initial after-tax cash outlay.

The PI is calculated as:

$$PI = PV(\text{cash inflows}) / PV(\text{cash outflows})$$

The PI equation suggests that projects with ratios greater than 1 should be accepted because, by definition, they have positive NPVs, whereas projects with ratios less than 1 should be rejected.

The same equation suggests that the greater the ratio the more attractive the project due to it having a higher NPV.

The discount rate used is the same as that used to calculate NPV. Usually, it is the project's cost of capital. Consequently, the relationship between NPV and PI is:

- If $NPV > 0$, then $PI > 1$
- If $NPV < 0$, then $PI < 1$

For example, consider the PI for a project with an initial outlay of HK\$12 million and subsequent end-of-year cash inflows of HK\$5, HK\$5 and HK\$8 million, respectively, if the discount rate is 15%. In this case, the following calculations are possible:

- Present value of cash inflows = HK\$4.34783 + 3.78072 + 5.26013 = HK\$13.38867 million
- Present value of cash outflows = HK\$12 million

Therefore:

- $PI = \text{HK\$13.38867 million} / 12 \text{ million} = 1.12 \text{ or } + 112\%$

Illustrative Example 4

ABC Investment Ltd's subsidiary, Ranger Corporation, has decided to invest in renewable energy sources to meet part of its energy needs for production. It is considering solar power versus wind power. After considering cost savings as well as incremental revenues from selling excess electricity into the power grid, it has determined the following.

	Solar	Wind
Present value of annual cash flows	HK\$52,580	HK\$128,450
Initial investment	HK\$39,500	HK\$105,300



Illustrative Example 4 (continued)

Determine the net present value and profitability index of each project. Explain which energy source it should choose based on profitability index.

	Solar	Wind
Present value of annual cash flows	HK\$52,580	HK\$128,450
Less: Initial investment	39,500	105,300
Net present value	HK\$13,080	HK\$23,150

Profitability index	Solar	Wind
Present value of annual cash flows	HK\$52,580	HK\$128,450
Divided by initial investment	÷ 39,500	105,300
	1.33	1.22

Ranger Corporation should choose the solar energy source as it has a higher profitability index.

Key Learning Point

Companies have multiple avenues to appraise their investments including the payback period, the discounted payback period, the accounting rate of return (ARR), the internal rate of return (IRR), the modified internal rate of return (MIRR), the net present value (NPV) and the profitability index (PI).

► Knowledge Check Questions

Question 8

An analyst has just completed estimating a project's NPV when she is told that its fixed capital outlay will be increased by HK\$100,000. Despite this, there will be no expected changes in its cash operating revenues, cash operating expenses or salvage value.

Given that the fixed capital outlay is depreciated straight-line over an 8-year life, that the tax rate is 30%, and that the required rate of return is 10%, calculate the effect on the project's NPV.

- A HK\$100,000 decrease
- B HK\$79,994 decrease
- C HK\$59,988 decrease
- D No change

► Knowledge Check Questions (*continued*)

Question 9

An investment of HK\$150,000 is projected to generate an after-tax cash flow of HK\$100,000 in one year and another HK\$120,000 in two years. Calculate the internal rate of return.

- A 28.19%
- B 28.39%
- C 28.59%
- D 28.79%

Question 10

Identify which of the following is true if a higher discount rate is used to calculate the net present value of a project.

- A It will be less likely that the project is accepted.
- B It will be more likely that the project is accepted.
- C Using a higher rate is appropriate if the project is seen as less risky than other projects.
- D Using a higher rate is appropriate if the project will have a shorter useful life than other projects being considered.

➤ 13.5 CAPITAL INVESTMENT APPRAISAL TECHNIQUES COMPARISON

Each of the seven investment appraisal techniques discussed in the previous section have both pros and cons. Some are more nuanced but harder to use to build projections, while others are relatively straightforward to use.

Exhibit 13.8 shows the advantages and disadvantage of the various methods.

	Advantages	Disadvantages
Payback period	• Easy to calculate. • A useful initial screening technique. • Measures the liquidity of a project (i.e. how fast cash inflows occur).	• Does not consider the time value of money. • Ignores cash flows beyond the payback period. • Has no objective criterion for what is a good payback period.
Discounted payback period	• Indicates whether a project will ever pay back. • Considers the time value of money of the project cash flows.	• Ignores cash flows beyond the payback period. • Has no objective criterion for what is a good discounted payback period.

► **EXHIBIT 13.8** Pros and cons of investment appraisal techniques (*Source: Booth 2014. Reproduced with permission of John Wiley & Sons.*)

	Advantages	Disadvantages
Accounting Rate of Return	• Easy to calculate and understand • Useful for a quick calculation of an investment's profitability and for comparison between multiple projects • Considers depreciation expense and any annual costs	• Does not consider the time value of money or risk factors for long term investments that can give misleading information when evaluating investments of different size • Ignores amount and timing of cash flows from investment and profit calculation can be affected by non-cash items • Can be easily manipulated with changes in accounting methods such as depreciation and stock valuation
Internal rate of return	• Considers the time value of money through the hurdle rate. • Considers all the project's cash flows. • Has an objective criterion (i.e. accept if IRR is greater than the project's cost of capital).	• Should not be used when deciding amongst mutually exclusive projects. • Cannot be used when the signs of cash flows change more than once. • Should not use when the capital budget is limited.
Modified internal rate of return	• Considers the time value of money through the hurdle rate. • Considers all of the project's cash flows. • Indicates the return on the project considering a realistic reinvestment of the project's cash inflows. • Has an objective criterion (i.e. accept if MIRR exceeds the project's cost of capital).	• Should not be used when deciding amongst mutually exclusive projects. • Should not use when the capital budget is limited.
Net present value	• Considers the time value of money through the discount rate. • Considers all of the project's cash flows. • Has an objective criterion (i.e. accept if NPV is positive).	• Result is in monetary terms, not as the more familiar return. • Sensitive to the project's cost of capital.
Profitability index	• Considers the time value of money through the discount rate. • Considers all the project's cash flows. • Has an objective criterion (i.e. accept if PI is greater than one). • Useful in ranking projects in terms of benefit/cost.	• Should not be used when deciding amongst mutually exclusive projects.

► **EXHIBIT 13.8** (continued)

13.5.1 Payback Period

The payback period method is easy to understand. At its heart is the idea that the faster a company can recover its investment outlay, the less risky the project. This means it is also sometimes used as an informal measure of project risk, particularly for short-term projects.

It does, however, have some major shortcomings. Since it treats a dollar of cash flow received in the next year the same as those received in current year, it ignores the time and risk

value of money. It also has no fixed rules for determining cut-off dates, which means they can be set arbitrarily. And it fails to account for cash flows received after a project's cut-off date.

Because payback period does not include calculations for the time value of money, it has limited value. However, it is a simple estimate of investment return, and can be used, notably for short-term projects or capital investments of limited dollar value. However, it should be used only as an initial estimate and not as a concluded description of appraisal for the capital investment.

13.5.2 Discounted Payback

The discounted payback period approach alleviates one of the shortcomings of the payback period technique by accounting for the **time value of money**. Unfortunately, just as was the case for the payback period criterion, the discounted payback period ignores cash flows beyond the cut-off date, and the cut-off date is somewhat arbitrary.

The discounted payback period is equal to or longer than the simple payback period because the future cash flows have less value due to being discounted for time and risk. If companies use discounting, they can just discount all future cash flows and instead calculate NPV.

Overall, the discounted payback period method is a compromise of the payback period and NPV methods. Generally, projects resulting in discounted payback period less than the targeted period will be accepted. In this method, the projected break-even date is calculated on a discounted payback basis.

13.5.3 Accounting Rate of Return

As with the payback period approach, the ARR is straightforward to calculate and understand. Many companies use it to gain a quick view of a project's profitability.

But ARR can also be problematic as a tool for evaluating capital expenditure. Although it uses accounting results rather than cash flows, it does not offer true accounting-based rate of return. Its output is a number based on average figures from the statement of profit or loss and the statement of financial position. As it does not discount a project's cash flows over time, it ignores the time value of money.

Moreover, as with the payback method, ARR has no mechanism linking the acceptance criterion to the maximisation of shareholder value.

Because of these major shortcomings, the ARR technique should not be used as the only method to evaluate the viability of capital projects.

13.5.4 Internal Rate of Return

Unconventional cash flows can lead to major discrepancies when using NPV and IRR methods. For example, difficulties can arise when a project's initial cash flows are positive and subsequent cash flows are negative.

For example, some types of insurance fall into this kind of category. When a company sells a lifetime annuity to a retired person, it typically receives a single cash payment upfront and



then makes monthly cash payments until that person dies. In this case, it is only necessary to reverse the IRR decision rule and accept the project if the IRR is less than the cost of capital to make the IRR and NPV methods agree. The intuition in this example is that the life insurance company is effectively borrowing money from the retiree and IRR is a measure of the cost of that money.

The cost of capital is the rate at which the life insurance company can borrow elsewhere. An IRR less than the cost of capital means that the lifetime annuity provides the insurance company with money at a lower cost than alternative sources.

Things become more complex when a project's future cash flows include both positive and negative figures. For example, this can happen if production facilities have to be closed periodically for renovation. A similar situation is when cash flow remains positive until shortly before the end of a project, when they turn negative. This could be due to having to conduct an environmental clean-up or because the equipment installed has no salvage value and must be removed.

The IRR method is an alternative to the NPV method. It accounts for the time value of money, unlike the payback period and ARR methods.

It also has great intuitive appeal, especially to investors or other businesspeople who think in terms of rates of return on their investments or the companies' capital expenditures. Indeed, IRR's popularity with business managers may result more from its intuitive appeal than from its merits.

13.5.5 Modified Internal Rate of Return (MIRR)

Internal rate of return calculations contain the assumption that all cash inflows are reinvested at the internal rate of return. However, this assumption is too aggressive and can lead to an overstatement of the profitability of a project. The calculation of the MIRR on the project has a more realistic assumption regarding the reinvestment of cash inflows.

With the MIRR, project managers are able to change the assumed rate of reinvested growth in a project based on their different stages. The most common way it is used is by using the average estimated cost of capital, but with a flexibility to add any specific anticipated reinvestment rate.

Different investments or projects of unequal size can be ranked by their MIRR. The criteria for accepting or rejecting a project or reinvestment opportunities based on the MIRR is similar to that of the IRR. If the MIRR is greater than the cost of capital, the project or opportunity is acceptable to proceed. However, unlike the IRR, the MIRR does not provide multiple solutions to the calculation for the same project. Only a single solution exists for each project.

There are a few downsides to the use of MIRR. First, the estimated cost of capital is required in order to reach a decision to be made. This calculation can be subjective and can vary based on the assumptions that are made by the manager.

Second, much like the IRR, the MIRR calculations could lead to sub-optimal decisions being made that do not maximise value when multiple investment options are being considered together. It is not possible for the MIRR to quantify the various impacts of different investments as a whole in absolute terms. A more effective method for selecting investments that are mutually exclusive would be to use NPV.

Third, in the case of capital rationing, the MIRR might not be successful in producing optimal results.

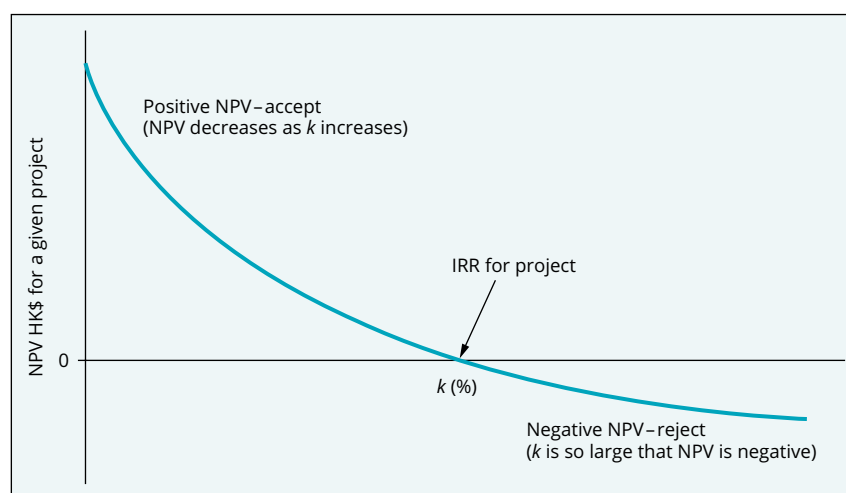
13.5.6 Net Present Value

The internal rate of return (IRR) method is often linked to the net present value (NPV) method. One reason for this is that both methods generate the same findings provided that they are evaluating independent projects that use conventional cash flows. Independent projects are those with no impact on any other of a company's projects, assuming the company has ability to access resources. For such projects, having a positive NPV will lead to it automatically having an IRR greater than the cost of capital.

Projects with a **conventional cash flow** are those with an initial cash outflow and subsequent cash inflows.

The relationship between the IRR and NPV methods can be visualised with a graph that plots NPV against the discount rate. Usually known as an NPV profile, such a graph shows the NPV of a project at different costs of capital.

Exhibit 13.9 offers an example of a basic NPV profile. The discount rate runs along the horizontal x-axis, with the NPV value running up vertical y-axis. As the discount rises, the NPV falls, with the curve intersecting the x-axis at where the NPV is zero.



► **EXHIBIT 13.9 NPV profile** (Source: Kidwell 2018. Reproduced with permission of John Wiley & Sons.)

Any independent project with conventional cash flows will produce a chart similar to the one in Exhibit 13.9.

Both the IRR and NPV methods will lead to the same investment decisions for independent capital projects with conventional cash flows. For projects that are not independent or which do not have conventional cash flows, the findings and investment decisions may be different. The two methods may result in conflicting accept/reject decisions when accepting one project means turning down another.

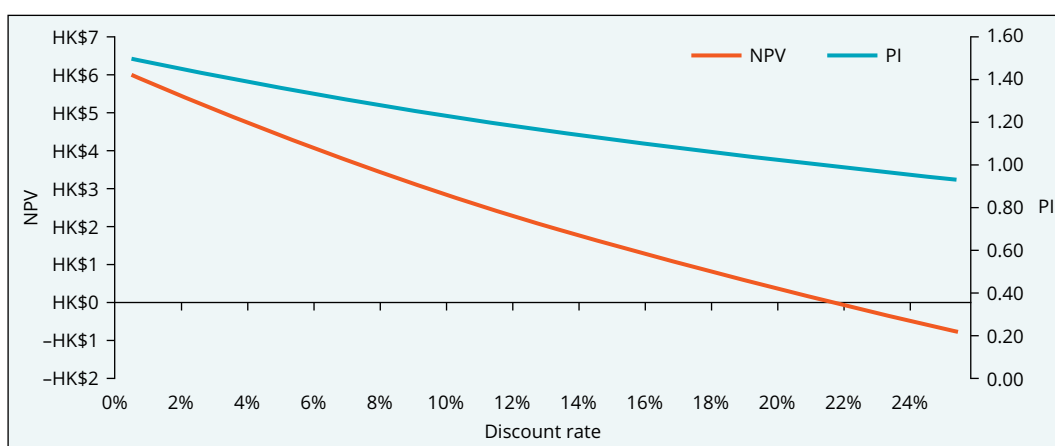
With mutually exclusive projects, the difficulty is selecting the best alternative. When using the IRR method, it seems logical to select the project with the highest IRR, but the logic is wrong. It is not possible to tell which mutually exclusive project to select just by looking at the projects' IRRs. With the NPV method, the answer is simple: selecting the project with the highest NPV will lead to the biggest increase in a company's value.

In certain circumstances, it is not possible to calculate an IRR. One instance of this is when initial cash flows, either positive or negative, are followed by cash flows involving more than one sign reversals. A contractor for a construction project, for example, may be paid in part in advance, it then uses this money to buy materials and pay labour costs, and finally at the end of the project it receives a final payment.

NPV should be the primary method for capital budgeting decisions as its methodology aligns with the more general objective of maximising the value of a company's shares. This is because its output is a figure of how much a project can be expected to increase a company's value.

13.5.7 Profitability Index

There is a clear link between the PI and the NPV, which is seen in Exhibit 13.10 with the NPV on the left vertical axis and the PI on the right vertical axis. When the NPV is negative, the PI is less than 1.0.



► **EXHIBIT 13.10** Sample relationship between PI and NPV (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

For independent projects, a PI should generate the same decision to accept or reject a project as the NPV and IRR methods.

As with the IRR, a PI can be expressed as a percentage: a PI of 1.2, for example, can be equally well stated as +20%.

The PI method has the advantages, unlike the IRR method, of using just a single consistent and reasonable discount rate and of working even if future cash flows switch between being positive and negative.

One weakness that the PI measure shares with IRR is that it is a relative measure and not an absolute measure of wealth, like NPV.

Although it is useful as a starting point for ranking projects when some projects must be rejected, final decisions should be based on which projects maximise the total NPV for the company. In this case, the PI is often used when companies are capital constrained.

Key Learning Point

The payback period method provides a useful intuitive measure of how long it takes to recover an investment. The discounted payback period approach alleviates one of the shortcomings of the payback period technique by accounting for the time value of money. The ARR is easy to compute and comprehend for people without a strong finance background. Unconventional cash flows can cause a conflict between the NPV and IRR decision rules. The profitability index produces the same accept/reject decisions as do the NPV and the IRR approaches in the case of independent projects and conventional cashflows.

► Knowledge Check Questions

Question 11

Identify the weakness of the cash payback technique.

- A It uses accrual-based accounting numbers.
- B It ignores the time value of money.
- C It ignores the useful life of alternative projects.
- D Both B and C are true.

Question 12

Identify what it means to have a positive net present value.

- A A project's rate of return is less than the cut-off rate.
- B A project's rate of return exceeds the required rate of return.
- C A project's rate of return equals the required rate of return.
- D A project is unacceptable.

Question 13

Identify at which internal rate of return a project should be accepted.

- A Zero.
- B The rate of return on a government bond.
- C The company's required rate of return.
- D The rate the company pays on borrowed funds.

13.6 INVESTMENT STRATEGY WHEN CAPITAL IS LIMITED OR RATIONED

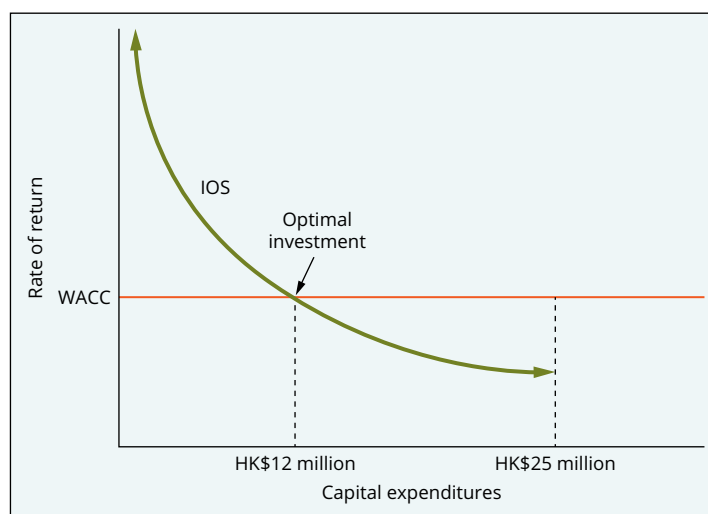
Theoretically, companies should accept all independent projects that generate positive NPVs, which enhance company value.

However, in practice, companies often face constraints that hold them back. These can be internal, such as management budget limits, or they can be due to market inefficiencies which limit the company's ability to raise funds a capital market.

In theory, such financial constraints should not exist in efficient markets, as a company should always be able to access new financing for any investment opportunity that would lead to a return above the cost of raising the required investment funds. Market inefficiencies, however, often create obstacles that hinder a company's ability to access capital markets.

When companies face financing constraints, **capital rationing** prevails, that is, investment capital must be rationed amongst available investment projects.

Exhibit 13.11 shows a sample company's investment decision environment. The company has HK\$25 million of available projects, which are ranked by internal rate of return and plotted in along the investment opportunity schedule (IOS).



► **EXHIBIT 13.11 Investment environment** (Source: Booth 2014. Reproduced with permission of John Wiley & Sons.)

A company should invest in opportunities along its IOS until either the IRR equals its weighted average cost of capital (WACC) or the last project accepted has an NPV of zero.

For the company above, despite having HK\$25 million available for new projects, it should only invest HK\$12 million, its optimal amount.

If the company's available funding totalled less than HK\$12 million, it would be in the position of being capital rationed. Under these circumstances, the cost of capital would no

longer be the appropriate opportunity cost. Instead, it would look at cash flows generated by projects and which could be reinvested at a rate higher than assumed by the cost of capital. Because of this, the reinvestment rate assumptions underlying IRR would be more valid, as the company could reinvest its cash flows from projects at the IRR of the marginal project, not its cost of capital.

Unfortunately, IOS is rarely as smooth as the one shown in Exhibit 13.11.

When companies with limited funds must choose between different projects, they have to consider how the cost of their investments is constrained by their capital budgets. In such circumstances, while NPVs and IRRs will provide useful information, such data has the weakness of ignoring investment costs and capital constraints.

The profitability index (PI) is often a useful starting point because it gives the highest present value relative to the initial cash outlay, which is constrained (i.e. similar to the contribution per limiting factor concept discussed in Module 2, Chapter 8). However, the final decision should be based on which combination of projects generates the highest total NPV while satisfying the capital budget constraint. This is because the PI, like the IRR, may lead to incorrect decisions because it is a relative measure.

Illustrative Example 5

Consider two mutually exclusive projects, Corvalis and Portland:

End-of-year cash flow		
Year	Corvalis	Portland
0	-HK\$9,500	-HK\$9,500
1	HK\$0	HK\$3,000
2	HK\$0	HK\$3,000
3	HK\$0	HK\$3,000
4	HK\$12,000	HK\$2,000

1. If the project discount rate for both projects is 5%, explain what the appropriate decision is.
2. If the project discount rate for both projects is 6%, explain what the appropriate decision is.
3. If the project discount rate for both projects is 7%, explain what the appropriate decision is.

		NPV of Portland	NPV of Corvalis
1. If the cost of capital is 5%	Invest in Corvalis	HK\$372.43	HK\$315.15
2. If the cost of capital is 6%	Invest in Portland	HK\$5.12	HK\$103.22
3. If the cost of capital is 7%	Invest in neither	-HK\$345.26	-HK\$101.26

Key Learning Point

Investment constraints may arise because of market inefficiencies, which restrict the company's ability to raise funds in the capital markets, or because they are imposed internally.

► Knowledge Check Questions

Question 14

A project requires a HK\$10 million initial investment and has a net present value of HK\$6 million. Identify which of the following statements is incorrect.

- A** The project's profitability index is 0.6.
- B** The project's investment offers a payback through its discounted cash flows.
- C** The project's internal rate of return is greater than its cost of capital.
- D** The investment is expected to increase the company's value by HK\$6 million.

Question 15

Define IRR and NPV and explain the difference between ranking according to IRR and NPV.

Question 16

Consider the following end-of-year cash flows for Projects X and Y:

Project X: $CF_0 = -2,000$; $CF_1 = 1,000$; $CF_2 = 1,000$; $CF_3 = \text{HK\$}1,000$

Project Y: $CF_0 = -2,000$; $CF_1 = 0$; $CF_2 = 500$; $CF_3 = \text{HK\$}3,000$

Identify, if the discount rate is 15%, what the decision regarding projects X and Y should be to accept.

	Independent	Mutually exclusive
A	Both	X
B	Neither	Y
C	Both	Y
D	Neither	X

13.7 IMPACT OF TAXATION, INFLATION, RISK AND UNCERTAINTY IN CAPITAL INVESTMENT APPRAISAL

Taxation, inflation, risk and uncertainty all have an impact on capital investment appraisal processes. Company decision makers can undertake different types of analysis to address these factors. The more rigorous the assessment process the more effective the decisions are likely to be.

13.7.1 Taxation

The taxable income of a company can be affected by many things. For example, by the launch of a new product, by the purchase of an asset it plans to sell in the future, or by replacing old equipment with newer, more productive machinery.

As changes in taxable income lead to changes in taxes payable and so on cash flows, it is necessary to estimate all the tax implications of such changes.

Companies can also benefit from tax credits governments offer for certain types of capital projects or for activities such as for research and development. An **investment tax credit** is a direct reduction of the company's tax bill at the time the investment is made, which can significantly affect the initial cash outlay for a project.

Therefore, when estimating cash flows, it is necessary to consider any tax effects and to estimate the after-tax cash flows of projects.

Consider a company that is thinking about introducing a new product line that is expected to generate HK\$100,000 per year in additional after-tax cash flows. The introduction of this new product line, however, will likely affect the sales of another product of the same company; more precisely, introducing the new product will cost the company HK\$40,000 in after-tax cash flows because of lost sales for an existing product.

The HK\$40,000 is the result of cannibalisation. Cannibalisation occurs when a new product competes with and affects the sales of another product of the same company. In this example, the incremental cash flows used in evaluating the new product line are HK\$60,000 per year (i.e. HK\$100,000 – HK\$40,000).

The investment climate in Hong Kong is favourable. It is a Special Administrative Region (SAR), meaning the local government controls its own legal and judicial system (including taxing policies), and the region has economic autonomy.

Much of Hong Kong's economic success is attributable to relatively simple taxing policies and simplicity in the rules. Although Hong Kong does not offer investment incentives to foreign entities, it does offer duty-free status, low tax rates and excellent infrastructure. Preferential taxing policies are offered, such as 50% reduction in the tax rate for some debt instruments, profits tax exemption for eligible onshore and offshore funds operating in Hong Kong and tax reduction for research and development expenditure.

Business taxation consists primarily of a tax on income. In Hong Kong, this is a profits tax. There are no capital gains taxes on capital assets, and no withholding tax on dividends or interest in Hong Kong.

In many countries, consolidating groups of companies and reporting income under a holding company or other unified format is popular. In Hong Kong, consolidated returns are not allowed. Losses from one company's operations cannot be applied to reduce taxable income on another company in the same group.

These tax rules may dictate how companies plan their investment strategies and allow for the impact of taxes. With the simplicity of the Hong Kong tax system, accompanied by low rates, the impact of taxation on investments is lower than in most developed country jurisdictions.



13.7.2 Inflation

Because capital expenditures typically involve the estimation of cash flows several years in the future, inflation plays an important role in determining these estimates and their accuracy. Inflation alters the level of interest rates and therefore affects the company's cost of capital that is to be used in discounting future cash flows. If inflation rises quickly, all estimates will be inaccurate; and the prospects of profiting from capital investments is worsened, especially if the purchase of those assets is financed with a variable interest rate loan.

The most successful investment appraisals are adjusted for inflation. As discussed in section 13.3, the easiest way to do this is to estimate nominal cash flows and discount them by using nominal discount rates.

As inflation rises, companies normally respond by reducing the level of capital expenditure investments. This means that opportunities are lost because a defensive posture is assumed in response to inflation. The NPV calculation is affected due to the higher inflation rate.

In a seeming contradiction, project revenues are most likely to increase as inflation rises. At first glance, higher capital investments are justified with higher inflation. However, expenses also rise and will offset any benefit gained due to inflationary pressures. It often is the case that net cash inflows are not enough to meet or surpass the inflationary effect on net cash outflows.

Inflation is probably built into the model being used to evaluate projects, but the projection might understate the inflation rate. The expected rate of inflation could be substantially lower than the realised rate, meaning future cash flows will suffer.

13.7.3 Risk and Uncertainty

A company's future is set by its long-term investment decisions. As such decisions typically involve major amounts of capital and are hard to change, they are usually the most important decisions a company has to make.

The preceding discussions of taxation and inflation represent forms of risk a company faces in evaluating and estimating its capital investment programme. However, other forms of risk also must be analysed and planned for.

The irrevocable nature of major investments makes them particularly risky. Whenever a major new project fails, it is almost impossible to recoup its investment costs. Indeed, in some cases, the survival of a company may turn out to be at stake. Therefore, it is critical that all capital investment decisions are made using the best-informed financial and economic reasoning.

Capital budgeting refers to the process through which a company makes such capital expenditure decisions.

It is a basic premise of finance that when a senior executive justifies a project on non-financial grounds, such as legal considerations, then the latter will prevail. There are situations where the government may pass a law and an industry has to satisfy the new regulations, such as when the government in Hong Kong passed a law (Ozone Layer

Protection Ordinance of 1987) that air conditioners could no longer use chlorofluorocarbons. As a result, the companies making air conditioners would have a non-financial need to change their manufacturing process.

Every project is amenable to analytical techniques. The inputs are more qualitative the more strategic the analysis.

Analysis can be divided into two kinds: bottom-up and top-down. Bottom-up analysis starts with the assumption that at its core a company is nothing more than a bundle of capital investment decisions.

An example of bottom-up analysis would be equipment replacement. This would begin with an engineer estimating what benefits in terms of labour costs, energy, materials and so on, would accrue from buying some new machinery. A financial analyst would then translate these savings into financial parameters to see whether the replacement savings are worth the cost of the equipment. However, at no point does either the engineer or the financial analyst consider whether the company should continue in this business.

In contrast, top-down analysis focuses on the strategic decisions about which industries or products the company should be involved in.

Both bottom-up and top-down analysis use the same capital budgeting framework. Where they vary is the quality of the estimates.

In the case of equipment replacement, for example, an analyst might be able to suggest that a company faced a 50% chance of saving a certain amount of money and a 50% chance of saving twice as much.

In contrast, top-down or strategic decisions always involve a situation of uncertainty where it is hard or even impossible to pair probabilities with possible outcomes. In such circumstances, analysis tends to generate different scenarios that a company's management can use to assess the risk of a scheme.

The flexibility in top-down decision-making is an application of the theory of option pricing to capital investment decisions, or real option valuation (ROV). ROV is the valuation of a capital investment considering the value of all the options that the investment provides, such as the option to expand, abandon or delay. ROV is very difficult to implement, but it is important in financial management because many capital investment decisions involve one or more of these options.

13.7.3.1 Risk-adjusted Discount Rate

Under efficient market conditions, the market value of a company is the present value of its expected incremental cash flows, discounted at an appropriate **risk-adjusted discount rate**.

The risk-adjusted discount rate is set according to the overall riskiness of that project, that is, the project's cost of capital. For companies, any project with a positive net present value that adds value to the company should be taken on if possible.

Having a positive NPV implies that the present value of expected future cash flows is greater than current cash outlays and so should increase the value of the company.



Because a company's creditors have a fixed claim on the company's income, regardless of the value of the project, NPV trickles down to the shareholders and increases the market value of the company's common shares. In this way, accepting positive NPV projects maximises the company's market value and creates shareholder value. In contrast, accepting negative NPV projects destroys company value and should be rejected because, by definition, the destruction of shareholder value is not in the best interest of the shareholders.

We would expect positive NPVs to arise only in situations in which a company has a competitive advantage. In today's competitive business environment, such opportunities are rare and do not persist for long. Therefore, projects that produce an NPV of zero will be the norm where companies operate in competitive markets. These projects should be accepted because they provide the appropriate return required to compensate for the **financing costs** and risks associated with the investment.

Illustrative Example 6

A company evaluates its capital investment programme based on the assumption that over the estimated five-year period inflation will average 3%. However, actual inflation begins at 4% and rises each year to end up at 7% with average annual inflation of 6% for the entire five years. This affects not only the period required to offset initial costs, but also impacts on cash flows through the entire period.

Key Learning Point

The market value of any company in an efficient market is the present value of its expected incremental cash flows, discounted at an appropriate risk-adjusted discount rate, which is based on the overall riskiness of the project.

► Knowledge Check Questions

Question 17

Explain the problem with adjusting a discount rate for inflation.

Question 18

If inflation is expected to increase in the future, explain how this would affect the profitability of a project.



13.8 IMPACT OF DIFFERENT CONSIDERATIONS ON INVESTMENT APPRAISAL

Beyond any considerations of cash flows or profitability, companies are increasingly expected to include subjective considerations before approving capital investment projects. These considerations include the impact on their decisions and actions on society and the environment. Such decisions are driven by the business strategy and plans of the firm reflecting the firm's approach to corporate social responsibility (CSR) or as articulated more recently in its approach to the management of environmental, social and governance (ESG) risk factors.

13.8.1 Environmental and Social Impacts of Firms' Activities

Over the last 50 years it has been observed that as businesses grew, significant inequalities occurred between developed and undeveloped countries and within countries with respect to environmental factors such as the threats of climate change due to greenhouse gas emissions, threats to the ozone layer, pollution, environmental degradation, and social factors such as uneven global economic development leading to widespread poverty and wealth concentrations, human rights breaches including access to food, water, healthcare, safety, and the need for investment in human capital and workplace diversity.

Environmental factors generate externalities such as the environmental damage arising from a capital project that must be considered when making investment decisions. Similarly, direct costs such as buying pollution control equipment or paying for other forms of abatement, must also be included in the cash flows of projects. These should be included either as known costs or as contingencies as part of the project assessment.

For some projects, due to various laws and/or regulations, firms may need to establish an environmental mitigation reserve account and set aside funds that might be needed for remediation activities at the end of the project. This is particularly the case in mining activities which must put funds aside for such activities following the mine shut-down or in the case of an environmental incident such as water pollution from effluent or involve reforestation. Often these effects and therefore their costs can be difficult to estimate. In some circumstances, significant negative environmental impacts could turn the positive cash flow of a project into negative cash flow if, for instance, it becomes necessary to implement remediation initiatives on a project site to satisfy environmental laws and regulations.

Illustrative Example 7

Your company has undertaken a project designed to generate substantial net profits over the next three years. It is a construction project but may involve several environmental impacts. Even though the initial approval has considered the known impacts, management is concerned about the potential of unknown future impacts during the project. Explain what you would recommend.



Illustrative Example 7 (continued)

The solution is to ensure that corporate policies conform as much as possible not only to regulatory requirements, but also to social impacts of a project. The solutions to unknown impacts may include setting aside land to create social benefits such as open space or parks, funding of schools or day care facilities.

Additionally, to plan for unknown future environmental impacts, a reserve fund should be established based on some formula. Identifying a percentage of future profits to set up this fund may involve periodic deposits of funds to be available for future impacts.

Societal factors can also generate externalities such as the benefit that arises if a firm initiates a project that generates employment in a depressed region. Although there are intangible benefits for the community and the firm, these benefits are not included in the project's cash flows.

Alternatively, negative externalities may arise from injury, death, and absenteeism of staff if there is an absence of strict workplace health and safety requirements, a lack of appropriate staff training and development, or a lack of diversity and anti-discrimination practices within the firm, or if there is a lack of community engagement about the impact of firm projects or engagement with suppliers in their supply chain.

Illustrative Example 8

Dark Void Enterprises (DVE) is a manufacturing company that has a long history of having poor workplace health and safety standards and has been served numerous breach notices by the Government Workplace Safety Regulator. DVE has a history of often using unsupervised uncertified process workers to perform technical procedures using machinery that requires staff with certified specialist skills. Errors in the use of this machinery may lead to serious injury or death. The firm pays the process workers a small daily allowance to operate the machinery, thereby lowering DVH's operating costs.

DVE's management is currently considering whether to continue this procedure of using the process workers considering possible future prosecution by the regulator. As part of their analysis, management wants to identify any possible negative externalities that may arise if that firm continues to use inappropriate workplace health and safety protocols.

Following an initial review, DVE identified the following long-term externalities that may arise from breaching its workplace health and safety requirements that will not be included in the firm's cash flows:

- loss of income for the family of the employee sustaining injury, thereby increasing reliance on social welfare;

Illustrative Example 8 (*continued*)

- the cost of specialist medical, hospitalisation, and rehabilitation costs to manage long-term injuries;
- loss to the broader community due to the loss of workers; these costs may be borne by the community and society at large; and
- reputational impacts whereby investors will not want to be associated with firms that cut costs in this way.

In response to the need for increased information about the firm's activities and governance by its stakeholders including investors, two primary trends have occurred – firstly in relation to the firm's corporate governance via its corporate social responsibility (CSR) and secondly in relation how the firm manages its exposure to the environment (e.g., climate change, energy and water use, waste management, and ecological impacts), social responsibility (e.g., fair trade principles, human rights, product safety, gender equality, health, and safety), and corporate governance (e.g., board independence, corruption, and bribery, reporting and disclosure, and shareholder protection). This emerging focus on how firms manage their exposure to environmental, social, and governance factors has come to be known by the acronym ESG.

13.8.2 Corporate Social Responsibility (CSR)

By the late 1970s, following long periods of global economic recession, inflation, unemployment, industrial shutdown, environmental destruction and damage, there was growing consensus that corporates needed to put into practice their economic responsibilities (to make profits), legal obligations (to not break the law), and ethical responsibilities (to be good corporate citizens). This unified framework is known as Corporate Social Responsibility (CSR) and over time became the cornerstone of expected corporate behaviour.

A company is said to practice CSR when it considers the ways in which its operations enhance society whilst balancing this requirement relative to the needs of shareholders who are also seeking a positive return on their investments. For example, consider a company is involved in efficient and successful manufacture and sale of electric motors for heavy duty machinery. The company operates on high standards of compliance with health and safety regulatory requirements, has an engaged and enthusiastic workforce, and has a well-functioning board that actively oversees the strategic direction of the firm. This firm may be considered as practicing corporate social responsibility because the motor vehicles will generate less greenhouse gas emissions than petrol or diesel machines. The company ensures that the investment in this firm meets the requirement to perform a positive social impact, while also generating net profits.

CSR is often driven within firms by the corporate governance position of the Board. Corporate governance is a set of rules, practices and procedures that guides company oversight and control by its Board of Directors and independent committees. In Hong Kong SAR, consistent with other jurisdictions, the Corporate Governance Code sets out the governance and reporting requirements and guidance to ensure that sound and effective



internal controls are in place to protect stakeholder's interests. It also specifies the mandatory requirements for the Corporate Governance Report that should be provided by the board for availability to all the business entities' stakeholders.

13.8.3 Sustainability and ESG

Sustainability

In response to the observed environmental and social problems as discussed above, the term Sustainable Development was used by the United Nations in 1987 that stated for society and the environment to be sustainable it was necessary for 'development that meets the needs of the present without compromising the ability of future generations to meet their own needs.' This desire to grow without damaging future generations' prospects gradually became central to business philosophies.

By defining sustainability in this way, pressure can be placed on both national governments and corporates to consider not only the economic outcomes associated with project developments that were parties to, but also the impact of their investment choices, activities, and project developments on both the environment and society. Hence, firms implementing a business strategy to create long-term value for the present shareholders while protecting the rights of future shareholders and other stakeholders are said to be operating in a sustainable manner.

Sustainability is similar to CSR in that it impacts firms' decision as to whether to undertake an investment or proceed with a project, in addition to project cash flows or its ultimate profitability. The sustainability of initiatives is intended to improve external relationships with all stakeholders while reducing costs and expenses in managing risks. It combines financial and non-financial policies and broadens the scope of corporate behaviour to include long-term interests beyond those of shareholders.

When used in business, economics, and finance, sustainability tends to share the following core elements:

- Environmental problems are typically presented in terms of their impact on both the economy and society. Neither economic growth nor social developments are permitted to take their ecological impacts of air, land, and water such as waste and pollution for granted.
- There is typically a strong focus on intergenerational equity and consideration of the impact on the environment, economy, and society in the long term, usually decades. This focus is much longer than what is considered in traditional business or even public planning cycles.
- There is an emphasis on achieving more than the minimum compliance with existing laws and regulations. This concept is captured for policymakers in terms of innovation that potentially confers competitive advantage.

Due to increasing global pressure via bodies such as the United Nations, academic research, and various activist groups, there was a growing awareness of the impact of firms' activities on the sustainability of the environment and society. Stakeholders, including investors, want firms to act in a more accountable and transparent manner for their environmental and social actions and governance decisions, rather than being focussed principally on profitability

and shareholder returns. Therefore, just because a project makes financial sense does not mean a firm should undertake it. Apart from the benefit of financial returns, investors also seek what has come to be known as non-financial benefit in the form of a 'social or environmental premium' with their investments.

Environmental, Social, and Governance

The term ESG is an acronym that describes environmental, social, and governance issues (known as ESG risk factors) that may potentially give rise to a risk exposure that can cause financial and/or non-financial risk to the firm (in some cases it may also provide opportunities). If the exposure should impact the firm or stakeholders in an unexpected way causing loss, then this may result in negative reputational and financial consequences for the firm and affect the relationship with its stakeholders.

Defining ESG factors in this way provides a framework to assess the risks and opportunities arising from these factors that may give rise to financial and non-financial impacts on corporates such as:

- ***Environmental factors*** – issues that may lead to serious problems affecting the ecology of the planet due to practices of firms. Examples include the impact of the firm due to its energy efficiency, carbon and greenhouse gas emissions, pollution, and waste management practices.
- ***Social factors*** – issues that may arise due to the business practices of the firm that cause negative consequences for the stakeholders of the firm and society in general. Examples include the firm's occupational health and safety practices; its levels of diversity, equal opportunity, and inclusiveness; and the social impacts arising from the quality of goods and services provided by the firm or its suppliers.
- ***Governance factors*** – corporate governance issues that may arise if there is a failure of board and senior management leadership due to violations of a code of conduct; a lack of accountability, transparency, and ethical breaches; or due to problems with the firm's existing internal controls framework. These issues typically involve a misalignment between the firm and its stakeholders' expectations such as breaching shareholders rights.

Each ESG factor plays an important role in measuring the future financial performance and the social influence of enterprises. The analysis of ESG factors is used by investors in equity and debt markets to evaluate corporate behaviour and future financial performance. As an investment principle, integrating ESG factors into financial analysis and investment decision making, has caused ESG to become an important driving force to trigger the sustainable development of enterprises.

The integration of ESG factors and risk management practices in business operations has transformed the strategy and business models of many companies. Examples include that some companies would:

- Counteract climate change by reducing greenhouse gas emissions, protecting biodiversity, treating wastewater, and implementing a circular economy by recycling waste.
- Invest in the human capital of the company to eliminate inequalities related to gender, ethnic origin, sexual orientation, and taking action to build or strengthen relations with customers or the local community.



- Take action to improve the organisational structure by increasing its transparency, openness to whistleblowing, detecting corruption, as well as improving investor relations and communication with shareholders.

ESG Reporting

In response to the increasingly severe global sustainable development problems in the environment, society, and in relation to corporate governance, financial markets, international organisations, and national regulators worldwide have put forward sustainable development action plans such as ESG Frameworks, Standards, and regulations regarding ESG reporting.

An ESG report provides an overview of an entity's commitment towards sustainable development, enhancing engagement with investors and other stakeholders by providing transparent information regarding the entity's ESG performance and the approach in achieving sustainable objectives. ESG report highlights the mix of quantitative and qualitative information required to assess the non-financial impacts of ESG factors on the firm's strategy and operational activities. It is a way for firms to demonstrate their business activities in relation to their sustainability practices and performance in achieving ESG targets via key performance indicators.

Hong Kong companies were first required to disclose ESG related information in 2014 as part of their Business Review requirement in the Directors Report under the new Companies Ordinance (Cap. 622). Subsequently, since 2016, it has been a condition of listing on the HKEX that companies are required to issue an ESG report in accordance with its Environmental, Social, and Governance Reporting Guide included in both the Main Board and GEM Board listing rules. This rule was amended in 2019 with effect from 2020 that listed companies were to produce a mandatory ESG report annually at the same time as their annual report.

When assessing performance using information in an ESG report, users should note that a significant amount of the information is qualitative in nature and presented in narrative description about a firm's ESG practices. The report contains forward-looking information explaining how a firm integrates ESG elements into business processes and how to create long-term value through its business operation.

Using this reported information, investors can assess the ESG factors relative to individual companies and industries when making investment decisions.

ESG Investing

The finance sector is a key driver in accelerating the transition to sustainability and climate neutrality. The growing importance of sustainability and ESG issues has heightened investor awareness of green finance instruments such as green bonds and green loans for capital intensive projects across a broad range of areas such as:

- **clean energy projects** like wind, geothermal, solar, small hydropower, biomass, waste-to-energy, and cogeneration;
- **green building projects** like building retrofits, new green buildings, energy audits/energy services, energy efficient equipment and building materials; and
- **pollution control and waste management projects** like air and water pollution control, soil remediation and mine rehabilitation, waste-to-energy, waste gasification, composting, scrubbers/filters, and recycling.

As firms transition to providing more accountability and transparency to stakeholders, including to investors regarding sustainability and ESG, so too have financial institutions and markets evolved to accommodate the impact of this transition. If investor preferences for sustainability are increasing, then firms will adjust to meet these expectations, and the capital markets will adjust to supply capital for this purpose.

Illustrative Example 9

The Global Sustainable Investment Alliance defines sustainable investment as an investment approach that considers environmental, social, and governance (ESG) factors in portfolio selection and management. This includes traditional sustainability responsible investment (SRI) approaches and ESG related strategies.

SRI strategies include:

- negative or exclusionary screenings to exclude potential projects that are specifically harmful to the environment or society; and
- positive screening which exclusively invest in particular types of ESG projects.

Other ESG related strategies used by fund managers include:

- impact investing where funds are targeted at investments aimed at solving social or environmental problems, including community investing, where capital is specifically directed to traditionally underserved individuals or communities;
- financing provided to businesses with a clear social or environmental purpose; and
- active ownership which involves the use of shareholder power through share ownership or proxy voting to influence corporate behaviour, including through direct corporate engagement (i.e., communicating with senior management and/or boards of companies) and using voting company shares on a variety of ESG issues to initiate changes in behaviour or in company policies and practices.

ESG standards have also become important benchmarks for the fund managers of many distinct types of funds. According to the Global Sustainable Investment Alliance based on its 2020 bi-annual survey, more than US\$35 trillion of professionally managed assets have been allocated according to such ESG standards. This represented 35.9% of all assets under management in the five major markets covered in their report.

Key Learning Point

Considerations of the impact on society and the environment of investments are often included in the social responsibility and corporate social responsibility (CSR) plans of a company.



► Knowledge Check Questions

Question 19

Explain what corporate policies and practices aimed at a broad range of social benefits are pursued by CSR.

Question 20

Explain how sustainability becomes a management philosophy.



13.9 NON-FINANCIAL CONSIDERATIONS FOR INVESTMENT APPRAISAL

Beyond the well-understood financial considerations for investment appraisal, companies must also consider more difficult to quantify non-financial issues when making decisions on specific investments. Beyond the considerations of CSR, sustainability and the environment, companies should also consider the impact of investments on their strategy, any legal issues that may arise, the culture impact of investments and any operational challenges.

Some key non-financial factors to consider when making investment decisions include:

- Future legislation;
- Industry standards;
- Staff morale and retention;
- Relationships with suppliers and customers;
- Reputation;
- Skills-building;
- Impact on a company's strategy.

These considerations should be included in any analysis, over and above any financial considerations.

13.9.1 Organisation Strategy

Investment appraisal is key to the organisation strategy of any company and the opposite is also true: how specific investments fit within the strategy of an organisation should also be considered by decision makers.

Investment appraisal and capital budgeting are vital to companies because they facilitate the creation of a structure process that makes it possible to:

- Develop long-term strategic goals and create a framework for long-term growth;
- Access to more projects;

- Determine the impact of project cash flows on company-wide cash flows;
- Assess whether a project works within the wider framework of a company structure;
- Develop decision-making parameters that facilitate the efficient running of a business.

A set of strategic objectives must include consideration of an organisation strategy. This includes objective statements or vision statements about the company's future. This is a guiding and defining step. Analysis of proposed investments in projects should be in line with a clearly understood organisation strategy. This is a long-term concern, but it also is used for short-term decisions. For example, why approve a short-term project today if it does not conform to the long-term organisation strategy?

Strategic objectives most often are expressed in financial terms. However, by including a range of non-financial objectives, the overall strategy is rounded out and encompasses more than profits and losses or cash flows. Thus, financial growth or efficiency may be combined with non-financial objectives and used to develop a comprehensive organisation strategy.

Non-financial objectives may include growth in a customer base or market; improvements in brand loyalty; development of new products based on CSR; and involvement in community projects or initiatives in which the organisation strategy is coordinated with a social cause.

A difference between financial and non-financial objectives does not mean the organisation strategy has to be separate. Financial objectives normally are designed to maximise shareholder profits. The non-financial objective tends to encompass interests beyond those of investors and generation of wealth through profits. The stakeholders include customers, lenders, suppliers and vendors.

The non-financial objectives amongst these stakeholders include improved customer services, vendor and supplier relations, and improved communications externally. These initiatives cost money, but may be viewed as improvements overall, which justify the initial costs involved.

13.9.2 Legal Issues

The legal issues associated with investment can be significant. Depending on the sector in which a company operates, legal costs could turn a profitable project into an unprofitable one.

Increasingly, management accountants are also expected to be aware of the legal requirements associated with investment decisions.

These issues span many areas. Product-specific exposure can arise from discovered flaws, recalls and class action lawsuits. Legal issues may also be generated internally if employees are discharged wrongfully or subjected to a hostile work environment. These human resource issues are mitigated through internal education and awareness programmes, but they cannot be prevented in all instances.

Once the company is advised of a pending legal action, it is not required to recognise expenses related to a possible judgement. However, the potential losses arising from legal action against the company must be disclosed by way of footnotes in the quarterly and annual



reports, in which contingent liabilities are listed. These are future expenses that might or might not be realised. Because litigation often takes years to conclude, disclosure of contingent liabilities is the only accounting disclosure required.

Some corporations with exposure to litigation above the average may establish reserve funds for both legal costs and the costs of future litigation, including possible judgements.

Investing in some projects also involves a legal concern. If environmental impact is a large concern, the impact on estimated profits and cash flows could be significant. As a matter of policy, a company may consider whether exposure to legal risk makes some projects reasonable, and which ones may be avoided or rejected.

13.9.3 Cultural Impacts

The cultural impact of an investment is also an important qualitative consideration for companies.

Investments can impact both the internal company culture or the culture of the locations where a company operates. A change has occurred in corporate culture over the past several decades. This may be called 'stakeholder engagement', a movement to consider the interests of all stakeholders, and to move away from past attitudes that the board of directors and top management have absolute power to make decisions. Today, the cultural changes resulting from CSR and sustainability have made management more responsive to a range of diverse interests.

It is not only investors who matter, but all stakeholders. This is the cultural impact of the CSR movement. A characteristic of companies successful in these efforts is success in sustainability, both financially and non-financially. Thus, sensitivity towards resource protection, social movements and even political matters often defines a company in terms of CSR and sustainability. Growing recognition reveals that younger, emerging investors are attracted to organisations with a culture based not only on profits, but also on CSR.

13.9.4 Operational Challenge

Investment projects should also be assessed considering the operational challenges they present for a company. A company that undertakes a project without any specific expertise is unlikely to be able to carry out the project effectively, even if the financial analysis of the project suggests it would have a positive cash flow.

Companies have at least four basic goals:

- Keep their customer base;
- Boost market share;
- Develop profitable products and services; and
- Lower expenses.

New investment projects should be considering these goals while taking into account the implementation challenges that may surface. In broad strokes, operational challenges may arise from working with partners, managing operations, sales and marketing, service delivery, customer care and perceptions, billing and infrastructure.

Another area of great interest within an operational challenge is identifying who works for companies. Are employees in high sustainability companies different than those in more traditional companies? Do they share a different set of values, or are the individual employees the same everywhere?

In many cases, the answer depends on how well the company has articulated its organisation strategy and cultural priorities. Employees can be trained to adopt a point of view encompassing financial and non-financial interests.

Some companies are highly focused areas, and this is reflected in the employees hired to work in these companies. Examples include companies whose main effort is focused on environmental impact, socially conscious product development, faith-based interests or even politically based efforts, either through products or services, or in the nature of CSR activities. In those instances, it is fair to assume that employees tend to be amongst the 'true believers' in the corporate structure, and that the operational challenges they face are approached with a single focus.

Beyond employees, what types of investors are involved with high sustainability companies? Are they likely to be more interested in CSR than in profits? Or do they balance the two interests? This question must be treated as an operational challenge because the day-to-day operational policies should reflect corporate objectives. Investors in conflict about operational policies may give management mixed messages. One group is interested in maximum profits, the other insists on CSR initiatives. The challenge to the company is to discover ways to merge these two interests into a single operational policy.

Illustrative Example 10

An organisation has identified its organisational strategy for merging non-financial and financial considerations. However, merging these has proven to be a greater problem. To succeed in this, the company needs to develop means for non-financial measurement. These include understanding of both cultural impacts and operational policies.

Amongst the challenges is determining how to respond to new cultural realities. For example, shareholders are not the only nor the most important stakeholders of an organisation. Many other interest groups, both external and internal, must be considered as parties of an investment appraisal process.

Illustrative Example 11

You have been given a challenging assignment by the CEO of your company. Describe examples of how non-financial considerations can be developed and pursued to generate financial advantages.

The greatest operational challenge is to merge financial and non-financial considerations. This is possible, but it requires strong leadership and internal training. You develop an example based on the following steps in non-financial initiatives that lead to financial benefits:

**Illustrative Example 11 (continued)**

1. Focused recruiting of high-calibre employees.
2. High morale amongst employees, growing from focused recruitment.
3. Value generation from a motivated employee base.
4. Responsive customers, generated from motivated employees.
5. Improved customer loyalty and buying patterns.

Key Learning Point

Non-financial considerations when appraising investments include organisational structure, legal issues, cultural impacts and operational challenges.

► Knowledge Check Questions**Question 21**

Identify which of the following are the possible legal issues included as non-financial considerations in investment appraisal.

- A** Lawsuits filed against the company by customers, possible lawsuits from employees and disputes between the company and some of its suppliers.
- B** Challenges to the company's legal standing to operate.
- C** Contingencies that probably will not be realised.
- D** Problems with attorneys hired by the company.

Question 22

Describe what non-financial objectives could be included in organisation strategy.



SUMMARY

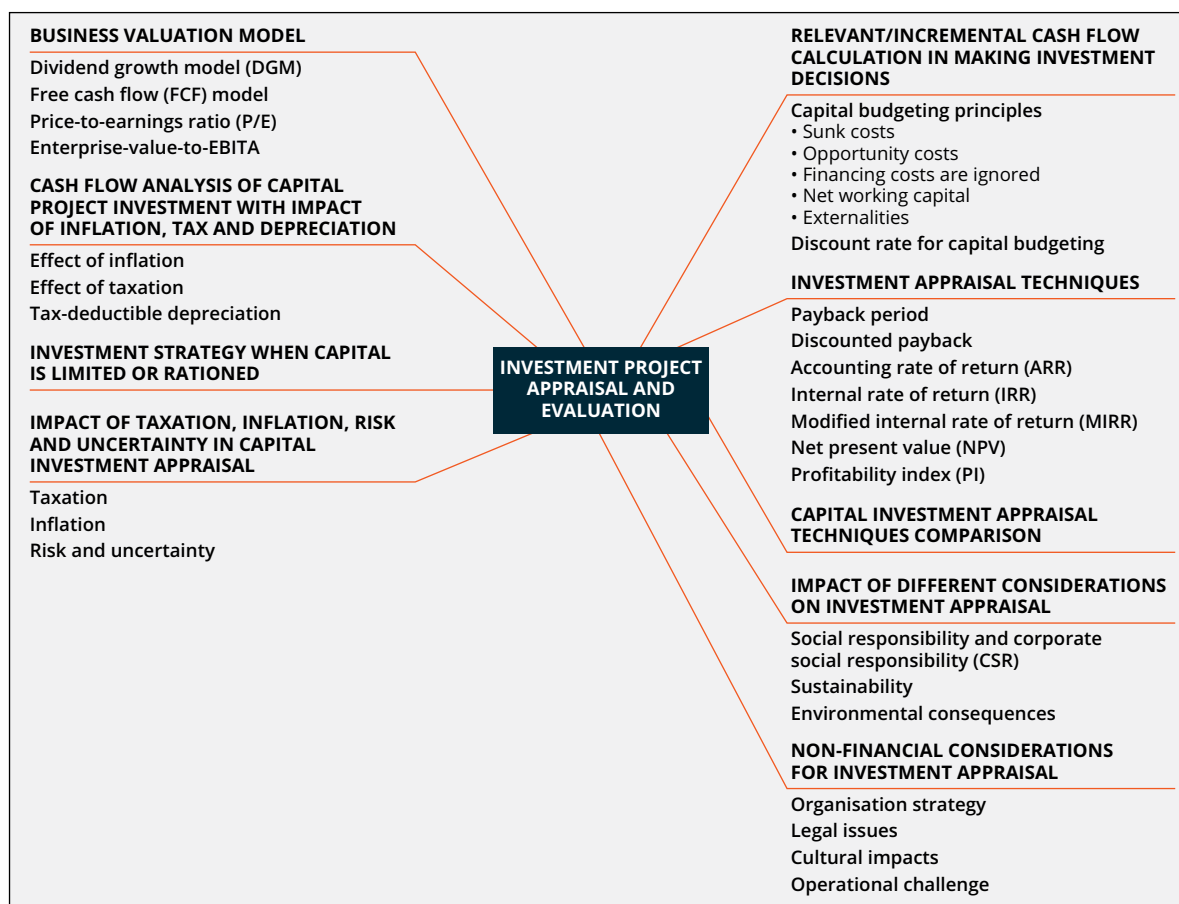
- Business valuation models are used to determine the economic value of a business and arrive at a fair value for the business.
- The dividend growth model is most commonly used to determine the value of the shares of a company by considering the growth of dividends over time.
- The free cash flow (FCF) model, as the name suggests, considers the cash flow that a company will generate to determine the value of shares. This approach is also useful in capital budgeting to determine the viability of a project based on the cash flows that the project will generate over time.
- The cash flows discounted in an NPV analysis are incremental after-tax free cash flows (FCF). The term incremental is used due to these cash flows measuring how much total after-tax free cash flows will change.
- The FCF equals the change in the company's cash income, excluding interest expense, that a project may generate, plus depreciation and amortisation for the project, minus all required capital expenditures and investments in working capital.
- It is possible to use relative valuation to estimate the value of common shares of a company by comparing the market values of similar companies, relative to a common variable, such as earnings, cash flow, book value or sales.
- The most commonly used relative valuation multiple is the price-to-earnings (P/E) ratio.
- It is possible to form multiples using earnings before interest and taxes (EBIT), or earnings before interest, taxes and amortisation (EBITA), or earnings before interest, depreciation and amortisation (EBITDA) instead of net income to eliminate a significant proportion of volatility caused in EPS figures using debt, and by depreciation and amortisation (which are non-cash expenses).
- Decisions involve choosing from alternative courses of action. The process of identifying the financial expenses that change between such alternative courses of action is incremental analysis.
- Capital budgeting decisions are important. They have a significant impact on a company's future profitability. Making poor capital budgeting decisions can cost the company a lot of money.
- Sunk cost are costs that have already been incurred and will not be changed or avoided by any future decision.
- Opportunity cost refers to the fact that, to take one action, companies must often give up the opportunity to benefit from some other action.
- The cash flows associated with financing a project (cash outflows or inflows to or from debtholders or shareholders) are not included in the calculation of FCFs because they are accounted for in the discount rate that is used in an NPV analysis.



- The dollar difference between current assets and current liabilities is called net working capital.
- Externalities are the side effects or consequences of an economic activity, such as an investment, that affects other parties in positive or negative ways, without being reflected in the cost of the goods or services involved.
- A question related to the application of net present value is to consider how a discount rate (or the required rate of return) is chosen in real capital budgeting decisions.
- Inflation can play an important role in determining these estimates because it will affect future levels of sales and expenses.
- Taxation and depreciation also have important implications in capital budgeting decisions.
- The planning process used to determine whether an organisation's long-term investments are worth funding is called capital budgeting or investment appraisal.
- The payback period is defined as the number of years required to fully recover the initial cash outlay. The discounted payback period considers the time value of money.
- The accounting rate of return (ARR) approach to appraisals of investments or capital projects calculates a project's return using accounting numbers – the project's net income (NI) and book value (BV) – instead of cash flow data.
- The internal rate of return (IRR) is defined as the discount rate that matches the present value of a project's expected cash inflows with the present value of its outflows. The modified internal rate of return (MIRR) considers that a project has positive cash flows and that those cash flows are reinvested at the same cost of capital of the firm.
- The net present value (NPV) decision rule states that a project proposal is acceptable if its net present value is zero or positive. If the net present value is negative, the project is unacceptable.
- The profitability index (PI) is another discounted cash flow (DCF) approach used to evaluate capital expenditure decisions.
- The payback period method provides a useful intuitive measure of how long it takes to recover an investment, and it is sometimes used as an informal measure of project risk, especially for short-lived projects.
- The discounted payback period approach alleviates one of the shortcomings of the payback period technique by accounting for the time value of money.
- Like the payback period approach, the ARR is easy to compute and comprehend for people without a strong finance background.
- Unconventional cash flows can cause a conflict between the NPV and IRR decision rules.
- A project with a conventional cash flow has an initial cash outflow followed by one or more future cash inflows, that is, after an initial investment is made (a cash outflow), all the following cash flows are positive (cash inflows).
- There is a clear link between the PI and the NPV.
- Theoretically, companies should accept all independent projects that generate positive NPVs, which enhance company value.

- Business decisions, such as to launch a new product, replace equipment or buy physical assets for sale in the future, will affect the taxable income of a company.
- As capital expenditures usually involve the estimation of cash flows several years in the future, inflation can play an important role in determining these estimates.
- The market value of any company in an efficient market is the present value of its expected incremental cash flows, discounted at an appropriate risk-adjusted discount rate, which is based on the overall riskiness of the project.
- One of the more subjective considerations for management when deciding whether to undertake an investment has to do with social responsibility and corporate social responsibility (CSR).
- Any direct costs, such as the cost of mitigation or pollution control devices, should be included in the cash flows of the project because they would be necessary costs related to the investment.
- Sustainability considerations are also important when making investment decisions.
- Some key non-financial factors to consider when making investment decisions include future legislation, industry standards; staff morale and retention, relationships with suppliers and customers; reputation, skills-building and impact on a company's strategy.
- Investment appraisal is key to the organisation strategy of any company and the opposite is also true: how specific investments fit within the strategy of an organisation should also be considered by decision makers.
- The legal issues associated with investment can be significant.
- The cultural impact of an investment is also an important qualitative consideration for companies.
- Investment projects should also be assessed considering the operational challenges they present for a company.

MIND MAP



LIST OF FORMULAS

Valuation under dividend growth model

$$P_0 = \frac{D_1}{R - g}$$

Free cash flow

$$FCF = [(Revenue - Op\ Ex - D \& A) \times (1 - t)] + D \& A - Cap\ Exp - Add\ WC$$

Valuation under price-to-earnings ratio

$$P_0 = Estimated\ EPS_1 \times Justified\ P/E$$

Accounting rate of return

$$ARR = Average\ net\ income / Average\ book\ value$$

Internal rate of return

$$\begin{aligned} \text{NPV} &= \text{NCF}_0 + \frac{\text{NCF}_1}{1 + \text{IRR}} + \frac{\text{NCF}_2}{(1 + \text{IRR})^2} + \dots + \frac{\text{NCF}_n}{(1 + \text{IRR})^n} \\ &= \sum_{t=0}^n \frac{\text{NCF}_t}{(1 + \text{IRR})^t} = 0 \end{aligned}$$

Modified internal rate of return

$$\text{MIRR} = \sqrt[n]{\frac{\text{FV}(\text{Positive cash flows} \times \text{Cost of capital})}{\text{PV}(\text{Initial outlays} \times \text{Financing cost})}} - 1$$

Profitability index

$$\text{PI} = \text{PV}(\text{cash inflows}) / \text{PV}(\text{cash outflows})$$

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. Dividends are allowed to show no growth.

Answer B is incorrect. Dividends are allowed to decline at a constant rate.

Answer C is correct. The equation is $P_0 = D_1 / (R - g)$. If the required rate of return is less than the growth rate, the denominator would become negative, and the computed share price would also become negative. Share price cannot be negative.

Dividends do not have to grow or could grow at less than the required rate of return. Dividends could also decline at constant rate. All these assumptions could still produce valid share price based on the formula.

Answer D is incorrect. The required rate of return must be greater than the growth rate.

Question 2

Answer D is correct.

Value = $15 \times \text{HK\$}3 \text{ million} = \text{HK\$}45 \text{ million}$. PE ratio times earnings would give you equity valuation.

Question 3

Answer A is incorrect. Revenues and costs may change.

Answer B is correct. In incremental analysis, variable costs may remain unchanged despite the adoption of an alternative course of action, and fixed costs may change even without the adoption of an alternative course of action. We are interested in financial data that would change whether they are originally variable or fixed. The data that do not change are ignored.

Answer C is incorrect. Revenues and costs may or may not change.

Answer D is incorrect. The magnitude of change is not preset to a particular percentage.

**Question 4**

Answer A is incorrect. The old equipment does not require new spending, therefore it is not an opportunity cost.

Answer B is correct. The old machine's book value is a sunk cost and so does not affect the decision. As sunk costs cannot be changed, they cannot increase the existing cost burden in anyway, therefore they are neither incremental nor marginal costs. Nor are they an opportunity cost as the money has been spent to acquire the machine in the past. The money spent is no longer available to finance other investment opportunity.

Answer C is incorrect. The old equipment does not require new spending, therefore it is not an incremental cost.

Answer D is incorrect. The old equipment does not require new spending, therefore it is not a marginal cost.

Question 5

Answer A is incorrect. Net income would not necessarily increase because the unprofitable segment may have shouldered some fixed overhead costs.

Answer B is incorrect. The variable expenses should disappear.

Answer C is correct. When deciding if they should eliminate an unprofitable segment, management should always choose the option which leads to the highest net income. Fixed costs allocated to the unprofitable segment must usually be absorbed by other segments. Only variable cost would be eliminated. It is possible, therefore, for net income to decrease when an unprofitable segment is eliminated. Hence there is no guarantee net income would always increase.

Answer D is incorrect. Net income may increase or decrease.

Question 6

Answer A is incorrect. Sales are affected by inflation.

Answer B is incorrect. Expenses are affected by inflation.

Answer C is correct. Inflation affects the different elements of a project's cash flows, except for depreciation, which is fixed in amount.

Answer D is incorrect. Cost of capital is affected by inflation.

Question 7

Answer A is correct. Depreciation is fixed, but the interest rate will reflect the increased inflation, the benefit from depreciation will be reduced.

Answer B is incorrect. The present value of tax shields would be reduced.

Answer C is incorrect. The present value of tax shields would be reduced.

Answer D is incorrect. The present value of tax shields would be reduced even if inflation remains below 5%.

Question 8

Answer B is correct.

The additional annual depreciation is HK\$100,000/8 = HK\$12,500. The depreciation tax savings is 0.30 (HK\$12,500) = HK\$3,750. The change in project NPV is:

$$-100,000 + \sum_{t=1}^8 \frac{3,750}{(1.10)^t} = -\text{HK\$}79,994$$

Question 9

Answer D is correct. The total PV is zero if the discount rate is 28.79%.

Year	Cash flow	Present value			
		28.19%	28.39%	28.59%	28.79%
0	-150,000	-150,000	-150,000	-150,000	-150,000
1	100,000	78,009	77,888	77,767	77,646
2	120,000	73,025	72,798	72,572	72,346
Total		1,034	686	338	-8

A more precise IRR of 28.7854% has a total PV closer to zero.

Question 10

Answer A is correct. A higher discount would lower NPV, hence make it less likely that the project will be accepted. Higher rate is generally used for more risky projects. The length of useful life has no impact on the levels of discount rate.

Answer B is incorrect. Higher discount rate would lower NPV, making it less likely for the project to be accepted.

Answer C is incorrect. A lower discount rate should be used for a less risky project.

Answer D is incorrect. The discount rate is not determined by the length of useful life.

Question 11

Answer A is incorrect. It uses cash-based accounting numbers.

Answer B is incorrect. This is not the only weakness.

Answer C is incorrect. This is not the only weakness.

Answer D is correct. Time value of money is ignored. It also ignores the useful life of alternative projects that may be more attractive in making the investment decision.

Question 12

Answer A is incorrect. This would cause a negative NPV.

Answer B is correct. It is only when the rate of return is higher than the required rate of return the NPV is positive, and the project would be acceptable.

Answer C is incorrect. This would cause zero NPV.

Answer D is incorrect. This would imply a negative NPV.

Question 13

Answer A is incorrect. The company's required rate of return should be higher than zero.
 Answer B is incorrect. The company's required rate of return should be higher than the rate on a government bond.

Answer C is correct. IRR must be sufficiently high to generate positive NPV for a project to be accepted. This would happen when IRR exceeds the company's required rate of return. The company's required rate of return should be higher than the return on government bond, and on the cost of its borrowed funds.

Answer D is incorrect. The company's required rate of return should be higher than its borrowing cost.

Question 14

Answer A is the correct. This is an incorrect statement. The profitability index should have been $1.6 \text{ (HK\$10 million + HK\$6 million) / HK\$10 million} = 1.6 \div \text{HK\$10} = 1.6$.

Question 15

NPV is the net present value of a project, while IRR is the return generated by the project that equates investment return with investment cost. Ranking by NPV means you accept projects based on the size of their NPV which may bear little resemblance to ranking based on the levels of IRR of projects.

Question 16

Answer A is incorrect. $\text{NPV}_x = \text{HK\$283}$; $\text{NPV}_y = \text{HK\$351}$. If only one is selected, project Y should be selected because it has larger NPV.

Answer B is incorrect. $\text{NPV}_x = \text{HK\$283}$; $\text{NPV}_y = \text{HK\$351}$. If investment decision is independent, both projects should be selected because they have positive NPV.

Answer C is correct. $\text{NPV}_x = \text{HK\$283}$; $\text{NPV}_y = \text{HK\$351}$. Both have positive NPV so could be accepted. If only one is selected, project Y should be selected because it has a larger NPV.

Answer D is incorrect. $\text{NPV}_x = \text{HK\$283}$; $\text{NPV}_y = \text{HK\$351}$. If investment decision is independent, both projects should be selected because they have positive NPV. If only one is selected, project Y should be selected because it has a larger NPV.

Question 17

The problem arises because some expenses (e.g. depreciation) do not change with inflation.

Question 18

Whether inflation makes a project more or less attractive depends on how inflation affects the company (e.g. income vs expenses).

Question 19

These include environmental concerns, consumer protection, human rights and workplace and social diversity. For many, this includes avoiding investments in organisations involved with the sale of alcohol or tobacco, and even extended to ensure that no funding finds its way to production of weapons, pornography, abortion, gambling, fossil fuel sales or military efforts.

Question 20

This concept becomes a management philosophy in which the company sets the objective of maximising economic benefits while also pursuing social and environmental values, in a way that can become a permanent and long-lasting set of policies.

Question 21

Answer A is correct. These items include some of the most likely legal issues that arise.

Answer B is incorrect. These challenges should be easily determined by attorneys early in the appraisal process.

Answer C is incorrect. Provisions for contingencies are normally part of financial considerations in investment appraisal.

Answer D is incorrect. Dismissal of attorneys is not part of investment appraisal.

Question 22

Non-financial objectives may include growth in a customer base or market; improvements in brand loyalty; development of new products based on CSR; and involvement in community projects or initiatives in which the organisation strategy is coordinated with a social cause.

EXAM PRACTICE

QUESTION 1

Identify which of the following is correct for the dividend growth model.

- A. Time horizon must be limited to a finite number of years.
- B. The further in the future a cash flow occurs, the greater its present value.
- C. Current price of a share is the next period dividend divided by the difference between the discount rate and the dividend growth rate.
- D. The model is valid when the growth rate is more than the discount rate or required rate of return.

QUESTION 2

Identify which of the following statements is correct about the investment appraisal technique.

- A. Conceptually and mathematically, payback period is as complicated as the net present value and the internal rate of return.
- B. The IRR can be defined as the discount rate that equates the present value of a project's expected cash inflows to the present value of the project's outflows.
- C. The accounting rate of return calculates the return on a capital project using cash flows data.
- D. Profitability index is defined as the ratio of a project's accounting net incremental after-tax cash inflows divided by the accounting cash outflows.



QUESTION 3

The Tsuen Mun Company Ltd is expected to pay a HK\$1.50 dividend next year. Dividends are expected to grow at 3% forever, and the required rate of return is 7%.

- (a) Calculate the following:
 - (i) The value of a share of Tsuen Mun stock today.
 - (ii) The expected dividend yield.
 - (iii) The expected capital gains yield.
- (b) Calculate the following: In one year, immediately after the dividend is paid,
 - (i) The value of a share of the stock.
 - (ii) The one-year holding period return.
 - (iii) The expected dividend and capital gains yields looking forward one year.
- (c) Calculate the following: In year 10, immediately after the dividend is paid,
 - (i) The value of a share of the stock.
 - (ii) The one-year holding period return (year 9 to 10).
 - (iii) The expected dividend and capital gains yields looking forward one year.

QUESTION 4

Bahrat Inc produces three separate products from a common process costing HK\$100,000. It has an option to sell each product at a split-off point or to process it further and then sell it for more, according to the following schedule:

	Sales value at split-off point	Cost to process further	Sales value after further processing
Product 12	HK\$50,000	HK\$100,000	HK\$190,000
Product 14	HK\$10,000	HK\$30,000	HK\$35,000
Product 16	HK\$60,000	HK\$150,000	HK\$220,000

- (a) Calculate the total net income if all products are sold at the split-off point.
- (b) Calculate the total net income if all products are sold after further processing.
- (c) Using incremental analysis, identify which products should Bahrat sell at the split-off point and which should it process further.
- (d) Use the results for (c) to calculate Bahrat's total net income; explain why its net income differs from that calculated for (b).

QUESTION 5

The Richmond Company has a project that requires the purchase of equipment costing HK\$90,000 and additional raw material inventory of HK\$2,000. Richmond's management estimated that the project would generate revenues of HK\$58,000 and operating expenses of HK\$30,000 each year. At the end of six years, the equipment will be worthless, and the company will need to pay HK\$5,000 for its disposal. Any additional inventory that was tied up will be released at the end of the project's life. The equipment annual depreciation

is 33.33%, 44.45%, 14.81%, and 7.41%. The tax rate = 40%, and the project's cost of capital is 15%.

If the revenues increase at a rate of 3% per year and the operating expenses increase at a rate of 5% each year, calculate the net present value of this project.

QUESTION 6

Unique Food Manufacturing Company has three new projects to consider. An equipment investment of HK\$25,000 is required for each project which will last for three years, and will produce the following net annual cash flows:

Year	Sweet	Sour	Spicy
1	HK\$7,000	HK\$9,600	HK\$13,000
2	9,000	9,600	9,000
3	12,000	9,600	11,000
Total	HK\$28,000	HK\$28,800	HK\$33,000

Unique uses straight-line depreciation, the equipment's salvage value is expected to fall to zero and will not accept any project with a payback period longer than two and a half years. The expected required rate of return of is 12%.

- Calculate the payback period for each project. Identify which project offers the best prospects, and which the worst. (Round your answer to two decimal points.)
- Calculate the net present value for each project. Identify which project looks to be the best and the worst. (Round your answer to the nearest dollar.)

QUESTION 7

The cash flows for the projects A through D are as follows:

Project end-of-year cash flows (in HK\$ millions)				
Year	A	B	C	D
0	-10	-10	-10	-10
1	2	0	0	5
2	2	0	0	0
3	2	0	6	2
4	2	0	0	0
5	4	13	6	5

The cost of capital for each project is 5%.

- Calculate the IRR for each project.
- Calculate the NPV for each project.
- Identify whether the project or projects should be accepted when the projects are independent.
- Identify whether the project or projects should be accepted when the projects are mutually exclusive.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. Perpetuity in dividend flow is allowed by the model.

Answer B is incorrect. IRR distant dividends have a small present value and contribute very little to the share price by the application of discount rate.

Answer C is correct. The difference between the discount rate and dividend growth rate is used in calculating current price.

Answer D is incorrect. The model is meaningless when the growth rate is more than the discount rate or required rate of return.

QUESTION 2

Answer A is incorrect. Payback period is much simpler to compute because it calculates cumulative cash flows over time. There is no discounting involved.

Answer B is correct. IRR accounts for the time value of money to equate the present value of cash inflow with the present value of cash outflow.

Answer C is incorrect. The accounting rate of return calculates the return using accounting results – the project's net income (NI) and book value (BV) – rather than cash flows data.

Answer D is incorrect. PI uses discounted net incremental after-tax cash inflows and discounted cash outflows in computation.

QUESTION 3

(a) Values today

Value today	HK\$37.50	←	$\text{HK\$1.50} / (0.07 - 0.03)$
Value next year	HK\$38.63	←	$\text{HK\$1.545} / (0.07 - 0.03)$
Dividend yield	4.00%	←	$\text{HK\$1.5} / \text{HK\$37.5}$
Capital yield	3.00%	←	$(\text{HK\$38.63} / \text{HK\$37.5}) - 1$
Year two dividend	HK\$1.545	←	$\text{HK\$1.50} \times 1.03$

(b) Values in one year

Value at end of one year	HK\$38.63	←	$\text{HK\$1.545} / (0.07 - 0.03)$
Holding period return	7.00%	←	capital yield 3% + dividend yield 4%
Year two end price	HK\$39.78	←	$\text{HK\$1.59135} / (0.07 - 0.03)$
Year three dividend	HK\$1.59135	←	$\text{HK\$1.50} \times 1.03^2$
Capital yield	3%	←	$(\text{HK\$39.78} / \text{HK\$38.63}) - 1$
Dividend yield	4%	←	$\text{HK\$1.59135} / \text{HK\$38.63}$

(c) Values at the end of year 10

Value at end of year 9	HK\$48.93	←	$\text{HK\$1.9572} / (0.07 - 0.03)$
Value at end of year 10	HK\$50.40	←	$\text{HK\$2.0159} / (0.07 - 0.03)$
Holding period return	7.00%	←	Capital yield 3% + Dividend yield 4%

Year 10 dividend	HK\$1.9572	←	$\text{HK\$1.50} \times 1.03^9$
Year 11 dividend	HK\$2.0159	←	$\text{HK\$1.50} \times 1.03^{10}$
Capital yield	3%	←	$(\text{HK\$50.40} / \text{HK\$48.93}) - 1$
Dividend yield	4%	←	$\text{HK\$2.0159} / \text{HK\$48.93}$

QUESTION 4

(a)	Sales ($\text{HK\$50,000} + \text{HK\$10,000} + \text{HK\$60,000}$)	HK\$120,000
	Less: Joint costs	100,000
	Net income	HK\$20,000

(b)	Sales ($\text{HK\$190,000} + \text{HK\$35,000} + \text{HK\$220,000}$)	HK\$445,000
	Joint costs	(100,000)
	Additional costs ($\text{HK\$100,000} + \text{HK\$30,000} + 150,000$)	(280,000)
	Net income	HK\$65,000

(c)	Product 12	Product 14	Product 16
Rev of final product	HK\$190,000	HK\$35,000	HK\$220,000
Revenue at split-off	50,000	10,000	60,000
Incremental revenue	140,000	25,000	160,000
Less: Incremental costs	100,000	30,000	150,000
Incremental profit (loss)	HK\$40,000	HK\$(5,000)	HK\$10,000

Products 12 and 16 should be processed further and product 14 should be sold at the split-off point.

(d)	Sales ($\text{HK\$190,000} + \text{HK\$10,000} + \text{HK\$220,000}$)	HK\$420,000
	Joint costs	(100,000)
	Additional costs ($\text{HK\$100,000} + \text{HK\$150,000}$)	(250,000)
	Net income	HK\$70,000

Net income is HK\$5,000 ($\text{HK\$70,000} - \text{HK\$65,000}$) higher in (d) than in (b) because product 14 is not processed further, thereby increasing overall profit HK\$5,000.

QUESTION 5

Initial cost:	HK\$90,000
Salvage:	-HK\$5,000
Useful life:	6
Tax rate:	40%
Project cost of capital:	15%

Year	0	1	2	3	4	5	6
Initial outlay	–HK\$90,000						–HK\$5,000
Tax on sale of asset							HK\$2,000
Change in working capital	–HK\$2,000						HK\$2,000
Investment cash flow	–HK\$92,000	HK\$0	HK\$0	HK\$0	HK\$0	HK\$0	–HK\$1,000
Change in revenues		HK\$58,000	HK\$59,740	HK\$61,532	HK\$63,378	HK\$65,280	HK\$67,238
Change in operating expenses		HK\$30,000	HK\$31,500	HK\$33,075	HK\$34,729	HK\$36,465	HK\$38,288
Change in depreciation		HK\$29,997	HK\$40,005	HK\$13,329	HK\$6,669	HK\$0	HK\$0
Change in taxable income		–HK\$1,997	–HK\$11,765	HK\$15,128	HK\$21,980	HK\$28,814	HK\$28,949
Change in taxes		–HK\$799	–HK\$4,706	HK\$6,051	HK\$8,792	HK\$11,526	HK\$11,580
Change in income after tax		–HK\$1,198	–HK\$7,059	HK\$9,077	HK\$13,188	HK\$17,289	HK\$17,370
Add: depreciation		HK\$29,997	HK\$40,005	HK\$13,329	HK\$6,669	HK\$0	HK\$0
Operating cash flow		HK\$28,799	HK\$32,946	HK\$22,406	HK\$19,857	HK\$17,289	HK\$17,370
Net cash flow	–HK\$92,000	HK\$28,799	HK\$32,946	HK\$22,406	HK\$19,857	HK\$17,289	HK\$16,370

Net present value is –HK\$287.41. In the case of higher inflation for operating expenses, compared to revenues, the project is not acceptable.

QUESTION 6

- (a) Based on average annual cash flows the project payback periods are:

$$\text{Sweet} - 2 + 9,000/12,000 = 2.75 \text{ years}$$

$$\text{Sour} - 2 + 5,800/9,600 = 2.60 \text{ years}$$

$$\text{Spicy} - 2 + 3,000/11,000 = 2.27 \text{ years}$$

The most desirable project is Spicy with the shortest payback period. The least desirable project is Sweet with the longest payback period. As shown, only Spicy is acceptable as it has a cash payback of less than 2.5 years.

- (b) Net present value of Sweet:

Year	Discount factor 12%	Amount	Present value
1	0.89286	HK\$7,000	HK\$6,250
2	0.79719	9,000	7,175
3	0.71178	12,000	8,541

Year	Discount factor 12%	Amount	Present value
Present value of cash inflows			HK\$21,966
Capital investment			25,000
Net present value			(HK\$3,034)

Net present value of Sour:

Year	Discount factor 12%	Amount	Present value
1-3	2.40183	HK\$9,600	HK\$23,057
Present value of cash inflows			23,057
Capital investment			25,000
Net present value			(HK\$1,943)

Net present value of Spicy:

Year	Discount factor 12%	Amount	Present value
1	0.89286	HK\$13,000	HK\$11,607
2	0.79719	9,000	7,175
3	0.71178	11,000	7,830
Present value of cash inflows			HK\$26,612
Capital investment			25,000
Net present value			HK\$1,612

Project Spicy is still the most desirable project. The other projects should be rejected because of the negative net present values.

QUESTION 7

Project end-of-year cash flows (in HK\$ millions)				
Year	A	B	C	D
0	-10	-10	-10	-10
1	2	0	0	5
2	2	0	0	0
3	2	0	6	2
4	2	0	0	0
5	4	13	6	5

- | | | | | | |
|-----|--------------------------------|------------|------------|-------------|------------|
| (a) | IRR | 5.732% | 5.387% | 4.691% | 6.500% |
| (b) | NPV | HK\$0.23 m | HK\$0.19 m | -HK\$0.12 m | HK\$0.41 m |
| (c) | If independent, accept? | Yes | Yes | No | Yes |
| (d) | If mutually exclusive, accept? | No | No | No | Yes |

14

Post-Appraisal Audit of Projects

CHAPTER TOPIC LIST

14.1 Post-Appraisal/Post-Completion Audit of the Success of a Project

14.1.1 Did This Project Exceed, Meet or Fall Short of the Goal?

14.1.2 Lessons Learned from This Project

14.2 Steps to Carry Out a Post-Completion Audit

14.2.1 Team Evaluation

14.2.2 Using Criteria Established at the Start of the Project to Ascertain if the Project Was a Success or Not

14.2.3 Project Scorecard

14.3 Management Behaviour of Post-Completion ('Neutrality' Until Done)

14.4 Establish Regular Times and Processes for Investments – Including Post-Completion Evaluation and Documentation



LEARNING OUTCOMES

PRINCIPAL LO4: ANALYSE THE STRATEGIC MANAGEMENT ACCOUNTING FRAMEWORK

LO4.03: Explain the results of a post-appraisal audit of projects
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4.03.01 Explain the role of a post-appraisal (or post-completion) audit in assessing the success of a project

4.03.02 Explain the steps in carrying out a post-appraisal audit of projects
--

4.03.03 Explain management behaviour in a post-appraisal audit of projects
--

4.03.04 Explain an entity's system for the post-appraisal audit of projects



OPENING CASE

ABC INVESTMENT LTD: A CRITICAL REVIEW

A BC Investment Ltd has completed several projects during the past year. These projects had mixed results. A post-project appraisal audit was revealing.

The purpose in performing a post-project audit has been to identify areas where projects succeeded and establish methods to repeat those results; and to also identify where goals were not met and the reasons why, to avoid similar problems happening again in the future.

The big question ABC asked was whether a project succeeded or failed (or came up short of goals), and what lessons were learned as a result. The failure to meet goals (financial and non-financial) led to team evaluation. The purpose in this procedure was to identify methods to improve internal project management and follow-through, and to improve team leadership. 

OVERVIEW

Identifying, analysing, approving and implementing are the first four of five steps involved in processing through capital investment projects. A fifth step is the follow-up or post-completion audit. Companies carry out these audits to determine whether their capital investment projects succeed or fail.

Following up is intended to identify whether a project succeeded or failed. The goal of these audits is to identify both positive and negative results, and to articulate the principles of how to best implement the objectives at the beginning of a project so that it can be monitored and evaluated. There are various steps to carry out a post-completion audit but the most effective of these audits include team evaluations and the use of established criteria to ascertain the success of a project.

This chapter also discusses the best possible approach for management to take while carrying out a post-completion project audit as well as the importance of setting up regular times and processes for such audits, including any post-completion evaluation and documentation that projects may generate.

14.1 POST-APPRAISAL/POST-COMPLETION
AUDIT OF THE SUCCESS OF A PROJECT

Companies use capital budgeting processes to make decisions on whether to invest in a **capital project**, such as a new plant or a new manufacturing line. Capital projects are typically long-term, lasting for more than a year.

Capital investments may represent significant resource, including cash outlays for companies and are crucial to the company's success. As a result, projects that have been completed need to be reviewed. It is also important to review these projects individually through a **post-completion appraisal** or **post-completion audit** to determine a project's overall success.

This post-completion audit is the final of five stages of any capital investment project. These five stages are:

1. Identification
2. Analysis
3. Selection
4. Implementation
5. Follow-up or post-audit



14.1.1 Did This Project Exceed, Meet or Fall Short of the Goal?

The aim of a post-completion audit – also known as a follow-up stage – is to determine if projects have met expectations.

In the unfortunate event that the results are disappointing, it may be necessary to end, modify or abandon projects. If an initial project introduced first in a new brand but it was not well received by consumers, this raises questions. Was the launch poorly prepared? Did the company fail to advertise and promote the product? Alternatively, was the product overpriced or did it fail to compete well?

In pricing, it may be that the factors of production at anticipated volume of output are simply too high to set prices competitively. In that case, abandoning future projects to expand the line would make the most sense.

Some problems such as this could be anticipated in advance. A typical capital budgeting process includes brainstorming for ideas, analysing the merits of the project, planning for the budget and financing and monitoring performance. The post-completion audit is part of this last step in the **capital project monitoring**.

As part of the post-completion audit, actual results are compared to the expected results. Differences are not unusual, but these differences must be explained. There are multiple areas for comparison. If assumptions were in error at the onset of the project, they must be revised to reflect the realities revealed by project variances. Assumptions are intended to accurately reflect the cost factors entering the project. If unexpected costs arise, or if actual costs are underestimated, this is valuable information. The company may be able to salvage a project by improving its assumption base, or it might conclude that there is no point in investing more capital if the assumptions are inaccurate.

For instance, actual expenses related to implementing the project must be compared to planned expenses. The same is true for revenues: is the project generating the expected revenues? What about **cash flows**?

The post-completion audit is necessary for multiple reasons, including:

- Monitoring the quality and accuracy of any forecasts, predictions and analyses, including the underlying assumptions used to develop these estimates.
- Any systematic errors, including overly optimistic forecasts, and the reason those forecasts were not corrected in advance.
- Tracking business operations associated with a capital investment project.
- Deciding whether sales or costs are out of line, and why.
- Generating new ideas for future projects.
- Providing a benchmark for the analysis of future projects.

Ultimately, a post-completion audit can be used to determine whether any pre-project assumptions were accurate and if outcomes were as expected.

14.1.2 Lessons Learned from This Project

Undertaking a post-completion audit tends to bring with it three main benefits.

The first is to find and identify lessons for future projects, including how to improve planning and execution and streamline strategic planning process.

The second benefit is to identify problems and find solutions. An audit done after the project could make it possible for a company to make changes and improve cash flows and revenues. Such information would also make it possible to abandon failing projects and limit losses.

A third benefit is to insert structure into the process of investment planning and control, particularly if managers know that such post-project audits will be done. Project managers are likely to be more careful in their project management if they know that their work will be carefully reviewed. This is especially relevant if the audit and review steps are planned at phases during project implementation. If a project is progressing away from clearly understood goals, interim audits will enable the company to make corrections or abandon projects immediately to avoid further losses.

By systematically reviewing and revising any initial forecasts with the help of actual outcomes, future estimates may improve. In turn, management can improve operations.

Illustrative Example 1

Your company has started a project to introduce a new brand. The belief in the beginning was that this brand would be accepted by the market, but revenues are far below expectations and you need to make possible recommendations.

Since this project has only recently been introduced, several considerations must be reviewed.

The company invested significant capital just to launch the project. However, this would not justify continuing the project if it appears to be failing. It makes more sense to cut costs immediately if the future does not appear likely to yield expected revenues and profits.

What are the initial assumptions about the new brand and how it was marketed? Did the company fail to inform the consumers well enough? Was initial product testing less than adequate? Is the pricing structure less competitive than it should be?

Your evaluation may lead to the conclusion that the project should be abandoned. Or it may lead to steps for improvement, which should be based on initial assumptions in comparison to actual results.

Key Learning Point

The aim of a post-completion audit – sometimes also known as a follow-up stage analysis – is to determine if projects meet expectations.

► Knowledge Check Questions

Question 1

Identify the possible reasons that projects may fail to meet expectations.

- A** Managers forget to audit during the project, leading to a lack of process.
- B** Project managers are poorly trained, project team members are unmotivated and management does not monitor adequately.
- C** The project was poorly prepared, marketing was not adequate, the price of a new product was not competitive and assumptions were too optimistic.
- D** Expectations were too high, and the business cycle was weakening, so most failures result from poor timing.

Question 2

Identify which of the following lists the three lessons to be learned from audit of projects.

- A** Future projects may be improved, problems and solutions are identified and structure is put into the process including interim audits.
- B** Success is difficult and does not usually occur, capital investment should be extremely limited and companies are likely to lose money on projects.
- C** Most marketing efforts fall short, budgeting and forecasting is difficult and most projects are not reviewed adequately.
- D** Capital investment is often too small to ensure profits, forecasting is often too optimistic and lack of follow-through is the most likely reason projects fail.

➤ 14.2 STEPS TO CARRY OUT A POST-COMPLETION AUDIT

Companies increasingly use post-completion audits to determine the success of their capital investment projects.

But some companies worry about the potential drawbacks of post-completion audits. One concern is that staff may be deterred from putting forward proposals for fear that the audit may show weaknesses in their proposals.

There is little evidence to back up these concerns. Rather, studies suggest post-completion audits are widely beneficial for companies.

A couple of considerations can strengthen these audits. One of them is the use of team evaluations as part of post-completion audits. Another is the use of consistent criteria to evaluate a project.

14.2.1 Team Evaluation

An effective budgeting process requires input from multiple departments in any business. The complexities of planning for a capital investment require involvement from multiple people inside and outside a company. A **team evaluation** of a project can be an effective approach to determining success.

A capital project typically requires input from marketing people, technical experts, regulatory staff, tax experts, finance experts and more. All these resources typically provide input to the board that makes the ultimate decision.

For smaller projects requiring less investment, lower-level managers may have discretion. Larger and more capital-intensive projects require input from top management and the board. In all cases, however, capital budgeting is an exercise in balancing costs and benefits.

Management and staff are both keenly and almost immediately interested in the success of projects. Anecdotal evidence is useful from tangible proof of the effectiveness of a new project. A post-completion audit can help make the determination of whether a project is successful.

The audit is a top-to-bottom evaluation of a project. For instance, such an appraisal can help determine whether a company is achieving its objectives and may identify lessons learned from the implementation of the project. Despite the benefits these evaluations provide, companies are often reluctant to carry out a **preliminary impact assessment (PIA)** because of the cost and human resources required.

Experience has helped companies identify some important factors in the success of a post-completion audit, including:

- performing team evaluations with the right people involved;
- timing of the audit; and
- quality of documentation.

Team evaluations are important components of post-completion audits.

Team evaluations may be performed by a committee or individuals, based on standards for the determination of results. These standards should include critical analysis of:

1. *Did the team understand goals and objectives?* At the beginning of the project the entire team should understand what they are expected to accomplish.
2. *Does the entire team agree on its goals and objectives?* Interviews may reveal that despite orientation, team members are not in full agreement.
3. *Was the team leadership effective and successful?* Interviews of team members may reveal flaws in leadership, including lack of giving clear directions, identifying end results and setting interim deadlines.
4. *Were team conflicts dealt with by leaders?* Conflicts may arise, and leaders must be able to deal with them to resolve differences. If conflicts can persist, the team cannot achieve its goals.



5. *Was leadership effective in identifying problems and making decisions?* Team members will be able to articulate whether the team leader recognised problems or was unaware of them; and whether the leader can make decisions.
6. *Was the overall process effective?* Do team members believe that leaders presented clear instructions, clear goals and positive incentives? Or was the process lacking due to poor leadership?
7. *Were procedures clearly expressed and enforced?* The team operates best when operating procedures are explained and enforced. The lack of procedures or enforcement may lead to poor results including project failure.
8. *Was team communication effective?* Teams work only if members can communicate with one another, and with the leader. Regular and direct communication leads to successful completion; minimum communication or lack of communication make poor team functions.
9. *Did the team use resources well?* Every team relies on internal resources and, in many instances, on external resources. Are these resources known by the team? Did they maximise those resources effectively?
10. *Was there trust among team members?* A team whose members trust one another will be effective, but low levels of trust prevent the team from operating efficiently or effectively.

14.2.2 Using Criteria Established at the Start of the Project to Ascertain if the Project Was a Success or Not

Was the project successful? What criteria are applied to make this determination?

It is necessary to establish early on the criteria that companies will use to determine whether a capital investment project is successful or not. Without these criteria, a company will not be able to determine whether the project met expectations.

At the time of the post-completion audit, these same criteria can and should be used to determine whether a project yields the expected benefits.

The process of determining the correct evaluation criteria at the start of the project makes it possible for companies to improve controls of projects as well as their decision-making and planning processes. Because of these benefits, more companies are using post-completion audits.

Measurements of a project's success can be determined in its most basic form by comparing initial assumptions going into a forecast, to actual outcomes. Did the project meet or exceed revenues and cash flows? Was it completed on time?

On a more detailed level, a list of key questions can be developed and responded to with a 'yes' or 'no' choice. However, this method is not conclusive. If you have 10 questions and 6 were 'yes' what does that mean? How crucial were the other four questions? It may be necessary to weigh each of the questions based on importance; even so, this simple technique may serve only as a starting point for post-project evaluation, but it will not be conclusive.

A second method for evaluation is continuous measurement. Under this system, criteria are limited in scope but clearly expressed and measured in some manner. Using a scale is effective. For example, if a 1-to-10 scale is used, with 10 the best result and 1 the worst, continuous measurement highlights areas needing improvement. As a monitoring system, this is preferred over end-of-project methods, because it points out emerging problems before they become critical.

You may also use a baseline performance system. By setting minimally acceptable results in a list of areas, results are compared to the baseline. This points out where improvement is needed, so this will be most effective if performed at junctures within the project. For example, when a two-year project is broken down into six segments and a baseline evaluation is performed at the end of each phase.

Post-project evaluation (or post-phase and continuous measurement) is based on several criteria. These are:

- Do the outcomes measure and quantify what matters to management and to stakeholders?
- What does success appear as, and how do you know when it has been achieved?
- Does everyone agree on the definition of a project's success?
- Have you employed continuous measurement to track and audit the project?
- Are you comparing outcomes to a baseline?

Illustrative Example 2

You have been assigned a project review for five current projects in your company. Using continuous measurement, you have to articulate the key elements to audit.

Continuous measurement requires monitoring and analysis during the project. This means that dividing it into identifiable phases makes the most sense. You can apply the standards against each phase to identify areas where improvement is needed.

Typical tests may include budgeting and forecasting, schedule of completion for each phase, team leadership and effectiveness and effective team communication.

Key Learning Point

An effective budgeting process requires input from multiple departments in any business. This is because the complexities of planning for a capital investment require involvement from multiple people inside and outside a company. A team evaluation of a project can be an effective approach to determining success.

► Knowledge Check Questions

Question 3

Identify which of the following explains why continuous measurement is an effective method for auditing a project.

- A** It ensures that team leaders are not redefining project goals.
- B** This allows the company to evaluate completed phases and determine whether it is meeting budget and time goals.
- C** Performed at the end of the project, it can be used to set new standards for future projects.
- D** It provides the means for monitoring performance of the team to discover whether they are working or idle.

Question 4

Success criteria evaluate the extent to which a project is successful or not for a range of stakeholders. In judging the outcome of a project, they can be important as its business outcome. Well-chosen criteria can avoid subjectivity by establishing quantifiable ways of measuring performance. Briefly describe the following three areas of success factors and state an example for each area:

- (a) Deliverable success
- (b) Process success
- (c) Stakeholder success

14.2.3 Project Scorecard

It is very common in the industry to have an evaluation scorecard for assessment of a project to measure the business impact of project management for a firm. Project Management Institute (PMI) also recommends developing a scorecard to track ongoing project management performance. (PMI, a non-profit project management professional association promoting project management best practices, was founded in 1969.) In measuring project performance, an enterprise can use diverse performance measurement to manage strategy, systems, and processes more effectively and more consistently for improvement, which finally creates value and competitive advantage to the company in return.

When developing a scorecard, there is no single set of measures that universally applies to all organizations and the appropriate set of measures depends on the company's strategy, and the particular industry and environment in which they compete. According to Pennypacker, 2005, there are 10 commonly used performance measures, either in terms of financial aspects and non-financial measurements, for an enterprise to benchmark target to achieve project management success (<https://www.pmi.org/learning/library/create-project-management-scorecard-7237>):

1. Return on Investment: The most appropriate formula for measuring project success is the percentage return for every dollar that has been invested.

2. **Productivity:** Productivity measures production efficiency as reflected by the number of the outputs produced per unit of input.
3. **Cost of Quality:** There are four categories of cost of quality – prevention, appraisal, internal failure and external failure – arising from tremendous possible sources of cost or money losses related to poor quality.
4. **Cost Performance Index:** This is an index measuring cost efficiency and is determined by dividing the value of the work actually performed (i.e. the earned value) by the actual costs.
5. **Schedule Performance Index:** This is the ratio of comparing total authorized project duration and total final project duration to determine the project management efficiency.
6. **Customer Satisfaction:** Customer satisfaction reflects the extent of the consumer expectations that are met. The customer satisfaction index comprises hard measures of the number of buying or consuming behaviour and soft measures of customer opinions or feelings.
7. **Cycle Time:** Cycle time defines the length of time period it takes to complete a project (i.e. the project life cycle).
8. **Requirement Performance:** Meeting requirements is one of the key factors for measuring the success of a project.
9. **Employee Satisfaction Index:** An employee satisfaction index comprises a mix of soft and hard measures that are each assigned a weight based on their importance to measure employee morale and satisfaction level.
10. **Alignment to Strategic Business Goals:** This means that a company has to evaluate whether it is doing the right projects.

In a nutshell, these evaluation metrics measure the value of the project management to organizations by comparing the results with a benchmark to see if changes have to be made for improvement and to conclude whether it can create value to the company as a whole.



14.3 MANAGEMENT BEHAVIOUR OF POST-COMPLETION ('NEUTRALITY' UNTIL DONE)

The fifth stage of the post-completion audit, the follow-up, is sometimes called the control phase. At this stage, management tracks spending and the results of any capital budgeting projects and compares with the predetermined objectives and goals.

During implementation, firms review their spending. Managers overseeing projects should stay neutral in their approach to the implementation criteria used to carry out the project and during the post-completion audit. At many firms, spending is reviewed on an ongoing basis



during the implementation stage of a project through quarterly reports and compared to budgeted amounts.

Through this type of monitoring, management can stay on top of spending and pre-empt cost overruns while identifying any potential problems in project and implementation as well as their approach to the management of projects. Neutral management behaviour throughout the implementation of the project and afterwards minimises the negative impacts of post-completion audits. And a key potentially negative impact of these audits is the possibility of staff refraining from putting forward proposals out of concern about management reaction to proposals that do not go as planned.

Neutral management behaviour develops effective and accurate control systems that can pinpoint personal responsibility without pre-assigning blame. If one department proposes a capital project that could help a company lower costs but, if those cost reductions do not come to pass, department heads can be questioned regarding the project.

Controls of this type can help companies reduce project 'padding' or overestimates and control of any intra-firm agency problems that may exist. In turn, this increases motivation to help companies meet their goals.

Management behaviour helps eliminate biases leading to overestimates of cash flows or the benefits of a project. Management behaviour may also benefit from experience.

Neutral management behaviour is based on good quality data from a variety of sources. Such data can facilitate objective and fair decision making.

At times, a post-completion audit may lead management to decide to terminate or abandon a failing project, if necessary. The possibility of abandoning an investment expands the options available to management and minimises risk. Contingency planning of this kind gives decision makers more options when dealing with uncertainties. A manager's neutrality tends to maximise effectiveness as a project leader.

The desirability of a neutral position may be difficult to accomplish. Project managers often are granted responsibility for the project but are not given adequate levels of authority to enact processes aimed at improving outcomes. Managers must control the actions and interactions among the team, senior management, other project managers and customers. Do all these groups share a common definition of the goals and objectives of the project? Chances are they do not.

Each of the project participants and stakeholders may have individual goals and priorities. If a project begins to fail, the participants may tend to set blame to protect themselves. This is an unproductive tendency, but it does occur. For this reason, the neutral project management concept can be used to offset an otherwise self-serving motivation if the project is not proceeding on time or on budget. The sharing of well-defined project goals is one step a project manager can set up impartial, fair and effective operating environments.

Maintaining a neutral policy may be invisible to team members. A project manager may follow the neutrality model without overtly imposing it on the team. This is achieved through managers' creating good communication within the team and from the team to external stakeholders; identifying and eliminating work slowdown or weak links; recognition of risks and taking steps to eliminate or mitigate them; and elimination of interruptions to the work flow such as excessive or frequent meetings.

Managers will generate efficient and smooth processes by following these steps:

1. *Know the goals.* Ensure that all team members agree on the goals of the project. Any disputes should be articulated in the context of how they affect the successful completion of those goals.
2. *Give team members a positive environment.* Encourage team members by trying to give each member a sense of accomplishment and participation.
3. *Let team members do their work.* An effective manager can be neutral on several levels, including allowing team members to work without interference.
4. *Emphasise the right matters.* The structure of the project should be consistent and all priorities based on the known goals. This enables workflow, fairness and positive relationships on the team.
5. *Protect team members with safety standards.* For projects in high-risk environments, such as a factory floor, reduce risk of injury or damage to equipment with basic, common-sense safety standards.

Project managers are expected to offer strong and effective leadership. Knowledge of the conditions addressed by the project is essential. However, the managers should also be authorised to consult with experts when problems arise beyond their experience. By continually expanding the knowledge base of a project, managers can aim for successful completion within budget and on time.

Neutrality in project management is based on many basic management skills, including self-confidence, focus, self-control, consistency and conflict resolution skills.

Apply and Analyse 1

Apply

You have been assigned as project manager to improve productivity and eliminate slowdowns due to:

1. defects;
2. shortages of parts; and
3. lack of training for some employees.

Your team includes a shift foreman, a worker on the line, a risk management supervisor, an accountant and a marketing employee. A problem has developed, a personal conflict between the shift foreman and the risk management supervisor. Explain how you, as project manager, can resolve this problem.

Analysis

The first step must be to bring the team members together and reiterate the original goals of the project. Allow each employee to explain their differences and propose solutions in line with the goals of the project. Avoid personal discussions and encourage team members to focus on the desired end results.

As leader, you also need to set boundaries. You expect every team member to communicate effectively and to abandon personal goals in favour of the project's goals. Invariably, a personal conflict will be based on a failure of one or more team members to fully understand or recognise the project's goals and objectives.

Apply and Analyse 1 (continued)

Finally, expect the need to have the situation monitored. Since team members must work together, encourage them to express any differences in the context of the project and not the personalities of individual team members. As leader, you may need to mediate the interaction of the team members.

Key Learning Point

Management behaviour throughout the implementation of the project and afterwards is important to minimise the negative impacts of post-completion audits.

► Knowledge Check Questions**Question 5**

Identify which of the following lists the components of neutral management behaviour in project management.

- A** Managers should explain their rules and swiftly dispel any differences in opinion among team members.
- B** The project manager must be neutral in resolving conflicts between employees, but probably will have to resolve disputes based on rank.
- C** Components include defining implementation criteria, managing expenses and schedules and eliminate budget and cash flows bias.
- D** Neutral management means focusing on process and not results, if the goals and objectives of the project will succeed if processes are managed well.

Question 6

Discuss how individual behaviour may impact project management.



14.4 ESTABLISH REGULAR TIMES AND PROCESSES FOR INVESTMENTS – INCLUDING POST-COMPLETION EVALUATION AND DOCUMENTATION

Setting up regular processes to identify, evaluate, approve, implement and audit capital spending projects includes any post-completion audits that companies undertake.

In audits, management should follow specific procedures while considering industry principles, regulations and accounting standards. Companies should consider all these when

setting up processes for every one of the five steps involved in implementing a capital project. These can help companies address some of the common problems likely to arise within a capital project.

The initial budgeting process for capital investment projects can be used to identify the criteria for regular formal reviews to help managers constantly evaluate and re-evaluate projects and priorities. Like any budgeting process, the purpose of this budget is to find variances and, when possible, take steps to correct them, or re-evaluate the initial assumptions and make changes.

An important process is to calculate **return on investment (ROI)** during the post-completion audit and to compare the performance of one project against another. While a thorough ROI analysis can drain resources, it may also help prevent future expensive mistakes.

One challenge that companies can avoid by setting up regular processes is the delays and extra costs that come from too many review cycles and too much time spent in analysis. Evaluation processes performed at regular times throughout the year can help companies avoid delays and extra costs while making it more likely that projects hit their proposed targets. This process includes streamlining decision authority and minimising the number of people involved in decisions.

Regular processes also make it easier for data to flow among decision makers and facilitate the management processes that are based on these data.

Management might suffer from operating in silos, with different sections communicating little if at all with one another. This hampers the effectiveness of capital project implementation and management processes. One way to break this silo mentality is to put in place regular times and processes to evaluate projects as well as any post-project data and documentation that may be available. However, if the silo mentality is largely cultural within the organisation, breaking it is not a simple task.

Companies must deal with several potential challenges associated with capital investment projects including:

- unrealistic assumptions;
- a lack of reference to life cycle costs and management costs;
- unsustainable debt;
- lack of management capacity;
- lack of funding; and
- abuses or conflicts.

Management may improve how projects are carried out by adopting carefully considered and regular processes to manage capital projects.

Effective management of capital investment projects may improve cash flows and ensure the success of such projects. Multiple avenues open to companies to improve the management of capital projects include:

- Prioritising capital projects, which can impact company value by driving growth and improving returns.



- Relying on institutional memory and knowledge, sourcing ideas from across the company and putting in place collaborative systems and best practices.
- Establishing clear investment objectives and comparing current and past projects. This can help management avoid pet projects that may lack value or fail to meet net present value (NPV) targets.
- Regularly reviewing projects to ensure they stay true to the original rationale.
- Tracking ROI throughout the implementation phase and later during the post-completion audit.
- Streamlining approvals and making decisions using the best data available.
- Developing frequent forecasts and be open to change tactics.
- Avoiding or eliminating management silos.
- Focusing on continuous improvement.

To maximise the value of capital investments, companies should identify errors and constantly adapt their strategies to fit.

Explicit investment objectives can facilitate a culture of success.

Illustrative Example 3

As part of your project management experience, you have noticed a lack of communication in the past between accounting, marketing and risk management departments. Your current project calls on all three of these resources and you have to prevent a silo mentality.

A first step is to bring all members of the team together in the same initial meeting, with the purpose of sharing goals and objectives. Next, give each team member assignments and deadlines. Individual assignments can be broken down into phases. Pay special attention to those phases requiring communication between departments. By encouraging effective communication and teamwork, the silos can be eliminated or at least reduced for the duration of the project.

Key Learning Point

One challenge that companies can avoid by setting up regular processes is the delays and extra costs that come from too many review cycles and too much time spent in analysis.

► Knowledge Check Questions

Question 7

Identify the challenges commonly encountered in project management processes.

- A** Conflicting budgets between projects, management favouritism towards some departments and unrealistic schedules.
- B** Unrealistic assumptions, lack of management capacity and lack of funding.
- C** Lack of cash flows to see a project through to the end, no support from the marketing department for accounting-based projects and poor understanding of the process on the part of top management.
- D** Personal animosity between team members from different departments, team member sabotage of the process and one manager being assigned too many projects at the same time.

Question 8

Your company has just spent a year developing a new quality management system. Now you want to make sure it will perform as desired, and that others will be able to benefit from the lessons learned developing the system when they run similar projects. Describe how the project's success can be properly measured and list out the works towards continuous improvement.

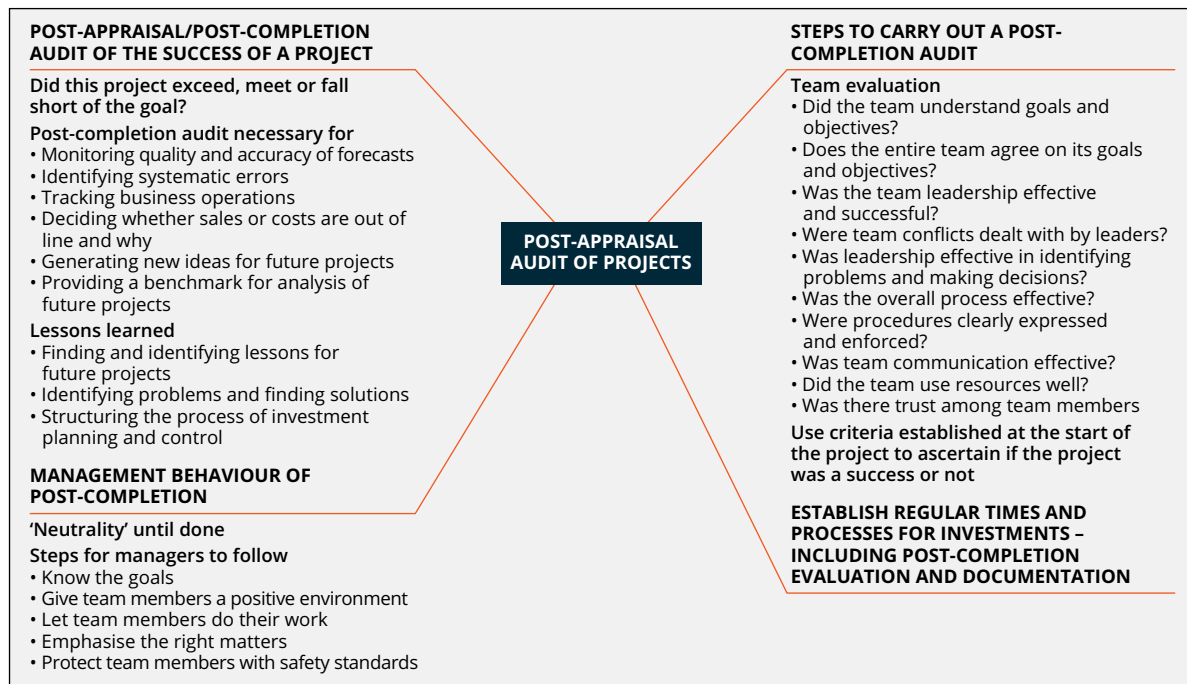


SUMMARY

- Companies use capital budgeting processes to make decisions on whether to invest in a capital project, such as a new plant or a new manufacturing line. Capital projects are typically long term, lasting for more than a year.
- A post-completion appraisal or post-completion audit can be used to determine the success of a project.
- The aim of a post-completion audit – sometimes also known as a follow-up stage analysis – is to determine if projects meet expectations.
- The post-completion audit is part of this last step in the capital project monitoring.
- As part of the post-completion audit, actual results are compared to the expected results. Differences are not unusual, but these differences must be explained. There are multiple areas for comparison.
- A post-completion audit can generate three main benefits: lessons for future projects, identifying problems and finding solutions and creating a structure for investment planning and control.
- Companies are increasingly using post-completion audits to determine the success of their capital investment projects.
- A team evaluation of a project can be an effective approach to determining success.
- A capital project typically requires input from marketing people, technical expertise, regulatory staff, tax experts, finance experts and more.
- The audit is a top-to-bottom evaluation of a project.
- Experience has helped companies identify some important factors in the success of a post-completion audit, including team evaluations, timing and quality of documentation.
- Consistency is important to both the success of the project and to evaluate whether a project is successful.
- The fifth stage of the post-completion audit, the follow-up, is sometimes called the control phase. At this stage, management tracks spending and the results of any capital budgeting projects.
- During implementation, firms review their spending. Managers overseeing projects should stay neutral in their approach to the implementation criteria used to carry out the project and during the post-completion audit.
- Neutral management behaviour throughout the implementation of the project and afterwards is important to minimise the negative impacts of post-completion audits.
- Also, important to ensure constant and neutral management behaviour is good quality data that comes from a variety of sources. Such data can facilitate objective and fair decision making.

- Setting up regular processes to identify, evaluate, approve, implement and audit capital spending projects is important. This includes any post-completion audits that companies undertake.
- The initial budgeting process for capital investment projects can be used to identify worthwhile projects while regular formal reviews help managers constantly evaluate and re-evaluate projects and priorities.
- Management may improve how projects are carried out by adopting carefully considered and regular processes to manage capital projects.
- To maximise the value of capital investments, companies should identify errors and constantly adapt their strategies to fit.

MIND MAP



► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. Lack of audit by itself is not a likely reason for project failure or results below expectations.

Answer B is incorrect. Training, motivation and monitor by themselves are not likely reasons for project failure or results below expectations.

Answer C is correct. Projects are likely to come in below expectations due to poor preparation, inadequate marketing, overpricing of products or optimistic but unrealistic assumptions.

Answer D is incorrect. Expectation management and business cycle by themselves are not likely reasons for project failure or results below expectations.

Question 2

Answer A is correct. The three lessons learned are how to improve in the future, identification of problems and solutions and the need for project structure and audit.

Answer B is incorrect. Capital allocation is not a main lesson learned from an audit.

Answer C is incorrect. Marketing and budgeting are not the main lessons learned from an audit.

Answer D is incorrect. Capital investment and forecasting are not the main lessons learned from an audit.

Question 3

Answer A is incorrect. Project goals could be redefined if necessary.

Answer B is correct. Continuous measurement is performed at the end-of-project phases and is used to compare progress with budget and timing standards.

Answer C is incorrect. It is performed at the end of each phase, not the end of the project.

Answer D is incorrect. Idle labour is not the main focus of continuous measurement.

Question 4

- (a) Deliverable success factors are focused on whether a project is able to solve the business problem it was set up to address.

Example: A project aimed at coming up with new ideas for internal innovation came up with ten schemes that participants felt were worth trialling.

- (b) Process success factors are focused on improving the way in which the project is managed, such as whether it meets its budget and timeline, initiates cross-department collaboration or results new management processes.

Example: A project aimed at encouraging collaboration between a company's business analysts and its R&D team led to monthly meetings to share ideas on improving financial efficiency.

- (c) Stakeholder success factors are focused on drawing support for projects from different groups with an interest in a business.

Example: A project aimed at finding new sources of customer loyalty by offering certain buyers of a company's goods or services with membership of a loyalty club with certain specified benefits.

Question 5

Answer A is incorrect. Neutrality does not require swift resolution.

Answer B is incorrect. Neutrality does not require resolution based on employee ranks.

Answer C is correct. Neutrality refers to managing the project based on budgets, schedules and well-articulated goals and objectives.

Answer D is incorrect. Neutrality is still concerned with results.

Question 6

A successful project is one that meets its original scope, time, cost and quality targets. Bad project management is detrimental to the success of a company and can pose significant challenges, especially when it affects large projects that involve significant amounts of investment.

Only a small percentage of the many projects undertaken around the world are really successful. While most projects are completed, many suffer from cost or schedule overruns. To avoid such problems, organisations should employ the most effective methodologies and frameworks to manage projects.

Managing behaviour is a key factor in successful project management. To get the most out of their teams, managers must monitor how different members react to each other emotionally and culturally as well as judging them on the nature of their work performance.

Question 7

Answer A is incorrect. The challenges listed are not typical of project management challenges.

Answer B is correct. The three challenges listed are among the most common project managers experience.

Answer C is incorrect. Lack of marketing support is not typical of management challenges.

Answer D is incorrect. Personal animosity and sabotage are not typical of management challenges.

Question 8

In order to measure the success of the new quality management system, post-completion audits can be helpful in allowing you to answer the following questions:

1. Did the project meet its initial goal(s)? – At the planning stage of the new quality management system, defined and quantifiable objects and targets (i.e. key performance indicators or KPIs) should have been defined (e.g. defective product rejection rate; late service delivery rate, etc.). These measurable KPIs can be measured against the actual outcomes after implementing the new quality management system, such as Total Quality Management System (TQM). Any significant deficiencies should be investigated, analysed and remedied in further developing the new quality management system. On the other hand, good performance above the planned targets can be further reinforced in future development.
2. How can its findings be applied in other areas or on a broader scale? – Both the positive and negative findings in the post-completion audits of the new quality management system (i.e. usually began with a pilot or smaller scale) will be applied to future quality management systems on a larger scale, such as expanding into other functions or aspects of the TQM system undertaking by the company (i.e. preventive quality assurance such as staff TQM training).
3. What lessons did we learn about our company's processes that can be applied to future projects? – Lessons learnt, especially the failures, during the implementation process of a new quality management system are as important as the final outcomes of the project as measured in the post-completion audits. They are the guidelines and revised or reasonable KPIs for further expanding the new quality management system or undertaking other company's projects related to TQM.

EXAM PRACTICE**QUESTION 1**

Identify which of the following is the primary purpose of the post-appraisal audit.

- A.** To finalise and revise the budget.
- B.** To ensure that cash flows meet projections.
- C.** To decide whether managers of the project performed as expected.
- D.** To assess the success of the project.

QUESTION 2

Identify which of the following describes the attributes of an effective team evaluation.

- A. Did team members communicate well and work towards the same goals and objectives?
- B. Did each member identify ways to save money?
- C. Which members out-performed others?
- D. Which members should be promoted, and which ones reassigned or fired?

QUESTION 3

Identify which of the following describes some attributes of management neutrality.

- A. Project managers should not favour one project over another.
- B. Managers do not show favouritism towards some team members.
- C. Managers should implement projects to minimise negative impacts of post-completion audits.
- D. Managers must adhere to budgets for the project, without exception.

QUESTION 4

Identify which of the following explains how the post-completion evaluation is used.

- A. It lists budget variances and assigns responsibility for them.
- B. It is a method for critically assessing project managers' performance.
- C. This enables the company to improve future project performance.
- D. It is only applied to the accounting portion of the project, specifically budgets and cash flows.

QUESTION 5

Explain the purpose of the post-completion audit.

QUESTION 6

Describe how a project is most effectively measured on completion.

QUESTION 7

List examples of manager neutrality until done.

QUESTION 8

Explain how a project manager will be judged in post-completion audit.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. Revision of the budget is not the primary purpose.

Answer B is incorrect. Cash flows meeting projections is not the primary purpose of the audit.

Answer C is incorrect. Manager performance review is not the primary purpose.

Answer D is correct. The purpose of this audit is to assess the success of the project.

**QUESTION 2**

Answer A is correct. Communication and working together to meet goals are primary examples of how teams should be evaluated.

Answer B is incorrect. Saving money is not an effective attribute in evaluation.

Answer C is incorrect. Outperformance is not an effective attribute in evaluation.

Answer D is incorrect. Decisions on promotion and dismissal are not effective attributes in evaluation.

QUESTION 3

Answer A is incorrect. Project prioritisation is not an attribute of neutrality.

Answer B is incorrect. Complimenting some team members is not an attribute of neutrality.

Answer C is correct. Managers should implement projects to minimise negative impacts of post-completion audits.

Answer D is incorrect. Strictly meeting budgets is not an attribute of neutrality.

QUESTION 4

Answer A is incorrect. It is not used for assigning responsibility for budget variances.

Answer B is incorrect. It is not used for manager performance review.

Answer C is correct. The purpose is to improve outcomes of future projects.

Answer D is incorrect. It is not strictly applied to accounting only.

QUESTION 5

The post-completion audit compares outcomes to expectations. There are many areas worthy of comparison, beginning with initial assumptions. How accurate were they? How much variance occurred? If unexpected costs were experienced, what were the underlying causes? Which assumptions were not articulated?

QUESTION 6

A series of five questions determine whether measurements were effective. These are:

1. Do the outcomes measure and quantify what matters to management and to stakeholders?
2. What does success appear as and how do you know when it has been achieved?
3. Does everyone agree on the definition of a project's success?
4. Have you employed continuous measurement to track and audit the project?
5. Are you comparing outcomes to a baseline?

QUESTION 7

Management neutrality can be defined as setting standards for efficient and smooth processes. This is accomplished with five primary actions: (1) know the goals of the project; (2) set up a positive environment for the team; (3) allow team members to work with minimum interference; (4) place emphasis on the most important issues; and (5) ensure safety of team members with risk management techniques to reduce exposure to injury or damage.

QUESTION 8

Project managers will be judged based on several tests. First, did they prioritise processes to promote company interests defined as goals and objectives? Were those goals and objectives clearly expressed and did the team follow them in completion of the project? Were interim reviews performed to ensure the project stayed on course? Were approvals and decisions enacted efficiently or did the team get mired down in meetings and indecision? Were forecasts used to examine the project and possibly to change tactics as needed? Did the project manager reduce the negative impact of management silos? And did the project remain flexible enough to improve the process as it proceeded?

15

Effective Performance Measurement



CHAPTER TOPIC LIST

15.1 Organisation Structure and Key Performance Indicators

- 15.1.1** Key Performance Indicators Basics
- 15.1.2** Determining Key Performance Indicators for the Entity
- 15.1.3** Charting Key Performance Indicators Using a Balanced Scorecard
- 15.1.4** Reviewing, Changing and Updating Key Performance Indicators

15.2 Types of Responsibility Centres

- 15.2.1** Cost Centres
- 15.2.2** Revenue Centres
- 15.2.3** Profit Centres
- 15.2.4** Investment Centres

15.3 Performance Evaluation of Responsibility Centres

- 15.3.1** Return on Investments
- 15.3.2** Residual Income
- 15.3.3** Economic Value Added

15.4 Limitations of Financial Measures

- 15.4.1** Return on Investment
- 15.4.2** Residual Income
- 15.4.3** Economic Value Added

15.5 Segment Reporting

15.6 Economic Value Added Model and Weighted Average Cost of Capital in Capital Management

- 15.6.1** Economic Value Added Model
- 15.6.2** Weighted Average Cost of Capital

15.7 Financial and Non-financial Measures in Performance Evaluation

15.8 Advantages and Disadvantages of Decentralisation

- 15.8.1** Advantages of Decentralisation
- 15.8.2** Disadvantages of Decentralisation



LEARNING OUTCOMES

PRINCIPAL LO6: CONSIDER AND APPLY INFORMATION FOR EFFECTIVE PERFORMANCE MEASUREMENT	
LO6.01: Apply information for effective performance management and consider the relationship between non-financial measures and financial measures	
6.01.01	Illustrate the organisational structure affecting performance measurement e.g. cascade down KPIs from the top
6.01.02	Identify different types of responsibility centres, including cost centres, revenue centres, profit centres and investment centres
6.01.03	Explain how to evaluate the performance of responsibility centres
6.01.04	Prepare segment reports showing performance or profitability of different segments
6.01.05	Analyse how to apply economic value added (EVA) as a performance measure, its linkage to weighted average cost of capital (WACC) and its roles in capital management
6.01.06	Analyse the limitations of financial measures such as return on investments (ROI), residual income (RI) and EVA
6.01.07	Consider the relationship between non-financial measures and financial measures
6.01.08	State the advantages and disadvantages of decentralisation in a sizable organisation



OPENING CASE

ABC INVESTMENT LTD: AT THE CENTRE OF ACTIVITY

ABC Investment Ltd has evaluated its performance in many ways. In the past, evaluation was limited to a study of financial results, comparison of forecasts and budgets to actual outcomes, tracking cash flows and taking steps to improve net return while growing revenues.

This effort was traditional. It was based on well-understood accounting principles. However, it lacked evaluation of performance by responsibility: cost, revenue, profit and investment. ABC's management realised that it is possible to have strong cost controls and policies; but if revenues are not expanded, no progress is possible. It is also possible for profits to grow every year but at the same time, to lack investment performance.

With these limitations in mind, ABC expanded its previous accounting-based analysis of outcomes and combined financial and non-financial measures as part of its overall performance evaluation. This 'holistic' approach vastly improved management's appreciation of the elements that go into development and application of key performance indicators (KPIs). 



OVERVIEW

Performance measurement is a key element of effective management and companies spend a significant amount of time and effort developing solid performance measurements that can help inform and review strategic decision making.

Most performance measurement begins with the development of key performance indicators (KPIs), which can help companies set goals and determine whether those goals were achieved. KPIs can be used to track the performance of responsibility centres, although they must be tailored to the type of centre that is being evaluated – including cost centres, revenue centres, profit centres and investment centres.

The performance of responsibility centres can be evaluated using some financial measures. These measures include return on investment (ROI), residual income (RI) and economic value added (EVA).

Finally, this chapter discusses financial and non-financial measures used to evaluate performance as well as the advantages and disadvantages of decentralisation.



15.1 ORGANISATION STRUCTURE AND KEY PERFORMANCE INDICATORS

Key performance indicators (KPIs) are values that can be identified, measured and used to track whether a company's implemented strategy successfully achieved its mission.

15.1.1 Key Performance Indicators Basics

Different companies and industries rely on different KPIs. For example, airlines often use on-time arrivals as a KPI because a poor showing on arrivals is a recipe for failure. Couriers such as FedEx and UPS track on-time deliveries as well. Wireless communications companies use coverage areas and quality of calls as KPIs.

A KPI can be a measure of a short-term operational process or a long-run strategic process.

Operational KPIs relate to the short-term and focus on the efficiency of utilising resources. There may be financial or non-financial operational KPIs. There are various examples of financial metrics including cost per unit and variances of purchase prices while examples of non-financial metrics are the average outgoing product quality level, the percentage of defects and the yield rate.



Strategic KPIs are long-term and firm-specific measures. Strategic KPIs are useful to monitor how successful strategies can be. For instance, a high-volume discount retailer may depend on customer traffic and measure sales turnover of products. A high-end retailer, on the other hand, may be more focused on measuring customer experience.

15.1.2 Determining Key Performance Indicators for the Entity

A KPI may be a defined operational goal that, when repeated generates success. Examples include achieving a low rate of defects in manufacturing, consistent on-time delivery of products or balancing a monthly budget.

A KPI also may be the process aimed at accomplishing a longer-term goal. For example, expanding a marketing territory, improving customer and vendor relations or improving an internal training programme all can be measured in terms of KPI at each phase.

The indicators developed as part of KPI processes are defined in two categories. Quantitative data are entirely objective and consist of raw data, often expressed in numerical form. Accounting reports are one example; marketing and sales statistics are another. A second category is qualitative data, developed from opinions, observations and tastes. These may be numerically presented or developed narratively. Examples include the results of customer surveys, internal audit of control systems and product test marketing based on interviews.

The test of performance may be expressed as input, output, activity, mechanism, control or time. At the very least, a KPI's data will consist of input and output, and this enables the study of how these two phases work. Activity is the work process involved in transforming input into output. These processes rely on mechanisms and any part of the process that makes the activity work. This may be a person (human) or a piece of equipment (system). Controls constrain or regulate how activities proceed. Time sets a standard for the span of time between input and output.

These elements are crucial in the development of a set of KPIs. Choosing KPIs to monitor processes is necessary for the successful implementation of a business strategy.

A good KPI is:

- *Easy to understand*: The indicators should, when possible, be simple metrics.
- *Readily quantifiable*: An indicator that cannot be measured with some ease may not be useful.
- *Easy to monitor*: Performance evaluation requires measurement of KPIs at regular time intervals. If KPIs are difficult or expensive to monitor, they may not be useful.
- *Linked to company strategy*: Indicators have to be related to company strategy, otherwise they may not serve a clear purpose.

The KPI defines expectations. However, regulating and critically assessing the success of KPIs may be costly and require dedicated oversight. Some KPIs may be set but will be difficult to quantify. For example, customer satisfaction, vendor trust or employee morale are intangible and not easily expressed on a benchmark for success. These intangible KPIs must be understood as guidelines only because no scaling method is reliable.

KPI monitoring also must be performed cautiously. Management must be aware that if too much emphasis is placed on meeting a defined KPI outcome, employees will be given a perverse incentive to meet the outcome even at the expense of work quality or accuracy.

For example, if a manufacturing plant is given the KPI to produce 5,000 units per shift, the incentive may overlook the potential for a higher level of defects. If line employees are not given a motive to stop progress upon recognising a defect (and correcting it before resuming) then no one will be willing to pursue quality. The units per shift goal becomes more important than quality of output.

Illustrative Example 1

Until the early 1970s, Toyota produced cars like other manufacturers. Line employees were afraid to call for a stoppage due to discovered defects, because other employees might resent them for failing to continue output.

That was until a new system, called Jidoka, was introduced. The new system combined with just-in-time (JIT) delivery to improve quality. Under this system, any line employee was authorised to stop the line immediately upon spotting a defect. This led to overall higher quality and lower defects in cars.

In this example, one KPI – high production levels – was replaced with another – employee incentive to improve quality. It had the secondary benefit of improving overall efficiency which, while counter-intuitive, succeeded in improved output and customer satisfaction.

15.1.3 Charting Key Performance Indicators Using a Balanced Scorecard

The **balanced scorecard** is a system used to measure performance. It links strategy, planning and control. The structure – format and components – of the balanced scorecard has, over the past three decades, become fairly standardised to allow organisations to choose amongst KPIs. Just as important, the balanced scorecard makes it possible for management to organise their measures to share strategic plans with stakeholders and consider how management decisions will affect performance.

The balanced scorecard includes performance measurement along with a few different dimensions, including:

- financial and non-financial measures;
- short-term and long-term objectives;
- measures of performance that look to the future and consider past outcomes;
- hard objective and soft subjective measures of performance; and
- external and internal performance measures.

Financial measures depict a specific type of performance (profitability and cash flows, for example), while non-financial measures may be reflective of future value or be reflective

of what drives future value. By managing these drivers, companies can increase profitability. For instance, they might reduce the rate of defective products or speed up the product cycle.

To implement a KPI strategy, companies need to consider critical success factors for the long and short term. Relying too much on short-term measures may have a negative impact on long-term interests. On the other hand, measures of past performance are useful to evaluate achievement and make management decisions while forward-looking measures are also important for the success of company.

'Hard' measures are both quantifiable and verifiable, such as profits and asset turnover, while 'soft' measures are qualitative, such as customer satisfaction, quality improvements and employee training. Some performance measures are linked to stakeholders outside the company itself. Return on equity (ROE), for example, takes in the views of equity stakeholders. Customer satisfaction is an external monitoring measure. Some measures, like the production cycle and on-time delivery, are internal.

It is necessary to have a clear vision and strategy to balance various dimensions when designing an effective balanced scorecard system. Free and frank communication are also necessary along with acceptance from across the organisation for the measures that are chosen.

Managers use a balanced scorecard to look at companies from four different perspectives:

1. *Financial perspective*: Financial measures show the impact on the bottom line of strategy and the success of implementing the strategy. Improving metrics like customer satisfaction or business processes makes sense only if they improve returns. The most common financial measures are ROE, sales growth and net profit.
2. *Customer perspective*: Companies have to be customer-focused. A balanced scorecard includes and reflect customer perceptions with measures such as surveys of customer satisfaction or number of complaints. Other indirect measures such as market shares and retention rates may also be used.
3. *Internal process perspective*: Companies use a collection of processes to turn inputs into outputs. Activity-based management suggests that business processes add value if they contribute to customer satisfaction. An effective company manages both effectiveness and efficiency of internal business processes like cycle times, process efficiencies or yield.
4. *Learning and growth perspective*: Organisational learning and innovation are important to remain competitive. For example, companies using a low-cost strategy have to reduce cost and improve efficiency through innovation. Companies using a product differentiation strategy have to develop new products through innovation.

15.1.4 Reviewing, Changing and Updating Key Performance Indicators

Companies decide how many KPIs to follow depending on its size and organisation, the nature of the industry and other factors. A good system to measure performance includes a few key measures. Any effort to track too many KPIs may provide detailed information but prove to be too expensive and cumbersome and create confusion.

Once KPIs are chosen, targets must be set, and processes must be monitored. There is no hard and fast rule to determine the level expected of an individual KPI and management has a lot of discretion in creating expectations. Hopefully, management will set realistic but challenging targets.

Apply and Analyse 1

Apply

Your organisation has set a KPI to produce and market 20,000 units of a certain product per month. The standard expectation is for 95% defect-free output, or 19,000.

Recently, the marketing vice president has suggested increasing output to 25,000 units per month, to be reached over the next three months. This was based on recent expansion of the marketing area, an expansion that added a considerable customer base. The KPI in this example was to add 25% to previous production levels.

Your analysis of the outcome for this new production level revealed interesting results. In the first four months, the results were:

Month 1 – 22,105 units sold, 21,200 defect-free

Month 2 – 24,034 units sold, 22,100 defect-free

Month 3 – 23,574 units sold, 20,750 defect-free

Month 4 – 24,700 units sold, 21,000 defect-free

Analysis

The number of units sold nearly reached the expressed KPI to produce 25,000 units in the fourth month. However, the level of factory labour was not adequately addressed in this KPI and you have concluded that for the programme to succeed, additional employees need to be added to each shift.

To support your conclusion, you point out the changes in rates of defects. Previously, the company had achieved its KPI related to defects, to produce 95% defect-free units. Since the new programme was implemented, however, the defect rate has worsened:

Month 1 – $21,200 \div 22,105 = 95.9\%$ defect-free, 4.1% defects

Month 2 – $22,100 \div 24,034 = 92.0\%$ defect-free, 8.0% defects

Month 3 – $20,750 \div 23,574 = 88.0\%$ defect-free, 12.0% defects

Month 4 – $21,000 \div 24,700 = 85.0\%$ defect-free, 15.0% defects

The two KPIs contradicted one another. The significant increase in defects demonstrated that increased output by itself was not enough. Your recommendation: Maintain the standard for 95% defect-free production but add labour and oversight to return to that standard with higher levels of monthly production.

Key Learning Point

Key performance indicators (KPIs) are values that can be identified, measured and used to track the ability of companies to implement a strategy successfully and achieve its mission.

► Knowledge Check Questions

Question 1

Identify the purpose of key performance indicators (KPIs).

- A To provide a means for smaller budget variances.
- B Enabling the company to take larger market share.
- C To establish the means for reaching clear goals.
- D For marketing purposes only.

Question 2

Identify which of the following is a potential problem with using KPIs.

- A Employees may be motivated to reach goals, even at the expense of quality.
- B Customers will invariably suffer because the company is less responsive.
- C No employees are given incentives to outperform.
- D Accounting systems are overlooked in favour of non-financial goals.

➤ 15.2 TYPES OF RESPONSIBILITY CENTRES

There are four types of **responsibility centres**: cost centres, revenue centres, profit centres and investment centres. These centres indicate the degree of responsibility the manager has for performance and accountability.

15.2.1 Cost Centres

A **cost centre** incurs costs and expenses but does not directly generate revenues. Examples include the purchasing department, management of direct labour and compensation or transportation.

Managers of cost centres have the authority to incur costs. They are assessed on their ability to manage those costs, by budgeting and budgetary control, as well as to ensure timely events (purchases of material for production or on-time delivery, for example). Cost centres are usually either production departments, service-line delivery departments or service departments, like accounting, supporting either of the prior two departments. Production

departments contribute directly in making a product; service-line delivery departments contribute directly in delivering a service-line (e.g. a parcel truck); service departments provide only support services to the prior two cost centres.

Only costs under the direct control of the manager are included in the responsibility report of cost centres.

In a cost centre, the performance of a particular manager is measured by the ability to meet goals for controllable costs. Responsibility reports in these cost centres compare controllable costs with budget data.

15.2.2 Revenue Centres

The primary aim of a **revenue centre** is to generate revenue.

Managers at revenue centres do not control costs or investments in assets but usually have control of expenses related to sales and marketing. The performance of a revenue centre may be measured by comparing actual and budgeted revenue or by comparing actual and budgeted marketing expenses. Ultimately, meeting revenue forecasts while controlling related costs is the test of effective revenue centre management.

The best example of a revenue centre is the sales department, which focuses on generating sales and not on items such as reducing costs or direct expenses.

15.2.3 Profit Centres

A **profit centre**, such as a department in a retail store, incurs costs and expenses and generates revenues. Managers are evaluated on the basis of profitability. The performance of a particular profit centre manager may be based on information about controllable revenues and costs. For example, operating revenues earned, and variable costs incurred by the profit centre are controllable by the manager.

To determine the controllability of fixed costs, direct and indirect fixed costs will need to be distinguished. A profit centre may include both.

Direct fixed costs are specific to a centre and may include salaries for supervisors or fixed costs that can be traced directly to the centre. Sometimes they are called traceable costs. The manager of the profit centre controls most direct fixed costs.

Indirect fixed costs, on the other hand, are related to overall operating activities and to more than one profit centre. Indirect fixed costs are allocated to a number of profit centres. Activity-based costing (ABC) is accepted as a management accounting method to accomplish this (see Module 2, Chapter 6). For example, using ABC a company's human resources department's activity costs may be allocated as indirect fixed cost to profit centres based on the number of employee hires, appraisals and terminations in each centre. As these fixed costs apply to more than one centre, they are also known as common costs.

Another example is the assigned expense of rent for part of a larger facility. This may be assigned on a square footage of use basis, for example. Profit centre management has no control over rent or lease costs. Like most indirect fixed costs, they cannot be controlled by the profit centre manager.



Without ABC the costs could be allocated based on the number of employees in each cost centre, but this would not be equitable to a manager with high turnover and recruiting compared to a manager with minimal or no employee turnover. The human resources department was providing more service to the former manager compared to the latter manager.

The responsibility report for a profit centre shows the budgeted and actual controllable revenues and costs. Controllable revenues less controllable costs equal **controllable profit margin**. Direct fixed costs are controllable, and therefore, are subtracted to compute controllable margin. Indirect fixed costs are not controllable, and therefore, are not included to compute controllable margin.

The controllable profit margin is the best measure of the manager's performance when it comes to controlling revenues and costs. Responsibility reports for profit centres may be prepared monthly and may include cumulative year-to-date results.

15.2.4 Investment Centres

An **investment centre** incurs costs and expenses and generates revenue but also controls funds used for investment. Investment centre managers are evaluated on profitability and the returns that the centre earns on its investments. Investment centres are always linked to product lines and subsidiaries.

The manager of an investment centre can control or considerably influence investment decisions for matters such as expansion of plant and access into new market areas. However, this involves not only effective management of assets but also conforming to a level of risk tolerance appropriate, and based on management's identification.

The manager of an investment centre has a lot of influence over investment funds. These managers are evaluated using **return on investment (ROI)** as a metric. ROI shows how effectively the manager uses investment centre assets.

The formula for calculating ROI for an investment centre is:

$$\text{ROI} = \text{Controllable margin} / \text{Average operating assets}$$

The investment centre manager can control both factors in the formula.

The evaluation of investment centre managers depends on the scope of their responsibilities. Most investment centres operate independently while the manager has control of all fixed costs. This also means managers can even set the basis for depreciation of real assets. This also means that – compared with performance reports for profit centre managers – more fixed costs are classified as controllable in investment centre performance reports. The report also shows the budgeted and actual ROI.

Return on investment includes two factors. The first is a valuation of operating assets which can be at acquisition cost, book value, appraised value or market value. The second is controllable margin, which is a valid basis for evaluating the performance of an investment centre manager.

Illustrative Example 2

An organisation that produces auto parts has identified its four responsibility centres and has focused on coordinating all four. They are related directly in many ways.

Profits are not possible without cost controls; and cost controls ensure profits based on levels of revenue generated. One insightful observation by the CFO was that responsibility in all four areas (revenue, cost, profit and investment) represented overall management and control over the company. Furthermore, this was not merely based on financial results or status. To be effective, responsibility centres had to merge both financial and non-financial aspects of their areas and ensure that the overall results conformed to goals and objectives.

Illustrative Example 3

Your company has established standards for revenue, costs, expenses and net profits.

The manager of your company's cost centre has incurred costs above the forecasted levels. This problem needs to be brought under control. However, it also has ramifications for both the revenue centre and the profit centre.

For revenues to produce profits effectively, the cost of goods sold must be maintained at the forecast level. If the cost centre overspends, the revenue centre's results will be judged negatively. The level of revenue is not the only test; that revenue should also yield an established level of net return. However, this is not possible if costs are not maintained at a level adequate to produce gross profit as expected.

The same may be said about the profit centre. As long as the cost centre does not conform to its forecast level, gross profit will be lower than expected; and so will net profit.

The responsibility centres affect one another and do not operate in isolation. In this example, the cost centre's management negatively affected two other responsibility centres.

Key Learning Point

There are four basic types of responsibility centres: cost centres, revenue centres, profit centres and investment centres. These centres indicate the degree of responsibility that the manager has for performance and accountability.

► Knowledge Check Questions

Question 3

Identify which of the following explains how the revenue, cost and profit centres are related to one another.

- A** Cost levels influence the effectiveness of revenue, and in turn the degree of profits it is possible to earn.
- B** The three centres are not related to one another in any way.
- C** Only revenue and cost centres are related; the profit centre operates separately.
- D** All three centres rely on the forecast and budget, but otherwise they do not operate interactively.

Question 4

Identify how the investment centre must also be concerned with risk.

- A** The investment manager is not a risk manager and is responsible only for selecting investments.
- B** The only way is that the investment manager consults with the risk manager occasionally.
- C** Risk is not associated directly with investment activities.
- D** The investment manager is concerned directly with market risk and liquidity risk of the portfolio and must monitor both while selecting investments.

Question 5

Define 'investment centre'.



15.3 PERFORMANCE EVALUATION OF RESPONSIBILITY CENTRES

Responsibility centres are company sub-units where managers are accountable for specific types of operating activities. The evaluation of the responsibility centres takes place on an ongoing basis.

Three financial measures used to evaluate the performance of responsibility centres are:

- return on investment (ROI);
- residual income (RI); and
- economic value added (EVA).

While each is calculated differently, they provide summaries of work-unit and organisational performance.

15.3.1 Return on Investments

Return on investment (ROI) is the ratio of net profit to average operating assets. It is an effective way to evaluate the performance of responsibility centres by comparing how much companies invest into their set up and operations, versus the income derived from them.

Operating assets include all assets used in the production of goods or services, such as cash, accounts receivable, inventory and plant and equipment. Non-operating assets (such as investments in other entities, or property and equipment currently rented to other entities) are excluded from this calculation.

When evaluating the entire entity's performance, all assets are included because owners want to evaluate their return based on the entire investment. But when evaluating the performance of a sub-unit, judgement is used to determine which assets should be included. Any assets included should be under the control of the manager being evaluated.

The average of beginning and ending operating assets is calculated for this component for two reasons. First, the measure is intended to capture operations over a period, not just at the end of the period. Second, the measure could be manipulated by temporarily decreasing investment at the time that the performance is measured.

ROI is used to evaluate responsibility centre performance. It can be compared across sub-units within a single organisation, amongst a group of entities within an industry and within a single organisation over time.

In addition, ROI can be divided into two components providing additional information about performance. ROI is broken up by multiplying both the numerator and denominator by revenue. This is known as the DuPont model.

ROI can be analysed using the DuPont model. The DuPont model allows management to break profitability into component parts to understand where it actually come from.

First, look at ROI:

$$\text{ROI} = \text{Net profit} / \text{Average operating assets}$$

For example, ABC Limited is a hardware company that generated HK\$113.5 million in net profit in 2019. The company has average operating assets of HK\$2,000 million. ROI would then be calculated as:

$$\text{ROI} = 113.5/2,000 = 5.7\%$$

It may then be possible to compare a company's ROI with the industry average, which is 9% for the hardware sector. The ROI for ABC Limited is below industry average. The question is why? The DuPont model may provide an answer as follows:

$$\text{ROI} = \text{Net profit} / \text{Sales} \times \text{Sales} / \text{Average operating assets}$$

Net profit/sales is the net profit margin and comes from the statement of profit or loss and other comprehensive income, and sales/average operating assets is the asset turnover and sales comes from the statement of profit or loss and other comprehensive income and average operating assets comes from the statement of financial position.

Hence, ROI has two parts. The profit margin and asset turnover, which includes the company's ability to generate profits. The next step is to determine the part of ROI that is causing the problem, whether it is the profit margin or asset turnover. For example, the net profit margin could be 3.8% and the asset turnover could be 1.5 times, then:

$$\text{ROI} = 3.8\% \times 1.5 = 5.7\%$$

Each part of the equation has an impact on ROI. ABC Limited earned 3.8% for every dollar of sales and turned over assets 1.5 times per year. If the sector has a net profit margin of 5% and a asset turnover of 1.8 times the company is low on both measures.

Illustrative Example 4

A company reported net profits of HK\$129,650 last year. Its average operating assets were HK\$463,009. Calculate the return on investment (ROI).

$$\text{ROI} = \text{Net profit} / \text{Average operating asset} \times 100\%$$

$$129,650 / 463,009 \times 100\% = 28\%$$

15.3.2 Residual Income

Residual income (RI) measures the dollar amount of profits more than a required rate of return, which is commonly referred to as a capital charge. It is calculated as follows:

$$\text{Residual income} = \text{Operating income} - (\text{Required rate of return} \times \text{Average operating assets})$$

Many companies have minimum expectations for their operations and investments.

Residual income is the difference between actual and required operating income, so it takes into account investments in operating assets and the required rate of return.

The size of an investment has less impact on residual income than ROI because it is used only to value the dollar amount of expected returns but is not used as a denominator. Changes in investment have less impact on residual income than on ROI.

15.3.3 Economic Value Added

Economic value added (EVA) is a type of residual income incorporating several adjustments to lower some of the disadvantages created by residual income.

The EVA calculation is:

$$\text{EVA} = \text{Adjusted after-tax operating income} - [\text{Weighted average cost of capital (WACC)} \times (\text{Adjusted total assets} - \text{Current liabilities})]$$

The weighted average cost of capital (WACC) is calculated by analysing sources of invested funds, including debt and equity financing. In effect, it represents the after-tax cost of long-term financing for a company or a division. With EVA, divisions can use their cost of capital, taking into account the industry and their own levels of risk.

Key Learning Point

Three financial measures used to evaluate the performance of responsibility centres are return on investment (ROI), residual income (RI) and economic value added (EVA).

► Knowledge Check Questions

Question 6

Identify which of the following describes how ROI is calculated.

- A** Net profit divided by total assets.
- B** Average operating assets divided by net profit.
- C** Net profit divided by average operating assets.
- D** Total assets divided by net profit.

Question 7

Define residual income.

➤ 15.4 LIMITATIONS OF FINANCIAL MEASURES

The various measures used to measure performance have their limitations. Return on investment (ROI), residual income (RI) and economic value added (EVA) rely on data that can guide management decisions but, at times, the use of these models can create misleading results.

15.4.1 Return on Investment

The ROI of one division or of a capital project can easily be compared with that of another division or project. One benefit of using ROI is that it may limit the tendency of the manager

to overinvest in a project. It also motivates the manager to work at boosting sales, decreasing costs and minimising asset investments.

ROI measurements to track performance also have their disadvantages, however.

It is not in the best interest of managers to undertake projects that lower ROI, even if they might actually improve the ROI for the whole company in the long run but have a short-term negative impact on the ROI of the division. Divisional ROI is an important part of incentive plans used to motivate divisional managers.

At the same time, ROI does not incorporate risk measurements. Managers may boost ROI by investing in projects with more risk. These projects with higher levels of risk may generate higher returns. If managers are rewarded just for boosting ROI, they may have an incentive to take on projects with higher levels of risk just to boost ROI.

This is a particularly significant problem when dealing with short time frames, such as when the manager is planning to retire or switch jobs. In these cases, managers may focus on projects that bring immediate improvement performance measures. Managers operating under short time horizons may also choose to cut costs inappropriately and without considering long-term impact. For instance, they might cut R&D or maintenance expenditures or forgo an investment based on short-term considerations.

Financial assets and income are used to calculate ROI. Financial accounting rules require that assets be recorded at their original cost and that intangible assets not be recognised. These intangible assets could be brand names. Accounting rules may lead to the understatement of an investment, especially when the value of assets goes up or the company has a lot of intangible assets. Although asset values may have higher market value, they may have been depreciated so that net book value is even lower than original cost.

The results of assets being understated, may be that ROI and investment turnover are overstated. At the same time, financial accounting rules may end up measuring revenues and costs in a fashion that distort ROI. For instance, overhead costs may be allocated to a particular division without reflecting how the division uses those resources and if costs are understated or overstated, ROI is then distorted.

15.4.2 Residual Income

The use of residual income to evaluate performance also has its limitations.

One positive aspect of using RI is its objectivity. RI does not penalise investment in projects with lower returns than current project returns.

In general, when RI is used as a performance measure, managers are willing to invest in any projects with returns equal to or greater than the required rate of return. With the use of RI, the current ROI of the division is not an influencing factor on decision making.

However, the use of RI has its own problems.

Because it is an absolute dollar value, larger companies or units are likely to have larger RIs. For example, a project to distribute retail goods in a sparsely populated area of the New Territories in Hong Kong is almost certain to have lower RI than one distributing goods in heavily populated Kowloon. As a result, managers may find it difficult to compare how different units perform.

RI and ROI have at least one disadvantage in common and that is the residual income increases when investment and costs go down, as long as sales are assumed to be constant. As a result, managers may be encouraged to minimise or cut costs for such items as R&D, maintenance or employee training. These are costs that could have long-term benefits for a company.

There is yet another problem linked to residual income. If managers from sub-units pick their own required rate of return, they may be incentivised to set a rate that is too low. This would make it easier for them to outperform the rate. At the same time, when a required rate of return is too low, managers may end up investing in projects that are less profitable or pass on riskier projects that could prove to be more profitable for the whole company.

Financial accounting information is used to measure operating income and RI suffers from the same problems as ROI.

15.4.3 Economic Value Added

Some of the advantages of using EVA stem from the adjustments made to personalise the measure to each entity. These adjustments provide incentives that align the goals of managers with owners.

However, EVA also has some disadvantages.

For instance, the appropriateness of the cost of capital for a division or entity is a matter of judgement, as is the level of risk that has been incorporated.

There is also judgement associated with the adjustments involved in using EVA. Managers may not know how to measure revenues, costs or assets or other acceptable methods that provide different incentives. EVA is complex. Consultants may help determine the right adjustments and this process can be expensive and lengthy. Because of this complexity, EVA may prove to be more suitable at higher levels of a company.

Illustrative Example 5

A production division attempts to evaluate performance of several projects but is concerned with the limitations for each method. They finally settle on using ROI for several reasons. First, the calculation is best understood in comparison to RI and EVA. Second, even though management is discouraged from selecting projects that may reduce ROI, the long view is that if the project is positive for the overall organisation, it should be pursued. Third, whereas ROI does not measure risk, management has resolved to incorporate standard risk management metrics to offset this disadvantage.

Illustrative Example 6

An organisation may use different methods to measure performance. ROI may provide incentives for managers to avoid some positive projects. RI does not work well for larger than average divisions. EVA is based at least partly on judgement rather than

Illustrative Example 6 (*continued*)

on actual financial information. However, each method has its own limitations. Explain which method you would use and what measures you would put in place to offset its drawbacks.

The selection of ROI has been determined. Like all methods of analysing feasibility, ROI has its drawbacks. However, it requires less explanation to stakeholders because it is widely understood. The drawbacks to ROI can be offset by counter-measures to eliminate the lack of risk factors. This may be the most sensible alternative compared to the other two methods.

Key Learning Point

There are some limitations to the use of ROI, RI and EVA. For instance, the use of ROI also discourages managers from investing in projects that reduce a division's ROI, even though they might improve the ROI for the overall entity. Because RI is an absolute dollar value, larger companies or units are more likely to have larger RIs. EVA can be complicated to use.

► Knowledge Check Questions

Question 8

Identify which of the following describes the limitations of each of the three methods of measuring performance: ROI, RI and EVA.

- A** ROI discourages some projects that would be positive for the entity. RI tends to become distorted when applied in larger than average divisions. EVA requires subjective judgement rather than relying on fact.
- B** ROI and RI are complex, and EVA is relatively simple to calculate. EVA also is widely understood and easily applied objectively.
- C** RI is preferred because of its overall simplicity. In comparison, ROI is less specific and EVA lacks risk metrics.
- D** ROI is based largely on individual judgement, so that different managers with similar sets of facts may draw different conclusions. This drawback is accounted for and eliminated with both RI and EVA.

Question 9

'One of the criticisms against the use of financial measures is that they are not consistent with today's business realities'. Explain this statement.

15.5 SEGMENT REPORTING

Segment reporting refers to the reporting of the operations of specific segments of a company, in particular in the disclosures that are attached to financial statements.

Publicly held entities are required to undertake segment reporting but privately held companies are not. The idea behind segment reporting is to provide investors and creditors with information regarding financial results and the financial position of the various and more significant operating units of a company. A summary of the entire company does not reveal the details of how each segment operated; so segment reporting provides stakeholders with worthwhile insights.

Operating segments undertake business activities that earn revenue and generate expenses and for which there may be financial information available. The information for those segments is typically regularly reviewed by top management to make decisions on how to assess performance and allocate resources.

It is not necessary to report specific financial information for all segments but some guidelines help companies determine which segments report added information and the best practices for how to report this information. These guidelines may include:

- Results for multiple segments that have similar products, services, processes, customers or distribution methods as well as regulatory environments may be aggregated. For example, in a diversified company, combining all the retail food products under one umbrella represents a key segment, separate from other lines of business in the same company.
- Companies should report information about segments that represent 10% of the company's revenues, profits or losses or assets. This means that relatively small segments are not reported separately; such reporting would demand time and effort and take up space in a large financial report, but without adding value. Defining segments worthy of added reporting at 10% or above solves this problem.
- If the total revenue of segments reported adds up to less than 75% of total company revenue, more segments should be added. In other words, the selected segments should represent most of the organisation's revenue, expenses and profits to add value in the details.
- It is always possible to add more segments but if companies report on more than 10 segments, a reduction may be in order. Excessive reporting may confuse stakeholders; and it often is possible to combine separate segments to reduce the volume of segment reporting.

Guidelines about the information that should be reported may include:

- The factors used to determine which segments to report;
- Information about each segment, including products and services sold;
- How information is organised;
- Revenues;

- Interest expenses;
- Depreciation and amortisation;
- Material expenses;
- Income tax expenses;
- Other material non-cash items;
- Profit or loss.

Illustrative Example 7

An organisation contains many segments and wants to be able to break down the financial statement with a specific guideline: Report most of the activities, but without cluttering the report with excessive detail.

The purpose of segment reporting must be kept in mind when combining different segments into a single segment report. However, there must be a rationale for how segments are combined. For example, related products may be placed into a single segment for the segment report. The standard should be to clarify the report and not to confuse stakeholders or obscure important trends and financial outcomes.

Illustrative Example 8

A diversified holding company has 15 distinct segments, each representing the following percentages of overall revenues:

Retail baby food	7%
Wholesale baby food	3
Medicine	6
Baby products	8
Child care publications	2
Jam and preserves	11
Cereal	14
Soft drinks	17
Canned vegetables	4
Pasta and pasta sauce	2
Canned fruit	3
Cookies and crackers	7
Candy	9
Nuts	2
Dairy	5

Illustrative Example 8 (continued)

Explain how these 15 segments can be reduced in a segment reporting system to represent more than 70% of overall revenue, but requiring reporting for fewer than 10 segments.

The segment reporting would not be effective if all of these were reported separately. Although management wants to track all 15 product lines separately, segment reporting has been reduced to the following combined lines of business:

Baby food and products	20%
Cereal	14
Jams and preserves	11
Snacks	18
Soft drinks	17

These five segments account for 80% of revenues and requires reporting for five segments versus a total of 15 segments.

Illustrative Example 9

Smart Computers Company Limited (SCCL) is a medium-sized company with two electrical divisions: Laptop Computer and Image Scanner. You are requested to prepare a segmented statement of profit or loss using contribution margin format for the year 2019, given the information below for the year 2019:

	Laptop Computer	Image Scanner
	HK\$	HK\$
Sales (in units)	1,040	3,280
Sales price (per unit)	8,200	650
Variable production costs (per unit)	800	280
Traceable fixed production costs	652,000	968,000
Variable selling expenses (per unit)	28	45
Traceable fixed selling expenses	854,000	68,000
Allocated portion of the corporate expenses	2,990,400	1,860,600

The segmented statement of profit or loss using contribution margin format is as follows:

	Laptop Computer	Image Scanner	Company
	HK\$	HK\$	HK\$
Sales	8,528,000	2,132,000	10,660,000
Variable production costs and selling expenses	861,120	1,066,000	1,927,120
Contribution margin	7,666,880	1,066,000	8,732,880
Fixed production costs and selling expenses	1,506,000	1,036,000	2,542,000

Illustrative Example 9 (continued)

	Laptop Computer	Image Scanner	Company
	HK\$	HK\$	HK\$
Segment margin	6,160,880	30,000	6,190,880
Corporate fixed expenses			4,851,000
Net Profit			1,339,880

Key Learning Point

Segment reporting refers to the concept of reporting the operations of specific segments of a company in the disclosures that go along with financial statements.

► Knowledge Check Questions**Question 10**

Identify which of the following explains the purpose of combining different segments into a single segment report.

- A** Related products can be combined to clarify results and to demonstrate significant trends and financial outcomes.
- B** The overall purpose is to make the financial report as brief as possible.
- C** By combining a profitable and an unprofitable segment, the overall results look favourable.
- D** The more detail in the report, the better; combinations should be used only when segments are individually minor in comparison to the overall results.

Question 11

Explain what segment reporting is.

➤ 15.6 ECONOMIC VALUE ADDED MODEL AND WEIGHTED AVERAGE COST OF CAPITAL IN CAPITAL MANAGEMENT

A pair of useful performance measures for the effectiveness of capital management are the economic value added (EVA) model and the weighted average cost of capital (WACC). Both offer a way to make capital management decisions by comparing the cost of capital against the expected outcomes of a capital project.

15.6.1 Economic Value Added Model

As discussed in Section 15.3.3, EVA refers to residual income with some adjustments to minimise the disadvantages such income can cause. In other words, EVA focuses on how much income a capital project will generate after the cost of capital is factored in. It provides a way to determine how much added value a capital project will generate for a company.

The basic formula for EVA calculations is:

$$\text{EVA} = \text{Adjusted after-tax operating income} - [\text{WACC} \times (\text{Adjusted total assets} - \text{Current liabilities})]$$

The adjustments necessary to develop an EVA calculation include substituting after-tax operating income for earnings before income tax (EBIT), which is consistent with using the (after-tax) weighted average cost of capital and gives managers incentives to reduce taxes.

Particularly relevant to considerations of capital investment, EVA reflects economic costs and benefits over time and facilitates decision making. For example, R&D costs are recognised immediately as an expense under most accounting standards but are often capitalised in EVA calculations. This adjustment encourages managers to invest in R&D projects that have long-term value for the entity.

EVA also applies to long-term leases, which are accounted for as operating leases under generally accepted accounting principles (GAAP) but are treated as capital leases in calculations of EVA. This adjustment cuts incentives for managers to use operating leases to understate investment in assets, an artificial understatement. Managers are then encouraged to make decisions on long-term asset acquisitions based on what is best for the company and not based on financial accounting treatments.

A key goal of adjusting financial accounting income and assets is to improve decision-making processes in capital management.

Measures used for internal purposes do not necessarily need to follow established accounting standards such as Hong Kong Financial Reporting Standards (HKFRS) or GAAP, which are typically used to prepare financial statements for external parties. Internal measures aim to reflect economic costs and benefits over time.

15.6.2 Weighted Average Cost of Capital

Calculations of WACC are linked to the company's after-tax cost of debt, cost of preference shares and cost of common shares. This is the rate a company should expect to pay (on average) to its investors, to finance its acquisition of assets. It is the minimum return on investment needed to meet obligations to creditors and investors.

The higher the WACC, the less value generated by capital. If the company has more than one type of debt outstanding or more than one type of preference or common share, the calculation is performed using a weighted average for each type of security. The equation for WACC is:

$$\text{WACC} = E/V \times R_e + D/V \times R_d \times (1 - T_c)$$

where:

- R_e = the cost of equity;
- R_d = the cost of debt;
- E = market value of a firm's equity;
- D = market value of a firm's debt;
- $V = E + D$;
- E/V = percentage of equity financing;
- D/V = percentage of debt financing;
- T_c = corporate tax rate.

Analysts use market values rather than the accounting book values of the various components of formula. The theory behind WACC, however, is that the underlying discounting process necessitates that the cost of different types of financing is given a weight which is relative to their market value. There is no place in these calculations for accounting book values, because book value often is outdated due to changes in price per share or increases in market value of assets versus depreciated value, both of which are drawbacks in the accounting rules.

Financial managers are likely to require that internal analysts use the company's cost of capital to discount cash flows for individual projects because it is difficult to estimate the discount rate for individual projects directly.

Using WACC to discount cash flows for a project makes sense in many cases. However, it can be very inaccurate as well.

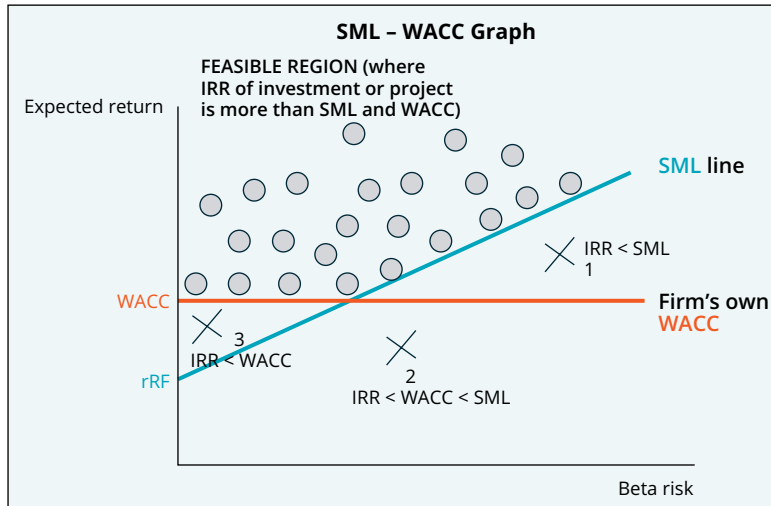
When an analyst forecasts cash flows for a project, the forecast represents incremental after-tax free cash flows for the whole company that show the difference between the cash flows for the company if a project is taken up and if it is not taken up.

According to financial theory, the rate used to discount incremental cash flows is a rate that adequately reflects systematic risk. WACC is appropriate to evaluate a project only when the systematic risks of the cash flows of a project are similar to the systematic risks of the whole company. This is not true for most projects. In fact, projects tend to have different degrees of risk.

When the discount rate is too low, the company runs the risk of accepting a project with a negative net present value (NPV).

Some projects may have an expected return above WACC but below the security market line (SML). In this case, using WACC as the discount rate may lead to the company accepting a project with negative NPV.

The estimated NPV will be positive despite the fact that the true NPV is negative. In these situations, the negative NPV projects that would be accepted have returns that, in an SML graph, would fall below the SML and above the WACC line and to the right of the company's beta at the intersection of the two lines.



► **EXHIBIT 15.1 SML and WACC**

Using WACC to discount expected cash flows for projects with low levels of risk may lead to management turning down positive NPV projects, a problem that is the opposite of when WACC is lower than the correct discount rate. These positive NPV projects that would be rejected are those below WACC but above SML and with betas less than that of the company.

If a company uses just one rate to evaluate all projects, it will be biased towards taking on more risky projects. Average risk of a company's assets should increase over time. Because some positive NPV projects may be rejected and negative NPV projects accepted, the shareholder value may drop. This would not be the case if the right discount rate is used to evaluate projects. This can create a disadvantage for companies and negatively impact the value of projects.

It makes sense to use WACC to discount cash flows only if these expected cash flows have a level of systematic risk that is similar to that of the entire company.

It is never possible to be sure whether this condition exists, but there are guidelines to assess when the systematic risk for a project matches the one of the company, while systematic risks depend on the nature of the business.

For some companies, changes in the broader economy have greater impact on revenues and expenses. Although total volatility is not the same as systematic volatility, businesses with more total volatility tend to have more systematic volatility. Beta measures systematic risk, which is a key factor to determine WACC.

This leads to a couple of conditions for the use of WACC:

- If systematic risk for a project is the same as that for the entire portfolio of projects that make up the company.
- If a project uses the same proportions of debt, preference and common shares as the entire company.

Illustrative Example 10

A company manufactures soft drinks. Managers worry about competition in the core soft drink business so are considering expanding into tropical beverages, a market that would allow for more differentiation and higher profits.

The beta for soft drink projects is 1.2 and the beta for tropical beverages is 1.5. The beta measures the responsiveness of cash flows to returns for the broad equity market. For now, the company is only in the business of soft drinks, so the beta for all cash flows is 1.2.

The beta for the tropical beverage project is larger than for the company as a whole and should be higher than the company's WACC.

If the company's WACC is used to discount expected cash flows for the tropical beverage project, the company may accept the project wrongly when the estimated NPV is positive while the true NPV is negative.

Key Learning Point

A pair of useful performance measures for the effectiveness of capital management are the economic value added (EVA) model and the weighted average cost of capital (WACC) approach. Both offer a way to make capital management decisions by comparing the cost of capital against the expected outcomes of a capital project.

► Knowledge Check Questions

Question 12

Identify what is accomplished in using EVA.

- A** It is the only reliable measure of income to be expected from a project.
- B** EVA presents outcomes regardless of the cost of capital.
- C** It calculates value of projects only on a company-wide basis and not for each project.
- D** EVA estimates income from a capital project after factoring in the cost of capital.

Question 13

Identify which of the following explains how WACC is related to tax liabilities.

- A** There is no correlation between the cost of capital and tax liabilities.
- B** The calculation is based on after-tax cost of debt, so the outcome represents the net payment to investors and creditors.
- C** WACC ignores tax liabilities because these are non-operating expenses.
- D** The method of calculation adjusts for different tax rates in each country where the company operates.



15.7 FINANCIAL AND NON-FINANCIAL MEASURES IN PERFORMANCE EVALUATION

A singular focus on financial results is often insufficient to help an organisation implement its strategy successfully. Most organisations need multiple measures. Even organisations with relatively simple strategies contend with many critical processes and these may be better quantified using non-financial measures.

ROI, RI and EVA are lagging measures reflecting past performance. Lagging performance measures contain limited information about an organisation's potential for future performance, especially when growth rates are rapid or when market circumstances have changed. Organisations also need to track measures such as customer satisfaction or return rates that can help them drive future performance.

Experts say that managing a company is a lot like flying a plane. A skilled pilot monitors multiple gauges to travel in the right direction and at the right speed and altitude. They also must account for weather, traffic and congestion at airports. Good managers, similarly, rely on more than one measure. A good manager knows that companies underperform in some areas and meet expectations in others and they want to have a comprehensive picture of their companies by using various measures.

Companies have to combine lagging and leading indicators and use both financial and non-financial measures to track performance. Financial measures have some problems. First, these measures tend to reflect aggregate performance of the entire company or division but without highlighting individual segment results. Second, financial measures provide historical as opposed to forward-looking data so the information derived from these measures is rarely timely or useful to take quick corrective action. Third, financial measures do not provide information about areas of potential concern. For example, weakening net profit may be gradual and, unless pointed out by analysts, may not be recognised. It is possible for net profits to rise even as net return declines.

Therefore, companies regularly use financial measures along with operational and strategic performance measures, despite the fact that many of these are subjective in terms of assessment. A key concern is identifying additional measures of the state of affairs of a company that can provide an idea of how to achieve future success. For example, four non-financial measures may enable measurement with exceptional value:

1. Company reputation – organisations rely on a positive perception amongst consumers, not only for its products or services, but also for its corporate governance, community involvement and ethical standards.
2. Customer value – the loyalty of customers to a brand or to any products produced by one company, has considerable value but it cannot be measured financially.
3. Competition – the degree of competitive success defines whether a company will be able to grow or has reached a growth plateau it will not be able to move beyond. For this reason, competition matters more but does not have a precise financial value.

4. Innovative initiatives – a company that can grab market share due to exceptional initiatives is likely to outperform its competitors. Companies that can develop intellectual property also gain significant advantages that cannot be measured.

Non-financial measures can track intellectual property or intangible assets to a certain degree. For example, balanced scorecards are another frequently used non-financial measurement.

Non-financial measures have four clear advantages:

1. They are closely linked to strategy.
2. Success drivers are often the intangibles that have no place in a statement of financial position, but may be tracked by non-financial measures.
3. Non-financial measures may provide a window into future performance.
4. Managers can choose non-financial measures with fewer variables that stand outside the control of management.

However, it is also worth noting that non-financial measures have several disadvantages.

One is the time and cost that goes into tracking them, which is often greater than the benefits it generates. Evaluating multiple measures can be time-consuming and get in the way of other, more tangible performance measures. A typical example is the use of environmental, economic and governance (ESG) reporting (i.e., non-finance measures) as discussed in Section 13.8.3.

There is also a danger that measuring can become mechanic and, consequently, useless. Finally, attempting to measure non-financial attributes may be worth evaluation on the executive level. But if the conclusion is less than satisfying to management, what short-term steps can be taken? Because non-financial aspects of an organisation cannot be controlled through internal controls, budgets or forecasts, measurement raises the question: What is the purpose to the effort?

This reality points to a second disadvantage of non-financial measures. They do not have a standard denominator. In other words, success or failure can be subjective.

A third problem is a lack of causality or causal links associated with most non-financial measures. Given their subjective measure it can be difficult, if not impossible, to say with any certainty that behaviour X led to outcome Y. This also makes it hard to articulate the importance of a particular measure.

A fourth problem is that non-financial measures are not statistically reliable. Most data are based on surveys, questions and opinions. The design of the surveys or the wording of the questions may alter the outcome of the measurement.

Finally, using too many non-financial measurements can lead to a disintegration of measurement itself, with managers tracking too many different types of performance.

Illustrative Example 11

A company has decided to combine financial and non-financial measures, notably to its intellectual property. How can it place value on patents developed and used to develop new products?

At first, it appeared obvious that increased revenue would be used to indicate the financial value of patents, even though these are non-financial assets. However, it turned out to be a difficult matter. Revenue generation involves many costs and direct or indirect expenses, so knowing the financial value of patents is not simple. Using net operating profit is equally elusive. Operating in several different countries, tax liabilities vary so that net profit is not consistent.

The only logical conclusion is that patent applications can be tracked and observed, and some degree of profits used to assign value. A second means for assigning value arises in cases of licensing agreements with other companies. However, if the company maintains its patents and applies them to product lines, it – like all non-financial tests – cannot be assigned financial value.

Illustrative Example 12

Your organisation has decided to track the value of customer loyalty. Even though this is a non-financial measure, management believe it can set up a tangible measure of this and determine whether loyalty is increasing or declining. Explain how this can be accomplished.

Amongst the metrics used are analysis of returned or repeated customers within a single 12-month period. Another is based on surveys of customers, asking for opinions concerning product quality and price. Third is brief survey questions asked at the end of customer contacts in the customer service department.

The problem with attempting to judge this non-financial metric is that it may not provide specific indicators of how to improve customer loyalty. In extreme cases, it can be recognised; but in general, it is elusive and difficult to measure. Attempts were made to reduce survey results to a numerical quality. However, you recognise that this does not provide any clear trends or indicators about whether customer service is improving or declining. Because it is non-financial, it is difficult to measure through statistical or accounting processes.

Key Learning Point

A singular focus on financial results is often insufficient to help an organisation implement its strategy successfully. Most organisations need multiple measures. Even organisations with relatively simple strategies tend to have many critical processes and these can often be better measured using non-financial measurements.

► Knowledge Check Questions

Question 14

Identify some of the better-known non-financial assets of a company.

- A** Signage, customer base, loyalty and advertising.
- B** Exceptional survey methods, goodwill, well-trained employees and successful management.
- C** Company reputation, customer value, competition and innovative initiatives.
- D** Quality control, streamlined supply chain, excellent transportation systems and just-in-time inventory.

Question 15

As one of the non-financial measures in performance evaluation, explain how a company measures its customer retention and churn.

➤ 15.8 ADVANTAGES AND DISADVANTAGES OF DECENTRALISATION

Accounting information is used in both centralised and decentralised entities to measure, monitor and reward performance.

Centralisation refers to the approach to management that puts most of the decision-making in the hands of a small group of managers who are typically located at headquarters. **Decentralisation** is the opposite, with decisions made by managers spread throughout the organisation.

Another way of describing decentralisation is the delegation of management authority to middle and lower-level management. Companies exhibit varying comfort levels with this version of decentralisation.

In entities that are decentralised, decision making happens throughout different management levels. Lower-level employees are responsible for their efforts and their decisions. Accounting systems provide information for decision making across the entire company, from management to the front line. Broader accounting measures of financial performance can measure and monitor performance.

15.8.1 Advantages of Decentralisation

In centralised entities, information from accounting systems is mostly used by top managers. They are responsible for effort and their decisions. Employees undertake tasks based on management decisions and are held responsible for their effort and compliance with the decisions. As a result, individual and team efforts are closely monitored and aligned to their contributions to success.

There are several key advantages to decentralisation.

At the top of the list is empowerment. In a decentralised organisation staff have more autonomy and decision-making power. They can also leverage their experience and knowledge while doing away with bureaucracy. This makes decision making more efficient.

Another advantage of a decentralised structure is that the operational burden is more widely distributed. This distribution frees up management to focus on strategy and big-picture decision making.

Decentralised organisations are often more effective at dealing with emergencies as staff have greater autonomy and may be more responsive to change and unexpected circumstances.

It is also easier to grow a decentralised business. New business units can be launched faster and with greater ease and may operate independently faster.

15.8.2 Disadvantages of Decentralisation

Decentralised companies and organisations also must overcome several disadvantages.

A first hurdle is cost. It is often more expensive to run a decentralised organisation because more skilled managers are required throughout the organisation. There is also the chance that skills and efforts may be duplicated, which increases costs.

A second disadvantage is the potential for conflict. Decentralisation puts pressure on every department and every manager to boost output and revenue, which often leads to every department or unit prioritising their own goals. Fragmentation is common.

A third hurdle can emerge from emergency situations. In some instances, decentralised companies are better at dealing with emergencies affecting top management but in other instances, for example, a natural disaster, decentralised units may not have the authority to make company-wide decisions.

A fourth disadvantage is increased risk. Because decision-making is often in the hands of lower-level managers, companies must face the possibility that the wrong decision may be taken, not necessarily through a lack of dedication but, perhaps, due to a lack of complete information.

A fifth and final disadvantage is that decentralised companies are not particularly effective for the provision of services that rely on a central decision-making management team. Accounting, for example, is not a profession that lends itself to a decentralised structure.

Illustrative Example 13

An organisation has selected a decentralised management style. The idea is that divisional managers across a broad geographic area can effectively enact positive change, such as expansion of a marketing plan and customer base. However, a typhoon shut down the divisional office. It was only then that headquarters discovered some of the serious flaws in their decentralised structure.



Illustrative Example 13 (*continued*)

For example, the division had not taken basic steps such as establishing a telephone tree to ensure continued communication amongst employees and management. Many employees remained in place but were not able to communicate with their families. Executives were not able to respond to family requests for information about whether family members were safe. They simply did not know because basic disaster risk management steps had not been put in place.

Illustrative Example 14

Your organisation is debating the pros and cons of centralised and decentralised management. You have been asked to evaluate both and to offer recommendations.

The advantages of decentralisation include empowerment and allowing managers to make their own decisions; wider distribution of operational burdens and duties; potentially greater efficiency in dealing with emergencies; and in the right circumstances an easier path to growth.

You also point out disadvantages to decentralisation. These include higher operational costs such as communication; the potential for conflict between departments; the possibility of poor response to emergencies; higher risks of making the wrong decisions; and poor implementation of services that are best provided from centralised management.

When a company or organisation has grown up or expanded to a certain scale or size in terms of business volume or workforce, it has to reconsider and revamp its organisational structure thoroughly, in order to strike a balance between decentralisation in duties and responsibilities, as well as centralisation in terms of control and accountability. There is no best model in fulfilling this balancing act but trying to maximise value creation to all stakeholders and the company as a whole.

Ethics in Practice

Decentralised management may offer a company greater efficiency, but this does not mean they cannot be subjected to oversight. If decentralised management is given unlimited freedom, they may expose the company to a range of risks:

- Disaster recovery systems;
- Product liability;
- Customer response and possible loss of loyalty;
- Failure to protect company assets.

For example, on 15 June 2018, the MTR Corporation put forward a report to the Hong Kong Highways Department on a scandal involving corner-cutting at the Hung Hom Station. The department referred the matter to the police. The 46-page report was submitted after MTRC chair, Frederick Ma, committed two days earlier to undertake an internal investigation.

Ethics in Practice (*continued*)

The MTR Corporation said workers cut steel bars at the Hung Hom Station expansion to make them look correctly installed. The expansion was part of the HK\$97.1 billion Sha Tin to Central Link (HK Free Press, 16 June 2018, <https://www.hongkongfp.com/2018/06/16/police-investigate-hung-hom-station-corner-cutting-scandal-based-latest-mtr-report/>).

The Hong Kong SAR government is the largest shareholder of MTR Corporation, who delegated or appointed Leighton Contractors (Asia) Ltd as the main contractor of the Sha Tin to Central Link project. Leighton delegated the construction works of the Hung Hom Station to another local subcontractor, who installed the steel bars.

The legal proceedings for this incident, which involved the supervisory duties and the chain of decentralisation of MTR Corporation and Leighton Contractors (Asia) Ltd and the impact this had on product quality, is still ongoing.

Management is responsible to investors and other stakeholders to ensure that divisional management, even when decentralised, is monitored and that basic oversight and auditing functions are in place. This is the responsibility of top management, which must ensure that stakeholder interests are protected.

Key Learning Point

Centralisation refers to the approach to management that puts most of the decision making in the hands of a small group of managers who are typically located at headquarters. Decentralisation is the delegation of management authority to middle and lower-level management. Companies may have different comfort levels with decentralisation.

► Knowledge Check Questions

Question 16

Explain the disadvantages or drawbacks of decentralisation from the following three aspects:

- (a) Increases expenditure
- (b) Creates conflict
- (c) Unsuitable for emergency situation

Question 17

Identify which of the following explains the advantages of decentralisation.

- A** It saves money, promotes better performance and allows remote management to generate higher profits.
- B** It empowers divisional managers to make their own decisions, provides wider distribution of duties, offers more efficient response in emergencies and may provide a better path for growth.
- C** It frees centralised management from responsibility.
- D** It removes the burdens of a top-heavy centralised and expensive management and shifts power to lower corporate levels.

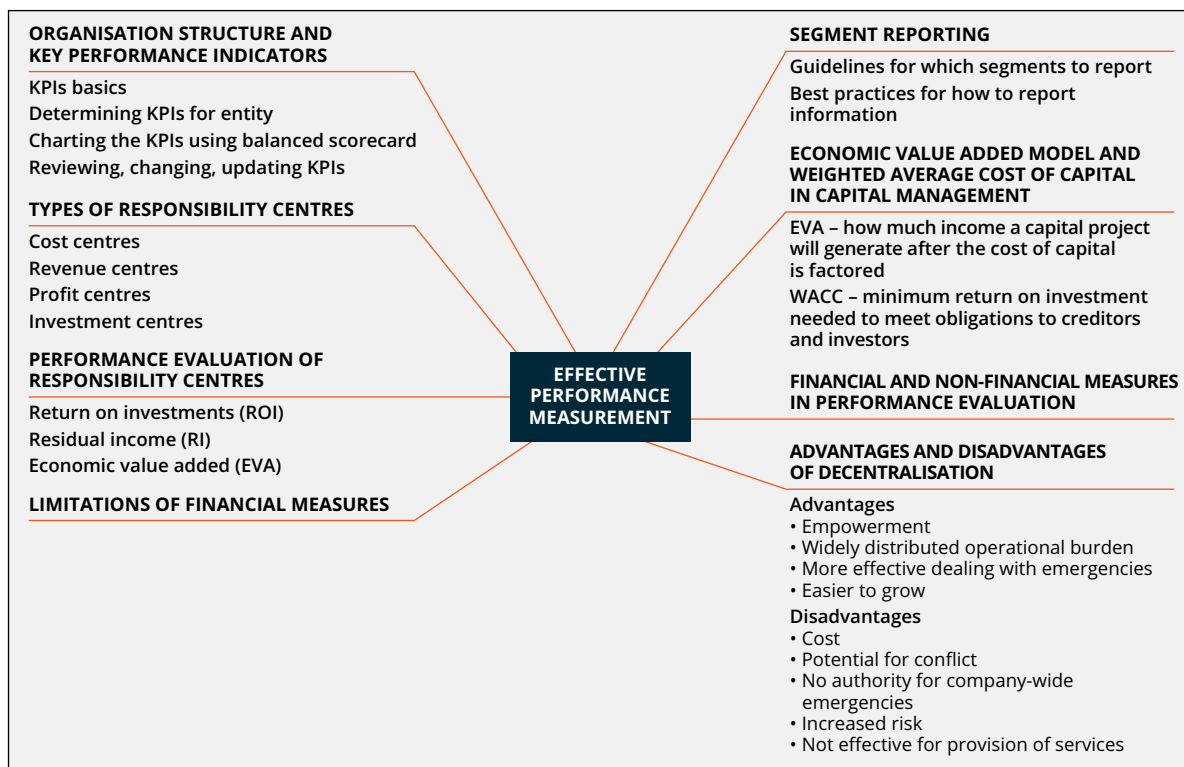


SUMMARY

- Key performance indicators (KPIs) are values that can be identified, measured and used to track the ability of companies to implement strategy successfully and achieve its mission.
- A KPI can be a measure of a short-term operational process or a long-run strategic process.
- Strategic KPIs are long-term, firm-specific measures that help companies monitor the success of their unique corporate and business strategies.
- Choosing KPIs to monitor is one of the most important steps in the successful implementation of a strategy.
- The balanced scorecard is a performance measurement system that includes a systematic approach for linking strategy to planning and control. The balanced scorecard includes performance measurement along several different dimensions.
- It is necessary for a company to use both short-term and long-term critical success factors to successfully implement a strategy.
- Using a balanced scorecard, management looks at firms from four perspectives: financial, customer, internal process, and learning and growth perspectives.
- There is no hard and fast rule on how many KPIs to monitor.
- There are four basic types of responsibility centres: cost centres, revenue centres, profit centres and investment centres.
- A cost centre incurs costs and expenses but does not directly generate revenues.
- A revenue centre is a segment of the organisation that is primarily responsible for generating revenue.
- A profit centre incurs costs and expenses and generates revenues.
- Direct fixed costs are costs that are specific to one centre and are incurred for the benefit of that centre alone. Indirect fixed costs are related to overall operating activities and to more than one profit centre.
- Like a profit centre, an investment centre incurs costs (and expenses) and generates revenues.
- Three financial measures commonly used to evaluate the performance of responsibility centres are: return on investment, residual income and economic value added.
- Return on investment (ROI) is the ratio of net profit to average operating assets.
- Residual income (RI) measures the dollar amount of profits more than a required rate of return, which is commonly referred to as a capital charge.
- Economic value added (EVA) is a type of residual income that incorporates several adjustments to reduce the disadvantages produced by residual income.

- Segment reporting refers to the concept of reporting the operations of specific segments of a company in the disclosures that go along with financial statements.
- Calculations of WACC are usually linked to the weighted average of a company's after-tax cost of debt, cost of preference shares and cost of ordinary shares.
- A singular focus on financial results is often insufficient to help an organisation implement its strategy successfully.
- Centralisation refers to the approach to management that puts most of the decision making in the hands of a small group of managers who are typically located at headquarters. Decentralisation is the opposite, with decisions made by managers spread throughout the organisation.
- In centralised entities, information produced by the accounting system for decision making is used primarily by top managers who are held responsible for both their effort and the quality of their decisions.
- There are several key advantages to decentralisation including empowerment, more widely distributed operational responsibilities, ability to deal with emergencies and easier growth.
- Decentralised companies and organisations must also overcome several disadvantages including expense, a potential for conflict, challenges dealing with emergencies, increased risk and ineffectiveness for provision of services.

MIND MAP



LIST OF FORMULAS

Return on investment (ROI)

$$\text{ROI} = \text{Net profit} / \text{Average operating assets}$$

Residual income

$$\text{Residual income} = \text{Operating income} - (\text{Required rate of return} \times \text{Average operating assets})$$

Economic value added (EVA)

$$\text{EVA} = \text{Adjusted after-tax operating income} - [\text{Weighted average cost of capital (WACC)} \times (\text{Adjusted total assets} - \text{Current liabilities})]$$

Weighted average cost of capital (WACC)

$$\text{WACC} = E/V \times R_e + D/V \times R_d \times (1 - T_c)$$

► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. This system is not restricted to budget tracking.

Answer B is incorrect. This is not a system for gaining greater market share.

Answer C is correct. This system is designed to create processes to reach goals.

Answer D is incorrect. This is not limited to marketing applications.

Question 2

Answer A is correct. A potential drawback is the unintended consequence of employees becoming focused on reaching the goal, but less so on quality results.

Answer B is incorrect. Customers do not suffer as the result of pursuing KPIs.

Answer C is incorrect. The KPI system should not affect employees and their motivation to perform well.

Answer D is incorrect. The accounting system should work well with KPIs when properly designed.

Question 3

Answer A is correct. The three efforts are directly related to one another.

Answer B is incorrect. These centres directly affect one another.

Answer C is incorrect. All three are related; profits cannot be separated from revenue and cost responsibilities.

Answer D is incorrect. These are not related solely to the budgeting process.

Question 4

Answer A is incorrect. Investment contains direct risk elements, and this cannot be separated.

Answer B is incorrect. Consultation is not the sole aspect of the investment centre responsibility.

Answer C is incorrect. Risk is directly associated with investment.

Answer D is correct. The investment centre manager must not only pick investments, but also be aware of how those decisions are affected by market and liquidity risks.

Question 5

An investment centre or investment division uses capital to contribute to the profitability of a company. Companies rely on revenues to determine how investment centres perform and evaluate the performance. These revenues come from investments in capital assets and are compared to overall expenses when the evaluation is done.

Question 6

Answer A is incorrect. Total assets are not used in this calculation.

Answer B is incorrect. This formula is reversed.

Answer C is correct. Net profit is divided by average operating assets to find ROI.

Answer D is incorrect. This calculation would not produce ROI.

Question 7

Residual income is the amount of net profit generated in excess of the minimum rate of return. Residual income concepts are used in a number of contexts, such as a measurement tool for internal corporate performance whereby a company's management team assesses the generated return relative to the company's minimum required return.

Question 8

Answer A is correct. ROI tends to distort the value of projects in some cases. RI may be less accurate when applied in larger divisions. And EVA may rely on judgement that could affect reliability.

Answer B is incorrect. The complexity for ROI and RI is not the primary issue for each.

Answer C is incorrect. RI is not advantageous due to simplicity; it may be overly complex.

Answer D is incorrect. ROI does not rely on judgement, whereas EVA does.

Question 9

Activities that create value throughout a company are not always included in the company's tangible fixed assets. There is often value to be found in ideas, relationships with customers and suppliers, databases with information, innovation and quality. For many of these, traditional financial measures may be tricky to use as they were designed to compare previous periods based on internal performance standards.

Question 10

Answer A is correct. The purpose of segment reporting is to preserve important outcomes while minimising the space required for the segment report.

Answer B is incorrect. The purpose is not to make reports brief, but to reduce the needed reporting space.

Answer C is incorrect. Segment reporting is not intended to obscure the fact that some segments may operate at a net loss.

Answer D is incorrect. A high level of detail is not desirable.

Question 11

Segment reporting refers to reporting related to different operating segments of a company. This reporting is done in the disclosures that go along with financial statements. Public companies are required to provide segment reporting. Private companies are not. The aim of segment reporting is to provide information about financial results and the performance and position of the important units of a company. This information can be used to make decisions related to the company.

Question 12

Answer A is incorrect. This is not the only reliable measure of income.

Answer B is incorrect. EVA takes the cost of capital into account.

Answer C is incorrect. This is designed for individual projects, not company-wide activity.

Answer D is correct. This specifically factors in the cost of capital.

Question 13

Answer A is incorrect. A direct correlation is part of the WACC calculation.

Answer B is correct. This is based on the after-tax cost of debt, removing incentives to time or adjust tax liabilities.

Answer C is incorrect. It does not ignore tax liabilities.

Answer D is incorrect. There is not a specific requirement to make tax liability adjustments for different jurisdictions.

Question 14

Answer A is incorrect. These are not typically listed as non-financial assets.

Answer B is incorrect. This is not a non-financial asset list.

Answer C is correct. These four assets represent non-financial value to a company.

Answer D is incorrect. These are not assets, either financial or non-financial.

Question 15

The metrics related to customer retention and churn are opposing metrics referring to the same variable. Marketing organisations often focus on customer acquisition. The ability to add customers is important but if companies are attracting new customers while old customers are exiting out the back door, they may be in trouble. Retention refers to the number of customers that continually buy from a company. Churn refers to the number of existing customers that stop buying products or services. The goal for companies is to increase retention and reduce churn. For example, if a company provides a subscription-based product, the company can decide churn is 60 days after renewal date.

Question 16

Decentralisation has the following disadvantages:

(a) Increases expenditure

Decentralisation required qualified, competent and skilled middle and lower-level managers and these managers have to be paid based on qualification and experience. A duplication of effort is another consideration that could lead to higher costs.

(b) Creates conflict

Top management in a decentralised organisation puts more pressure on department managers to boost output and revenue. This means every department puts more emphasis on their own department goals rather than corporate goals. This may lead to conflicts between departments and fragmentation.

(c) Unsuitable for emergency situation

Lower and middle-level managers have authority to deal with routine decisions and may not be able to handle more complex problems.

Question 17

Answer A is incorrect. Decentralisation does not save money.

Answer B is correct. These items are typically listed as advantages to decentralisation.

Answer C is incorrect. The issue is not one of responsibility and this would not accurately describe how decentralisation works.

Answer D is incorrect. Decentralisation is not intended as a power shift or even as a means for reducing management costs.

 EXAM PRACTICE**QUESTION 1**

Identify an example of KPIs in an organisation.

- A.** Standards for creating profits matching budgets.
- B.** On-time delivery of products.
- C.** Reducing inventory in storage to save on insurance costs.
- D.** Expanding market area to set up higher revenues.

QUESTION 2

Identify the three forms of financial measurement in popular use for responsibility centres.

- A.** Return on investment, residual income and economic value added.
- B.** Net return, gross profit and year-over-year revenue.
- C.** Dividend yield, P/E ratio and earnings per share.
- D.** Debt ratio, current ratio, quick assets ratio.

QUESTION 3

Identify which of the following are examples of lagging indicators in performance measurement.

- A.** Future performance tests or customer surveys.
- B.** Return on equity and marketing forecasts.
- C.** Stock price performance and plans to acquire and retire shares of stock.
- D.** ROI, RI and EVA.

QUESTION 4

The area manager of Pasta Village Restaurants is considering two possible expansion alternatives. The required investment, expected operating profit and the ROIs of each are as follows:

Project	Investment	Operating profit	ROI
Winnipeg	HK\$300,000	HK\$100,000	33.33%
Regina	HK\$700,000	HK\$200,000	28.57%

Currently, Pasta Village Restaurants has \$HK5,000,000 of capital invested in the company and operating profit of HK\$1,500,000. Identify which of the following projects will increase the Pasta Village Restaurants' ROI.

- A.** Both the Winnipeg and Regina alternatives.
- B.** Only the Winnipeg alternative.
- C.** Only the Regina alternative.
- D.** Neither the Winnipeg nor the Regina alternatives.

QUESTION 5

Describe how cost, revenue and profit centres are related to one another.

QUESTION 6

Explain how a company with many segments can reduce the number reported.

QUESTION 7

List the limitations for each type of financial measure.

QUESTION 8

State the disadvantages of decentralisation.

QUESTION 9

Wimmer Division's operating results include:

Operating profit,	HK\$150,000
Sales revenue,	HK\$1,200,000
Operating assets,	HK\$500,000

Wimmer is considering a project with sales of HK\$120,000, expenses of HK\$84,000 and an investment of HK\$180,000. Wimmer's required rate of return (RRR) is 15%. Analyse the data to determine whether Wimmer should accept this project considering the change to return on investment (ROI) that will occur. Identify and explain what other metrics of profitability can be used.

QUESTION 10

Mary's Hamburger Barn is planning to develop a balanced scorecard. Her strategy is a low-cost leadership strategy that focuses on reducing costs and scrap while maintaining high quality.

Identify a measure for each of the four perspectives of the balanced scorecard that Mary's Hamburger Barn might use to help track performance.

QUESTION 11

Belk's Pizza has purchased a small scooter to offer delivery to customers. An analysis of financial metrics suggests that the scooter is not a good decision for the business. However, the owner suggests that there are several non-financial considerations that make purchasing the scooter a good decision. Determine three non-financial considerations that Belk may have addressed in making the decision to purchase the new scooter.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. It would not represent KPIs.

Answer B is correct. It is an example of KPIs.

Answer C is incorrect. It would not represent KPIs.

Answer D is incorrect. It would not represent KPIs.

QUESTION 2

Answer A is correct. These three are the three forms of financial measurement for responsibility centres.

Answer B is incorrect. These are not performance measures for responsibility centres.

Answer C is incorrect. These are not performance measures for responsibility centres.

Answer D is incorrect. These are not performance measures for responsibility centres.

QUESTION 3

Answer A is incorrect. Events that happen in the future cannot be used for lagging indicators, which use current signs used to confirm long-term trends.

Answer B is incorrect. A lagging indicator is not a predictive tool, hence not a marketing forecast.

Answer C is incorrect. Stock price performance and prospective plans are not indicators used to confirm the strength of a trend.

Answer D is correct. These are the three lagging indicators for performance measurement.

QUESTION 4

Answer B is correct.

Calculation:

Operating profit	HK\$1,500,000
Divide by: Invested capital	HK\$5,000,000
Current ROI	30%

The Winnipeg alternative will increase company ROI because the project's ROI is 33.33%, which is higher than the company's ROI of 30%. In contrast, Regina will decrease ROI because the project's ROI is 28.57%, which is lower than the company's ROI of 30%.

QUESTION 5

These three centres affect one another directly. Costs affect both gross and net profits.

Even if revenues rise during a period, uncontrolled costs have a negative effect on profitability. Profits are not possible if either revenue or cost centres are not controlled and managed well.

QUESTION 6

The goal in combining segments is twofold: to reduce the number of segments that must be reported, and to represent most of the overall activity. For example, an organisation with 15 or 20 segments may set a goal to combine segments so that fewer than 10 must be reported. At the same time, the combination may be designed to capture 75% or more of overall results.

QUESTION 7

ROI is well understood and compared but may discourage managers from investing in some projects when ROI would be low. It also tends to understate or ignore risks, or to misstate investment turnover. RI may lead to cost cutting beyond what is needed; and management may have incentives to set required rates of return too low. EVA requires individual judgement rather than objective measures; and its complexity can be expensive and time consuming.

QUESTION 8

A decentralised segment is likely to be expensive compared to the centralised approach. Conflicts may occur between managers or departments that could be better controlled in a centralised system. In emergencies, decentralised management may not have the authority to take necessary recovery steps. In decentralisation, lower-level managers or remotely located managers could increase risk exposure for the company. Some services are better provided centrally. Examples of this are accounting or human resources.

QUESTION 9

Determining if this project is accepted or rejected should be based on whether the project's ROI is greater than the division's required rate of return. Since the project's ROI of 20% exceeds the division's RRR of 15%, this project should be accepted. Even though division ROI decreases, this project will increase the overall value of the division.

$$\text{Current division ROI} = \text{HK\$}150,000 / \text{HK\$}500,000 = 30\%$$

$$\text{ROI of new project} = \text{HK\$}36,000 / \text{HK\$}180,000 = 20\%$$

$$\text{New ROI with project} = (\text{HK\$}150,000 + \text{HK\$}36,000) / (\text{HK\$}500,000 + \text{HK\$}180,000) = 27.4\%$$

Computing residual income suggests this project should be accepted as this project increases division residual income by HK\$9,000, from HK\$75,000 up to HK\$84,000.

$$\text{Current division RI} = 150,000 - (500,000 \times 15\% \text{RRR}) = \text{HK\$}75,000$$

$$\text{New RI with project} = (150,000 + 36,000) - [(500,000 + 180,000) \times 15\% \text{RRR}] = \text{HK\$}84,000$$

QUESTION 10

(There are multiple acceptable measures.)

Perspective	Measure
Financial	Gross profit margin
	Profit margin
Customer	% increase in customer volume
	Customer satisfaction survey
Internal processes	Time from order to delivery
	% of scrapped raw materials
Learning and growth	% of employees trained in food sanitation
	Employee turnover
	Employee satisfaction survey

QUESTION 11

Possible non-financial considerations:

Competition: If competition has moved into the area, Belk may have chosen to purchase the scooter in anticipation of lost customers, if it does not meet the competition.

Customer service: Belk may have purchased the vehicle to deliver pizza because of a desire to satisfy customers who have requested delivered pizza.

Location: Belk may be in a location with little parking and purchasing a scooter to deliver will better serve customers.

Advertising: Belk will be able to put advertisements on the scooter making its name better known.



Part F

Financial Risk Framework



Chapter 16 Financial Risks and Their Management

16

Financial Risks and Their Management



CHAPTER TOPIC LIST

16.1 Types of Risk

- 16.1.1** Financial Risks
- 16.1.2** Business Risks
- 16.1.3** Operating Risks
- 16.1.4** Sales Risks

16.2 Key Risk Indicator

16.3 Financial Risk Management Approaches

- 16.3.1** Avoidance of Risk
- 16.3.2** Risk Mitigation
- 16.3.3** Transfer of Risk
- 16.3.4** Risk Acceptance



LEARNING OUTCOMES

PRINCIPAL LO2: APPLY FINANCIAL RISK FRAMEWORK AND DIFFERENT METHODS OF MANAGING FINANCIAL RISKS
--

LO2.01: Account for the nature and types of risk

2.01.01 Illustrate financial risk with their respective measurements

2.01.02 Explain and compare different methods of managing key financial risks
--



OPENING CASE

ABC INVESTMENT LTD

ABC Investment Ltd has developed a risk management programme as part of its investment portfolio.

In articulating a range of financial risks requiring management, ABC also has acknowledged that many related risks also apply. These come in many forms: credit risk, foreign investment risk, liquidity risk and, of course, market risk.

Beyond financial risks as direct concerns, ABC has had to determine how its risk management programme can and should combine four risk management ideas: avoidance, mitigation, transfer and acceptance.

A combination of these four risk management approaches applies to every aspect of ABC's operations. For its financial portfolio, for example, avoidance includes not investing in securities of companies with volatile stock price history, working capital problems or financial weakness revealed in financial statements. Mitigation is accomplished in some market sectors through diversification, not only amongst companies but using specialised sector mutual funds or exchange-traded funds. Transfer has involved the use of financial experts including financial planners and brokers, to take responsibility for managing a portion of specialised products in the portfolio. And risk acceptance includes investing in shares of some companies even while understanding that certain risks, notably market risk, are facts of life in investing. 

A large, stylized grey 'V' shape pointing downwards, with the word 'OVERVIEW' in orange capital letters positioned to its right.

OVERVIEW

Risks exist in many forms. An organisation that recognises this as an aspect of doing business is well suited to begin the process of risk management. If an understanding of risk is limited to only some types of risk, unexpected and costly surprises may arise.

For example, an accountant may be aware of the risks associated with working capital and cash flows; spending controls; forecasting and budgeting; and other strictly financial risks relating to internal control and reporting. However, the accountant may be less aware of credit, foreign investment and liquidity risks. These forms of risk may be addressed in a treasury department or in some cases overlooked altogether.

This chapter addresses financial risks and their management in three major sections: types of risk, key risk indicators (KRI) and financial risk management approaches.

In addition to examining attributes in these three areas of interest, a secondary purpose is to demonstrate that risk and risk management are not merely operational functions apart from other activities in an organisation. Risk exists everywhere and cannot be separated or treated as a function like accounting, marketing or operations.

Risk equates to exposure to loss in one form or another. However, along with risk comes opportunity. Future growth and profitability grow out of risk exposure and the uncertainties in the market. Success is not possible without exposure to risk; so rather than thinking of risk as a problem to be overcome or avoided in every case, it is more productive and realistic to view risk as an aspect of business to be managed and controlled; and to accept risk exposure as necessary to pursue future profits and growth of the organisation.

A large, stylized grey arrow pointing to the right, with the text '16.1 TYPES OF RISKS' in orange and black capital letters positioned to its right.

16.1 TYPES OF RISKS

Risk encompasses several different forms. These include financial, business, operating and sales risks.

Focus on financial risk is a problem in organisations. Great emphasis is placed on the financial realm, as the method for scorekeeping in most organisations is based on financial matters – profits, budgets and forecasts, marketing success or failure, cash flows and capitalisation.

Beyond these appropriate concerns, however, are other forms of risk. These contain importance to the overall health of a company, and exposure to this broader range of risks may threaten a company's health and survival as much as the strictly financial risks well known to most financial professionals.



16.1.1 Financial Risks

Risk is a constant factor of doing business. A well understood form of risk is *financial risk*, which is any risk involved with financial transactions, lending, capitalisation and in the extreme, the possibility of default in loans or even bankruptcy in the case of insolvency.

Financial risk extends well beyond the well-known market **downside risk** of stock prices.

Although the risk of loss in stock investments is a financial risk, it is only one kind of risk faced by every organisation.

16.1.1.1 Credit Risk

The first variation of financial risk is **credit risk**, which can be summarised in a general definition as the risk of default of borrowed funds. For a company carrying loans or bonds on its books as liabilities (acting as a borrower), the risk of default is related to unexpected decline in profitability and cash flows, translating to an inability to make timely payments of principal and interest.

For an organisation granting loans to others or investing in bonds (acting as a lender such as banks), the risk is concerned with default by the borrower. Both – borrower and lender risk – are part of the broader classification of credit risk.

Credit risk is not limited to companies carrying debt or lending; it comes in many forms within these broad types. These include failure to make payments on any form of loan (mortgage, credit card, line of credit, for example); failure to make secured debt payments when due; late payment of invoices or wages; failure by a bond issuer to make payments of principal or interest by a due date; insolvency leading to a company not being able to meet contractual obligation or to refund a customer's assets (for example, by a broker or bank); the results of fraud or theft by an employee or company; or filing bankruptcy and returning only a small portion of claims to creditors.

Credit risks can be reduced by a lender. Performing a credit check for a proposed borrower is a first step. A lender may also require borrowers to pay for **insurance** (for example, mortgage insurance for granting of a home loan) or a **third-party guarantee** (often called a co-signer).

The higher a lender's perceived risk in lending money, the higher the interest rate will be. In this manner, the lender requires many borrowers to share in the risk of default, understanding that some high-risk borrowers will default on their repayments.

These practices and methods for assessing credit risk are best known when applied to consumer borrowing. However, the same standards apply to business credit risk as well.

Several types of credit risks are possible. These include:

- **Default risk:** The risk of loss because a borrower is unable to make timely payments on a loan. The default may be partial or full. In a partial default, notably in a secured loan, the lender will acquire **collateral** in partial settlement of the remaining debt.
- **Country risk:** The risk of loss resulting from freezing of a company's assets located in a foreign country, when that country defaults on its loans or freezes assets. In extreme cases, a country may take over the assets of a company with assets within its borders. This risk is associated with countries suffering from political uncertainty and instability.

- **Concentration risk:** The risk of loss derived from exposure in a single company or industry. The 2008–2009 credit debacle (i.e. financial tsunami) in the financial world is an example of exposure in one industry (financial). Years earlier, the bankruptcy of Enron in 2000 represented concentration risk in one company. Individuals and companies invested in Enron stock suffered big losses resulting from reliance on falsified financial reports.

Another form of credit risk is **sovereign risk**, which is the risk that a government will be unable (or unwilling) to make loan repayments in a timely manner, or which defaults on its loan guarantees. This is most likely to occur during periods of severe recession.

Another variation of credit risk is **counterparty risk** (a form of default risk), in which a related or counterparty does not meet its obligations to pay on contracted loans, bonds, derivatives or insurance policies. This often results from liquidity problems with the counterparty.

Lenders can mitigate credit risks in several ways.

One way is by adopting risk-based pricing strategy. By increasing the interest rate for higher-risk borrowers, lenders spread credit risk amongst a population of borrowers. This risk-based pricing may be based on credit history, payment history, the purpose of a loan, amount borrowed and the borrower's credit rating. Lenders also change interest rate based on the **loan-to-value ratio**. This is the percentage of the amount borrowed to the total value of an asset. The lower the ratio, the lower the risk.

For example, a borrower is purchasing capital equipment worth HK\$400,000. The amount borrowed is HK\$310,000. The loan-to-value ratio is: $\text{HK\$310,000} \div \text{HK\$400,000} = 77.5\%$

A second way is through **credit insurance**. A lender hedges their **credit risk** by purchasing credit insurance. In exchange for a premium, the insurer assumes the credit risk.

A third way is through **credit derivatives**. A form of option known as a credit derivative hedges the risk in a different manner. A **credit default swap (CDS)** involves a seller who pays the buyer if a debt default occurs. The seller is insuring the buyer against credit default.

A fourth way is through **covenants**. A lender includes contractual covenants (stipulations) required of the borrower. These include financial reports to be given periodically, or restrictions such as payment of dividends, repurchasing shares, or placing caps on further borrowing. If other changes occur, such as increases in the debt-to-equity ratio or interest coverage ratio, it may also lead to a demand for full repayment as another form of covenant.

A fifth way is through **diversification**. Lenders reduce exposure to credit risk by offsetting concentration risk. This means expanding the pool of borrowers to avoid a catastrophic loss on a single debt.

A sixth way is **tightening**. A lender reduces exposure to credit risk by limiting or reducing maximum credit, either to a pool of similar borrowers or to a single borrower. Another form of tightening involves a change in repayment terms, such as requiring payment within 15 days rather than 30 days.

16.1.1.2 Foreign Investment Risk

Many companies lend money to those based in other countries. This **foreign investment risk** refers to changes in taxation imposed by the host country; rapid inflation; economic conflict



or expropriation of assets; nationalisation of assets belonging to companies invested in the country; and a volatile political climate.

Less severe forms of foreign investment risk include difference between a home country and a foreign country in regulatory requirements, accounting standards, reporting requirements and rules governing external audits.

Valuation of currency is another potentially serious form of foreign investment risk. This could affect liquidity or even the ability to conduct business at a profit. If customers in a foreign country cannot afford to buy products due to severe inflation and currency valuation changes, the market could become impractical for an exporter.

16.1.1.3 Liquidity Risk

In terms of financial risk, **liquidity risk** refers to problems in finding a **ready market** to trade an asset or commodity. In some markets, the time required to complete a trade may affect the profitability of a trade.

This is a different form of liquidity than the accounting-related use of the term in describing working capital or comparisons of cash flows between current and long-term assets and liabilities.

In financial risk, two forms of liquidity risk are of concern: **asset liquidity** refers to lack of a ready market due to a too wide bid-ask spread or other inhibiting factors. **Funding liquidity** is the risk that liabilities may not be paid when due because of illiquid conditions (too little cash).

A ready market is defined as 'a recognised established securities market in which there exists independent bona fide offers to buy and sell so that a price reasonably related to the last sale price or current bona fide competitive bid and offer quotations can be determined for a particular security almost instantaneously and where payment will be received in settlement of a sale at such price within a relatively short time conforming to trade custom'.

Liquidity risk occurs when a company or individual wishes to sell an asset, but no buyer can be located to pay the price the seller asks; or when no buyer can be found at any price. In terms of liquidity, the lack of a market at any price indicates that the asset is worthless. However, in some cases the asset is not worthless; the problem is in matching seller and buyer. In over-the-counter markets, where assets are not listed or priced on a public exchange, liquidity (and value) are uncertain unless seller and buyer locate one another and agree on price.

16.1.1.4 Market Risk

A final type of financial risk is **market risk**, the risk of loss due to movement in the price of assets. It may take one of several forms. **Equity risk** is commonly known and represents the risk that investments in stocks, indices or mutual funds will lose value as prices change. A similar risk is **commodity risk**, which occurs when a company is invested in a commodity futures position or an option and the commodity moves in an unfavourable direction.

A closely related range of risks include **interest rate risk**, the risk of unfavourable changes in rates; and **currency risk**, when a company is invested in a currency exchange product, such as those offered on the FOREX.

In short positions, investors are required to post collateral in a margin account, notably on commodity futures and **options** trades. After movement in the underlying, margin collateral

requirements may change, leading to the possibility of *margin risk*, when funds are not available to satisfy a margin call.

Market risk is measured in many ways, but it can be summarised as the risk of losing money in an invested position due to adverse movement in the product or in a derivative of a product.

16.1.2 Business Risk

Business risk is the risk of not earning profits or meeting profit goals; or of suffering a net loss instead of a profit. This may be the consequence of changing consumer preferences, fierce price competition, overall economic conditions such as inflation and high unemployment and obsolescence. In the extreme of this risk, a business could become insolvent and be forced into bankruptcy.

Business risk may take one of two forms.

Internal risks include any event occurring within the organisation. This may include labour strikes, uninsured disaster losses or loss of vendors for essential raw materials.

External risks include any event outside of the organisation, including work stoppage due to natural disasters, terrorist attacks, loss of a large customer, compliance and regulatory changes and price competition.

Overall, business risk may be classified in five ways. First, strategic risks arise from the business environment, investor relations or transactions such as mergers or joint venture failures. Second, the risk can be purely financial, involving failures due to cash flows or low profitability. Third, they may involve compliance risks, such as failure to follow rules and regulations set by regulatory and governmental agencies. Fourth, risks may be operational, those associated with the process of running the business and its segments or departments. Finally, they may be caused by natural disasters (fire, flood, volcano) or human-made events (terrorism, embezzlement, sabotage).

16.1.3 Operating Risk

Operating risk is the risk derived from losses and resulting reduction in value of assets. These are incurred by failed internal controls or systems, or from external risks.

Typical forms of operational risks include fraud, sabotage or embezzlement as well as decisions leading to environmental damage. This risk has been termed 'quality control' in a broad sense. It encompasses losses resulting from poor customer relations, shareholder relations and lack of internal controls during periods of volatility. Operating risk may also result from poor training of employees. This range of risks is not associated directly with profitability, that level of risk familiar to accounting employees and management. For that reason, it is easily overlooked or underestimated in importance.

Operating risk is broad and encompasses many disciplines. It was defined in the Basel II Accords in June 2004 as: 'The risk of loss resulting from inadequate or failed internal processes, people and systems or from external events.'

The **Basel Accords** are a series of banking regulation recommendations, issued by the Basel Committee on Banking Supervision (BCBS), based in Basel, Switzerland.



Basel II contains three principles, or pillars: minimum capital requirements, supervisory review and market discipline. The third pillar (market discipline) includes a series of internal policies and controls aimed at improved operational management (thus, reduction of operating risk). The Basel Accords have been further strengthened with the introduction and implementation of Basel III since 2013.

A purpose in operating risk, beyond the obvious demand for improved internal monitoring of operations, routines and decisions, is to maintain levels of risk tolerance. This is complex because it is not revenue-driven, meaning measuring operating threats of loss is less tangible or recognisable than better-understood risks that are revenue-driven.

Under the Basel definition of operating risk, examples include internal and external fraud, workplace safety, employment practices, business practices, risks of damage to capital assets, business disruption and failures, process management and data or accounting errors and loss of client assets. This range of operating risks often is articulated as the major forms of risks addressed in a risk management programme.

16.1.4 Sales Risk

The term **sales risk** is more accurately called the potential for sales failures. For example, the loss of future business due to sales failures eventually affects revenue and profits.

Types of sales risk include poor lead generation or qualification; loss of opportunities because a sales team spends excessive time pursuing low-probability customer leads; lack of product or service knowledge on the part of inadequately trained sales employees; poor basic skills such as training in how to manage objections by potential customers, leading to a failure to close the sale.

Another form of sales risk occurs when competitors recruit the best salespeople from a company's sales force, offering more attractive commissions or improved sales support and training. To offset this risk, companies may consider specific steps such as sales incentives and rewards for meeting clearly expressed quotas; or analysis of existing quotas to remove unfair or unreasonable expectations.

A sales force must be trained at several levels. This includes understanding the customer base and shifting customer expectations or requirements. When demand for a product or service declines, the company needs to maintain its competitive position with improved offerings, so that sales risk does not increase due to inaction by the company.

Finally, sales risk is directly associated with competition. This refers not only to higher quality in products offered by a competitor, but also in better policies related to the sales force: training, commissions, incentives and interaction. When marketing management consults with sales teams to discover what is changing in the market, the range of sales risks is reduced.

Illustrative Example 1

There are a number of differences between operating risk and business risk.

Risk represents a change or deviation in actual returns from the returns that were expected. In a business, risk typically refers to a variation in earnings, a variation that may be caused by any of a number of reasons. These variations may in broad strokes, be classified in two groups.

Illustrative Example 1 (*continued*)

The first set of reasons are linked to the uneconomical use of operating expenses, which are classified as operating risk. The second group includes inefficient financing decisions, which are classified as financial risk.

Operating risk is associated with day-to-day operations. Companies operate in specific internal and external operating environments and the impacts of environments are reflected in operating costs. Operating costs include fixed costs and variable costs. Too high fixed costs are disadvantageous. When a company has high fixed costs and revenues decline, operating profit will drop proportionately more.

Business risk emerges from operating conditions and the potential changes in operating income that may be caused by changes in operating conditions. Business risk is linked to the possibility of losses or inadequate profits caused by uncertainties or unexpected events outside of management control. Business risk is related to operations and is independent of the company's capital structure.

Apply and Analyse 1

Apply

The risk management consultant your investment company has hired prepares monthly assessment reports. However, this is usually restricted to analysis of internal safety summaries and status of insurance the company holds. Your task as an accountant reporting to the CEO is to evaluate what financial risk should be included to expand the scope of the risk management effort.

Analysis

You have studied the broad range of financial risks and drawn several conclusions.

Credit risk from a lender's perspective is of immediate concern. Your company's investment portfolio includes several bonds, some of which pay rates well above current market rate. However, the rating on these bonds is below investment grade and you conclude that these are 'junk bonds' (likely to default). Your first recommendation is to sell these bonds and avoid exposure to default risk.

Foreign investment risk would be of concern if your company investment is in overseas ventures. Although many customers are out of the region, the scope of this risk is not great. However, the monthly analysis should monitor steps that might be taken resulting in higher risk.

Liquidity risk is not of immediate concern, as the investment portfolio does not include any assets that would be difficult to dispose of. However, you do think the monthly risk assessment should consider the dangers of some assets that do not have a liquid secondary market; these should be avoided.

Market risk is a constant threat, and you recommend that margin requirement be monitored continuously as part of the portfolio management effort.

Key Learning Point

Both borrowers and lenders face and must deal with credit risk, which exists on both sides of a debt transaction.

► Knowledge Check Questions

Question 1

Identify the forms of financial risk faced, especially in the investment sector.

- A** These risks include liquidity risk, business risk and sales risk.
- B** The major types of risk relate to operating risk and credit risk.
- C** Four types of risk apply: credit, foreign investment, liquidity and market risk.
- D** Foreign investment risk is the most serious and the only type of immediate concern in a risk programme.

Question 2

Identify how business risk represents a threat to a company's continued operations.

- A** Failing to make revenue and profit goals threatens cash flows.
- B** Both internal and external business risks must be monitored.
- C** The impact of competition or changing economic conditions should be part of the risk analysis.
- D** All of the above.

Question 3

Identify how operating risk functions and the threat it poses.

- A** This risk is generally minor and does not require a great deal of attention.
- B** It describes problems with internal staff and their lack of training, as a primary form of operating risk.
- C** This may include events such as fraud, sabotage and embezzlement, which may all severely impact a company's profitability.
- D** Operating risk relates to decisions made by sector management and not reported to headquarters.

➤ 16.2 KEY RISK INDICATOR

The **key risk indicator (KRI)** is a valuable tool employed by risk managers to assign degrees of risk to specific activities or threats. It estimates potential for future losses and consequences if those risks occur.

The KRI metric is designed to provide management with warnings of emerging risk exposure. However, it is designed to monitor known risks and cannot be expected to anticipate a broad range of risks beyond the most likely forms of risk to which the organisation is exposed.

In comparison, a **key performance indicator (KPI)** measures the level of performance of internal routines in terms of risk exposure and is not a metric of potential future risk events. Both metrics may be used in conjunction as part of a company's operating risk analysis and control process.

The value to a KRI is that it provides information about the probability of a risk occurring and potential costs associated with that risk. A company cannot prevent all forms of risk but use of the KRI should identify risks that exceed the company's *risk appetite*.

Risk appetite is the degree or level of risk the company accepts as either affordable or unavoidable. No initiative is possible without risk, and risk appetite should be well defined.

For example, in pursuing a new marketing strategy to expand a customer base, a company invests money in promotion, opening new branch offices, employee training and product development. This is a substantial investment. The risk is that the expansion may not be profitable. Investing money to expand should fit with the company's risk appetite.

In describing and defining risk appetite, the KRI comes into play and helps to narrow down the definition of how risk may be accepted or managed. This definition phase may result in several classifications of the company:

Averse to risk means the company wants to not take any steps that could lead to exposure.

Minimal profile describes a company willing to take only safe steps to minimise likely exposure.

Open is a company able and ready to consider all avenues and willing to accept levels of risk if those risks are matched to potential profits.

Cautious is like minimal, because the company prefers safe courses of action and associated limited rewards.

Risk appetite is part of the risk management process, not a substitute for it nor for employing the KRI to articulate levels of risk exposure. Risk management too often overlooks the collateral negative outcomes of a risk event, even when insuring against the event itself. For example, a natural disaster leading to a shutdown of headquarters may be recovered through a highly detailed programme; but key employees may find employment elsewhere and those resources lost permanently even after the operation is resumed.

A company may consider several steps in determining the most effective KRI. These steps include:

- Who are the stakeholders? These include employees, investors or shareholders, vendors, suppliers and regulatory agencies. How will specific forms of risk affect stakeholders?
- Risk monitoring indicators. What indicators should be used to track risk exposure?
- What are the underlying causes of risk exposure? By understanding the causes, many risks can be avoided or mitigated. A hazardous situation in a manufacturing plant can

be corrected with safety equipment; the danger of flooding in a facility is overcome by relocation or by mitigation steps, such as creation of water barriers to prevent internal damage.

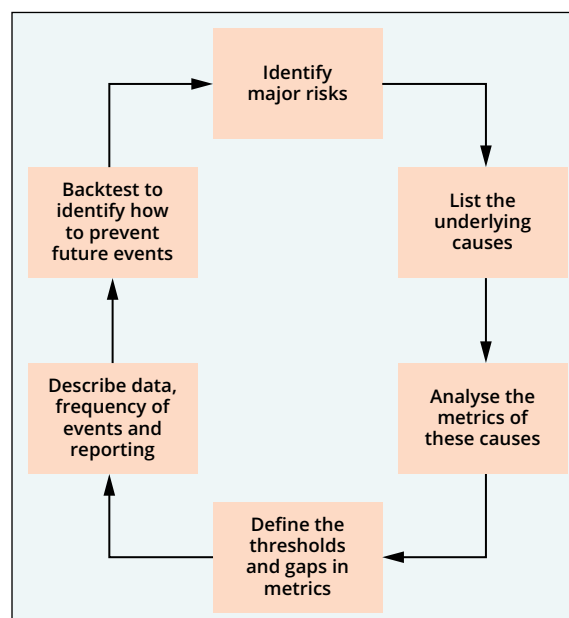
- How are risks classified? Many systems can be used to define risks. These may include dividing them into four classifications: High impact with high likelihood, high impact with low likelihood, low impact with high likelihood and low impact with low likelihood. Emphasis in selection of a KRI programme should be on risks with high impact and likelihood (which usually also means high cost).
- What are the thresholds to risks? By focusing on the most urgent risks (likelihood and cost levels), a company can also focus its KRI selection.

The purpose in identifying these factors is to select KRIs most likely to provide early notice to management of risk exposure, a means to learn from past risk events, identification of how and when risk appetite levels and tolerances are met, and insights for action points management needs to take.

KRIs should be able to measure risk exposure accurately; identify possible losses in financial terms; and establish levels of confidence that the company will have in how it has measured and anticipated risk.

This is a circular process, beginning with identification of major risks; listing the known underlying causes; analysis of the metrics in these causes; defining the risk appetite (thresholds and gaps in metrics); describing relevant data, frequency of events and reporting systems; and finally, backtesting to determine how experience could prevent future occurrences. Finally, the circle concludes by returning to the beginning and again monitoring major risks.

This series of phases is shown in Exhibit 16.1.



► **EXHIBIT 16.1** Steps in developing a KRI programme

From this exercise, the KRI is developed. It may take the form of ratios or trends, or just a series of numerical values intended to provide comparative data of financial impacts, probability of occurrence and other metrics that may apply. The purpose to development of KRI data is to track long-term trends and converting those trends to advance warnings for possible events.

For example, relaxing safety rules for a plant floor may lead to higher likelihood of injury. Tracking the trends is revealing. If trends reveal lower incidence of injury to employees when safety equipment is operational, it is significant. If safety equipment is disengaged or used improperly, the backtested trend reveals a higher incidence of injury.

Illustrative Example 2

A production facility tracks delays in the line. As delays increase, the causes are investigated. Amongst those discovered is running out of parts, defects in one or more stages, workers not at their posts or poor training.

Each of these contributed to delay increases. The KRI addresses all the causes so that plant management can determine how to reduce delays by solving the contributing problems.

In this example, the KRI developed to analyse delays contained three attributes required to identify solutions. It was easily measured (time of delays and the trend it represented); it was significant (because delays cost the company money, notably in labour costs when the production line is delayed); and it makes sense (delays are well understood when the underlying causes become known, and this leads to deciding how to reduce future delays).

These attributes also define minimum thresholds a company should expect and can tolerate. This may be based on risk appetite, tolerance or well-known norms in the industry.

Apply and Analyse 2

Apply

Your organisation has experienced problems in its production facility. Specifically, the incidence of employee injury has been trending higher. This has led to higher insurance and medical costs, increased production delays and higher defects due to employees needing to work longer hours to cover for those injured and away from work. What KRI can mitigate this problem?

Analysis

Employee injuries invariably occur due to unsafe working conditions. These include hazards on the shop floor, improper use of safety equipment and even a lack of signage warning about dangers.

Apply and Analyse 2 (continued)

The solution is to generate a training update focused on safety. This includes reminders about steps employees must take in exposure to risk (everything from proper lifting to adequate eye protection and headgear); use of safety equipment and reasons why this equipment should not be disabled or not installed; and other basic but essential safety reminders.

For example, your study of the programme reveals that the cutting saw used on the shop floor comes with a safety shield; but the shield has been removed with an explanation that it is cumbersome to use and not necessary. Several injuries have resulted from cuts to workers' hands and shards of sawdust in eyes.

The KRI includes not only instalment and use of the shield, but wearing safety goggles at all times when working with electrical equipment. A secondary concern is that many employees are not wearing hardhats on the shop floor, and this is also required. The KRI lists these and other steps. For example, safety warning signs should not be removed or covered up. They serve a purpose of reminding employees about hazards.

Key Learning Point

The KRI is not exclusively meant to prevent all risks, but is also used to monitor known risks and to mitigate their impact. The purpose of the KRI is to measure risk exposure and enable the company to measure and anticipate risk events.

► Knowledge Check Questions

Question 4

Describe the KRI of a risk management programme.

Question 5

Identify which of the following is the purpose of developing KRIs in an organisation.

- A** This is a bureaucratic exercise of little value.
- B** It provides information about the probability of a risk occurring and potential costs associated with that risk.
- C** It helps prevent all future risks when applied properly.
- D** The KRI programme focuses on responsibility for risks after they occur and assignment of blame.



16.3 FINANCIAL RISK MANAGEMENT APPROACHES

There are four primary methods for dealing with risk. Financial risk management is a reference to creating economic value through risk avoidance, mitigation or transfer. In some instances, a company will decide to accept risk as an affordable consequence of doing business, meaning the cost is minimal versus the cost and effort required to take other steps.

Theory and reality are not identical in terms of risk management. In theory, a **perfect market** enjoys perfect competition and enjoys equilibrium between supply and demand at all times. In reality, companies do not exist in a perfect market. Within the realistic framework of a market with degrees of chaos, management must set priorities on addressing risks, and must determine how to address those risks.

16.3.1 Avoidance of Risk

The first method for addressing identified risks is **risk avoidance**. This is the removal of exposure to hazards that could cause losses to the company's assets, cash flows or reputation.

Risk avoidance is not responsive to impending or realised losses. It is an action designed not to be exposed in any sense to a known potential loss. Unlike other forms of risk management designed to control the financial cost of loss or to mitigate losses when they occur, avoidance is a means for removing the exposure altogether.

Avoidance requires knowledge of risk before steps can be taken to prevent exposure from occurring. When a company suffers from **risk ignorance**, meaning it is unaware or unconcerned with some forms of risk, it cannot avoid those risks. Risk ignorance assumes that a company does not know about its exposure.

Forms of risk avoidance include not constructing a plant facility in a flood plain or on the edge of a cliff; wearing seatbelts in vehicles; and wearing safety goggles and helmets when working in areas with hazardous conditions. There is a significant difference between avoidance and ignorance. In avoidance, workers wear safety goggles to prevent injury to their eyes. In ignorance, workers fail to wear safety goggles because they do not know about the risk.

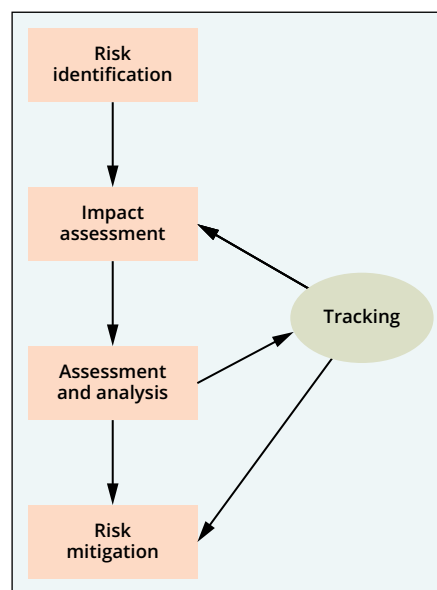
Risk avoidance does not require that remediation steps be enacted to prevent known events. It means the risks have been evaluated and understood; and that decisions have been based on what was learned about the risk. For example, when working with wood or metal, the risk of having damage to the eyes is avoided by understanding the risk and instructing employees to wear properly rated goggles. An unsatisfactory remediation step could be to allow employees to wear cheap plastic goggles that will not prevent injury; or cautioning employees to be careful when in the work area.

16.3.2 Risk Mitigation

A second form of risk management is **risk mitigation**. This is identification and enactment of processes and actions to reduce threats. Mitigation is not avoidance or removal, but reduction in threat levels. This involves tracking known risks and identifying new risks, and then evaluating processes to prevent larger losses.

For example, in a plant environment, management knows that hardhats will not prevent injury if falling objects hit a worker; but the helmet reduces (mitigates) the degree of injury likely to occur.

Mitigation is divided into four steps: risk identification, impact assessment, analysis and mitigation. Tracking amongst these four steps is essential as part of the risk management programme. This is summarised in Exhibit 16.2.



► **EXHIBIT 16.2** Risk mitigation process

Mitigation involves a series of options. These include:

- **Acceptance.** Once a risk is known and has been mitigated, management may determine to accept the mitigated risk.
- **Avoidance.** The second option reverts to risk avoidance. A mitigated risk does not require acceptance; it may still be avoided even after mitigation.
- **Control.** Management may take further steps to further mitigate the risk or to reduce the financial loss associated with it.
- **Transfer.** Management may deal with a mitigated risk by assigning it to a vendor or supplier, different division or other stakeholder who is willing to assume the risk.
- **Monitoring.** Finally, the company may identify the risk impact and determine whether it has changed; and then determine to continue relying on mitigation or take other steps when required.

16.3.3 Transfer of Risk

The third form of risk management is **transfer of risk**. This means assigning the risk to another group or company.

The most common form of transfer is *insurance*. By purchasing insurance against an event occurring, the risk (or part of the risk) is assigned to the insurer in exchange for payment of a premium. For example, chances for a fire are small but it would not be affordable. A company transfers the risk to a casualty insurance company and pays a premium. In the event of a fire, the company will be reimbursed for its losses (minus identified deductibles, co-pays and policy limits).

A second form of transfer is from purchase of **derivatives**. These hedge against financial risk. In the market, options are a form of derivative that is used to offset or reduce market risk. Another form of derivative can be purchased to hedge currency exchange rate risks.

Outsourcing of risk is another form of transfer. These are accomplished by contract, often between a producer and a supplier or vendor. For example, failure to deliver on the terms of a contract imposes a monetary penalty on the producer. Since the producer relies on a vendor to manufacture the goods involved, its contract transfers the penalty to the vendor in the event the vendor does not meet agreed upon delivery schedules.

16.3.4 Risk Acceptance

Beyond the three forms of risk management (avoidance, mitigation and transfer) a company may also decide to adopt a policy of **risk acceptance**. This is assumption of a known risk without attempting to remove it.

When the cost of managing the risk in some form is too high given the potential loss, risk acceptance is a logical and rational choice.

Many companies practice risk acceptance in many ways. For example, policies in management of an investment portfolio is based largely on risk acceptance. By analysing and studying investments and seeking conservative levels of risk, the potential for loss is mitigated to a degree. Even so, all investing involves risk, and the portfolio risks are accepted based on the initial research conducted.

Also, in a portfolio, some derivatives can be employed to hedge against market risk. In this use of derivatives, a combination of transfer and mitigation reduces the level of risk that must be accepted by the investor.

All forms of risk acceptance are entered with awareness of the risk levels, and with a willingness to manage those risks. Acceptance can be limited through insurance or the use of derivatives; but the remaining risk is accepted after these mitigation and transfer decisions.

As already explained in this chapter, risk appetite is the degree or level of risk the company accepts as either affordable or unavoidable. Therefore, final decisions on risk acceptance have to fit parameters of risk appetite. Remaining risk and associated costs of risk mitigation and transfer must be assessed thoughtfully in their entirety. Total cost and benefit to the company through risk taking and risk acceptance must be finely balanced to stakeholders' advantage within company risk appetite.



A company should take steps to ensure that its exposure to loss is well understood. Risk acceptance should be made only after the full range of possible exposure is well understood, and management has decided it is acceptable. Risk acceptance resulting from inaction or a lack of knowledge is a form of risk ignorance, which is the lack of risk management and not a conscious decision based on financial analysis.

Risk ignorance can become a serious problem in two instances. First, known levels of risk are not explained to management. As a result, a decision to take no action is based on faulty or missing information. Second, risks are simply not known. This occurs when risk assessment is not performed or when the nature of risk was not anticipated. For example, a small company suffers a substantial case of embezzlement on the part of a trusted employee. Management had no indication in advance of its vulnerability to this risk.

Illustrative Example 3

Your company offers health insurance to its employees. Full coverage is very expensive, so the company provides basic coverage (physical exams, minor illness coverage, inoculations, etc.) without charge to employees but subject to a small deductible and a 20% co-pay agreement. It provides catastrophic coverage for incidents above a ceiling provided for in basic coverage.

This decision was based on analysis of the cost of full coverage provided by an insurer, versus levels of claims by employees in past years. The company determined that it would be ahead financially by accepting risks up to the limits of basic coverage. Potential losses were reduced by deductible and co-pay agreements in the coverage your employer provides.

Ethics in Practice

Risk and ethics are not separate but may involve both considerations.

An employee in the accounting department has been told by a marketing executive that one line of products marketed by the company is potentially hazardous to children. Incidents of small children choking on removable parts have yet to be examined or reported.

The marketing executive proposed that no action needed to be taken because no lawsuits had been filed against the company, and because liability insurance would cover any future losses. This has created a dilemma for the accountant.

From a purely financial point of view, the product (which has been sold only for the past six months) has been profitable beyond projections. However, from an ethical standpoint, is it fair to the consumer to continue marketing a product that could lead to injury or even death?

In this case, 'risk' exists on several levels. Financially, the company is exposed to lawsuits and even with insurance coverage, the overall cost could exceed coverage. On this basis alone, the accountant is compelled to recommend withdrawing the product from the market.

Ethics in Practice (continued)

On an ethical basis, it is improper to continue selling a product if the hazards are known and incidents have occurred. This basis is an even stronger reason to recommend recalling all sold products.

On a risk basis, the flawed product is now known to marketing and financial employees. Exposure to the risk on the part of the company is secondary to the known hazards to the consumer. The accountant is ethically required to make a recommendation, with a cost and benefit analysis, to immediately stop selling the product and to issue a recall for those products already sold.

Apply and Analyse 3

Apply

Your accounting department manager is working on annual budgets. This includes attempting to budget for a small staff in a newly formed 'risk management' department.

Without a history to draw upon, explain how this budget should be prepared. Your manager has asked you to contribute ideas for this.

Analysis

Your task is complicated by two factors. First is the lack of history, as this is a new department. Second is an understanding of exactly what routines will be pursued in the department and what the related costs should be. The objective statement does not address this adequately.

You meet with the manager of the department and determine that amongst the areas of activity are: insurance review, risk analysis and identification of how some risks can be mitigated or transferred through recommendations.

Your budget is broken down by type of risk. In each of the three areas (avoidance, mitigation and transfer) you address costs as specifically as possible. Many risks can be avoided by actions and you attempt to set up a budget to aid in the scope of recommendations the risk management department can offer. Risk mitigation is more complex and may involve identification of known risks and discovery of ways to reduce both exposure and cost. Risk transfer is focused primarily on existing and proposed new insurance coverage.

The most challenging part of the budget involves examination of risk acceptance. What forms and levels of risk can or should be accepted by the company? For this, you develop a budget to perform cost analysis of identified risks versus estimated costs of risk management actions. The purpose is to identify areas where risk acceptance makes the most sense. This requires a separate budget line for estimated expenses that may result from acceptance of relatively minor risks (or for risks that would be too expensive to deal with through other means).



Key Learning Point

Risk management has the purpose of planning for, preventing or mitigating losses. Risk avoidance – taking steps to not be at risk – is not the same as risk ignorance – being unaware of the risks faced – and knowing the difference is crucial in a risk management programme.

► Knowledge Check Questions

Question 6

Identify the differences between risk avoidance and risk acceptance.

- A** They result in the same outcome, low cost and relatively low exposure.
- B** Avoidance requires no specific actions to overcome risk, but risk acceptance is, in comparison, more proactive.
- C** Avoidance is based on actions designed to move away from exposure to known risks; acceptance results from an analysis concluding that other steps are not justified by limited exposure.
- D** Avoidance is usually illegal or unethical, but acceptance is undertaken within the law.

Question 7

Identify how risk mitigation works.

- A** A company takes steps to reduce exposure to a known risk or risks but cannot be expected to avoid it entirely.
- B** The budget is reduced for specific risks.
- C** The company mitigates risks by abandoning overly expensive markets.
- D** All of the above.

Question 8

Identify how risk transfer is most often undertaken.

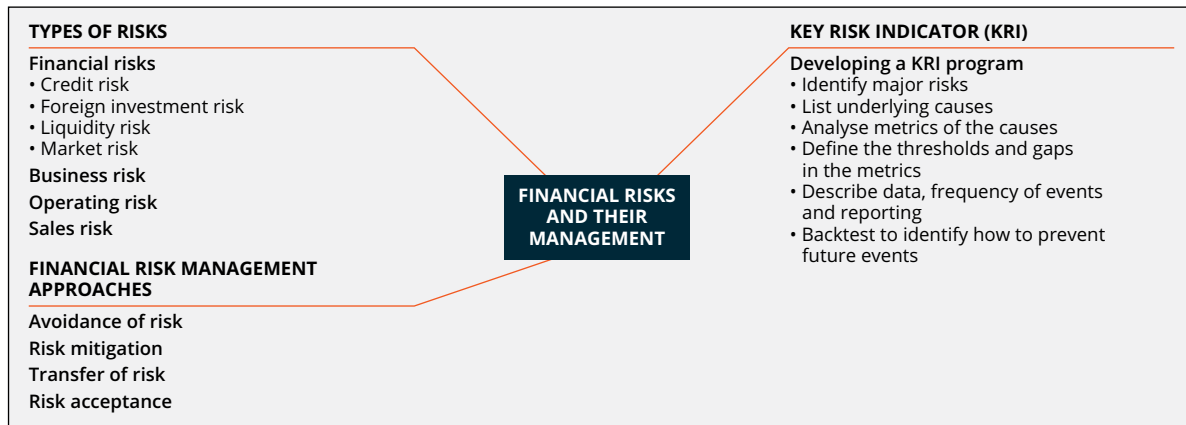
- A** The most common form is to accept some risks and assume that shareholders are willing to be exposed to the outcome.
- B** The most popular form of risk transfer is to select consumers less likely to be litigious.
- C** Insurance is the most common transfer form.
- D** Risks cannot be transferred without considerable costs, so it is the most unusual form of risk management.



SUMMARY

- Risk is found in several forms, including financial, business, operating and sales risk.
- Credit risk involves forms of risk to both borrower and lender.
- Foreign investment risk occurs in many ways: currency exchange, political change, nationalisation of assets or changes in import laws, for example.
- Liquidity risk refers to problems in finding markets to trade an asset or commodity.
- Market risk usually refers to the risk of loss in asset value.
- Business risk is the risk of not earning profits or meeting profit goals, or of suffering a net loss instead of a profit.
- Operating risk is the risk derived from losses and resulting reduction in value of assets.
- Sales risk is the potential for sales failures.
- A key risk indicator (KRI) is a conclusion drawn from analysis of trends and conditions and is used to decide how to manage an identified risk.
- Financial risk management takes one of four forms: avoidance, mitigation, transfer or acceptance.
- Avoidance of risk refers to steps taken to remove activity from known exposure.
- Risk mitigation occurs when steps are taken to reduce exposure, but the entire scope of risk cannot be eliminated completely.
- Transfer of risk usually takes the form of insurance.
- Risk acceptance should result from analysis of risk management costs and a conclusion that the cost of acceptance is lower than the cost of risk management.

MIND MAP



► Answers to Knowledge Check Questions

Question 1

Answer A is incorrect. Financial risk does not include business or sales risk.

Answer B is incorrect. Operating risk is separate from financial risk.

Answer C is correct. These four encompass the scope of financial risk.

Answer D is incorrect. Foreign investment risk is one form, but it is not the only one, nor is it always accurate to describe it as the most serious.

Question 2

Answer A is incorrect. This is a form of business risk, but not the only one.

Answer B is incorrect. It is true that both internal and external risks need to be monitored, but this is not the sole description of business risk.

Answer C is incorrect. These external factors are only a part of business risk.

Answer D is correct. All of the descriptions in choices A, B and C are part of the broader scope of business risk.

Question 3

Answer A is incorrect. Operating risk is not minor and it should be monitored.

Answer B is incorrect. This is not a problem related to training.

Answer C is correct. Operating risk includes events such as fraud, sabotage and embezzlement, all of which may represent major threats.

Answer D is incorrect. This is not a form of risk relating to remote operational lack of oversight.

Question 4

The KRI is a key risk indicator, a conclusion drawn from analysis of trends and conditions and is used to decide how to manage an identified risk in a risk management program.

Question 5

Answer A is incorrect. This is of value in defining the scope of risk management.

Answer B is correct. This exercise identifies probability and cost of potential risks.

Answer C is incorrect. Risks cannot be prevented in every instance.

Answer D is incorrect. This is not a means for assigning responsibility or blame.

Question 6

Answer A is incorrect. Avoidance and acceptance are not the same.

Answer B is incorrect. Risk acceptance is not more proactive than risk avoidance.

Answer C is correct. Avoidance consists of actions to remove activity from hazards, and acceptance is the result of analysing the costs involved and deciding to live with relatively small risks.

Answer D is incorrect. Avoidance is not illegal or unethical.

Question 7

Answer A is correct. Mitigation is the result of specific steps taken to reduce exposure.

Answer B is incorrect. Mitigation is not a reference to the risk management budget.

Answer C is incorrect. This is not the result of abandoning high-risk markets exclusively.

Answer D is incorrect. Options B and C are not correct.

Question 8

Answer A is incorrect. Risk transfer is not a reference to passing risks to the shareholder.

Answer B is incorrect. This is not related to selection of one group of consumers over another.

Answer C is correct. Risk transfer most often refers to one or more forms of insurance.

Answer D is incorrect. Risk transfer is not necessarily high-cost and it is not an unusual selection.

EXAM PRACTICE

QUESTION 1

Identify which of the following describes financial risk.

- A.** This is a risk that costs investment money to counter.
- B.** Financial risk is any risk that impacts the company on a financial basis.
- C.** This is the risk associated with paying to manage risks that probably will not occur.
- D.** All of the above.

QUESTION 2

Identify which of the following describes operating risk.

- A.** This encompasses internal and external risks to the organisation.
- B.** The risk that operating management will make errors that could be avoided with checks and balances.

- C. This is a risk associated with employee actions contrary to stated goals and objectives.
- D. The risk apart and separate from marketing risk.

QUESTION 3

Identify the purpose of a key risk indicator (KRI).

- A. This is meant to place risks on a spreadsheet in terms of priority.
- B. KRI is a metric to identify risks and methods to manage them.
- C. This is a warning received from a regulatory agency that the company is exposed to foreign intrusion, usually of a cyber nature.
- D. KRI is not relevant to industry but affects governments.

QUESTION 4

Identify the major types of financial risk management.

- A. Proactive, reactive, passive and unknown.
- B. Financial, non-financial, brand-specific and marketing.
- C. Profit-based, cash flow-based, expansion-based and capital investment-based.
- D. Avoidance, mitigation, transfer and acceptance.

QUESTION 5

Illustrate and explain how credit risk may impact a company's portfolio.

QUESTION 6

Explain the dangers in sales risk in terms of (a) lost sales and (b) loss of a sales force.

QUESTION 7

Describe how risk mitigation functions.

QUESTION 8

Summarise the attributes of risk acceptance.

QUESTION 9

EC company is a Hong Kong company which has excessive cash in Hong Kong dollars on hand. Three projects have been proposed to the CFO.

1. Lending to an unrelated company in Hong Kong.
2. Investment in a water purification plant in Nigeria and receive Nigerian naira in return.
3. More investment in an existing subsidiary in Hong Kong to increase sales as it faces severe sales pressure.

For each project, identify the risks involved and propose mitigating steps.

ANSWERS TO EXAM PRACTICE

QUESTION 1

Answer A is incorrect. Financial risk management is not limited to the cost of mitigation.

Answer B is correct. This is any risk impacting the company financially.

Answer C is incorrect. It is not associated with identifying risks unlikely to occur.

Answer D is incorrect. Only option B is correct.

QUESTION 2

Answer A is correct. Operating risk may be internal or external.

Answer B is incorrect. Operating risk is not limited to management errors.

Answer C is incorrect. This is not limited to employee actions or errors.

Answer D is incorrect. This is not a risk separate from marketing-based risks.

QUESTION 3

Answer A is incorrect. The KRI is not simply a spreadsheet used to study risks.

Answer B is correct. KRI is the means for identifying and managing risks.

Answer C is incorrect. KRI is not a warning from regulatory agencies.

Answer D is incorrect. This is not limited to government activities.

QUESTION 4

Answer A is incorrect. They are not the major types of financial risk management.

Answer B is incorrect. They are not the major types of financial risk management.

Answer C is incorrect. They are not the major types of financial risk management.

Answer D is correct. These four classifications describe the four forms of financial risk management.

QUESTION 5

Credit risk is associated with both borrower and lender. In a portfolio, the company may be invested in notes or bonds issued by countries or government agencies, or by other companies. The default risk to any long-term debt is a constant factor; however, credit risk is mitigated by selecting high-rated debts and avoiding those paying exceptionally high rates of interest but rated poorly by rating agencies.

QUESTION 6

Sales risk may take more than one form. The most apparent of these is failing to meet quotas or goals expressed in a sales forecast. The consequences of this include lower net profit and potential damage to cash flows. A second form of sales risk is having successful sales employees recruited by competitors, perhaps offering higher commissions. This is most likely to occur when marketing management does not address or support a sales staff's requirements.

QUESTION 7

Mitigation refers to reduction of risk exposure but not to its complete removal. For example, the risk of injury in a production facility is reduced by requiring employees to wear helmets and safety goggles; and to ensure that safety equipment is used and not disabled. Mitigation also involves taking steps to avoid exposure to a degree of risk; however, unlike risk avoidance, mitigation does not eliminate a risk, it only reduces the level of exposure and cost.

QUESTION 8

Risk acceptance means living with a risk rather than spending money to mitigate, transfer or avoid it. This makes the most sense when the cost of acceptance is lower than the cost of managing the risk. For example, a company decides to self-insure employee health benefits

with conditions. These include requirement for payment of premiums and deductibles and imposition of a co-pay. It also limits maximum liability per event; losses above that event are insured by a catastrophic insurance policy. The risk acceptance portion is the net cost to the company of health care under these terms.

QUESTION 9

1. This lending incurs credit risk, which is generally the risk of default of borrowed funds. There is risk of loss because a borrower may be unable to make timely payments of principal or interest on the borrowed fund. The default may be partial or full.

Credit risks can be reduced by a lender in a few ways. A credit check for a proposed borrower is a first step. A lender may also require borrowers to pay for credit default insurance, or a third-party guarantee (often called a co-signer). The borrower may also be asked to provide *collateral* to secure the borrowing. The collateral would be sold for repayment in case of default.

2. Investment in another country incurs foreign investment risk which refers to changes in taxation imposed by the host country; rapid inflation; economic conflict or expropriation of assets; nationalisation of assets belonging to companies invested in the country; and a volatile political climate. Less severe forms of foreign investment risk include difference between a home country and a foreign country in regulatory requirements, accounting standards, reporting requirements and rules governing external audits.

Valuation of currency is another foreign investment risk. This could affect liquidity or even the ability to conduct business at a profit. If Nigerian naira valuation experiences severe changes, the market could become impractical to operate in.

Investment risk can be mitigated by purchasing political risk insurance for Nigeria. *Insurance* helps businesses to protect their overseas investments in situations such as political violence or confiscation of assets. It covers risks from business disruption to outright loss of the overseas business.

Currency risk can be mitigated through the use of derivatives against depreciation of Nigerian naira. The company can also borrow in local currency naira in Nigeria to finance assets in Nigeria as natural hedge against naira depreciation.

3. This situation involves sales risk which is more accurately called the potential for sales failures. The loss of future business due to sales failures eventually affects revenue and profits. Types of sales risk include poor lead generation or qualification; loss of opportunities because a sales team spends excessive time pursuing low-probability customer leads; lack of product or service knowledge on the part of inadequately trained sales employees; poor basic skills such as training in how to manage objections by potential customers, leading to a failure to close the sale.

Steps the company should take to reduce sales risk include understanding the customer base and shifting customer expectations or requirements, improving product offers in light of competition and improving training, support, incentives and interaction with sales teams.



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GLOSSARY OF TERMS

ability to pay an approach to pricing based on the idea that buyers should not pay more than they are able or may face an unnecessary burden.

accounting rate of return (ARR) an approach to appraise investments or capital projects that calculates the return on a capital project using net income and book value rather than cash flow data.

activity-based costing an accounting method that, based on the consumption of resources, allocates indirect expenses to multiple activity cost pools and then uses cost drivers to assign those costs to the products and service lines that a company produces and delivers.

agency relationship a relationship in which one party, known as the principal, delegates decision-making authority to another party, known as the agent.

agency theory a financial theory that attempts to explain the dynamic between principals and agents in business transactions, and aims to avoid and solve problems that may arise in any such agency relationship.

annual percentage rate (APR) annualised effective interest rate without compounding on loans that are for the fraction of a year.

annuity a series of equal dollar amounts to be received or paid at equal intervals over a limited number of periods.

arbitrage the concurrent purchase and sale of an asset in order to profit from an imbalance in its price in different markets or forms.

asset liquidity a lack of a ready market due to a too-wide bid-ask spread or other inhibiting factors.

averse describes that a company wants to not take any steps that could lead to risk exposure.

balanced scorecard a performance measurement system that includes a systematic approach for linking strategy to planning and control.

bankers' acceptance (BA) a promissory note issued by a foreign importer and guaranteed by a bank in return for the export of goods in cross-border trading, issued at a discount from their face value with maturities of between 1 to 180 days and typically 90 days.

bank overdrafts flexible borrowing facilities on a bank account that are repayable on demand.

bankruptcy legally declared inability of an individual or a company to pay creditors.

bankruptcy costs costs of financial distress associated with financial difficulties a firm might experience because it uses debt financing.

Basel Accords a series of banking regulation recommendations issued by the Basel Committee on Banking Supervision (BCBS), based in Basel, Switzerland.

basis risk risk associated with a hedged position that is attributable to the fact that the asset to be hedged is not identical to the asset used as the hedge.

beta is the gradient of the line of best fit between the returns of the stock and the general market and the method by which to quantify the relation between the returns on a stock and the general market.

bond a fixed-income investment in which an investor lends money to an entity, such as a corporation or government, for a defined period of time at a variable or fixed rate of interest.

bond agreement sometimes called bond indenture, it is a legal document that specifies the payment requirements, other salient matters relating to issues, any protective provisions and other additional features.

bond indenture see **bond agreement**.

bond theorems a collective term to describe the relations that exist between bond prices and interest rates.

book value per share (BVPS) a measure of a company's relative value. It is the ratio of the ending shareholders' equity to the ending number of shares.

brokers market specialists who bring buyers and sellers together for a sale to take place and charge a fee from both buyers and sellers for their services, but do not bear the risk of ownership of the securities during the transaction.

bullet payment (or balloon payment) when the principal payment is made in one lump sum at maturity.

bundle pricing an approach to pricing that allows entities to sell packages or sets of goods or services for a lower price than they would charge if each product or service was bought separately.

burn rate the rate that a company uses cash.

business finance companies companies that sell short-term debt to investors in the direct credit markets, where these funds are used to make a variety of short- and intermediate-term loans and leases to small and large businesses.

business plan a written document that describes in detail how a business will achieve its goals.

business risk 1. Risk associated with the specific characteristics of particular projects or assets.
2. Risk of not earning profits or meeting profit goals; or of suffering a net loss instead of a profit.

business valuation models used to determine the economic value of a business to arrive at a fair value for the business.

callable bonds bonds that give the issuer the option to 'call', or repurchase, outstanding bonds at predetermined call prices (generally at a premium over par) at specified times.

call option an agreement that gives a buyer the right, but not the obligation, to buy an underlying asset at a specified price within a specific period of time.

call premium the amount that a purchaser of a call option must pay to the writer of the option.

capital asset pricing model (CAPM) a financial model that describes the relationship between systematic risk and expected return for an asset.

capital budgeting the decision-making process through which the company purchases productive assets.

capital intensity ratio a measure of the amount of capital needed per dollar of sales. It is calculated by dividing total assets of a company by its sales. It is reciprocal of total asset turnover ratio.

capital market financial markets with a focus on long-term funds.

capital project a long-term, capital-intensive project with the goal of building on or improving a capital asset. Capital projects are typically large and costly.

capital project monitoring the process used to track the evolution of a capital investment project.

capital rationing when a firm does not have enough capital to invest in all attractive projects and must therefore ration investment capital among investment projects.

capital structure the approach that a company takes to finance its overall operations and growth by using different sources of funds, including debt (bonds, notes or loans) and equity (common or preferred stock or retained earnings).

cash money in hand, including money in bank accounts, cheques or other forms of currency that is easily accessible and can be quickly turned into physical cash.

cash conversion cycle the period of time from when a company actually pays for raw materials until it receives cash from the sale of finished goods.

cash coverage ratio the ratio of the sum of EBIT and depreciation and/or amortisation to the interest.

cash dividends dividends paid in cash.

cash flow the amount of money transferred in and out of a business or capital project.

casualty insurance companies insurance companies that sell protection against loss or property from fire, theft, accidents, negligence and other causes.

cautious a risk profile when the company prefers safe courses of action and associated limited rewards.

centralisation an approach to management putting most of the decision-making in the hands of a small group of managers who are typically located at headquarters.

chief executive officer (CEO) the top company executive reporting directly to the board of directors and has the final decision-making authority among all the company's executives.

chief financial officer (CFO) the most senior financial manager in a company.

chief risk officer (CRO) a key company executive whose job is to identify, analyse and mitigate risks, internal and external, that could threaten the operations of a company.

clean price the clean price of a bond does not include any accrued interest or interest that will be accrued on the next couple of payment dates.

clearing float the delay in transferring funds between payer and payee that arises from the banking system clearing processes.

collateral assets held as guarantee of debt repayment.

combined leverage a combination of operating and financial leverage.

commercial banks the most prominent and largest financial intermediaries in the economy and that offer the widest range of financial services to businesses.

commercial paper (CP) a promissory note with credit risk.

commodity risk the risk in a commodity futures position or an option that the commodity might move in an unfavourable direction.

common stock or ordinary shares, the most prevalent type of equity security, represents the basic ownership claim in a corporation.

competition the effort of enterprises to be a leader in its industry and increase market share.

complementary product pricing strategy an approach to pricing in which a company may set the price of one product so as to achieve the largest possible volume of sales regardless of cost or profits in order to increase demand for another product that may have higher profit margins.

compounding converting an initial amount into future value.

compound interest interest earned on the principal amount invested and on the future interest payments.

concentration risk the risk of loss derived from exposure in a single company or industry.

consumer surplus the difference between what a customer pays and what a customer is willing to pay.

contingent convertible (coco) bond a fixed-income instrument that can be converted into equity when a trigger event happens.

controllable profit margin controllable revenues less controllable costs.

conventional cash flow where an initial cash outflow gives way to subsequent cash inflows.

convertible bond a type of debt security or bond that can be converted into a predetermined amount of the underlying company's equity (stock) of the issuing company at certain times during the life of the bond.

corporate social responsibility (CSR) refers to the impact that any company has on society and the environment and to a self-regulating business model that companies use to be more socially accountable.

corporate strategic plans form of plans that consider internal factors as well as external considerations that could impact a company.

correlation coefficient a statistical measure that identifies how security returns move in relation to one another.

cost accounting a field of accounting focused on recording all the costs incurred in a business to improve management.

cost-based pricing strategy a strategy relying on costs to determine the right price.

cost centre a responsibility centre that incurs costs and expenses but does not directly generate revenues.

cost of capital the required rate of return for a capital investment.

cost of production costs incurred by a business when manufacturing a good or providing a service.

cost of quality (COQ) a methodology that allows an organisation to determine the extent to which its resources are used for activities that prevent poor quality, that appraise the quality of the organisation's products or services, and that result from internal and external failures.

counterparty risk a form of default risk when a related or counterparty does not meet its obligations to pay on contracted loans, bonds, derivatives or insurance policies.

country risk the risk of loss resulting from freezing of a company's assets located in a foreign country when that country defaults on its loans or freezes assets.

coupon (or regular interest payments) refers to the interest payment paid to the holder of a bond.

coupon payment an interest payment made to bondholders.

coupon rate a bond's annual coupon payment divided by its face value.

covariance a measure of how the returns on two assets covary or move together.

covenants contractual terms (stipulations) required of the borrower to track solvency and ability to repay the debt.

covering also known as hedging, refers to the act of removing a naked financial position.

creative accounting the process of applying accounting rules in a distorted manner to exaggerate results favourably so that profits may be misrepresented.

credit default swap (CDS) insurance taken by a seller to insure the buyer against credit default.

credit derivatives options used to hedge lending risks.

credit insurance a form of insurance taken out by a lender to ensure repayment of a debt.

credit policy a company's policy of whether or not to extend credit, and how it decides to which customers to extend credit.

credit risk the risk of default of borrowed funds.

cumulative dividend a feature of preference share that stipulates that preference shareholders must be paid both current-year dividends and any unpaid prior-year dividends before common stock holders are paid dividends.

currency risk the risk when a company is invested in a currency exchange product.

current asset cash and other assets that a firm expects to convert into cash in a year or less.

current liability obligations that a firm expects to repay in a year or less.

current ratio estimated by dividing current assets by current liabilities.

current yield the ratio of the annual coupon interest divided by the current market price.

dealers market specialists who bring buyers and sellers together for a sale to take place and charge a fee from both buyers and sellers for their services, and also bear the risk of ownership of the securities during the transaction.

debentures similar to bonds but are generally unsecured, or are secured by a general floating charge over the company's unencumbered assets.

debt money borrowed by one party from another.

debt ratings ratings mainly assigned by the three debt-rating services, which perform detailed analyses of bond issuers to determine their ability to sustain their required payments of interest and principal.

debt-to-equity ratio the ratio of debt a company has to every dollar of its equity.

decentralisation the delegation of management authority to middle and lower level management.

declaration date the date when the board of directors of a corporation authorises a dividend payout.

declared dividend a dividend payout that has been 'declared' by the board of directors of a company.

default risk the possibility that the borrower may default on the promised debt payments.

degree of combined leverage (DCL) a measure to evaluate how a company's income (earnings per share) changes with respect to a percentage change in sales.

degree of financial leverage (DFL) a measure of how sensitive a company's earnings per share are to changes in EBIT.

degree of operating leverage (DOL) a measure of the sensitivity of accounting operating profits (EBIT) to changes in sales.

depreciation a method of allocating the value of a tangible asset over the course of its useful life.

derivative options and other contracts used to hedge against financial risk.

derivative securities markets financial markets for financial contracts or instruments that derive

their values from underlying debt and equity securities.

differential cost the difference in cost that would arise between two or more alternative decisions or from a change in output volume. This concept is used to help a company choose between two or more possible options.

direct costs costs representing a consistent percentage of the cost of producing a good or service.

direct fixed costs costs that are specific to one centre and are incurred for the benefit of that centre alone.

dirty price a price quote for a bond that includes the current value of all cash flows, including the interest on the next coupon payment.

discount bonds bonds that sell at a price below par value.

discounted payback period the number of years required to fully recover an initial cash outlay in an investment in terms of discounted cash flows.

discounting the process by which the present value of future cash flows is obtained.

discount rate the interest rate or required rate of return of investment used to determine the present value of future cash flows.

diversification 1. Method for reducing a portfolio's risk by spreading investment funds across several assets, with the key being to choose assets whose returns are less than perfectly positively correlated. 2. Method for reducing a lender's exposure to credit risk by offsetting concentration risk and expanding the pool of borrowers.

dividend a distribution of cash or stock to stockholders of a company on a proportional basis.

dividend growth model sometimes called constant dividend growth model, it is a valuation model used to calculate the fair value of a stock and makes an assumption that dividends grow at either a stable rate forever or at a different rate in the current period.

dividend payout ratio a ratio used to analyse a company's ability to sustain dividend payments. It is the ratio of dividend per share to earnings per share.

dividend yield a basic ratio used to analyse a company's valuation. It is the ratio of dividend per share to the current share price.

downside risk the risk of decline in asset values due to market price movement.

DuPont equation a fundamental performance measurement framework that is useful to decompose what drives return on equity (ROE).

earnings per share (EPS) net income divided by the number of common shares outstanding.

EBIT earnings before interest and taxes.

EBIT return on assets (EROA) a measure of how efficiently management uses its assets. It is the ratio of EBIT to total assets.

EBITA earnings before interest, taxes and amortisation.

EBITDA earnings before interest, taxes, depreciation and amortisation.

EBITDA margin it is the ratio of EBITDA to net sales.

economic order quantity (EOQ) a model that determines the optimal inventory level that minimises shortage costs and carrying costs incurred to order and hold inventory.

economic value added (EVA) a type of residual income that incorporates several adjustments to reduce the disadvantages produced by residual income.

effective annual rate (EAR) also known as the effective annual yield (EAY), this is the annual percentage rate with compounding on loans that are for a fraction of a year.

effective annual yield (EAY) also known as the effective annual rate (EAR), this is the annual percentage rate with compounding on loans that are for a fraction of a year.

Efficient Market Hypothesis (EMH) a market theory stating that stock prices are immediately discounting for all known information about a company.

efficient portfolio portfolio that provides the highest expected return for a particular level of risk or the lowest risk for a particular return.

enterprise value also known as takeover value, it is the market value of an entity's capital, less cash and cash equivalents.

environmental costs those costs affecting the environment and resulting from a company's activities.

equity a stock or any other security representing an ownership interest in a company.

equity multiplier the ratio of the amount of assets a company has to every dollar of equity.

equity risk the risk that investments in stocks, indices or mutual funds will lose value as prices change.

ex-dividend date the first date on which the stock will trade without rights to the dividend.

expected return a measure that represents the sum of the products of the possible outcomes and the probabilities that those outcomes will be realised.

extendible bonds bonds that allow the bondholder to extend the maturity date of the bond.

external financing requirements (EFR) the amount of funding a company needs for the rest of the year based on pro forma financial statements.

externalities side effects or consequences of an economic activity, such as an investment, that affects other parties in positive or negative ways, without being reflected in the cost of the goods or services involved.

external risks the risk of any adverse event outside of the organisation, including work stoppage due to natural disasters, terrorist attacks, loss of a large customer, compliance and regulatory changes and price competition.

face value the principal amount owed to the bondholder at maturity.

fair value an alternative approach to measurement that seeks to capture changes in value of assets and liabilities over time.

finance study of how money flows between lenders and borrowers.

financial accounting a specialised branch of accounting that keeps track of a company's financial transactions. It focuses on the aggregation of accounting information into the financial statements of a company.

financial leverage the amount of debt that an entity uses to acquire assets.

financial management refers to the management of financial resources of a company, which includes assets and the financing of those assets.

financial market a broad concept that covers all forms of markets that deal with short-term and long-term funds.

financial planning model a series of equations that is used to generate projected financial statements for a company, such as a statement of profit or loss and other comprehensive income or a statement of financial position.

financial risk refers to any risk that companies face from their financing decisions or transactions. This includes risk from activities like lending, capitalisation, financing and default in loans and bankruptcy in the case of insolvency.

financial strategy outlines the short- and long-term goals of an organisation and can have

an impact on the liquidity and solvency of a company.

financing costs the costs associated with the financing of a project.

financing plan also known as a financial plan, it describes a company's plan to determine how much external funding it may need.

fixed costs costs that do not vary in the short term, regardless of changes in levels of production or sales.

fixed-income bond a bond in which the interest paid to investors is fixed for the life of the contract.

float the delay that occurs when funds are sent by a payer to a payee.

floating-rate bond a bond that pays a rate of interest that fluctuates over the life of the instrument.

foreign exchange markets (FOREX markets) electronic markets in which banks and institutional traders buy and sell various currencies on behalf of businesses and other clients.

foreign investment risk the risk of changes in taxation imposed by the host country, rapid inflation, economic conflict or expropriation of assets, nationalisation of assets belonging to companies invested in the country and a volatile political climate.

forward a contract similar to a future contract but is standardised and is done between two parties to buy or sell a particular asset in the future.

free cash flow (FCF) a measure of how much cash a business generates after accounting for capital expenditures such as buildings or equipment.

free cash flow (FCF) model a model that considers the cash flow that a company will generate to determine the value of shares.

funding liquidity the risk that liabilities may not be paid when due because of illiquid conditions (too little cash).

future contracts a legal agreement to buy or sell a particular asset (typically stocks, commodities or foreign currency) at a predetermined price at some point in the future.

future value a measure of how much an investment will be after it has earned interest for one or more time periods.

gross profit margin measures the percentage of net sales remaining after the cost of goods sold is paid.

hedging the act of entering into a contract with the goal of lowering existing exposure to risk, such as price fluctuations.

incremental after-tax free cash flows total after-tax free cash flows a company or project generates.

incremental analysis the process used to identify the financial expenses and future earnings that change under alternative courses of action.

incremental cost is the relevant cost for making a short-run decision between two alternatives, such as accepting a special order.

indirect costs those costs that change as a percentage of total costs as the volume of production grows.

indirect fixed costs fixed costs for overall operating activities and are incurred for the benefit of more than one profit centre.

industry group analysis an analysis of a group of companies with similar size and product lines competing in the same market.

inflation the increase in the general level of prices of goods and services.

initial public offering (IPO) the process of raising capital by offering stock to the public for the first time.

insurance a contract to transfer risk, obtained through payment of premiums; a limitation or removal of identified risks.

interest the cost of funds loaned to an entity by a lender.

interest rate risk the risk of unfavourable changes in rates.

interest rate swap an agreement between two counterparties to exchange streams of future interest payments.

internal growth rate (IGR) the maximum growth rate that a firm can achieve without external funding.

internal rate of return (IRR) an investment appraisal technique that considers the rate of return associated with a project to determine whether the rate is higher or lower than the project's cost of capital.

internal risks include any event occurring within the organisation.

inventory turnover the number of times inventory is sold during a particular period. This is used to measure the liquidity of inventory.

inventory turnover ratio the average number of times a typical inventory level is replaced. It is represented by the ratio of the cost of goods sold to the inventory ending balance.

investment appraisal the process used to determine whether an organisation's long-term investments are worth funding.

investment centres responsibility centres that incur costs (and expenses) and generates revenues. In addition, an investment centre has control over the investment funds that are available for use.

investment funds sell shares to investors and use the funds to purchase securities.

investment tax credit a direct reduction of the company's tax bill at the time an investment is made, which can significantly affect the initial cash outlay for a project.

junk bonds also known as high-yield bonds, bonds with higher risk but offer higher expected returns.

just-in-time (JIT) inventory an inventory system in which production inputs for a manufacturing system are delivered or made just before they are used.

key performance indicators (KPIs) these refer to the values that can be identified, measured and used to track the ability of companies to implement strategy successfully and achieve its mission.

key risk indicator (KRI) a tool employed by risk managers to assign degrees of risk to specific activities or threats and to estimate potential for future losses and consequences if those risks occur.

lease a contract that outlines the terms under which one party receives the rights to use the assets of another party for a specified period of time.

leverage ratios the measures of how much debt a firm uses as opposed to equity financing. Also known as the long-term solvency ratios.

life insurance companies companies that sell life insurance policies that protect individuals against the loss of income from a family member's premature death.

lines of credit (LC) arrangements between a financial institution, such as a bank, and a customer, such as a corporation, that establishes how much a customer can borrow.

liquidity the ability to convert assets into cash quickly without financial loss.

liquidity risk risks related to problems in finding a ready market to trade an asset or commodity.

loan-to-value ratio the percentage of the amount borrowed to the total value of an asset.

long-term funding strategy financing strategy that relies on long-term debt and equity to finance both fixed assets and working capital.

management accounting the analysis of financial data from internal processes that are used by managers to track and account for business transactions and in the development of the business.

marketability the ease with which a security can be sold and converted into cash.

market penetration strategy an approach to pricing in which the price is set lower than competing products.

market risk a risk of loss due to movement in the price of assets.

mark-up amount added to a cost base to find the target selling price.

materials requirement planning (MRP) a computerised system to order inventory in conjunction with production schedules, intended to determine the exact level of raw materials and work-in-process that must be on hand to meet finished goods demand.

maturity matching strategy financing strategy in which all seasonal working capital needs are funded with short-term borrowing and long-term financing is used to fund permanent working capital and fixed assets.

memberships privilege offered to allow members to access a good or service at special prices or included in the membership.

method of multiples valuation method of comparing the price of an asset to the market value of similar assets.

minimal a profile describes a company willing to take only safe steps to minimise likely risk exposure.

mixed cost contains both fixed and variable components.

Modigliani-Miller (M&M) propositions two propositions developed by Franco Modigliani and Merton Miller that provide insight into how financing decisions affect the value of a firm.

money a means of exchanging value between parties.

money market financial markets with a focus on short-term funds.

multiple common variables used in relative valuation.

multistage growth dividend model a valuation model used to calculate the value of stocks that identifies different phases of growth and projects dividends per share for each period before discounting them to a valuation date.

naked position a financial position, such as an investment in a security or foreign currency, that does not have a hedging associated with it.

net operating profits after tax (NOPAT) a measure of profit that excludes the cost and tax benefits of debt financing.

net present value (NPV) the primary discounted cash flow technique used to appraise investments, it is the difference between the present value of cash inflows and cash outflows over time.

net profit margin the percentage of sales that remains after all expenses, including interest and taxes, have been paid.

net working capital (NWC) the difference between current assets and current liabilities.

nominal yield interest paid divided by the face value of the bond.

normal distribution a symmetric frequency distribution that is completely described by its mean (average) and standard deviation.

open descriptive of a company able and ready to consider all avenues and willing to accept levels of risk if those risks are matched to potential profits.

operating cycle the average period of time a firm takes from purchasing and selling inventory to collecting cash from its customers.

operating leverage a measure of the relative amounts of fixed and variable costs in a project's cost structure.

operating profit generally measured as earnings before interest and taxes.

operating risk the risk of loss that results from internal processes that are not adequate or fail, from people and systems or from external events.

opportunity costs the costs that a company faces by taking up one opportunity and missing out the benefits from an alternative opportunity.

option payoff function the relation between the value of an option and the price (value) of the underlying asset.

options forms of contractual agreements to hedge known risks.

ordinary shares also known as common stock, shares that represent a basic equity ownership claim in a corporation.

outsourcing a form of transfer accomplished by contract, often between a producer and a supplier or vendor.

overtrading when a company buys or sells stocks excessively, to the point when the trades become counter-productive to the financial strategy and investment objectives of the company.

par value also known as face value, the principal amount owed to the bondholder at maturity.

par value bonds bonds that sell at par value.

payable date the date when investors receive dividends, generally a couple of weeks after the record date.

payback period the number of years required to fully recover the initial outlay associated with a capital expenditure.

pecking order theory a capital structure theory that says that different types of capital come at different costs. These differences lead to a pecking order in financing choices. Managers typically choose the least expensive capital first and move on to more expensive choices in order after the least-expensive option is no longer available.

pension funds provide retirement programmes for businesses as part of their employee benefits programmes.

percentage of sales method the most basic forecasting technique that involves:

- (1) Determining which financial policy variables you are interested in.
- (2) Setting all the non-financial policy variables as a percentage of sales.
- (3) Using percentage of sales to extrapolate the statement of financial position.
- (4) Estimating of future retained earnings.
- (5) Re-examining and refining the forecast until it makes sense.

perfect market a theoretical market with perfect competition and equilibrium between supply and demand at all times.

perpetuity a perpetual annuity. An asset that generates returns for an indefinite period of time.

planning process incorporating predictions as well as examination of possible outcomes ranging from a worst-case to a best-case scenario.

portfolio a collection of securities, such as stocks and bonds.

post-completion appraisal see **post-completion audit**.

post-completion audit sometimes called post-completion appraisal, it refers to the process of auditing capital investment projects to determine their success or failure.

preference shares also known as preferred stock, represent an ownership interest in the corporation that receives preferential treatment over common stock, including in areas such as

payment of dividends and the distribution of assets in the event of liquidation.

preference stock see **preference shares**.

preferred stock see **preference shares**.

preliminary impact assessment (PIA) the audit of an in-process project to determine whether goals and objectives are being achieved.

premium bonds bonds that sell at prices above par.

present value the current value of a single future amount.

price discrimination a pricing strategy based on charging different customers different prices for the same product or service.

price-earnings (P/E) ratio see **price-to-earnings (P/E) ratio**.

price takers those entities that cannot set prices on their own, because prices are set by market forces.

price-to-book (P/B) ratio a comparison of the market value of a company's investments to its book value—sometimes called market-to-book ratio.

price-to-earnings (P/E) ratio a ratio that represents the number of times investors are willing to pay for a company's earnings, as expressed in the share price, or the share price divided by the earnings per share.

primary market the part of the capital market that deals with issuing new securities.

principal an initial amount of money borrowed in a loan or put into an investment.

processing float delays in processing incoming payments from customers by the receiving firm.

productive assets assets of a company that generate revenues and profits.

product life cycle costing a production costing technique that considers all life cycle costs.

product line pricing strategy an approach to pricing to maximise sales by offering complementary—as opposed to competitive—products.

profitability index (PI) a measure of the value a project generates for each dollar invested in that project, which can be used to evaluate capital expenditure decisions in the DCF approach.

profitability ratios measures of how efficiently management uses assets to generate sales and manage operations.

profit centre a responsibility centre that incurs costs and expenses and generates revenues.

pro forma a method by which financial results are calculated based on certain assumptions and projections.

pro forma financial statements financial statements based on certain assumptions and projections.

psychological pricing the practice of setting prices slightly lower than rounded numbers.

purchase fund provisions provisions that require the repurchase of a certain amount of debt only if it can be repurchased at or below a given price.

put options option contracts that give owners the right, but not the obligation, to sell a certain amount of an underlying asset at a specific price and a specific time frame.

quality costing a method that allows companies to determine how their resources are used to prevent poor quality and appraise quality.

quick ratio an indicator of whether a company is able to pay off all of its current liabilities from its most liquid assets. It is the sum of cash, marketable securities and accounts receivable, divided by current liabilities.

ready market existence of adequate levels of buyers to sellers or vice versa, facilitating exchanges.

realised yield the return on a bond given the cash flows actually received by the investor.

record date the date when an investor must be officially recorded as a shareholder to receive the dividend.

regular interest payment also known as **coupon**, refers to the coupon payment or interest paid to the holder of a bond.

relative valuation a generic term that refers to the notion of comparing the price of an asset to the market value of similar assets.

relevant cost cost associated with a particular management decision and that is likely to change in the future.

required yield the yield investors are looking for.

residual cash flows cash left over after a company pays operating expenses, creditors and taxes.

residual income (RI) a measure of the dollar amount of profits more than a required rate of return, which is commonly referred to as a capital charge.

responsibility centres centres that indicate the degree of responsibility a manager has for performance and accountability.

retained earnings accumulated net income of a company that is retained by the company.

retractable (or putable) bonds bonds that allow the bondholders to sell the bonds back to the issuer at predetermined prices at specified times earlier than the maturity date.

return on equity (ROE) measures net income as a percentage of the stockholders' investment in the company.

return on investment (ROI) the main basis for evaluating the performance of a manager of an investment centre.

revenue centre a segment of the organisation that is primarily responsible for generating revenue.

risk any potential event that could damage company resources or that the actual return on an investment will be different from expected.

risk acceptance assumption of a known risk without attempting to remove it.

risk-adjusted discount rate a rate set according to the overall riskiness of a project.

risk appetite the degree or level of risk the company accepts as either affordable or unavoidable.

risk avoidance removal of exposure to hazards that could cause losses to the company's assets, cash flows or reputation.

risk-free interest rate a reflection of the reward anticipated for the deferred current consumption that is incurred when an investment is made. It is the theoretical rate of return of an investment with no risk.

risk-free rate see **risk-free interest rate**.

risk ignorance unawareness or lack of concern of some forms of risk, meaning those risks cannot be avoided.

risk mitigation identification and enactment of processes and actions to reduce threats.

risk premium the excess return of an asset above the risk-free rate (R_f).

sales risk the risk of sales failures, including loss of future business affecting revenue and profits; or loss of sales force members due to competitive recruitment.

seasoned public offerings sometimes known as secondary offerings, it is a new issue of shares or equity by a company that already has shares that are publicly traded.

secondary market a market where owners of securities can sell them to other investors.

security market line (SML) illustrates what the CAPM predicts the expected total return should be for various values of beta.

segment reporting the concept of reporting the operations of specific segments of a company in the disclosures that go along with financial statements.

short-term funding strategy financing strategy that relies on short-term debt to finance all seasonal working capital and a portion of permanent working capital and fixed assets.

signalling theory a theory that links the future performance of a company's stock with announcements that are made with regard to its debt. It is sometimes also called debt signalling theory.

simple interest interest paid or received on principal.

sinking fund provisions provisions that require the issuer to set money aside each year so that funds are available at maturity to pay off the debt.

skimming an approach to pricing in which the initial price of a product is set higher than competing products.

sovereign risk the risk that a government will be unable (or unwilling) to make loan repayments in a timely manner, or which defaults on its loan guarantees.

speculation the practice of trading a financial instrument involving high risk in the expectation of generating returns. The goal is to take advantage from fluctuations in the market.

spontaneous liabilities accruals and payables generated by companies.

stakeholder a party that has interest or concern in a company, such as employees, suppliers, creditors and the government.

stand-alone principle it is possible to treat the project as if it were a stand-alone company that has its own revenue, expenses and investment requirements.

stock dividends dividends paid in stock.

stock split a division of each share into multiple shares.

strategic management setting objectives, analysing the competitive environment, internal organisation and evaluating strategies for management.

strategic management accounting (SMA) a specific field of accounting that brings together the strategic business objectives of a business with the information generated by management accounting.

sunk costs costs that have already been incurred and will not be changed or avoided by future decisions.

- supply and demand** the total amount of a product available for purchase at any specified price, and consumers' desire and willingness to pay a price for a specific good or service.
- sustainability** refers to the internal needs of a company to operate over a long period of time, the sustainability over time of a project or the impact of a project on social initiatives, environmental concerns and corporate governance.
- sustainable growth rate (SGR)** the maximum rate of growth that a company can sustain without changing its capital structure.
- systematic risk** risk based on external factors to a specific investment that cannot be diversified away.
- target cost** an approach to determine a product's life cycle cost enough to develop functionality and quality while ensuring a desired profit level.
- target selling price** the cost base plus mark-up.
- taxation** the amount of tax a company has to pay.
- team evaluation** an assessment carried out by multiple people with a stake or view of a project, process or company.
- term loan** a loan with a fixed maturity and a requirement for repayment to be made on a fixed schedule.
- theory of constraints (TOC)** a methodology for identifying and dealing with the main constraints holding back a company's production of a good or service.
- third-party guarantee** an agreement by a co-signer to ensure repayment of a debt to reduce exposure to credit risk.
- throughput analysis** an approach that requires looking at investment decisions in terms of their impact on the entire company and not just on the specific area of investment.
- tightening** when a lender reduces exposure to credit risk by limiting or reducing maximum credit, either to a pool of similar borrowers or to a single borrower.
- times-interest-earned ratio** ratio of earnings before interest and tax (EBIT) to interest.
- time value of money** the idea that the value of money available today is worth more than the value of the same amount of money to be received in the future due to the potential earning capacity of the money in hand.
- total cost-plus pricing** the practice of determining a price based on costs and then adding a mark-up.
- total debt ratio** a measure of how much of the finances of the company derive from sources other than the stockholders. It is calculated by dividing total liabilities by total assets.
- total quality management (TQM)** systems implemented to reduce defects in finished products with the goal of achieving zero defects.
- trade-off theory** a theory of capital structure. It states managers choose capital structures based on trade-offs between benefits and costs of debt.
- transfer of risk** assigning the risk to another group or company.
- transmission float** the delay in transferring the means of payment from the payer or customer to the payee or provider of goods or services.
- treasury bills** (T-bills) short-term bonds issued by governments.
- treasury bond** (T-bond) a debt security typically with a maturity of more than ten years issued by a government.
- treasury note** a debt instrument or security typically with a maturity of two to ten years issued by a government.
- trend analysis** analysis of trends in financial data over time.
- trust company** entities that make sure that covenant provisions within an indenture are observed by the issuer.
- unsystematic risk** risk that can be diversified away and is related to a specific investment, not the market environment.
- upside risk** the possibility that returns will exceed expectations.
- variable costs** costs that vary in relation to changes in the volume of an activity.
- variance (σ^2)** a measure of possible performance that squares the difference between each possible occurrence and the mean while multiplying each difference by its corresponding probability before summing everything.
- versioning** an approach to pricing that relies on businesses using different models of the same product and charging different prices for each model.
- volatility** the degree of variation in a particular value such as income or the price of a security.
- volume discounting** an approach to pricing in which a customer purchases a lot of a particular product to get a lower price per unit.
- voluntary pricing** an approach to pricing that gives stakeholders the choice on what to pay for a particular good or service.

warrant a security that provides the investor with the option to convert or exchange the warrant into another security, such as a common stock.

weighted average cost of capital (WACC) a calculation of a firm's cost of capital in which each category of capital is proportionally weighted according to their relative importance as sources of finance.

working capital capital a business uses in daily operations. It is defined in terms of current assets.

yield spread the difference between the yield on CP and on T-bills.

yield to maturity (YTM) the yield that the investor would realise if he or she bought a bond at the current price, held it to maturity, received all the promised payments on their scheduled dates and reinvested all the cash flows received at the YTM.

zero coupon bond a debt security that does not pay interest during its life.

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